impossible because such a process could not conserve both energy and momentum. It's easiest to analyze this annihilation process in the frame of reference called the *center-of-momentum system,* in which the total momentum is zero. It is the relativistic generalization of the center-of-mass system that we discussed in Section 8.5.

EXAMPLE 38.6 Pair annihilation

WITH VARIATION PROBLEMS

An electron and a positron, initially far apart, move toward each other with the same speed. They collide head-on, annihilating each other and producing two photons. Find the energies, wavelengths, and frequencies of the photons if the initial kinetic energies of the electron and positron are (a) both negligible and (b) both 5.000 MeV. The rest energy of an electron or a positron is 0.511 MeV.

IDENTIFY and SET UP Just as in the elastic collisions we studied in Chapter 8, both momentum and energy are conserved in pair annihilation. The electron and positron are initially far apart, so the initial electric potential energy is zero and the initial energy is the sum of the particle kinetic and rest energies. The final energy is the sum of the photon energies. The total initial momentum is zero; the total momentum of the two photons must likewise be zero. We find the photon energy E by using conservation of energy, conservation of momentum, and the relationship $E = pc$ (see Section 38.1). We then calculate the wavelengths and frequencies from $E = hc/\lambda = hf$.

EXECUTE If the total momentum of the two photons is to be zero, their momenta must have equal magnitudes p and opposite directions. From $E = pc = hc/\lambda = hf$, the two photons must also have the same energy E, wavelength λ, and frequency f.

Before the collision the electron and the positron each have energy $K + mc^2$, where K is its kinetic energy and $mc^2 = 0.511$ MeV. Conservation of energy then gives

$$(K + mc^2) + (K + mc^2) = E + E$$

Hence the energy of each photon is $E = K + mc^2$.

(a) In this case the electron or positron kinetic energy K is negligible compared to its rest energy mc^2, so each photon has energy $E = mc^2 = 0.511$ MeV. The corresponding photon wavelength and frequency are

$$\lambda = \frac{hc}{E} = \frac{(4.136 \times 10^{-15}\ \text{eV}\cdot\text{s})(3.00 \times 10^{8}\ \text{m/s})}{0.511 \times 10^{6}\ \text{eV}}$$
$$= 2.43 \times 10^{-12}\ \text{m} = 2.43\ \text{pm}$$

$$f = \frac{E}{h} = \frac{0.511 \times 10^{6}\ \text{eV}}{4.136 \times 10^{-15}\ \text{eV}\cdot\text{s}} = 1.24 \times 10^{20}\ \text{Hz}$$

(b) In this case $K = 5.000$ MeV, so each photon has energy $E = 5.000\ \text{MeV} + 0.511\ \text{MeV} = 5.511$ MeV. Proceeding as in part (a), you can show that the photon wavelength is 0.225 pm and the frequency is 1.33×10^{21} Hz.

EVALUATE As a check, recall from Example 38.1 that a 650 nm visible-light photon has energy 1.91 eV and frequency 4.62×10^{14} Hz. The photon energy in part (a) is about 2.5×10^5 times greater. As expected, the photon's wavelength is shorter and its frequency higher than those for a visible-light photon by the same factor. You can check the results for part (b) in the same way.

KEYCONCEPT In *pair annihilation,* an electron and a positron collide and disappear, and they are replaced by two or more photons. Energy and momentum are conserved in pair annihilation.

TEST YOUR UNDERSTANDING OF SECTION 38.3 If you used visible-light photons in the experiment shown in Fig. 38.11, would the photons undergo a wavelength shift due to the scattering? If so, is it possible to detect the shift with the human eye?

ANSWER

yes, no Equation (38.7) shows that the wavelength shift $\Delta\lambda = \lambda' - \lambda$ depends only on the photon scattering angle ϕ, not on the wavelength of the incident photon. So a visible-light photon scattered through an angle ϕ undergoes the same wavelength shift as an x-ray photon. Equation (38.7) also shows that this shift is of the order of $h/mc = 2.426 \times 10^{-12}\ \text{m} = 0.002426$ nm. This is a few percent of the wavelength of x rays (see Example 38.5), so the effect is noticeable in x-ray scattering. However, h/mc is a tiny fraction of the wavelength of visible light (between 380 and 750 nm). The human eye cannot distinguish such minuscule differences in wavelength (that is, differences in color).

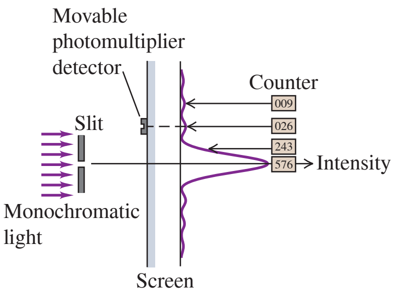
38.4 WAVE–PARTICLE DUALITY, PROBABILITY, AND UNCERTAINTY

We have studied many examples of the behavior of light and other electromagnetic radiation. Some, including the interference and diffraction effects described in Chapters 35 and 36, demonstrate conclusively the *wave* nature of light. Others, the subject of the present chapter, point with equal force to the *particle* nature of light. This *wave–particle duality* means that light has two aspects that seem to be in direct conflict. How can light be a wave and a particle at the same time?

We can find the answer to this apparent wave–particle conflict in the **principle of complementarity,** first stated by the Danish physicist Niels Bohr in 1928. The wave descriptions and the particle descriptions are complementary. That is, we need both to complete our model of nature, but we'll never need to use both at the same time to describe a single part of an occurrence.

Diffraction and Interference in the Photon Picture

Figure 38.15 Single-slit diffraction pattern of light observed with a movable photomultiplier. The curve shows the intensity distribution predicted by the wave picture. The photon distribution is shown by the numbers of photons counted at various positions.

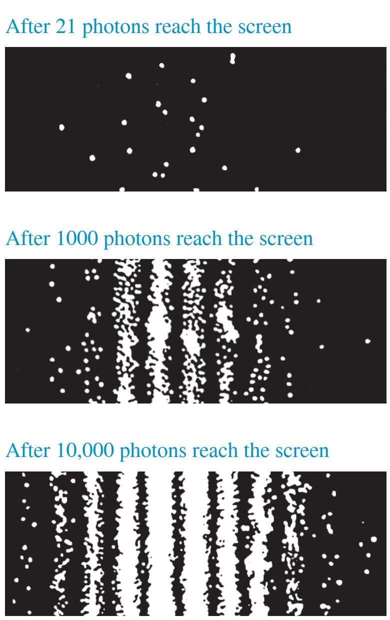


Let's start by considering again the diffraction pattern for a single slit, which we analyzed in Sections 36.2 and 36.3. Instead of recording the pattern on a digital camera chip or photographic film, we use a detector called a *photomultiplier* that can actually detect individual photons. Using the setup shown in **Fig. 38.15**, we place the photomultiplier at various positions for equal time intervals, count the photons at each position, and plot out the intensity distribution.

We find that, on average, the distribution of photons agrees with our predictions from Section 36.3. At points corresponding to the maxima of the pattern, we count many photons; at minimum points, we count almost none; and so on. The graph of the counts at various points gives the same diffraction pattern that we predicted with Eq. (36.7).

But suppose we now reduce the intensity to such a low level that only a few photons per second pass through the slit. We now record a series of discrete strikes, each representing a single photon. While we *cannot predict* where any given photon will strike, over time the accumulating strikes build up the familiar diffraction pattern we expect for a wave. To reconcile the wave and particle aspects of this pattern, we have to regard the pattern as a *statistical* distribution that tells us how many photons, on average, go to each spot. Equivalently, the pattern tells us the *probability* that any individual photon will land at a given spot. If we shine our faint light beam on a two-slit apparatus, we get an analogous result (**Fig. 38.16**). Again we can't predict exactly where an individual photon will go; the interference pattern is a statistical distribution.

Figure 38.16 These images record the positions where individual photons in a two-slit interference experiment strike the screen. As more photons reach the screen, a recognizable interference pattern appears.



How does the principle of complementarity apply to these diffraction and interference experiments? The wave description, not the particle description, explains the single- and double-slit patterns. But the particle description, not the wave description, explains why the photomultiplier records discrete packages of energy. The two descriptions complete our understanding of the results. For instance, suppose we consider an individual photon and ask how it knows "which way to go" when passing through the slit. This question seems like a conundrum, but that is because it is framed in terms of a *particle* description—whereas it is the *wave* nature of light that determines the distribution of photons. Conversely, the fact that the photomultiplier detects faint light as a sequence of individual "spots" can't be explained in wave terms.

Probability and Uncertainty

Although photons have energy and momentum, they are nonetheless very different from the particle model we used for Newtonian mechanics in Chapters 4 through 8. The Newtonian particle model treats an object as a point mass. We can describe the location and state of motion of such a particle at any instant with three spatial coordinates and three components of momentum, and we can then predict the particle's future motion. This model doesn't work at all for photons, however: We *cannot* treat a photon as a point object. This is because there are fundamental limitations on the precision with which we can simultaneously determine the position and momentum of a photon. Many aspects of a photon's behavior can be stated only in terms of *probabilities*. (In Chapter 39 we'll find that the non-Newtonian ideas we develop for photons in this section also apply to particles such as electrons.)

To get more insight into the problem of measuring a photon's position and momentum simultaneously, let's look again at the single-slit diffraction of light. Suppose the wavelength λ is much less than the slit width a (**Fig. 38.17**). Then most (85%) of the photons go into the central maximum of the diffraction pattern, and the remainder go into other parts of the pattern. We use θ_1 to denote the angle between the central maximum and the first minimum. Using Eq. (36.2) with $m = 1$, we find that θ_1 is given by $\sin\theta_1 = \lambda/a$.

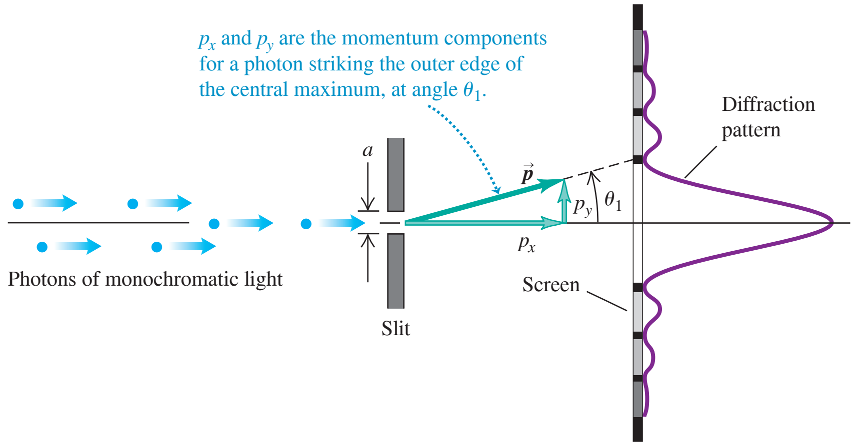


Figure 38.17 Interpreting single-slit diffraction in terms of photon momentum.

Since we assume $\lambda \ll a$, it follows that θ_1 is very small, $\sin\theta_1$ is very nearly equal to θ_1 (in radians), and

$$\theta_1 = \frac{\lambda}{a} \tag{38.12}$$

Even though the photons all have the same initial state of motion, they don't all follow the same path. We can't predict the exact trajectory of any individual photon from knowledge of its initial state; we can only describe the *probability* that an individual photon will strike a given spot on the screen. This fundamental indeterminacy has no counterpart in Newtonian mechanics.

Furthermore, there are fundamental *uncertainties* in both the position and the momentum of an individual particle, and these uncertainties are related inseparably. To clarify this point, let's go back to Fig. 38.17. A photon that strikes the screen at the outer edge of the central maximum, at angle θ_1, must have a component of momentum p_y in the y-direction, as well as a component p_x in the x-direction, despite the fact that initially the beam was directed along the x-axis. From the geometry of the situation the two components are related by $p_y/p_x = \tan\theta_1$. Since θ_1 is small, we may use the approximation $\tan\theta_1 = \theta_1$, and

$$p_y = p_x\theta_1 \tag{38.13}$$

Substituting Eq. (38.12), $\theta_1 = \lambda/a$, into Eq. (38.13) gives

$$p_y = p_x\frac{\lambda}{a} \tag{38.14}$$

Equation (38.14) says that for the 85% of the photons that strike the detector within the central maximum (that is, at angles between $-\lambda/a$ and $+\lambda/a$), the y-component of momentum is spread out over a range from $-p_x\lambda/a$ to $+p_x\lambda/a$. Now let's consider *all* the photons that pass through the slit and strike the screen. Again, they may hit above or below the center of the pattern, so their component p_y may be positive or negative. However the symmetry of the diffraction pattern shows us the average value $(p_y)_{av} = 0$. There will be an *uncertainty* Δp_y in the y-component of momentum at least as great as $p_x\lambda/a$. That is,

$$\Delta p_y \geq p_x\frac{\lambda}{a} \tag{38.15}$$

The narrower the slit width a, the broader is the diffraction pattern and the greater is the uncertainty in the y-component of momentum p_y.

The photon wavelength λ is related to the momentum p_x by Eq. (38.5), which we can rewrite as $\lambda = h/p_x$. Using this relationship in Eq. (38.15) and simplifying, we find

$$\Delta p_y \geq p_x\frac{h}{p_x a} = \frac{h}{a}$$

$$\Delta p_y a \geq h \tag{38.16}$$

What does Eq. (38.16) mean? The slit width a represents an uncertainty in the y-component of the *position* of a photon as it passes through the slit. We don't know exactly *where* in the slit each photon passes through. So both the y-position and the y-component of momentum have uncertainties, and the two uncertainties are related by Eq. (38.16). We can reduce the *momentum* uncertainty Δp_y only by reducing the width of the diffraction pattern. To do this, we have to increase the slit width a, which increases the *position* uncertainty. Conversely, when we *decrease* the position uncertainty by narrowing the slit, the diffraction pattern broadens and the corresponding momentum uncertainty *increases.*

You may protest that it doesn't seem to be consistent with common sense for a photon not to have a definite position and momentum. But what we call *common sense* is based on familiarity gained through experience. Our usual experience includes very little contact with the microscopic behavior of particles like photons. Sometimes we have to accept conclusions that violate our intuition when we are dealing with areas that are far removed from everyday experience.

The Uncertainty Principle

In more general discussions of uncertainty relationships, the uncertainty of a quantity is usually described in terms of the statistical concept of *standard deviation,* which is a measure of the spread or dispersion of a set of numbers around their average value. Suppose we now begin to describe uncertainties in this way [neither Δp_y nor a in Eq. (38.16) is a standard deviation]. If a coordinate x has an uncertainty Δx and if the corresponding momentum component p_x has an uncertainty Δp_x, then we find that in general

CAUTION ***h* versus *h*-bar** It's common for students to plug in the value of h when what they wanted was $\hbar = h/2\pi$, or vice versa. Don't make the same mistake, or your answer will be off by a factor of 2π!

Heisenberg uncertainty principle for position and momentum:

Uncertainty in coordinate x

Planck's constant divided by 2π

$$\Delta x \Delta p_x \geq \hbar/2 \qquad (38.17)$$

Uncertainty in corresponding momentum component p_x

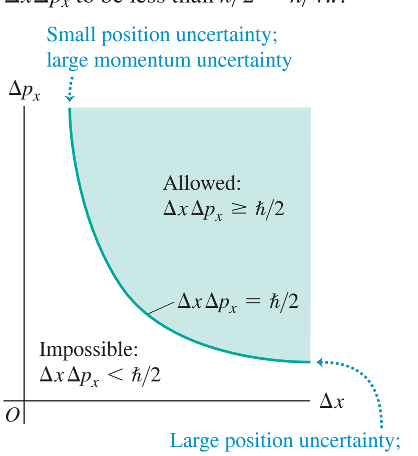
The quantity $\hbar$ (pronounced "h-bar") is Planck's constant divided by 2π. To four significant figures,

$$\hbar = \frac{h}{2\pi} = 1.055 \times 10^{-34}\ \text{J} \cdot \text{s} = 6.582 \times 10^{-16}\ \text{eV} \cdot \text{s}$$

We'll use $\hbar$ frequently to avoid writing a lot of factors of 2π in later equations.

Equation (38.17) is one form of the **Heisenberg uncertainty principle,** first discovered by the German physicist Werner Heisenberg (1901–1976). It states that, in general, it is impossible to simultaneously determine both the position and the momentum of a particle with arbitrarily great precision, as classical physics would predict. Instead, the uncertainties in the two quantities play complementary roles, as we have described. **Figure 38.18** shows the relationship between the two uncertainties. Our derivation of Eq. (38.16), a less refined form of the uncertainty principle given by Eq. (38.17), shows that this principle has its roots in the wave aspect of photons. We'll see in Chapter 39 that electrons and other subatomic particles also have a wave aspect, and the same uncertainty principle applies to them as well.

Figure 38.18 The Heisenberg uncertainty principle for position and momentum components. It is impossible for the product $\Delta x \Delta p_x$ to be less than $\hbar/2 = h/4\pi$.



It is tempting to suppose that we could get greater precision by using more sophisticated detectors of position and momentum. This turns out not to be possible. To detect a particle, the detector must *interact* with it, and this interaction unavoidably changes the state of motion of the particle, introducing uncertainty about its original state. For example, we could imagine placing an electron at a certain point in the middle of the slit in Fig. 38.17. If the photon passes through the middle, we would see the electron recoil. We would then know that the photon passed through that point in the slit, and we would be much more certain about the x-coordinate of the photon. However, the collision between the photon and the electron would change the photon momentum, giving us greater uncertainty in the value of that momentum. A more detailed analysis of such hypothetical experiments shows that the uncertainties we have described are fundamental and intrinsic. They *cannot* be circumvented *even in principle* by any experimental technique, no matter how sophisticated.

There is nothing special about the x-axis. In a three-dimensional situation with coordinates (x, y, z) there is an uncertainty relationship for each coordinate and its corresponding momentum component: $\Delta x \Delta p_x \geq \hbar/2$, $\Delta y \Delta p_y \geq \hbar/2$, and $\Delta z \Delta p_z \geq \hbar/2$. However, the uncertainty in one coordinate is *not* related to the uncertainty in a different component of momentum. For example, Δx is not related directly to Δp_y.

Waves and Uncertainty

Here's an alternative way to understand the Heisenberg uncertainty principle in terms of the properties of waves. Consider a sinusoidal electromagnetic wave propagating in the positive x-direction with its electric field polarized in the y-direction. If the wave has wavelength λ, frequency f, and amplitude A, we can write the wave function as

$$E_y(x, t) = A \sin(kx - \omega t) \tag{38.18}$$

In this expression the wave number is $k = 2\pi/\lambda$ and the angular frequency is $\omega = 2\pi f$. We can think of the wave function in Eq. (38.18) as a description of a photon with a definite wavelength and a definite frequency. In terms of k and ω we can express the momentum and energy of the photon as

$$p_x = \frac{h}{\lambda} = \frac{h}{2\pi}\frac{2\pi}{\lambda} = \hbar k \quad \text{(photon momentum in terms of wave number)} \tag{38.19a}$$

$$E = hf = \frac{h}{2\pi}2\pi f = \hbar\omega \quad \text{(photon energy in terms of angular frequency)} \tag{38.19b}$$

Using Eqs. (38.19) in Eq. (38.18), we can rewrite our photon wave equation as

$$E_y(x, t) = A \sin[(p_x x - Et)/\hbar] \quad \text{(wave function for a photon with } x\text{-momentum } p_x \text{ and energy } E) \tag{38.20}$$

Since this wave function has a definite value of x-momentum p_x, there is *no* uncertainty in the value of this quantity: $\Delta p_x = 0$. The Heisenberg uncertainty principle, Eq. (38.17), says that $\Delta x \Delta p_x \geq \hbar/2$. If Δp_x is zero, then Δx must be infinite. Indeed, the wave described by Eq. (38.20) extends along the entire x-axis and has the same amplitude everywhere. The price we pay for knowing the photon's momentum precisely is that we have no idea *where* the photon is!

In practical situations we always have *some* idea where a photon is. To describe this situation, we need a wave function that is more localized in space. We can create one by superimposing two or more sinusoidal functions. To keep things simple, we'll consider only waves propagating in the positive x-direction. For example, let's add together two sinusoidal wave functions like those in Eqs. (38.18) and (38.20), but with slightly different wavelengths and frequencies and hence slightly different values p_{x1} and p_{x2} of x-momentum and slightly different values E_1 and E_2 of energy. The total wave function is

$$E_y(x, t) = A_1 \sin[(p_{1x}x - E_1 t)/\hbar] + A_2 \sin[(p_{2x}x - E_2 t)/\hbar] \tag{38.21}$$

Consider what this wave function looks like at a particular instant of time, say, $t = 0$. At this instant Eq. (38.21) becomes

$$E_y(x, t = 0) = A_1 \sin(p_{1x}x/\hbar) + A_2 \sin(p_{2x}x/\hbar) \tag{38.22}$$

APPLICATION Butterfly Hunting with Heisenberg Because $\hbar$ has such a small value, the Heisenberg uncertainty principle comes into play only for objects on the scale of atoms or smaller. To visualize what this principle means, imagine that we could make the value of $\hbar$ larger by a factor of 10^{34} so that $\hbar = 1.05\ \text{J}\cdot\text{s}$. If you trap a butterfly in a butterfly net, you know the butterfly's position to within the 0.25 m diameter of the net. Then the uncertainty in the butterfly's position is approximately $\Delta x = 0.25$ m. The minimum uncertainty in its momentum is then $\Delta p_x = (\hbar/2\Delta x) = (1.05\ \text{J}\cdot\text{s})/2(0.25\ \text{m}) = 2.1\ \text{kg}\cdot\text{m/s}$, so just by trapping the butterfly you could impart this much momentum to it. A typical butterfly has a mass of about 3×10^{-4} kg. With this much momentum, the butterfly's speed would be about 7000 m/s (about 20 times the speed of sound!) and its kinetic energy about 7000 J (that of a baseball traveling at about 300 m/s, just under the speed of sound). By confining the butterfly in the net, you could give it so much momentum and kinetic energy that it could burst out of the net!

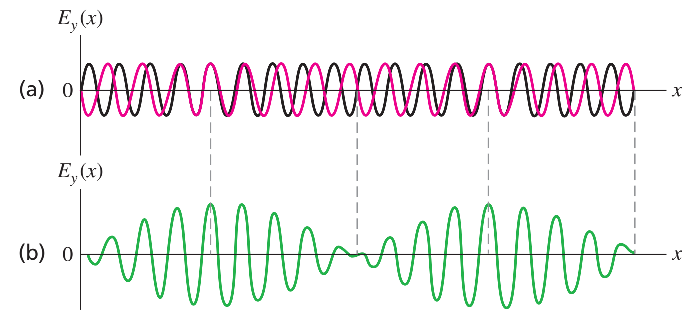


Figure 38.19 (a) Two sinusoidal waves with slightly different wave numbers k and hence slightly different values of momentum $p_x = \hbar k$ shown at one instant of time. (b) The superposition of these waves has a momentum equal to the average of the two individual values of momentum. The amplitude varies, giving the total wave a lumpy character not possessed by either individual wave.

Figure 38.19a is a graph of the individual wave functions at $t = 0$ for the case $A_2 = -A_1$, and Fig. 38.19b graphs the combined wave function $E_y(x, t = 0)$ given by Eq. (38.22). We saw something very similar to Fig. 38.19b in our discussion of beats in Section 16.7: When we superimposed two sinusoidal waves with slightly different frequencies (see Fig. 16.25), the resulting wave exhibited amplitude variations not present in the original waves. In the same way, a photon represented by the wave function in Eq. (38.21) is most likely to be found in the regions where the wave function's amplitude is greatest. That is, the photon is *localized*. However, the photon's momentum no longer has a definite value because we began with two different x-momentum values, p_{x1} and p_{x2}. This agrees with the Heisenberg uncertainty principle: By decreasing the uncertainty in the photon's position, we have increased the uncertainty in its momentum.

Uncertainty in Energy

Our discussion of combining waves also shows that there is an uncertainty principle that involves *energy* and *time*. To see why this is so, imagine measuring the combined wave function described by Eq. (38.21) at a certain position, say $x = 0$, over a period of time. At $x = 0$, the wave function from Eq. (38.21) becomes

$$\begin{aligned} E_y(x, t) &= A_1 \sin(-E_1 t/\hbar) + A_2 \sin(-E_2 t/\hbar) \\ &= -A_1 \sin(E_1 t/\hbar) - A_2 \sin(E_2 t/\hbar) \end{aligned} \tag{38.23}$$

What we measure at $x = 0$ is a combination of two oscillating electric fields with slightly different angular frequencies $\omega_1 = E_1/\hbar$ and $\omega_2 = E_2/\hbar$. This is exactly the phenomenon of beats that we discussed in Section 16.7 (compare Fig. 16.25). The amplitude of the combined field rises and falls, so the photon described by this field is localized in *time* as well as in position. The photon is most likely to be found at the times when the amplitude is large. The price we pay for localizing the photon in time is that the wave does not have a definite energy. By contrast, if the photon is described by a sinusoidal wave like that in Eq. (38.20) that *does* have a definite energy E but that has the same amplitude at all times, we have no idea when the photon will appear at $x = 0$. So the better we know the photon's energy, the less certain we are of when we'll observe the photon.

Just as for the momentum–position uncertainty principle, we can write a mathematical expression for the uncertainty principle that relates energy and time. In fact, except for an overall minus sign, Eq. (38.23) is identical to Eq. (38.22) if we replace the x-momentum p_x by energy E and the position x by time t. This tells us that in the momentum–position uncertainty relationship, Eq. (38.17), we can replace the momentum uncertainty Δp_x with the energy uncertainty ΔE and replace the position uncertainty Δx with the time uncertainty Δt. The result is

Heisenberg uncertainty principle for energy and time:

Time uncertainty of a phenomenon

Planck's constant divided by 2π

$$\Delta t \, \Delta E \geq \hbar/2 \tag{38.24}$$

Energy uncertainty of same phenomenon

In practice, any real photon has a limited spatial extent and hence passes any point in a limited amount of time. The following example illustrates how this affects the momentum and energy of the photon.

EXAMPLE 38.7 Ultrashort laser pulses and the uncertainty principle

WITH VARIATION PROBLEMS

Many varieties of lasers emit light in the form of pulses rather than a steady beam. A tellurium–sapphire laser can produce light at a wavelength of 800 nm in ultrashort pulses that last only 4.00×10^{-15} s (4.00 femtoseconds, or 4.00 fs). The energy in a single pulse produced by one such laser is $2.00\ \mu\text{J} = 2.00 \times 10^{-6}$ J, and the pulses propagate in the positive x-direction. Find (a) the frequency of the light; (b) the energy and minimum energy uncertainty of a single photon in the pulse; (c) the minimum frequency uncertainty of the light in the pulse; (d) the spatial length of the pulse, in meters and as a multiple of the wavelength; (e) the momentum and minimum momentum uncertainty of a single photon in the pulse; and (f) the approximate number of photons in the pulse.

IDENTIFY and SET UP It's important to distinguish between the light pulse as a whole (which contains a very large number of photons) and an individual photon within the pulse. The 4.00 fs pulse duration represents the time it takes the pulse to emerge from the laser; it is also the time *uncertainty* for an individual photon within the pulse, since we don't know when during the pulse that photon emerges. Similarly, the position uncertainty of a photon is the spatial length of the pulse, since a given photon could be found anywhere within the pulse. To find our target variables, we'll use the relationships for photon energy and momentum from Section 38.1 and the two Heisenberg uncertainty principles, Eqs. (38.17) and (38.24).

EXECUTE (a) From the relationship $c = \lambda f$, the frequency of 800 nm light is

$$f = \frac{c}{\lambda} = \frac{3.00 \times 10^{8}\ \text{m/s}}{8.00 \times 10^{-7}\ \text{m}} = 3.75 \times 10^{14}\ \text{Hz}$$

(b) From Eq. (38.2) the energy of a single 800 nm photon is

$$\begin{aligned} E &= hf = (6.626 \times 10^{-34}\ \text{J}\cdot\text{s})(3.75 \times 10^{14}\ \text{Hz}) \\ &= 2.48 \times 10^{-19}\ \text{J} \end{aligned}$$

The time uncertainty equals the pulse duration, $\Delta t = 4.00 \times 10^{-15}$ s. From Eq. (38.24) the minimum uncertainty in energy corresponds to the case $\Delta t\, \Delta E = \hbar/2$, so

$$\Delta E = \frac{\hbar}{2\Delta t} = \frac{1.055 \times 10^{-34}\ \text{J}\cdot\text{s}}{2(4.00 \times 10^{-15}\ \text{s})} = 1.32 \times 10^{-20}\ \text{J}$$

This is 5.3% of the photon energy $E = 2.48 \times 10^{-19}$ J, so the energy of a given photon is uncertain by at least 5.3%. The uncertainty could be greater, depending on the shape of the pulse.

(c) From the relationship $f = E/h$, the minimum frequency uncertainty is

$$\Delta f = \frac{\Delta E}{h} = \frac{1.32 \times 10^{-20}\ \text{J}}{6.626 \times 10^{-34}\ \text{J}\cdot\text{s}} = 1.99 \times 10^{13}\ \text{Hz}$$

This is 5.3% of the frequency $f = 3.75 \times 10^{14}$ Hz we found in part (a). Hence these ultrashort pulses do not have a definite frequency; the average frequency of many such pulses will be 3.75×10^{14} Hz, but the frequency of any individual pulse can be anywhere from 5.3% higher to 5.3% lower.

(d) The spatial length Δx of the pulse is the distance that the front of the pulse travels during the time $\Delta t = 4.00 \times 10^{-15}$ s it takes the pulse to emerge from the laser:

$$\begin{aligned} \Delta x &= c\Delta t = (3.00 \times 10^{8}\ \text{m/s})(4.00 \times 10^{-15}\ \text{s}) \\ &= 1.20 \times 10^{-6}\ \text{m} \\ \Delta x &= \frac{1.20 \times 10^{-6}\ \text{m}}{8.00 \times 10^{-7}\ \text{m/wavelength}} = 1.50\ \text{wavelengths} \end{aligned}$$

This justifies the term *ultrashort*. The pulse is less than two wavelengths long!

(e) From Eq. (38.5), the momentum of an average photon in the pulse is

$$p_x = \frac{E}{c} = \frac{2.48 \times 10^{-19}\ \text{J}}{3.00 \times 10^{8}\ \text{m/s}} = 8.28 \times 10^{-28}\ \text{kg}\cdot\text{m/s}$$

The spatial uncertainty is $\Delta x = 1.20 \times 10^{-6}$ m. From Eq. (38.17) the minimum momentum uncertainty corresponds to $\Delta x\, \Delta p_x = \hbar/2$, so

$$\Delta p_x = \frac{\hbar}{2\Delta x} = \frac{1.055 \times 10^{-34}\ \text{J}\cdot\text{s}}{2(1.20 \times 10^{-6}\ \text{m})} = 4.40 \times 10^{-29}\ \text{kg}\cdot\text{m/s}$$

This is 5.3% of the average photon momentum p_x. An individual photon within the pulse can have a momentum that is 5.3% greater or less than the average.

(f) To estimate the number of photons in the pulse, we divide the total pulse energy by the average photon energy:

$$\frac{2.00 \times 10^{-6}\ \text{J/pulse}}{2.48 \times 10^{-19}\ \text{J/photon}} = 8.06 \times 10^{12}\ \text{photons/pulse}$$

The energy of an individual photon is uncertain, so this is the *average* number of photons per pulse.

EVALUATE The percentage uncertainties in energy and momentum are large because this laser pulse is so short. If the pulse were longer, both Δt and Δx would be greater and the corresponding uncertainties in photon energy and photon momentum would be smaller.

Our calculation in part (f) shows an important distinction between photons and other kinds of particles. In principle it is possible to make an exact count of the number of electrons, protons, and neutrons in an object such as this book. If you repeated the count, you would get the same answer as the first time. By contrast, if you counted the number of photons in a laser pulse you would *not* necessarily get the same answer every time! The uncertainty in photon energy means that on each count there could be a different number of photons whose individual energies sum to 2.00×10^{-6} J. That's yet another of the many strange properties of photons.

KEYCONCEPT The Heisenberg uncertainty principle takes two forms. First, the smaller the uncertainty in the *location* of a physical system, the greater the uncertainty in the *momentum* of the system. Second, the smaller the uncertainty in the *time* of a physical phenomenon, the greater the uncertainty in the *energy* of the phenomenon.

TEST YOUR UNDERSTANDING OF SECTION 38.4 Through which of the following angles is a photon of wavelength λ most likely to be deflected after passing through a slit of width a? Assume that λ is much less than a. (i) $\theta = \lambda/a$; (ii) $\theta = 3\lambda/2a$; (iii) $\theta = 2\lambda/a$; (iv) $\theta = 3\lambda/a$; (v) not enough information given to decide.

ANSWER

(ii) There is *zero* probability that a photon will be deflected by one of the angles where the diffraction pattern has zero intensity. These angles are given by $a \sin\theta = m\lambda$ with $m = \pm 1, \pm 2, \pm 3, \ldots$. Since λ is much less than a, we can write these angles as $\theta = m\lambda/a = \pm\lambda/a, \pm 2\lambda/a, \pm 3\lambda/a, \ldots$. These values include answers (i), (iii), and (iv), so it is impossible for a photon to be deflected through any of these angles. The intensity is not zero at $\theta = 3\lambda/2a$ (located between two zeros in the diffraction pattern), so there is some probability that a photon will be deflected through this angle.

CHAPTER 38 SUMMARY

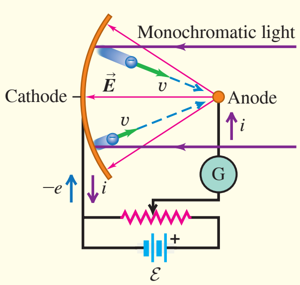
Photons: Electromagnetic radiation behaves as both waves and particles. The energy in an electromagnetic wave is carried in units called photons. The energy E of one photon is proportional to the wave frequency f and inversely proportional to the wavelength λ, and is proportional to a universal quantity h called Planck's constant. The momentum of a photon has magnitude E/c. (See Example 38.1.)

$$E = hf = \frac{hc}{\lambda} \tag{38.2}$$

$$p = \frac{E}{c} = \frac{hf}{c} = \frac{h}{\lambda} \tag{38.5}$$

The photoelectric effect: In the photoelectric effect, a surface can eject an electron by absorbing a photon whose energy hf is greater than or equal to the work function ϕ of the material. The stopping potential V_0 is the voltage required to stop a current of ejected electrons from reaching an anode. (See Examples 38.2 and 38.3.)

$$eV_0 = hf - \phi \tag{38.4}$$



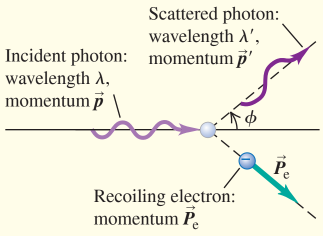
Photon production, photon scattering, and pair production: X rays can be produced when electrons accelerated to high kinetic energy across a potential increase V_{AC} strike a target. The photon model explains why the maximum frequency and minimum wavelength produced are given by Eq. (38.6). (See Example 38.4.) In Compton scattering a photon transfers some of its energy and momentum to an electron with which it collides. For free electrons (mass m), the wavelengths of incident and scattered photons are related to the photon scattering angle ϕ by Eq. (38.7). (See Example 38.5.) In pair production a photon of sufficient energy can disappear and be replaced by an electron–positron pair. In the inverse process, an electron and a positron can annihilate and be replaced by a pair of photons. (See Example 38.6.)

$$eV_{AC} = hf_{max} = \frac{hc}{\lambda_{min}} \tag{38.6}$$

(bremsstrahlung)

$$\lambda' - \lambda = \frac{h}{mc}(1 - \cos\phi) \tag{38.7}$$

(Compton scattering)



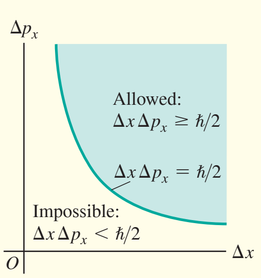
The Heisenberg uncertainty principle: It is impossible to determine both a photon's position and its momentum at the same time to arbitrarily high precision. The precision of such measurements for the x-components is limited by the Heisenberg uncertainty principle, Eq. (38.17); there are corresponding relationships for the y- and z-components. The uncertainty ΔE in the energy of a state that is occupied for a time Δt is given by Eq. (38.24). In these expressions, $\hbar = h/2\pi$. (See Example 38.7.)

$$\Delta x \, \Delta p_x \geq \hbar/2 \tag{38.17}$$

(Heisenberg uncertainty principle for position and momentum)

$$\Delta t \, \Delta E \geq \hbar/2 \tag{38.24}$$

(Heisenberg uncertainty principle for energy and time)



Chapter 38 Media Assets

GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review EXAMPLES 38.1, 38.2, and 38.3 (Section 38.1) before attempting these problems.

VP38.3.1 A laser emits 2.13×10^{16} photons per second, each of which has wavelength 625 nm. Find (a) the energy and momentum of a single photon and (b) the power output of the laser.

VP38.3.2 A certain metal has a work function of 4.55 eV. For this metal, find (a) the minimum frequency that light must have to produce photoelectrons and (b) the frequency that light must have to produce photoelectrons with maximum kinetic energy 1.53 eV.

VP38.3.3 In a photoelectric-effect experiment, the stopping potential is 1.37 V if the light used to illuminate the cathode has wavelength 475 nm. Find (a) the work function (in eV) of the cathode material and (b) the stopping potential if the wavelength is decreased to 425 nm.

VP38.3.4 Your results from a photoelectric-effect experiment show that the maximum speed of the emitted photoelectrons is 6.95×10^5 m/s when the cathode is illuminated with ultraviolet light of wavelength 306 nm. Find (a) the maximum photoelectron kinetic energy in eV, (b) the photon energy in eV, and (c) the work function.

Be sure to review EXAMPLES 38.5 and 38.6 (Section 38.3) before attempting these problems.

VP38.6.1 You use x-ray photons of wavelength 0.251 nm in a Compton-scattering experiment. At a scattering angle of 14.5°, what is the increase in the wavelength of the scattered x rays compared to the incident x rays? Express your results (a) in nm and (b) as a percentage of the wavelength of the incident x rays.

VP38.6.2 You use x rays of *frequency* 1.260×10^{18} Hz in a Compton-scattering experiment. Find the frequency of the scattered x rays if the scattering angle is (a) 90.00° and (b) 180.0°.

VP38.6.3 Compton scattering can also occur if a photon collides with a proton (mass 1.673×10^{-27} kg) at rest. If you do an experiment of this kind using gamma-ray photons with wavelength 2.50×10^{-12} m = 2.50 pm, for what scattering angle is the wavelength of the scattered gamma rays longer than the wavelength of the incident gamma rays by (a) 0.0100% and (b) 0.0800%?

VP38.6.4 An antiproton has the same mass and rest energy (938.3 MeV) as a proton, but has a negative change $-e$ instead of the positive charge $+e$ of the proton. A proton and an antiproton, initially far apart, move toward each other with the same speed and collide head-on, annihilating each other and producing two photons. Find the energies and wavelengths of the photons if the initial kinetic energies of the proton and antiproton are (a) both negligible and (b) both 545 MeV.

Be sure to review EXAMPLE 38.7 (Section 38.4) before attempting these problems.

VP38.7.1 A laser produces light of wavelength 633 nm in pulses that propagate in the $+x$-direction and that are 7.00×10^{-6} m in length. For an average photon in this pulse, find (a) the momentum p, (b) the minimum momentum uncertainty in kg · m/s and as a percentage of p, (c) the energy E, and (d) the minimum energy uncertainty in J and as a percentage of E.

VP38.7.2 Weather radar systems emit radio waves in pulses. For a typical system the frequency used is 3.00 GHz (1 GHz = 10^9 Hz) and the minimum energy uncertainty of the radio photons is 5.50×10^{-29} J. Find (a) the time duration of each pulse, (b) the length of each pulse, (c) the energy of an average photon, and (d) the minimum frequency uncertainty.

VP38.7.3 In a pulsed laser the minimum energy uncertainty of a photon is 2.33×10^{-20} J, which is 6.70% of the average energy of a photon in the pulse. Find (a) the wavelength and frequency of the laser light, (b) the minimum frequency uncertainty, and (c) the time duration of the pulse.

VP38.7.4 The average energy of a photon in a pulsed laser beam is 2.39 eV, with a minimum uncertainty of 0.0155 eV. Each pulse has an average of 5.00×10^{12} photons. Find (a) the time duration of each pulse, (b) the wavelength of the light, and (c) the energy per pulse in J.

BRIDGING PROBLEM Compton Scattering and Electron Recoil

An incident x-ray photon is scattered from a free electron that is initially at rest. The photon is scattered straight back at an angle of 180° from its initial direction. The wavelength of the scattered photon is 0.0830 nm. (a) What is the wavelength of the incident photon? (b) What are the magnitude of the momentum and the speed of the electron after the collision? (c) What is the kinetic energy of the electron after the collision?

SOLUTION GUIDE

IDENTIFY and SET UP

1. In this problem a photon is scattered by an electron initially at rest. In Section 38.3 you learned how to relate the wavelengths of the incident and scattered photons; in this problem you must also find the momentum, speed, and kinetic energy of the recoiling electron. You can find these because momentum and energy are conserved in the collision. Draw a diagram showing the momentum vectors of the photon and electron before and after the scattering.
2. Which key equation can be used to find the incident photon wavelength? What is the photon scattering angle ϕ in this problem?

EXECUTE

3. Use the equation you selected in step 2 to find the wavelength of the incident photon.
4. Use momentum conservation and your result from step 3 to find the momentum of the recoiling electron. (*Hint:* All of the momentum vectors are along the same line, but not all point in the same direction. Be careful with signs.)
5. Find the speed of the recoiling electron from your result in step 4. (*Hint:* Assume that the electron is nonrelativistic, so you can use the relationship between momentum and speed from Chapter 8. This is acceptable if the speed of the electron is less than about 0.1c. Is it?)
6. Use your result from step 4 or step 5 to find the electron kinetic energy.

EVALUATE

7. You can check your answer in step 6 by finding the difference between the energies of the incident and scattered photons. Is your result consistent with conservation of energy?

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q38.1 In what ways do photons resemble other particles such as electrons? In what ways do they differ? Do photons have mass? Do they have electric charge? Can they be accelerated? What mechanical properties do they have?

Q38.2 There is a certain probability that a single electron may simultaneously absorb *two* identical photons from a high-intensity laser. How would such an occurrence affect the threshold frequency and the equations of Section 38.1? Explain.

Q38.3 According to the photon model, light carries its energy in packets called quanta or photons. Why then don't we see a series of flashes when we look at things?

Q38.4 Would you expect effects due to the photon nature of light to be generally more important at the low-frequency end of the electromagnetic spectrum (radio waves) or at the high-frequency end (x rays and gamma rays)? Why?

Q38.5 During the photoelectric effect, light knocks electrons out of metals. So why don't the metals in your home lose their electrons when you turn on the lights?

Q38.6 Most black-and-white photographic film (with the exception of some special-purpose films) is less sensitive to red light than blue light and has almost no sensitivity to infrared. How can these properties be understood on the basis of photons?

Q38.7 Human skin is relatively insensitive to visible light, but ultraviolet radiation can cause severe burns. Does this have anything to do with photon energies? Explain.

Q38.8 Explain why Fig. 38.4 shows that most photoelectrons have kinetic energies less than $hf - \phi$, and also explain how these smaller kinetic energies occur.

Q38.9 In a photoelectric-effect experiment, the photocurrent i for large positive values of V_{AC} has the same value no matter what the light frequency f (provided that f is higher than the threshold frequency f_0). Explain why.

Q38.10 In an experiment involving the photoelectric effect, if the intensity of the incident light (having frequency higher than the threshold frequency) is reduced by a factor of 10 without changing anything else, which (if any) of the following statements about this process will be true? (a) The number of photoelectrons will most likely be reduced by a factor of 10. (b) The maximum kinetic energy of the ejected photoelectrons will most likely be reduced by a factor of 10. (c) The maximum speed of the ejected photoelectrons will most likely be reduced by a factor of 10. (d) The maximum speed of the ejected photoelectrons will most likely be reduced by a factor of $\sqrt{10}$. (e) The time for the first photoelectron to be ejected will be increased by a factor of 10.

Q38.11 The materials called *phosphors* that coat the inside of a fluorescent lamp convert ultraviolet radiation (from the mercury-vapor discharge inside the tube) into visible light. Could one also make a phosphor that converts visible light to ultraviolet? Explain.

Q38.12 In a photoelectric-effect experiment, which of the following will increase the maximum kinetic energy of the photoelectrons? (a) Use light of greater intensity; (b) use light of higher frequency; (c) use light of longer wavelength; (d) use a metal surface with a larger work function. In each case justify your answer.

Q38.13 A photon of frequency f undergoes Compton scattering from an electron at rest and scatters through an angle ϕ. The frequency of the scattered photon is f'. How is f' related to f? Does your answer depend on ϕ? Explain.

Q38.14 Can Compton scattering occur with protons as well as electrons? For example, suppose a beam of x rays is directed at a target of liquid hydrogen. (Recall that the nucleus of hydrogen consists of a single proton.) Compared to Compton scattering with electrons, what similarities and differences would you expect? Explain.

Q38.15 Why must engineers and scientists shield against x-ray production in high-voltage equipment?

Q38.16 In attempting to reconcile the wave and particle models of light, some people have suggested that the photon rides up and down on the crests and troughs of the electromagnetic wave. What things are *wrong* with this description?

Q38.17 Some lasers emit light in pulses that are only 10^{-12} s in duration. The length of such a pulse is $(3 \times 10^8 \text{ m/s})(10^{-12} \text{ s}) = 3 \times 10^{-4} \text{ m} = 0.3$ mm. Can pulsed laser light be as monochromatic as light from a laser that emits a steady, continuous beam? Explain.

EXERCISES

Section 38.1 Light Absorbed as Photons: The Photoelectric Effect

38.1 • A photon of green light has a wavelength of 520 nm. Find the photon's frequency, magnitude of momentum, and energy. Express the energy in both joules and electron volts.

38.2 • **BIO** **Response of the Eye.** The human eye is most sensitive to green light of wavelength 505 nm. Experiments have found that when people are kept in a dark room until their eyes adapt to the darkness, a *single* photon of green light will trigger receptor cells in the rods of the retina. (a) What is the frequency of this photon? (b) How much energy (in joules and electron volts) does it deliver to the receptor cells? (c) To appreciate what a small amount of energy this is, calculate how fast a typical bacterium of mass 9.5×10^{-12} g would move if it had that much energy.

38.3 • A 75 W light source consumes 75 W of electrical power. Assume all this energy goes into emitted light of wavelength 600 nm. (a) Calculate the frequency of the emitted light. (b) How many photons per second does the source emit? (c) Are the answers to parts (a) and (b) the same? Is the frequency of the light the same thing as the number of photons emitted per second? Explain.

38.4 • **BIO** A laser used to weld detached retinas emits light with a wavelength of 652 nm in pulses that are 20.0 ms in duration. The average power during each pulse is 0.600 W. (a) How much energy is in each pulse in joules? In electron volts? (b) What is the energy of one photon in joules? In electron volts? (c) How many photons are in each pulse?

38.5 • A photon has momentum of magnitude 8.24×10^{-28} kg · m/s. (a) What is the energy of this photon? Give your answer in joules and in electron volts. (b) What is the wavelength of this photon? In what region of the electromagnetic spectrum does it lie?

38.6 •• A clean nickel surface is exposed to light of wavelength 235 nm. What is the maximum speed of the photoelectrons emitted from this surface? Use Table 38.1.

38.7 •• When ultraviolet light with a wavelength of 400.0 nm falls on a certain metal surface, the maximum kinetic energy of the emitted photoelectrons is measured to be 1.10 eV. What is the maximum kinetic energy of the photoelectrons when light of wavelength 300.0 nm falls on the same surface?

38.8 •• The photoelectric work function of potassium is 2.3 eV. If light that has a wavelength of 190 nm falls on potassium, find (a) the stopping potential in volts; (b) the kinetic energy, in electron volts, of the most energetic electrons ejected; (c) the speed of these electrons.

38.9 • When ultraviolet light with a wavelength of 254 nm falls on a clean copper surface, the stopping potential necessary to stop emission of photoelectrons is 0.181 V. (a) What is the photoelectric threshold wavelength for this copper surface? (b) What is the work function for this surface, and how does your calculated value compare with that given in Table 38.1?

Section 38.2 Light Emitted as Photons: X-Ray Production

38.10 • The cathode-ray tubes that generated the picture in early color televisions were sources of x rays. If the acceleration voltage in a television tube is 15.0 kV, what are the shortest-wavelength x rays produced by the television?

38.11 • An electron accelerates through a potential difference of 50.0 kV in an x-ray tube. When the electron strikes the target, 70.0% of its kinetic energy is imparted to a single photon. Find the photon's frequency, wavelength, and magnitude of momentum.

38.12 •• (a) What is the minimum potential difference between the filament and the target of an x-ray tube if the tube is to produce x rays with a wavelength of 0.150 nm? (b) What is the shortest wavelength produced in an x-ray tube operated at 30.0 kV?

Section 38.3 Light Scattered as Photons: Compton Scattering and Pair Production

38.13 • An x ray with a wavelength of 0.100 nm collides with an electron that is initially at rest. The x ray's final wavelength is 0.110 nm. What is the final kinetic energy of the electron?

38.14 • X rays are produced in a tube operating at 24.0 kV. After emerging from the tube, x rays with the minimum wavelength produced strike a target and undergo Compton scattering through an angle of 45.0°. (a) What is the original x-ray wavelength? (b) What is the wavelength of the scattered x rays? (c) What is the energy of the scattered x rays (in electron volts)?

38.15 •• X rays with initial wavelength 0.0665 nm undergo Compton scattering. What is the longest wavelength found in the scattered x rays? At which scattering angle is this wavelength observed?

38.16 •• A photon with wavelength $\lambda = 0.1385$ nm scatters from an electron that is initially at rest. What must be the angle between the direction of propagation of the incident and scattered photons if the speed of the electron immediately after the collision is 8.90×10^6 m/s?

38.17 •• If a photon of wavelength 0.04250 nm strikes a free electron and is scattered at an angle of 35.0° from its original direction, find (a) the change in the wavelength of this photon; (b) the wavelength of the scattered light; (c) the change in energy of the photon (is it a loss or a gain?); (d) the energy gained by the electron.

38.18 •• A photon scatters in the backward direction ($\phi = 180°$) from a free proton that is initially at rest. What must the wavelength of the incident photon be if it is to undergo a 10.0% change in wavelength as a result of the scattering?

38.19 • A beam of photons with just barely enough individual photon energy to create electron–positron pairs undergoes Compton scattering from free electrons before any pair production occurs. (a) Can the Compton-scattered photons create electron–positron pairs? (b) Calculate the momentum magnitude of the photons found at a 20.0° angle from the direction of the original beam.

38.20 • An electron and a positron are moving toward each other and each has speed $0.500c$ in the lab frame. (a) What is the kinetic energy of each particle? (b) The e^+ and e^- meet head-on and annihilate. What is the energy of each photon that is produced? (c) What is the wavelength of each photon? How does the wavelength compare to the photon wavelength when the initial kinetic energy of the e^+ and e^- is negligibly small (see Example 38.6)?

Section 38.4 Wave–Particle Duality, Probability, and Uncertainty

38.21 • An ultrashort pulse has a duration of 9.00 fs and produces light at a wavelength of 556 nm. What are the momentum and momentum uncertainty of a single photon in the pulse?

38.22 • A horizontal beam of laser light of wavelength 585 nm passes through a narrow slit that has width 0.0620 mm. The intensity of the light is measured on a vertical screen that is 2.00 m from the slit. (a) What is the minimum uncertainty in the vertical component of the momentum of each photon in the beam after the photon has passed through the slit? (b) Use the result of part (a) to estimate the width of the central diffraction maximum that is observed on the screen.

38.23 • A laser produces light of wavelength 625 nm in an ultrashort pulse. What is the minimum duration of the pulse if the minimum uncertainty in the energy of the photons is 1.0%?

PROBLEMS

38.24 •• (a) If the average frequency emitted by a 120 W light bulb is 5.00×10^{14} Hz and 10.0% of the input power is emitted as visible light, approximately how many visible-light photons are emitted per second? (b) At what distance would this correspond to 1.00×10^{11} visible-light photons per cm^2 per second if the light is emitted uniformly in all directions?

38.25 •• CP BIO **Removing Vascular Lesions.** A pulsed dye laser emits light of wavelength 585 nm in 450 μs pulses. Because this wavelength is strongly absorbed by the hemoglobin in the blood, the method is especially effective for removing various types of blemishes due to blood, such as port-wine–colored birthmarks. To get a reasonable estimate of the power required for such laser surgery, we can model the blood as having the same specific heat and heat of vaporization as water ($4190\ J/kg \cdot K$, $2.256 \times 10^6\ J/kg$). Suppose that each pulse must remove 2.0 μg of blood by evaporating it, starting at 33°C. (a) How much energy must each pulse deliver to the blemish? (b) What must be the power output of this laser? (c) How many photons does each pulse deliver to the blemish?

38.26 • A 2.50 W beam of light of wavelength 124 nm falls on a metal surface. You observe that the maximum kinetic energy of the ejected electrons is 4.16 eV. Assume that each photon in the beam ejects a photoelectron. (a) What is the work function (in electron volts) of this metal? (b) How many photoelectrons are ejected each second from this metal? (c) If the power of the light beam, but not its wavelength, were reduced by half, what would be the answer to part (b)? (d) If the wavelength of the beam, but not its power, were reduced by half, what would be the answer to part (b)?

38.27 •• CP We can estimate the number of photons in a room using the following reasoning: The intensity of the sun's rays at the earth's surface is roughly 1000 W/m^2. A typical room is illuminated by indirect light or by light bulbs so that its light intensity is some fraction of that value. (a) Estimate the intensity of the light that enters your room. (b) Model the light as entering uniformly at the ceiling and exiting uniformly at the floor. Estimate the area A of the floor and ceiling, and the height H of your room. (c) Estimate how long it takes light to travel from the ceiling to the floor of your room by dividing H by the speed of light. (d) Estimate the total power of the light that enters your room P by multiplying your estimated intensity I by the area of your ceiling A. (e) Estimate the total light energy in your room E_{room} by multiplying the power P by the length of time it takes light to travel from ceiling to floor. (f) An average wavelength for light is in the middle of the visible spectrum, at roughly 500 nm. What is the energy of a 500 nm photon? (g) The total number of photons in your room N is the ratio of the energy E_{room} to the energy per photon. What is your estimate for N?

38.28 •• CP A photon with wavelength $\lambda = 0.0980$ nm is incident on an electron that is initially at rest. If the photon scatters in the backward direction, what is the magnitude of the linear momentum of the electron just after the collision with the photon?

38.29 •• CP A photon with wavelength $\lambda = 0.1050$ nm is incident on an electron that is initially at rest. If the photon scatters at an angle of 60.0° from its original direction, what are the magnitude and direction of the linear momentum of the electron just after it collides with the photon?

38.30 •• CP All of the nutritional energy contained in fruits and vegetables is supplied by solar photons through photosynthesis reactions. (a) Estimate the amount of energy stored in a tomato by considering the number of Calories in that fruit. (*Note:* 1 Cal = 1 kcal = 4186 J, and you can look up the Calorie content of a tomato.) (b) Estimate the total leaf area on an average tomato plant. (c) If the solar intensity in your garden is 800 W/m^2, then what is the power supplied by solar photons to those leaves? (d) Fruit leaves have approximately 5% efficiency. The majority, 95%, of photons do not land on chloroplasts or lie outside the frequency range for photosynthesis. Use this information and an average wavelength of 600 nm to determine the number of solar photons involved in photosynthesis reactions each second on a tomato plant. (e) If an average tomato plant has ten tomatoes, and if half of the photon energy enters the fruit while the rest is used to grow roots and maintain plant cells, then how many photons supply energy to each tomato each second? (f) How many photons are needed to supply the energy content of each tomato? (g) Using these results and assuming the sun shines on the tomatoes 12 hours per day, estimate how long it should take to grow a tomato.

38.31 •• Nuclear fusion reactions at the center of the sun produce gamma-ray photons with energies of about 1 MeV (10^6 eV). By contrast, what we see emanating from the sun's surface are visible-light photons with wavelengths of about 500 nm. A simple model that explains this difference in wavelength is that a photon undergoes Compton scattering many times—in fact, about 10^{26} times, as suggested by models of the solar interior—as it travels from the center of the sun to its surface. (a) Estimate the increase in wavelength of a photon in an average Compton-scattering event. (b) Find the angle in degrees through which the photon is scattered in the scattering event described in part (a). (*Hint:* A useful approximation is $\cos\phi \approx 1 - \phi^2/2$, which is valid for $\phi \ll 1$. Note that ϕ is in radians in this expression.) (c) It is estimated that a photon takes about 10^6 years to travel from the core to the surface of the sun. Find the average distance that light can travel within the interior of the sun without being scattered. (This distance is roughly equivalent to how far you could see if you were inside the sun and could survive the extreme temperatures there. As your answer shows, the interior of the sun is *very* opaque.)

38.32 ••• CP A positron with speed v impinges horizontally upon an electron at rest. These particles annihilate and produce two photons with the same wavelength λ, each of which travels at angle ϕ with respect to the horizontal. (a) What is the condition for energy conservation in terms of m, γ, c, and h, where m is the electron mass and $\gamma = (1 - v^2/c^2)^{-1/2}$? (b) What is the condition for momentum conservation in terms of the same parameters and the angle ϕ? (c) Solve these equations and determine the energy of each photon E_{photon} and the angle ϕ, the latter as a function of only γ. (d) If the positron had 5.11 MeV of kinetic energy, then what is the energy of each photon, and what is the angle ϕ? (e) What positron speed would result in the photons traveling off perpendicular to each other? (f) If the positron with speed v was traveling in the $+x$-direction, then with what speed u should we perform a Lorentz velocity transformation to bring us to the "center of momentum" frame, where the total momentum is zero?

38.33 •• A photon with wavelength 0.1100 nm collides with a free electron that is initially at rest. After the collision the wavelength is 0.1132 nm. (a) What is the kinetic energy of the electron after the collision? What is its speed? (b) If the electron is suddenly stopped (for example, in a solid target), all of its kinetic energy is used to create a photon. What is the wavelength of this photon?

38.34 •• An x-ray photon is scattered from a free electron (mass m) at rest. The wavelength of the scattered photon is λ', and the final speed of the struck electron is v. (a) What was the initial wavelength λ of the photon? Express your answer in terms of λ', v, and m. (*Hint:* Use the relativistic expression for the electron kinetic energy.) (b) Through what angle ϕ is the photon scattered? Express your answer in terms of λ, λ', and m. (c) Evaluate your results in parts (a) and (b) for a wavelength of 5.10×10^{-3} nm for the scattered photon and a final electron speed of 1.80×10^8 m/s. Give ϕ in degrees.

38.35 •• DATA In developing night-vision equipment, you need to measure the work function for a metal surface, so you perform a photoelectric-effect experiment. You measure the stopping potential V_0 as a function of the wavelength λ of the light that is incident on the surface. You get the results in the table.

λ (nm)	100	120	140	160	180	200
V_0 (V)	7.53	5.59	3.98	2.92	2.06	1.43

In your analysis, you use $c = 2.998 \times 10^8$ m/s and $e = 1.602 \times 10^{-19}$ C, which are values obtained in other experiments. (a) Select a way to plot your results so that the data points fall close to a straight line. Using that plot, find the slope and y-intercept of the best-fit straight line to the data. (b) Use the results of part (a) to calculate Planck's constant h (as a test of your data) and the work function (in eV) of the surface. (c) What is the longest wavelength of light that will produce photoelectrons from this surface? (d) What wavelength of light is required to produce photoelectrons with kinetic energy 10.0 eV?

38.36 •• DATA While analyzing smoke detector designs that rely on the photoelectric effect, you are evaluating surfaces made from each of the materials listed in Table 38.1. One particular application uses ultraviolet light with wavelength 270 nm. (a) For which of the materials in Table 38.1 will this light produce photoelectrons? (b) Which material will result in photoelectrons of the greatest kinetic energy? What will be the maximum speed of the photoelectrons produced as they leave this material's surface? (c) What is the longest wavelength that will produce photoelectrons from a gold surface, if the surface has a work function equal to the value given for gold in Table 38.1? (d) For the wavelength calculated in part (c), what will be the maximum kinetic energy of the photoelectrons produced from a sodium surface that has a work function equal to the value given in Table 38.1 for sodium?

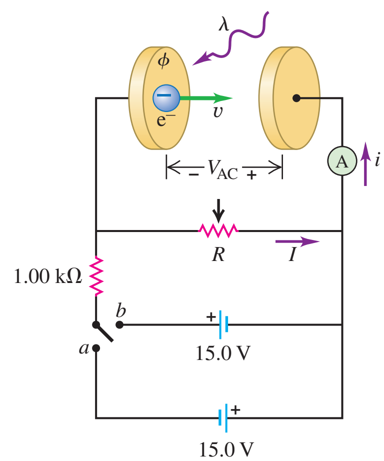
38.37 •• DATA To test the photon concept, you perform a Compton-scattering experiment in a research lab. Using photons of very short wavelength, you measure the wavelength λ' of scattered photons as a function of the scattering angle ϕ, the angle between the direction of a scattered photon and the incident photon. You obtain these results.

φ (deg)	30.6	58.7	90.2	119.2	151.3
λ′ (pm)	5.52	6.40	7.60	8.84	9.69

Your analysis assumes that the target is a free electron at rest. (a) Graph your data as λ' versus $1 - \cos\phi$. What are the slope and y-intercept of the best-fit straight line to your data? (b) The Compton wavelength λ_C is defined as $\lambda_C = h/mc$, where m is the mass of an electron. Use the results of part (a) to calculate λ_C. (c) Use the results of part (a) to calculate the wavelength λ of the incident light.

38.38 ••• **CP** The work function of a material is examined using the apparatus shown schematically in **Fig. P38.38**. The ammeter measures the photocurrent. The voltage of the anode relative to the cathode is V_{AC}. (a) The switch is in position a while radiation with wavelength 140 nm is incident on the cathode. The potentiometer is set at $R = 3.20\ \text{k}\Omega$, and there is no photocurrent. What is the potential difference V_{AC}? (b) The resistance R is slowly adjusted to lower values until the first hint of a photocurrent is detected in the ammeter. That happens when $R = 334\ \Omega$. What is the value of V_{AC} at that point? (c) What is the work function ϕ of the cathode material? (d) The switch is then thrown to position b and the ammeter measures a photocurrent $i = 3.71$ mA. What is the current I in this case? (e) What is the value of V_{AC}? (f) The switch is returned to position a, and the wavelength of the light is reduced to 65.0 nm. The photocurrent increases. What is the minimum value of R for which the photocurrent will cease?

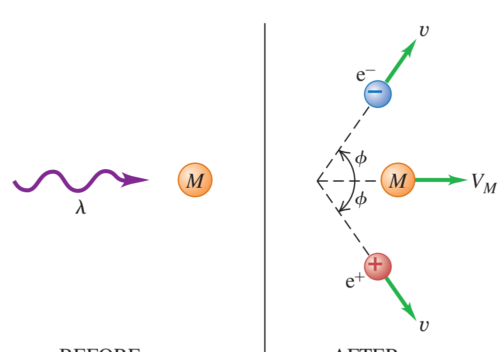
Figure **P38.38**



CHALLENGE PROBLEMS

38.39 ••• **CP** A photon with wavelength λ is incident on a stationary particle with mass M, as shown in **Fig. P38.39**. The photon is annihilated while an electron–positron pair is produced. The target particle moves off in the original direction of the photon with speed V_M. The electron travels with speed v at angle ϕ with respect to that direction. Owing to momentum conservation, the positron has the same speed as the electron. (a) Using the notation $\gamma = (1 - v^2/c^2)^{-1/2}$ and $\gamma_M = (1 - V_M^2/c^2)^{-1/2}$, write a relativistic expression for energy conservation in this process. Use the symbol m for the electron mass. (b) Write an analogous expression for momentum conservation.

Figure **P38.39**



(c) Eliminate the ratio h/λ between your energy and momentum equations to derive a relationship between v/c, V_M/c, and ϕ in terms of the ratio m/M, keeping γ and γ_M as a useful shorthand notation. (d) Consider the case where $V_M \ll c$, and use a binomial expansion to derive an expression for γ_M to the first order in V_M. Use that result to rewrite your previous result as an expression for V_M in terms of c, v, m/M, and ϕ. (e) Are there choices of v and ϕ for which $V_M = 0$? (f) Suppose the target particle is a proton. If the electron and positron remain stationary, so that $v = 0$, then with what speed does the proton move, in km/s? (g) If the electron and positron each have total energy 5.00 MeV and move with $\phi = 60°$, then what is the speed of the proton? (*Hint:* First solve for γ and v.) (h) What is the energy of the incident photon in this case?

38.40 ••• Consider Compton scattering of a photon by a *moving* electron. Before the collision the photon has wavelength λ and is moving in the $+x$-direction, and the electron is moving in the $-x$-direction with total energy E (including its rest energy mc^2). The photon and electron collide head-on. After the collision, both are moving in the $-x$-direction (that is, the photon has been scattered by 180°). (a) Derive an expression for the wavelength λ' of the scattered photon. Show that if $E \gg mc^2$, where m is the rest mass of the electron, your result reduces to

$$\lambda' = \frac{hc}{E}\left(1 + \frac{m^2c^4\lambda}{4hcE}\right)$$

(b) A beam of infrared radiation from a CO_2 laser ($\lambda = 10.6\ \mu\text{m}$) collides head-on with a beam of electrons, each of total energy $E = 10.0$ GeV ($1\ \text{GeV} = 10^9$ eV). Calculate the wavelength λ' of the scattered photons, assuming a 180° scattering angle. (c) What kind of scattered photons are these (infrared, microwave, ultraviolet, etc.)? Can you think of an application of this effect?

MCAT-STYLE PASSAGE PROBLEMS

BIO Radiation Therapy for Tumors. Malignant tumors are commonly treated with targeted x-ray radiation therapy. To generate these medical x rays, a linear accelerator directs a high-energy beam of electrons toward a metal target—typically tungsten. As they near the tungsten nuclei, the electrons are deflected and accelerated, emitting high-energy photons via bremsstrahlung. The resulting x rays are collimated into a beam that is directed at the tumor. The photons can deposit energy in the tumor through Compton and photoelectric interactions. A typical tumor has 10^8 cells/cm^3, and in a full treatment, 4 MeV photons may produce a dose of 70 Gy in 35 fractional exposures on different days. The *gray* (Gy) is a measure of the absorbed energy dose of radiation per unit mass of tissue: 1 Gy = 1 J/kg.

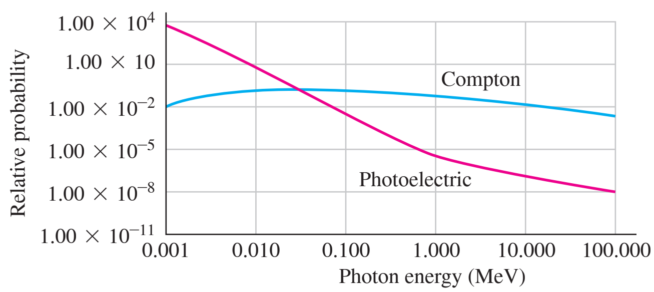
38.41 How much energy is imparted to one cell during one day's treatment? Assume that the specific gravity of the tumor is 1 and that $1\ \text{J} = 6 \times 10^{18}$ eV. (a) 120 keV; (b) 12 MeV; (c) 120 MeV; (d) 120×10^3 MeV.

38.42 While interacting with molecules (mainly water) in the tumor tissue, each Compton electron or photoelectron causes a series of ionizations, each of which takes about 40 eV. Estimate the maximum number of ionizations that one photon generated by this linear accelerator can produce in tissue. (a) 100; (b) 1000; (c) 10^4; (d) 10^5.

38.43 The high-energy photons can undergo Compton scattering off electrons in the tumor. The energy imparted by a photon is a maximum when the photon scatters straight back from the electron. In this process, what is the maximum energy that a photon with the energy described in the passage can give to an electron? (a) 3.8 MeV; (b) 2.0 MeV; (c) 0.40 MeV; (d) 0.23 MeV.

38.44 The probability of a photon interacting with tissue via the photoelectric effect or the Compton effect depends on the photon energy. Use **Fig. P38.44** to determine the best description of how the photons from the linear accelerator described in the passage interact with a tumor. (a) Via the Compton effect only; (b) mostly via the photoelectric effect until they have lost most of their energy, and then mostly via the Compton effect; (c) mostly via the Compton effect until they have lost most of their energy, and then mostly via the photoelectric effect; (d) via the Compton effect and the photoelectric effect equally.

Figure **P38.44**



38.45 Higher-energy photons might be desirable for the treatment of certain tumors. Which of these actions would generate higher-energy photons in this linear accelerator? (a) Increasing the number of electrons that hit the tungsten target; (b) accelerating the electrons through a higher potential difference; (c) both (a) and (b); (d) none of these.

ANSWERS

Chapter Opening Question ?

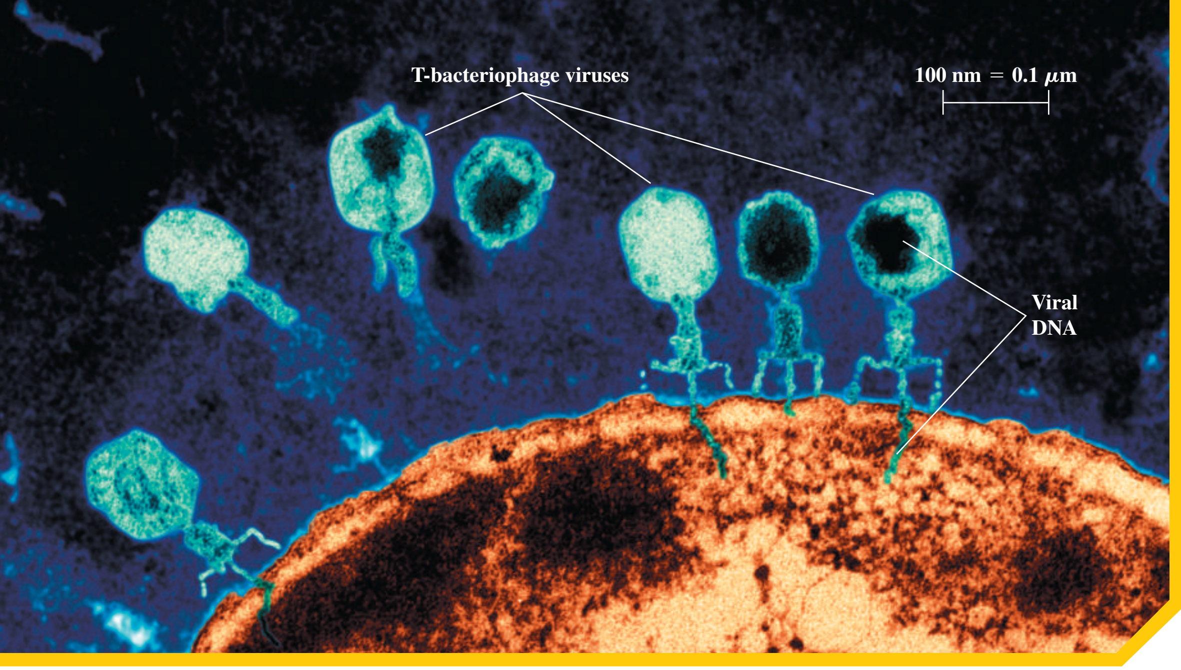
(i) The energy of a photon E is inversely proportional to its wavelength λ: The shorter the wavelength, the more energetic is the photon. Since visible light has shorter wavelengths than infrared light, the headlamp emits photons of greater energy. However, the light from the infrared laser is far more *intense* (delivers much more energy per second per unit area to the patient's skin) because it emits many more photons per second than does the headlamp and concentrates them onto a very small spot.

Key Example VARIATION Problems

VP38.3.1 (a) $E = 3.18 \times 10^{-19}$ J, $p = 1.06 \times 10^{-27}$ kg·m/s (b) 6.77×10^{-3} W
VP38.3.2 (a) 1.10×10^{15} Hz (b) 1.47×10^{15} Hz
VP38.3.3 (a) 1.24 eV (b) 1.68 V
VP38.3.4 (a) 1.37 eV (b) 4.05 eV (c) 2.68 eV
VP38.6.1 (a) 7.73×10^{-5} nm (b) 0.0308%
VP38.6.2 (a) 1.247×10^{18} Hz (b) 1.235×10^{18} Hz
VP38.6.3 (a) 35.8° (b) 121°
VP38.6.4 (a) $E = 938.3$ MeV, $\lambda = 1.32 \times 10^{-15}$ m (b) $E = 1483.3$ MeV, $\lambda = 8.36 \times 10^{-16}$ m
VP38.7.1 (a) 1.05×10^{-27} kg·m/s (b) 7.54×10^{-30} kg·m/s, 0.720% (c) 3.14×10^{-19} J (d) 2.26×10^{-21} J, 0.720%
VP38.7.2 (a) 9.59×10^{-7} s (b) 288 m (c) 1.99×10^{-24} J (d) 8.30×10^4 Hz $= 8.30 \times 10^{-5}$ GHz
VP38.7.3 (a) $\lambda = 572$ nm, $f = 5.25 \times 10^{14}$ Hz (b) 3.52×10^{13} Hz (c) 2.26×10^{-15} s
VP38.7.4 (a) 2.12×10^{-14} s (b) 519 nm (c) 1.91×10^{-6} J $= 1.91\ \mu$J

Bridging Problem

(a) 0.0781 nm
(b) 1.65×10^{-23} kg·m/s, 1.81×10^7 m/s
(c) 1.49×10^{-16} J



? Viruses (shown in blue) have landed on an *E. coli* bacterium and injected their DNA, converting the bacterium into a virus factory. This false-color image was made by using a beam of electrons rather than a light beam. Electrons are used for imaging such fine details because, compared to visible-light photons, (i) electrons can have much shorter wavelengths; (ii) electrons can have much longer wavelengths; (iii) electrons can have much less momentum; (iv) electrons have more total energy for the same momentum; (v) more than one of these.

39 Particles Behaving as Waves

In Chapter 38 we discovered one aspect of nature's wave–particle duality: Light and other electromagnetic radiation act sometimes like waves and sometimes like particles. Interference and diffraction demonstrate wave behavior, while emission and absorption of photons demonstrate particle behavior.

If light waves can behave like particles, can the particles of matter behave like waves? The answer is a resounding yes. Electrons can interfere and diffract just like other kinds of waves. The wave nature of electrons is not merely a laboratory curiosity: It is the fundamental reason why atoms, which according to classical physics should be unstable, are able to exist. In this chapter the wave nature of matter will help us understand the structure of atoms, the operating principles of a laser, and the curious properties of the light emitted by a heated, glowing object. Without the wave picture of matter, there would be no way to explain these phenomena.

In Chapter 40 we'll introduce an even more complete wave picture of matter called *quantum mechanics*. Through the remainder of this book we'll use the ideas of quantum mechanics to understand the nature of molecules, solids, atomic nuclei, and the fundamental particles that are the building blocks of our universe.

LEARNING OUTCOMES

In this chapter, you'll learn...

39.1 De Broglie's proposal that electrons and other particles can behave like waves, and the experimental evidence for de Broglie's ideas.
39.2 How physicists discovered the atomic nucleus.
39.3 How Bohr's model of electron orbits explained the spectra of hydrogen and hydrogenlike atoms.
39.4 How a laser operates.
39.5 How the idea of energy levels, coupled with the photon model of light, explains the spectrum of light emitted by a hot, opaque object.
39.6 What the uncertainty principle tells us about the nature of the atom.

You'll need to review...

13.4 Satellites.
17.7 Stefan–Boltzmann law.
18.4, 18.5 Equipartition principle; Maxwell–Boltzmann distribution function.
32.1, 32.5 Radiation from an accelerating charge; electromagnetic standing waves.
36.5–36.7 Light diffraction, x-ray diffraction, resolution.
38.1, 38.4 Photoelectric effect, photons, interference.

39.1 ELECTRON WAVES

In 1924 a French physicist, Louis de Broglie (pronounced "de broy"; **Fig. 39.1**, next page), made a remarkable proposal about the nature of matter. His reasoning, freely paraphrased, went like this: Nature loves symmetry. Light is dualistic in nature, behaving in some situations like waves and in others like particles. If nature is symmetric, this duality should also hold for matter. Electrons, which we usually think of as *particles,* may in some situations behave like *waves.*

If a particle acts like a wave, it should have a wavelength and a frequency. De Broglie postulated that a free particle with rest mass m, moving with nonrelativistic speed v, should have a wavelength λ related to its momentum $p = mv$ in exactly the same way

Figure 39.1 Louis-Victor de Broglie, the seventh Duke de Broglie (1892–1987), broke with family tradition by choosing a career in physics rather than as a diplomat. His revolutionary proposal that particles have wave characteristics—for which de Broglie won the 1929 Nobel Prize in physics—was published in his doctoral thesis.

as for a photon, as expressed by Eq. (38.5) from Section 38.1: $\lambda = h/p$. The **de Broglie wavelength** of a particle is then

De Broglie wavelength of a particle; Planck's constant; Particle's momentum; Particle's speed; Particle's mass

$$\lambda = \frac{h}{p} = \frac{h}{mv} \qquad (39.1)$$

If the particle's speed is an appreciable fraction of the speed of light c, we replace mv in Eq. (39.1) with $\gamma mv = mv/\sqrt{1 - v^2/c^2}$ [Eq. (37.27) from Section 37.7]. The frequency f, according to de Broglie, is also related to the particle's energy E in exactly the same way as for a photon:

Energy of a particle; Planck's constant; Frequency

$$E = hf \qquad (39.2)$$

CAUTION **Not all photon equations apply to particles with mass** Be careful when applying $E = hf$ to particles with nonzero rest mass, such as electrons. Unlike a photon, they do *not* travel at speed c, so the equations $f = c/\lambda$ and $E = pc$ do *not* apply to them!

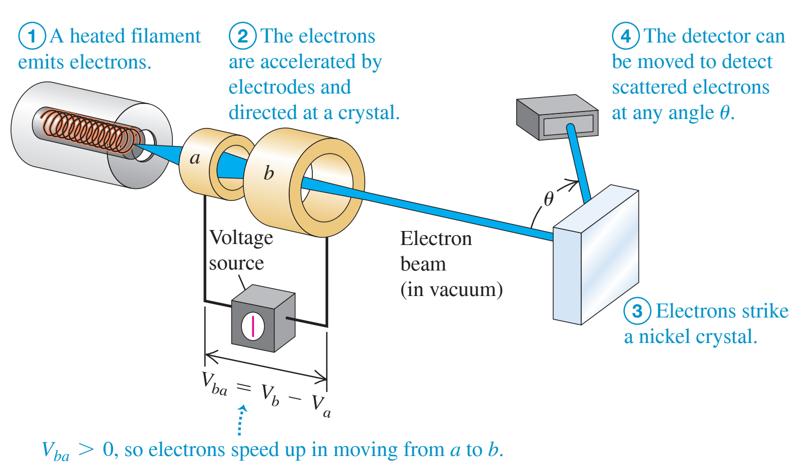
Observing the Wave Nature of Electrons

De Broglie's proposal was a bold one, made at a time when there was no direct experimental evidence that particles have wave characteristics. But within a few years, his ideas were resoundingly verified by a diffraction experiment with electrons. This experiment was analogous to those we described in Section 36.6, in which atoms in a crystal act as a three-dimensional diffraction grating for x rays. An x-ray beam is strongly reflected when it strikes a crystal at an angle that gives constructive interference among the waves scattered from the various atoms in the crystal. These interference effects demonstrate the *wave* nature of x rays.

In 1927 the American physicists Clinton Davisson and Lester Germer, working at the Bell Telephone Laboratories, were studying the surface of a piece of nickel by directing a beam of *electrons* at the surface and observing how many electrons bounced off at various angles. **Figure 39.2** shows an experimental setup like theirs. Like many ordinary metals, the sample was *polycrystalline:* It consisted of many randomly oriented microscopic crystals bonded together. As a result, the electron beam reflected diffusely, like light bouncing off a rough surface (see Fig. 33.6b), with a smooth distribution of intensity as a function of the angle θ.

During the experiment an accident occurred that permitted air to enter the vacuum chamber, and an oxide film formed on the metal surface. To remove this film, Davisson and Germer baked the sample in a high-temperature oven. Unknown to them, this had the effect of creating large regions within the nickel with crystal planes that were continuous

Figure 39.2 An apparatus similar to that used by Davisson and Germer to discover electron diffraction.



over the width of the electron beam. From the perspective of the electrons, the sample looked like a *single* crystal of nickel.

When the observations were repeated with this sample, the results were quite different. Now strong maxima in the intensity of the reflected electron beam occurred at specific angles (**Fig. 39.3a**), in contrast to the smooth variation of intensity with angle that Davisson and Germer had observed before the accident. The angular positions of the maxima depended on the accelerating voltage V_{ba} used to produce the electron beam. Davisson and Germer were familiar with de Broglie's hypothesis, and they noticed the similarity of this behavior to x-ray diffraction. This was not the effect they had been looking for, but they immediately recognized that the electron beam was being *diffracted*. They had discovered a very direct experimental confirmation of the wave hypothesis.

Davisson and Germer could determine the speeds of the electrons from the accelerating voltage, so they could compute the de Broglie wavelength from Eq. (39.1). If an electron is accelerated from rest at point a to point b through a potential increase $V_{ba} = V_b - V_a$ as shown in Fig. 39.2, the work done on the electron eV_{ba} equals its kinetic energy K. Using $K = \left(\frac{1}{2}\right)mv^2 = p^2/2m$ for a nonrelativistic particle, we have

$$eV_{ba} = \frac{p^2}{2m} \qquad p = \sqrt{2meV_{ba}}$$

We substitute this into Eq. (39.1) for the de Broglie wavelength of the electron:

$$\underbrace{\lambda}_{\text{De Broglie wavelength of an electron}} = \frac{\overbrace{h}^{\text{Planck's constant}}}{\underbrace{p}_{\text{Electron momentum}}} = \frac{h}{\sqrt{2meV_{ba}}} \tag{39.3}$$

Planck's constant; Accelerating voltage; Magnitude of electron charge; Electron mass

The greater the accelerating voltage V_{ba}, the shorter the wavelength of the electron.

To predict the angles at which strong reflection occurs, note that the electrons were scattered primarily by the planes of atoms near the surface of the crystal. Atoms in a surface plane are arranged in rows, with a distance d that can be measured by x-ray diffraction techniques. These rows act like a reflecting diffraction grating; the angles at which strong reflection occurs are the same as for a grating with center-to-center distance d between its slits (Fig. 39.3b). From Eq. (36.13) the angles of maximum reflection are given by

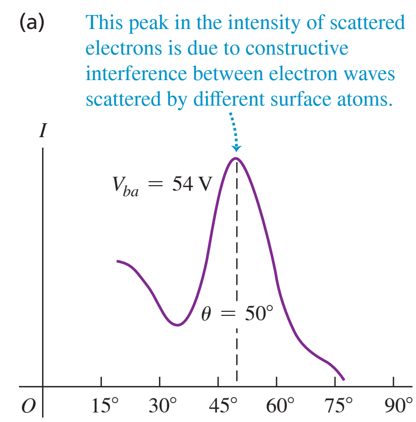
$$d\sin\theta = m\lambda \qquad (m = 1, 2, 3, \ldots) \tag{39.4}$$

where θ is the angle shown in Fig. 39.2. [Note that the geometry in Fig. 39.3b is different from that for Fig. 36.22, so Eq. (39.4) is different from Eq. (36.16).] Davisson and Germer found that the angles predicted by this equation, with the de Broglie wavelength given by Eq. (39.3), agreed with the observed values (Fig. 39.3a). Thus the accidental discovery of **electron diffraction** was the first direct evidence confirming de Broglie's hypothesis.

In 1928, just a year after the Davisson–Germer discovery, the English physicist G. P. Thomson carried out electron-diffraction experiments using a thin, polycrystalline, metallic foil as a target. Debye and Sherrer had used a similar technique several years earlier to study x-ray diffraction from polycrystalline specimens. In these experiments the beam passes *through* the target rather than being reflected from it. Because of the random orientations of the individual microscopic crystals in the foil, the diffraction pattern consists of intensity maxima forming rings around the direction of the incident beam. Thomson's results again confirmed the de Broglie relationship. **Figure 39.4** shows both electron and x-ray diffraction patterns for polycrystalline metal foils. (G. P. Thomson was the son of J. J. Thomson, who 31 years earlier discovered the electron. Davisson and the younger Thomson shared the 1937 Nobel Prize in physics for their discoveries.)

Additional diffraction experiments were soon carried out in many laboratories using not only electrons but also various ions and low-energy neutrons. All of these are in agreement with de Broglie's bold predictions. Thus the wave nature of particles, so strange in 1924, became firmly established in the years that followed.

Figure **39.3** (a) Intensity of the scattered electron beam in Fig. 39.2 as a function of the scattering angle θ. (b) Electron waves scattered from two adjacent atoms interfere constructively when $d\sin\theta = m\lambda$. In the case shown here, $\theta = 50°$ and $m = 1$.



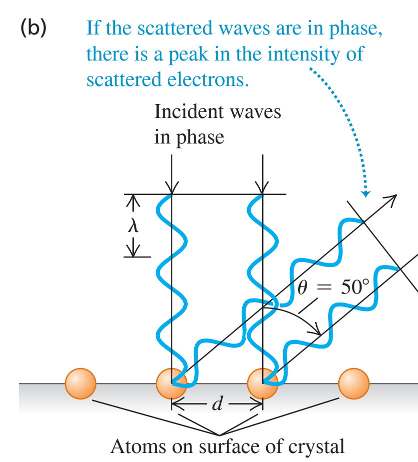
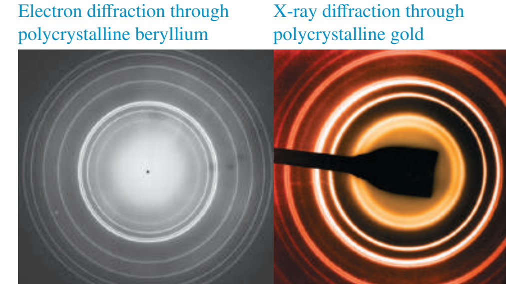


Figure **39.4** When electrons of a single momentum are sent through a polycrystalline metal foil, they produce the same kind of diffraction pattern as do x rays (shown here in enhanced color) sent through the same kind of foil. For both kinds of diffraction, the angles of the maxima depend on the wavelengths of the electrons or photons used.



The similarity of the two patterns shows that electrons undergo the same kind of diffraction as x rays.

PROBLEM-SOLVING STRATEGY 39.1 Wavelike Properties of Particles

IDENTIFY *the relevant concepts:* Particles have wavelike properties. A particle's (de Broglie) wavelength is inversely proportional to its momentum, and its frequency is proportional to its energy.

SET UP *the problem:* Identify the target variables and decide which equations you'll use to calculate them.

EXECUTE *the solution* as follows:

1. Use Eq. (39.1) to relate a particle's momentum p to its wavelength λ; use Eq. (39.2) to relate its energy E to its frequency f.
2. Nonrelativistic kinetic energy may be expressed as either $K = \frac{1}{2}mv^2$ or (because $p = mv$) $K = p^2/2m$. The latter form is useful in calculations involving the de Broglie wavelength.
3. You may express energies in either joules or electron volts, using $h = 6.626 \times 10^{-34}\ \text{J}\cdot\text{s}$ or $h = 4.136 \times 10^{-15}\ \text{eV}\cdot\text{s}$ as appropriate.

EVALUATE *your answer:* To check numerical results, it helps to remember some approximate orders of magnitude. Here's a partial list:

Size of an atom: $10^{-10}\ \text{m} = 0.1\ \text{nm}$

Mass of an atom: $10^{-26}\ \text{kg}$

Mass of an electron: $m = 10^{-30}\ \text{kg}$; $mc^2 = 0.511\ \text{MeV}$

Electron charge magnitude: $10^{-19}\ \text{C}$

kT at room temperature: $\frac{1}{40}\ \text{eV}$

Difference between energy levels of an atom (to be discussed in Section 39.3): 1 to 10 eV

Speed of an electron in the Bohr model of a hydrogen atom (to be discussed in Section 39.3): $10^6\ \text{m/s}$

EXAMPLE 39.1 An electron-diffraction experiment

WITH VARIATION PROBLEMS

In an electron-diffraction experiment using an accelerating voltage of 54 V, an intensity maximum occurs for $\theta = 50°$ (see Fig. 39.3a). X-ray diffraction indicates that the atomic spacing in the target is $d = 2.18 \times 10^{-10}\ \text{m} = 0.218\ \text{nm}$. The electrons have negligible kinetic energy before being accelerated. Find the electron wavelength.

IDENTIFY, SET UP, and EXECUTE We'll determine λ from both de Broglie's equation, Eq. (39.3), and the diffraction equation, Eq. (39.4). From Eq. (39.3),

$$\lambda = \frac{6.626 \times 10^{-34}\ \text{J}\cdot\text{s}}{\sqrt{2(9.109 \times 10^{-31}\ \text{kg})(1.602 \times 10^{-19}\ \text{C})(54\ \text{V})}}$$
$$= 1.7 \times 10^{-10}\ \text{m} = 0.17\ \text{nm}$$

Alternatively, using Eq. (39.4) and assuming $m = 1$, we get

$$\lambda = d\sin\theta = (2.18 \times 10^{-10}\ \text{m})\sin 50° = 1.7 \times 10^{-10}\ \text{m}$$

EVALUATE The two numbers agree within the accuracy of the experimental results, which gives us an excellent check on our calculations. Note that this electron wavelength is less than the spacing between the atoms.

KEYCONCEPT Electrons and other subatomic particles have both particle properties (kinetic energy and momentum) and wave properties (frequency and wavelength). Like a photon, the wavelength of a particle is inversely proportional to its momentum; unlike a photon, the wavelength of a nonrelativistic particle is inversely proportional to the square root of its kinetic energy.

EXAMPLE 39.2 Energy of a thermal neutron

WITH VARIATION PROBLEMS

Find the speed and kinetic energy of a neutron ($m = 1.675 \times 10^{-27}\ \text{kg}$) with de Broglie wavelength $\lambda = 0.200\ \text{nm}$, a typical interatomic spacing in crystals. Compare this energy with the average translational kinetic energy of an ideal-gas molecule at room temperature ($T = 20°\text{C} = 293\ \text{K}$).

IDENTIFY and SET UP This problem uses the relationships between particle speed and wavelength, between particle speed and kinetic energy, and between gas temperature and the average kinetic energy of a gas molecule. We'll find the neutron speed v by using Eq. (39.1) and from that calculate the neutron kinetic energy $K = \frac{1}{2}mv^2$. We'll use Eq. (18.16) to find the average kinetic energy of a gas molecule.

EXECUTE From Eq. (39.1), the neutron speed is

$$v = \frac{h}{\lambda m} = \frac{6.626 \times 10^{-34}\ \text{J}\cdot\text{s}}{(0.200 \times 10^{-9}\ \text{m})(1.675 \times 10^{-27}\ \text{kg})} = 1.98 \times 10^3\ \text{m/s}$$

The neutron kinetic energy is

$$K = \tfrac{1}{2}mv^2 = \tfrac{1}{2}(1.675 \times 10^{-27}\ \text{kg})(1.98 \times 10^3\ \text{m/s})^2$$
$$= 3.28 \times 10^{-21}\ \text{J} = 0.0205\ \text{eV}$$

From Eq. (18.16), the average translational kinetic energy of an ideal-gas molecule at $T = 293\ \text{K}$ is

$$\tfrac{1}{2}m(v^2)_{\text{av}} = \tfrac{3}{2}kT = \tfrac{3}{2}(1.38 \times 10^{-23}\ \text{J/K})(293\ \text{K})$$
$$= 6.07 \times 10^{-21}\ \text{J} = 0.0379\ \text{eV}$$

The two energies are comparable in magnitude, which is why a neutron with kinetic energy in this range is called a *thermal neutron*. Diffraction of thermal neutrons is used to study crystal and molecular structure in the same way as x-ray diffraction. Neutron diffraction has proved to be especially useful in the study of large organic molecules.

EVALUATE Note that the calculated neutron speed is much less than the speed of light. This justifies our use of the nonrelativistic form of Eq. (39.1).

KEYCONCEPT For a nonrelativistic particle, the speed is proportional to the momentum and hence inversely proportional to the particle's wavelength.

De Broglie Waves and the Macroscopic World

If the de Broglie picture is correct and matter has wave aspects, you might wonder why we don't see these aspects in everyday life. As an example, we know that waves diffract when sent through a single slit. Yet when we walk through a doorway (a kind of single slit), we don't worry about our body diffracting!

The main reason we don't see these effects on human scales is that Planck's constant h has such a minuscule value. As a result, the de Broglie wavelengths of even the smallest ordinary objects that you can see are extremely small, and the wave effects are unimportant. For instance, what is the wavelength of a falling grain of sand? If the grain's mass is 5×10^{-10} kg and its diameter is 0.07 mm $= 7 \times 10^{-5}$ m, it will fall in air with a terminal speed of about 0.4 m/s. The magnitude of its momentum is $p = mv = (5 \times 10^{-10}\ \text{kg})(0.4\ \text{m/s}) = 2 \times 10^{-10}\ \text{kg} \cdot \text{m/s}$. The de Broglie wavelength of this falling sand grain is then

$$\lambda = \frac{h}{p} = \frac{6.626 \times 10^{-34}\ \text{J} \cdot \text{s}}{2 \times 10^{-10}\ \text{kg} \cdot \text{m/s}} = 3 \times 10^{-24}\ \text{m}$$

Not only is this wavelength far smaller than the diameter of the sand grain, but it's also far smaller than the size of a typical atom (about 10^{-10} m). A more massive, faster-moving object would have an even larger momentum and an even smaller de Broglie wavelength. The effects of such tiny wavelengths are so small that they are never noticed in daily life.

The Electron Microscope

The **electron microscope** offers an important and interesting example of the interplay of wave and particle properties of electrons. An electron beam can be used to form an image of an object in much the same way as a light beam. A ray of light can be bent by reflection or refraction, and an electron trajectory can be bent by an electric or magnetic field. Rays of light diverging from a point on an object can be brought to convergence by a converging lens or concave mirror, and electrons diverging from a small region can be brought to convergence by electric and/or magnetic fields.

The analogy between light rays and electrons goes deeper. The *ray* model of geometric optics is an approximate representation of the more general *wave* model. Geometric optics (ray optics) is valid whenever interference and diffraction effects can be ignored. Similarly, the model of an electron as a point particle following a line trajectory is an approximate description of the actual behavior of the electron; this model is useful when we can ignore effects associated with the wave nature of electrons.

How is an electron microscope superior to an optical microscope? The *resolution* of an optical microscope is limited by diffraction effects, as we discussed in Section 36.7. Since an optical microscope uses wavelengths around 500 nm, it can't resolve objects smaller than a few hundred nanometers, no matter how carefully its lenses are made. The resolution of an electron microscope is similarly limited by the wavelengths of the electrons, but these wavelengths may be many thousands of times smaller than wavelengths of visible light. As a result, the useful magnification of an electron microscope can be thousands of times greater than that of an optical microscope.

Note that the ability of the electron microscope to form a magnified image *does not* depend on the wave properties of electrons. Within the limitations of the Heisenberg uncertainty principle (which we'll discuss in Section 39.6), we can compute the electron trajectories by treating them as classical charged particles under the action of electric and magnetic forces. Only when we talk about *resolution* do the wave properties become important.

EXAMPLE 39.3 An electron microscope

In an electron microscope, the nonrelativistic electron beam is formed by a setup similar to the electron gun used in the Davisson–Germer experiment (see Fig. 39.2). The electrons have negligible kinetic energy before they are accelerated. What accelerating voltage is needed to produce electrons with wavelength 10 pm = 0.010 nm (roughly 50,000 times smaller than typical visible-light wavelengths)?

IDENTIFY, SET UP, and EXECUTE We can use the same concepts we used to understand the Davisson–Germer experiment. The accelerating voltage is the quantity V_{ba} in Eq. (39.3). Rewrite this equation to solve for V_{ba}:

$$V_{ba} = \frac{h^2}{2me\lambda^2}$$

$$= \frac{(6.626 \times 10^{-34}\ \mathrm{J \cdot s})^2}{2(9.109 \times 10^{-31}\ \mathrm{kg})(1.602 \times 10^{-19}\ \mathrm{C})(10 \times 10^{-12}\ \mathrm{m})^2}$$

$$= 1.5 \times 10^4\ \mathrm{V} = 15{,}000\ \mathrm{V}$$

EVALUATE It is easy to attain 15 kV accelerating voltages from 120 V or 240 V line voltage by using a step-up transformer (Section 31.6) and a rectifier (Section 31.1). The accelerated electrons have kinetic energy 15 keV; since the electron rest energy is 0.511 MeV = 511 keV, these electrons are indeed nonrelativistic.

KEYCONCEPT The kinetic energy K of an electron equals the magnitude e of its charge multiplied by the voltage V_{ba} required to accelerate it from rest. If the electron is nonrelativistic, both K and V_{ba} are inversely proportional to the square of the particle's wavelength.

Types of Electron Microscope

Figure 39.5 Schematic diagram of a transmission electron microscope (TEM).

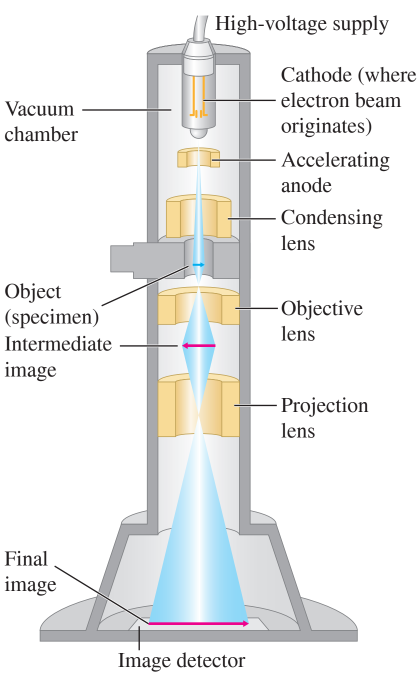


Figure 39.5 shows the design of a *transmission electron microscope,* in which electrons actually pass through the specimen being studied. The specimen to be viewed can be no more than 10 to 100 nm thick so the electrons are not slowed appreciably as they pass through. The electrons used in a transmission electron microscope are emitted from a hot cathode and accelerated by a potential difference, typically 40 to 400 kV. They then pass through a condensing "lens" that uses magnetic fields to focus the electrons into a parallel beam before they pass through the specimen. The beam then passes through two more magnetic lenses: an objective lens that forms an intermediate image of the specimen and a projection lens that produces a final real image of the intermediate image. The objective and projection lenses play the roles of the objective and eyepiece lenses, respectively, of a compound optical microscope (see Section 34.8). The final image is projected onto a fluorescent screen for viewing or photographing. The entire apparatus, including the specimen, must be enclosed in a vacuum container; otherwise, electrons would scatter off air molecules and muddle the image. The image that opens this chapter was made with a transmission electron microscope.

We might think that when the electron wavelength is 0.01 nm (as it is in Example 39.3), the resolution would also be about 0.01 nm. In fact, it is seldom better than 0.1 nm, in part because the focal length of a magnetic lens depends on the electron speed, which is never exactly the same for all electrons in the beam.

An important variation is the *scanning electron microscope*. The electron beam is focused to a very fine line and scanned across the specimen. The beam knocks additional electrons off the specimen wherever it hits. These ejected electrons are collected by an anode that is kept at a potential a few hundred volts positive with respect to the specimen. The current of ejected electrons flowing to the collecting anode varies as the microscope beam sweeps across the specimen. The varying strength of the current is then used to create a "map" of the scanned specimen, and this map forms a greatly magnified image of the specimen.

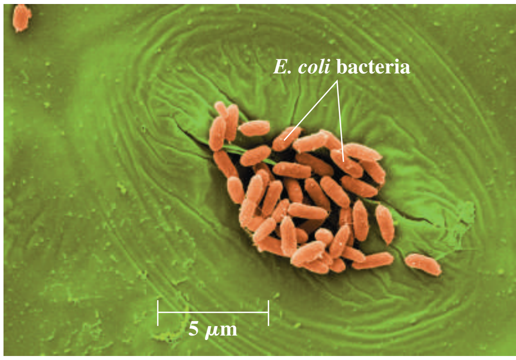
This scheme has several advantages. The specimen can be thick because the beam does not need to pass through it. Also, the knock-off electron production depends on the *angle* at which the beam strikes the surface. Thus scanning electron micrographs have an appearance that is much more three-dimensional than conventional visible-light micrographs (**Fig. 39.6**). The resolution is typically of the order of 10 nm, not as good as a transmission electron microscope but still much finer than the best optical microscopes.

TEST YOUR UNDERSTANDING OF SECTION 39.1 (a) A proton has a slightly smaller mass than a neutron. Compared to the neutron described in Example 39.2, would a proton of the same wavelength have (i) more kinetic energy; (ii) less kinetic energy; or (iii) the same kinetic energy? (b) Example 39.1 shows that to give electrons a wavelength of 1.7×10^{-10} m, they must be accelerated from rest through a voltage of 54 V and so acquire a kinetic energy of 54 eV. Does a photon of this same energy also have a wavelength of 1.7×10^{-10} m?

ANSWER

(a) **(i)**, (b) **no** From Example 39.2, the speed of a particle is $v = h/\lambda m$ and the kinetic energy is $K = \frac{1}{2}mv^2 = (m/2)(h/\lambda m)^2 = h^2/2\lambda^2 m$. This shows that for a given wavelength, the kinetic energy is inversely proportional to the mass. Hence the proton, with a smaller mass, has more kinetic energy than the neutron. For part (b), the energy of a photon is $E = hf$, and the frequency of a photon is $f = c/\lambda$. Hence $E = hc/\lambda$ and $\lambda = hc/E = (4.136 \times 10^{-15}\ \text{eV}\cdot\text{s})(2.998 \times 10^{8}\ \text{m/s})/(54\ \text{eV}) = 2.3 \times 10^{-8}$ m. This is more than 100 times greater than the wavelength of an electron of the same energy. While both photons and electrons have wavelike properties, they have different relationships between their energy and momentum and hence between their frequency and wavelength.

Figure **39.6** This scanning electron microscope image shows *Escherichia coli* bacteria crowded into a stoma, or respiration opening, on the surface of a lettuce leaf. (False color has been added.) If not washed off before the lettuce is eaten, these bacteria can be a health hazard. The *transmission* electron micrograph that opens this chapter shows a greatly magnified view of the surface of an *E. coli* bacterium.



39.2 THE NUCLEAR ATOM AND ATOMIC SPECTRA

Every neutral atom contains at least one electron. How does the wave aspect of electrons affect atomic structure? As we'll see, it is crucial for understanding not only the structure of atoms but also how they interact with light. Historically, the quest to understand the nature of the atom was intimately linked with both the idea that electrons have wave characteristics and the notion that light has particle characteristics. Before we explore how these ideas shaped atomic theory, it's useful to look at what was known about atoms—as well as what remained mysterious—by the first decade of the 20th century.

Line Spectra

Heated materials emit light, and different materials emit different kinds of light. The coils of a toaster glow red when in operation, the flame of a match has a characteristic yellow color, and the flame from a gas range is a distinct blue. To analyze these different types of light, we can use a prism or a diffraction grating to separate the various wavelengths in a beam of light into a spectrum. If the light source is a hot solid (such as the filament of an incandescent light bulb) or liquid, the spectrum is *continuous;* light of all wavelengths is present (**Fig. 39.7a**). But if the source is a heated *gas,* such as the neon in a sign or the sodium vapor formed when table salt is thrown into a campfire, the spectrum includes only a few colors in the form of isolated sharp parallel lines (Fig. 39.7b). (Each "line" is an image of the spectrograph slit, deviated through an angle that depends on the wavelength of the light forming that image; see Section 36.5.) A spectrum of this sort is called an **emission line spectrum,** and the lines are called **spectral lines.** Each spectral line corresponds to a definite wavelength and frequency.

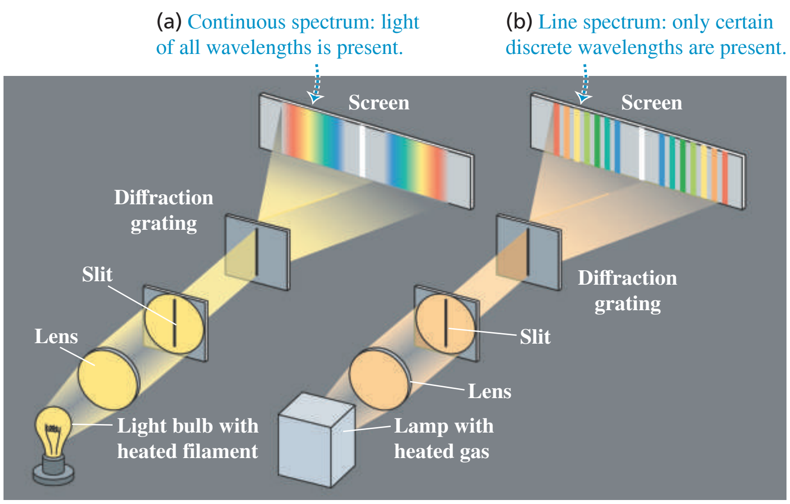
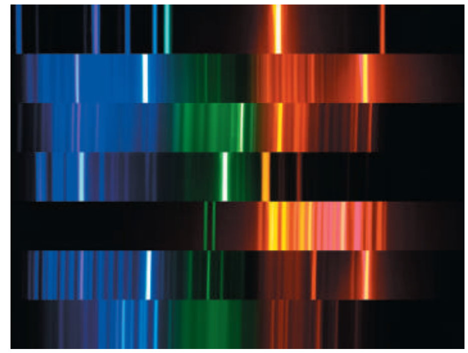


Figure **39.7** (a) Continuous spectrum produced by a glowing light bulb filament. (b) Emission line spectrum emitted by a lamp containing a heated gas.

APPLICATION Using Spectra to Analyze an Interstellar Gas Cloud The light from this glowing gas cloud—located in the Small Magellanic Cloud, a small satellite galaxy of the Milky Way some 200,000 light-years (1.9×10^{18} km) from earth—has an emission line spectrum. Despite its immense distance, astronomers can tell that this cloud is composed mostly of hydrogen because its spectrum is dominated by red light at a wavelength of 656.3 nm, a wavelength emitted by hydrogen and no other element.

Figure **39.8** The emission line spectra of several kinds of atoms and molecules. No two are alike. Note that the spectrum of water vapor (H_2O) is similar to that of hydrogen (H_2), but there are important differences that make it straightforward to distinguish these two spectra.



It was discovered early in the 19th century that each element in its gaseous state has a unique set of wavelengths in its line spectrum. The spectrum of hydrogen always contains a certain set of wavelengths; mercury produces a different set, neon still another, and so on (**Fig. 39.8**). Scientists find the use of spectra to identify elements and compounds to be an invaluable tool. For instance, astronomers have detected the spectra from more than 100 different molecules in interstellar space, including some that are not found naturally on earth.

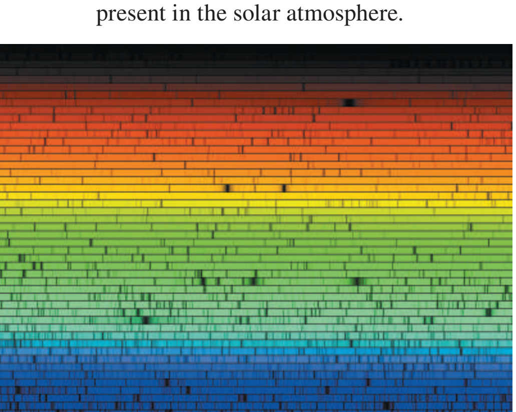
While a *heated* gas selectively *emits* only certain wavelengths, a *cool* gas selectively *absorbs* certain wavelengths. If we pass white (continuous-spectrum) light through a gas and look at the *transmitted* light with a spectrometer, we find a series of dark lines corresponding to the wavelengths that have been absorbed (**Fig. 39.9**). This is called an **absorption line spectrum.** What's more, a given kind of atom or molecule absorbs the *same* characteristic set of wavelengths when it's cool as it emits when heated. Hence scientists can use absorption line spectra to identify substances in the same manner that they use emission line spectra.

As useful as emission line spectra and absorption line spectra are, they presented a quandary to scientists: *Why* does a given kind of atom emit and absorb only certain very specific wavelengths? To answer this question, we need to have a better idea of what the inside of an atom is like. We know that atoms are much smaller than the wavelengths of visible light, so there is no hope of actually using that light to *see* an atom. But we can still describe how the mass and electric charge are distributed throughout the volume of the atom.

Here's where things stood in 1910. In 1897 the English physicist J. J. Thomson had discovered the electron and measured its charge-to-mass ratio e/m. By 1909, the American physicist Robert Millikan had made the first measurements of the electron charge $-e$. These and other experiments showed that almost all the mass of an atom had to be associated with the *positive* charge, not with the electrons. It was also known that the overall size of atoms is of the order of 10^{-10} m and that all atoms except hydrogen contain more than one electron.

In 1910 the best available model of atomic structure was one developed by Thomson. He envisioned the atom as a sphere of some as yet unidentified positively charged substance, within which the electrons were embedded like raisins in cake. This model offered an explanation for line spectra. If the atom collided with another atom, as in a heated gas, each electron would oscillate around its equilibrium position with a characteristic frequency and emit electromagnetic radiation with that frequency. If the atom were illuminated with light of many frequencies, each electron would selectively absorb only light whose frequency matched the electron's natural oscillation frequency. (This is the phenomenon of resonance that we discussed in Section 14.8.)

Figure **39.9** The absorption line spectrum of the sun. (The spectrum "lines" read from left to right and from top to bottom, like text on a page.) The spectrum is produced by the sun's relatively cool atmosphere, which absorbs photons from deeper, hotter layers. The absorption lines thus indicate what kinds of atoms are present in the solar atmosphere.



Rutherford's Exploration of the Atom

The first experiments designed to test Thomson's model by probing the interior structure of the atom were carried out in 1910–1911 by Ernest Rutherford (**Fig. 39.10**) and two of his students, Hans Geiger and Ernest Marsden, at the University of Manchester in England. These experiments consisted of shooting a beam of charged particles at thin foils of various elements and observing how the foils deflected the particles.

The particle accelerators now in common use in laboratories had not yet been invented, and Rutherford's projectiles were *alpha particles* emitted from naturally radioactive elements. The nature of these alpha particles was not completely understood, but it was known that they are ejected from unstable nuclei with speeds of the order of 10^7 m/s, are positively charged, and can travel several centimeters through air or 0.1 mm or so through solid matter before they are brought to rest by collisions.

Figure 39.11 is a schematic view of Rutherford's experimental setup. A radioactive substance at the left emits alpha particles. Thick lead screens stop all particles except those in a narrow beam. The beam passes through the foil target (consisting of gold, silver, or copper) and strikes screens coated with zinc sulfide, creating a momentary flash, or *scintillation*. Rutherford and his students counted the numbers of particles deflected through various angles.

The atoms in a metal foil are packed together like marbles in a box (not spaced apart). Because the particle beam passes through the foil, the alpha particles must pass through the interior of atoms. Within an atom, the charged alpha particle will interact with the electrons and the positive charge. (Because the *total* charge of the atom is zero, alpha particles feel little electric force outside an atom.) An electron has about 7300 times less mass than an alpha particle, so momentum considerations indicate that the atom's electrons cannot appreciably deflect the alpha particle—any more than a swarm of gnats deflects a tossed pebble. Any deflection will be due to the positively charged material that makes up almost all of the atom's mass.

In the Thomson model, the positive charge and the negative electrons are distributed through the whole atom. Hence the electric field inside the atom should be quite small, and the electric force on an alpha particle that enters the atom should be quite weak. The maximum deflection to be expected is then only a few degrees (**Fig. 39.12a**, next page). The results of the Rutherford experiments were *very* different from the Thomson prediction.

Figure 39.10 Born in New Zealand, Ernest Rutherford (1871–1937) spent his professional life in England and Canada. Before carrying out the experiments that established the existence of atomic nuclei, he shared (with Frederick Soddy) the 1908 Nobel Prize in chemistry for showing that radioactivity results from the disintegration of atoms. Rutherford is shown here on the New Zealand $100 banknote.

Figure 39.11 The Rutherford scattering experiments investigated what happens to alpha particles fired at a thin gold foil. The results of this experiment helped reveal the structure of atoms.

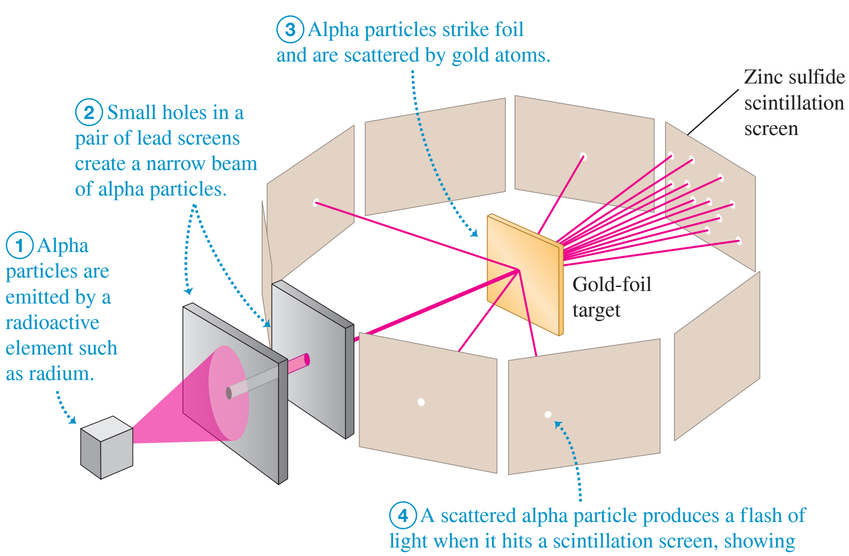
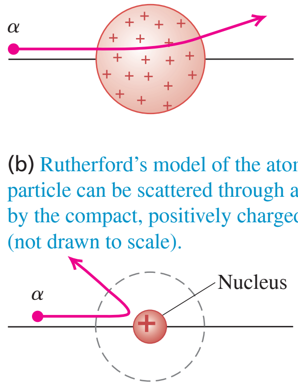


Figure **39.12** A comparison of Thomson's and Rutherford's models of the atom.

(a) Thomson's model of the atom: An alpha particle is scattered through only a small angle.

(b) Rutherford's model of the atom: An alpha particle can be scattered through a large angle by the compact, positively charged nucleus (not drawn to scale).



Some alpha particles were scattered by nearly 180°—that is, almost straight backward (Fig. 39.12b). Rutherford later wrote:

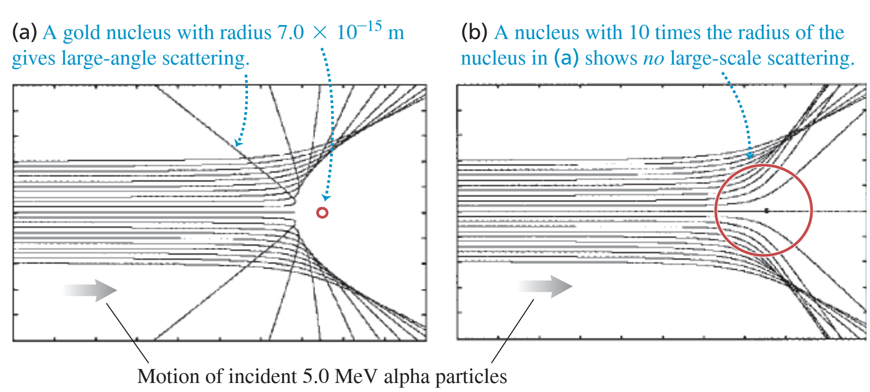
It was quite the most incredible event that ever happened to me in my life. It was almost as incredible as if you had fired a 15-inch shell at a piece of tissue paper and it came back and hit you.

Clearly the Thomson model was wrong and a new model was needed. Suppose the positive charge, instead of being distributed through a sphere with atomic dimensions (of the order of 10^{-10} m), is all concentrated in a much *smaller* volume. Then it would act like a point charge down to much smaller distances. The maximum electric field repelling the alpha particle would be much larger, and the amazing large-angle scattering that Rutherford observed could occur. Rutherford developed this model and called the concentration of positive charge the **nucleus.** He again computed the numbers of particles expected to be scattered through various angles. Within the accuracy of his experiments, the computed and measured results agreed, down to distances of the order of 10^{-14} m. His experiments therefore established that the atom does have a nucleus—a very small, very dense structure, no larger than 10^{-14} m in diameter. The nucleus occupies only about 10^{-12} of the total volume of the atom or less, but it contains *all* the positive charge and at least 99.95% of the total mass of the atom.

Figure 39.13 shows a computer simulation of alpha particles with a kinetic energy of 5.0 MeV being scattered from a gold nucleus of radius 7.0×10^{-15} m (the actual value) and from a nucleus with a hypothetical radius ten times larger. In the second case there is *no* large-angle scattering. The presence of large-angle scattering in Rutherford's experiments thus attested to the small size of the nucleus.

Later experiments showed that all nuclei are composed of positively charged protons (discovered in 1918) and electrically neutral neutrons (discovered in 1930). For example, the gold atoms in Rutherford's experiments have 79 protons and 118 neutrons. In fact, an alpha particle is itself the nucleus of a helium atom, with two protons and two neutrons. It is much more massive than an electron but only about 2% as massive as a gold nucleus, which helps explain why alpha particles are scattered by gold nuclei but not by electrons.

Figure **39.13** Computer simulation of scattering of 5.0 MeV alpha particles from a gold nucleus. Each curve shows a possible alpha-particle trajectory. (a) The scattering curves match Rutherford's experimental data if a radius of 7.0×10^{-15} m is assumed for a gold nucleus. (b) A model with a much larger radius for the gold nucleus does not match the data.



EXAMPLE 39.4 A Rutherford experiment

An alpha particle (charge $2e$) is aimed directly at a gold nucleus (charge $79e$). What minimum initial kinetic energy must the alpha particle have to approach within 5.0×10^{-14} m of the center of the gold nucleus before reversing direction? Assume that the gold nucleus, which has about 50 times the mass of an alpha particle, remains at rest.

IDENTIFY The repulsive electric force exerted by the gold nucleus makes the alpha particle slow to a halt as it approaches, then reverse direction. This force is conservative, so the total mechanical energy (kinetic energy of the alpha particle plus electric potential energy of the system) is conserved.

SET UP Let point 1 be the initial position of the alpha particle, very far from the gold nucleus, and let point 2 be 5.0×10^{-14} m from the center of the gold nucleus. Our target variable is the kinetic energy K_1 of the alpha particle at point 1 that allows it to reach point 2 with $K_2 = 0$. To find this we'll use the law of conservation of energy and Eq. (23.9) for electric potential energy, $U = qq_0/4\pi\epsilon_0 r$.

EXECUTE At point 1 the separation r of the alpha particle and gold nucleus is effectively infinite, so from Eq. (23.9) $U_1 = 0$. At point 2 the potential energy is

$$U_2 = \frac{1}{4\pi\epsilon_0}\frac{qq_0}{r}$$
$$= (9.0 \times 10^9\ \mathrm{N \cdot m^2/C^2})\frac{(2)(79)(1.60 \times 10^{-19}\ \mathrm{C})^2}{5.0 \times 10^{-14}\ \mathrm{m}}$$
$$= 7.3 \times 10^{-13}\ \mathrm{J} = 4.6 \times 10^6\ \mathrm{eV} = 4.6\ \mathrm{MeV}$$

In accordance with energy conservation, $K_1 + U_1 = K_2 + U_2$, so $K_1 = K_2 + U_2 - U_1 = 0 + 4.6\ \mathrm{MeV} - 0 = 4.6\ \mathrm{MeV}$. Thus, to approach within 5.0×10^{-14} m, the alpha particle must have initial kinetic energy $K_1 = 4.6$ MeV.

EVALUATE Alpha particles emitted from naturally occurring radioactive elements typically have energies in the range 4 to 6 MeV. For example, the common isotope of radium, ^{226}Ra, emits an alpha particle with energy 4.78 MeV.

Was it valid to assume that the gold nucleus remains at rest? To find out, note that when the alpha particle stops momentarily, all of its initial momentum has been transferred to the gold nucleus. An alpha particle has a mass $m_\alpha = 6.64 \times 10^{-27}$ kg; if its initial kinetic energy $K_1 = \frac{1}{2}mv_1^2$ is 7.3×10^{-13} J, you can show that its initial speed is $v_1 = 1.5 \times 10^7$ m/s and its initial momentum is $p_1 = m_\alpha v_1 = 9.8 \times 10^{-20}\ \mathrm{kg \cdot m/s}$. A gold nucleus (mass $m_{\mathrm{Au}} = 3.27 \times 10^{-25}$ kg) with this much momentum has a much slower speed $v_{\mathrm{Au}} = 3.0 \times 10^5$ m/s and kinetic energy $K_{\mathrm{Au}} = \frac{1}{2}mv_{\mathrm{Au}}^2 = 1.5 \times 10^{-14}\ \mathrm{J} = 0.092$ MeV. This *recoil kinetic energy* of the gold nucleus is only 2% of the total mechanical energy in this situation, so we are justified in ignoring it.

KEYCONCEPT In Rutherford scattering, a positively charged alpha particle is scattered by a positively charged atomic nucleus. You can find the distance of closest approach using conservation of total mechanical energy (the kinetic energy plus the electric potential energy of the interaction between the alpha particle and the nucleus).

The Failure of Classical Physics

Rutherford's discovery of the atomic nucleus raised a serious question: What prevented the negatively charged electrons from falling into the positively charged nucleus due to the strong electrostatic attraction? Rutherford suggested that perhaps the electrons *revolve* in orbits about the nucleus, just as the planets revolve around the sun.

But according to classical electromagnetic theory, any accelerating electric charge (either oscillating or revolving) radiates electromagnetic waves. An example is the radiation from an oscillating point charge that we depicted in Fig. 32.3 (Section 32.1). An electron orbiting inside an atom would always have a centripetal acceleration toward the nucleus, and so should be emitting radiation *at all times*. The energy of an orbiting electron should therefore decrease continuously, its orbit should become smaller and smaller, and it should spiral into the nucleus within a fraction of a second (**Fig. 39.14**). Even worse, according to classical theory the *frequency* of the electromagnetic waves emitted should equal the frequency of revolution. As the electrons radiated energy, their angular speeds would change continuously, and they would emit a *continuous* spectrum (a mixture of all frequencies), not the *line* spectrum actually observed.

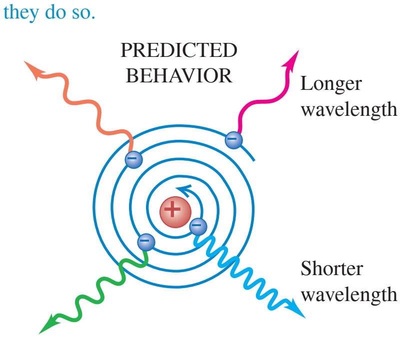
Thus Rutherford's model of electrons orbiting the nucleus, which is based on Newtonian mechanics and classical electromagnetic theory, makes three entirely *wrong* predictions about atoms: They should emit light continuously, they should be unstable, and the light they emit should have a continuous spectrum. Clearly a radical reappraisal of physics on the scale of the atom was needed. In the next section we'll see the bold idea that led to a new understanding of the atom, and see how this idea meshes with de Broglie's no less bold notion that electrons have wave attributes.

Figure **39.14** Classical physics makes predictions about the behavior of atoms that do not match reality.

ACCORDING TO CLASSICAL PHYSICS:

- An orbiting electron is accelerating, so it should radiate electromagnetic waves.
- The waves would carry away energy, so the electron should lose energy and spiral inward.
- The electron's angular speed would increase as its orbit shrank, so the frequency of the radiated waves should increase.

Thus, classical physics says that atoms should collapse within a fraction of a second and should emit light with a continuous spectrum as they do so.



IN FACT:

- Atoms are stable.
- They emit light only when excited, and only at specific frequencies (as a line spectrum).

TEST YOUR UNDERSTANDING OF SECTION 39.2 Suppose you repeated Rutherford's scattering experiment with a thin sheet of solid hydrogen in place of the gold foil. (Hydrogen is a solid at temperatures below 14.0 K.) The nucleus of a hydrogen atom is a single proton, with about one-fourth the mass of an alpha particle. Compared to the original experiment with gold foil, would you expect the alpha particles in this experiment to undergo (i) more large-angle scattering; (ii) the same amount of large-angle scattering; or (iii) less large-angle scattering?

ANSWER

(iii) Because the alpha particle is more massive, it won't bounce back in even a head-on collision with a proton that's initially at rest, any more than a bowling ball would when colliding with a Ping-Pong ball at rest (see Fig. 8.23b). Thus there would be *no* large-angle scattering in this case. Rutherford saw large-angle scattering in his experiment because gold nuclei are more massive than alpha particles (see Fig. 8.23a).

39.3 ENERGY LEVELS AND THE BOHR MODEL OF THE ATOM

In 1913 a young Danish physicist working with Ernest Rutherford at the University of Manchester made a revolutionary proposal to explain both the stability of atoms and their emission and absorption line spectra. The physicist was Niels Bohr (**Fig. 39.15**), and his innovation was to combine the photon concept that we introduced in Chapter 38 with a fundamentally new idea: The energy of an atom can have only certain particular values. His hypothesis represented a clean break from 19th-century ideas.

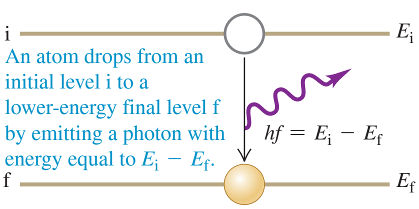
Figure 39.15 Niels Bohr (1885–1962) was a young postdoctoral researcher when he proposed the novel idea that the energy of an atom could have only certain discrete values. He won the 1922 Nobel Prize in physics for these ideas. Bohr went on to make seminal contributions to nuclear physics and to become a passionate advocate for the free exchange of scientific ideas among all nations.

Photon Emission and Absorption by Atoms

Bohr's reasoning went like this. The emission line spectrum of an element tells us that atoms of that element emit photons with only certain specific frequencies f and hence certain specific energies $E = hf$. During the emission of a photon, the internal energy of the atom changes by an amount equal to the energy of the photon. Therefore, said Bohr, each atom must be able to exist with only certain specific values of internal energy. Each atom has a set of possible **energy levels.** An atom can have an amount of internal energy equal to any one of these levels, but it *cannot* have an energy *intermediate* between two levels. All isolated atoms of a given element have the same set of energy levels, but atoms of different elements have different sets.

Suppose an atom is raised, or *excited,* to a high energy level. (In a hot gas this happens when fast-moving atoms undergo inelastic collisions with each other or with the walls of the gas container. In an electric discharge tube, such as those used in a neon light fixture, atoms are excited by collisions with fast-moving electrons.) According to Bohr, an excited atom can make a *transition* from one energy level to a lower level by emitting a photon with energy equal to the energy *difference* between the initial and final levels (**Fig. 39.16**):

Figure 39.16 An excited atom emitting a photon.



Energy of emitted photon

Speed of light in vacuum

$$hf = \frac{hc}{\lambda} = E_i - E_f \tag{39.5}$$

Final energy of atom after transition

Planck's constant

Photon frequency

Photon wavelength

Initial energy of atom before transition

For example, an excited lithium atom emits red light with wavelength $\lambda = 671$ nm. The corresponding photon energy is

$$E = \frac{hc}{\lambda} = \frac{(6.63 \times 10^{-34}\ \text{J}\cdot\text{s})(3.00 \times 10^{8}\ \text{m/s})}{671 \times 10^{-9}\ \text{m}}$$

$$= 2.96 \times 10^{-19}\ \text{J} = 1.85\ \text{eV}$$

This photon is emitted during a transition like that shown in Fig. 39.16 between two levels of the atom that differ in energy by $E_i - E_f = 1.85$ eV.

Emission line spectra (Fig. 39.8) show that many different wavelengths are emitted by each atom. Hence each kind of atom must have a number of energy levels, with different spacings in energy between them. Each wavelength in the spectrum corresponds to a transition between two specific atomic energy levels.

The observation that atoms are stable means that each atom has a *lowest* energy level, called the **ground level.** Levels with energies greater than the ground level are called **excited levels.** An atom in an excited level, called an *excited atom,* can make a transition into the ground level by emitting a photon as in Fig. 39.16. But since there are no levels below the ground level, an atom in the ground level cannot lose energy and so cannot emit a photon.

CAUTION Producing a line spectrum The lines of an emission line spectrum, such as the helium spectrum shown at the top of Fig. 39.8, are *not* all produced by a single atom. The sample of helium gas that produced the spectrum in Fig. 39.8 contained a large number of helium atoms; these were excited in an electric discharge tube to various energy levels. The spectrum of the gas shows the light emitted from all the different transitions that occurred in different atoms of the sample.

Collisions are not the only way that an atom's energy can be raised from one level to a higher level. If an atom initially in the lower energy level in Fig. 39.16 is struck by a photon with just the right amount of energy, the photon can be *absorbed* and the atom will end up in the higher level (**Fig. 39.17**). As an example, we previously mentioned two levels in the lithium atom with an energy difference of 1.85 eV. For a photon to be absorbed and excite the atom from the lower level to the higher one, the photon must have an energy of 1.85 eV and a wavelength of 671 nm. In other words, an atom *absorbs* the same wavelengths that it *emits*. This explains the correspondence between an element's emission line spectrum and its absorption line spectrum that we described in Section 39.2.

Note that a lithium atom *cannot* absorb a photon with a slightly longer wavelength (say, 672 nm) or one with a slightly shorter wavelength (say, 670 nm). That's because these photons have, respectively, slightly too little or slightly too much energy to raise the atom's energy from one level to the next, and an atom cannot have an energy that's intermediate between levels. This explains why absorption line spectra have distinct dark lines (see Fig. 39.9): Atoms can absorb only photons with specific wavelengths.

An atom that's been excited into a high energy level, either by photon absorption or by collisions, does not stay there for long. After a short time, called the *lifetime* of the level (typically around 10^{-8} s), the excited atom will emit a photon and make a transition into a lower excited level or the ground level. A cool gas that's illuminated by white light to make an *absorption* line spectrum thus also produces an *emission* line spectrum when viewed from the side, since when the atoms de-excite they emit photons in all directions (**Fig. 39.18**). To keep a gas of atoms glowing, you have to continually provide energy to the gas in order to re-excite atoms so that they can emit more photons. If you turn off the energy supply (for example, by turning off the electric current through a neon light fixture, or by shutting off the light source in Fig. 39.18), the atoms drop back into their ground levels and cease to emit light.

By working backward from the observed emission line spectrum of an element, physicists can deduce the arrangement of energy levels in an atom of that element. As an example, **Fig. 39.19a** (next page) shows some of the energy levels for a sodium atom. You may have noticed the yellow-orange light emitted by sodium vapor street lights. Sodium atoms emit this characteristic yellow-orange light with wavelengths 589.0 and 589.6 nm when they make transitions from the two closely spaced levels labeled *lowest excited levels* to the ground level. A standard test for the presence of sodium compounds is to look for this yellow-orange light from a sample placed in a flame (Fig. 39.19b).

Figure **39.17** An atom absorbing a photon. (Compare with Fig. 39.16.)

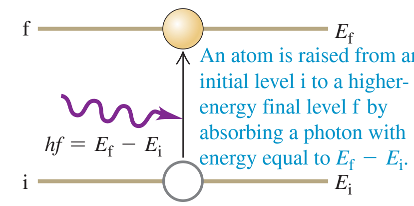


Figure **39.18** When a beam of white light with a continuous spectrum passes through a cool gas, the transmitted light has an absorption spectrum. The absorbed light energy excites the gas and causes it to emit light of its own, which has an emission spectrum.

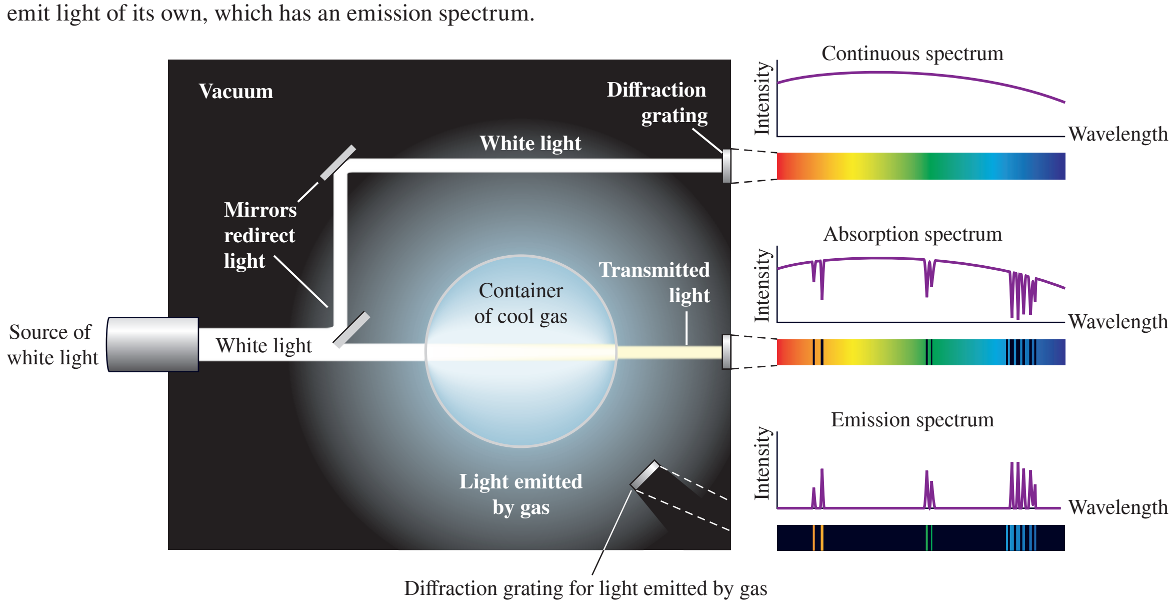
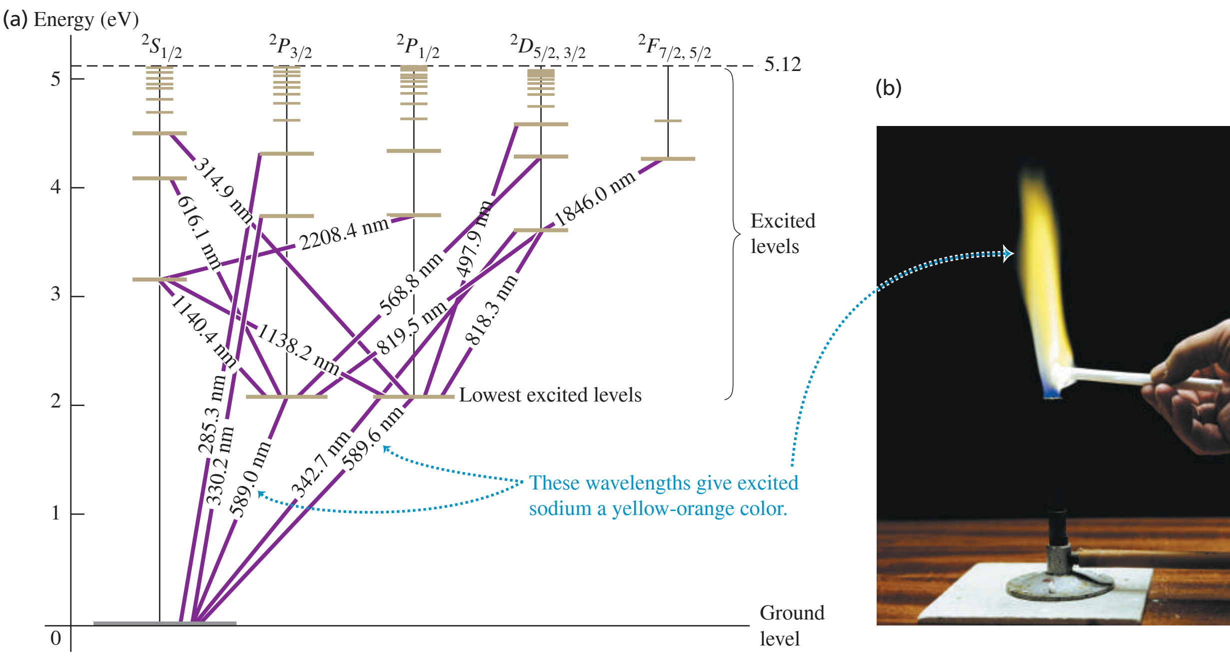


Figure **39.19** (a) Energy levels of the sodium atom relative to the ground level. Numbers on the lines between levels are wavelengths of the light emitted or absorbed during transitions between those levels. The column labels, such as $^2S_{1/2}$, refer to certain quantum states of the atom. (b) When a sodium compound is placed in a flame, sodium atoms are excited into the lowest excited levels. As they drop back to the ground level, the atoms emit photons of yellow-orange light with wavelengths 589.0 and 589.6 nm.



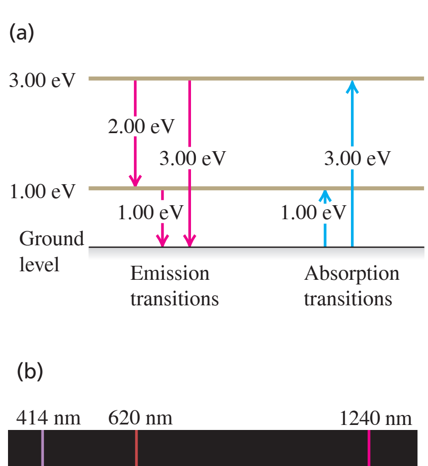
EXAMPLE 39.5 Emission and absorption spectra

WITH VARIATION PROBLEMS

A hypothetical atom (**Fig. 39.20a**) has energy levels at 0.00 eV (the ground level), 1.00 eV, and 3.00 eV. (a) What are the frequencies and wavelengths of the spectral lines this atom can emit when excited? (b) What wavelengths can this atom absorb if it is in its ground level?

IDENTIFY and SET UP Energy is conserved when a photon is emitted or absorbed. In each transition the photon energy is equal to the difference between the energies of the levels involved in the transition.

Figure **39.20** (a) Energy-level diagram for the hypothetical atom, showing the possible transitions for emission from excited levels and for absorption from the ground level. (b) Emission spectrum of this hypothetical atom.



EXECUTE (a) The possible energies of emitted photons are 1.00 eV, 2.00 eV, and 3.00 eV. For 1.00 eV, Eq. (39.2) gives

$$f = \frac{E}{h} = \frac{1.00\ \text{eV}}{4.136 \times 10^{-15}\ \text{eV} \cdot \text{s}} = 2.42 \times 10^{14}\ \text{Hz}$$

For 2.00 eV and 3.00 eV, $f = 4.84 \times 10^{14}$ Hz and 7.25×10^{14} Hz, respectively. For 1.00 eV photons,

$$\lambda = \frac{c}{f} = \frac{3.00 \times 10^8\ \text{m/s}}{2.42 \times 10^{14}\ \text{Hz}} = 1.24 \times 10^{-6}\ \text{m} = 1240\ \text{nm}$$

This is in the infrared region of the spectrum (Fig. 39.20b). For 2.00 eV and 3.00 eV, the wavelengths are 620 nm (red) and 414 nm (violet), respectively.

(b) From the ground level, only a 1.00 eV or a 3.00 eV photon can be absorbed (Fig. 39.20a); a 2.00 eV photon cannot be absorbed because the atom has no energy level 2.00 eV above the ground level. Passing light from a hot solid through a gas of these hypothetical atoms (almost all of which would be in the ground level if the gas were cool) would yield a continuous spectrum with dark absorption lines at 1240 nm and 414 nm.

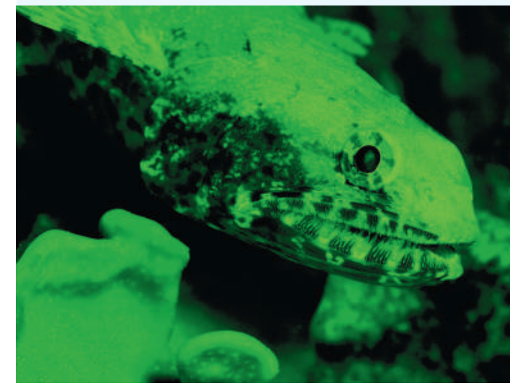
EVALUATE Note that if a gas of these atoms were at a sufficiently high temperature, collisions would excite a number of atoms into the 1.00 eV energy level. Such excited atoms *can* absorb 2.00 eV photons, as Fig. 39.20a shows, and an absorption line at 620 nm would appear in the spectrum. Thus the observed spectrum of a given substance depends on its energy levels and its temperature.

KEYCONCEPT When an atom makes a transition between two energy levels, it either emits a photon (if the atom loses energy) or absorbs a photon (if the atom gains energy). The energy of the photon that is emitted or absorbed equals the difference in energy between the two levels. The greater the photon energy, the shorter its wavelength.

Suppose we take a gas of the hypothetical atoms in Example 39.5 and illuminate it with violet light of wavelength 414 nm. Atoms in the ground level can absorb this photon and make a transition to the 3.00 eV level. Some of these atoms will make a transition back to the ground level by emitting a 414 nm photon. But other atoms will return to the ground level in two steps, first emitting a 620 nm photon to transition to the 1.00 eV level, then a 1240 nm photon to transition back to the ground level. Thus this gas will emit longer-wavelength radiation than it absorbs, a phenomenon called *fluorescence*. For example, the electric discharge in a fluorescent lamp causes the mercury vapor in the tube to emit ultraviolet radiation. This radiation is absorbed by the atoms of the coating on the inside of the tube. The coating atoms then re-emit light in the longer-wavelength, visible portion of the spectrum. Fluorescent lamps are more efficient than incandescent lamps in converting electrical energy to visible light because they do not waste as much energy producing (invisible) infrared photons.

Our discussion of energy levels and spectra has concentrated on *atoms,* but the same ideas apply to *molecules*. Figure 39.8 shows the emission line spectra of two molecules, hydrogen (H_2) and water (H_2O). Just as for sodium or other atoms, physicists can work backward from these molecular spectra and deduce the arrangement of energy levels for each kind of molecule. We'll return to molecules and molecular structure in Chapter 42.

BIO APPLICATION Fish Fluorescence When illuminated by blue light, this tropical lizardfish (family Synodontidae) fluoresces and emits longer-wavelength green light. The fluorescence may be a sexual signal or a way for the fish to camouflage itself among coral (which also have a green fluorescence).



The Franck–Hertz Experiment: Are Energy Levels Real?

Are atomic energy levels real or just a convenient fiction that helps us to explain spectra? In 1914, the German physicists James Franck and Gustav Hertz answered this question when they found direct experimental evidence for the existence of atomic energy levels.

Franck and Hertz studied the motion of electrons through mercury vapor under the action of an electric field. They found that when the electron kinetic energy was 4.9 eV or greater, the vapor emitted ultraviolet light of wavelength 250 nm. Suppose mercury atoms have an excited energy level 4.9 eV above the ground level. An atom can be raised to this level by collision with an electron; it later decays back to the ground level by emitting a photon. From the photon formula $E = hc/\lambda$, the wavelength of the photon should be

$$\lambda = \frac{hc}{E} = \frac{(4.136 \times 10^{-15}\ \text{eV}\cdot\text{s})(3.00 \times 10^{8}\ \text{m/s})}{4.9\ \text{eV}}$$
$$= 2.5 \times 10^{-7}\ \text{m} = 250\ \text{nm}$$

This is equal to the wavelength that Franck and Hertz measured, which demonstrates that this energy level actually exists in the mercury atom. Similar experiments with other atoms yield the same kind of evidence for atomic energy levels. Franck and Hertz shared the 1925 Nobel Prize in physics for their research.

Electron Waves and the Bohr Model of Hydrogen

Bohr's hypothesis established the relationship between atomic spectra and energy levels. By itself, however, it provided no general principles for *predicting* the energy levels of a particular atom. Bohr addressed this problem for the case of the simplest atom, hydrogen, which has just one electron.

In the **Bohr model,** Bohr postulated that each energy level of a hydrogen atom corresponds to a specific *stable* circular orbit of the electron around the nucleus. In a break with classical physics, Bohr further postulated that an electron in such an orbit does *not* radiate. Instead, an atom radiates energy only when an electron makes a transition from an orbit of energy E_i to a different orbit with lower energy E_f, emitting a photon of energy $hf = E_i - E_f$ in the process.

As a result of a rather complicated argument that related the angular frequency of the light emitted to the angular speed of the electron in highly excited energy levels, Bohr found that the magnitude of the electron's angular momentum is *quantized;* that is, this magnitude must be an integral multiple of $h/2\pi$. (Because $1\ \text{J} = 1\ \text{kg}\cdot\text{m}^2/\text{s}^2$, the SI units of Planck's constant h, $\text{J}\cdot\text{s}$, are the same as the SI units of angular momentum, usually

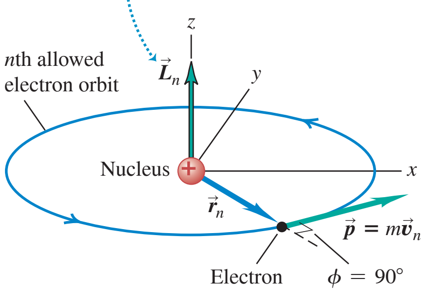


Figure 39.21 Calculating the angular momentum of an electron in a circular orbit around an atomic nucleus.

Angular momentum $\vec{L}_n$ of orbiting electron is perpendicular to plane of orbit (since we take origin to be at nucleus) and has magnitude $L = mv_n r_n \sin\phi = mv_n r_n \sin 90° = mv_n r_n$.

written as kg · m^2/s.) Let's number the orbits by an integer n, where $n = 1, 2, 3, \ldots$, and call the radius of orbit n r_n and the speed of the electron in that orbit v_n. The value of n for each orbit is called the **principal quantum number** for the orbit. From Section 10.5, Eq. (10.25), the magnitude of the angular momentum of an electron of mass m in such an orbit is $L_n = mv_n r_n$ (**Fig. 39.21**). So Bohr's argument led to

Quantization of angular momentum:

$$\overset{\text{Orbital angular momentum}}{L_n} = \underset{\text{Electron mass}}{m}\,\underset{\text{Electron speed}}{v_n}\,\underset{\text{Electron orbital radius}}{r_n} = \overset{\text{Principal quantum number } (n = 1, 2, 3, \ldots)}{n}\,\frac{\overset{\text{Planck's constant}}{h}}{2\pi} \qquad (39.6)$$

Instead of going through Bohr's argument to justify Eq. (39.6), we can use de Broglie's picture of electron waves. Rather than visualizing the orbiting electron as a particle moving around the nucleus in a circular path, think of it as a sinusoidal *standing wave* with wavelength λ that extends around the circle. A standing wave on a string transmits no energy (see Section 15.7), and electrons in Bohr's orbits radiate no energy. For the wave to "come out even" and join onto itself smoothly, the circumference of this circle must include some *whole number* of wavelengths, as **Fig. 39.22** suggests. Hence for an orbit with radius r_n and circumference $2\pi r_n$, we must have $2\pi r_n = n\lambda_n$, where λ_n is the wavelength and $n = 1, 2, 3, \ldots$. According to the de Broglie relationship, Eq. (39.1), the wavelength of a particle with rest mass m moving with nonrelativistic speed v_n is $\lambda_n = h/mv_n$. Combining $2\pi r_n = n\lambda_n$ and $\lambda_n = h/mv_n$, we find $2\pi r_n = nh/mv_n$ or

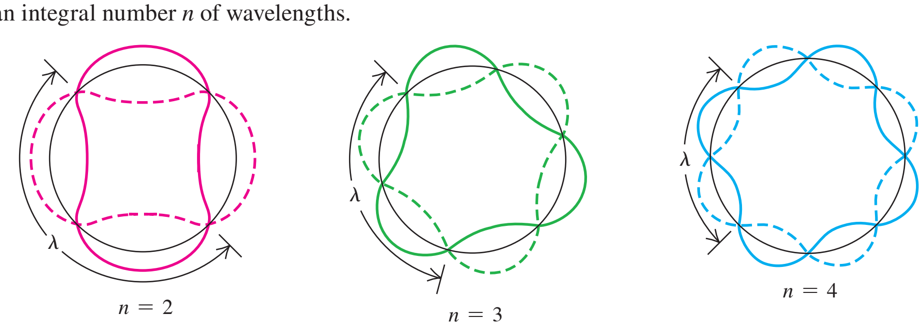
$$mv_n r_n = n\frac{h}{2\pi}$$

This is the same as Bohr's result, Eq. (39.6). Thus a wave picture of the electron leads naturally to the quantization of the electron's angular momentum.

Now let's consider a model of the hydrogen atom that is Newtonian in spirit but incorporates this quantization assumption (**Fig. 39.23**). This atom consists of a single electron with mass m and charge $-e$ in a circular orbit around a single proton with charge $+e$. The proton is nearly 2000 times as massive as the electron, so we can assume that the proton does not move. We learned in Section 5.4 that when a particle with mass m moves with speed v_n in a circular orbit with radius r_n, its centripetal (inward) acceleration is v_n^2/r_n. According to Newton's second law, a radially inward net force with magnitude $F = mv_n^2/r_n$ is needed to cause this acceleration. We discussed in Section 13.4 how the gravitational attraction provides that inward force for satellite orbits. In hydrogen the force is provided by the electrical attraction between the proton (charge $+e$) and the electron (charge $-e$). From Coulomb's law, Eq. (21.2),

$$F = \frac{1}{4\pi\epsilon_0}\frac{|(+e)(-e)|}{r_n^2} = \frac{1}{4\pi\epsilon_0}\frac{e^2}{r_n^2}$$

Figure 39.22 The idea of fitting a standing electron wave around a circular orbit. For the wave to join onto itself smoothly, the circumference of the orbit must be an integral number n of wavelengths.



Hence Newton's second law states that

$$\frac{1}{4\pi\epsilon_0}\frac{e^2}{r_n^2} = \frac{mv_n^2}{r_n} \tag{39.7}$$

When we solve Eqs. (39.6) and (39.7) simultaneously for r_n and v_n, we get

Radius of *n*th orbit in the Bohr model

$$r_n = \epsilon_0 \frac{n^2h^2}{\pi m e^2} \tag{39.8}$$

Principal quantum number ($n = 1, 2, 3, \dots$)
Planck's constant
Magnitude of electron charge
Electric constant
Electron mass

Orbital speed in *n*th orbit in the Bohr model

$$v_n = \frac{1}{\epsilon_0}\frac{e^2}{2nh} \tag{39.9}$$

Magnitude of electron charge
Planck's constant
Electric constant
Principal quantum number ($n = 1, 2, 3, \dots$)

Equation (39.8) shows that the orbital radius r_n is proportional to n^2, so the smallest orbital radius corresponds to $n = 1$. We'll denote this minimum radius, called the *Bohr radius,* as a_0:

$$a_0 = \epsilon_0 \frac{h^2}{\pi m e^2} \quad \text{(Bohr radius)} \tag{39.10}$$

Then we can rewrite Eq. (39.8) as

Radius of *n*th orbit in the Bohr model

$$r_n = n^2 a_0 \tag{39.11}$$

Bohr radius
Principal quantum number ($n = 1, 2, 3, \dots$)

The permitted orbits have radii a_0, $4a_0$, $9a_0$, and so on.

You can find the numerical values of the quantities on the right-hand side of Eq. (39.10) in Appendix F. Using these values, we find that the radius a_0 of the smallest Bohr orbit is

$$a_0 = \frac{(8.854 \times 10^{-12}\ \text{C}^2/\text{N}\cdot\text{m}^2)(6.626 \times 10^{-34}\ \text{J}\cdot\text{s})^2}{\pi(9.109 \times 10^{-31}\ \text{kg})(1.602 \times 10^{-19}\ \text{C})^2} = 5.29 \times 10^{-11}\ \text{m}$$

This gives an atomic diameter of about $10^{-10}\ \text{m} = 0.1\ \text{nm}$, which is consistent with atomic dimensions estimated by other methods.

Equation (39.9) shows that the orbital speed v_n is proportional to $1/n$. Hence the greater the value of n, the larger the orbital radius of the electron and the slower its orbital speed. (We saw the same relationship between orbital radius and speed for satellite orbits in Section 13.4.) We leave it to you to calculate the speed in the $n = 1$ orbit, which is the greatest possible speed of the electron in the hydrogen atom (see Exercise 39.23); the result is $v_1 = 2.19 \times 10^6\ \text{m/s}$. This is less than 1% of the speed of light, so relativistic considerations aren't significant.

Hydrogen Energy Levels in the Bohr Model

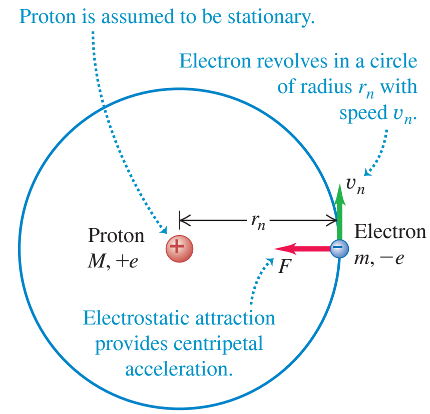
We can now use Eqs. (39.8) and (39.9) to find the kinetic and potential energies K_n and U_n when the electron is in the orbit with quantum number n:

$$K_n = \tfrac{1}{2}mv_n^2 = \frac{1}{\epsilon_0^2}\frac{me^4}{8n^2h^2} \quad \text{(kinetic energies in the Bohr model)} \tag{39.12}$$

$$U_n = -\frac{1}{4\pi\epsilon_0}\frac{e^2}{r_n} = -\frac{1}{\epsilon_0^2}\frac{me^4}{4n^2h^2} \quad \text{(potential energies in the Bohr model)} \tag{39.13}$$

The electric potential energy is negative because we have taken its value to be zero when the electron is infinitely far from the nucleus. We are interested only in the *differences* in

Figure 39.23 The Bohr model of the hydrogen atom.



energy between orbits, so the reference position doesn't matter. The total mechanical energy E_n is the sum of the kinetic and potential energies:

$$E_n = K_n + U_n = -\frac{1}{\epsilon_0^2}\frac{me^4}{8n^2h^2} \qquad \text{(total energies in the Bohr model)} \qquad (39.14)$$

Since E_n in Eq. (39.14) has a different value for each n, you can see that this equation gives the *energy levels* of the hydrogen atom in the Bohr model. Each distinct orbit corresponds to a distinct energy level.

Figure 39.24 depicts the orbits and energy levels. We label the possible energy levels of the atom by values of the quantum number n. For each value of n there are corresponding values of orbital radius r_n, speed v_n, angular momentum $L_n = nh/2\pi$, and total mechanical energy E_n. The energy of the atom is least when $n = 1$ and E_n has its most negative value. This is the *ground level* of the hydrogen atom; it is the level with the smallest orbit, of radius a_0. For $n = 2, 3, \ldots$, the absolute value of E_n is smaller and the total mechanical energy is progressively larger (less negative).

Figure 39.24 also shows some of the possible transitions from one electron orbit to an orbit of lower energy. Consider a transition from orbit n_U (for "upper") to a smaller orbit n_L (for "lower"), with $n_L < n_U$—or, equivalently, from *level* n_U to a lower *level* n_L. Then the energy hc/λ of the emitted photon of wavelength λ is equal to $E_{n_U} - E_{n_L}$. Before we use this relationship to solve for λ, it's convenient to rewrite Eq. (39.14) for the energies as

Planck's constant
Speed of light in vacuum
Electron mass
Magnitude of electron charge
Total energy for nth orbit in the Bohr model

$$E_n = -\frac{hcR}{n^2}, \quad \text{where} \quad R = \frac{me^4}{8\epsilon_0^2h^3c} \qquad (39.15)$$

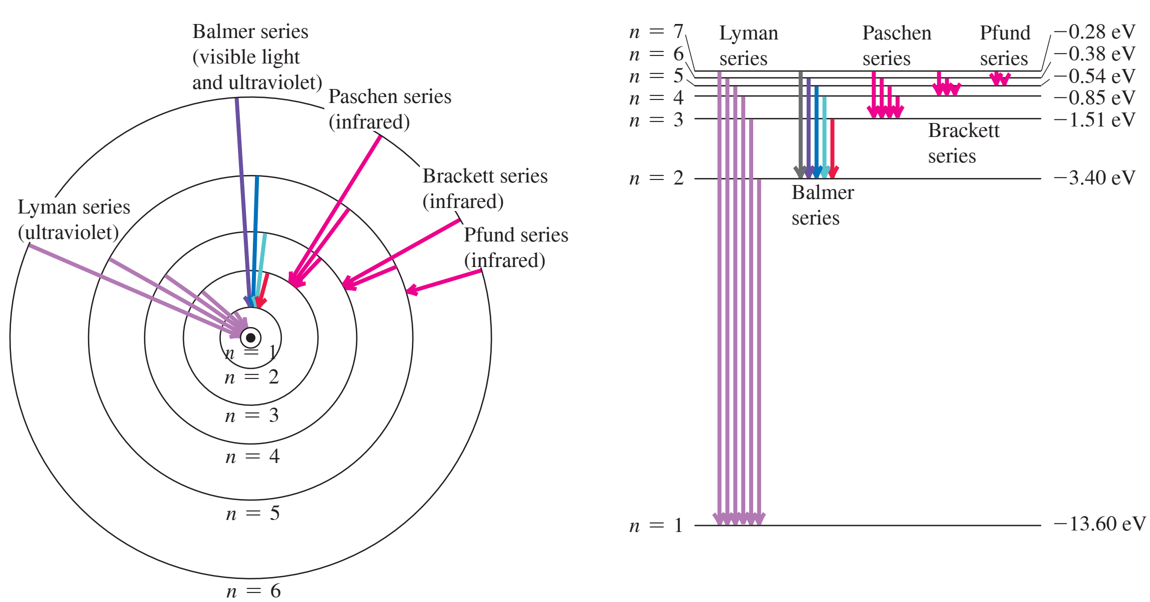
Principal quantum number ($n = 1, 2, 3, \ldots$)
Rydberg constant
Electric constant

The quantity R in Eq. (39.15) is called the **Rydberg constant** (named for the Swedish physicist Johannes Rydberg, who did pioneering work on the hydrogen spectrum). When

Figure **39.24** Two ways to represent the energy levels of the hydrogen atom and the transitions between them. Note that the radius of the nth permitted orbit is actually n^2 times the radius of the $n = 1$ orbit.

(a) Permitted orbits of an electron in the Bohr model of a hydrogen atom (not to scale). Arrows indicate the transitions responsible for some of the lines of various series.

(b) Energy-level diagram for hydrogen, showing some transitions corresponding to the various series



we substitute the numerical values of the fundamental physical constants m, c, e, h, and ϵ_0, all of which can be determined quite independently of the Bohr theory, we find that $R = 1.097 \times 10^7\ \text{m}^{-1}$. Now we solve for the wavelength of the photon emitted in a transition from level n_U to level n_L:

$$\frac{hc}{\lambda} = E_{n_U} - E_{n_L} = \left(-\frac{hcR}{n_U{}^2}\right) - \left(-\frac{hcR}{n_L{}^2}\right) = hcR\left(\frac{1}{n_L{}^2} - \frac{1}{n_U{}^2}\right)$$

$$\frac{1}{\lambda} = R\left(\frac{1}{n_L{}^2} - \frac{1}{n_U{}^2}\right) \quad \text{(hydrogen wavelengths in the Bohr model, } n_L < n_U\text{)} \tag{39.16}$$

CAUTION "Skipping over" energy levels is allowed An atom can make transitions between nonconsecutive energy levels, such as from $n = 5$ to $n = 3$ or from $n = 1$ to $n = 4$, by emitting or absorbing a single photon. (Figure 39.24 shows several transitions of this kind.) The atom does *not* have to move only one level at a time (e.g., emit a photon and transition from $n = 5$ to $n = 4$, then emit another photon and transition from $n = 4$ to $n = 3$).

Equation (39.16) is a *theoretical prediction* of the wavelengths found in the *emission* line spectrum of hydrogen atoms. When a hydrogen atom *absorbs* a photon, an electron makes a transition from a level n_L to a *higher* level n_U. This can happen only if the photon energy hc/λ is equal to $E_{n_U} - E_{n_L}$, which is the same condition expressed by Eq. (39.16). So this equation also predicts the wavelengths found in the *absorption* line spectrum of hydrogen.

How does this prediction compare with experiment? If $n_L = 2$, corresponding to transitions to the second energy level in Fig. 39.24, the wavelengths predicted by Eq. (39.16)—collectively called the *Balmer series* (**Fig. 39.25**)—are all in the visible and ultraviolet parts of the electromagnetic spectrum. If we let $n_L = 2$ and $n_U = 3$ in Eq. (39.16) we obtain the wavelength of the H_α line:

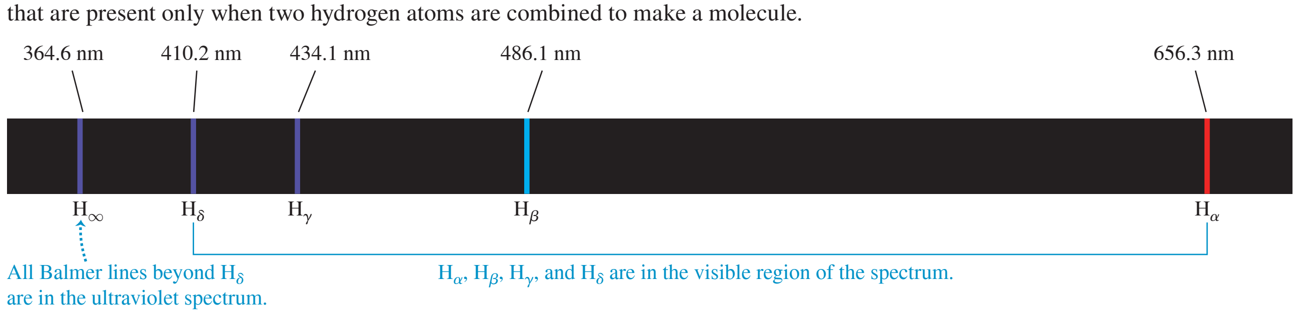
$$\frac{1}{\lambda} = (1.097 \times 10^7\ \text{m}^{-1})\left(\tfrac{1}{4} - \tfrac{1}{9}\right) \qquad \text{or} \qquad \lambda = 656.3\ \text{nm}$$

With $n_L = 2$ and $n_U = 4$ we obtain the wavelength of the H_β line, and so on. With $n_L = 2$ and $n_U = \infty$ we obtain the shortest wavelength in the series, $\lambda = 364.6$ nm. These theoretical predictions are within 0.1% of the observed hydrogen wavelengths! This close agreement provides very strong and direct confirmation of Bohr's theory.

The Bohr model also predicts many other wavelengths in the hydrogen spectrum, as Fig. 39.24 shows. The observed wavelengths of all of these series, each of which is named for its discoverer, match the predicted values with the same percent accuracy as for the Balmer series. The *Lyman series* of spectral lines is caused by transitions between the ground level and the excited levels, corresponding to $n_L = 1$ and $n_U = 2, 3, 4, \ldots$ in Eq. (39.16). The energy difference between the ground level and any of the excited levels is large, so the emitted photons have wavelengths in the ultraviolet part of the electromagnetic spectrum. Transitions among the higher energy levels involve a much smaller energy difference, so the photons emitted in these transitions have little energy and long, infrared wavelengths. That's the case for both the *Brackett series* ($n_L = 3$ and $n_U = 4, 5, 6, \ldots$, corresponding to transitions between the third and higher energy levels) and the *Pfund series* ($n_L = 4$ and $n_U = 5, 6, 7, \ldots$, with transitions between the fourth and higher energy levels).

Figure 39.24 shows only transitions in which a hydrogen atom emits a photon. But as we discussed previously, the wavelengths of those photons that an atom can *absorb* are the

Figure **39.25** The Balmer series of spectral lines for atomic hydrogen. You can see these same lines in the spectrum of *molecular* hydrogen (H_2) shown in Fig. 39.8, as well as additional lines that are present only when two hydrogen atoms are combined to make a molecule.



same as those that it can emit. For example, a hydrogen atom in the $n = 2$ level can absorb a 656.3 nm photon and end up in the $n = 3$ level.

One additional test of the Bohr model is its predicted value of the *ionization energy* of the hydrogen atom. This is the energy required to remove the electron completely from the atom. Ionization corresponds to a transition from the ground level $(n = 1)$ to an infinitely large orbital radius $(n = \infty)$, so the energy that must be added to the atom is $E_\infty - E_1 = 0 - E_1 = -E_1$ (recall that E_1 is negative). Substituting the constants from Appendix F into Eq. (39.15) gives an ionization energy of 13.606 eV. The ionization energy can also be measured directly; the result is 13.60 eV. These two values agree within 0.1%.

EXAMPLE 39.6 Exploring the Bohr model

WITH VARIATION PROBLEMS

Find the kinetic, potential, and total mechanical energies of the hydrogen atom in the first excited level, and find the wavelength of the photon emitted in a transition from that level to the ground level.

IDENTIFY and SET UP This problem uses the ideas of the Bohr model. We use simplified versions of Eqs. (39.12), (39.13), and (39.14) to find the energies of the atom, and Eq. (39.16), $hc/\lambda = E_{n_U} - E_{n_L}$, to find the photon wavelength λ in a transition from $n_U = 2$ (the first excited level) to $n_L = 1$ (the ground level).

EXECUTE We could evaluate Eqs. (39.12), (39.13), and (39.14) for the nth level by substituting the values of m, e, ϵ_0, and h. But we can simplify the calculation by comparing with Eq. (39.15), which shows that the constant $me^4/8\epsilon_0^2h^2$ that appears in Eqs. (39.12), (39.13), and (39.14) is equal to hcR:

$$\begin{aligned}\frac{me^4}{8\epsilon_0^2h^2} &= hcR\\ &= (6.626 \times 10^{-34}\ \text{J}\cdot\text{s})(2.998 \times 10^8\ \text{m/s})(1.097 \times 10^7\ \text{m}^{-1})\\ &= 2.179 \times 10^{-18}\ \text{J} = 13.60\ \text{eV}\end{aligned}$$

This allows us to rewrite Eqs. (39.12), (39.13), and (39.14) as

$$K_n = \frac{13.60\ \text{eV}}{n^2} \qquad U_n = \frac{-27.20\ \text{eV}}{n^2} \qquad E_n = \frac{-13.60\ \text{eV}}{n^2}$$

For the first excited level $(n = 2)$, we have $K_2 = 3.40\ \text{eV}$, $U_2 = -6.80\ \text{eV}$, and $E_2 = -3.40\ \text{eV}$. For the ground level $(n = 1)$, $E_1 = -13.60\ \text{eV}$. The energy of the emitted photon is then $E_2 - E_1 = -3.40\ \text{eV} - (-13.60\ \text{eV}) = 10.20\ \text{eV}$, and

$$\begin{aligned}\lambda &= \frac{hc}{E_2 - E_1} = \frac{(4.136 \times 10^{-15}\ \text{eV}\cdot\text{s})(3.00 \times 10^8\ \text{m/s})}{10.20\ \text{eV}}\\ &= 1.22 \times 10^{-7}\ \text{m} = 122\ \text{nm}\end{aligned}$$

This is the wavelength of the Lyman-alpha (L_α) line, the longest-wavelength line in the Lyman series of ultraviolet lines in the hydrogen spectrum (see Fig. 39.24).

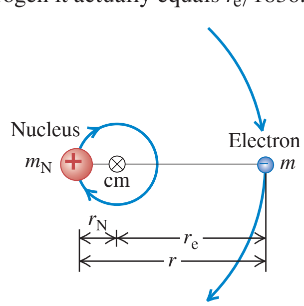
EVALUATE The total mechanical energy for any level is negative and is equal to one-half the potential energy. We found the same energy relationship for Newtonian satellite orbits in Section 13.4. The situations are similar because both the electrostatic and gravitational forces are inversely proportional to $1/r^2$.

KEYCONCEPT In the Bohr model of the hydrogen atom, the electron moves around the nucleus in a circular Newtonian orbit. However, only certain orbital radii are allowed. The nth allowed orbit has a distinct kinetic energy K_n, electric potential energy U_n, and total mechanical energy $E_n = K_n + U_n$. When the electron makes a transition between two allowed orbits, a photon is emitted or absorbed with an energy equal to the difference between the values of E_n for the two orbits.

Nuclear Motion and the Reduced Mass of an Atom

The Bohr model is so successful that we can justifiably ask why its predictions for the wavelengths and ionization energy of hydrogen differ from the measured values by about 0.1%. The explanation is that we assumed that the nucleus (a proton) remains at rest. However, as **Fig. 39.26** shows, the proton and electron *both* orbit about their common center of mass (see Section 8.5). It turns out that we can take this motion into account by using in Bohr's equations not the electron rest mass m but a quantity called the **reduced mass** m_r of the system. For a system composed of two objects of masses m_1 and m_2, the reduced mass is

Figure 39.26 The nucleus and the electron both orbit around their common center of mass. The distance r_N has been exaggerated for clarity; for ordinary hydrogen it actually equals $r_e/1836.2$.



$$m_r = \frac{m_1m_2}{m_1 + m_2} \tag{39.17}$$

For ordinary hydrogen we let m_1 equal m and m_2 equal the proton mass, $m_p = 1836.2m$. Thus ordinary hydrogen has a reduced mass of

$$m_r = \frac{m(1836.2m)}{m + 1836.2m} = 0.99946m$$

When this value is used instead of the electron mass m in the Bohr equations, the predicted values agree very well with the measured values.

In an atom of deuterium, also called *heavy hydrogen,* the nucleus is not a single proton but a proton and a neutron bound together to form a composite object called the *deuteron*. The reduced mass of the deuterium atom turns out to be $0.99973m$. Equations (39.15) and (39.16) (with m replaced by m_r) show that all wavelengths are inversely proportional to m_r. Thus the wavelengths of the deuterium spectrum should be those of hydrogen divided by $(0.99973m)/(0.99946m) = 1.00027$. This is a small effect but well within the precision of modern spectrometers. This small wavelength shift led the American scientist Harold Urey to the discovery of deuterium in 1932, an achievement that earned him the 1934 Nobel Prize in chemistry.

Hydrogenlike Atoms

We can extend the Bohr model to other one-electron atoms, such as singly ionized helium (He^+), doubly ionized lithium (Li^{2+}), and so on. Such atoms are called *hydrogenlike* atoms. In such atoms, the nuclear charge is not e but Ze, where Z is the *atomic number,* equal to the number of protons in the nucleus. The effect in the previous analysis is to replace e^2 everywhere by Ze^2. You should verify that the orbital radii r_n given by Eq. (39.8) become smaller by a factor of Z, and the energy levels E_n given by Eq. (39.14) are multiplied by Z^2. The reduced-mass correction in these cases is even less than 0.1% because the nuclei are more massive than the single proton of ordinary hydrogen. **Figure 39.27** compares the energy levels for H and for He^+, which has $Z = 2$.

Atoms of the alkali metals (at the far left-hand side of the periodic table; see Appendix D) have one electron outside a core consisting of the nucleus and the inner electrons, with net core charge $+e$. These atoms are approximately hydrogenlike, especially in excited levels. Physicists have studied alkali atoms in which the outer electron has been excited into a very large orbit with $n = 1000$. From Eq. (39.8), the radius of such a *Rydberg atom* with $n = 1000$ is $n^2 = 10^6$ times the Bohr radius, or about 0.05 mm—about the size of a small grain of sand.

Although the Bohr model predicted the energy levels of the hydrogen atom correctly, it raised as many questions as it answered. It combined elements of classical physics with new postulates that were inconsistent with classical ideas. The model provided no insight into what happens during a transition from one orbit to another; the angular speeds of the electron motion were not in general the angular frequencies of the emitted radiation, a result that is contrary to classical electrodynamics. Attempts to extend the model to atoms with two or more electrons were not successful. An electron moving in one of Bohr's circular orbits forms a current loop and should produce a magnetic dipole moment (see Section 27.7). However, a hydrogen atom in its ground level has *no* magnetic moment due to orbital motion. In Chapters 40 and 41 we'll find that an even more radical departure from classical concepts was needed before the understanding of atomic structure could progress further.

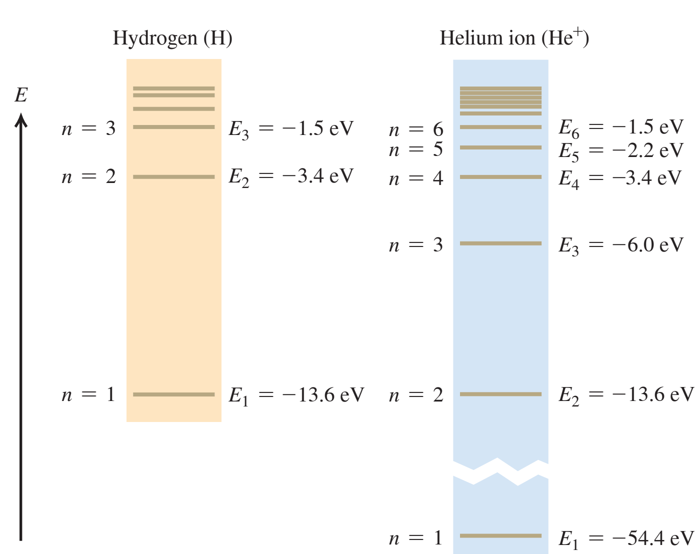


Figure **39.27** Energy levels of H and He^+. The energy expression, Eq. (39.14), is multiplied by $Z^2 = 4$ for He^+, so the energy of an He^+ ion with a given n is almost exactly four times that of an H atom with the same n. (There are small differences of the order of 0.05% because the reduced masses are slightly different.)

TEST YOUR UNDERSTANDING OF SECTION 39.3 Consider the possible transitions between energy levels in a He^+ ion. For which of these transitions in He^+ will the wavelength of the emitted photon be nearly the same as one of the wavelengths emitted by excited H atoms? (i) $n = 2$ to $n = 1$; (ii) $n = 3$ to $n = 2$; (iii) $n = 4$ to $n = 3$; (iv) $n = 4$ to $n = 2$; (v) more than one of these; (vi) none of these.

ANSWER

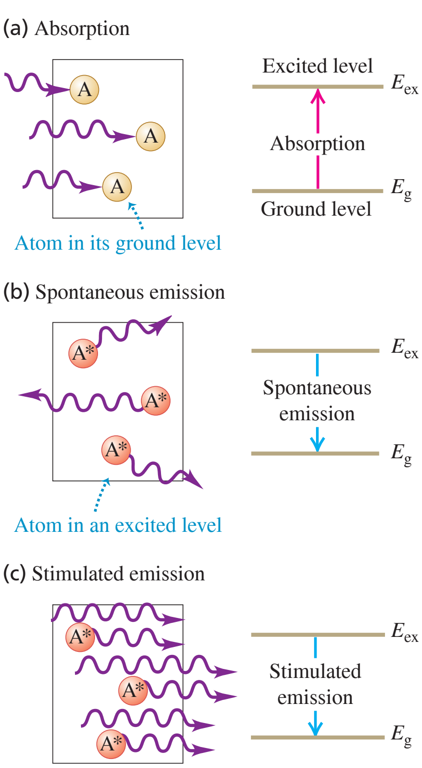
▮ (iv) Figure 39.27 shows that many (though *not* all) of the energy levels of He^+ are the same as those of H. Hence photons emitted during transitions between corresponding pairs of levels in He^+ and H have the same energy E and the same wavelength $\lambda = hc/E$. An H atom that drops from the $n = 2$ level to the $n = 1$ level emits a photon of energy 10.20 eV and wavelength 122 nm (see Example 39.6); a He^+ ion emits a photon of the same energy and wavelength when it drops from the $n = 4$ level to the $n = 2$ level. Inspecting Fig. 39.27 will show you that every even-numbered level in He^+ matches a level in H, while none of the odd-numbered He^+ levels do. The first three He^+ transitions given in the question ($n = 2$ to $n = 1$, $n = 3$ to $n = 2$, and $n = 4$ to $n = 3$) all involve an odd-numbered level, so none of their wavelengths match a wavelength emitted by H atoms.

39.4 THE LASER

The **laser** is a light source that produces a beam of highly coherent and very nearly monochromatic light as a result of cooperative emission from many atoms. The name "laser" is an acronym for "light amplification by stimulated emission of radiation." We can understand the principles of laser operation from what we have learned about atomic energy levels and photons. To do this we'll have to introduce two new concepts: *stimulated emission* and *population inversion*.

Spontaneous and Stimulated Emission

Figure **39.28** Three processes in which atoms interact with light.



Consider a gas of atoms in a transparent container. Each atom is initially in its ground level of energy E_g and also has an excited level of energy E_{ex}. If we shine light of frequency f on the container, an atom can absorb one of the photons provided the photon energy $E = hf$ equals the energy difference $E_{ex} - E_g$ between the levels. **Figure 39.28a** shows this process, in which three atoms A each absorb a photon and go into the excited level. Some time later, the excited atoms (which we denote as A*) return to the ground level by each emitting a photon with the same frequency as the one originally absorbed (Fig. 39.28b). This process is called **spontaneous emission.** The direction and phase of each spontaneously emitted photon are random.

In **stimulated emission** (Fig. 39.28c), each incident photon encounters a previously excited atom. A kind of resonance effect induces each atom to emit a second photon with the same frequency, direction, phase, and polarization as the incident photon, which is not changed by the process. For each atom there is one photon before a stimulated emission and two photons after—thus the name *light amplification*. Because the two photons have the same phase, they emerge together as *coherent* radiation. The laser makes use of stimulated emission to produce a beam consisting of a large number of such coherent photons.

To discuss stimulated emission from atoms in excited levels, we need to know something about how many atoms are in each of the various energy levels. First, we need to make the distinction between the terms *energy level* and *state*. A system may have more than one way to attain a given energy level; each different way is a different **state.** For instance, there are two ways of putting an ideal unstretched spring in a given energy level. Remembering that the spring potential energy is $U = \frac{1}{2}kx^2$, we could compress the spring by $x = -b$ or we could stretch it by $x = +b$ to get the same $U = \frac{1}{2}kb^2$. The Bohr model had only one state in each energy level, but we'll find in Chapter 41 that the hydrogen atom (Fig. 39.24b) actually has two *ground states* in its -13.60 eV ground level, eight *excited states* in its -3.40 eV first excited level, and so on.

The Maxwell–Boltzmann distribution function (see Section 18.5) determines the number of atoms in a given state in a gas. The function tells us that when the gas is in thermal equilibrium at absolute temperature T, the number n_i of atoms in a state with energy E_i equals $Ae^{-E_i/kT}$, where k is the Boltzmann constant and A is another constant determined by the total number of atoms in the gas. (In Section 18.5, E was the kinetic energy $\frac{1}{2}mv^2$ of a gas molecule; here we're talking about the internal energy of an atom.) Because of the negative exponent, fewer atoms are in higher-energy states. If E_g is a ground-state energy and E_{ex} is the energy of an excited state, then the ratio of numbers of atoms in the two states is

$$\frac{n_{ex}}{n_g} = \frac{Ae^{-E_{ex}/kT}}{Ae^{-E_g/kT}} = e^{-(E_{ex}-E_g)/kT} \tag{39.18}$$

For example, suppose $E_{ex} - E_g = 2.0\text{ eV} = 3.2 \times 10^{-19}\text{ J}$, the energy of a 620 nm visible-light photon. At $T = 3000\text{ K}$ (roughly the temperature of the filament in an incandescent light bulb or restaurant heat lamp),

$$\frac{E_{ex} - E_g}{kT} = \frac{3.2 \times 10^{-19}\text{ J}}{(1.38 \times 10^{-23}\text{ J/K})(3000\text{ K})} = 7.73$$

and

$$e^{-(E_{ex}-E_g)/kT} = e^{-7.73} = 0.00044$$

That is, the fraction of atoms in a state 2.0 eV above a ground state is extremely small, even at this high temperature. The point is that at any reasonable temperature there aren't enough atoms in excited states for any appreciable amount of stimulated emission from these states to occur. Rather, a photon emitted by one of the rare excited atoms will almost certainly be absorbed by an atom in the ground state rather than encountering another excited atom.

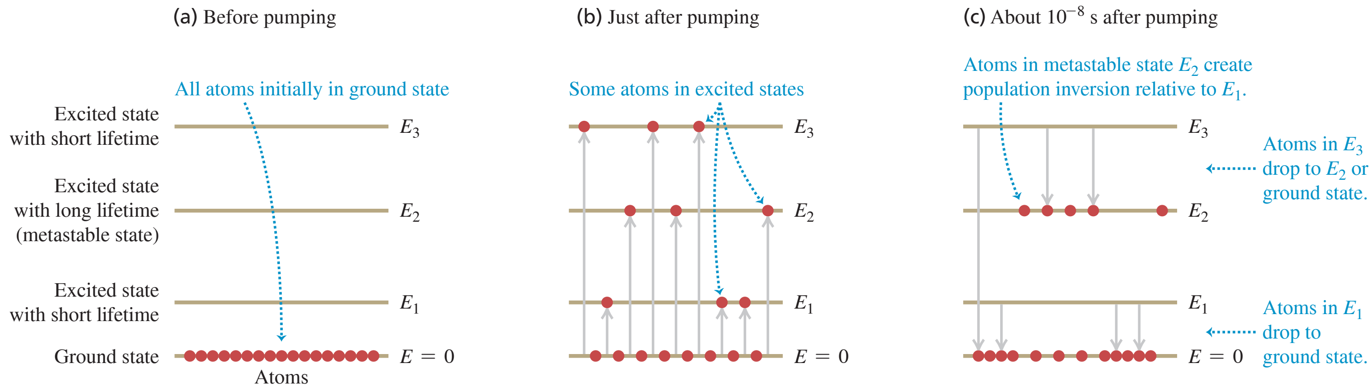
Enhancing Stimulated Emission: Population Inversions

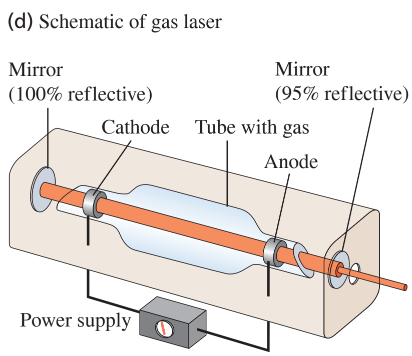
To make a laser, we need to promote stimulated emission by increasing the number of atoms in excited states. Can we do that simply by illuminating the container with radiation of frequency $f = E/h$ corresponding to the energy difference $E = E_{ex} - E_g$, as in Fig. 39.28a? Some of the atoms absorb photons of energy E and are raised to the excited state, and the population ratio n_{ex}/n_g momentarily increases. But because n_g is originally so much larger than n_{ex}, an enormously intense beam of light would be required to momentarily increase n_{ex} to a value comparable to n_g. The rate at which energy is *absorbed* from the beam by the n_g ground-state atoms far exceeds the rate at which energy is added to the beam by stimulated emission from the relatively rare (n_{ex}) excited atoms.

We need to create a *nonequilibrium* situation in which there are more atoms in a higher-energy state than in a lower-energy state. Such a situation is called a **population inversion.** Then the rate of energy radiation by stimulated emission can *exceed* the rate of absorption, and the system will act as a net *source* of radiation with photon energy E. We can achieve a population inversion by starting with atoms that have the right kinds of excited states. **Figure 39.29a** (next page) shows an energy-level diagram for such an atom with a ground state and *three* excited states of energies E_1, E_2, and E_3. A laser that uses a material with energy levels like these is called a *four-level laser.* For the laser action to work, the states of energies E_1 and E_3 must have ordinary short lifetimes of about 10^{-8} s, while the state of energy E_2 must have an unusually long lifetime of 10^{-3} s or so. Such a long-lived **metastable state** can occur if, for instance, there are restrictions imposed by conservation of angular momentum that hinder photon emission from this state. (We'll discuss these restrictions in Chapter 41.) The metastable state is the one that we want to populate.

To produce a population inversion, we *pump* the material to excite the atoms out of the ground state into the states of energies E_1, E_2, and E_3 (Fig. 39.29b). If the atoms are in a gas, we can do this by inserting two electrodes into the gas container. When a burst of sufficiently high voltage is applied to the electrodes, an electric discharge occurs. Collisions between ionized atoms and electrons carrying the discharge current then excite the atoms to various energy states. Within about 10^{-8} s the atoms that are excited to states E_1 and E_3 undergo spontaneous photon emission, so these states end up depopulated. But atoms "pile up" in the metastable state with energy E_2. The number of atoms in the metastable state is *less* than the number in the ground state, but is *much greater* than in the nearly unoccupied state of energy E_1. Hence there is a population inversion of state E_2 relative

Figure 39.29 (a), (b), (c) Stages in the operation of a four-level laser. (d) The light emitted by atoms making spontaneous transitions from state E_2 to state E_1 is reflected between mirrors, so it continues to stimulate emission and gives rise to coherent light. One mirror is partially transmitting and allows the high-intensity light beam to escape.





to state E_1 (Fig. 39.29c). You can see why we need the two levels E_1 and E_3: Atoms that undergo spontaneous emission from the E_3 level help to populate the E_2 level, and the presence of the E_1 level makes a population inversion possible.

Over the next 10^{-3} s, some of the atoms in the long-lived metastable state E_2 transition to state E_1 by spontaneous emission. The emitted photons of energy $hf = E_2 - E_1$ are sent back and forth through the gas many times by a pair of parallel mirrors (Fig. 39.29d), so that they can *stimulate* emission from as many of the atoms in state E_2 as possible. The net result of all these processes is a beam of light of frequency f that can be quite intense, has parallel rays, is highly monochromatic, and is spatially *coherent* at all points within a given cross section—that is, a laser beam. One of the mirrors is partially transparent, so a portion of the beam emerges.

What we've described is a *pulsed* laser that produces a burst of coherent light every time the atoms are pumped. Pulsed lasers are used in LASIK eye surgery (an acronym for *laser-assisted in situ keratomileusis*) to reshape the cornea and correct for nearsightedness, farsightedness, or astigmatism. In a *continuous* laser, such as those found in the barcode scanners used at retail checkout counters, energy is supplied to the atoms continuously (for instance, by having the power supply in Fig. 39.29d provide a steady voltage to the electrodes) and a steady beam of light emerges from the laser. For such a laser the pumping must be intense enough to sustain the population inversion, so that the rate at which atoms are added to level E_2 through pumping equals the rate at which atoms in this level emit a photon and transition to level E_1.

Since a special arrangement of energy levels is needed for laser action, it's not surprising that only certain materials can be used to make a laser. Some types of laser use a solid, transparent material such as neodymium glass rather than a gas. The most common kind of laser—used in laser printers (Section 21.1), laser pointers, and to read the data on the disc in a DVD player or Blu-ray player—is a *semiconductor laser,* which doesn't use atomic energy levels at all. As we'll discuss in Chapter 42, these lasers instead use the energy levels of electrons that are free to roam throughout the volume of the semiconductors.

TEST YOUR UNDERSTANDING OF SECTION 39.4 An ordinary neon light fixture like those used in advertising signs emits red light of wavelength 632.8 nm. Neon atoms are also used in a helium–neon laser (a type of gas laser). The light emitted by a neon light fixture is (i) spontaneous emission; (ii) stimulated emission; (iii) both spontaneous and stimulated emission.

ANSWER

❙ (i) In a neon light fixture, a large potential difference is applied between the ends of a neon-filled glass tube. This ionizes some of the neon atoms, allowing a current of electrons to flow through the gas. Some of the neon atoms are struck by fast-moving electrons, making them transition to an excited level. From this level the atoms undergo *spontaneous* emission, as depicted in Fig. 39.28b, and emit 632.8 nm photons in the process. No population inversion occurs and the photons are not trapped by mirrors as shown in Fig. 39.29d, so there is no stimulated emission. Hence there is no laser action.

39.5 CONTINUOUS SPECTRA

Emission line spectra come from matter in the gaseous state, in which the atoms are so far apart that interactions between them are negligible and each atom behaves as an isolated system. By contrast, a heated solid or liquid (in which atoms are close to each other) nearly always emits radiation with a *continuous* distribution of wavelengths rather than a line spectrum.

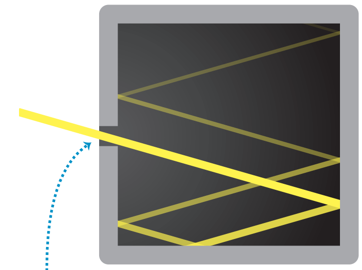
Here's an analogy that suggests why there is a difference. A tuning fork emits sound waves of a single definite frequency (a pure tone) when struck. But if you tightly pack a suitcase full of tuning forks and then shake the suitcase, the proximity of the tuning forks to each other affects the sound that they produce. What you hear is mostly noise, which is sound with a continuous distribution of all frequencies. In the same manner, isolated atoms in a gas emit light of certain distinct frequencies when excited, but if the same atoms are crowded together in a solid or liquid they produce a continuous spectrum of light.

In this section we'll study an idealized case of continuous-spectrum radiation from a hot, dense object. Just as was the case for the emission line spectrum of light from an atom, we'll find that we can understand the continuous spectrum only if we use the ideas of energy levels and photons.

In the same way that an atom's emission spectrum has the same lines as its absorption spectrum, the ideal surface for *emitting* light with a continuous spectrum is one that also *absorbs* all wavelengths of electromagnetic radiation. Such an ideal surface is called a *blackbody* because it would appear perfectly black when illuminated; it would reflect no light at all. The continuous-spectrum radiation that a blackbody emits is called **blackbody radiation.** Like a perfectly frictionless incline or a massless rope, a perfect blackbody does not exist but is nonetheless a useful idealization.

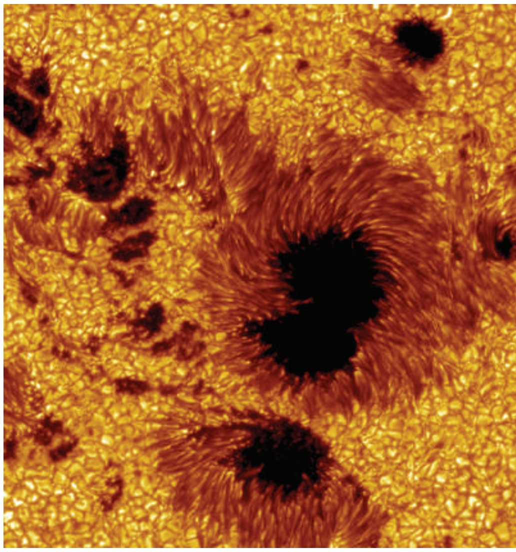
A good approximation to a blackbody is a hollow box with a small aperture in one wall (**Fig. 39.30**). Light that enters the aperture will eventually be absorbed by the walls of the box, so the box is a nearly perfect absorber. Conversely, when we heat the box, the light that emanates from the aperture is nearly ideal blackbody radiation with a continuous spectrum.

Figure 39.30 A hollow box with a small aperture behaves like a blackbody. When the box is heated, the electromagnetic radiation that emerges from the aperture has a blackbody spectrum.



By 1900 blackbody radiation had been studied extensively, and three characteristics had been established. First, the total intensity I (the average rate of radiation of energy per unit surface area or average power per area) emitted from the surface of an ideal radiator is proportional to the fourth power of the absolute temperature (**Fig. 39.31**). This is the **Stefan–Boltzmann law:**

Figure 39.31 This close-up view of the sun's surface shows dark sunspots. Their temperature is about 4000 K, while the surrounding solar material is at $T =$ 5800 K. From the Stefan–Boltzmann law, the intensity from a given area of sunspot is only $(4000\text{ K}/5800\text{ K})^4 = 0.23$ as great as the intensity from the same area of the surrounding material—which is why sunspots appear dark.



Stefan–Boltzmann law for a blackbody:

Intensity of radiation from blackbody

$$I = \sigma T^4 \qquad (39.19)$$

Absolute temperature of blackbody

Stefan–Boltzmann constant

We encountered a version of this relationship in Section 17.7 during our study of heat transfer. To nine significant figures, the value of the Stefan–Boltzmann constant σ is

$$\sigma = 5.67037442 \times 10^{-8}\ \text{W/m}^2 \cdot \text{K}^4$$

Second, the intensity is not uniformly distributed over all wavelengths. Its distribution can be measured and described by the intensity per wavelength interval $I(\lambda)$, called the *spectral emittance*. Thus $I(\lambda)\,d\lambda$ is the intensity corresponding to wavelengths in the interval from λ to $\lambda + d\lambda$. The *total* intensity I, given by Eq. (39.19), is the *integral* of the distribution function $I(\lambda)$ over all wavelengths, which equals the area under the $I(\lambda)$-versus-λ curve:

$$I = \int_0^\infty I(\lambda)\,d\lambda \qquad (39.20)$$

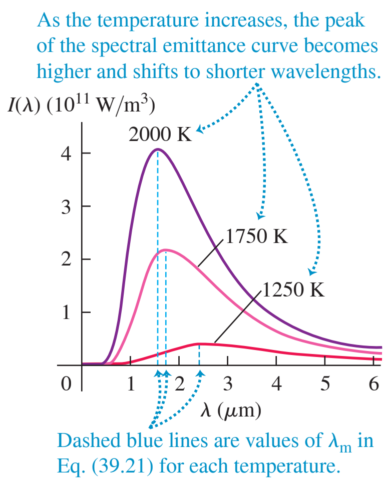


Figure 39.32 These graphs show the spectral emittance $I(\lambda)$ for radiation from a blackbody at three different temperatures.

CAUTION **Spectral emittance vs. intensity** Although we use the symbol $I(\lambda)$ for spectral emittance, keep in mind that spectral emittance is *not* the same thing as intensity I. Intensity is power per unit area, with units W/m^2. Spectral emittance is power per unit area *per unit wavelength interval*, with units W/m^3.

Figure 39.32 shows the measured spectral emittances $I(\lambda)$ for blackbody radiation at three different temperatures. Each has a peak wavelength λ_m at which the emitted intensity per wavelength interval is largest. Experiment shows that λ_m is inversely proportional to T, so their product is constant and equal to 2.90×10^{-3} m·K. This observation is called the **Wien displacement law:**

Wien displacement law for a blackbody:

$$\underbrace{\lambda_m}_{\text{Peak wavelength in spectral emittance curve}} \underbrace{T}_{\text{Absolute temperature of blackbody}} = 2.90 \times 10^{-3}\ \text{m}\cdot\text{K} \qquad (39.21)$$

As the temperature rises, the peak of $I(\lambda)$ becomes higher and shifts to shorter wavelengths. Yellow light has shorter wavelengths than red light, so an object that glows yellow is hotter and brighter than one of the same size that glows red.

Third, experiments show that the *shape* of the distribution function is the same for all temperatures. We can make a curve for one temperature fit any other temperature by simply changing the scales on the graph.

Rayleigh and the "Ultraviolet Catastrophe"

During the last decade of the 19th century, many attempts were made to derive these empirical results about blackbody radiation from basic principles. In one attempt, the English physicist Lord Rayleigh considered the light enclosed within a rectangular box like that shown in Fig. 39.30. Such a box, he reasoned, has a series of possible *normal modes* for electromagnetic waves, as we discussed in Section 32.5. It also seemed reasonable to assume that the distribution of energy among the various modes would be given by the equipartition principle (see Section 18.4), which had been used successfully in the analysis of heat capacities.

Including both the electric- and magnetic-field energies, Rayleigh assumed that the total energy of each normal mode was equal to kT. Then by computing the *number* of normal modes corresponding to a wavelength interval $d\lambda$, Rayleigh calculated the expected distribution of wavelengths in the radiation within the box. Finally, he computed the predicted intensity distribution $I(\lambda)$ for the radiation emerging from the hole. His result was quite simple:

$$I(\lambda) = \frac{2\pi ckT}{\lambda^4} \qquad \text{(Rayleigh's calculation)} \qquad (39.22)$$

At large wavelengths this formula agrees quite well with the experimental results shown in Fig. 39.32, but there is serious disagreement at small wavelengths. The experimental curves in Fig. 39.32 fall toward zero at small λ. By contrast, Rayleigh's prediction in Eq. (39.22) goes in the opposite direction, approaching infinity as $1/\lambda^4$, a result that was called in Rayleigh's time the "ultraviolet catastrophe." Even worse, the integral of Eq. (39.22) over all λ is infinite, indicating an infinitely large *total* radiated intensity. Clearly, something is wrong.

Planck and the Quantum Hypothesis

Finally, in 1900, the German physicist Max Planck succeeded in deriving a function, now called the **Planck radiation law,** that agreed very well with experimental intensity distribution curves. In his derivation he made what seemed at the time to be a crazy assumption: that electromagnetic oscillators (electrons) in the walls of Rayleigh's box vibrating at a frequency f could have only certain values of energy equal to nhf, where $n = 0, 1, 2, 3, \ldots$ and h is the constant that now bears Planck's name. These oscillators were in equilibrium with the electromagnetic waves in the box, so they both emitted and absorbed light. His assumption gave quantized energy levels and said that the energy in each normal mode was also a multiple of hf. This was in sharp contrast to Rayleigh's point of view that each normal mode could have any amount of energy.

Planck was not comfortable with this quantum hypothesis; he regarded it as a calculational trick rather than a fundamental principle. In a letter to a friend, he called it "an act of desperation" into which he was forced because "a theoretical explanation had to be found at any cost, whatever the price." But five years later, Einstein identified the energy change hf between levels as the energy of a photon (see Section 38.1), and other evidence quickly mounted. By 1915 there was little doubt about the validity of the quantum concept and the existence of photons. By discussing atomic spectra *before* continuous spectra, we have departed from the historical order of things. The credit for inventing the concept of quantization of energy levels goes to Planck, even though he didn't believe it at first. He received the 1918 Nobel Prize in physics for his achievements.

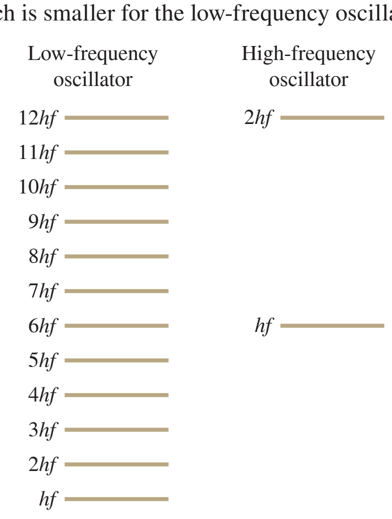
Figure 39.33 shows energy-level diagrams for two of the oscillators that Planck envisioned in the walls of the rectangular box, one with a low frequency and the other with a high frequency. The spacing in energy between adjacent levels is hf. This spacing is small for the low-frequency oscillator that emits and absorbs photons of low frequency f and long wavelength $\lambda = c/f$. The energy spacing is greater for the high-frequency oscillator, which emits high-frequency photons of short wavelength.

According to Rayleigh's picture, both of these oscillators have the same amount of energy kT and are equally effective at emitting radiation. In Planck's model, however, the high-frequency oscillator is very ineffective as a source of light. To see why, we can use the ideas from Section 39.4 about the populations of various energy states. If we consider all the oscillators of a given frequency f in a box at temperature T, the number of oscillators that have energy nhf is $Ae^{-nhf/kT}$. The ratio of the number of oscillators in the first excited state ($n = 1$, energy hf) to the number of oscillators in the ground state ($n = 0$, energy zero) is

$$\frac{n_1}{n_0} = \frac{Ae^{-hf/kT}}{Ae^{-(0)/kT}} = e^{-hf/kT} \tag{39.23}$$

Let's evaluate Eq. (39.23) for $T = 2000$ K, one of the temperatures shown in Fig. 39.32. At this temperature $kT = 2.76 \times 10^{-20}$ J $= 0.172$ eV. For an oscillator that emits photons

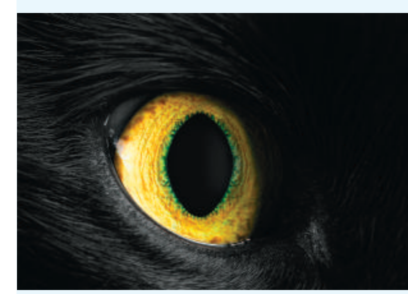
Figure **39.33** Energy levels for two of the oscillators that Planck envisioned in the walls of a blackbody like that shown in Fig. 39.30. The spacing between adjacent energy levels for each oscillator is hf, which is smaller for the low-frequency oscillator.



BIO APPLICATION Blackbody Eyes The interior of (a) a human eye, (b) a cat eye, or (c) a fish eye looks black, even though the tissue inside the eye is *not* black. That's because each eye acts like a blackbody, akin to the hollow box in Fig. 39.30: Light entering the eye is eventually absorbed after several reflections from the interior surfaces. Each eye also radiates like a blackbody, although the temperature is so low (around 300 K) that this radiation is principally at invisible infrared wavelengths.

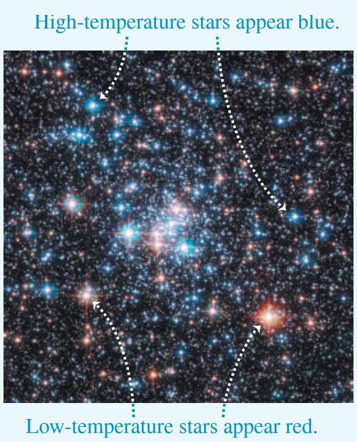
(a)

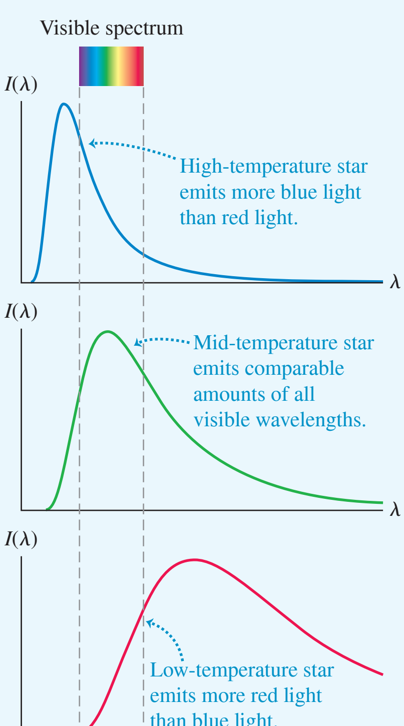
(b)



(c)

APPLICATION Star Colors and the Planck Radiation Law Stars (with radiation very similar to that of a blackbody) have a broad range of surface temperatures, from lower than 2500 K to higher than 30,000 K. The Wien displacement law and the shape of the Planck spectral emittance curve explain why these stars have different colors. From Eq. (39.21), a star with a high surface temperature of, say, 12,000 K has a short peak wavelength λ_m in the ultraviolet. Hence such a star emits more blue light than red light and appears blue to the eye. A star with a low surface temperature of, say, 3000 K has a long peak wavelength λ_m in the infrared, emits more red light than blue light, and appears red to the eye. For a star like the sun, which has a surface temperature of 5800 K, λ_m lies in the visible spectrum and the star appears white.





of wavelength $\lambda = 3.00\ \mu\text{m}$, $hf = hc/\lambda = 0.413$ eV; for a higher-frequency oscillator that emits photons of wavelength $\lambda = 0.500\ \mu\text{m}$, $hf = hc/\lambda = 2.48$ eV. For these two cases Eq. (39.23) gives

$$\frac{n_1}{n_0} = e^{-hf/kT} = 0.0909 \text{ for } \lambda = 3.00\ \mu\text{m}$$

$$\frac{n_1}{n_0} = e^{-hf/kT} = 5.64 \times 10^{-7} \text{ for } \lambda = 0.500\ \mu\text{m}$$

The value for $\lambda = 3.00\ \mu\text{m}$ means that of all the oscillators that can emit light at this wavelength, 0.0909 of them—about one in 11—are in the first excited state. These excited oscillators can each emit a 3.00 μm photon and contribute it to the radiation inside the box. Hence we would expect that this radiation would be rather plentiful in the spectrum of radiation from a 2000 K blackbody. By contrast, the value for $\lambda = 0.500\ \mu\text{m}$ means that only 5.64×10^{-7} (about one in two million) of the oscillators that can emit this wavelength are in the first excited state. An oscillator can't emit if it's in the ground state, so the amount of radiation in the box at this wavelength is *tremendously* suppressed compared to Rayleigh's prediction. That's why the spectral emittance curve for 2000 K in Fig. 39.32 has such a low value at $\lambda = 0.500\ \mu\text{m}$ and shorter wavelengths. So Planck's quantum hypothesis provided a natural way to suppress the spectral emittance of a blackbody at short wavelengths, and hence averted the ultraviolet catastrophe that plagued Rayleigh's calculations.

We won't go into all the details of Planck's derivation of the spectral emittance. Here is his result:

Planck radiation law:

Spectral emittance of blackbody · Planck's constant · Speed of light in vacuum

$$I(\lambda) = \frac{2\pi hc^2}{\lambda^5(e^{hc/\lambda kT} - 1)} \qquad (39.24)$$

Wavelength · Boltzmann constant · Absolute temperature of blackbody

This function turns out to agree well with experimental emittance curves such as those in Fig. 39.32.

The Planck radiation law also contains the Wien displacement law and the Stefan–Boltzmann law as consequences. To derive the Wien law, we find the value of λ at which $I(\lambda)$ is maximum by taking the derivative of Eq. (39.24) and setting it equal to zero. We leave it to you to fill in the details; the result is

$$\lambda_m = \frac{hc}{4.965kT} \qquad (39.25)$$

To obtain this result, you have to solve the equation

$$5 - x = 5e^{-x} \qquad (39.26)$$

The root of this equation, found by trial and error or more sophisticated means, is 4.965 to four significant figures. You should evaluate the constant $hc/4.965k$ and show that it agrees with the experimental value of $2.90 \times 10^{-3}\ \text{m} \cdot \text{K}$ given in Eq. (39.21).

We can obtain the Stefan–Boltzmann law for a blackbody by integrating Eq. (39.24) over all λ to find the *total* radiated intensity (see Problem 39.59). This is not a simple integral; the result is

$$I = \int_0^\infty I(\lambda)\, d\lambda = \frac{2\pi^5 k^4}{15c^2h^3} T^4 = \sigma T^4 \qquad (39.27)$$

in agreement with Eq. (39.19). Our result in Eq. (39.27) also shows that the constant σ in that law can be expressed in terms of other fundamental constants:

$$\sigma = \frac{2\pi^5 k^4}{15c^2h^3} \qquad (39.28)$$

Substitute the values of k, c, and h from Appendix F and verify that you obtain the value $\sigma = 5.6704 \times 10^{-8}\ \mathrm{W/m^2 \cdot K^4}$ for the Stefan–Boltzmann constant.

The Planck radiation law, Eq. (39.24), looks so different from the unsuccessful Rayleigh expression, Eq. (39.22), that it may seem unlikely that they would agree for any value of λ. But when λ is large, the exponent in the denominator of Eq. (39.24) is very small. We can then use the approximation $e^x \approx 1 + x$ (for x much less than 1). You should verify that when this is done, the result approaches Eq. (39.22), showing that the two expressions do agree in the limit of very large λ. We also note that the Rayleigh expression does not contain h. At very long wavelengths (very small photon energies), quantum effects become unimportant.

EXAMPLE 39.7 Light from the sun

WITH VARIATION PROBLEMS

To a good approximation, the sun's surface is a blackbody with a surface temperature of 5800 K. (We are ignoring the absorption produced by the sun's atmosphere, shown in Fig. 39.9.) (a) At what wavelength does the sun emit most strongly? (b) What is the total radiated power per unit surface area?

IDENTIFY and SET UP Our target variables are the peak-intensity wavelength λ_m and the radiated power per area I. Hence we'll use the Wien displacement law, Eq. (39.21) (which relates λ_m to the blackbody temperature T), and the Stefan–Boltzmann law, Eq. (39.19) (which relates I to T).

EXECUTE (a) From Eq. (39.21),

$$\lambda_m = \frac{2.90 \times 10^{-3}\ \mathrm{m \cdot K}}{T} = \frac{2.90 \times 10^{-3}\ \mathrm{m \cdot K}}{5800\ \mathrm{K}}$$
$$= 0.500 \times 10^{-6}\ \mathrm{m} = 500\ \mathrm{nm}$$

(b) From Eq. (39.19),

$$I = \sigma T^4 = (5.67 \times 10^{-8}\ \mathrm{W/m^2 \cdot K^4})(5800\ \mathrm{K})^4$$
$$= 6.42 \times 10^7\ \mathrm{W/m^2} = 64.2\ \mathrm{MW/m^2}$$

EVALUATE The 500 nm wavelength found in part (a) is near the middle of the visible spectrum. This should not be a surprise: The human eye evolved to take maximum advantage of natural light.

The enormous value $I = 64.2\ \mathrm{MW/m^2}$ that we obtained in part (b) is the intensity at the *surface* of the sun, which is a sphere of radius 6.96×10^8 m. When this radiated energy reaches the earth, 1.50×10^{11} m away, the intensity has decreased by the factor $[(6.96 \times 10^8\ \mathrm{m})/(1.50 \times 10^{11}\ \mathrm{m})]^2 = 2.15 \times 10^{-5}$ to the still-impressive $1.4\ \mathrm{kW/m^2}$.

KEYCONCEPT The continuous spectrum of light emitted by a blackbody is maximum at a wavelength λ_{max} given by the Wien displacement law: The greater the absolute temperature T of the blackbody, the smaller the value of λ_{max}. The total power radiated from a unit surface area of the blackbody is proportional to T^4.

EXAMPLE 39.8 A slice of sunlight

WITH VARIATION PROBLEMS

Find the power per unit area radiated from the sun's surface in the wavelength range 600.0 to 605.0 nm.

IDENTIFY and SET UP This question concerns the power emitted by a blackbody over a narrow range of wavelengths, and so involves the spectral emittance $I(\lambda)$ given by the Planck radiation law, Eq. (39.24). This requires that we find the area under the $I(\lambda)$ curve between 600.0 and 605.0 nm. We'll *approximate* this area as the product of the height of the curve at the median wavelength $\lambda = 602.5$ nm and the width of the interval, $\Delta\lambda = 5.0$ nm. From Example 39.7, $T = 5800$ K.

EXECUTE To obtain the height of the $I(\lambda)$ curve at $\lambda = 602.5\ \mathrm{nm} = 6.025 \times 10^{-7}$ m, we first evaluate the quantity $hc/\lambda kT$ in Eq. (39.24) and then substitute the result into Eq. (39.24):

$$\frac{hc}{\lambda kT} = \frac{(6.626 \times 10^{-34}\ \mathrm{J \cdot s})(2.998 \times 10^8\ \mathrm{m/s})}{(6.025 \times 10^{-7}\ \mathrm{m})(1.381 \times 10^{-23}\ \mathrm{J/K})(5800\ \mathrm{K})} = 4.116$$

$$I(\lambda) = \frac{2\pi(6.626 \times 10^{-34}\ \mathrm{J \cdot s})(2.998 \times 10^8\ \mathrm{m/s})^2}{(6.025 \times 10^{-7}\ \mathrm{m})^5(e^{4.116} - 1)}$$
$$= 7.81 \times 10^{13}\ \mathrm{W/m^3}$$

The intensity in the 5.0 nm range from 600.0 to 605.0 nm is then approximately

$$I(\lambda)\Delta\lambda = (7.81 \times 10^{13}\ \mathrm{W/m^3})(5.0 \times 10^{-9}\ \mathrm{m})$$
$$= 3.9 \times 10^5\ \mathrm{W/m^2} = 0.39\ \mathrm{MW/m^2}$$

EVALUATE In part (b) of Example 39.7, we found the power radiated per unit area by the sun at *all* wavelengths to be $I = 64.2\ \mathrm{MW/m^2}$; here we have found that the power radiated per unit area in the wavelength range from 600 to 605 nm is $I(\lambda)\Delta\lambda = 0.39\ \mathrm{MW/m^2}$, about 0.6% of the total.

KEYCONCEPT The Planck radiation law gives the spectral emittance (power per unit area per unit wavelength interval) $I(\lambda)$ for an ideal blackbody. To find the power per unit area emitted over a range of wavelengths, integrate $I(\lambda)$ over that range.

TEST YOUR UNDERSTANDING OF SECTION 39.5 (a) Does a blackbody at 2000 K emit x rays? (b) Does it emit radio waves?

ANSWER

❙ (a) **yes,** (b) **yes** The Planck radiation law, Eq. (39.24), shows that an ideal blackbody emits radiation at *all* wavelengths: The spectral emittance $I(\lambda)$ is equal to zero only for $\lambda = 0$ and in the limit $\lambda \rightarrow \infty$. So a blackbody at 2000 K does indeed emit both x rays and radio waves. However, Fig. 39.32 shows that the spectral emittance for this temperature is very low for wavelengths much shorter than 1 μm (including x rays) and for wavelengths much longer than a few μm (including radio waves). Hence such a blackbody emits very little in the way of x rays or radio waves.

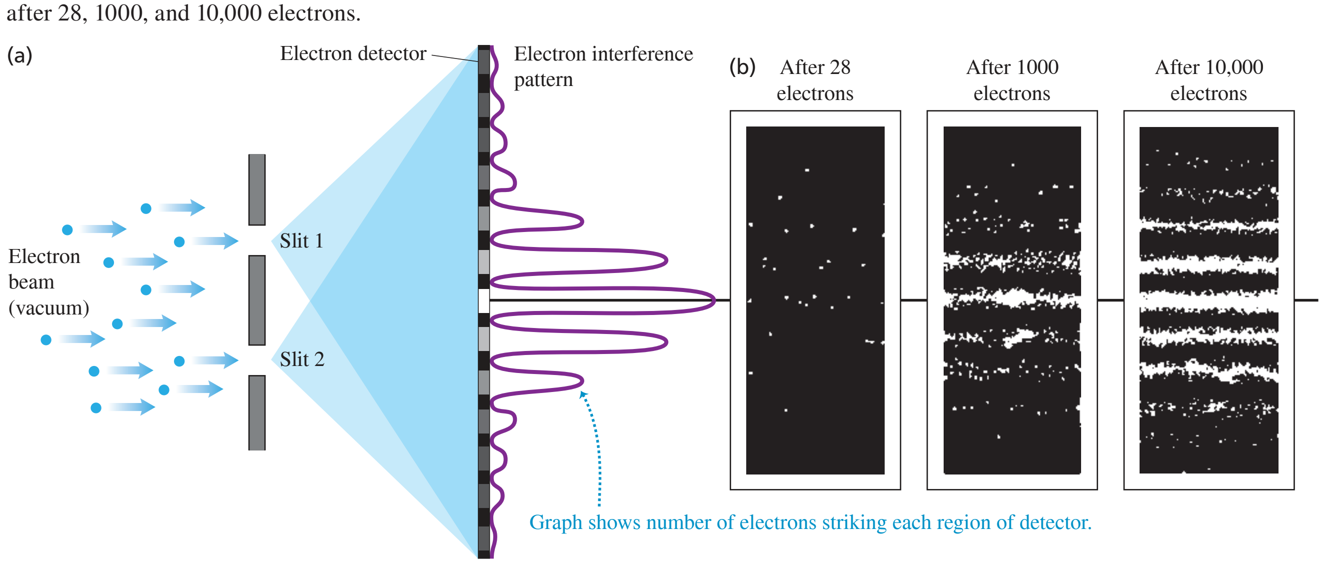
39.6 THE UNCERTAINTY PRINCIPLE REVISITED

The discovery of the dual wave–particle nature of matter forces us to reevaluate the kinematic language we use to describe the position and motion of a particle. In classical Newtonian mechanics we think of a particle as a point. We can describe its location and state of motion at any instant with three spatial coordinates and three components of velocity. But because matter also has a wave aspect, when we look at the behavior on a small enough scale—comparable to the de Broglie wavelength of the particle—we can no longer use the Newtonian description. Certainly no Newtonian particle would undergo diffraction like electrons do (Section 39.1).

To demonstrate just how non-Newtonian the behavior of matter can be, let's look at an experiment involving the two-slit interference of electrons (**Fig. 39.34**). We aim an electron beam at two parallel slits, as we did for light in Section 38.4. (The electron experiment has to be done in vacuum so that the electrons don't collide with air molecules.) What kind of pattern appears on the detector on the other side of the slits? The answer is: *exactly the same* kind of interference pattern we saw for photons in Section 38.4! Moreover, the principle of complementarity, which we introduced in Section 38.4, tells us that we cannot apply the wave and particle models simultaneously to describe any single element of this experiment. Thus we *cannot* predict exactly where in the pattern (a wave phenomenon) any individual electron (a particle) will land. We can't even ask which slit an individual electron passes through. If we tried to look at where the electrons were going by shining a light on them—that is, by scattering photons off them—the electrons would recoil, which would modify their motions so that the two-slit interference pattern would not appear.

CAUTION Electron two-slit interference is not interference between two electrons It's a common misconception that the pattern in Fig. 39.34b is due to the interference between *two* electron waves, each representing an electron passing through one slit. To show that this cannot be the case, we can send just one electron at a time through the apparatus. It makes no difference; we end up with the same interference pattern. In a sense, each electron wave interferes with itself. ❙

Figure **39.34** (a) A two-slit interference experiment for electrons. (b) The interference pattern after 28, 1000, and 10,000 electrons.



The Heisenberg Uncertainty Principles for Matter

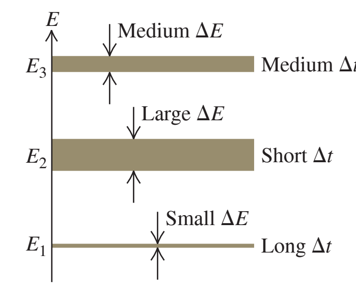
Just as electrons and photons show the same behavior in a two-slit interference experiment, electrons and other forms of matter obey the same Heisenberg uncertainty principles as photons do:

$$\Delta x \Delta p_x \geq \hbar/2 \quad \Delta y \Delta p_y \geq \hbar/2 \quad \Delta z \Delta p_z \geq \hbar/2 \qquad \text{(Heisenberg uncertainty principle for position and momentum)} \tag{39.29}$$

$$\Delta t\, \Delta E \geq \hbar/2 \qquad \text{(Heisenberg uncertainty principle for energy and time interval)} \tag{39.30}$$

In these equations $\hbar = h/2\pi = 1.055 \times 10^{-34}\ \text{J} \cdot \text{s}$. The uncertainty principle for energy and time interval has a direct application to energy levels. We have assumed that each energy level in an atom has a very definite energy. However, Eq. (39.30) says that this is not true for all energy levels. A system that remains in a metastable state for a very long time (large Δt) can have a very well-defined energy (small ΔE), but if it remains in a state for only a short time (small Δt) the uncertainty in energy must be correspondingly greater (large ΔE). **Figure 39.35** illustrates this idea.

Figure **39.35** The longer the lifetime Δt of a state, the smaller is its spread in energy (shown by the width of the energy levels).



EXAMPLE 39.9 The uncertainty principle: position and momentum

An electron is confined within a region of width 5.000×10^{-11} m (roughly the Bohr radius). (a) Estimate the minimum uncertainty in the x-component of the electron's momentum. (b) What is the kinetic energy of an electron with this magnitude of momentum? Express your answer in both joules and electron volts.

IDENTIFY and SET UP This problem uses the Heisenberg uncertainty principle for position and momentum and the relationship between a particle's momentum and its kinetic energy. The electron could be anywhere within the region, so we take $\Delta x = 5.000 \times 10^{-11}$ m as its position uncertainty. We then find the momentum uncertainty Δp_x from Eq. (39.29) and the kinetic energy from the relationships $p = mv$ and $K = \frac{1}{2}mv^2$.

EXECUTE (a) From Eqs. (39.29), for a given value of Δx, the uncertainty in momentum is minimum when the product $\Delta x\, \Delta p_x$ equals $\hbar/2$. Hence

$$\Delta p_x = \frac{\hbar}{2\Delta x} = \frac{1.055 \times 10^{-34}\ \text{J} \cdot \text{s}}{2(5.000 \times 10^{-11}\ \text{m})} = 1.055 \times 10^{-24}\ \text{J} \cdot \text{s/m}$$
$$= 1.055 \times 10^{-24}\ \text{kg} \cdot \text{m/s}$$

(b) We can rewrite the nonrelativistic expression for kinetic energy as

$$K = \tfrac{1}{2}mv^2 = \frac{(mv)^2}{2m} = \frac{p^2}{2m}$$

Hence an electron with a magnitude of momentum equal to Δp_x from part (a) has kinetic energy

$$K = \frac{p^2}{2m} = \frac{(1.055 \times 10^{-24}\ \text{kg} \cdot \text{m/s})^2}{2(9.11 \times 10^{-31}\ \text{kg})}$$
$$= 6.11 \times 10^{-19}\ \text{J} = 3.81\ \text{eV}$$

EVALUATE This energy is typical of electron energies in atoms. This agreement suggests that the uncertainty principle is deeply involved in atomic structure.

A similar calculation explains why electrons in atoms do not fall into the nucleus. If an electron were confined to the interior of a nucleus, its position uncertainty would be $\Delta x \approx 10^{-14}$ m. This would give the electron a momentum uncertainty about 5000 times greater than that of the electron in this example, and a kinetic energy so great that the electron would immediately be ejected from the nucleus.

KEYCONCEPT The Heisenberg uncertainty principle for position and momentum applies to particles as well as to photons. The smaller the uncertainty in the position of a particle, the greater the uncertainty in the particle's momentum. Consequently, a particle confined to a small region of space could have a very large magnitude of momentum and a very large kinetic energy.

EXAMPLE 39.10 The uncertainty principle: energy and time

A sodium atom in one of the states labeled "Lowest excited levels" in Fig. 39.19a remains in that state, on average, for 1.6×10^{-8} s before it makes a transition to the ground state, emitting a photon with wavelength 589.0 nm and energy 2.105 eV. What is the uncertainty in energy of that excited state? What is the wavelength spread of the corresponding spectral line?

IDENTIFY and SET UP We use the Heisenberg uncertainty principle for energy and time interval and the relationship between photon energy and wavelength. The average time that the atom spends in this excited state is equal to Δt in Eq. (39.30). We find the minimum uncertainty in the energy of the excited state by replacing the $\geq$ sign in Eq. (39.30) with an equals sign and solving for ΔE.

EXECUTE From Eq. (39.30),

$$\Delta E = \frac{\hbar}{2\Delta t} = \frac{1.055 \times 10^{-34}\ \text{J} \cdot \text{s}}{2(1.6 \times 10^{-8}\ \text{s})}$$
$$= 3.3 \times 10^{-27}\ \text{J} = 2.1 \times 10^{-8}\ \text{eV}$$

Continued

The atom remains in the ground state indefinitely, so that state has *no* associated energy uncertainty. The fractional uncertainty of the *photon* energy is therefore

$$\frac{\Delta E}{E} = \frac{2.1 \times 10^{-8}\ \text{eV}}{2.105\ \text{eV}} = 1.0 \times 10^{-8}$$

You can use some simple calculus and the relationship $E = hc/\lambda$ to show that $\Delta\lambda/\lambda \approx \Delta E/E$, so that the corresponding spread in wavelength, or "width," of the spectral line is approximately

$$\Delta\lambda = \lambda\frac{\Delta E}{E} = (589.0\ \text{nm})(1.0 \times 10^{-8}) = 0.0000059\ \text{nm}$$

EVALUATE This irreducible uncertainty $\Delta\lambda$ is called the *natural line width* of this particular spectral line. Though very small, it is within the limits of resolution of present-day spectrometers. Ordinarily, the natural line width is much smaller than the line width arising from other causes such as the Doppler effect and collisions among the rapidly moving atoms.

KEYCONCEPT The Heisenberg uncertainty principle for energy and time interval states that the shorter the duration of an excited state of a system, the greater the minimum uncertainty in the energy of that state.

The Uncertainty Principle and the Limits of the Bohr Model

We saw in Section 39.3 that the Bohr model of the hydrogen atom was tremendously successful. However, the Heisenberg uncertainty principle for position and momentum shows that this model *cannot* be a correct description of how an electron in an atom behaves. Figure 39.22 shows that in the Bohr model as interpreted by de Broglie, an electron wave moves in a plane around the nucleus. Let's call this the xy-plane, so the z-axis is perpendicular to the plane. Hence the Bohr model says that an electron is always found at $z = 0$, and its z-momentum p_z is always zero (the electron does not move out of the xy-plane). But this implies that there are *no* uncertainties in either z or p_z, so $\Delta z = 0$ and $\Delta p_z = 0$. This directly contradicts Eq. (39.29), which says that the product $\Delta z \Delta p_z$ must be greater than or equal to $\hbar/2$.

This conclusion isn't too surprising, since the electron in the Bohr model is a mix of particle and wave ideas (the electron moves in an orbit like a miniature planet, but has a wavelength). To get an accurate picture of how electrons behave inside an atom and elsewhere, we need a description that is based *entirely* on the electron's wave properties. Our goal in Chapter 40 will be to develop this description, which we call *quantum mechanics.* To do this we'll introduce the *Schrödinger equation,* the fundamental equation that describes the dynamics of matter waves. This equation, as we'll see, is as fundamental to quantum mechanics as Newton's laws are to classical mechanics or as Maxwell's equations are to electromagnetism.

TEST YOUR UNDERSTANDING OF SECTION 39.6 Rank the following situations according to the uncertainty in x-momentum, from largest to smallest. The mass of the proton is 1836 times the mass of the electron. (i) An electron whose x-coordinate is known to within 2×10^{-15} m; (ii) an electron whose x-coordinate is known to within 4×10^{-15} m; (iii) a proton whose x-coordinate is known to within 2×10^{-15} m; (iv) a proton whose x-coordinate is known to within 4×10^{-15} m.

ANSWER

(i) and (iii) (tie), (ii) and (iv) (tie) According to the Heisenberg uncertainty principle, the smaller the uncertainty Δx in the x-coordinate, the greater the uncertainty Δp_x in the x-momentum. The relationship between Δx and Δp_x does not depend on the mass of the particle, and so is the same for a proton as for an electron.

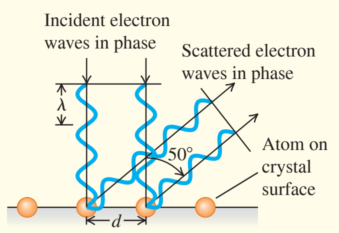
CHAPTER 39 SUMMARY

De Broglie waves and electron diffraction: Electrons and other particles have wave properties. A particle's wavelength depends on its momentum in the same way as for photons. A nonrelativistic electron accelerated from rest through a potential difference V_{ba} has a wavelength given by Eq. (39.3). Electron microscopes use the very small wavelengths of fast-moving electrons to make images with resolution thousands of times finer than is possible with visible light. (See Examples 39.1–39.3.)

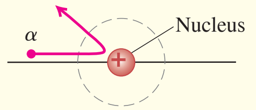
$$\lambda = \frac{h}{p} = \frac{h}{mv} \tag{39.1}$$

$$E = hf \tag{39.2}$$

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2meV_{ba}}} \tag{39.3}$$

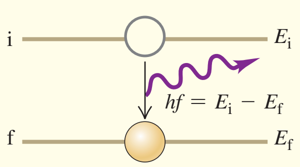


The nuclear atom: The Rutherford scattering experiments show that most of an atom's mass and all of its positive charge are concentrated in a tiny, dense nucleus at the center of the atom. (See Example 39.4.)



Atomic line spectra and energy levels: The energies of atoms are quantized: They can have only certain definite values, called energy levels. When an atom makes a transition from an energy level E_i to a lower level E_f, it emits a photon of energy $E_i - E_f$. The same photon can be absorbed by an atom in the lower energy level, which excites the atom to the upper level. (See Example 39.5.)

$$hf = \frac{hc}{\lambda} = E_i - E_f \tag{39.5}$$



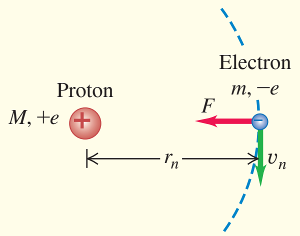
The Bohr model: In the Bohr model of the hydrogen atom, the permitted values of angular momentum are integral multiples of $h/2\pi$. The integer multiplier n is called the principal quantum number for the level. The orbital radii are proportional to n^2. The energy levels of the hydrogen atom are given by Eq. (39.15), where R is the Rydberg constant. (See Example 39.6.)

$$L_n = mv_nr_n = n\frac{h}{2\pi} \tag{39.6}$$
$$(n = 1, 2, 3, \ldots)$$

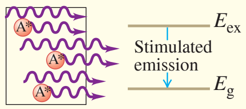
$$r_n = \epsilon_0\frac{n^2h^2}{\pi me^2} = n^2a_0 \tag{39.8), (39.11}$$

$$v_n = \frac{1}{\epsilon_0}\frac{e^2}{2nh} \tag{39.9}$$

$$E_n = -\frac{hcR}{n^2} = -\frac{13.60\text{ eV}}{n^2} \tag{39.15}$$
$$(n = 1, 2, 3, \ldots)$$



The laser: The laser operates on the principle of stimulated emission, by which many photons with identical wavelength and phase are emitted. Laser operation requires a nonequilibrium condition called a population inversion, in which more atoms are in a higher-energy state than are in a lower-energy state.

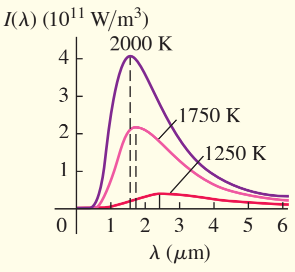


Blackbody radiation: The total radiated intensity (average power radiated per area) from a blackbody surface is proportional to the fourth power of the absolute temperature T. The quantity $\sigma = 5.67 \times 10^{-8}$ W/m$^2 \cdot$ K^4 is called the Stefan–Boltzmann constant. The wavelength λ_m at which a blackbody radiates most strongly is inversely proportional to T. The Planck radiation law gives the spectral emittance $I(\lambda)$ (intensity per wavelength interval in blackbody radiation). (See Examples 39.7 and 39.8.)

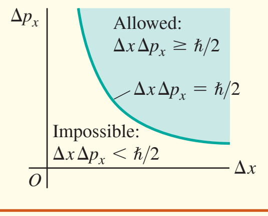
$$I = \sigma T^4$$
(Stefan–Boltzmann law) (39.19)

$$\lambda_m T = 2.90 \times 10^{-3}\text{ m} \cdot \text{K}$$
(Wien displacement law) (39.21)

$$I(\lambda) = \frac{2\pi hc^2}{\lambda^5(e^{hc/\lambda kT} - 1)}$$
(Planck radiation law) (39.24)



The Heisenberg uncertainty principle for particles: The same uncertainty considerations that apply to photons also apply to particles such as electrons. The uncertainty ΔE in the energy of a state that is occupied for a time Δt is given by Eq. (39.30), $\Delta t\, \Delta E \geq \hbar/2$. (See Examples 39.9 and 39.10.)



Chapter 39 Media Assets

GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review EXAMPLES 39.1 and 39.2 (Section 39.1) before attempting these problems.

VP39.2.1 In an electron-diffraction experiment, the spacing between atoms in the target is 0.172 nm and the electrons have negligible kinetic energy before being accelerated by an accelerating voltage of 69.0 V. Find (a) the de Broglie wavelength of each electron and (b) the smallest scattering angle for which there is a maximum in the diffraction pattern.

VP39.2.2 The $m = 1$ intensity maximum in an electron-diffraction experiment occurs at an angle of 48.0°. The accelerating voltage of the electrons is 36.5 V. Find (a) the kinetic energy of each electron in joules, (b) the electron de Broglie wavelength, and (c) the spacing between atoms in the target.

VP39.2.3 Using a target with a spacing between atoms of 0.218 nm, you want to produce an electron-diffraction pattern for which the $m = 2$ intensity maximum occurs at an angle of 75.0°. Find (a) the required electron de Broglie wavelength and (b) the required accelerating voltage.

VP39.2.4 A proton (charge e, mass 1.67×10^{-27} kg) is accelerated from rest to 2.38×10^5 m/s. Find (a) the proton de Broglie wavelength and (b) the accelerating voltage.

Be sure to review EXAMPLES 39.5 and 39.6 (Section 39.3) before attempting these problems.

VP39.6.1 A hypothetical atom has energy levels at 0.00 eV, 2.00 eV, and 5.00 eV. If the atom starts in the 5.00 eV level, find all the wavelengths that the atom could possibly emit in the process of returning to the ground level.

VP39.6.2 A hypothetical atom emits a photon of 674 nm when it transitions from the second excited level to the first excited level, and a photon of 385 nm when it transitions from the first excited level to the ground level. Find (a) the energies of the first and second excited levels relative to the ground level and (b) the wavelength of the photon emitted when the atom transitions from the second excited energy level to the ground level.

VP39.6.3 Calculate the energy and the wavelength of the photon emitted by a hydrogen atom when it makes a transition from (a) the $n = 5$ level to the $n = 3$ level, (b) the $n = 4$ level to the $n = 2$ level, and (c) the $n = 3$ level to the $n = 1$ level.

VP39.6.4 A hydrogen atom makes a transition from the $n = 2$ level to the $n = 6$ level. In the Bohr model, find (a) the change in electron kinetic energy, (b) the change in electric potential energy, and (c) the wavelength of the photon that the atom absorbs to cause the transition.

Be sure to review EXAMPLES 39.7 and 39.8 (Section 39.5) before attempting these problems.

VP39.8.1 The star Betelgeuse has surface temperature 3590 K and can be regarded as a blackbody. (a) Find the wavelength at which Betelgeuse emits most strongly. Is this visible, ultraviolet, or infrared? (b) Find the amount of power radiated per unit area of the surface of Betelgeuse.

VP39.8.2 The total power radiated per unit area from a blackbody is 78.0 MW/m^2. Find (a) the temperature of the blackbody and (b) the wavelength at which the blackbody emits most strongly. (c) Is the wavelength in part (b) visible, ultraviolet, or infrared?

VP39.8.3 The spectral emittance curve for the star Rigel A is a good approximation of the curve for a blackbody. The maximum of this curve is at 239 nm. Find (a) the surface temperature of Rigel A and (b) the power that Rigel A radiates per unit surface area.

VP39.8.4 Proxima Centauri, the nearest star to our sun, has a surface temperature of 3040 K. Find (a) the wavelength at which this star emits most strongly and (b) the power that this star radiates per unit surface area in a range of wavelengths 12.0 nm wide centered on the wavelength in part (a).

BRIDGING PROBLEM Hot Stars and Hydrogen Clouds

Figure 39.36 shows a cloud, or *nebula,* of glowing hydrogen in interstellar space. The atoms in this cloud are excited by short-wavelength radiation emitted by the bright blue stars at the center of the nebula.

(a) The blue stars act as blackbodies and emit light with a continuous spectrum. What is the wavelength at which a star with a surface temperature of 15,100 K (about $2\frac{1}{2}$ times the surface temperature of the sun) has the maximum spectral emittance? In what region of the electromagnetic spectrum is this?

(b) Figure 39.32 shows that most of the energy radiated by a blackbody is at wavelengths between about one half and three times the wavelength of maximum emittance. If a hydrogen atom near the star in part (a) is initially in the ground level, what is the principal quantum number of the highest energy level to which it could be excited by a photon in this wavelength range?

(c) The red color of the nebula is primarily due to hydrogen atoms making a transition from $n = 3$ to $n = 2$ and emitting photons of wavelength 656.3 nm. In the Bohr model as interpreted by de Broglie, what are the *electron* wavelengths in the $n = 2$ and $n = 3$ levels?

Figure 39.36 The Rosette Nebula.

SOLUTION GUIDE

IDENTIFY and SET UP

1. To solve this problem you need to use your knowledge of both blackbody radiation (Section 39.5) and the Bohr model of the hydrogen atom (Section 39.3).
2. In part (a) the target variable is the wavelength at which the star emits most strongly; in part (b) the target variable is a principal quantum number, and in part (c) it is the de Broglie wavelength of an electron in the $n = 2$ and $n = 3$ Bohr orbits (see Fig. 39.24). Select the equations you'll need to find the target variables. (*Hint:* In Section 39.5 you learned how to find the energy change involved in a transition between two given levels of a hydrogen atom. Part (b) is a variation on this: You are to find the final level in a transition that starts in the $n = 1$ level and involves the absorption of a photon of a given wavelength and hence a given energy.)

EXECUTE

3. Use the Wien displacement law to find the wavelength at which the star has maximum spectral emittance. In what part of the electromagnetic spectrum is this wavelength?
4. Use your result from step 3 to find the range of wavelengths in which the star radiates most of its energy. Which end of this range corresponds to a photon with the greatest energy?
5. Write an expression for the wavelength of a photon that must be absorbed to cause an electron transition from the ground level $(n = 1)$ to a higher level n. Solve for the value of n that corresponds to the highest-energy photon in the range you calculated in step 4. (*Hint:* Remember that n must be an integer.)
6. Find the electron wavelengths that correspond to the $n = 2$ and $n = 3$ orbits shown in Fig. 39.22.

EVALUATE

7. Check your result in step 5 by calculating the wavelength needed to excite a hydrogen atom from the ground level into the level *above* the highest-energy level that you found in step 5. Is it possible for light in the range of wavelengths you found in step 4 to excite hydrogen atoms from the ground level into this level?
8. How do the electron wavelengths you found in step 6 compare to the wavelength of a *photon* emitted in a transition from the $n = 3$ level to the $n = 2$ level?

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q39.1 If a proton and an electron have the same speed, which has the longer de Broglie wavelength? Explain.

Q39.2 If a proton and an electron have the same kinetic energy, which has the longer de Broglie wavelength? Explain.

Q39.3 Does a photon have a de Broglie wavelength? If so, how is it related to the wavelength of the associated electromagnetic wave? Explain.

Q39.4 When an electron beam goes through a very small hole, it produces a diffraction pattern on a screen, just like that of light. Does this mean that an electron spreads out as it goes through the hole? What does this pattern mean?

Q39.5 Galaxies tend to be strong emitters of Lyman-α photons (from the $n = 2$ to $n = 1$ transition in atomic hydrogen). But the intergalactic medium—the very thin gas between the galaxies—tends to *absorb* Lyman-α photons. What can you infer from these observations about the temperature in these two environments? Explain.

Q39.6 A doubly ionized lithium atom (Li^{++}) is one that has had two of its three electrons removed. The energy levels of the remaining single-electron ion are closely related to those of the hydrogen atom. The nuclear charge for lithium is $+3e$ instead of just $+e$. How are the energy levels related to those of hydrogen? How is the *radius* of the ion in the ground level related to that of the hydrogen atom? Explain.

Q39.7 The emission of a photon by an isolated atom is a recoil process in which momentum is conserved. Thus Eq. (39.5) should include a recoil kinetic energy K_r for the atom. Why is this energy negligible in that equation?

Q39.8 How might the energy levels of an atom be measured directly—that is, without recourse to analysis of spectra?

Q39.9 Elements in the gaseous state emit line spectra with well-defined wavelengths. But hot solid objects always emit a continuous spectrum—that is, a continuous smear of wavelengths. Can you account for this difference?

Q39.10 As an object is heated to a very high temperature and becomes self-luminous, the apparent color of the emitted radiation shifts from red to yellow and finally to blue as the temperature increases. Why does the color shift? What other changes in the character of the radiation occur?

Q39.11 Do the planets of the solar system obey a distance law $(r_n = n^2 r_1)$ as the electrons of the Bohr atom do? Should they? Why (or why not)? (Consult Appendix F for the appropriate distances.)

Q39.12 You have been asked to design a magnet system to steer a beam of 54 eV electrons like those described in Example 39.1 (Section 39.1). The goal is to be able to direct the electron beam to a specific target location with an accuracy of ± 1.0 mm. In your design, do you need to take the wave nature of electrons into account? Explain.

Q39.13 Why go through the expense of building an electron microscope for studying very small objects such as organic molecules? Why not just use extremely short electromagnetic waves, which are much cheaper to generate?

Q39.14 Which has more total energy: a hydrogen atom with an electron in a high shell (large n) or in a low shell (small n)? Which is moving faster: the high-shell electron or the low-shell electron? Is there a contradiction here? Explain.

Q39.15 Does the uncertainty principle have anything to do with marksmanship? That is, is the accuracy with which a bullet can be aimed at a target limited by the uncertainty principle? Explain.

Q39.16 Suppose a two-slit interference experiment is carried out using an electron beam. Would the same interference pattern result if one slit at a time is uncovered instead of both at once? If not, why not? Doesn't each electron go through one slit or the other? Or does every electron go through both slits? Discuss the latter possibility in light of the principle of complementarity.

Q39.17 Equation (39.30) states that the energy of a system can have uncertainty. Does this mean that the principle of conservation of energy is no longer valid? Explain.

Q39.18 Laser light results from transitions from long-lived metastable states. Why is it more monochromatic than ordinary light?

Q39.19 Could an electron-diffraction experiment be carried out using three or four slits? Using a grating with many slits? What sort of results would you expect with a grating? Would the uncertainty principle be violated? Explain.

Q39.20 As the lower half of Fig. 39.4 shows, the diffraction pattern made by electrons that pass through aluminum foil is a series of concentric rings. But if the aluminum foil is replaced by a single crystal of aluminum, only certain points on these rings appear in the pattern. Explain.

Q39.21 Why can an electron microscope have greater magnification than an ordinary microscope?

Q39.22 When you check the air pressure in a tire, a little air always escapes; the process of making the measurement changes the quantity being measured. Think of other examples of measurements that change or disturb the quantity being measured.

EXERCISES

Section 39.1 Electron Waves

39.1 • (a) An electron moves with a speed of 4.70×10^6 m/s. What is its de Broglie wavelength? (b) A proton moves with the same speed. Determine its de Broglie wavelength.

39.2 •• For crystal diffraction experiments (discussed in Section 39.1), wavelengths on the order of 0.20 nm are often appropriate. Find the energy in electron volts for a particle with this wavelength if the particle is (a) a photon; (b) an electron; (c) an alpha particle $(m = 6.64 \times 10^{-27}\ \text{kg})$.

39.3 • An electron has a de Broglie wavelength of 2.80×10^{-10} m. Determine (a) the magnitude of its momentum and (b) its kinetic energy (in joules and in electron volts).

39.4 •• **Wavelength of an Alpha Particle.** An alpha particle $(m = 6.64 \times 10^{-27}\ \text{kg})$ emitted in the radioactive decay of uranium-238 has an energy of 4.20 MeV. What is its de Broglie wavelength?

39.5 • An electron is moving with a speed of 8.00×10^6 m/s. What is the speed of a proton that has the same de Broglie wavelength as this electron?

39.6 • (a) A nonrelativistic free particle with mass m has kinetic energy K. Derive an expression for the de Broglie wavelength of the particle in terms of m and K. (b) What is the de Broglie wavelength of an 800 eV electron?

39.7 • Calculate the de Broglie wavelength of (a) a 50.0 kg woman jogging leisurely at 2.0 m/s, (b) a free electron with kinetic energy 2.0 MeV, and (c) a free electron with kinetic energy 20 eV. Use the proper relativistic expression when necessary.

39.8 •• What is the de Broglie wavelength for an electron with speed (a) $v = 0.480c$ and (b) $v = 0.960c$? (*Hint:* Use the correct relativistic expression for linear momentum if necessary.)

39.9 • **Wavelength of a Bullet.** Calculate the de Broglie wavelength of a 5.00 g bullet that is moving at 340 m/s. Will the bullet exhibit wavelike properties?

39.10 •• Through what potential difference must electrons be accelerated if they are to have (a) the same wavelength as an x ray of wavelength 0.220 nm and (b) the same energy as the x ray in part (a)?

39.11 •• (a) What accelerating potential is needed to produce electrons of wavelength 5.00 nm? (b) What would be the energy of photons having the same wavelength as these electrons? (c) What would be the wavelength of photons having the same energy as the electrons in part (a)?

39.12 •• **CP** A beam of electrons is accelerated from rest through a potential difference of 0.100 kV and then passes through a thin slit. When viewed far from the slit, the diffracted beam shows its first diffraction minima at $\pm 14.6°$ from the original direction of the beam. (a) Do we need to use relativity formulas? How do you know? (b) How wide is the slit?

39.13 •• A beam of neutrons that all have the same energy scatters from atoms that have a spacing of 0.0910 nm in the surface plane of a crystal. The $m = 1$ intensity maximum occurs when the angle θ in Fig. 39.2 is 28.6°. What is the kinetic energy (in electron volts) of each neutron in the beam?

39.14 • (a) In an electron microscope, what accelerating voltage is needed to produce electrons with wavelength 0.0600 nm? (b) If protons are used instead of electrons, what accelerating voltage is needed to produce protons with wavelength 0.0600 nm? (*Hint:* In each case the initial kinetic energy is negligible.)

39.15 • A photon and a free electron each have an energy of 6.00 eV. (a) What is the wavelength of the photon if it is traveling in air? (b) What is the de Broglie wavelength of the electron? (c) Which wavelength is longer?

39.16 • A photon traveling in air has a wavelength of 500 nm, and a free electron has a de Broglie wavelength of 500 nm. (a) What is the energy of each, in eV? (b) Which energy is greater?

Section 39.2 The Nuclear Atom and Atomic Spectra

39.17 • A beam of alpha particles is incident on a target of lead. A particular alpha particle comes in "head-on" to a particular lead nucleus and stops 6.50×10^{-14} m away from the center of the nucleus. (This point is well outside the nucleus.) Assume that the lead nucleus, which has 82 protons, remains at rest. The mass of the alpha particle is 6.64×10^{-27} kg. (a) Calculate the electrostatic potential energy at the instant that the alpha particle stops. Express your result in joules and in MeV. (b) What initial kinetic energy (in joules and in MeV) did the alpha particle have? (c) What was the initial speed of the alpha particle?

39.18 •• **CP** A 4.78 MeV alpha particle from a ^{226}Ra decay makes a head-on collision with a uranium nucleus. A uranium nucleus has 92 protons. (a) What is the distance of closest approach of the alpha particle to the center of the nucleus? Assume that the uranium nucleus remains at rest and that the distance of closest approach is much greater than the radius of the uranium nucleus. (b) What is the force on the alpha particle at the instant when it is at the distance of closest approach?

Section 39.3 Energy Levels and the Bohr Model of the Atom

39.19 •• A hydrogen atom is in a state with energy -1.51 eV. In the Bohr model, what is the angular momentum of the electron in the atom, with respect to an axis at the nucleus?

39.20 • A hydrogen atom initially in its ground level absorbs a photon, which excites the atom to the $n = 3$ level. Determine the wavelength and frequency of the photon.

39.21 • A triply ionized beryllium ion, Be^{3+} (a beryllium atom with three electrons removed), behaves very much like a hydrogen atom except that the nuclear charge is four times as great. (a) What is the ground-level energy of Be^{3+}? How does this compare to the ground-level energy of the hydrogen atom? (b) What is the ionization energy of Be^{3+}? How does this compare to the ionization energy of the hydrogen atom? (c) For the hydrogen atom, the wavelength of the photon emitted in the $n = 2$ to $n = 1$ transition is 122 nm (see Example 39.6). What is the wavelength of the photon emitted when a Be^{3+} ion undergoes this transition? (d) For a given value of n, how does the radius of an orbit in Be^{3+} compare to that for hydrogen?

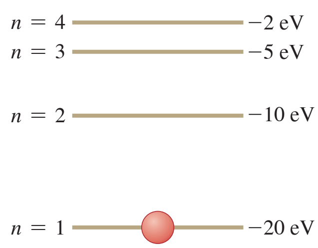
39.22 •• Consider the Bohr-model description of a hydrogen atom. (a) Calculate $E_2 - E_1$ and $E_{10} - E_9$. As n increases, does the energy separation between adjacent energy levels increase, decrease, or stay the same? (b) Show that $E_{n+1} - E_n$ approaches $(27.2\ \text{eV})/n^3$ as n becomes large. (c) How does $r_{n+1} - r_n$ depend on n? Does the radial distance between adjacent orbits increase, decrease, or stay the same as n increases?

39.23 • (a) Using the Bohr model, calculate the speed of the electron in a hydrogen atom in the $n = 1, 2,$ and 3 levels. (b) Calculate the orbital period in each of these levels. (c) The average lifetime of the first excited level of a hydrogen atom is 1.0×10^{-8} s. In the Bohr model, how many orbits does an electron in the $n = 2$ level complete before returning to the ground level?

39.24 • An electron is in a bound state of a hydrogen atom. The energy state of the atom is labeled with principal quantum number n. In the Bohr model description of this bound state, the electron has linear momentum $p = 6.65 \times 10^{-25}\ \text{kg} \cdot \text{m/s}$. In the Bohr model description, what are (a) the kinetic energy of the electron, (b) the angular momentum of the electron, and (c) the quantum number n?

39.25 • **CP** The energy-level scheme for the hypothetical one-electron element Searsium is shown in **Fig. E39.25**. The potential energy is taken to be zero for an electron at an infinite distance from the nucleus. (a) How much energy (in electron volts) does it take to ionize an electron from the ground level? (b) An 18 eV photon is absorbed by a Searsium atom in its ground level. As the atom returns to its ground level, what possible energies can the emitted photons have? Assume that there can be transitions between all pairs of levels. (c) What will happen if a photon with an energy of 8 eV strikes a Searsium atom in its ground level? Why? (d) Photons emitted in the Searsium transitions $n = 3 \rightarrow n = 2$ and $n = 3 \rightarrow n = 1$ will eject photoelectrons from an unknown metal, but the photon emitted from the transition $n = 4 \rightarrow n = 3$ will not. What are the limits (maximum and minimum possible values) of the work function of the metal?

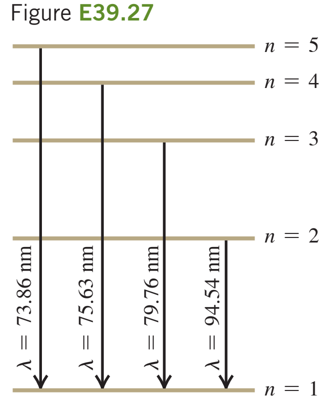
Figure **E39.25**



39.26 •• A positronium atom consists of a positron and an electron. In a Bohr-like model, the two particles rotate in circles about their common center of mass. (a) Calculate the reduced mass of a positronium atom in terms of the mass of an electron. (b) Determine the orbital radius of its ground-state electron. (c) Find its ground-state energy. (d) The longest visible-light emission wavelength for ordinary hydrogen is 656.3 nm in air and is for the $n = 3$ to $n = 2$ transition. Calculate the wavelength for the same transition in positronium.

39.27 • In a set of experiments on a hypothetical one-electron atom, you measure the wavelengths of the photons emitted from transitions ending in the ground level ($n = 1$), as shown in the energy-level diagram in **Fig. E39.27**. You also observe that it takes 17.50 eV to ionize this atom. (a) What is the energy of the atom in each of the levels ($n = 1$, $n = 2$, etc.) shown in the figure? (b) If an electron made a transition from the $n = 4$ to the $n = 2$ level, what wavelength of light would it emit?

Figure **E39.27**



39.28 • Find the longest and shortest wavelengths in the Lyman and Paschen series for hydrogen. In what region of the electromagnetic spectrum does each series lie?

39.29 • (a) An atom initially in an energy level with $E = -6.52$ eV absorbs a photon that has wavelength 860 nm. What is the internal energy of the atom after it absorbs the photon? (b) An atom initially in an energy level with $E = -2.68$ eV emits a photon that has wavelength 420 nm. What is the internal energy of the atom after it emits the photon?

39.30 •• Use Balmer's formula to calculate (a) the wavelength, (b) the frequency, and (c) the photon energy for the H_γ line of the Balmer series for hydrogen.

Section 39.4 The Laser

39.31 • **BIO Laser Surgery.** Using a mixture of CO_2, N_2, and sometimes He, CO_2 lasers emit a wavelength of 10.6 μm. At power outputs of 0.100 kW, such lasers are used for surgery. How many photons per second does a CO_2 laser deliver to the tissue during its use in an operation?

39.32 • **BIO Removing Birthmarks.** Pulsed dye lasers emit light of wavelength 585 nm in 0.45 ms pulses to remove skin blemishes such as birthmarks. The beam is usually focused onto a circular spot 5.0 mm in diameter. Suppose that the output of one such laser is 20.0 W. (a) What is the energy of each photon, in eV? (b) How many photons per square millimeter are delivered to the blemish during each pulse?

39.33 • How many photons per second are emitted by a 7.50 mW CO_2 laser that has a wavelength of 10.6 μm?

39.34 • **BIO PRK Surgery.** Photorefractive keratectomy (PRK) is a laser-based surgical procedure that corrects near- and farsightedness by removing part of the lens of the eye to change its curvature and hence focal length. This procedure can remove layers 0.25 μm thick using pulses lasting 12.0 ns from a laser beam of wavelength 193 nm. Low-intensity beams can be used because each individual photon has enough energy to break the covalent bonds of the tissue. (a) In what part of the electromagnetic spectrum does this light lie? (b) What is the energy of a single photon? (c) If a 1.50 mW beam is used, how many photons are delivered to the lens in each pulse?

39.35 • Figure 39.19a shows the energy levels of the sodium atom. The two lowest excited levels are shown in columns labeled $^2P_{3/2}$ and $^2P_{1/2}$. Find the ratio of the number of atoms in a $^2P_{3/2}$ state to the number in a $^2P_{1/2}$ state for a sodium gas in thermal equilibrium at 500 K. In which state are more atoms found?

Section 39.5 Continuous Spectra

39.36 • The temperature of a blackbody is changed so that the intensity I of radiation from the blackbody increases by a factor of 16. By what factor does the peak wavelength λ_m change?

39.37 •• A 100 W incandescent light bulb has a cylindrical tungsten filament 30.0 cm long, 0.40 mm in diameter, and with an emissivity of 0.26. (a) What is the temperature of the filament? (b) For what wavelength does the spectral emittance of the bulb peak? (c) Incandescent light bulbs are not very efficient sources of visible light. Explain why this is so.

39.38 • Determine λ_m, the wavelength at the peak of the Planck distribution, and the corresponding frequency f, at these temperatures: (a) 3.00 K; (b) 300 K; (c) 3000 K.

39.39 • Radiation has been detected from space that is characteristic of an ideal radiator at $T = 2.728$ K. (This radiation is a relic of the Big Bang at the beginning of the universe.) For this temperature, at what wavelength does the Planck distribution peak? In what part of the electromagnetic spectrum is this wavelength?

39.40 • The wavelength 10.0 μm is in the infrared region of the electromagnetic spectrum, whereas 600 nm is in the visible region and 100 nm is in the ultraviolet. What is the temperature of an ideal blackbody for which the peak wavelength λ_m is equal to each of these wavelengths?

39.41 •• Two stars, both of which behave like ideal blackbodies, radiate the same total energy per second. The cooler one has a surface temperature T and a diameter 3.0 times that of the hotter star. (a) What is the temperature of the hotter star in terms of T? (b) What is the ratio of the peak-intensity wavelength of the hot star to the peak-intensity wavelength of the cool star?

39.42 • **Sirius B.** The brightest star in the sky is Sirius, the Dog Star. It is actually a binary system of two stars, the smaller one (Sirius B) being a white dwarf. Spectral analysis of Sirius B indicates that its surface temperature is 24,000 K and that it radiates energy at a total rate of 1.0×10^{25} W. Assume that it behaves like an ideal blackbody. (a) What is the total radiated intensity of Sirius B? (b) What is the peak-intensity wavelength? Is this wavelength visible to humans? (c) What is the radius of Sirius B? Express your answer in kilometers and as a fraction of our sun's radius. (d) Which star radiates more *total* energy per second, the hot Sirius B or the (relatively) cool sun with a surface temperature of 5800 K? To find out, calculate the ratio of the total power radiated by our sun to the power radiated by Sirius B.

Section 39.6 The Uncertainty Principle Revisited

39.43 • (a) The x-coordinate of an electron is measured with an uncertainty of 0.30 mm. What is the x-component of the electron's velocity, v_x, if the minimum percent uncertainty in a simultaneous measurement of v_x is 1.0%? (b) Repeat part (a) for a proton.

39.44 • A pesky 1.5 mg mosquito is annoying you as you attempt to study physics in your room, which is 5.0 m wide and 2.5 m high. You decide to swat the bothersome insect as it flies toward you, but you need to estimate its speed to make a successful hit. (a) What is the maximum uncertainty in the horizontal position of the mosquito? (b) What limit does the Heisenberg uncertainty principle place on your ability to know the horizontal velocity of this mosquito? Is this limitation a serious impediment to your attempt to swat it?

39.45 • (a) The uncertainty in the y-component of a proton's position is 2.0×10^{-12} m. What is the minimum uncertainty in a simultaneous measurement of the y-component of the proton's velocity? (b) The uncertainty in the z-component of an electron's velocity is 0.250 m/s. What is the minimum uncertainty in a simultaneous measurement of the z-coordinate of the electron?

39.46 • A 10.0 g marble is gently placed on a horizontal tabletop that is 1.75 m wide. (a) What is the maximum uncertainty in the horizontal position of the marble? (b) According to the Heisenberg uncertainty principle, what is the minimum uncertainty in the horizontal velocity of the marble? (c) In light of your answer to part (b), what is the longest time the marble could remain on the table? Compare this time to the age of the universe, which is approximately 14 billion years. (*Hint:* Can you know that the horizontal velocity of the marble is *exactly* zero?)

39.47 • A scientist has devised a new method of isolating individual particles. He claims that this method enables him to detect simultaneously the position of a particle along an axis with a standard deviation of 0.12 nm and its momentum component along this axis with a standard deviation of 3.0×10^{-25} kg · m/s. Use the Heisenberg uncertainty principle to evaluate the validity of this claim.

PROBLEMS

39.48 • An atom with mass m emits a photon of wavelength λ. (a) What is the recoil speed of the atom? (b) What is the kinetic energy K of the recoiling atom? (c) Find the ratio K/E, where E is the energy of the emitted photon. If this ratio is much less than unity, the recoil of the atom can be neglected in the emission process. Is the recoil of the atom more important for small or large atomic masses? For long or short wavelengths? (d) Calculate K (in electron volts) and K/E for a hydrogen atom (mass 1.67×10^{-27} kg) that emits an ultraviolet photon of energy 10.2 eV. Is recoil an important consideration in this emission process?

39.49 •• The negative muon has a charge equal to that of an electron but a mass that is 207 times as great. Consider a hydrogenlike atom consisting of a proton and a muon. (a) What is the reduced mass of the atom? (b) What is the ground-level energy (in electron volts)? (c) What is the wavelength of the radiation emitted in the transition from the $n = 2$ level to the $n = 1$ level?

39.50 •• A hydrogen atom in an excited bound state labeled with principal quantum number $n = 3$ absorbs a photon that has wavelength λ. The atom is ionized and the electron has kinetic energy 8.00 eV after it has left the atom. What was the wavelength λ of the photon?

39.51 •• The wavelengths λ in the Pickering emission series are given by $\frac{1}{\lambda} = (1.097 \times 10^7\ \text{m}^{-1})\left[\frac{1}{4} - \frac{1}{(n/2)^2}\right]$ for $n = 5, 6, 7, \ldots$ and were attributed to hydrogen by some scientists. However, Bohr realized that this was *not* a hydrogen series, but rather belonged to another element, ionized so that it has only one electron. (a) What are the shortest and longest wavelengths in the Pickering series? (b) Which element gives rise to this series, and what is the common final-state quantum number n_L for each transition in the series?

39.52 •• In the Bohr model of the hydrogen atom, what is the de Broglie wavelength of the electron when it is in (a) the $n = 1$ level and (b) the $n = 4$ level? In both cases, compare the de Broglie wavelength to the circumference $2\pi r_n$ of the orbit.

39.53 ••• A sample of hydrogen atoms is irradiated with light with wavelength 85.5 nm, and electrons are observed leaving the gas. (a) If each hydrogen atom were initially in its ground level, what would be the maximum kinetic energy in electron volts of these photo-electrons? (b) A few electrons are detected with energies as much as 10.2 eV greater than the maximum kinetic energy calculated in part (a). How can this be?

39.54 •• **CALC** Consider two energy levels, 1 and 2, of an atom that have nearly the same energies, so that their energies E_1 and E_2 relative to the ground state are close in value. When an atom undergoes a transition from one of these states to the ground state, the wavelength of the photon emitted is λ_1 or λ_2, respectively. Since $\Delta E = |E_1 - E_2|$ is small, $\Delta\lambda = |\lambda_1 - \lambda_2|$ is small. (a) Use the fact that $\Delta\lambda$ is small to derive an expression for ΔE in terms of $\Delta\lambda$ and $\lambda \approx \lambda_1 \approx \lambda_2$. (b) The transitions that produce the 589.0 nm and 589.6 nm wavelengths in the atomic emission spectrum of sodium are shown in Fig. 39.19. Use the result from part (a) to calculate the energy difference, in eV, between the two initial states for these two transitions.

39.55 •• **The Red Supergiant Betelgeuse.** The star Betelgeuse has a surface temperature of 3000 K and is 600 times the diameter of our sun. (If our sun were that large, we would be inside it!) Assume that it radiates like an ideal blackbody. (a) If Betelgeuse were to radiate all of its energy at the peak-intensity wavelength, how many photons per second would it radiate? (b) Find the ratio of the power radiated by Betelgeuse to the power radiated by our sun (at 5800 K).

39.56 •• **CP** Light from an ideal spherical blackbody 15.0 cm in diameter is analyzed by using a diffraction grating that has 3850 lines/cm. When you shine this light through the grating, you observe that the peak-intensity wavelength forms a first-order bright fringe at $\pm 14.4°$ from the central bright fringe. (a) What is the temperature of the blackbody? (b) How long will it take this sphere to radiate 12.0 MJ of energy at constant temperature?

39.57 •• **CP** The moon has a mass of 7.35×10^{22} kg, and the length of a sidereal day is 27.3 days. (a) Estimate the de Broglie wavelength of the moon in its orbit around the earth. (b) Using M_{earth} for the mass of the earth and M_{moon} for the mass of the moon, we can use Newton's law of gravitation to determine the radius of the moon's orbit in terms of an integer-valued quantum number m as $R_m = m^2 a_{\text{moon}}$, where a_{moon} is the analog of the Bohr radius for the earth–moon gravitational system. Determine a_{moon} in terms of Newton's constant G, Planck's constant h, and the masses M_{earth} and M_{moon}. (c) The mass of the earth is $M_{\text{earth}} = 5.97 \times 10^{24}$ kg. Estimate the numerical value of a_{moon}. (d) The radius of the moon's orbit is 3.84×10^8 m. Estimate the moon's quantum number m. (e) The quantized energy levels of the moon are given by $E = -E_0/m^2$. Estimate the quantum ground-state energy E_0 of the moon.

39.58 •• An Ideal Blackbody. A large cavity that has a very small hole and is maintained at a temperature T is a good approximation to an ideal radiator or blackbody. Radiation can pass into or out of the cavity only through the hole. The cavity is a perfect absorber, since any radiation incident on the hole becomes trapped inside the cavity. Such a cavity at 400°C has a hole with area 4.00 mm^2. How long does it take for the cavity to radiate 100 J of energy through the hole?

39.59 •• CALC (a) Write the Planck distribution law in terms of the frequency f, rather than the wavelength λ, to obtain $I(f)$. (b) Show that

$$\int_0^\infty I(\lambda)\,d\lambda = \frac{2\pi^5 k^4}{15c^2h^3}T^4$$

where $I(\lambda)$ is the Planck distribution formula of Eq. (39.24). *Hint:* Change the integration variable from λ to f. You'll need to use the following tabulated integral:

$$\int_0^\infty \frac{x^3}{e^{\alpha x}-1}dx = \frac{1}{240}\left(\frac{2\pi}{\alpha}\right)^4$$

(c) The result of part (b) is I and has the form of the Stefan–Boltzmann law, $I = \sigma T^4$ (Eq. 39.19). Evaluate the constants in part (b) to show that σ has the value given in Section 39.5.

39.60 •• CP A beam of 40 eV electrons traveling in the $+x$-direction passes through a slit that is parallel to the y-axis and 5.0 μm wide. The diffraction pattern is recorded on a screen 2.5 m from the slit. (a) What is the de Broglie wavelength of the electrons? (b) How much time does it take the electrons to travel from the slit to the screen? (c) Use the width of the central diffraction pattern to calculate the uncertainty in the y-component of momentum of an electron just after it has passed through the slit. (d) Use the result of part (c) and the Heisenberg uncertainty principle [(Eq. 39.29) for y] to estimate the minimum uncertainty in the y-coordinate of an electron just after it has passed through the slit. Compare your result to the width of the slit.

39.61 •• If you could keep utterly motionless, your de Broglie wavelength would be infinite. As soon as you make the slightest motion, however, your wavelength collapses. (a) Estimate the lowest speed you can perceive. (b) Estimate your wavelength if you moved with that slowest perceptible speed. (c) A grain of sand has a mass of about 0.5 mg. Estimate the wavelength of a grain of sand moving at your slowest perceptible speed. (It should be clear that the wave aspects of macroscopic material things are hidden from us by our size.) (d) If nature were to alter her laws so that Planck's constant became $h = 1\ \text{J}\cdot\text{s}$, then what would be the wavelength of a grain of sand moving at 1 m/s? (e) Under these same circumstances, estimate your own wavelength if you ran at 2.5 m/s. (f) A baseball has a mass of 145 g. Estimate the speed that a baseball would need to have a perceptible diffraction, meaning a central maximum subtending 10°, when thrown through a doorway, if h were 1 J·s.

39.62 • CP Electrons go through a single slit 300 nm wide and strike a screen 24.0 cm away. At angles of $\pm 20.0°$ from the center of the diffraction pattern, no electrons hit the screen, but electrons hit at all points closer to the center. (a) How fast were these electrons moving when they went through the slit? (b) What will be the next pair of larger angles at which no electrons hit the screen?

39.63 •• CP A beam of electrons is accelerated from rest and then passes through a pair of identical thin slits that are 1.25 nm apart. You observe that the first double-slit interference dark fringe occurs at $\pm 18.0°$ from the original direction of the beam when viewed on a distant screen. (a) Are these electrons relativistic? How do you know? (b) Through what potential difference were the electrons accelerated?

39.64 • CP Coherent light is passed through two narrow slits whose separation is 20.0 μm. The second-order bright fringe in the interference pattern is located at an angle of 0.0300 rad. If electrons are used instead of light, what must the kinetic energy (in electron volts) of the electrons be if they are to produce an interference pattern for which the second-order maximum is also at 0.0300 rad?

39.65 •• CP An electron beam and a photon beam pass through identical slits. On a distant screen, the first dark fringe occurs at the same angle for both of the beams. The electron speeds are much slower than that of light. (a) Express the energy of a photon in terms of the kinetic energy K of one of the electrons. (b) Which is greater, the energy of a photon or the kinetic energy of an electron?

39.66 • BIO What is the de Broglie wavelength of a red blood cell, with mass 1.00×10^{-11} g, that is moving with a speed of 0.400 cm/s? Do we need to be concerned with the wave nature of the blood cells when we describe the flow of blood in the body?

39.67 • High-speed electrons are used to probe the interior structure of the atomic nucleus. For such electrons the expression $\lambda = h/p$ still holds, but we must use the relativistic expression for momentum, $p = mv/\sqrt{1 - v^2/c^2}$. (a) Show that the speed of an electron that has de Broglie wavelength λ is

$$v = \frac{c}{\sqrt{1 + (mc\lambda/h)^2}}$$

(b) The quantity h/mc equals 2.426×10^{-12} m. (As we saw in Section 38.3, this same quantity appears in Eq. (38.7), the expression for Compton scattering of photons by electrons.) If λ is small compared to h/mc, the denominator in the expression found in part (a) is close to unity and the speed v is very close to c. In this case it is convenient to write $v = (1 - \Delta)c$ and express the speed of the electron in terms of Δ rather than v. Find an expression for Δ valid when $\lambda \ll h/mc$. [*Hint:* Use the binomial expansion $(1 + z)^n = 1 + nz + [n(n-1)z^2/2] + \cdots$, valid for the case $|z| < 1$.] (c) How fast must an electron move for its de Broglie wavelength to be 1.00×10^{-15} m, comparable to the size of a proton? Express your answer in the form $v = (1 - \Delta)c$, and state the value of Δ.

39.68 • Suppose that the uncertainty of position of an electron is equal to the radius of the $n = 1$ Bohr orbit for hydrogen. Calculate the simultaneous minimum uncertainty of the corresponding momentum component, and compare this with the magnitude of the momentum of the electron in the $n = 1$ Bohr orbit. Discuss your results.

39.69 • CP (a) A particle with mass m has kinetic energy equal to three times its rest energy. What is the de Broglie wavelength of this particle? (*Hint:* You must use the relativistic expressions for momentum and kinetic energy: $E^2 = (pc)^2 + (mc^2)^2$ and $K = E - mc^2$.) (b) Determine the numerical value of the kinetic energy (in MeV) and the wavelength (in meters) if the particle in part (a) is (i) an electron and (ii) a proton.

39.70 • Proton Energy in a Nucleus. The radii of atomic nuclei are of the order of 5.0×10^{-15} m. (a) Estimate the minimum uncertainty in the momentum of a proton if it is confined within a nucleus. (b) Take this uncertainty in momentum to be an estimate of the magnitude of the momentum. Use the relativistic relationship between energy and momentum, Eq. (37.39), to obtain an estimate of the kinetic energy of a proton confined within a nucleus. (c) For a proton to remain bound within a nucleus, what must the magnitude of the (negative) potential energy for a proton be within the nucleus? Give your answer in eV and in MeV. Compare to the potential energy for an electron in a hydrogen atom, which has a magnitude of a few tens of eV. (This shows why the interaction that binds the nucleus together is called the "strong nuclear force.")

39.71 • **Electron Energy in a Nucleus.** The radii of atomic nuclei are of the order of 5.0×10^{-15} m. (a) Estimate the minimum uncertainty in the momentum of an electron if it is confined within a nucleus. (b) Take this uncertainty in momentum to be an estimate of the magnitude of the momentum. Use the relativistic relationship between energy and momentum, Eq. (37.39), to obtain an estimate of the kinetic energy of an electron confined within a nucleus. (c) Compare the energy calculated in part (b) to the magnitude of the Coulomb potential energy of a proton and an electron separated by 5.0×10^{-15} m. On the basis of your result, could there be electrons within the nucleus? (*Note:* It is interesting to compare this result to that of Problem 39.70.)

39.72 • The neutral pion (π^0) is an unstable particle produced in high-energy particle collisions. Its mass is about 264 times that of the electron, and it exists for an average lifetime of 8.4×10^{-17} s before decaying into two gamma-ray photons. Using the relationship $E = mc^2$ between rest mass and energy, find the uncertainty in the mass of the particle and express it as a fraction of the mass.

39.73 • **Doorway Diffraction.** If your wavelength were 1.0 m, you would undergo considerable diffraction in moving through a doorway. (a) What must your speed be for you to have this wavelength? (Assume that your mass is 60.0 kg.) (b) At the speed calculated in part (a), how many years would it take you to move 0.80 m (one step)? Will you notice diffraction effects as you walk through doorways?

39.74 • **Atomic Spectra Uncertainties.** A certain atom has an energy level 2.58 eV above the ground level. Once excited to this level, the atom remains in this level for 1.64×10^{-7} s (on average) before emitting a photon and returning to the ground level. (a) What is the energy of the photon (in electron volts)? What is its wavelength (in nanometers)? (b) What is the smallest possible uncertainty in energy of the photon? Give your answer in electron volts. (c) Show that

$$|\Delta E/E| = |\Delta\lambda/\lambda|$$

if $|\Delta\lambda/\lambda| \ll 1$. Use this to calculate the magnitude of the smallest possible uncertainty in the wavelength of the photon. Give your answer in nanometers.

39.75 •• For x rays with wavelength 0.0300 nm, the $m = 1$ intensity maximum for a crystal occurs when the angle θ in Fig. 39.2 is 35.8°. At what angle θ does the $m = 1$ maximum occur when a beam of 4.50 keV electrons is used instead? Assume that the electrons also scatter from the atoms in the surface plane of this same crystal.

39.76 •• A certain atom has an energy state 3.50 eV above the ground state. When excited to this state, the atom remains for 2.0 μs, on average, before it emits a photon and returns to the ground state. (a) What are the energy and wavelength of the photon? (b) What is the smallest possible uncertainty in energy of the photon?

39.77 •• BIO **Structure of a Virus.** To investigate the structure of extremely small objects, such as viruses, the wavelength of the probing wave should be about one-tenth the size of the object for sharp images. But as the wavelength gets shorter, the energy of a photon of light gets greater and could damage or destroy the object being studied. One alternative is to use electron matter waves instead of light. Viruses vary considerably in size, but 50 nm is not unusual. Suppose you want to study such a virus, using a wave of wavelength 5.00 nm. (a) If you use light of this wavelength, what would be the energy (in eV) of a single photon? (b) If you use an electron of this wavelength, what would be its kinetic energy (in eV)? Is it now clear why matter waves (such as in the electron microscope) are often preferable to electromagnetic waves for studying microscopic objects?

39.78 •• CALC **Zero-Point Energy.** Consider a particle with mass m moving in a potential $U = \frac{1}{2}kx^2$, as in a mass–spring system. The total energy of the particle is $E = (p^2/2m) + \frac{1}{2}kx^2$. Assume that p and x are approximately related by the Heisenberg uncertainty principle, so $px \approx h$. (a) Calculate the minimum possible value of the energy E, and the value of x that gives this minimum E. This lowest possible energy, which is not zero, is called the *zero-point energy*. (b) For the x calculated in part (a), what is the ratio of the kinetic to the potential energy of the particle?

39.79 •• CALC A particle with mass m moves in a potential energy $U(x) = A|x|$, where A is a positive constant. In a simplified picture, quarks (the constituents of protons, neutrons, and other particles, as will be described in Chapter 44) have a potential energy of interaction of approximately this form, where x represents the separation between a pair of quarks. Because $U(x) \rightarrow \infty$ as $x \rightarrow \infty$, it's not possible to separate quarks from each other (a phenomenon called *quark confinement*). (a) Classically, what is the force acting on this particle as a function of x? (b) Using the uncertainty principle as in Problem 39.78, determine approximately the zero-point energy of the particle.

39.80 •• Imagine another universe in which the value of Planck's constant is 0.0663 J · s, but in which the physical laws and all other physical constants are the same as in our universe. In this universe, two physics students are playing catch. They are 12 m apart, and one throws a 0.25 kg ball directly toward the other with a speed of 6.0 m/s. (a) What is the uncertainty in the ball's horizontal momentum, in a direction perpendicular to that in which it is being thrown, if the student throwing the ball knows that it is located within a cube with volume 125 cm^3 at the time she throws it? (b) By what horizontal distance could the ball miss the second student?

39.81 •• DATA For your work in a mass spectrometry lab, you are investigating the absorption spectrum of one-electron ions. To maintain the atoms in an ionized state, you hold them at low density in an ion trap, a device that uses a configuration of electric fields to confine ions. The majority of the ions are in their ground state, so that is the initial state for the absorption transitions that you observe. (a) If the longest wavelength that you observe in the absorption spectrum is 13.56 nm, what is the atomic number Z for the ions? (b) What is the next shorter wavelength that the ions will absorb? (c) When one of the ions absorbs a photon of wavelength 6.78 nm, a free electron is produced. What is the kinetic energy (in electron volts) of the electron?

39.82 •• DATA In the crystallography lab where you work, you are given a single crystal of an unknown substance to identify. To obtain one piece of information about the substance, you repeat the Davisson–Germer experiment to determine the spacing of the atoms in the surface planes of the crystal. You start with electrons that are essentially stationary and accelerate them through a potential difference of magnitude V_{ac}. The electrons then scatter off the atoms on the surface of the crystal (as in Fig. 39.3b). Next you measure the angle θ that locates the first-order diffraction peak. Finally, you repeat the measurement for different values of V_{ac}. Your results are given in the table.

V_{ac} (V)	106.3	69.1	49.9	25.2	16.9	13.6
θ(°)	20.4	24.8	30.2	45.5	59.1	73.1

(a) Graph your data in the form $\sin\theta$ versus $1/\sqrt{V_{ac}}$. What is the slope of the straight line that best fits the data points when plotted in this way? (b) Use your results from part (a) to calculate the value of d for this crystal.

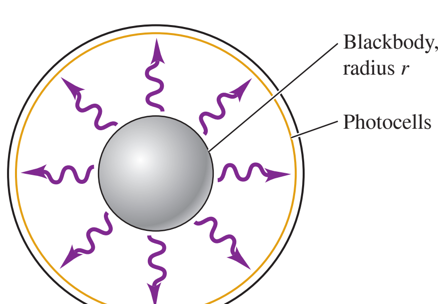
39.83 •• **DATA** As an amateur astronomer, you are studying the apparent brightness of stars. You know that a star's apparent brightness depends on its distance from the earth and also on the fraction of its radiated energy that is in the visible region of the electromagnetic spectrum. But, as a first step, you search the Internet for information on the surface temperatures and radii of some selected stars so that you can calculate their total radiated power. You find the data given in the table.

Star	Polaris	Vega	Antares	α Centauri B
Surface temperature (K)	6015	9602	3400	5260
Radius relative to that of the sun (R_{sun})	46	2.73	883	0.865

The radius is given in units of the radius of the sun, $R_{sun} = 6.96 \times 10^8$ m. The surface temperature is the effective temperature that gives the measured photon luminosity of the star if the star is assumed to radiate as an ideal blackbody. The photon luminosity is the power emitted in the form of photons. (a) Which star in the table has the greatest radiated power? (b) For which of these stars, if any, is the peak wavelength λ_m in the visible range (380–750 nm)? (c) The sun has a total radiated power of 3.85×10^{26} W. Which of these stars, if any, have a total radiated power less than that of our sun?

39.84 ••• **CP CALC** An advanced civilization uses a fusion reactor to maintain a high temperature on a spherical blackbody whose radiation supplies energy for use on a spacecraft. The blackbody has radius $r = 1.23$ m and is surrounded by a larger sphere whose inside surface is densely lined with photocells (**Fig. P39.84**). These capture all photons that have energy within 1.00% of a central value E_0. The photocells are designed so that E_0 corresponds to the peak energy radiated from the blackbody. The range ΔE is sufficiently small that we can approximate the spectral emittance as constant over the range $E_0 \pm \Delta E$. This device generates an emf across a motor of resistance 1.00 kΩ. (a) The spectral emittance $I = \int I(\lambda)\,d\lambda$ can be structured as $\int I(E)\,dE$ by parameterizing the integral using $E = hc/\lambda$. Use Eq. (39.24) to determine $I(E)$. Be careful with the differential factor. (b) Determine the energy E_0 that maximizes $I(E)$, and determine $I_{max} = I(E_0)$ in terms of the temperature T. (*Hint:* You have to solve the equation $3 - x = 3e^{-x}$. Use the solution $x = 2.821$, which is accurate to four significant figures.) (c) If the device is designed for $T = 1000$ K, then how much power is supplied by the motor, and what is the current I? (*Hint:* Integrate the spectral emittance around the blackbody.) (d) If the device is designed for $T = 5000$ K, then how much power is supplied and what is the current?

Figure **P39.84**



39.85 ••• **CP** An alpha particle is incident with kinetic energy K on a gold nucleus at rest. The aim is direct. (a) If m is the mass of an alpha particle and M is the mass of a gold nucleus, solve the classical conditions for energy and momentum conservation to determine the recoil speed V of the nucleus after the collision. (b) Determine an expression for the fractional energy lost to the nucleus. (c) Is your result independent of the initial kinetic energy? (d) An alpha particle has mass $m = 6.64 \times 10^{-27}$ kg, and a gold nucleus has mass $M = 1.32 \times 10^{-25}$ kg. If $K = 5.00$ MeV, then what is the speed V as a fraction of c, and what proportion of the original energy is transferred to the gold nucleus? (e) According to the classical analysis, what speed of the incident alpha particle would result in a nuclear speed V of $0.10c$? (f) Is that possible?

CHALLENGE PROBLEMS

39.86 ••• **CP CALC** You have entered a contest in which the contestants drop a marble with mass 20.0 g from the roof of a building onto a small target 25.0 m below. From uncertainty considerations, what is the typical distance by which you'll miss the target, given that you aim with the highest possible precision? (*Hint:* The uncertainty Δx_f in the x-coordinate of the marble when it reaches the ground comes in part from the uncertainty Δx_i in the x-coordinate initially and in part from the initial uncertainty in v_x. The latter gives rise to an uncertainty Δv_x in the horizontal motion of the marble as it falls. The values of Δx_i and Δv_x are related by the uncertainty principle. A small Δx_i gives rise to a large Δv_x, and vice versa. Find the value of Δx_i that gives the smallest total uncertainty in x at the ground. Ignore any effects of air resistance.)

39.87 ••• (a) Show that in the Bohr model, the frequency of revolution of an electron in its circular orbit around a stationary hydrogen nucleus is $f = me^4/4\epsilon_0^2 n^3 h^3$. (b) In classical physics, the frequency of revolution of the electron is equal to the frequency of the radiation that it emits. Show that when n is very large, the frequency of revolution does indeed equal the radiated frequency calculated from Eq. (39.5) for a transition from $n_1 = n + 1$ to $n_2 = n$. (This illustrates Bohr's *correspondence principle,* which is often used as a check on quantum calculations. When n is small, quantum physics gives results that are very different from those of classical physics. When n is large, the differences are not significant, and the two methods then "correspond." In fact, when Bohr first tackled the hydrogen atom problem, he sought to determine f as a function of n such that it would correspond to classical results for large n.)

MCAT-STYLE PASSAGE PROBLEMS

BIO **Ion Microscopes.** Whereas electron microscopes make use of the wave properties of electrons, ion microscopes make use of the wave properties of atomic ions, such as helium ions (He^+), to image materials. A helium ion has a mass 7300 times that of an electron. In a typical helium-ion microscope, helium ions are accelerated by a high voltage of 10–50 kV and focused into a beam onto the sample to be imaged. At these energies, the ions don't travel very far into the sample, so this type of microscope is used primarily for the surface imaging of biological structures. The use of helium ions with much greater energies (in the MeV range) has been proposed as a way to image the entire thickness of a sample, because these faster helium ions can pass all the way through biological samples such as cells. In this type of ion microscope, the energy lost as the ion beam passes through different parts of a cell can be measured and related to the distribution of material in the cell, with thicker parts of the cell causing greater energy loss. [Source: "Whole-Cell Imaging at Nanometer Resolutions Using Fast and Slow Focused Helium Ions," by Xiao Chen et al., *Biophysical Journal* 101(7): 1788–1793, Oct. 5, 2011.]

39.88 How does the wavelength of a helium ion compare to that of an electron accelerated through the same potential difference? (a) The helium ion has a longer wavelength, because it has greater mass. (b) The helium ion has a shorter wavelength, because it has greater mass. (c) The wavelengths are the same, because the kinetic energy is the same. (d) The wavelengths are the same, because the electric charge is the same.

39.89 Can the first type of helium-ion microscope, used for surface imaging, produce helium ions with a wavelength of 0.1 pm? (a) Yes; the voltage required is 21 kV. (b) Yes; the voltage required is 42 kV. (c) No; a voltage higher than 50 kV is required. (d) No; a voltage lower than 10 kV is required.

39.90 Why is it easier to use helium ions rather than neutral helium atoms in such a microscope? (a) Helium atoms are not electrically charged, and only electrically charged particles have wave properties. (b) Helium atoms form molecules, which are too large to have wave properties. (c) Neutral helium atoms are more difficult to focus with electric and magnetic fields. (d) Helium atoms have much larger mass than helium ions do and thus are more difficult to accelerate.

39.91 In the second type of helium-ion microscope, a 1.2 MeV ion passing through a cell loses 0.2 MeV per μm of cell thickness. If the energy of the ion can be measured to 6 keV, what is the smallest difference in thickness that can be discerned? (a) 0.03 μm; (b) 0.06 μm; (c) 3 μm; (d) 6 μm.

ANSWERS

Chapter Opening Question ?

(i) The smallest detail visible in an image is comparable to the wavelength used to make the image. Electrons can easily be given a large momentum p and hence a short wavelength $\lambda = h/p$, and so can be used to resolve extremely fine details. (See Section 39.1.)

Key Example √ARIATION Problems

VP39.2.1 (a) 0.148 nm (b) 59.1°
VP39.2.2 (a) 5.85×10^{-18} J (b) 0.203 nm (c) 0.273 nm
VP39.2.3 (a) 0.105 nm (b) 136 V
VP39.2.4 (a) 1.67×10^{-12} m (b) 295 V
VP39.6.1 620 nm, 414 nm, 248 nm
VP39.6.2 (a) 3.22 eV for first excited level, 5.06 eV for second excited level (b) 245 nm
VP39.6.3 (a) $E = 0.967$ eV, $\lambda = 1.28$ μm (b) $E = 2.55$ eV, $\lambda = 487$ nm (c) $E = 12.1$ eV, $\lambda = 103$ nm
VP39.6.4 (a) -3.02 eV (b) $+6.04$ eV (c) 411 nm
VP39.8.1 (a) 808 nm, infrared (b) 9.42 MW/m^2
VP39.8.2 (a) 6.09×10^3 K (b) 476 nm (c) visible
VP39.8.3 (a) 1.21×10^4 K (b) 1.23×10^9 W/m^2
VP39.8.4 (a) 954 nm (b) 40.1 kW/m^2

Bridging Problem

(a) 192 nm; ultraviolet
(b) $n = 4$
(c) $\lambda_2 = 0.665$ nm, $\lambda_3 = 0.997$ nm

? These containers hold solutions of microscopic semiconductor particles of different sizes. The particles glow when exposed to ultraviolet light; the smallest particles glow blue and the largest particles glow red. This is because the energy levels of electrons (i) are spaced farther apart in smaller particles; (ii) are spaced farther apart in larger particles; (iii) have the same spacing in all particles but have higher energies in smaller particles; (iv) have the same spacing in all particles but have higher energies in larger particles; (v) depend on the wavelength of ultraviolet light used.

40 Quantum Mechanics I: Wave Functions

In Chapter 39 we found that particles can behave like waves. In fact, it turns out that we can use the wave picture to completely describe the behavior of a particle. This approach, called *quantum mechanics,* is the key to understanding the behavior of matter on the molecular, atomic, and nuclear scales. In this chapter we'll see how to find the *wave function* of a particle by solving the *Schrödinger equation,* which is as fundamental to quantum mechanics as Newton's laws are to mechanics or as Maxwell's equations are to electromagnetism.

We'll begin with a quantum-mechanical analysis of a *free particle* that moves along a straight line without being acted on by forces of any kind. We'll then consider particles that are acted on by forces and are trapped in *bound states,* just as electrons are bound within an atom. We'll see that solving the Schrödinger equation automatically gives the possible energy levels for the system.

Besides energies, solving the Schrödinger equation gives us the probabilities of finding a particle in various regions. One surprising result is that there is a nonzero probability that microscopic particles will pass through thin barriers, even though such a process is forbidden by Newtonian mechanics.

In this chapter we'll consider the Schrödinger equation for one-dimensional motion only. In Chapter 41 we'll see how to extend this equation to three-dimensional problems such as the hydrogen atom. The hydrogen-atom wave functions will in turn form the foundation for our analysis of more complex atoms, of the periodic table of the elements, of x-ray energy levels, and of other properties of atoms.

LEARNING OUTCOMES

In this chapter, you'll learn...

40.1 The wave function that describes the behavior of a particle and the Schrödinger equation that this function must satisfy.
40.2 How to calculate the wave functions and energy levels for a particle confined to a box.
40.3 How to analyze the quantum-mechanical behavior of a particle in a potential well.
40.4 How quantum mechanics makes it possible for particles to go where Newtonian mechanics says they cannot.
40.5 How to use quantum mechanics to analyze a harmonic oscillator.
40.6 How measuring a quantum-mechanical system can change that system's state.

You'll need to review...

7.5 Potential wells.
14.2, 14.3 Harmonic oscillators.
15.3, 15.7, 15.8 Wave functions for waves on a string; standing waves.
32.3 Wave functions for electromagnetic waves.
38.1, 38.4 Work function; photon interpretation of interference and diffraction.
39.1, 39.3, 39.5, 39.6 De Broglie relationships; Bohr model for hydrogen; Planck radiation law; Heisenberg uncertainty principle.

40.1 WAVE FUNCTIONS AND THE ONE-DIMENSIONAL SCHRÖDINGER EQUATION

We have now seen compelling evidence that on an atomic or subatomic scale, an object such as an electron cannot be described simply as a classical, Newtonian point particle. Instead, we must take into account its *wave* characteristics. In the Bohr model

Figure **40.1** These children are talking over a cup-and-string "telephone." The displacement of the string is completely described by a wave function $y(x, t)$. In an analogous way, a particle is completely described by a quantum-mechanical wave function $\Psi(x, y, z, t)$.

of the hydrogen atom (Section 39.3) we tried to have it both ways: We pictured the electron as a classical particle in a circular orbit around the nucleus, and used the de Broglie relationship between particle momentum and wavelength to explain why only orbits of certain radii are allowed. As we saw in Section 39.6, however, the Heisenberg uncertainty principle tells us that a hybrid description of this kind can't be wholly correct. In this section we'll explore how to describe the state of a particle by using *only* the language of waves. This new description, called **quantum mechanics,** replaces the classical scheme of describing the state of a particle by its coordinates and velocity components.

Our new quantum-mechanical scheme for describing a particle has a lot in common with the language of classical wave motion. In Section 15.3 of Chapter 15, we described transverse waves on a string by specifying the position of each point in the string at each instant of time by means of a *wave function* $y(x, t)$ that represents the displacement from equilibrium, at time t, of a point on the string at a distance x from the origin (**Fig. 40.1**). Once we know the wave function for a particular wave motion, we know everything there is to know about the motion. For example, we can find the velocity and acceleration of any point on the string at any time. We worked out specific forms for these functions for *sinusoidal* waves, in which each particle undergoes simple harmonic motion.

We followed a similar pattern for sound waves in Chapter 16. The wave function $p(x, t)$ for a wave traveling along the x-direction represented the pressure variation at any point x and any time t. In Section 32.3 we used *two* wave functions to describe the $\vec{E}$ and $\vec{B}$ fields in an electromagnetic wave.

Thus it's natural to use a wave function as the central element of our new language of quantum mechanics. The customary symbol for this wave function is the Greek letter psi, Ψ or ψ. In general, we'll use an uppercase Ψ to denote a function of all the space coordinates and time, and a lowercase ψ for a function of the space coordinates only—*not* of time. Just as the wave function $y(x, t)$ for mechanical waves on a string provides a complete description of the motion, so the wave function $\Psi(x, y, z, t)$ for a particle contains all the information that can be known about the particle.

CAUTION Particle waves vs. mechanical waves Unlike waves on a string or sound waves in air, the wave function for a particle is *not* a mechanical wave that needs a material medium in order to propagate. The wave function describes the particle, but we cannot define the function itself in terms of anything material. We can describe only how it is related to physically observable effects.

Waves in One Dimension: Waves on a String

The wave function of a particle depends in general on all three dimensions of space. For simplicity, however, we'll begin our study of these functions by considering *one-dimensional* motion, in which a particle of mass m moves parallel to the x-axis and the wave function Ψ depends on the coordinate x and the time t only. (In the same way, we studied one-dimensional kinematics in Chapter 2 before going on to study two- and three-dimensional motion in Chapter 3.)

What does a one-dimensional quantum-mechanical wave look like, and what determines its properties? We can answer this question by first recalling the properties of a wave on a string. We saw in Section 15.3 that any wave function $y(x, t)$ that describes a wave on a string must satisfy the *wave equation:*

$$\frac{\partial^2 y(x, t)}{\partial x^2} = \frac{1}{v^2}\frac{\partial^2 y(x, t)}{\partial t^2} \qquad \text{(wave equation for waves on a string)} \qquad (40.1)$$

In Eq. (40.1) v is the speed of the wave, which is the same no matter what the wavelength. As an example, consider the following wave function for a wave of wavelength λ and frequency f moving in the positive x-direction along a string:

$$y(x, t) = A\cos(kx - \omega t) + B\sin(kx - \omega t) \qquad \text{(sinusoidal wave on a string)} \qquad (40.2)$$

Here $k = 2\pi/\lambda$ is the *wave number* and $\omega = 2\pi f$ is the *angular frequency.* (We used these same quantities for mechanical waves in Chapter 15 and electromagnetic waves in Chapter 32.) The quantities A and B are constants that determine the amplitude and phase of the wave. The expression in Eq. (40.2) is a valid wave function if and only if it satisfies the wave equation, Eq. (40.1). To check this, take the first and second derivatives of $y(x, t)$ with respect to x and take the first and second derivatives with respect to t:

$$\frac{\partial y(x, t)}{\partial x} = -kA\sin(kx - \omega t) + kB\cos(kx - \omega t) \tag{40.3a}$$

$$\frac{\partial^2 y(x, t)}{\partial x^2} = -k^2A\cos(kx - \omega t) - k^2B\sin(kx - \omega t) \tag{40.3b}$$

$$\frac{\partial y(x, t)}{\partial t} = \omega A\sin(kx - \omega t) - \omega B\cos(kx - \omega t) \tag{40.3c}$$

$$\frac{\partial^2 y(x, t)}{\partial t^2} = -\omega^2A\cos(kx - \omega t) - \omega^2B\sin(kx - \omega t) \tag{40.3d}$$

If we substitute Eqs. (40.3b) and (40.3d) into the wave equation, Eq. (40.1), we get

$$\begin{aligned} &-k^2A\cos(kx - \omega t) - k^2B\sin(kx - \omega t) \\ &\quad = \frac{1}{v^2}[-\omega^2A\cos(kx - \omega t) - \omega^2B\sin(kx - \omega t)] \end{aligned} \tag{40.4}$$

For Eq. (40.4) to be satisfied at all coordinates x and all times t, the coefficients of $\cos(kx - \omega t)$ must be the same on both sides of the equation, and likewise for the coefficients of $\sin(kx - \omega t)$. Both of these conditions will be satisfied if

$$k^2 = \frac{\omega^2}{v^2} \quad \text{or} \quad \omega = vk \quad \text{(waves on a string)} \tag{40.5}$$

Since $\omega = 2\pi f$ and $k = 2\pi/\lambda$, Eq. (40.5) is equivalent to

$$2\pi f = v\frac{2\pi}{\lambda} \quad \text{or} \quad v = \lambda f \quad \text{(waves on a string)}$$

This equation is just the familiar relationship among wave speed, wavelength, and frequency for waves on a string. So our calculation shows that Eq. (40.2) is a valid wave function for waves on a string for any values of A and B, provided that ω and k are related by Eq. (40.5).

Waves in One Dimension: Particle Waves

What we need is a quantum-mechanical version of the wave equation, Eq. (40.1), valid for particle waves. We expect this equation to involve partial derivatives of the wave function $\Psi(x, t)$ with respect to x and with respect to t. However, this new equation *cannot* be the same as Eq. (40.1) for waves on a string because the relationship between ω and k is different. We can show this by considering a **free particle,** one that experiences no force at all as it moves along the x-axis. For such a particle the potential energy $U(x)$ has the same value for all x (recall from Chapter 7 that $F_x = -dU(x)/dx$, so zero force means the potential energy has zero derivative). For simplicity let $U = 0$ for all x. Then the energy of the free particle is equal to its kinetic energy, which we can express in terms of its momentum p:

$$E = \tfrac{1}{2}mv^2 = \frac{m^2v^2}{2m} = \frac{(mv)^2}{2m} = \frac{p^2}{2m} \quad \text{(energy of a free particle)} \tag{40.6}$$

The de Broglie relationships (Section 39.1) tell us that the energy E is proportional to the angular frequency ω and the momentum p is proportional to the wave number k:

$$E = hf = \frac{h}{2\pi}2\pi f = \hbar\omega \tag{40.7a}$$

$$p = \frac{h}{\lambda} = \frac{h}{2\pi}\frac{2\pi}{\lambda} = \hbar k \tag{40.7b}$$

Remember that $\hbar = h/2\pi$. If we substitute Eqs. (40.7) into Eq. (40.6), we find that the relationship between ω and k for a free particle is

$$\hbar\omega = \frac{\hbar^2 k^2}{2m} \qquad \text{(free particle)} \tag{40.8}$$

Equation (40.8) is *very* different from the corresponding relationship for waves on a string, Eq. (40.5): The angular frequency ω for particle waves is proportional to the *square* of the wave number, while for waves on a string ω is directly proportional to k. Our task is therefore to construct a quantum-mechanical version of the wave equation whose free-particle solutions satisfy Eq. (40.8).

We'll attack this problem by assuming a sinusoidal wave function $\Psi(x, t)$ of the same form as Eq. (40.2) for a sinusoidal wave on a string. For a wave on a string, Eq. (40.2) represents a wave of wavelength $\lambda = 2\pi/k$ and frequency $f = \omega/2\pi$ propagating in the positive x-direction. By analogy, our sinsuoidal wave function $\Psi(x, t)$ represents a free particle of mass m, momentum $p = \hbar k$, and energy $E = \hbar\omega$ moving in the positive x-direction:

$$\Psi(x, t) = A\cos(kx - \omega t) + B\sin(kx - \omega t) \qquad \begin{array}{l}\text{(sinusoidal wave}\\ \text{function representing}\\ \text{a free particle)}\end{array} \tag{40.9}$$

The wave number k and angular frequency ω in Eq. (40.9) must satisfy Eq. (40.8). If you look at Eq. (40.3b), you'll see that taking the second derivative of $\Psi(x, t)$ in Eq. (40.9) with respect to x gives us $\Psi(x, t)$ multiplied by $-k^2$. Hence if we multiply $\partial^2\Psi(x, t)/\partial x^2$ by $-\hbar^2/2m$, we get

$$\begin{aligned} -\frac{\hbar^2}{2m}\frac{\partial^2\Psi(x, t)}{\partial x^2} &= -\frac{\hbar^2}{2m}\left[-k^2 A\cos(kx - \omega t) - k^2 B\sin(kx - \omega t)\right] \\ &= \frac{\hbar^2 k^2}{2m}\left[A\cos(kx - \omega t) + B\sin(kx - \omega t)\right] \\ &= \frac{\hbar^2 k^2}{2m}\Psi(x, t) \end{aligned} \tag{40.10}$$

Equation (40.10) suggests that $(-\hbar^2/2m)\partial^2\Psi(x, t)/\partial x^2$ should be one side of our quantum-mechanical wave equation, with the other side equal to $\hbar\omega\Psi(x, t)$ in order to satisfy Eq. (40.8). If you look at Eq. (40.3c), you'll see that taking the *first* time derivative of $\Psi(x, t)$ in Eq. (40.9) brings out a factor of ω. So we'll make the educated guess that the right-hand side of our quantum-mechanical wave equation involves $\hbar = h/2\pi$ times $\partial\Psi(x, t)/\partial t$. So our tentative equation is

$$-\frac{\hbar^2}{2m}\frac{\partial^2\Psi(x, t)}{\partial x^2} = C\hbar\frac{\partial\Psi(x, t)}{\partial t} \tag{40.11}$$

At this point we include a constant C as a "fudge factor" to make sure that everything turns out right. Now let's substitute the wave function from Eq. (40.9) into Eq. (40.11). From Eq. (40.10) and Eq. (40.3c), we get

$$\begin{aligned} &\frac{\hbar^2 k^2}{2m}\left[A\cos(kx - \omega t) + B\sin(kx - \omega t)\right] \\ &\quad = C\hbar\omega\left[A\sin(kx - \omega t) - B\cos(kx - \omega t)\right] \end{aligned} \tag{40.12}$$

From Eq. (40.8), $\hbar\omega = \hbar^2 k^2/2m$, so we can cancel these factors on the two sides of Eq. (40.12). What remains is

$$\begin{aligned} &A\cos(kx - \omega t) + B\sin(kx - \omega t) \\ &\quad = CA\sin(kx - \omega t) - CB\cos(kx - \omega t) \end{aligned} \tag{40.13}$$

As in our discussion above of the wave equation for waves on a string, in order for Eq. (40.13) to be satisfied for all values of x and all values of t, the coefficients of $\cos(kx - \omega t)$ must be the same on both sides of the equation, and likewise for the coefficients of $\sin(kx - \omega t)$. Hence we have the following relationships among the coefficients A, B, and C in Eqs. (40.9) and (40.11):

$$A = -CB \tag{40.14a}$$

$$B = CA \tag{40.14b}$$

If we use Eq. (40.14b) to eliminate B from Eq. (40.14a), we get $A = -C^2 A$, which means that $C^2 = -1$. Thus C is equal to the *imaginary* number $i = \sqrt{-1}$, and Eq. (40.11) becomes

$$-\frac{\hbar^2}{2m}\frac{\partial^2\Psi(x,t)}{\partial x^2} = i\hbar\frac{\partial\Psi(x,t)}{\partial t} \quad \text{(one-dimensional Schrödinger equation for a free particle)} \tag{40.15}$$

Equation (40.15) is the one-dimensional **Schrödinger equation** for a free particle, developed in 1926 by the Austrian physicist Erwin Schrödinger (**Fig. 40.2**). The presence of the imaginary number i in Eq. (40.15) means that the solutions to the Schrödinger equation are complex quantities, with a real part and an imaginary part. (The imaginary part of $\Psi(x, t)$ is a real function multiplied by the imaginary number $i = \sqrt{-1}$.) An example is our free-particle wave function from Eq. (40.9). Since we found that $C = i$ in Eqs. (40.14), it follows from Eq. (40.14b) that $B = iA$. Then Eq. (40.9) becomes

$$\Psi(x,t) = A[\cos(kx - \omega t) + i\sin(kx - \omega t)] \quad \text{(sinusoidal wave function representing a free particle)} \tag{40.16}$$

The real part of $\Psi(x, t)$ is $\mathrm{Re}\Psi(x,t) = A\cos(kx - \omega t)$ and the imaginary part is $\mathrm{Im}\Psi(x,t) = A\sin(kx - \omega t)$. **Figure 40.3** graphs the real and imaginary parts of $\Psi(x, t)$ at $t = 0$, so $\Psi(x, 0) = A\cos kx + iA\sin kx$.

We can rewrite Eq. (40.16) with *Euler's formula,* which states that for any angle θ,

$$\begin{aligned} e^{i\theta} &= \cos\theta + i\sin\theta \\ e^{-i\theta} &= \cos(-\theta) + i\sin(-\theta) = \cos\theta - i\sin\theta \end{aligned} \tag{40.17}$$

Thus our sinusoidal free-particle wave function becomes

$$\Psi(x,t) = Ae^{i(kx-\omega t)} = Ae^{ikx}e^{-i\omega t} \quad \text{(sinusoidal wave function representing a free particle)} \tag{40.18}$$

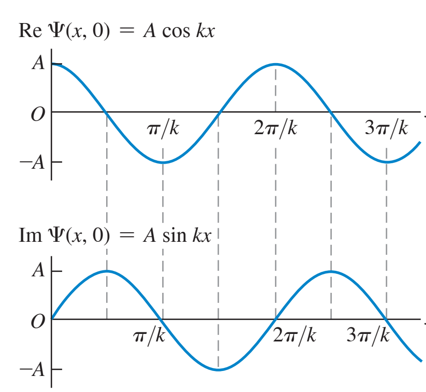
If k is positive in Eq. (40.16), the wave function represents a free particle moving in the positive x-direction with momentum $p = \hbar k$ and energy $E = \hbar\omega = \hbar^2 k^2/2m$. If k is negative, the momentum and hence the motion are in the negative x-direction. (With a negative value of k, the wavelength is $\lambda = 2\pi/|k|$.)

Interpreting the Wave Function

The complex nature of the wave function for a free particle makes this function challenging to interpret. (We certainly haven't needed imaginary numbers before this point to describe real physical phenomena.) Here's how to think about this function: $\Psi(x, t)$ describes the *distribution* of a particle in space, just as the wave functions for an electromagnetic wave describe the distribution of the electric and magnetic fields. When we worked out interference and diffraction patterns in Chapters 35 and 36, we found that the intensity I of the radiation at any point in a pattern is proportional to the square of the electric-field magnitude—that is, to E^2. In the photon interpretation of interference and diffraction (see Section 38.4), the intensity at each point is proportional to the number of photons striking around that point or, alternatively, to the *probability* that any individual

Figure **40.2** Erwin Schrödinger (1887–1961) developed the equation that bears his name in 1926, an accomplishment for which he shared (with the British physicist P. A. M. Dirac) the 1933 Nobel Prize in physics. His grave marker is adorned with a version of Eq. (40.15).

Figure **40.3** The spatial wave function $\Psi(x,t) = Ae^{i(kx-\omega t)}$ for a free particle of definite momentum $p = \hbar k$ is a complex function: It has both a real part and an imaginary part. These are graphed here as functions of x for $t = 0$.



photon will strike around the point. Thus the square of the electric-field magnitude at each point is proportional to the probability of finding a photon around that point.

In exactly the same way, the square of the wave function of a particle at each point tells us about the probability of finding the particle around that point. More precisely, we should say the square of the *absolute value* of the wave function, $|\Psi|^2$. This is necessary because, as we have seen, the wave function is a complex quantity with real and imaginary parts.

For a particle that can move only along the x-direction, the quantity $|\Psi(x, t)|^2 dx$ is the probability that the particle will be found at time t at a coordinate in the range from x to $x + dx$. The particle is most likely to be found in regions where $|\Psi|^2$ is large, and so on. This interpretation, first made by the German physicist Max Born (**Fig. 40.4**), requires that the wave function Ψ be *normalized*. That is, the integral of $|\Psi(x, t)|^2 dx$ over all possible values of x must equal exactly 1. In other words, the probability is exactly 1, or 100%, that the particle is *somewhere*.

Figure 40.4 In 1926, the German physicist Max Born (1882–1970) devised the interpretation that $|\Psi|^2$ is the probability distribution function for a particle that is described by the wave function Ψ. He also coined the term "quantum mechanics" (in the original German, *Quantenmechanik*). For his contributions, Born shared (with Walther Bothe) the 1954 Nobel Prize in physics.

CAUTION **Interpreting $|\Psi|^2$** Note that $|\Psi(x, t)|^2$ itself is *not* a probability. Rather, $|\Psi(x, t)|^2 dx$ is the probability of finding the particle between position x and position $x + dx$ at time t. If the length dx is made smaller, it becomes less likely that the particle will be found within that length, so the probability decreases. A better name for $|\Psi(x, t)|^2$ is the **probability distribution function,** since it describes how the probability of finding the particle at different locations is distributed over space. Another common name for $|\Psi(x, t)|^2$ is the *probability density.*

We can use the probability interpretation of $|\Psi|^2$ to get a better understanding of Eq. (40.18), the wave function for a free particle. This function describes a particle that has a definite momentum $p = \hbar k$ in the x-direction and *no* uncertainty in momentum: $\Delta p_x = 0$. The Heisenberg uncertainty principle for position and momentum, Eqs. (39.29), says that $\Delta x \Delta p_x \geq \hbar/2$. If Δp_x is zero, then Δx must be infinite, and we have no idea where along the x-axis the particle can be found. (We saw a similar result for photons in Section 38.4.) We can show this by calculating the probability distribution function $|\Psi(x, t)|^2$: the product of Ψ and its *complex conjugate* Ψ^*. To find the complex conjugate of a complex number, we replace all i with $-i$. For example, the complex conjugate of $c = a + ib$, where a and b are real, is $c^* = a - ib$, so $|c|^2 = c^*c = (a + ib)(a - ib) = a^2 + b^2$ (recall that $i^2 = -1$). The complex conjugate of Eq. (40.18) is

$$\Psi^*(x, t) = A^* e^{-i(kx - \omega t)} = A^* e^{-ikx} e^{i\omega t}$$

(We have to allow for the possibility that the coefficient A is itself a complex number.) Hence the probability distribution function is

$$\begin{aligned} |\Psi(x, t)|^2 &= \Psi^*(x, t)\Psi(x, t) = (A^* e^{-ikx} e^{i\omega t})(A e^{ikx} e^{-i\omega t}) \\ &= A^* A e^0 = |A|^2 \end{aligned}$$

The probability distribution function doesn't depend on position, which says that we are equally likely to find the particle *anywhere* along the x-axis! Mathematically, this is because the sinusoidal wave function $\Psi(x, t) = Ae^{i(kx - \omega t)} = A[\cos(kx - \omega t) + i\sin(kx - \omega t)]$ extends all the way from $x = -\infty$ to $x = +\infty$ with the same amplitude A. This also means that the wave function can't be normalized: The integral of $|\Psi(x, t)|^2$ over all space is infinite for any value of A.

Note also that the wave function in Eq. (40.18) describes a particle with a definite energy $E = \hbar\omega$, so there is zero uncertainty in energy: $\Delta E = 0$. The Heisenberg uncertainty principle for energy and time interval, $\Delta t \Delta E \geq \hbar/2$ [Eq. (39.30)], tells us that the time uncertainty Δt for this particle is infinite. In other words, we can have no idea *when* the particle will pass a given point on the x-axis. That also agrees with our result $|\Psi(x, t)|^2 = |A|^2$; the probability distribution function has the same value at all times.

Since we always have some idea of where a particle is, the wave function given in Eq. (40.18) isn't a realistic description. In our study of light in Section 38.4, we saw that we can make a wave function that's more *localized* in space by superposing two or more sinusoidal functions. (This would be a good time to review that section.) As an illustration, let's calculate $|\Psi(x, t)|^2$ for a wave function of this kind.

EXAMPLE 40.1 A localized free-particle wave function

The wave function $\Psi(x,t) = Ae^{i(k_1x-\omega_1t)} + Ae^{i(k_2x-\omega_2t)}$ is a superposition of *two* free-particle wave functions of the form given by Eq. (40.18). Both k_1 and k_2 are positive. (a) Show that this wave function satisfies the Schrödinger equation for a free particle of mass m. (b) Find the probability distribution function for $\Psi(x,t)$.

IDENTIFY and SET UP Both wave functions $Ae^{i(k_1x-\omega_1t)}$ and $Ae^{i(k_2x-\omega_2t)}$ represent a particle moving in the positive x-direction, but with different momenta and kinetic energies: $p_1 = \hbar k_1$ and $E_1 = \hbar\omega_1 = \hbar^2k_1^2/2m$ for the first function, $p_2 = \hbar k_2$ and $E_2 = \hbar\omega_2 = \hbar^2k_2^2/2m$ for the second function. To test whether a superposition of these is also a valid wave function for a free particle, we'll see whether our function $\Psi(x,t)$ satisfies the free-particle Schrödinger equation, Eq. (40.15). We'll use the derivatives of the exponential function: $(d/du)e^{au} = ae^{au}$ and $(d^2/du^2)e^{au} = a^2e^{au}$. The probability distribution function $|\Psi(x,t)|^2$ is the product of $\Psi(x,t)$ and its complex conjugate.

EXECUTE (a) If we substitute $\Psi(x,t)$ into Eq. (40.15), the left-hand side of the equation is

$$
\begin{aligned}
-\frac{\hbar^2}{2m}\frac{\partial^2\Psi(x,t)}{\partial x^2} &= -\frac{\hbar^2}{2m}\frac{\partial^2\left(Ae^{i(k_1x-\omega_1t)} + Ae^{i(k_2x-\omega_2t)}\right)}{\partial x^2}\\
&= -\frac{\hbar^2}{2m}\left[(ik_1)^2Ae^{i(k_1x-\omega_1t)} + (ik_2)^2Ae^{i(k_2x-\omega_2t)}\right]\\
&= \frac{\hbar^2k_1^2}{2m}Ae^{i(k_1x-\omega_1t)} + \frac{\hbar^2k_2^2}{2m}Ae^{i(k_2x-\omega_2t)}
\end{aligned}
$$

The right-hand side is

$$
\begin{aligned}
i\hbar\frac{\partial\Psi(x,t)}{\partial t} &= i\hbar\frac{\partial\left(Ae^{i(k_1x-\omega_1t)} + Ae^{i(k_2x-\omega_2t)}\right)}{\partial t}\\
&= i\hbar\left[(-i\omega_1)Ae^{i(k_1x-\omega_1t)} + (-i\omega_2)Ae^{i(k_2x-\omega_2t)}\right]\\
&= \hbar\omega_1Ae^{i(k_1x-\omega_1t)} + \hbar\omega_2Ae^{i(k_2x-\omega_2t)}
\end{aligned}
$$

The two sides *are* equal, provided that $\hbar\omega_1 = \hbar^2k_1^2/2m$ and $\hbar\omega_2 = \hbar^2k_2^2/2m$. These are just the relationships that we noted above. So we conclude that $\Psi(x,t) = Ae^{i(k_1x-\omega_1t)} + Ae^{i(k_2x-\omega_2t)}$ is a valid free-particle wave function. In general, if we take any two wave functions that are solutions of the Schrödinger equation and then make a superposition of these to create a third wave function $\Psi(x,t)$, then $\Psi(x,t)$ is also a solution of the Schrödinger equation.

(b) The complex conjugate of $\Psi(x,t)$ is

$$\Psi^*(x,t) = A^*e^{-i(k_1x-\omega_1t)} + A^*e^{-i(k_2x-\omega_2t)}$$

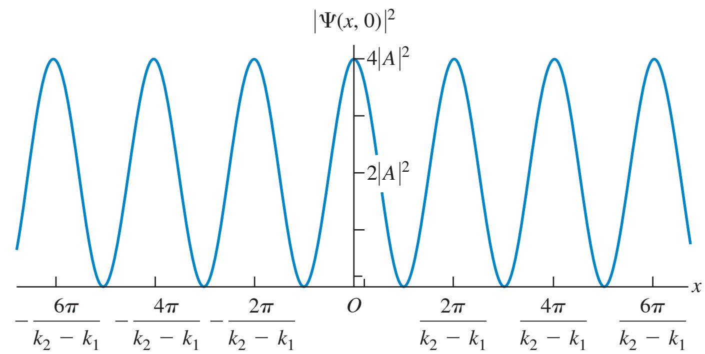
Hence

$$
\begin{aligned}
&|\Psi(x,t)|^2\\
&= \Psi^*(x,t)\Psi(x,t)\\
&= \left(A^*e^{-i(k_1x-\omega_1t)} + A^*e^{-i(k_2x-\omega_2t)}\right)\left(Ae^{i(k_1x-\omega_1t)} + Ae^{i(k_2x-\omega_2t)}\right)\\
&= A^*A\left[\begin{aligned}&e^{-i(k_1x-\omega_1t)}e^{i(k_1x-\omega_1t)} + e^{-i(k_2x-\omega_2t)}e^{i(k_2x-\omega_2t)}\\ &+ e^{-i(k_1x-\omega_1t)}e^{i(k_2x-\omega_2t)} + e^{-i(k_2x-\omega_2t)}e^{i(k_1x-\omega_1t)}\end{aligned}\right]\\
&= |A|^2\left[e^0 + e^0 + e^{i[(k_2-k_1)x-(\omega_2-\omega_1)t]} + e^{-i[(k_2-k_1)x-(\omega_2-\omega_1)t]}\right]
\end{aligned}
$$

To simplify this expression, recall that $e^0 = 1$. From Euler's formula, $e^{i\theta} = \cos\theta + i\sin\theta$ and $e^{-i\theta} = \cos\theta - i\sin\theta$, so $e^{i\theta} + e^{-i\theta} = 2\cos\theta$. Hence

$$
\begin{aligned}
|\Psi(x,t)|^2 &= |A|^2\{2 + 2\cos[(k_2-k_1)x - (\omega_2-\omega_1)t]\}\\
&= 2|A|^2\{1 + \cos[(k_2-k_1)x - (\omega_2-\omega_1)t]\}
\end{aligned}
$$

Figure 40.5 The probability distribution function at $t = 0$ for $\Psi(x,t) = Ae^{i(k_1x-\omega_1t)} + Ae^{i(k_2x-\omega_2t)}$.



EVALUATE **Figure 40.5** is a graph of the probability distribution function $|\Psi(x,t)|^2$ at $t = 0$. The value of $|\Psi(x,t)|^2$ varies between 0 and $4|A|^2$; probabilities can never be negative! The particle has become *somewhat* localized: The particle is most likely to be found near a point where $|\Psi(x,t)|^2$ is maximum (where the functions $Ae^{i(k_1x-\omega_1t)}$ and $Ae^{i(k_2x-\omega_2t)}$ interfere constructively) and is very unlikely to be found near a point where $|\Psi(x,t)|^2 = 0$ (where $Ae^{i(k_1x-\omega_1t)}$ and $Ae^{i(k_2x-\omega_2t)}$ interfere destructively).

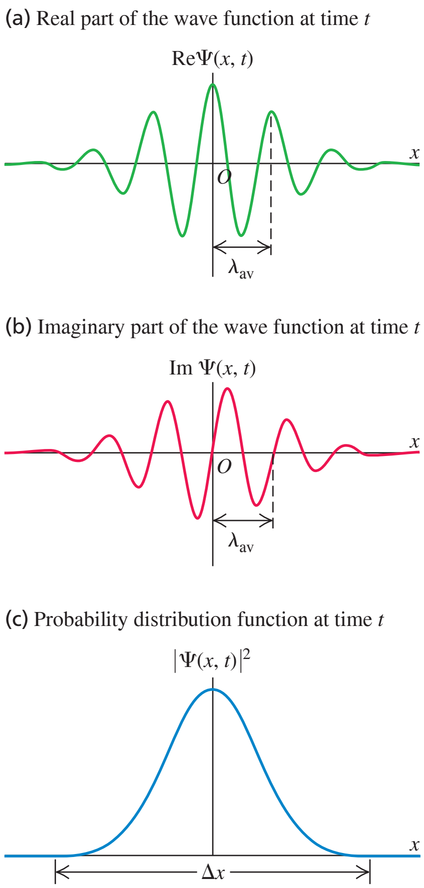
Note also that the probability distribution function is not stationary: It moves in the positive x-direction like the particle that it represents. To see this, recall from Section 15.3 that a sinusoidal wave given by $y(x,t) = A\cos(kx - \omega t)$ moves in the positive x-direction with speed $v = \omega/k$; since $|\Psi(x,t)|^2$ includes a term $\cos[(k_2-k_1)x - (\omega_2-\omega_1)t]$, the probability distribution moves at a speed $v_{av} = (\omega_2-\omega_1)/(k_2-k_1)$. The subscript "av" reminds us that v_{av} represents the *average* value of the particle's speed.

The price we pay for localizing the particle somewhat is that, unlike a particle represented by Eq. (40.18), it no longer has either a definite momentum or a definite energy. That's consistent with the Heisenberg uncertainty principles: If we decrease the uncertainties about where a particle is and when it passes a certain point, the uncertainties in its momentum and energy must increase.

The average momentum of the particle is $p_{av} = (\hbar k_2 + \hbar k_1)/2$, which is the average of the momenta associated with the free-particle wave functions we added to create $\Psi(x,t)$. This corresponds to the particle having an average speed $v_{av} = p_{av}/m = (\hbar k_2 + \hbar k_1)/2m$. Can you show that this is equal to the expression $v_{av} = (\omega_2-\omega_1)/(k_2-k_1)$ that we found above?

KEYCONCEPT A free particle (one on which no forces act) that moves in the x-direction only is described by a wave function $\Psi(x,t)$ that obeys the free-particle Schrödinger equation [Eq. (40.15)]. A wave function with definite momentum and energy is completely delocalized; to write a wave function that is localized in space, you must superpose wave functions with different values of momentum and energy.

Figure 40.6 Superposing a large number of sinusoidal waves with different wave numbers and appropriate amplitudes can produce a wave pulse that has a wavelength $\lambda_{av} = 2\pi/k_{av}$ and is localized within a region of space of length Δx. This localized pulse has aspects of both particle and wave.



Wave Packets

The wave function that we examined in Example 40.1 is not very well localized: The probability distribution function still extends from $x = -\infty$ to $x = +\infty$. Hence this wave function can't be normalized. To make a wave function that's more highly localized, imagine superposing two additional sinusoidal waves with different wave numbers and amplitudes so as to reinforce alternate maxima of $|\Psi(x, t)|^2$ in Fig. 40.5 and cancel out the in-between ones. If we continue this process and superpose a very large number of waves with different wave numbers, we can construct a wave with only *one* maximum of $|\Psi(x, t)|^2$ (**Fig. 40.6**). Then we have something that begins to look like both a particle and a wave. It is a particle in the sense that it is localized in space; if we look from a distance, it may look like a point. But it also has a periodic structure that is characteristic of a wave.

A localized wave pulse like that shown in Fig. 40.6 is called a **wave packet.** We can represent a wave packet by an expression such as

$$\Psi(x, t) = \int_{-\infty}^{\infty} A(k) e^{i(kx - \omega t)}\, dk \qquad (40.19)$$

This integral represents a superposition of a very large number of waves, each with a different wave number k and angular frequency $\omega = \hbar k^2/2m$, and each with an amplitude $A(k)$ that depends on k.

There is an important relationship between the two functions $\Psi(x, t)$ and $A(k)$, which we show qualitatively in **Fig. 40.7**. If the function $A(k)$ is sharply peaked, as in Fig. 40.7a, we are superposing only a narrow range of wave numbers. The resulting wave pulse is then relatively broad (Fig. 40.7b). But if we use a wider range of wave numbers, so that the function $A(k)$ is broader (Fig. 40.7c), then the wave pulse is more narrowly localized (Fig. 40.7d). This is simply the uncertainty principle in action. A narrow range of k means a narrow range of $p_x = \hbar k$ and thus a small Δp_x; the result is a relatively large Δx. A broad range of k corresponds to a large Δp_x, and the resulting Δx is smaller. You can see that the uncertainty principle for position and momentum, $\Delta x \Delta p_x \geq \hbar/2$, is really just a consequence of the properties of integrals like Eq. (40.19).

CAUTION Matter waves versus light waves in vacuum We can regard both a wave packet that represents a particle and a short pulse of light from a laser as superpositions of waves of different wave numbers and angular frequencies. An important difference is that the speed of light in vacuum is the same for all wavelengths λ and hence all wave numbers $k = 2\pi/\lambda$, but the speed of a matter wave is *different* for different wavelengths. You can see this from the formula for the speed of the wave crests in a periodic wave, $v = \lambda f = \omega/k$. For a matter wave, $\omega = \hbar k^2/2m$, so $v = \hbar k/2m = h/2m\lambda$. Hence matter waves with longer wavelengths and smaller wave numbers travel more slowly than those with short wavelengths and large wave numbers. (This shouldn't be too surprising. The de Broglie relationships that we learned in Section 39.1 tell us that shorter wavelength corresponds to greater momentum and hence a greater speed.) Since the individual sinusoidal waves that make up a wave packet travel at different speeds, the shape of the packet changes as it moves. That's why we've specified the time for which the wave packets in Figs. 40.6 and 40.7 are drawn; at later times, the packets become more spread out. By contrast, a pulse of light waves in vacuum retains the same shape at all times because all of its constituent sinusoidal waves travel together at the same speed.

The One-Dimensional Schrödinger Equation with Potential Energy

The one-dimensional Schrödinger equation that we presented in Eq. (40.15) is valid only for free particles, for which the potential-energy function is zero: $U(x) = 0$. But for an electron within an atom, a proton within an atomic nucleus, and many other real situations, the potential energy plays an important role. To study the behavior of matter waves in these situations, we need a version of the Schrödinger equation that describes a particle moving in the presence of a nonzero potential-energy function $U(x)$. This equation is

General one-dimensional Schrödinger equation:

Planck's constant divided by 2π · Particle's wave function

$$-\frac{\hbar^2}{2m}\frac{\partial^2\Psi(x,t)}{\partial x^2}+U(x)\Psi(x,t)=i\hbar\frac{\partial\Psi(x,t)}{\partial t} \qquad (40.20)$$

Particle's mass · Potential-energy function

Note that if $U(x) = 0$, Eq. (40.20) reduces to the free-particle Schrödinger equation given in Eq. (40.15).

Here's the motivation behind Eq. (40.20). If $\Psi(x, t)$ is a sinusoidal wave function for a free particle, $\Psi(x, t) = Ae^{i(kx-\omega t)} = Ae^{ikx}e^{-i\omega t}$, the derivative terms in Eq. (40.20) become

$$\begin{aligned}
-\frac{\hbar^2}{2m}\frac{\partial^2\Psi(x,t)}{\partial x^2} &= -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}(Ae^{ikx}e^{-i\omega t}) = -\frac{\hbar^2}{2m}(ik)^2(Ae^{ikx}e^{-i\omega t})\\
&= \frac{\hbar^2k^2}{2m}\Psi(x,t)\\
i\hbar\frac{\partial\Psi(x,t)}{\partial t} &= i\hbar\frac{\partial}{\partial t}(Ae^{ikx}e^{-i\omega t}) = i\hbar(-i\omega)(Ae^{ikx}e^{-i\omega t}) = \hbar\omega\Psi(x,t)
\end{aligned}$$

In these expressions $(\hbar^2k^2/2m)\Psi(x, t)$ is just the kinetic energy $K = p^2/2m = \hbar^2k^2/2m$ multiplied by the wave function, and $\hbar\omega\Psi(x, t)$ is the total energy $E = \hbar\omega$ multiplied by the wave function. So for a wave function of this kind, Eq. (40.20) says that kinetic energy times $\Psi(x, t)$ plus potential energy times $\Psi(x, t)$ equals total energy times $\Psi(x, t)$. That's equivalent to the statement in classical physics that the sum of kinetic energy and potential energy equals total mechanical energy: $K + U = E$.

The observations we've just made certainly aren't a *proof* that Eq. (40.20) is correct. The real reason we know this equation *is* correct is that it works: Predictions made with this equation agree with experimental results. Later in this chapter we'll apply Eq. (40.20) to several physical situations, each with a different form of the function $U(x)$.

Stationary States

We saw in our discussion of wave packets that any free-particle wave function can be built up as a superposition of sinusoidal wave functions of the form $\Psi(x, t) = Ae^{ikx}e^{-i\omega t}$. Each such sinusoidal wave function corresponds to a state of definite energy $E = \hbar\omega = \hbar^2k^2/2m$ and definite angular frequency $\omega = E/\hbar$, so we can rewrite these functions as $\Psi(x, t) = Ae^{ikx}e^{-iEt/\hbar}$. If the potential-energy function $U(x)$ is nonzero, these sinusoidal wave functions do not satisfy the Schrödinger equation, Eq. (40.20), and so these functions cannot be the basic "building blocks" of more complicated wave functions. However, we can still write the wave function for a state of definite energy E in the form

Time-dependent wave function for a state of definite energy

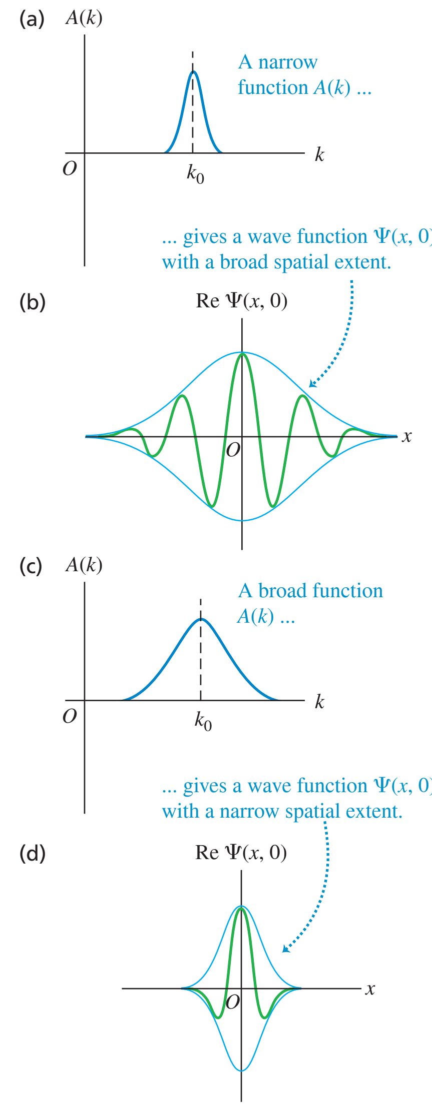
Time-independent wave function

$$\Psi(x,t) = \psi(x)e^{-iEt/\hbar} \qquad (40.21)$$

Energy of state · Planck's constant divided by 2π

That is, the wave function $\Psi(x, t)$ for a state of definite energy is the product of a *time-independent* wave function $\psi(x)$ and a factor $e^{-iEt/\hbar}$. (For the free-particle sinusoidal wave function, $\psi(x) = Ae^{ikx}$.) States of definite energy are of tremendous importance in quantum mechanics. For example, for each energy level in a hydrogen atom (Section 39.3) there is a specific wave function. It is possible for an atom to be in a state that does not have a definite energy. The wave function for any such state can be written as a combination of definite-energy wave functions, in precisely the same way that a free-particle wave packet can be written as a superposition of sinusoidal wave functions of definite energy as in Eq. (40.19).

Figure 40.7 How varying the function $A(k)$ in the wave-packet expression, Eq. (40.19), changes the character of the wave function $\Psi(x, t)$ (shown here at a specific time $t = 0$).



A state of definite energy is commonly called a **stationary state.** To see where this name comes from, let's multiply Eq. (40.21) by its complex conjugate to find the probability distribution function $|\Psi|^2$:

$$\begin{aligned} |\Psi(x,t)|^2 &= \Psi^*(x,t)\Psi(x,t) = [\psi^*(x)e^{+iEt/\hbar}][\psi(x)e^{-iEt/\hbar}] \\ &= \psi^*(x)\psi(x)e^{(+iEt/\hbar)+(-iEt/\hbar)} = |\psi(x)|^2 e^0 \\ &= |\psi(x)|^2 \end{aligned} \tag{40.22}$$

Since $|\psi(x)|^2$ does not depend on time, Eq. (40.22) shows that the same must be true for the probability distribution function $|\Psi(x,t)|^2$. This justifies the term "stationary state" for a state of definite energy.

CAUTION A stationary state does not mean a stationary particle Saying that a particle is in a stationary state does *not* mean that the particle is at rest. It's the *probability distribution* (that is, the relative likelihood of finding the particle at various positions), not the particle itself, that's stationary.

The Schrödinger equation, Eq. (40.20), becomes quite a bit simpler for stationary states. To see this, we substitute Eq. (40.21) into Eq. (40.20):

$$-\frac{\hbar^2}{2m}\frac{\partial^2[\psi(x)e^{-iEt/\hbar}]}{\partial x^2} + U(x)\psi(x)e^{-iEt/\hbar} = i\hbar\frac{\partial[\psi(x)e^{-iEt/\hbar}]}{\partial t}$$

The derivative in the first term on the left-hand side is with respect to x, so the factor $e^{-iEt/\hbar}$ comes outside of the derivative. Now we take the derivative with respect to t on the right-hand side of the equation:

$$\begin{aligned} -\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2}e^{-iEt/\hbar} + U(x)\psi(x)e^{-iEt/\hbar} &= i\hbar\left(\frac{-iE}{\hbar}\right)[\psi(x)e^{-iEt/\hbar}] \\ &= E\psi(x)e^{-iEt/\hbar} \end{aligned}$$

If we divide both sides of this equation by $e^{-iEt/\hbar}$, we get

Time-independent one-dimensional Schrödinger equation:

Planck's constant divided by 2π

Time-independent wave function

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + U(x)\psi(x) = E\psi(x) \tag{40.23}$$

Particle's mass

Potential-energy function

Energy of state

This is called the **time-independent one-dimensional Schrödinger equation.** The time-dependent factor $e^{-iEt/\hbar}$ does not appear, and Eq. (40.23) involves only the time-independent wave function $\psi(x)$. We'll devote much of this chapter to solving this equation to find the definite-energy, stationary-state wave functions $\psi(x)$ and the corresponding values of E—that is, the energies of the allowed levels—for different physical situations.

EXAMPLE 40.2 A stationary state

Consider the wave function $\psi(x) = A_1e^{ikx} + A_2e^{-ikx}$, where k is positive. Is this a valid time-independent wave function for a free particle in a stationary state? What is the energy corresponding to this wave function?

IDENTIFY and SET UP A valid stationary-state wave function for a free particle must satisfy the time-independent Schrödinger equation, Eq. (40.23), with $U(x) = 0$. To test the given function $\psi(x)$, we simply substitute it into the left-hand side of the equation. If the result is a constant times $\psi(x)$, then the wave function is indeed a solution and the constant is equal to the particle energy E.

EXECUTE Substituting $\psi(x) = A_1e^{ikx} + A_2e^{-ikx}$ and $U(x) = 0$ into Eq. (40.23), we obtain

$$\begin{aligned} -\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} &= -\frac{\hbar^2}{2m}\frac{d^2(A_1e^{ikx} + A_2e^{-ikx})}{dx^2} \\ &= -\frac{\hbar^2}{2m}[(ik)^2A_1e^{ikx} + (-ik)^2A_2e^{-ikx}] \\ &= \frac{\hbar^2k^2}{2m}(A_1e^{ikx} + A_2e^{-ikx}) = \frac{\hbar^2k^2}{2m}\psi(x) \end{aligned}$$

The result is a constant times $\psi(x)$, so this $\psi(x)$ is indeed a valid stationary-state wave function for a free particle. Comparing with Eq. (40.23) shows that the constant on the right-hand side is the particle energy: $E = \hbar^2 k^2/2m$.

EVALUATE Note that $\psi(x)$ is a *superposition* of two different wave functions: one function $(A_1 e^{ikx})$ that represents a particle with magnitude of momentum $p = \hbar k$ moving in the positive x-direction, and one function $(A_2 e^{-ikx})$ that represents a particle with the same magnitude of momentum moving in the negative x-direction. So while the combined wave function $\psi(x)$ represents a stationary state with a definite energy, this state does *not* have a definite momentum. We'll see in Section 40.2 that such a wave function can represent a *standing wave*, and we'll explore situations in which such standing matter waves can arise.

KEYCONCEPT A quantum-mechanical state of definite energy E has a wave function of the form $\Psi(x, t) = \psi(x)e^{-iEt/\hbar}$. It is called a stationary state because the associated probability distribution function $|\Psi(x, t)|^2 = |\psi(x)|^2$ does not depend on time.

TEST YOUR UNDERSTANDING OF SECTION 40.1 Does a wave packet given by Eq. (40.19) represent a stationary state?

ANSWER

no Equation (40.19) represents a superposition of wave functions with different values of wave number k and hence different values of energy $E = \hbar^2 k^2/2m$. The state that this combined wave function represents is not a state of definite energy, and therefore not a stationary state. Another way to see this is to note that there is a factor $e^{-iEt/\hbar}$ inside the integral in Eq. (40.19), with a different value of E for each value of k. This wave function therefore has a very complicated time dependence, and the probability distribution function $|\Psi(x, t)|^2$ does depend on time.

40.2 PARTICLE IN A BOX

An important problem in quantum mechanics is how to use the time-independent Schrödinger equation, Eq. (40.23), to determine the possible energy levels and the corresponding wave functions for various systems. That is, for a given potential-energy function $U(x)$, what are the possible stationary-state wave functions $\psi(x)$, and what are the corresponding energies E?

In Section 40.1 we solved this problem for the case $U(x) = 0$, corresponding to a *free* particle. The allowed wave functions and corresponding energies are

$$\psi(x) = Ae^{ikx} \qquad E = \frac{\hbar^2 k^2}{2m} \qquad \text{(free particle)} \tag{40.24}$$

The wave number k is equal to $2\pi/\lambda$, where λ is the wavelength. We found that k can have any real value, so the energy E of a free particle can have any value from zero to infinity. Furthermore, the particle can be found with equal probability at any value of x from $-\infty$ to $+\infty$.

Now let's look at a simple model in which a particle is *bound* so that it cannot escape to infinity, but rather is confined to a restricted region of space. Our system consists of a particle confined between two rigid walls separated by a distance L (**Fig. 40.8**). The motion is purely one dimensional, with the particle moving along the x-axis only and the walls at $x = 0$ and $x = L$. The potential energy corresponding to the rigid walls is infinite, so the particle cannot escape; between the walls, the potential energy is zero (**Fig. 40.9**). This situation is often described as a **"particle in a box."** This model might represent an electron that is free to move within a long, straight molecule or along a very thin wire.

Figure **40.8** The Newtonian view of a particle in a box.

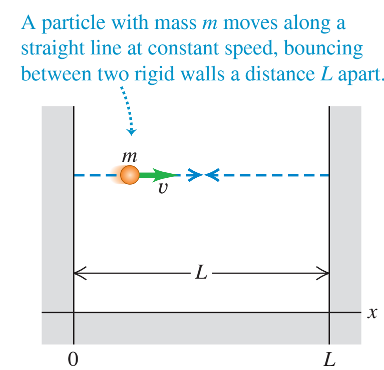
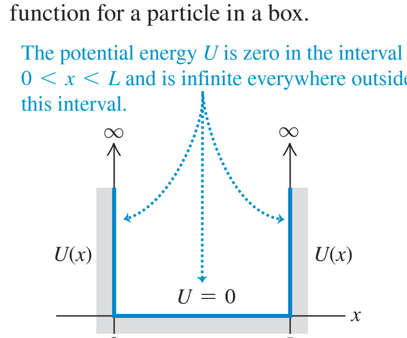


Figure **40.9** The potential-energy function for a particle in a box.

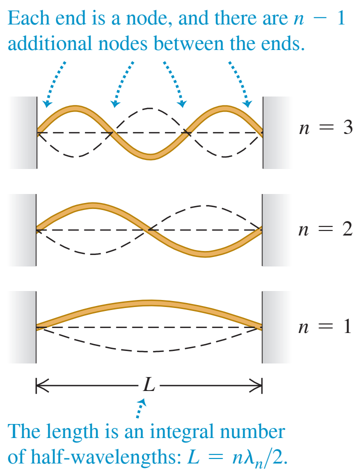


Wave Functions for a Particle in a Box

To solve the Schrödinger equation for this system, we begin with some restrictions on the particle's stationary-state wave function $\psi(x)$. Because the particle is confined to the region $0 \le x \le L$, we expect the probability distribution function $|\Psi(x, t)|^2 = |\psi(x)|^2$ and the wave function $\psi(x)$ to be zero outside that region. This agrees with the Schrödinger equation: If the term $U(x)\psi(x)$ in Eq. (40.23) is to be finite, then $\psi(x)$ must be zero where $U(x)$ is infinite.

Furthermore, $\psi(x)$ must be a *continuous* function to be a mathematically well-behaved solution to the Schrödinger equation. This implies that $\psi(x)$ must be zero at the region's boundary, $x = 0$ and $x = L$. These two conditions serve as *boundary conditions* for the

Figure **40.10** Normal modes of vibration for a string with length L, held at both ends.



problem. They should look familiar, because they are the same conditions that we used to find the normal modes of a vibrating string in Section 15.8 (**Fig. 40.10**); you should review that discussion.

An additional condition is that to calculate the second derivative $d^2\psi(x)/dx^2$ in Eq. (40.23), the *first* derivative $d\psi(x)/dx$ must also be continuous except at points where the potential energy becomes infinite (as it does at the walls of the box). This is analogous to the requirement that a vibrating string, like those shown in Fig. 40.10, can't have any kinks in it (which would correspond to a discontinuity in the first derivative of the wave function) except at the ends of the string.

We now solve for the wave functions in the region $0 \leq x \leq L$ subject to the above conditions. In this region $U(x) = 0$, so $\psi(x)$ in this region must satisfy

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} = E\psi(x) \qquad \text{(particle in a box)} \tag{40.25}$$

Equation (40.25) is the *same* Schrödinger equation as for a free particle, so it is tempting to conclude that the wave functions and energies are given by Eq. (40.24). It is true that $\psi(x) = Ae^{ikx}$ satisfies the Schrödinger equation with $U(x) = 0$, is continuous, and has a continuous first derivative $d\psi(x)/dx = ikAe^{ikx}$. However, this wave function does *not* satisfy the boundary conditions that $\psi(x)$ must be zero at $x = 0$ and $x = L$: At $x = 0$ the wave function in Eq. (40.24) is equal to $Ae^0 = A$, and at $x = L$ it is equal to Ae^{ikL}. (These would be equal to zero if $A = 0$, but then the wave function would be zero and there would be no particle at all!)

The way out of this dilemma is to recall Example 40.2 (Section 40.1), in which we found that a more general stationary-state solution to the time-independent Schrödinger equation with $U(x) = 0$ is

$$\psi(x) = A_1e^{ikx} + A_2e^{-ikx} \tag{40.26}$$

This wave function is a superposition of two waves: one traveling in the $+x$-direction of amplitude A_1, and one traveling in the $-x$-direction with the same wave number but amplitude A_2. This is analogous to a standing wave on a string (Fig. 40.10), which we can regard as the superposition of two sinusoidal waves propagating in opposite directions (see Section 15.7). The energy that corresponds to Eq. (40.26) is $E = \hbar^2k^2/2m$, just as for a single wave.

To see whether the wave function given by Eq. (40.26) can satisfy the boundary conditions, let's first rewrite it in terms of sines and cosines by using Euler's formula, Eq. (40.17):

$$\begin{aligned}\psi(x) &= A_1(\cos kx + i\sin kx) + A_2[\cos(-kx) + i\sin(-kx)]\\ &= A_1(\cos kx + i\sin kx) + A_2(\cos kx - i\sin kx)\\ &= (A_1 + A_2)\cos kx + i(A_1 - A_2)\sin kx\end{aligned} \tag{40.27}$$

At $x = 0$ this is equal to $\psi(0) = A_1 + A_2$, which must equal zero to satisfy the boundary condition at that point. Hence $A_2 = -A_1$, and Eq. (40.27) becomes

$$\psi(x) = 2iA_1\sin kx = C\sin kx \tag{40.28}$$

We have simplified the expression by introducing the constant $C = 2iA_1$. (We'll come back to this constant later.) We can also satisfy the second boundary condition that $\psi = 0$ at $x = L$ by choosing values of k such that $kL = n\pi$ $(n = 1, 2, 3, \ldots)$. Hence Eq. (40.28) does indeed give the stationary-state wave functions for a particle in a box in the region $0 \leq x \leq L$. (Outside this region, $\psi(x) = 0$.) The possible values of k and the wavelength $\lambda = 2\pi/k$ are

$$k = \frac{n\pi}{L} \quad \text{and} \quad \lambda = \frac{2\pi}{k} = \frac{2L}{n} \quad (n = 1, 2, 3, \ldots) \tag{40.29}$$

Just as for the string in Fig. 40.10, the length L of the region is an integral number of half-wavelengths.

Energy Levels for a Particle in a Box

The possible energy levels for a particle in a box are given by $E = \hbar^2 k^2/2m = p^2/2m$, where $p = \hbar k = (h/2\pi)(2\pi/\lambda) = h/\lambda$ is the magnitude of momentum of a free particle with wave number k and wavelength λ. This makes sense, since inside the region $0 \le x \le L$ the potential energy is zero and the energy is all kinetic. For each value of n, there are corresponding values of p, λ, and E; let's call them p_n, λ_n, and E_n. Putting the pieces together, we get

$$p_n = \frac{h}{\lambda_n} = \frac{nh}{2L} \tag{40.30}$$

and so the energy levels for a particle in a box are

Magnitude of momentum; Planck's constant; Planck's constant divided by 2π

Energy levels for a particle in a box

$$E_n = \frac{p_n^2}{2m} = \frac{n^2h^2}{8mL^2} = \frac{n^2\pi^2\hbar^2}{2mL^2} \quad (n = 1, 2, 3, \ldots) \tag{40.31}$$

Particle's mass; Width of box; Quantum number

Each energy level has its own value of the quantum number n and a corresponding wave function, which we denote by ψ_n. When we replace k in Eq. (40.28) by $n\pi/L$ from Eq. (40.29), we find

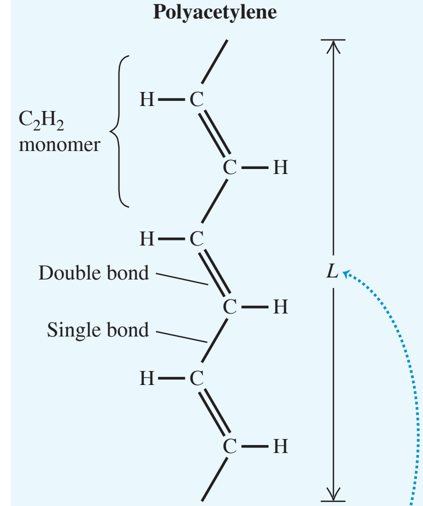
$$\psi_n(x) = C\sin\frac{n\pi x}{L} \quad (n = 1, 2, 3, \ldots) \tag{40.32}$$

The energy-level diagram in **Fig. 40.11a** shows the five lowest levels for a particle in a box. The energy levels are proportional to n^2, so successively higher levels are spaced farther and farther apart. There are an infinite number of levels because the walls are perfectly rigid; even a particle of infinitely great kinetic energy is confined within the box. Figure 40.11b shows graphs of the wave functions $\psi_n(x)$ for $n = 1, 2, 3, 4$, and 5. Note that these functions look identical to those for a standing wave on a string (see Fig. 40.10).

CAUTION A particle in a box cannot have zero energy The energy of a particle in a box *cannot* be zero. Equation (40.31) shows that $E = 0$ would require $n = 0$, but substituting $n = 0$ into Eq. (40.32) gives a zero wave function. Since a particle is described by a *nonzero* wave function, there cannot be a particle with $E = 0$. This is a consequence of the Heisenberg uncertainty principle: A particle in a zero-energy state would have a definite value of momentum (precisely zero), so its position uncertainty would be infinite and the particle could be found anywhere along the x-axis. But this is impossible, since a particle in a box can be found only between $x = 0$ and $x = L$. Hence $E = 0$ is not allowed. By contrast, the allowed stationary-state wave functions with $n = 1, 2, 3, \ldots$ do not represent states of definite momentum (each is an equal mixture of a state of x-momentum $+p_n = nh/2L$ and a state of x-momentum $-p_n = -nh/2L$). Hence each stationary state has a nonzero momentum uncertainty, consistent with having a finite position uncertainty.

APPLICATION Particles in a Polymer "Box" Polyacetylene is one of a class of long-chain organic molecules that conduct electricity along their length. The molecule is made up of a large number of (C_2H_2) units, called *monomers* (only three monomers are shown here). Electrons can move freely along the length of the molecule but not perpendicular to the length, so the molecule is like a one-dimensional "box" for electrons. The length L of the molecule depends on the number of monomers. Experiment shows that the allowed energy levels agree well with Eq. (40.31): The greater the number of monomers and the greater the length L, the lower the energy levels and the smaller the spacing between these levels.

Polyacetylene



Electrons are confined to the length L of the molecule (like particles in a box).

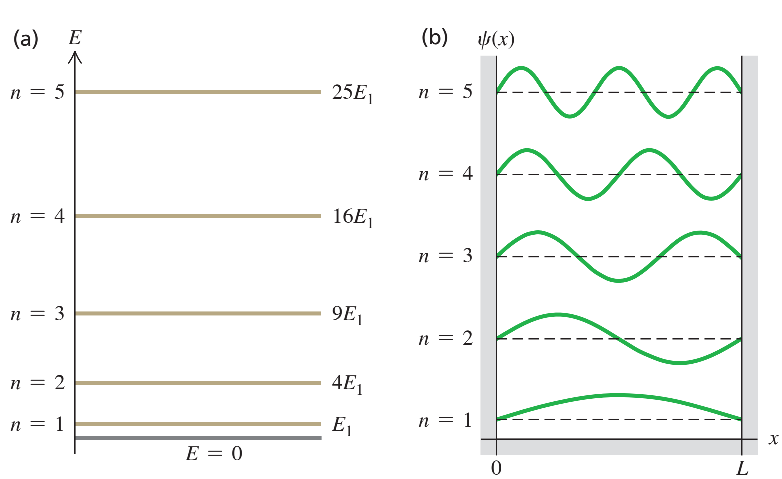


Figure 40.11 (a) Energy-level diagram for a particle in a box. Each energy is n^2E_1, where E_1 is the ground-level energy. (b) Wave functions for a particle in a box, with $n = 1, 2, 3, 4$, and 5. **CAUTION:** The five graphs have been displaced vertically for clarity, as in Fig. 40.10. Each of the horizontal dashed lines represents $\psi = 0$ for the respective wave function.

EXAMPLE 40.3 Electron in an atom-size box

WITH VARIATION PROBLEMS

Find the first two energy levels for an electron confined to a one-dimensional box 5.0×10^{-10} m across (about the diameter of an atom).

IDENTIFY and SET UP This problem uses what we have learned in this section about a particle in a box. The first two energy levels correspond to $n = 1$ and $n = 2$ in Eq. (40.31).

EXECUTE From Eq. (40.31),

$$E_1 = \frac{h^2}{8mL^2} = \frac{(6.626 \times 10^{-34}\ \mathrm{J \cdot s})^2}{8(9.109 \times 10^{-31}\ \mathrm{kg})(5.0 \times 10^{-10}\ \mathrm{m})^2}$$

$$= 2.4 \times 10^{-19}\ \mathrm{J} = 1.5\ \mathrm{eV}$$

$$E_2 = \frac{2^2h^2}{8mL^2} = 4E_1 = 9.6 \times 10^{-19}\ \mathrm{J} = 6.0\ \mathrm{eV}$$

EVALUATE The difference between the first two energy levels is $E_2 - E_1 = 4.5$ eV. An electron confined to a box is different from an electron bound in an atom, but it is reassuring that this result is of the same order of magnitude as the difference between actual atomic energy levels.

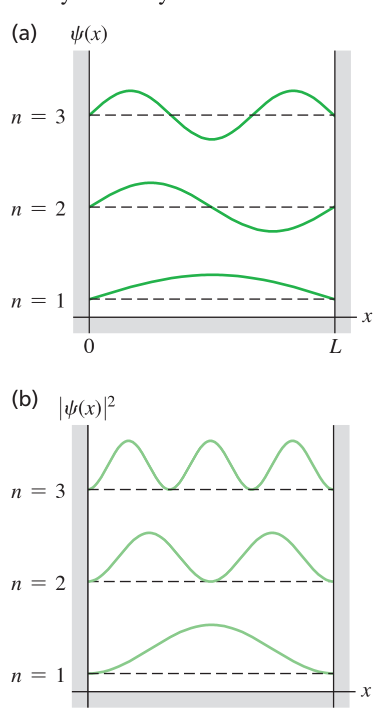
You can show that for a proton or neutron $(m = 1.67 \times 10^{-27}\ \mathrm{kg})$ confined to a box 1.1×10^{-14} m across (the width of a medium-sized atomic nucleus), the energies of the first two levels are about a million times larger: $E_1 = 1.7 \times 10^6\ \mathrm{eV} = 1.7\ \mathrm{MeV}$, $E_2 = 4E_1 = 6.8\ \mathrm{MeV}$, $E_2 - E_1 = 5.1\ \mathrm{MeV}$. This suggests why nuclear reactions (which involve transitions between energy levels in nuclei) release so much more energy than chemical reactions (which involve transitions between energy levels of electrons in atoms).

Finally, you can show (see Exercise 40.9) that the energy levels of a billiard ball $(m = 0.2\ \mathrm{kg})$ confined to a box 1.3 m across—the width of a billiard table—are separated by about 5×10^{-67} J. Quantum effects won't disturb a game of billiards.

KEYCONCEPT The wave functions for the stationary states of a particle confined to a one-dimensional box are sinusoidal standing waves inside the box with a node at each end; the wave function is zero outside the box. The nth stationary state has a wave function with n half-wavelengths within the length of the box, and has an energy equal to n^2 times the energy of the lowest-energy ($n = 1$) state. There are an infinite number of these stationary states.

Probability and Normalization

Figure 40.12 Graphs of (a) $\psi(x)$ and (b) $|\psi(x)|^2$ for the first three wave functions ($n = 1, 2, 3$) for a particle in a box. The horizontal dashed lines represent $\psi(x) = 0$ and $|\psi(x)|^2 = 0$ for each of the three levels. The value of $|\psi(x)|^2\,dx$ at each point is the probability of finding the particle in a small interval dx about the point. As in Fig. 40.11b, the three graphs in each part have been displaced vertically for clarity.



Let's look a bit more closely at the wave functions for a particle in a box, keeping in mind the *probability* interpretation of the wave function ψ that we discussed in Section 40.1. In our one-dimensional situation the quantity $|\psi(x)|^2\,dx$ is proportional to the probability that the particle will be found within a small interval dx about x. For a particle in a box,

$$|\psi(x)|^2\,dx = C^2 \sin^2 \frac{n\pi x}{L}\,dx$$

Figure 40.12 shows graphs of both $\psi(x)$ and $|\psi(x)|^2$ for $n = 1$, 2, and 3. Note that not all positions are equally likely. By contrast, in classical mechanics the particle is equally likely to be found at any position between $x = 0$ and $x = L$. We see from Fig. 40.12b that $|\psi(x)|^2 = 0$ at some points, so there is zero probability of finding the particle at exactly these points. Don't let that bother you; the uncertainty principle has already shown us that we can't measure position exactly. The particle is localized only to be somewhere between $x = 0$ and $x = L$.

The particle must be *somewhere* on the x-axis—that is, somewhere between $x = -\infty$ and $x = +\infty$. So the *sum* of the probabilities for all the dx's everywhere (the *total* probability of finding the particle) must equal 1. That's the normalization condition that we discussed in Section 40.1:

Normalization condition, time-independent wave function:

$$\overbrace{\int_{-\infty}^{\infty}}^{\text{Integral over all } x} \underbrace{|\overset{\text{Time-independent wave function}}{\psi(x)}|^2}_{\text{Probability distribution function}}\,dx = \underset{\text{Probability that particle is } \textit{somewhere} \text{ on } x\text{-axis}}{1} \qquad (40.33)$$

A wave function is said to be *normalized* if it has a constant such as C in Eq. (40.32) that is calculated to make the total probability equal 1 in Eq. (40.33). For a normalized wave function, $|\psi(x)|^2\,dx$ is not merely proportional to, but *equals,* the probability of finding the particle between the coordinates x and $x + dx$. That's why we call $|\psi(x)|^2$ the probability distribution function. (In Section 40.1 we called $|\Psi(x, t)|^2$ the probability distribution function. For the case of a stationary-state wave function, however, $|\Psi(x, t)|^2$ is equal to $|\psi(x)|^2$.)

Let's normalize the particle-in-a-box wave functions $\psi_n(x)$ given by Eq. (40.32). Since $\psi_n(x)$ is zero except between $x = 0$ and $x = L$, Eq. (40.33) becomes

$$\int_0^L C^2 \sin^2 \frac{n\pi x}{L}\,dx = 1 \tag{40.34}$$

You can evaluate this integral by using the trigonometric identity $\sin^2\theta = \frac{1}{2}(1 - \cos 2\theta)$; the result is $C^2L/2$. Thus our probability interpretation of the wave function demands that $C^2L/2 = 1$, or $C = (2/L)^{1/2}$; the constant C is *not* arbitrary. (This is in contrast to the classical vibrating string problem, in which C represents an amplitude that depends on initial conditions.) Thus the normalized stationary-state wave functions for a particle in a box are

Stationary-state wave functions for a particle in a box

$$\psi_n(x) = \sqrt{\frac{2}{\underbrace{L}_{\text{Width of box}}}}\sin\frac{\overbrace{n}^{\text{Quantum number}}\pi x}{L} \qquad (n = 1, 2, 3, \ldots) \tag{40.35}$$

EXAMPLE 40.4 A nonsinusoidal wave function?

WITH VARIATION PROBLEMS

(a) Show that $\psi(x) = Ax + B$, where A and B are constants, is a solution of the Schrödinger equation for an $E = 0$ energy level of a particle in a box. (b) What constraints do the boundary conditions at $x = 0$ and $x = L$ place on the constants A and B?

IDENTIFY and SET UP To be physically reasonable, a wave function must satisfy both the Schrödinger equation and the appropriate boundary conditions. In part (a) we'll substitute $\psi(x)$ into the Schrödinger equation for a particle in a box, Eq. (40.25), to determine whether it is a solution. In part (b) we'll see what restrictions on $\psi(x)$ arise from applying the boundary conditions that $\psi(x) = 0$ at $x = 0$ and $x = L$.

EXECUTE (a) From Eq. (40.25), the Schrödinger equation for an $E = 0$ energy level of a particle in a box is

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} = E\psi(x) = 0$$

in the region $0 \le x \le L$. Differentiating $\psi(x) = Ax + B$ twice with respect to x gives $d^2\psi(x)/dx^2 = 0$, so the left side of the equation is zero, and so $\psi(x) = Ax + B$ is a solution of this Schrödinger equation for $E = 0$. (Note that both $\psi(x)$ and its derivative $d\psi(x)/dx = A$ are continuous functions, as they must be.)

(b) Applying the boundary condition at $x = 0$ gives $\psi(0) = B = 0$, and so $\psi(x) = Ax$. Applying the boundary condition at $x = L$ gives $\psi(L) = AL = 0$, so $A = 0$. Hence $\psi(x) = 0$ both inside the box $(0 \le x \le L)$ *and outside:* There is *zero* probability of finding the particle anywhere with this wave function, and so $\psi(x) = Ax + B$ is *not* a physically valid wave function.

EVALUATE The moral is that there are many functions that satisfy the Schrödinger equation for a given physical situation, but most of these—including the function considered here—have to be rejected because they don't satisfy the appropriate boundary conditions.

KEYCONCEPT To be physically reasonable, a stationary-state wave function for a given situation must satisfy both the Schrödinger equation and any boundary conditions that apply to that situation.

Time Dependence

The wave functions $\psi_n(x)$ in Eq. (40.35) depend only on the *spatial* coordinate x. Equation (40.21) shows that if $\psi(x)$ is the wave function for a state of definite energy E, the full time-dependent wave function is $\Psi(x, t) = \psi(x)e^{-iEt/\hbar}$. Hence the *time-dependent* stationary-state wave functions for a particle in a box are

$$\Psi_n(x, t) = \sqrt{\frac{2}{L}}\sin\left(\frac{n\pi x}{L}\right)e^{-iE_nt/\hbar} \qquad (n = 1, 2, 3, \ldots) \tag{40.36}$$

In this expression the energies E_n are given by Eq. (40.31). The higher the quantum number n, the greater the angular frequency $\omega_n = E_n/\hbar$ at which the wave function oscillates. Note that since $|e^{-iE_nt/\hbar}|^2 = e^{+iE_nt/\hbar}e^{-iE_nt/\hbar} = e^0 = 1$, the probability distribution function $|\Psi_n(x, t)|^2 = (2/L)\sin^2(n\pi x/L)$ is independent of time and does *not* oscillate. (Remember, this is why we say that these states of definite energy are *stationary*.)

TEST YOUR UNDERSTANDING OF SECTION 40.2 If a particle in a box is in the nth energy level, what is the average value of its x-component of momentum p_x? (i) $nh/2L$; (ii) $(\sqrt{2}/2)nh/L$; (iii) $(1/\sqrt{2})nh/L$; (iv) $[1/(2\sqrt{2})]nh/L$; (v) zero.

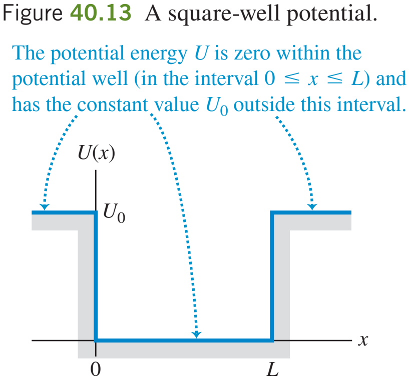
ANSWER (v) Our derivation of the stationary-state wave functions for a particle in a box shows that they are superpositions of waves propagating in opposite directions, just like a standing wave on a string. One wave has momentum in the positive x-direction, while the other wave has an equal magnitude of momentum in the negative x-direction. The *total* x-component of momentum is zero.

40.3 POTENTIAL WELLS

A **potential well** is a potential-energy function $U(x)$ that has a minimum. We introduced this term in Section 7.5, and we also used it in our discussion of periodic motion in Chapter 14. In Newtonian mechanics a particle trapped in a potential well can vibrate back and forth with periodic motion. Our first application of the Schrödinger equation, the particle in a box, involved a rudimentary potential well with a function $U(x)$ that is zero within a certain interval and infinite everywhere else. As we mentioned in Section 40.2, this function corresponds to a few situations found in nature, but the correspondence is only approximate.

A better approximation to several physical situations is a **finite well,** which is a potential well that has straight sides but *finite* height. **Figure 40.13** shows a potential-energy function that is zero in the interval $0 \le x \le L$ and has the value U_0 outside this interval. This function is often called a **square-well potential.** It could serve as a simple model of an electron within a metallic sheet with thickness L, moving perpendicular to the surfaces of the sheet. The electron can move freely inside the metal but has to climb a potential-energy barrier with height U_0 to escape from either surface of the metal. The energy U_0 is related to the *work function* that we discussed in Section 38.1 in connection with the photoelectric effect. In three dimensions, a spherical version of a finite well gives an approximate description of the motions of protons and neutrons within a nucleus.

Figure 40.13 A square-well potential.



Bound States of a Square-Well Potential

In Newtonian mechanics, the particle is trapped (localized) in a well if the total mechanical energy E is less than U_0. In quantum mechanics, such a trapped state is often called a **bound state.** All states are bound for an infinitely deep well like the one we described in Section 40.2. For a finite well like that shown in Fig. 40.13, if E is greater than U_0, the particle is *not* bound.

Let's see how to solve the Schrödinger equation for the bound states of a square-well potential. Our goal is to find the energies and wave functions for which $E < U_0$. It's easiest to consider separately the regions where $U = 0$ and where $U = U_0$. Inside the square well $(0 \le x \le L)$, where $U = 0$, the time-independent Schrödinger equation is

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} = E\psi(x) \quad \text{or} \quad \frac{d^2\psi(x)}{dx^2} = -\frac{2mE}{\hbar^2}\psi(x) \qquad (40.37)$$

This is the same as Eq. (40.25) from Section 40.2, which describes a particle in a box. As in Section 40.2, we can express the solutions of this equation as combinations of $\cos kx$ and $\sin kx$, where $E = \hbar^2k^2/2m$. We can rewrite the relationship between E and k as $k = \sqrt{2mE}/\hbar$. Hence inside the square well we have

$$\psi(x) = A\cos\left(\frac{\sqrt{2mE}}{\hbar}x\right) + B\sin\left(\frac{\sqrt{2mE}}{\hbar}x\right) \qquad \text{(inside the well)} \quad (40.38)$$

where A and B are constants. So far, this looks a lot like the particle-in-a-box analysis in Section 40.2. The difference is that for the square-well potential, the potential energy outside the well is not infinite, so the wave function $\psi(x)$ outside the well is *not* zero.

For the regions outside the well ($x < 0$ and $x > L$) the potential-energy function in the time-independent Schrödinger equation is $U = U_0$:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + U_0\psi(x) = E\psi(x) \quad \text{or} \quad \frac{d^2\psi(x)}{dx^2} = \frac{2m(U_0 - E)}{\hbar^2}\psi(x) \qquad (40.39)$$

The quantity $U_0 - E$ is positive, so the solutions of this equation are exponential functions rather than sines or cosines. Using κ (the Greek letter kappa) to represent the quantity $[2m(U_0 - E)]^{1/2}/\hbar$ and taking κ as positive, we can write the solutions as

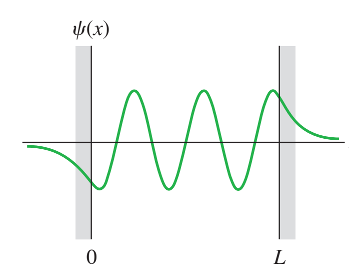
$$\psi(x) = Ce^{\kappa x} + De^{-\kappa x} \qquad \text{(outside the well)} \qquad (40.40)$$

where C and D are constants with different values in the two regions $x < 0$ and $x > L$. Note that ψ can't be allowed to approach infinity as $x \to +\infty$ or $x \to -\infty$. [If it did, we wouldn't be able to satisfy the normalization condition, Eq. (40.33).] This means that in Eq. (40.40), we must have $D = 0$ for $x < 0$ and $C = 0$ for $x > L$.

Our calculations so far show that the bound-state wave functions for a finite well are sinusoidal inside the well [Eq. (40.38)] and exponential outside it [Eq. (40.40)]. We have to *match* the wave functions inside and outside the well so that they satisfy the boundary conditions that we mentioned in Section 40.2: $\psi(x)$ and $d\psi(x)/dx$ must be continuous at the boundary points $x = 0$ and $x = L$. If the wave function $\psi(x)$ or the slope $d\psi(x)/dx$ were to change discontinuously at a point, the second derivative $d^2\psi(x)/dx^2$ would be *infinite* at that point. That would violate the time-independent Schrödinger equation, Eq. (40.23), which says that at every point $d^2\psi(x)/dx^2$ is proportional to $U - E$. For a finite well $U - E$ is finite everywhere, so $d^2\psi(x)/dx^2$ must also be finite everywhere.

Matching the sinusoidal and exponential functions at the boundary points so that they join smoothly is possible only for certain specific values of the total energy E, so this requirement determines the possible energy levels of the finite square well. There is no simple formula for the energy levels as there was for the infinitely deep well. Finding the levels is a fairly complex mathematical problem that requires solving a transcendental equation by numerical approximation; we won't go into the details. **Figure 40.14** shows the general shape of a possible wave function. The most striking features of this wave function are the "exponential tails" that extend outside the well into regions that are forbidden by Newtonian mechanics (because in those regions the particle would have negative kinetic energy). We see that there is some probability for finding the particle *outside* the potential well, which would be impossible in classical mechanics. In Section 40.4 we'll discuss an amazing result of this effect.

Figure **40.14** A possible wave function for a particle in a finite potential well. The function is sinusoidal inside the well ($0 \le x \le L$) and exponential outside it. It approaches zero asymptotically at large $|x|$. The functions must join smoothly at $x = 0$ and $x = L$; the wave function and its derivative must be continuous.



EXAMPLE 40.5 Outside a finite well

(a) Show that Eq. (40.40), $\psi(x) = Ce^{\kappa x} + De^{-\kappa x}$, is indeed a solution of the time-independent Schrödinger equation outside a finite well of height U_0. (b) What happens to $\psi(x)$ in the limit $U_0 \to \infty$?

IDENTIFY and SET UP In part (a), we try the given function $\psi(x)$ in the time-independent Schrödinger equation for $x < 0$ and for $x > L$, Eq. (40.39). In part (b), we note that in the limit $U_0 \to \infty$ the finite well becomes an *infinite* well, like that for a particle in a box (Section 40.2). So in this limit the wave functions outside a finite well must reduce to the wave functions outside the box.

EXECUTE (a) We must show that $\psi(x) = Ce^{\kappa x} + De^{-\kappa x}$ satisfies $d^2\psi(x)/dx^2 = [2m(U_0 - E)/\hbar^2]\psi(x)$. We recall that $(d/du)e^{au} = ae^{au}$ and $(d^2/du^2)e^{au} = a^2e^{au}$; the left-hand side of the Schrödinger equation is then

$$\begin{aligned}\frac{d^2\psi(x)}{dx^2} &= \frac{d^2}{dx^2}(Ce^{\kappa x}) + \frac{d^2}{dx^2}(De^{-\kappa x})\\ &= C\kappa^2e^{\kappa x} + D(-\kappa)^2e^{-\kappa x}\\ &= \kappa^2(Ce^{\kappa x} + De^{-\kappa x}) = \kappa^2\psi(x)\end{aligned}$$

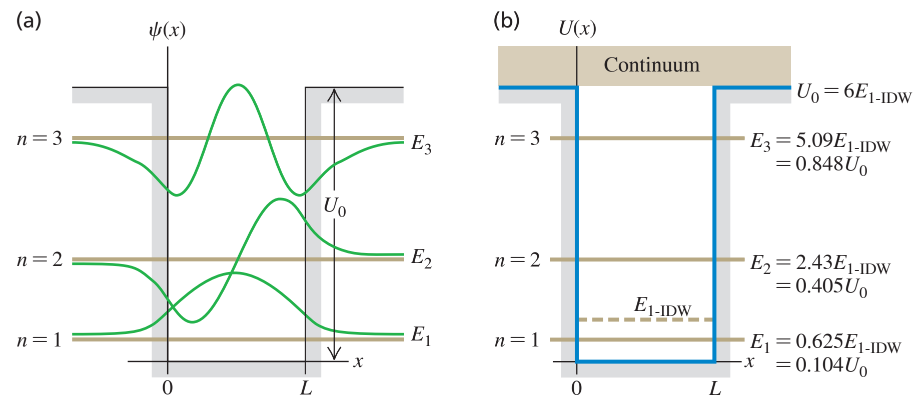
Since from Eq. (40.40) $\kappa^2 = 2m(U_0 - E)/\hbar^2$, this is equal to the right-hand side of the equation. The equation is satisfied, and $\psi(x)$ is a solution.

(b) As U_0 approaches infinity, κ also approaches infinity. In the region $x < 0$, $\psi(x) = Ce^{\kappa x}$; as $\kappa \to \infty$, $\kappa x \to -\infty$ (since x is negative) and $e^{\kappa x} \to 0$, so the wave function approaches zero for all $x < 0$. Likewise, we can show that the wave function also approaches zero for all $x > L$. This is just what we found in Section 40.2; the wave function for a particle in a box must be zero outside the box.

EVALUATE Our result in part (b) shows that the infinite square well is a *limiting case* of the finite well. We've seen many cases in Newtonian mechanics where it's important to consider limiting cases [such as Examples 5.11 (Section 5.2) and 5.13 (Section 5.3)]. Limiting cases are no less important in quantum mechanics.

KEYCONCEPT A finite potential well is a region where the potential energy is constant and less than the constant value outside that region. If a particle is in a bound state, so that in Newtonian mechanics it would not have enough energy to escape from the well, its stationary-state wave function is sinusoidal within the well and exponential outside the well.

Figure **40.15** (a) Wave functions for the three bound states for a particle in a finite potential well of depth $U_0 = 6E_{1-\mathrm{IDW}}$. (Here $E_{1-\mathrm{IDW}}$ is the ground-level energy for an infinite well of the same width.) The horizontal brown line for each wave function corresponds to $\psi = 0$; the vertical placement of these lines indicates the energy of each bound state (compare Fig. 40.11). (b) Energy-level diagram for this system. All energies greater than U_0 are possible; states with $E > U_0$ form a continuum.



Comparing Finite and Infinite Square Wells

Let's continue the comparison of the finite-depth potential well with the infinitely deep well, which we began in Example 40.5. First, because the wave functions for the finite well don't go to zero at $x = 0$ and $x = L$, the wavelength of the sinusoidal part of each wave function is *longer* than it would be with an infinite well. This increase in λ corresponds to a reduced magnitude of momentum $p = h/\lambda$ and therefore a reduced energy. Thus each energy level, including the ground level, is *lower* for a finite well than for an infinitely deep well with the same width.

Second, a well with finite depth U_0 has only a *finite* number of bound states and corresponding energy levels, compared to the *infinite* number for an infinitely deep well. How many levels there are depends on the magnitude of U_0 in comparison with the ground-level energy for the infinitely deep well (IDW), which we call $E_{1-\mathrm{IDW}}$. From Eq. (40.31),

$$E_{1-\mathrm{IDW}} = \frac{\pi^2\hbar^2}{2mL^2} \quad \text{(ground-level energy, infinitely deep well)} \tag{40.41}$$

When the well is very deep so U_0 is much larger than $E_{1-\mathrm{IDW}}$, there are many bound states and the energies of the lowest few are nearly the same as the energies for the infinitely deep well. When U_0 is only a few times as large as $E_{1-\mathrm{IDW}}$ there are only a few bound states. (There is always at least *one* bound state, no matter how shallow the well.) As with the infinitely deep well, there is no state with $E = 0$; such a state would violate the uncertainty principle.

Figure 40.15 shows the case $U_0 = 6E_{1-\mathrm{IDW}}$, for which there are three bound states. In the figure, we express the energy levels both as fractions of the well depth U_0 and as multiples of $E_{1-\mathrm{IDW}}$. If the well were infinitely deep, the lowest three levels, as given by Eq. (40.31), would be $E_{1-\mathrm{IDW}}$, $4E_{1-\mathrm{IDW}}$, and $9E_{1-\mathrm{IDW}}$. Figure 40.15 also shows the wave functions for the three bound states.

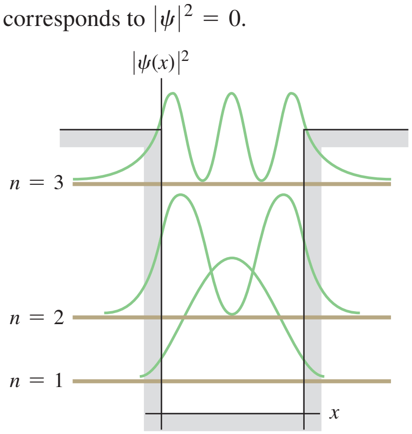
It turns out that when U_0 is less than $E_{1-\mathrm{IDW}}$, there is only one bound state. In the limit when U_0 is *much smaller* than $E_{1-\mathrm{IDW}}$ (a very shallow well), the energy of this single state is approximately $E = 0.68U_0$.

Figure 40.16 shows graphs of the probability distributions—that is, the values of $|\psi|^2$—for the wave functions shown in Fig. 40.15a. As with the infinite well, not all positions are equally likely. Unlike the infinite well, there is some probability of finding the particle outside the well in the classically forbidden regions.

There are also states for which E is *greater* than U_0. In these *free-particle states* the particle is not bound but is free to move through all values of x. *Any* energy E greater than U_0 is possible, so the free-particle states form a *continuum* rather than a discrete set

CAUTION Misconceptions about the finite potential well Note that in our notation the energy of a particle in a finite potential well is given relative to the *bottom* of the well, which we take to be $E = 0$. (We did the same for the particle in a box, or infinite potential well, in Section 40.2.) If the depth of a finite square well is U_0, a bound state has an energy that is greater than zero but less than U_0. Note also that how far the exponential part of a bound state wave function extends outside a finite potential well depends on the energy of the bound state. The greater the bound state energy, the farther outside the well the wave function extends and the farther outside the particle is likely to be found (Fig. 40.15).

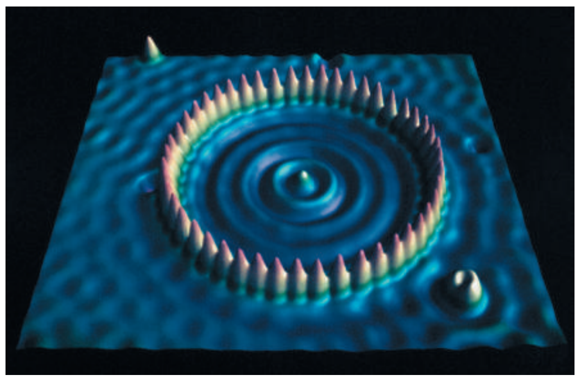
Figure **40.16** Probability distribution functions $|\psi(x)|^2$ for the square-well wave functions shown in Fig. 40.15. The horizontal brown line for each wave function corresponds to $|\psi|^2 = 0$.



of states with definite energy levels. The free-particle wave functions are sinusoidal both inside and outside the well. The wavelength is shorter inside the well than outside, corresponding to greater kinetic energy inside the well than outside it.

Figure 40.17 shows particles in a *two*-dimensional finite potential well. Example 40.6 describes another application of the square-well potential.

Figure **40.17** To make this image, 48 iron atoms (shown as yellow peaks) were placed in a circle on a copper surface. The "elevation" at each point inside the circle indicates the electron density within the circle. The standing-wave pattern is very similar to the probability distribution function for a particle in a one-dimensional finite potential well. (This image was made with a scanning tunneling microscope, discussed in Section 40.4.)



EXAMPLE 40.6 An electron in a finite well

WITH VARIATION PROBLEMS

An electron is trapped in a square well 0.50 nm across (roughly five times a typical atomic diameter). (a) Find the ground-level energy $E_{1-\mathrm{IDW}}$ if the well is infinitely deep. (b) Find the energy levels if the actual well depth U_0 is six times the ground-level energy found in part (a). (c) Find the wavelength of the photon emitted when the electron makes a transition from the $n = 2$ level to the $n = 1$ level. In what region of the electromagnetic spectrum does the photon wavelength lie? (d) If the electron is in the $n = 1$ (ground) level and absorbs a photon, what is the minimum photon energy that will free the electron from the well? In what region of the spectrum does the wavelength of this photon lie?

IDENTIFY and SET UP Equation (40.41) gives the ground-level energy $E_{1-\mathrm{IDW}}$ for an infinitely deep well, and Fig. 40.15b shows the energies for a square well with $U_0 = 6E_{1-\mathrm{IDW}}$. The energy of the photon emitted or absorbed in a transition is equal to the difference in energy between two levels involved in the transition; the photon wavelength is given by $E = hc/\lambda$ (see Chapter 38).

EXECUTE (a) From Eq. (40.41),

$$E_{1-\mathrm{IDW}} = \frac{\pi^2\hbar^2}{2mL^2} = \frac{\pi^2(1.055 \times 10^{-34}\ \mathrm{J \cdot s})^2}{2(9.11 \times 10^{-31}\ \mathrm{kg})(0.50 \times 10^{-9}\ \mathrm{m})^2}$$
$$= 2.4 \times 10^{-19}\ \mathrm{J} = 1.5\ \mathrm{eV}$$

(b) We have $U_0 = 6E_{1-\mathrm{IDW}} = 6(1.5\ \mathrm{eV}) = 9.0\ \mathrm{eV}$. We can read off the energy levels from Fig. 40.15b:

$$E_1 = 0.625E_{1-\mathrm{IDW}} = 0.625(1.5\ \mathrm{eV}) = 0.94\ \mathrm{eV}$$
$$E_2 = 2.43E_{1-\mathrm{IDW}} = 2.43(1.5\ \mathrm{eV}) = 3.6\ \mathrm{eV}$$
$$E_3 = 5.09E_{1-\mathrm{IDW}} = 5.09(1.5\ \mathrm{eV}) = 7.6\ \mathrm{eV}$$

(c) The photon energy and wavelength for the $n = 2$ to $n = 1$ transition are

$$E_2 - E_1 = 3.6\ \mathrm{eV} - 0.94\ \mathrm{eV} = 2.7\ \mathrm{eV}$$
$$\lambda = \frac{hc}{E} = \frac{(4.136 \times 10^{-15}\ \mathrm{eV \cdot s})(3.00 \times 10^8\ \mathrm{m/s})}{2.7\ \mathrm{eV}} = 460\ \mathrm{nm}$$

in the blue region of the visible spectrum.

(d) We see from Fig. 40.15b that the minimum energy needed to free the electron from the well from the $n = 1$ level is $U_0 - E_1 = 9.0\ \mathrm{eV} - 0.94\ \mathrm{eV} = 8.1\ \mathrm{eV}$, which is three times the 2.7 eV photon energy found in part (c). Hence the corresponding photon wavelength is one-third of 460 nm, or (to two significant figures) 150 nm, which is in the ultraviolet region of the spectrum.

EVALUATE As a check, you can calculate the bound-state energies by using the formulas $E_1 = 0.104U_0$, $E_2 = 0.405U_0$, and $E_3 = 0.848U_0$ given in Fig. 40.15b. As an additional check, note that the first three energy levels of an infinitely deep well of the same width are $E_{1-\mathrm{IDW}} = 1.5\ \mathrm{eV}$, $E_{2-\mathrm{IDW}} = 4E_{1-\mathrm{IDW}} = 6.0\ \mathrm{eV}$, and $E_{3-\mathrm{IDW}} = 9E_{1-\mathrm{IDW}} = 13.5\ \mathrm{eV}$. The energies we found in part (b) are less than these values: As we mentioned earlier, the finite depth of the well lowers the energy levels compared to the levels for an infinitely deep well.

One application of these ideas is *quantum dots,* which are nanometer-sized particles of a semiconductor such as cadmium selenide (CdSe). An electron within a quantum dot behaves much like a particle in a finite potential well of width L equal to the size of the dot. When quantum dots are illuminated with ultraviolet light, the electrons absorb the ultraviolet photons and are excited into high energy levels, such as the $n = 3$ level described in this example. If the electron returns to the ground level ($n = 1$) in two or more steps (for example,

Continued

from $n = 3$ to $n = 2$ and from $n = 2$ to $n = 1$), one step will involve emitting a visible-light photon, as we have calculated here. (We described this process of *fluorescence* in Section 39.3.) Increasing the value of L decreases the energies of the levels and hence the spacing between them, and thus decreases the energy and increases the wavelength of the emitted photons. The photograph that opens this chapter shows quantum dots of different sizes in solution: Each emits a characteristic wavelength that depends on the dot size. Quantum dots can be injected into living tissue and their fluorescent glow used as a tracer for biological research and for medicine. They may also be the key to a new generation of lasers and ultrafast computers.

KEYCONCEPT A finite square well has a finite number of bound states. The energies of these bound states are determined by the boundary conditions that the wave function and its derivative are both continuous at the edges of the well.

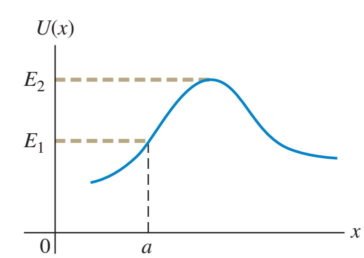
TEST YOUR UNDERSTANDING OF SECTION 40.3 Suppose that the width of the finite potential well shown in Fig. 40.15 is reduced by one-half. How must the value of U_0 change so that there are still just three bound energy levels whose energies are the fractions of U_0 shown in Fig. 40.15b? U_0 must (i) increase by a factor of four; (ii) increase by a factor of two; (iii) remain the same; (iv) decrease by a factor of one-half; (v) decrease by a factor of one-fourth.

ANSWER

(i) The energy levels are arranged as shown in Fig. 40.15b if $U_0 = 6E_{1\text{-IDW}}$, where $E_{1\text{-IDW}} = \pi^2\hbar^2/2mL^2$ is the ground-level energy of an infinite well. If the well width L is reduced to one-half of its initial value, $E_{1\text{-IDW}}$ increases by a factor of four and so U_0 must also increase by a factor of four. The energies E_1, E_2, and E_3 shown in Fig. 40.15b are all specific fractions of U_0, so they will also increase by a factor of four.

40.4 POTENTIAL BARRIERS AND TUNNELING

Figure 40.18 A potential-energy barrier. According to Newtonian mechanics, if the total energy of the system is E_1, a particle to the left of the barrier can go no farther than $x = a$. If the total energy is greater than E_2, the particle can pass over the barrier.



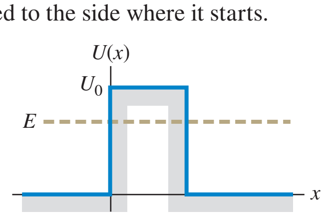
A **potential barrier** is the opposite of a potential well; it is a potential-energy function with a *maximum*. **Figure 40.18** shows an example. In classical Newtonian mechanics, if a particle (such as a roller coaster) is located to the left of the barrier (which might be a hill), and if the total mechanical energy of the system is E_1, the particle cannot move farther to the right than $x = a$. If it did, the potential energy U would be greater than the total mechanical energy E and the kinetic energy $K = E - U$ would be negative. This is impossible in classical mechanics since $K = \frac{1}{2}mv^2$ can never be negative.

A quantum-mechanical particle behaves differently: If it encounters a barrier like the one in Fig. 40.18 and has energy less than E_2, it *may* appear on the other side. This phenomenon is called *tunneling*. In quantum-mechanical tunneling, unlike macroscopic, mechanical tunneling, the particle does not actually push through the barrier and loses no energy in the process.

Tunneling Through a Rectangular Barrier

To understand how tunneling can occur, let's look at the potential-energy function $U(x)$ shown in **Fig. 40.19**. It's like Fig. 40.13 turned upside-down; the potential energy is zero everywhere except in the range $0 \le x \le L$, where it has the value U_0. This might represent a simple model for the potential energy of an electron in the presence of two slabs of metal separated by an air gap of thickness L. The potential energy is lower within either slab than in the gap between them.

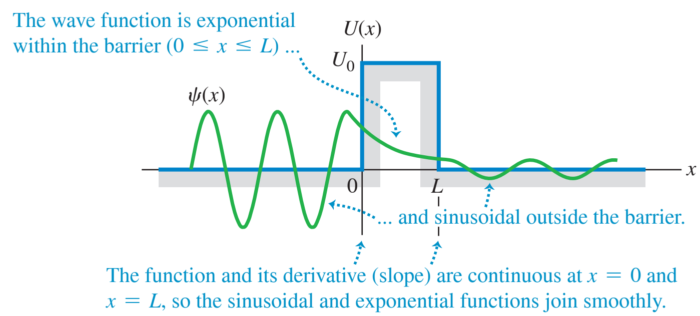
Figure 40.19 A rectangular potential-energy barrier with width L and height U_0. According to Newtonian mechanics, if the total energy E is less than U_0, a particle cannot pass over this barrier but is confined to the side where it starts.



Let's consider solutions of the Schrödinger equation for this potential-energy function for the case in which E is less than U_0. We can use our results from Section 40.3. In the regions $x < 0$ and $x > L$, where $U = 0$, the solution is sinusoidal and is given by Eq. (40.38). Within the barrier $(0 \le x \le L)$, $U = U_0$ and the solution is exponential as in Eq. (40.40). Just as with the finite potential well, the functions have to join smoothly at the boundary points $x = 0$ and $x = L$, which means that both $\psi(x)$ and $d\psi(x)/dx$ have to be continuous at these points.

These requirements lead to a wave function like the one shown in **Fig. 40.20**. The function is *not* zero inside the barrier (the region forbidden by Newtonian mechanics). Even more remarkable, a particle that is initially to the *left* of the barrier has some probability of being found to the *right* of the barrier. How great this probability is depends on the width L of the barrier and the particle's energy E in comparison with the barrier height U_0.

Figure **40.20** A possible wave function for a particle tunneling through the potential-energy barrier shown in Fig. 40.19.



The **tunneling probability** T that the particle gets through the barrier is proportional to the square of the ratio of the amplitudes of the sinusoidal wave functions on the two sides of the barrier. These amplitudes are determined by matching wave functions and their derivatives at the boundary points, a fairly involved mathematical problem. When T is much smaller than unity, it is given approximately by

$$T = Ge^{-2\kappa L} \text{ where } G = 16\frac{E}{U_0}\left(1 - \frac{E}{U_0}\right) \text{ and } \kappa = \frac{\sqrt{2m(U_0 - E)}}{\hbar} \qquad (40.42)$$

(probability of tunneling)

The probability decreases rapidly with increasing barrier width L. It also depends critically on the energy difference $U_0 - E$, which in Newtonian physics is the additional kinetic energy the particle would need to be able to climb over the barrier.

EXAMPLE 40.7 Tunneling through a barrier

WITH VARIATION PROBLEMS

A 2.0 eV electron encounters a barrier 5.0 eV high. What is the probability that it will tunnel through the barrier if the barrier width is (a) 1.00 nm and (b) 0.50 nm?

IDENTIFY and SET UP This problem uses the ideas of tunneling through a rectangular barrier, as in Figs. 40.19 and 40.20. Our target variable is the tunneling probability T in Eq. (40.42), which we evaluate for the given values $E = 2.0$ eV (electron energy), $U = 5.0$ eV (barrier height), $m = 9.11 \times 10^{-31}$ kg (mass of the electron), and $L = 1.00$ nm or 0.50 nm (barrier width).

EXECUTE First we evaluate G and κ in Eq. (40.42), using $E = 2.0$ eV:

$$G = 16\left(\frac{2.0\text{ eV}}{5.0\text{ eV}}\right)\left(1 - \frac{2.0\text{ eV}}{5.0\text{ eV}}\right) = 3.8$$

$$U_0 - E = 5.0\text{ eV} - 2.0\text{ eV} = 3.0\text{ eV} = 4.8 \times 10^{-19}\text{ J}$$

$$\kappa = \frac{\sqrt{2(9.11 \times 10^{-31}\text{ kg})(4.8 \times 10^{-19}\text{ J})}}{1.055 \times 10^{-34}\text{ J}\cdot\text{s}} = 8.9 \times 10^{9}\text{ m}^{-1}$$

(a) When $L = 1.00\text{ nm} = 1.00 \times 10^{-9}$ m, we have $2\kappa L = 2(8.9 \times 10^{9}\text{ m}^{-1})(1.00 \times 10^{-9}\text{ m}) = 17.8$ and $T = Ge^{-2\kappa L} = 3.8e^{-17.8} = 7.1 \times 10^{-8}$.

(b) When $L = 0.50$ nm, one-half of 1.00 nm, $2\kappa L$ is one-half of 17.8, or 8.9. Hence $T = 3.8e^{-8.9} = 5.2 \times 10^{-4}$.

EVALUATE Halving the width of this barrier increases the tunneling probability T by a factor of $(5.2 \times 10^{-4})/(7.1 \times 10^{-8}) = 7.3 \times 10^{3}$, or nearly ten thousand. The tunneling probability is an *extremely* sensitive function of the barrier width. Note that our results for T in both part (a) and (b) are much less than 1, which is consistent with the assumptions that go into Eq. (40.42).

KEYCONCEPT Quantum-mechanical particles can be found in places where Newtonian mechanics would forbid them to be. An example is tunneling, in which a particle can pass through a barrier where its total mechanical energy is less than the potential energy. The probability that tunneling occurs depends on the width of the barrier and the additional energy the particle would need to "climb" over the barrier.

Applications of Tunneling

Tunneling has a number of practical applications, some of considerable importance. When you twist two copper wires together or close the contacts of a switch, current passes from one conductor to the other despite a thin layer of nonconducting copper oxide between them. The electrons tunnel through this thin insulating layer. A *tunnel diode* is a semiconductor device in which electrons tunnel through a potential barrier. The current can be switched on and off very quickly (within a few picoseconds) by varying the height of the barrier.

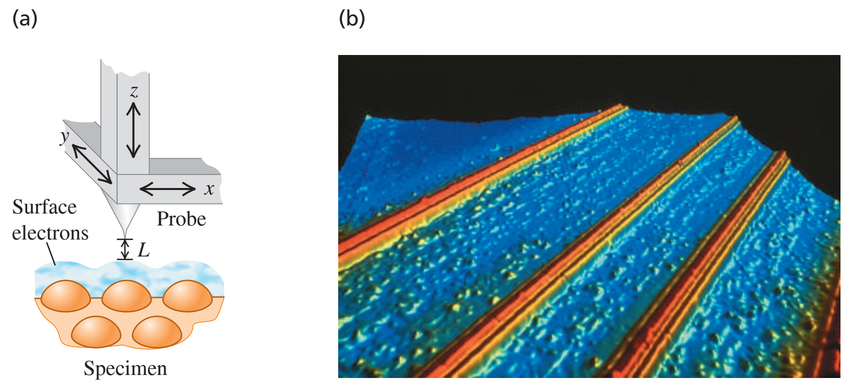
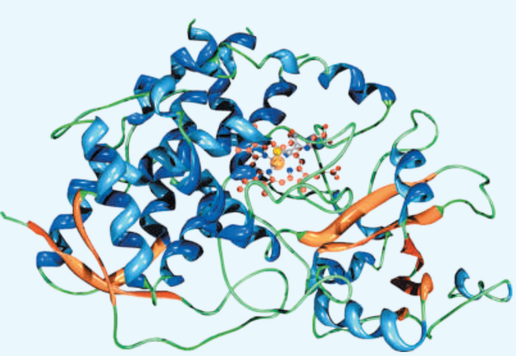


Figure 40.21 (a) Schematic diagram of the probe of a scanning tunneling microscope (STM). As the sharp conducting probe is scanned across the surface in the x- and y-directions, it is also moved in the z-direction to maintain a constant tunneling current. The changing position of the probe is recorded and used to construct an image of the surface. (b) This colored STM image shows "quantum wires": thin strips, just 10 atoms wide, of a conductive rare-earth silicide atop a silicon surface. Such quantum wires may one day be the basis of ultraminiaturized circuits.

BIO APPLICATION Electron Tunneling in Enzymes Protein molecules play essential roles as enzymes in living organisms. Enzymes like the one shown here are large molecules, and in many cases their function depends on the ability of electrons to tunnel across the space that separates one part of the molecule from another. Without tunneling, life as we know it would be impossible!

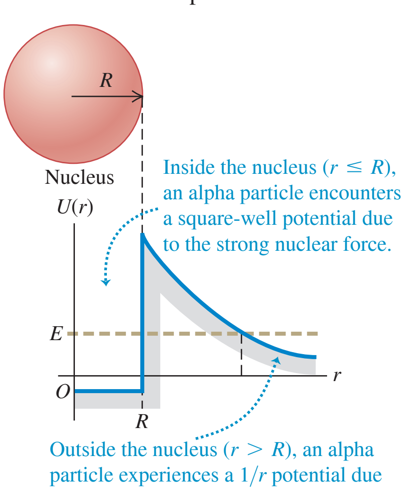


A *Josephson junction* consists of two superconductors separated by an oxide layer a few atoms (1 to 2 nm) thick. Electron pairs in the superconductors can tunnel through the barrier layer, giving such a device unusual circuit properties. Josephson junctions are useful for establishing precise voltage standards and measuring tiny magnetic fields, and they play a crucial role in the developing field of quantum computing.

The *scanning tunneling microscope* (STM) uses electron tunneling to create images of surfaces down to the scale of individual atoms. An extremely sharp conducting needle is brought very close to the surface, within 1 nm or so (**Fig. 40.21a**). When the needle is at a positive potential with respect to the surface, electrons can tunnel through the surface potential-energy barrier and reach the needle. As Example 40.7 shows, the tunneling probability and hence the tunneling current are very sensitive to changes in the width L of the barrier (the distance between the surface and the needle tip). In one mode of operation the needle is scanned across the surface while being moved perpendicular to the surface to maintain a constant tunneling current. The needle motion is recorded; after many parallel scans, an image of the surface can be reconstructed. Extremely precise control of needle motion, including isolation from vibration, is essential. Figure 40.21b shows an STM image. (Figure 40.17 is also an STM image.)

Tunneling is also of great importance in nuclear physics. A fusion reaction can occur when two nuclei tunnel through the barrier caused by their electrical repulsion and approach each other closely enough for the attractive nuclear force to cause them to fuse. Fusion reactions occur in the cores of stars, including the sun; without tunneling, the sun wouldn't shine. The emission of alpha particles from unstable nuclei such as radium also involves tunneling. An alpha particle is a cluster of two protons and two neutrons (the same as a nucleus of the most common form of helium). Such clusters form naturally within larger atomic nuclei. An alpha particle trying to escape from a nucleus encounters a potential barrier that results from the combined effect of the attractive nuclear force and the electrical repulsion of the remaining part of the nucleus (**Fig. 40.22**). To escape, the alpha particle must tunnel through this barrier. Depending on the barrier height and width for a given kind of alpha-emitting nucleus, the tunneling probability can be low or high, and the alpha-emitting material will have low or high radioactivity. Recall from Section 39.2 that Ernest Rutherford used alpha particles from a radioactive source to discover the atomic nucleus. Although Rutherford did not know it, tunneling allowed these alpha particles to escape from their parent nuclei, which made his experiments possible! We'll learn more about alpha decay in Chapter 43.

Figure 40.22 Approximate potential-energy function for an alpha particle interacting with a nucleus of radius R. If an alpha particle inside the nucleus has energy E greater than zero, it can tunnel through the barrier and escape from the nucleus.



TEST YOUR UNDERSTANDING OF SECTION 40.4 Is it possible for a particle undergoing tunneling to be found *within* the barrier rather than on either side of it?

ANSWER | **yes** Figure 40.20 shows a possible wave function $\psi(x)$ for tunneling. Since $\psi(x)$ is not zero within the barrier ($0 \leq x \leq L$), there is some probability that the particle can be found there.

40.5 THE HARMONIC OSCILLATOR

Systems that *oscillate* are of tremendous importance in the physical world, from the oscillations of your eardrums in response to a sound wave to the vibrations of the ground caused by an earthquake. Oscillations are equally important on the microscopic scale where quantum effects dominate. The molecules of the air around you can be set into vibration when they collide with each other, the protons and neutrons in an excited atomic nucleus can oscillate in opposite directions, and a microwave oven transfers energy to food by making water molecules in the food flip back and forth. In this section we'll look at the solutions of the Schrödinger equation for the simplest kind of vibrating system, the quantum-mechanical harmonic oscillator.

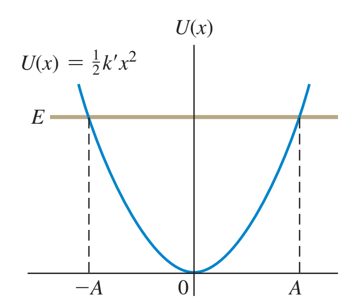
As we learned in Section 14.2, a **harmonic oscillator** is a particle with mass m that moves along the x-axis under the influence of a conservative force $F_x = -k'x$. The constant k' is called the *force constant*. (In Section 14.2 we used the symbol k for the force constant. In this section we'll use the symbol k' instead to minimize confusion with the wave number $k = 2\pi/\lambda$.) The force is proportional to the particle's displacement x from its equilibrium position, $x = 0$. The corresponding potential-energy function is $U = \frac{1}{2}k'x^2$ (**Fig. 40.23**). In Newtonian mechanics, when the particle is displaced from equilibrium, it undergoes sinusoidal motion with frequency $f = (1/2\pi)(k'/m)^{1/2}$ and angular frequency $\omega = 2\pi f = (k'/m)^{1/2}$. The amplitude (that is, the maximum displacement from equilibrium) of these Newtonian oscillations is A, which is related to the energy E of the oscillator by $E = \frac{1}{2}k'A^2$.

Let's make an enlightened guess about the energy levels of a quantum-mechanical harmonic oscillator. In classical physics an electron oscillating with angular frequency ω emits electromagnetic radiation with that same angular frequency. It's reasonable to guess that when an excited quantum-mechanical harmonic oscillator with angular frequency $\omega = (k'/m)^{1/2}$ (according to Newtonian mechanics, at least) makes a transition from one energy level to a lower level, it would emit a photon with this same angular frequency ω. The energy of such a photon is $hf = (2\pi\hbar)(\omega/2\pi) = \hbar\omega$. So we would expect that the spacing between adjacent energy levels of the harmonic oscillator would be

$$hf = \hbar\omega = \hbar\sqrt{\frac{k'}{m}} \tag{40.43}$$

That's the same spacing between energy levels that Planck assumed in deriving his radiation law (see Section 39.5). It was a good assumption; as we'll see, the energy levels are in fact half-integer $\left(\frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \ldots\right)$ multiples of $\hbar\omega$.

Figure **40.23** Potential-energy function for the harmonic oscillator. In Newtonian mechanics the amplitude A is related to the total energy E by $E = \frac{1}{2}k'A^2$, and the particle is restricted to the range from $x = -A$ to $x = A$. In quantum mechanics the particle can be found at $x > A$ or $x < -A$.



Wave Functions, Boundary Conditions, and Energy Levels

We'll begin our quantum-mechanical analysis of the harmonic oscillator by writing down the one-dimensional time-independent Schrödinger equation, Eq. (40.23), with $\frac{1}{2}k'x^2$ in place of U:

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + \frac{1}{2}k'x^2\psi(x) = E\psi(x) \quad \text{(Schrödinger equation for the harmonic oscillator)} \tag{40.44}$$

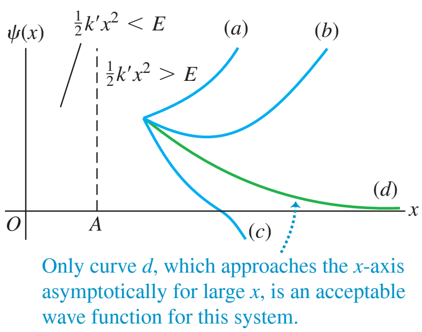
The solutions of this equation are wave functions for the physically possible states of the system.

In the discussion of square-well potentials in Section 40.2 we found that the energy levels are determined by boundary conditions at the walls of the well. However, the harmonic-oscillator potential has no walls as such; what, then, are the appropriate boundary conditions? Classically, $|x|$ cannot be greater than the amplitude A given by $E = \frac{1}{2}k'A^2$. Quantum mechanics does allow some penetration into classically forbidden regions, but the probability decreases as that penetration increases. Thus the wave functions must approach zero as $|x|$ grows large.

Satisfying the requirement that $\psi(x) \rightarrow 0$ as $|x| \rightarrow \infty$ is not as trivial as it may seem. To see why this is, let's rewrite Eq. (40.44) in the form

$$\frac{d^2\psi(x)}{dx^2} = \frac{2m}{\hbar^2}\left(\tfrac{1}{2}k'x^2 - E\right)\psi(x) \tag{40.45}$$

Figure **40.24** Possible behaviors of harmonic-oscillator wave functions in the region $\frac{1}{2}k'x^2 > E$. In this region, $\psi(x)$ and $d^2\psi(x)/dx^2$ have the same sign. The curve is concave upward when $d^2\psi(x)/dx^2$ is positive and concave downward when $d^2\psi(x)/dx^2$ is negative.



Equation (40.45) shows that when x is large enough (either positive or negative) to make the quantity $\left(\frac{1}{2}k'x^2 - E\right)$ positive, the function $\psi(x)$ and its second derivative $d^2\psi(x)/dx^2$ have the same sign. **Figure 40.24** shows four possible kinds of behavior of $\psi(x)$ beginning at a point where x is greater than the classical amplitude A, so that $\frac{1}{2}k'x^2 - \frac{1}{2}k'A^2 = \frac{1}{2}k'x^2 - E > 0$. Let's look at these four cases more closely. If $\psi(x)$ is positive as shown in Fig. 40.24, Eq. (40.45) tells us that $d^2\psi(x)/dx^2$ is also positive and the function is *concave upward*. Note also that $d^2\psi(x)/dx^2$ is the rate of change of the *slope* of $\psi(x)$; this will help us understand how our four possible wave functions behave.

- *Curve a:* The slope of $\psi(x)$ is positive at point x. Since $d^2\psi(x)/dx^2 > 0$, the function curves upward increasingly steeply and goes to infinity. This violates the boundary condition that $\psi(x) \rightarrow 0$ as $|x| \rightarrow \infty$, so this isn't a viable wave function.
- *Curve b:* The slope of $\psi(x)$ is negative at point x, and $d^2\psi(x)/dx^2$ has a large positive value. Hence the slope changes rapidly from negative to positive and keeps on increasing—so, again, the wave function goes to infinity. This wave function isn't viable either.
- *Curve c:* As for curve b, the slope is negative at point x. However, $d^2\psi(x)/dx^2$ now has a *small* positive value, so the slope increases only gradually as $\psi(x)$ decreases to zero and crosses over to negative values. Equation (40.45) tells us that once $\psi(x)$ becomes negative, $d^2\psi(x)/dx^2$ also becomes negative. Hence the curve becomes concave *downward* and heads for *negative* infinity. This wave function, too, fails to satisfy the requirement that $\psi(x) \rightarrow 0$ as $|x| \rightarrow \infty$ and thus isn't viable.
- *Curve d:* If the slope of $\psi(x)$ at point x is negative, and the positive value of $d^2\psi(x)/dx^2$ at this point is neither too large nor too small, the curve bends just enough to glide in asymptotically to the x-axis. In this case $\psi(x)$, $d\psi(x)/dx$, and $d^2\psi(x)/dx^2$ all approach zero at large x. This case offers the only hope of satisfying the boundary condition that $\psi(x) \rightarrow 0$ as $|x| \rightarrow \infty$, and it occurs only for certain very special values of the energy E.

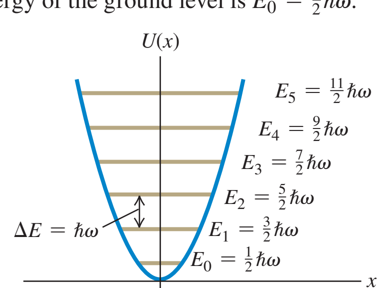
This qualitative discussion suggests how the boundary conditions as $|x| \rightarrow \infty$ determine the possible energy levels for the quantum-mechanical harmonic oscillator. It turns out that these boundary conditions are satisfied only if the energy E is equal to one of the values E_n:

Energy levels for a harmonic oscillator

$$E_n = \left(n + \tfrac{1}{2}\right)\hbar\sqrt{\frac{k'}{m}} = \left(n + \tfrac{1}{2}\right)\hbar\omega \quad (n = 0, 1, 2, \ldots) \tag{40.46}$$

Quantum number; Planck's constant divided by 2π; Particle's mass; Force constant; Oscillation angular frequency

Figure **40.25** Energy levels for the harmonic oscillator. The spacing between any two adjacent levels is $\Delta E = \hbar\omega$. The energy of the ground level is $E_0 = \frac{1}{2}\hbar\omega$.



Note that the ground level of energy $E_0 = \frac{1}{2}\hbar\omega$ is denoted by the quantum number $n = 0$, *not* $n = 1$.

Equation (40.46) confirms our guess [Eq. (40.43)] that adjacent energy levels are separated by a constant interval of $\hbar\omega = hf$, as Planck assumed in 1900. There are infinitely many levels; this shouldn't be surprising because we are dealing with an infinitely deep potential well. As $|x|$ increases, $U = \frac{1}{2}k'x^2$ increases without bound.

Figure 40.25 shows the lowest six energy levels and the potential-energy function $U(x)$. For each level n, the value of $|x|$ at which the horizontal line representing the total energy E_n intersects $U(x)$ gives the amplitude A_n of the corresponding Newtonian oscillator.

EXAMPLE 40.8 Vibration in a crystal

A sodium atom of mass 3.82×10^{-26} kg vibrates within a crystal. The potential energy increases by 0.0075 eV when the atom is displaced 0.014 nm from its equilibrium position. Treat the atom as a harmonic oscillator. (a) Find the angular frequency of the oscillations according to Newtonian mechanics. (b) Find the spacing (in electron volts) of adjacent vibrational energy levels according to quantum mechanics. (c) What is the wavelength of a photon emitted as the result of a transition from one level to the next lower level? In what region of the electromagnetic spectrum does this lie?

IDENTIFY and SET UP We'll find the force constant k' from the expression $U = \frac{1}{2}k'x^2$ for potential energy. We'll then find the angular frequency $\omega = (k'/m)^{1/2}$ and use this in Eq. (40.46) to find the spacing between adjacent energy levels. We'll calculate the wavelength of the emitted photon as in Example 40.6.

EXECUTE We are given that $U = 0.0075\ \text{eV} = 1.2 \times 10^{-21}$ J when $x = 0.014 \times 10^{-9}$ m, so we can solve $U = \frac{1}{2}k'x^2$ for k':

$$k' = \frac{2U}{x^2} = \frac{2(1.2 \times 10^{-21}\ \text{J})}{(0.014 \times 10^{-9}\ \text{m})^2} = 12.2\ \text{N/m}$$

(a) The Newtonian angular frequency is

$$\omega = \sqrt{\frac{k'}{m}} = \sqrt{\frac{12.2\ \text{N/m}}{3.82 \times 10^{-26}\ \text{kg}}} = 1.79 \times 10^{13}\ \text{rad/s}$$

(b) From Eq. (40.46) and Fig. 40.25, the spacing between adjacent energy levels is

$$\begin{aligned}\hbar\omega &= (1.055 \times 10^{-34}\ \text{J} \cdot \text{s})(1.79 \times 10^{13}\ \text{s}^{-1})\\ &= 1.89 \times 10^{-21}\ \text{J}\left(\frac{1\ \text{eV}}{1.602 \times 10^{-19}\ \text{J}}\right) = 0.0118\ \text{eV}\end{aligned}$$

(c) The energy E of the emitted photon is equal to the energy lost by the oscillator in the transition, 0.0118 eV. Then

$$\begin{aligned}\lambda &= \frac{hc}{E} = \frac{(4.136 \times 10^{-15}\ \text{eV} \cdot \text{s})(3.00 \times 10^{8}\ \text{m/s})}{0.0118\ \text{eV}}\\ &= 1.05 \times 10^{-4}\ \text{m} = 105\ \mu\text{m}\end{aligned}$$

This photon wavelength is in the infrared region of the spectrum.

EVALUATE This example shows us that interatomic force constants are a few newtons per meter, about the same as those of household springs or spring-based toys such as the Slinky®. It also suggests that we can learn about the vibrations of molecules by measuring the radiation that they emit in transitioning to a lower vibrational state. We'll explore this idea further in Chapter 42.

KEYCONCEPT As in Newtonian physics, in a quantum-mechanical harmonic oscillator a particle of mass m moves under the influence of the potential-energy function of an ideal spring with force constant k'. The lowest-energy stationary state of the harmonic oscillator has energy $\frac{1}{2}\hbar\omega$, where $\omega = \sqrt{k'/m}$ is the Newtonian angular frequency; each excited state has $\hbar\omega$ more energy than the state below it.

Comparing Quantum and Newtonian Oscillators

The wave functions for the levels $n = 0, 1, 2, \ldots$ of the harmonic oscillator are called *Hermite functions;* they aren't encountered in elementary calculus courses but are well known to mathematicians. Each Hermite function is an exponential function multiplied by a polynomial in x. The harmonic-oscillator wave function corresponding to $n = 0$ and $E = E_0$ (the ground level) is

$$\psi(x) = Ce^{-\sqrt{mk'}\,x^2/2\hbar} \qquad (40.47)$$

The constant C is chosen to normalize the function—that is, to make $\int_{-\infty}^{\infty}|\psi|^2\,dx = 1$. (We're using C rather than A as a normalization constant in this section, since we've already appropriated the symbol A to denote the Newtonian amplitude of a harmonic oscillator.) You can find C by using the following result from integral tables:

$$\int_{-\infty}^{\infty} e^{-a^2x^2}dx = \frac{\sqrt{\pi}}{a}$$

To confirm that $\psi(x)$ as given by Eq. (40.47) really *is* a solution of the Schrödinger equation for the harmonic oscillator, you can calculate the second derivative of this wave function, substitute it into Eq. (40.44), and verify that the equation is satisfied if the energy E is equal to $E_0 = \frac{1}{2}\hbar\omega$. It's a little messy, but the result is satisfying and worth the effort.

Figure 40.26 (next page) shows the the first four harmonic-oscillator wave functions. Each graph also shows the amplitude A of a Newtonian harmonic oscillator with the same energy—that is, the value of A determined from

$$\tfrac{1}{2}k'A^2 = \left(n + \tfrac{1}{2}\right)\hbar\omega \qquad (40.48)$$

In each case there is some penetration of the wave function into the regions $|x| > A$ that are forbidden by Newtonian mechanics. This is similar to the effect that we noted in Section 40.3 for a particle in a finite square well.

Figure **40.26** The first four wave functions for the harmonic oscillator. The amplitude A of a Newtonian oscillator with the same total energy is shown for each. Each wave function penetrates somewhat into the classically forbidden regions $|x| > A$. The total number of finite maxima and minima for each function is $n + 1$, one more than the quantum number.

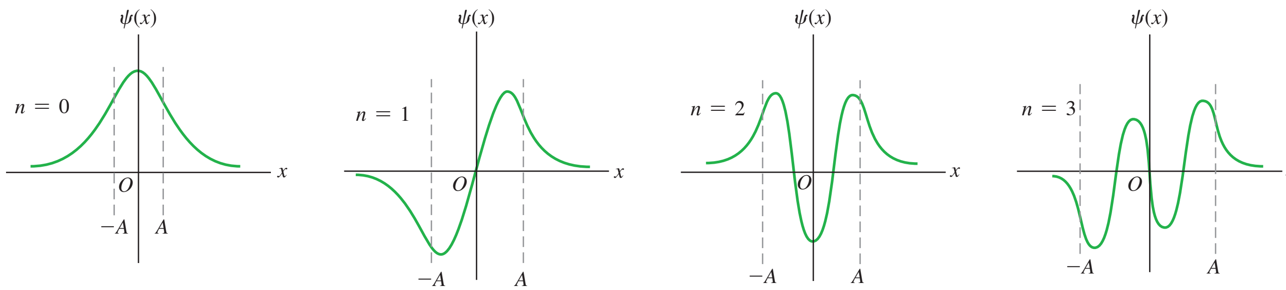


Figure **40.27** Probability distribution functions $|\psi(x)|^2$ for the harmonic-oscillator wave functions shown in Fig. 40.26. The amplitude A of the Newtonian motion with the same energy is shown for each. The blue lines show the corresponding probability distributions for the Newtonian motion. As n increases, the averaged-out quantum-mechanical functions resemble the Newtonian curves more and more.

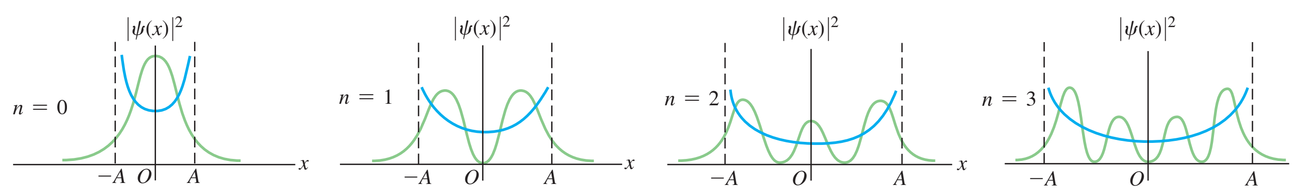


Figure **40.28** Newtonian and quantum-mechanical probability distribution functions for a harmonic oscillator for the state $n = 10$. The Newtonian amplitude A is also shown.

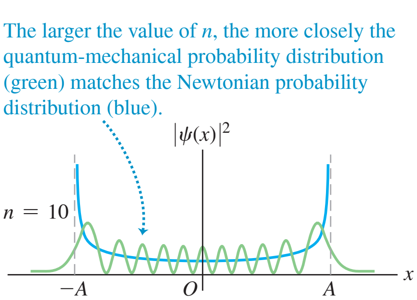


Figure 40.27 shows the probability distributions $|\psi(x)|^2$ for these states. Each graph also shows the probability distribution determined from Newtonian physics, in which the probability of finding the particle near a randomly chosen point is inversely proportional to the particle's speed at that point. If we average out the wiggles in the quantum-mechanical probability curves, the results for $n > 0$ resemble the Newtonian predictions. This agreement improves with increasing n; **Fig. 40.28** shows the classical and quantum-mechanical probability functions for $n = 10$. Notice that the spacing between zeros of $|\psi(x)|^2$ in Fig. 40.28 increases with increasing distance from $x = 0$. This makes sense from the Newtonian perspective: As a particle moves away from $x = 0$, its kinetic energy K and the magnitude p of its momentum both decrease. Thinking quantum-mechanically, this means that the wavelength $\lambda = h/p$ increases, so the spacing between zeros of $\psi(x)$ (and hence of $|\psi(x)|^2$) also increases.

In the Newtonian analysis of the harmonic oscillator the minimum energy is zero, with the particle at rest at its equilibrium position $x = 0$. This is not possible in quantum mechanics; no solution of the Schrödinger equation has $E = 0$ and satisfies the boundary conditions. Furthermore, such a state would violate the Heisenberg uncertainty principle because there would be no uncertainty in either position or momentum. The energy must be at least $\frac{1}{2}\hbar\omega$ for the system to conform to the uncertainty principle. To see qualitatively why this is so, consider a Newtonian oscillator with total energy $\frac{1}{2}\hbar\omega$. We can find the amplitude A and the maximum velocity just as we did in Section 14.3. When the particle is at its maximum displacement ($x = \pm A$) and instantaneously at rest, $K = 0$ and $E = U = \frac{1}{2}k'A^2$. When the particle is at equilibrium ($x = 0$) and moving at its maximum speed, $U = 0$ and $E = K = \frac{1}{2}mv_{\max}{}^2$. Setting $E = \frac{1}{2}\hbar\omega$, we find

$$E = \tfrac{1}{2}k'A^2 = \tfrac{1}{2}\hbar\omega = \tfrac{1}{2}\hbar\left(\frac{k'}{m}\right)^{1/2} \quad \text{so} \quad A = \frac{\hbar^{1/2}}{k'^{1/4}m^{1/4}}$$

$$E = \tfrac{1}{2}mv_{\max}{}^2 = \tfrac{1}{2}k'A^2 \quad \text{so} \quad v_{\max} = A\left(\frac{k'}{m}\right)^{1/2} = \frac{\hbar^{1/2}k'^{1/4}}{m^{3/4}}$$

The maximum *momentum* of the particle is

$$p_{\max} = mv_{\max} = \hbar^{1/2}k'^{1/4}m^{1/4}$$

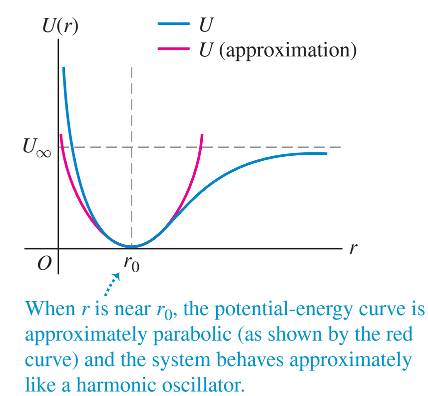
Here's where the Heisenberg uncertainty principle comes in. It turns out that the uncertainties in the particle's position and momentum (calculated as standard deviations) are, respectively, $\Delta x = A/\sqrt{2} = A/2^{1/2}$ and $\Delta p_x = p_{\max}/\sqrt{2} = p_{\max}/2^{1/2}$. Then the product of the two uncertainties is

$$\Delta x\, \Delta p_x = \left(\frac{\hbar^{1/2}}{2^{1/2}\, k'^{1/4}\, m^{1/4}}\right)\left(\frac{\hbar^{1/2}\, k'^{1/4}\, m^{1/4}}{2^{1/2}}\right) = \frac{\hbar}{2}$$

This product equals the minimum value allowed by Eq. (39.29), $\Delta x\, \Delta p_x \geq \hbar/2$, and thus satisfies the uncertainty principle. If the energy had been less than $\frac{1}{2}\hbar\omega$, the product $\Delta x\, \Delta p_x$ would have been less than $\hbar/2$, and the uncertainty principle would have been violated.

Even when a potential-energy function isn't precisely parabolic in shape, we may be able to approximate it by the harmonic-oscillator potential for sufficiently small displacements from equilibrium. **Figure 40.29** shows a typical potential-energy function for an interatomic force in a molecule. At large separations the curve of $U(r)$ versus r levels off, corresponding to the absence of force at great distances. But the curve is approximately parabolic near the minimum of $U(r)$ (the equilibrium separation of the atoms). Near equilibrium the molecular vibration is approximately simple harmonic with energy levels given by Eq. (40.46), as we assumed in Example 40.8.

Figure **40.29** A potential-energy function describing the interaction of two atoms in a diatomic molecule. The distance r is the separation between the centers of the atoms, and the equilibrium separation is $r = r_0$. The energy needed to dissociate the molecule is U_∞.



TEST YOUR UNDERSTANDING OF SECTION 40.5 A quantum-mechanical system initially in its ground level absorbs a photon and ends up in the first excited state. The system then absorbs a second photon and ends up in the second excited state. For which of the following systems does the second photon have a longer wavelength than the first one? (i) A harmonic oscillator; (ii) a hydrogen atom; (iii) a particle in a box.

ANSWER

❙ **(ii)** If the second photon has a longer wavelength and hence lower energy than the first photon, the difference in energy between the first and second excited levels must be less than the difference between the ground level and the first excited level. This is the case for the hydrogen atom, for which the energy difference between levels decreases as the energy increases (see Fig. 39.24). By contrast, the energy difference between successive levels increases for a particle in a box (see Fig. 40.11b) and is constant for a harmonic oscillator (see Fig. 40.25).

40.6 MEASUREMENT IN QUANTUM MECHANICS

We've seen how to use the Schrödinger equation to calculate the stationary-state wave functions and energy levels for various potential-energy functions $U(x)$. We've also seen how to interpret the wave function $\Psi(x, t)$ of a particle in terms of the probability distribution function $|\Psi(x, t)|^2$. We'll conclude with a brief discussion of what happens when we try to *measure* the properties of a quantum-mechanical particle. As we'll see, the consequences of such a measurement can be startlingly different from what happens when we measure the properties of a familiar Newtonian particle, such as a marble or billiard ball.

Let's consider a "particle in a box"—that is, a particle in an infinite square well of width L, as described in Section 40.2. This particle of mass m is free to move along the x-axis in the region $0 \leq x \leq L$ but cannot move beyond this region. Let's suppose the particle is in a stationary state with definite energy E, equal to one of the energy levels E_n given by Eq. (40.31). If we measure the x-component of momentum of this particle, what is the result?

First let's consider the answer to that question for a Newtonian particle in a box (see Fig. 40.8). This could be a hockey puck sliding on frictionless ice and bouncing back and forth between two parallel walls. The energy E of the puck is equal to its kinetic energy $p^2/2m$, so the magnitude of its momentum is $p = \sqrt{2mE}$. The x-component of its momentum p_x is therefore

$$p_x = +\sqrt{2mE} \quad \text{or} \quad p_x = -\sqrt{2mE} \tag{40.49}$$

Whether p_x is positive or negative depends on whether the hockey puck is moving in the $+x$-direction (then $p_x = +\sqrt{2mE}$) or the $-x$-direction (then $p_x = -\sqrt{2mE}$). To determine which value of p_x is correct at a given time, we need only look at the puck to see in which direction it's moving.

We can't make such an observation in the dark; we need to shine some light on the hockey puck. We know from Section 38.1 that light comes in the form of photons and that a photon of wavelength λ has momentum $p = h/\lambda$. When we shine light on the puck to observe it, the photons collide with the puck and *change* its momentum. The mere act of measuring the puck's momentum can affect the quantity that we're trying to measure! The good news is that this change is minuscule: A hockey puck of mass $m = 0.165$ kg moving at speed $v = 1.00$ m/s has momentum $p = mv = 0.165$ kg·m/s, while a visible-light photon of wavelength 500 nm has momentum $p = h/\lambda = 1.33 \times 10^{-27}$ kg·m/s. Even if we directed all of the photons from a 100 W light source onto the puck for a 1.00 s burst of light, the total momentum in this burst would be only 3.33×10^{-7} kg·m/s, and the resulting change in the momentum of the puck would be negligible. In general, we can measure any of the properties of a Newtonian particle—its momentum, position, energy, and so on—without appreciably changing the quantity that we are measuring.

The situation is very different for a quantum-mechanical particle in a box. From Eq. (40.21) the state of such a particle with energy $E = E_n$ is described by the wave function

$$\Psi(x, t) = \psi_n(x)e^{-iE_nt/\hbar} = \psi_n(x)e^{-i\omega_nt} \tag{40.50}$$

In Eq. (40.50) the angular frequency is $\omega_n = E_n/\hbar$ and the time-independent, stationary-state wave function $\psi_n(x)$ is given by Eq. (40.35):

$$\psi_n(x) = \sqrt{\frac{2}{L}}\sin\frac{n\pi x}{L} \qquad (n = 1, 2, 3, \dots) \tag{40.51}$$

This is a state of definite *energy,* but it is *not* a state of definite momentum: It represents a standing wave with equal amounts of momentum in the $+x$-direction and the $-x$-direction. To make this more explicit, recall from Eqs. (40.30) and (40.31) that the magnitude of momentum in a state of energy E_n is $p_n = \sqrt{2mE_n} = nh/2L = n\pi\hbar/L$, and the corresponding wave number is $k_n = p_n/\hbar = n\pi/L$. So we can replace $n\pi/L$ in Eq. (40.51) with k_n:

$$\psi_n(x) = \sqrt{\frac{2}{L}}\sin k_nx$$

Recall also Euler's formula from Eq. (40.17): $e^{i\theta} = \cos\theta + i\sin\theta$ and $e^{-i\theta} = \cos\theta - i\sin\theta$. Hence $\sin\theta = (e^{i\theta} - e^{-i\theta})/2i$, and we can write

$$\psi_n(x) = \sqrt{\frac{2}{L}}\left(\frac{e^{ik_nx} - e^{-ik_nx}}{2i}\right) = \frac{1}{i\sqrt{2L}}(e^{ik_nx} - e^{-ik_nx}) \tag{40.52}$$

Now we substitute Eq. (40.52) into Eq. (40.50) and distribute the factors $1/i\sqrt{2L}$ and $e^{-i\omega_nt}$:

$$\begin{aligned}\Psi(x, t) &= \frac{1}{i\sqrt{2L}}(e^{ik_nx} - e^{-ik_nx})e^{-i\omega_nt} \\ &= \frac{1}{i\sqrt{2L}}e^{ik_nx}e^{-i\omega_nt} - \frac{1}{i\sqrt{2L}}e^{-ik_nx}e^{-i\omega_nt}\end{aligned} \tag{40.53}$$

In Eq. (40.53) the $e^{ik_nx}e^{-i\omega_nt}$ term is a wave function for a free particle with energy $E_n = \hbar\omega_n$ and a *positive* x-component of momentum $p_x = p_n = \hbar k_n$. In the $e^{-ik_nx}e^{-i\omega_nt}$ term, k_n is replaced by $-k_n$, so this term is a wave function for a free particle with the same energy $E_n = \hbar\omega_n$ but a *negative* x-component of momentum $p_x = -p_n = -\hbar k_n$. These two possible values of p_x are the same as for a Newtonian particle in a box, Eq. (40.49). The difference is that as the Newtonian particle bounces back and forth between the walls of the box, it has positive p_x half of the time and negative p_x half of the time. Only its time-averaged value of p_x is zero. But because both terms for positive p_x and negative p_x are present in Eq. (40.53), the quantum-mechanical particle has *both* signs of the x-component of momentum present at *all* times. As we stated earlier, this stationary state for a quantum-mechanical particle in a box has a definite energy [both terms in Eq. (40.53) have the same value of ω_n

and hence the same value of $E_n = \hbar\omega_n$] but does not have a definite momentum. Because the $e^{ik_nx}e^{-i\omega_nt}$ and $e^{-ik_nx}e^{-i\omega_nt}$ terms in Eq. (40.53) have coefficients of the same magnitude, $1/\sqrt{2L}$, the *instantaneous* average value of p_x for the quantum-mechanical particle is zero (the average of $\hbar k_n$ and $-\hbar k_n$) at *all* times.

What value do we get if we *measure* the momentum of the quantum-mechanical particle in a box? As for the Newtonian particle, we can measure the momentum by shining a light on it. Let's fire a single photon, moving in the $-y$-direction, at the particle and allow the photon and particle to collide (**Fig. 40.30**). Before the collision the total x-component of momentum of the system of photon and particle is zero. Momentum is conserved in the collision, so the same is true after the collision. After the collision, whichever sign of p_x the photon has, the x-component of momentum of the particle will have the opposite sign. If the photon is detected in detector A, we conclude that the particle has $p_x = \hbar k_n$; if instead the photon is detected in detector B, we conclude that the particle has $p_x = -\hbar k_n$.

In this experiment, we need to be even more concerned about how the photon changes the momentum of the particle than in the Newtonian case. For an electron in a box of width $L = 1.00 \times 10^{-6}$ m $= 1.00\ \mu$m, the electron momentum has a minimum magnitude of $p = 3.31 \times 10^{-28}$ kg $\cdot$ m/s (corresponding to the $n = 1$ energy level), which is only about one-quarter that of a visible-light photon of wavelength 500 nm. To minimize the change in the electron's magnitude of momentum due to the collision, we should use a photon of much longer wavelength (say, a radio-wave photon) and hence much smaller momentum.

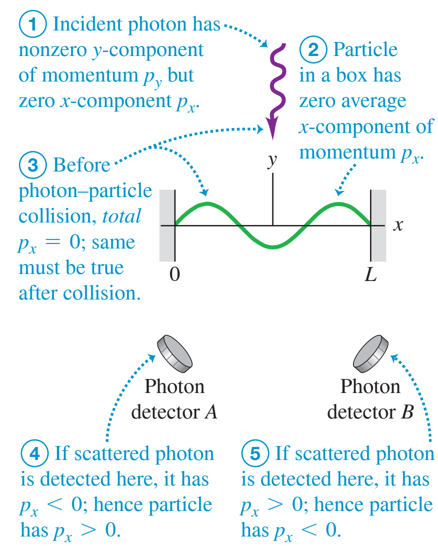
Even when we use a photon with the lowest possible momentum, however, we find that the state of the particle in the box *must* change as a result of the experiment. Here's a summary of the results:

1. If the measurement shows that the particle has positive $p_x = \hbar k_n$, the wave function *changes* from that given in Eq. (40.53) to one with an $e^{ik_nx}e^{-i\omega_nt}$ term *only*. The other term, which corresponds to $p_x = -\hbar k_n$, disappears. We say that the wave function, which was a combination of two terms with different values of p_x, has undergone *wave-function collapse*—it has collapsed to one term with $p_x = \hbar k_n$ as a consequence of measuring the value of p_x. To test this result, we fire a second photon immediately after the first. The second photon scatters from the particle as we would expect if the particle had the value $p_x = \hbar k_n$.
2. If the measurement shows that the particle has negative $p_x = -\hbar k_n$, the wave function collapses in the opposite way: It changes to one with an $e^{-ik_nx}e^{-i\omega_nt}$ term only. The $p_x = \hbar k_n$ term disappears.
3. If we repeat the experiment many times, each time starting with the particle described by the wave function in Eq. (40.53), 50% of the time we measure the particle to have $p_x = \hbar k_n$ and 50% of the time we measure the particle to have $p_x = -\hbar k_n$. For any given time that we try the experiment, there is no way to predict which outcome will occur. We can state only that there is equal probability of either outcome.

These results reveal a fact of quantum-mechanical life: *Measuring a physical property of a system can change the wave function of that system.* By measuring the value of p_x for a particle in a box, we changed the wave function from one that was a combination of two wave functions, one for $p_x = \hbar k_n$ and one for $p_x = -\hbar k_n$, to one with a definite value of p_x. This change in the wave function is not described by the time-dependent Schrödinger equation [Eq. (40.20)] but is a consequence of the measurement process. It is also independent of how the measurement is carried out: No matter how small the momentum of the incident photon shown in Fig. 40.30, the same collapse of the wave function takes place. Indeed, *any* experiment to measure p_x for a particle in a box in a steady state, no matter how the experiment is designed, will have the results that we described earlier.

(*After* the measurement, the wave function will undergo further change that *is* described by the Schrödinger equation. Neither $e^{ik_nx}e^{-i\omega_nt}$ nor $e^{-ik_nx}e^{-i\omega_nt}$ by itself satisfies the boundary conditions for a particle in a box—namely, that the wave function vanishes at $x = 0$ and $x = L$. The wave function must evolve to satisfy these conditions.)

Figure **40.30** Using photon scattering to measure the x-component of momentum of a particle in a box.



CAUTION Quantum measurement misconceptions If we measure the particle to have $p_x = \hbar k_n$, does that mean it had $p_x = \hbar k_n$ before the measurement? No; the particle acquired that value as a result of the measurement. If we measure the particle to have $p_x = \hbar k_n$ instead of $p_x = -\hbar k_n$, does that mean there was some bias in the way we did the measurement? Again, no; the result of any given experiment is random. All quantum mechanics can do is predict the probability that this experiment will give us a certain result.

Note that not every measurement of a quantum-mechanical system causes a change in the wave function. If we perform an experiment that measures only the *energy* of a particle given by the wave function in Eq. (40.53), the wave function does *not* change. That's because the wave function already corresponds to a state of definite energy $E_n = \hbar\omega_n$, so there is a 100% probability that we'll measure that value of energy.

You may ask, Does the wave function really collapse? Many physicists would answer yes, but some theorists have devised alternative models of what happens in a quantum-mechanical measurement. One model, called the *many-worlds interpretation,* asserts that there is a *universal* wave function that describes all particles in the universe. Whenever a measurement of any sort takes place, whether of human origin (like our experiment) or natural origin (for example, a photon of sunlight scattering from an electron in an atom in the atmosphere), this universal wave function does not collapse. Instead, every measurement causes the universe to branch into alternative timelines. So, when we carry out the experiment depicted in Fig. 40.30, the universe splits into one timeline in which the photon goes into detector A and a second timeline in which the photon goes into detector B. These two timelines then no longer communicate.

As weird as these aspects of quantum mechanics are, others are far weirder. We'll investigate these in Chapter 41 after we have learned more about the nature of the electron.

TEST YOUR UNDERSTANDING OF SECTION 40.6 A particle in a box is described by a wave function that is a combination of the $n = 1$ and $n = 2$ stationary states: $\Psi(x, t) = C\psi_1(x)e^{-iE_1t/\hbar} + D\psi_2(x)e^{-iE_2t/\hbar}$, where $\psi_1(x)$ and $\psi_2(x)$ are given by Eq. (40.35), E_1 and E_2 are given by Eq. (40.31), and C and D are nonzero constants. If you carry out an experiment to measure the energy of this particle, the result is *guaranteed* to be (i) E_1; (ii) E_2; (iii) $(E_1 + E_2)/2$; (iv) intermediate between E_1 and E_2, with a value that depends on the values of C and D; (v) none of these.

ANSWER

(v) The value of the energy of a particle in a box must be equal to one of the allowed energy levels, so the measured value will be either E_1 or E_2. *Neither* of these results is guaranteed. If $|C| = |D|$, then E_1 and E_2 are of equal probability. E_1 is the more likely result if $|C| > |D|$, and E_2 is the more likely result if $|C| < |D|$.

CHAPTER 40 SUMMARY

Wave functions: The wave function for a particle contains all of the information about that particle. If the particle moves in one dimension in the presence of a potential-energy function $U(x)$, the wave function $\Psi(x, t)$ obeys the one-dimensional Schrödinger equation. (For a *free* particle on which no forces act, $U(x) = 0$.) The quantity $|\Psi(x, t)|^2$, called the probability distribution function, determines the relative probability of finding a particle near a given position at a given time. If the particle is in a state of definite energy, called a stationary state, $\Psi(x, t)$ is a product of a function $\psi(x)$ that depends on only spatial coordinates and a function $e^{-iEt/\hbar}$ that depends on only time. For a stationary state, the probability distribution function is independent of time.

A spatial stationary-state wave function $\psi(x)$ for a particle that moves in one dimension in the presence of a potential-energy function $U(x)$ satisfies the time-independent Schrödinger equation. More complex wave functions can be constructed by superposing stationary-state wave functions. These can represent particles that are localized in a certain region, thus representing both particle and wave aspects. (See Examples 40.1 and 40.2.)

$$-\frac{\hbar^2}{2m}\frac{\partial^2\Psi(x,t)}{\partial x^2} + U(x)\Psi(x,t) = i\hbar\frac{\partial\Psi(x,t)}{\partial t} \quad (40.20)$$

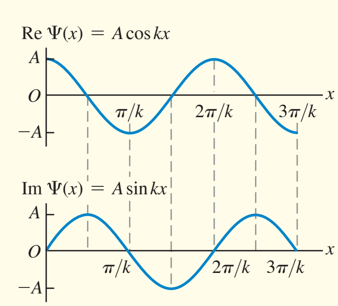
(general 1-D Schrödinger equation)

$$\Psi(x,t) = \psi(x)e^{-iEt/\hbar} \quad (40.21)$$

(time-dependent wave function for a state of definite energy)

$$-\frac{\hbar^2}{2m}\frac{d^2\psi(x)}{dx^2} + U(x)\psi(x) = E\psi(x) \quad (40.23)$$

(time-independent 1-D Schrödinger equation)



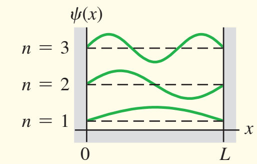
Particle in a box: The energy levels for a particle of mass m in a box (an infinitely deep square potential well) with width L are given by Eq. (40.31). The corresponding normalized stationary-state wave functions of the particle are given by Eq. (40.35). (See Examples 40.3 and 40.4.)

$$E_n = \frac{p_n{}^2}{2m} = \frac{n^2h^2}{8mL^2} = \frac{n^2\pi^2\hbar^2}{2mL^2} \quad (40.31)$$

$(n = 1, 2, 3, \ldots)$

$$\psi_n(x) = \sqrt{\frac{2}{L}}\sin\frac{n\pi x}{L} \quad (40.35)$$

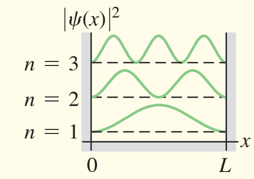
$(n = 1, 2, 3, \ldots)$



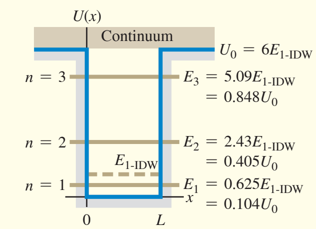
Wave functions and normalization: To be a solution of the Schrödinger equation, the wave function $\psi(x)$ and its derivative $d\psi(x)/dx$ must be continuous everywhere except where the potential-energy function $U(x)$ has an infinite discontinuity. Wave functions are usually normalized so that the total probability of finding the particle somewhere is unity.

$$\int_{-\infty}^{\infty} |\psi(x)|^2\,dx = 1 \quad (40.33)$$

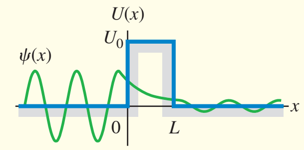
(normalization condition)



Finite potential well: In a potential well with finite depth U_0, the energy levels are lower than those for an infinitely deep well with the same width, and the number of energy levels corresponding to bound states is finite. The levels are obtained by matching wave functions at the well walls to satisfy the continuity of $\psi(x)$ and $d\psi(x)/dx$. (See Examples 40.5 and 40.6.)



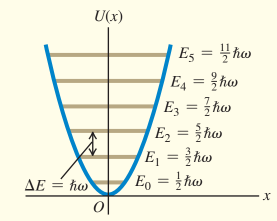
Potential barriers and tunneling: There is a certain probability that a particle will penetrate a potential-energy barrier even though its initial energy is less than the barrier height. This process is called tunneling. (See Example 40.7.)



Quantum harmonic oscillator: The energy levels for the harmonic oscillator (for which $U(x) = \frac{1}{2}k'x^2$) are given by Eq. (40.46). The spacing between any two adjacent levels is $\hbar\omega$, where $\omega = \sqrt{k'/m}$ is the oscillation angular frequency of the corresponding Newtonian harmonic oscillator. (See Example 40.8.)

$$E_n = \left(n + \tfrac{1}{2}\right)\hbar\sqrt{\frac{k'}{m}} = \left(n + \tfrac{1}{2}\right)\hbar\omega \quad (40.46)$$

$(n = 0, 1, 2, 3, \ldots)$



Measurement in quantum mechanics: If the wave function of a particle does not correspond to a definite value of a certain physical property (such as momentum or energy), the wave function changes when we measure that property. This phenomenon is called wave-function collapse.

Chapter 40 Media Assets

GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review EXAMPLES 40.3 and 40.4 (Section 40.2) before attempting these problems.

VP40.4.1 The lowest energy level for an electron confined to a one-dimensional box is 2.00×10^{-19} J. Find (a) the width of the box and (b) the energies of the $n = 2$ and $n = 3$ energy levels.

VP40.4.2 A proton (mass 1.67×10^{-27} kg) is confined to a one-dimensional box of width 5.00×10^{-15} m. Find the energy difference between (a) the $n = 2$ and $n = 1$ energy levels and (b) the $n = 3$ and $n = 2$ energy levels.

VP40.4.3 A photon is emitted when an electron in a one-dimensional box transitions from the $n = 2$ energy level to the $n = 1$ energy level.

The wavelength of this photon is 655 nm. Find (a) the energy of this photon, (b) the width of the box, and (c) the wavelength of the photon emitted when the electron transitions from the $n = 3$ level to the $n = 2$ level.

VP40.4.4 A wave function for a particle in a box with energy E is

$$\psi(x) = A\cos\left(\frac{x\sqrt{2mE}}{\hbar} + \phi\right)$$

(a) Find the second derivative of $\psi(x)$ and show that this function satisfies the Schrödinger equation. (b) Use the boundary condition at $x = 0$ to determine the value of ϕ. (c) Use the boundary condition at $x = L$ to determine the value of E.

Be sure to review EXAMPLE 40.6 (Section 40.3) before attempting these problems.

VP40.6.1 An electron is confined to a potential well of width 0.350 nm. (a) Find the ground-level energy $E_{1-\text{IDW}}$ if the well is infinitely deep. If instead the depth U_0 of the well is six times the value of $E_{1-\text{IDW}}$, find (b) the ground-level energy and (c) the minimum energy required to free the electron from the well.

VP40.6.2 A finite potential well has a depth U_0 that is six times greater than the ground-level energy $E_{1-\text{IDW}}$ of an electron in an infinitely deep well of the same width. The photon emitted when an electron in the finite potential well transitions from the $n = 3$ level to the $n = 2$ level has energy 2.50×10^{-19} J. Find (a) the value of $E_{1-\text{IDW}}$, (b) the value of U_0, and (c) the width of the potential well.

VP40.6.3 An electron in a finite potential well can emit a photon of only one of three different wavelengths when it makes a transition from one energy level to a lower energy level. The depth of this potential well is six times greater than the ground-level energy $E_{1-\text{IDW}}$ that the electron would have if the well had the same width but was infinitely deep. (a) If the shortest of the three wavelengths is 355 nm, find the quantum numbers n for the initial and final electron energy levels for this transition. (b) Find the other two wavelengths and the quantum numbers n of the initial and final energy levels for each transition.

VP40.6.4 An electron is in a potential well that is 0.400 nm wide. (a) Find the ground-level energy of the electron $E_{1-\text{IDW}}$, in eV, if the potential well is infinitely deep. (b) If the depth of the potential is only 0.015 eV, much less than $E_{1-\text{IDW}}$, there is only one bound state. Find the energy, in eV, of this bound state.

Be sure to review EXAMPLE 40.7 (Section 40.4) before attempting these problems.

VP40.7.1 An electron of energy 3.75 eV encounters a potential barrier 6.10 eV high. Find the probability that the electron will tunnel through the barrier if the barrier width is (a) 0.750 nm and (b) 0.500 nm.

VP40.7.2 A 3.50 eV electron encounters a barrier with height 4.00 eV. Find (a) the probability that the electron will tunnel through the barrier if its width is 0.800 nm and (b) the barrier width that gives twice the tunneling probability you found in part (a).

VP40.7.3 A potential barrier is 5.00 eV high and 0.900 nm wide. Find the probability that an electron will tunnel through this barrier if its energy is (a) 4.00 eV, (b) 4.30 eV, and (c) 4.60 eV.

VP40.7.4 There is a 1-in-417 probability that an electron with energy 3.00 eV will tunnel through a barrier with height 5.00 eV. Find the width of the barrier.

BRIDGING PROBLEM A Packet in a Box

A particle of mass m in an infinitely deep well (see Fig. 40.9) has the following wave function in the region from $x = 0$ to $x = L$:

$$\Psi(x, t) = \frac{1}{\sqrt{2}}\psi_1(x)e^{-iE_1t/\hbar} + \frac{1}{\sqrt{2}}\psi_2(x)e^{-iE_2t/\hbar}$$

Here $\psi_1(x)$ and $\psi_2(x)$ are the normalized stationary-state wave functions for the first two levels ($n = 1$ and $n = 2$), given by Eq. (40.35). E_1 and E_2, given by Eq. (40.31), are the energies of these levels. The wave function is zero for $x < 0$ and for $x > L$. (a) Find the probability distribution function for this wave function. (b) Does $\Psi(x, t)$ represent a stationary state of definite energy? How can you tell? (c) Show that the wave function $\Psi(x, t)$ is normalized. (d) Find the angular frequency of oscillation of the probability distribution function. What is the interpretation of this oscillation? (e) Suppose instead that $\Psi(x, t)$ is a combination of the wave functions of the two lowest levels of a finite well of length L and height U_0 equal to six times the energy of the lowest-energy bound state of an infinite well of length L. What would be the angular frequency of the probability distribution function in this case?

SOLUTION GUIDE

IDENTIFY and SET UP

1. In Section 40.1 we saw how to interpret a combination of two free-particle wave functions of different energies. In this problem you need to apply these same ideas to a combination of wave functions for the infinite well (Section 40.2) and the finite well (Section 40.3).

EXECUTE

2. Write down the full time-dependent wave function $\Psi(x, t)$ and its complex conjugate $\Psi^*(x, t)$ by using the functions $\psi_1(x)$ and $\psi_2(x)$ from Eq. (40.35). Use these to calculate the probability distribution function, and decide whether or not this function depends on time.
3. To check for normalization, you'll need to verify that when you integrate the probability distribution function from step 2 over all values of x, the integral is equal to 1. [*Hint:* The trigonometric identities $\sin^2\theta = \frac{1}{2}(1 - \cos 2\theta)$ and $\sin\theta\sin\phi = \cos(\theta - \phi) - \cos(\theta + \phi)$ may be helpful.]
4. To find the answer to part (d) you'll need to identify the oscillation angular frequency ω_{osc} in your expression from step 2 for the probability distribution function. To interpret the oscillations, draw graphs of the probability distribution functions at times $t = 0$, $t = T/4$, $t = T/2$, and $t = 3T/4$, where $T = 2\pi/\omega_{\text{osc}}$ is the oscillation period of the probability distribution function.
5. For the finite well you do not have simple expressions for the first two stationary-state wave functions $\psi_1(x)$ and $\psi_2(x)$. However, you can still find the oscillation angular frequency ω_{osc}, which is related to the energies E_1 and E_2 in the same way as for the infinite-well case. (Can you see why?)

EVALUATE

6. Why are the factors of $1/\sqrt{2}$ in the wave function $\Psi(x, t)$ important?
7. Why do you suppose the oscillation angular frequency for a finite well is lower than for an infinite well of the same width?

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q40.1 If quantum mechanics replaces the language of Newtonian mechanics, why don't we have to use wave functions to describe the motion of macroscopic objects such as baseballs and cars?

Q40.2 A student remarks that the relationship of ray optics to the more general wave picture is analogous to the relationship of Newtonian mechanics, with well-defined particle trajectories, to quantum mechanics. Comment on this remark.

Q40.3 As Eq. (40.21) indicates, the time-dependent wave function for a stationary state is a complex number having a real part and an imaginary part. How can this function have any physical meaning, since part of it is *imaginary*?

Q40.4 Why must the wave function of a particle be normalized?

Q40.5 If a particle is in a stationary state, does that mean that the particle is not moving? If a particle moves in empty space with constant momentum $\vec{p}$ and hence constant energy $E = p^2/2m$, is it in a stationary state? Explain your answers.

Q40.6 For the particle in a box, we chose $k = n\pi/L$ with $n = 1, 2, 3, \ldots$ to fit the boundary condition that $\psi = 0$ at $x = L$. However, $n = 0, -1, -2, -3, \ldots$ also satisfy that boundary condition. Why didn't we also choose those values of n?

Q40.7 If ψ is normalized, what is the physical significance of the area under a graph of $|\psi|^2$ versus x between x_1 and x_2? What is the total area under the graph of $|\psi|^2$ when all x are included? Explain.

Q40.8 For a particle in a box, what would the probability distribution function $|\psi|^2$ look like if the particle behaved like a classical (Newtonian) particle? Do the actual probability distributions approach this classical form when n is very large? Explain.

Q40.9 In Chapter 15 we represented a standing wave as a superposition of two waves traveling in opposite directions. Can the wave functions for a particle in a box also be thought of as a combination of two traveling waves? Why or why not? What physical interpretation does this representation have? Explain.

Q40.10 A particle in a box is in the ground level. What is the probability of finding the particle in the right half of the box? (Refer to Fig. 40.12, but don't evaluate an integral.) Is the answer the same if the particle is in an excited level? Explain.

Q40.11 The wave functions for a particle in a box (see Fig. 40.12a) are zero at certain points. Does this mean that the particle can't move past one of these points? Explain.

Q40.12 For a particle confined to an infinite square well, is it correct to say that each state of definite energy is also a state of definite wavelength? Is it also a state of definite momentum? Explain. (*Hint:* Remember that momentum is a vector.)

Q40.13 For a particle in a finite potential well, is it correct to say that each bound state of definite energy is also a state of definite wavelength? Is it a state of definite momentum? Explain.

Q40.14 In Fig. 40.12b, the probability function is zero at the points $x = 0$ and $x = L$, the "walls" of the box. Does this mean that the particle never strikes the walls? Explain.

Q40.15 A particle is confined to a finite potential well in the region $0 < x < L$. How does the area under the graph of $|\psi|^2$ in the region $0 < x < L$ compare to the total area under the graph of $|\psi|^2$ when including all possible x?

Q40.16 Compare the wave functions for the first three energy levels for a particle in a box of width L (see Fig. 40.12a) to the corresponding wave functions for a finite potential well of the same width (see Fig. 40.15a). How does the wavelength in the interval $0 \le x \le L$ for the $n = 1$ level of the particle in a box compare to the corresponding wavelength for the $n = 1$ level of the finite potential well? Use this to explain why E_1 is less than $E_{1-\mathrm{IDW}}$ in the situation depicted in Fig. 40.15b.

Q40.17 It is stated in Section 40.3 that a finite potential well always has at least one bound level, no matter how shallow the well. Does this mean that as $U_0 \to 0$, $E_1 \to 0$? Does this violate the Heisenberg uncertainty principle? Explain.

Q40.18 Figure 40.15a shows that the higher the energy of a bound state for a finite potential well, the more the wave function extends outside the well (into the intervals $x < 0$ and $x > L$). Explain why this happens.

Q40.19 In classical (Newtonian) mechanics, the total energy E of a particle can never be less than the potential energy U because the kinetic energy K cannot be negative. Yet in barrier tunneling (see Section 40.4) a particle passes through regions where E is less than U. Is this a contradiction? Explain.

Q40.20 Figure 40.17 shows the scanning tunneling microscope image of 48 iron atoms placed on a copper surface, the pattern indicating the density of electrons on the copper surface. What can you infer about the potential-energy function inside the circle of iron atoms?

Q40.21 Qualitatively, how would you expect the probability for a particle to tunnel through a potential barrier to depend on the height of the barrier? Explain.

Q40.22 The wave function shown in Fig. 40.20 is nonzero for both $x < 0$ and $x > L$. Does this mean that the particle splits into two parts when it strikes the barrier, with one part tunneling through the barrier and the other part bouncing off the barrier? Explain.

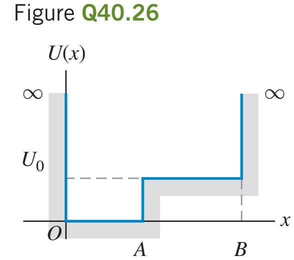
Q40.23 The probability distributions for the harmonic-oscillator wave functions (see Figs. 40.27 and 40.28) begin to resemble the classical (Newtonian) probability distribution when the quantum number n becomes large. Would the distributions become the same as in the classical case in the limit of very large n? Explain.

Q40.24 In Fig. 40.28, how does the probability of finding a particle in the center half of the region $-A < x < A$ compare to the probability of finding the particle in the outer half of the region? Is this consistent with the physical interpretation of the situation?

Q40.25 Compare the allowed energy levels for the hydrogen atom, the particle in a box, and the harmonic oscillator. What are the values of the quantum number n for the ground level and the second excited level of each system?

Q40.26 Sketch the wave function for the potential-energy well shown in **Fig. Q40.26** when E_1 is less than U_0 and when E_3 is greater than U_0.

Figure **Q40.26**



Q40.27 (a) A particle in a box has wave function $\Psi(x, t) = \psi_2(x)e^{-iE_2t/\hbar}$, where ψ_n and E_n are given by Eqs. (40.35) and (40.31), respectively. If the energy of the particle is measured, what is the result? (b) If instead the particle has wave function $\Psi(x, t) = (1/\sqrt{2})(\psi_1(x)e^{-iE_1t/\hbar} + \psi_2(x)e^{-iE_2t/\hbar})$ and the energy of the particle is measured, what is the result? (c) If we had many identical particles with the wave function of part (b) and measured the energy of each, what would be the average value of all of the measurements? Can we say that, before the measurement was made, each particle had this average energy? Explain.

EXERCISES

Section 40.1 Wave Functions and the One-Dimensional Schrödinger Equation

40.1 • An electron is moving as a free particle in the $-x$-direction with momentum that has magnitude $4.50 \times 10^{-24}\ \mathrm{kg \cdot m/s}$. What is the one-dimensional time-dependent wave function of the electron?

40.2 • A free particle moving in one dimension has wave function

$$\Psi(x, t) = A\left[e^{i(kx-\omega t)} - e^{i(2kx-4\omega t)}\right]$$

where k and ω are positive real constants. (a) At $t = 0$ what are the two smallest positive values of x for which the probability function $|\Psi(x, t)|^2$ is a maximum? (b) Repeat part (a) for time $t = 2\pi/\omega$. (c) Calculate v_{av} as the distance the maxima have moved divided by the elapsed time. Compare your result to the expression $v_{\mathrm{av}} = (\omega_2 - \omega_1)/(k_2 - k_1)$ from Example 40.1.

40.3 • Consider the free-particle wave function of Example 40.1. Let $k_2 = 3k_1 = 3k$. At $t = 0$ the probability distribution function $|\Psi(x, t)|^2$ has a maximum at $x = 0$. (a) What is the smallest positive value of x for which the probability distribution function has a maximum at time $t = 2\pi/\omega$, where $\omega = \hbar k^2/2m$? (b) From your result in part (a), what is the average speed with which the probability distribution is moving in the $+x$-direction? Compare your result to the expression $v_{\mathrm{av}} = (\omega_2 - \omega_1)/(k_2 - k_1)$ from Example 40.1.

40.4 • A particle is described by a wave function $\psi(x) = Ae^{-\alpha x^2}$, where A and α are real, positive constants. If the value of α is increased, what effect does this have on (a) the particle's uncertainty in position and (b) the particle's uncertainty in momentum? Explain your answers.

40.5 • Consider a wave function given by $\psi(x) = A\sin kx$, where $k = 2\pi/\lambda$ and A is a real constant. (a) For what values of x is there the highest probability of finding the particle described by this wave function? Explain. (b) For which values of x is the probability *zero*? Explain.

40.6 • CALC Let ψ_1 and ψ_2 be two solutions of Eq. (40.23) with energies E_1 and E_2, respectively, where $E_1 \neq E_2$. Is $\psi = A\psi_1 + B\psi_2$, where A and B are nonzero constants, a solution to Eq. (40.23)? Explain your answer.

Section 40.2 Particle in a Box

40.7 • An electron is in a one-dimensional box. When the electron is in its ground state, the longest-wavelength photon it can absorb is 420 nm. What is the next longest-wavelength photon it can absorb, again starting in the ground state?

40.8 •• CALC A particle moving in one dimension (the x-axis) is described by the wave function

$$\psi(x) = \begin{cases} Ae^{-bx}, & \text{for } x \geq 0 \\ Ae^{bx}, & \text{for } x < 0 \end{cases}$$

where $b = 2.00\ \mathrm{m}^{-1}$, $A > 0$, and the $+x$-axis points toward the right. (a) Determine A so that the wave function is normalized. (b) Sketch the graph of the wave function. (c) Find the probability of finding this particle in each of the following regions: (i) within 50.0 cm of the origin, (ii) on the left side of the origin (can you first guess the answer by looking at the graph of the wave function?), (iii) between $x = 0.500$ m and $x = 1.00$ m.

40.9 • **Ground-Level Billiards.** (a) Find the lowest energy level for a particle in a box if the particle is a billiard ball ($m = 0.20$ kg) and the box has a width of 1.3 m, the size of a billiard table. (Assume that the billiard ball slides without friction rather than rolls; that is, ignore the *rotational* kinetic energy.) (b) Since the energy in part (a) is all kinetic, to what speed does this correspond? How much time would it take at this speed for the ball to move from one side of the table to the other? (c) What is the difference in energy between the $n = 2$ and $n = 1$ levels? (d) Are quantum-mechanical effects important for the game of billiards?

40.10 • A proton is in a box of width L. What must the width of the box be for the ground-level energy to be 5.0 MeV, a typical value for the energy with which the particles in a nucleus are bound? Compare your result to the size of a nucleus—that is, on the order of 10^{-14} m.

40.11 •• Find the width L of a one-dimensional box for which the ground-state energy of an electron in the box equals the absolute value of the ground state of a hydrogen atom.

40.12 •• When a hydrogen atom undergoes a transition from the $n = 2$ to the $n = 1$ level, a photon with $\lambda = 122$ nm is emitted. (a) If the atom is modeled as an electron in a one-dimensional box, what is the width of the box in order for the $n = 2$ to $n = 1$ transition to correspond to emission of a photon of this energy? (b) For a box with the width calculated in part (a), what is the ground-state energy? How does this correspond to the ground-state energy of a hydrogen atom? (c) Do you think a one-dimensional box is a good model for a hydrogen atom? Explain. (*Hint:* Compare the spacing between adjacent energy levels as a function of n.)

40.13 •• A certain atom requires 3.0 eV of energy to excite an electron from the ground level to the first excited level. Model the atom as an electron in a box and find the width L of the box.

40.14 • An electron in a one-dimensional box has ground-state energy 2.00 eV. What is the wavelength of the photon absorbed when the electron makes a transition to the second excited state?

40.15 • A particle with mass m is in a one-dimensional box with width L. If the energy of the particle is $9\pi^2\hbar^2/2mL^2$, (a) what is the linear momentum of the particle and (b) what is the ratio of the width of the box to the de Broglie wavelength λ of the particle?

40.16 • Recall that $|\psi|^2 dx$ is the probability of finding the particle that has normalized wave function $\psi(x)$ in the interval x to $x + dx$. Consider a particle in a box with rigid walls at $x = 0$ and $x = L$. Let the particle be in the ground level and use ψ_n as given in Eq. (40.35). (a) For which values of x, if any, in the range from 0 to L is the probability of finding the particle zero? (b) For which values of x is the probability highest? (c) In parts (a) and (b) are your answers consistent with Fig. 40.12? Explain.

40.17 • Consider two different particles in two different boxes. Particle A has nine times the mass of particle B, while particle B's box has twice the width of particle A's box. What are the two lowest quantum numbers for the two systems for which particles A and B have equal energies?

40.18 • (a) Find the excitation energy from the ground level to the third excited level for an electron confined to a box of width 0.360 nm. (b) The electron makes a transition from the $n = 1$ to $n = 4$ level by absorbing a photon. Calculate the wavelength of this photon.

40.19 • An electron is in a box of width 3.0×10^{-10} m. What are the de Broglie wavelength and the magnitude of the momentum of the electron if it is in (a) the $n = 1$ level; (b) the $n = 2$ level; (c) the $n = 3$ level? In each case how does the wavelength compare to the width of the box?

40.20 •• When an electron in a one-dimensional box makes a transition from the $n = 1$ energy level to the $n = 2$ level, it absorbs a photon of wavelength 426 nm. What is the wavelength of that photon when the electron undergoes a transition (a) from the $n = 2$ to the $n = 3$ energy level and (b) from the $n = 1$ to the $n = 3$ energy level? (c) What is the width L of the box?

40.21 • A particle is in a box with infinitely rigid walls. The walls are at $x = -L/2$ and $x = +L/2$. (a) Show that $\psi_n = A\cos k_n x$ is a possible solution. What must k_n equal? (b) Show that $\psi_n = A\sin k_n x$ is a possible solution. What must k_n equal? (c) What are the allowed energy levels for $\psi_n = A\sin k_n x$ and for $\psi_n = A\cos k_n x$? (d) How does the set of energy levels you found in part (c) compare to the energy levels given by Eq. (40.31)?

Section 40.3 Potential Wells

40.22 •• An electron is moving past the square well shown in Fig. 40.13. The electron has energy $E = 3U_0$. What is the ratio of the de Broglie wavelength of the electron in the region $x > L$ to the wavelength for $0 < x < L$?

40.23 • An electron is bound in a square well of depth $U_0 = 6E_{1-\text{IDW}}$. What is the width of the well if its ground-state energy is 2.00 eV?

40.24 •• An electron is in the ground state of a square well of width $L = 4.00 \times 10^{-10}$ m. The depth of the well is six times the ground-state energy of an electron in an infinite well of the same width. What is the kinetic energy of this electron after it has absorbed a photon of wavelength 72 nm and moved away from the well?

40.25 •• An electron is bound in a square well of width 1.50 nm and depth $U_0 = 6E_{1-\text{IDW}}$. If the electron is initially in the ground level and absorbs a photon, what maximum wavelength can the photon have and still liberate the electron from the well?

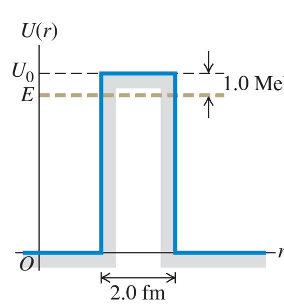
40.26 •• An electron is bound in a square well that has a depth equal to six times the ground-level energy $E_{1-\text{IDW}}$ of an infinite well of the same width. The longest-wavelength photon that is absorbed by this electron has a wavelength of 582 nm. Determine the width of the well.

40.27 •• A proton is bound in a square well of width 4.0 fm $= 4.0 \times 10^{-15}$ m. The depth of the well is six times the ground-level energy $E_{1-\text{IDW}}$ of the corresponding infinite well. If the proton makes a transition from the level with energy E_1 to the level with energy E_3 by absorbing a photon, find the wavelength of the photon.

Section 40.4 Potential Barriers and Tunneling

40.28 •• Alpha Decay. In a simple model for a radioactive nucleus, an alpha particle ($m = 6.64 \times 10^{-27}$ kg) is trapped by a square barrier that has width 2.0 fm and height 30.0 MeV. (a) What is the tunneling probability when the alpha particle encounters the barrier if its kinetic energy is 1.0 MeV below the top of the barrier (**Fig. E40.28**)? (b) What is the tunneling probability if the energy of the alpha particle is 10.0 MeV below the top of the barrier?

Figure **E40.28**



40.29 •• (a) An electron with initial kinetic energy 32 eV encounters a square barrier with height 41 eV and width 0.25 nm. What is the probability that the electron will tunnel through the barrier? (b) A proton with the same kinetic energy encounters the same barrier. What is the probability that the proton will tunnel through the barrier?

40.30 • An electron with initial kinetic energy 5.0 eV encounters a barrier with height U_0 and width 0.60 nm. What is the transmission coefficient if (a) $U_0 = 7.0$ eV; (b) $U_0 = 9.0$ eV; (c) $U_0 = 13.0$ eV?

40.31 • An electron with initial kinetic energy 6.0 eV encounters a barrier with height 11.0 eV. What is the probability of tunneling if the width of the barrier is (a) 0.80 nm and (b) 0.40 nm?

40.32 •• An electron is moving past the square barrier shown in Fig. 40.19, but the energy of the electron is *greater* than the barrier height. If $E = 2U_0$, what is the ratio of the de Broglie wavelength of the electron in the region $x > L$ to the wavelength for $0 < x < L$?

Section 40.5 The Harmonic Oscillator

40.33 • A wooden block with mass 0.250 kg is oscillating on the end of a spring that has force constant 110 N/m. Calculate the ground-level energy and the energy separation between adjacent levels. Express your results in joules and in electron volts. Are quantum effects important?

40.34 •• A harmonic oscillator with mass m and force constant k' is in an excited state that has quantum number n. (a) Let $p_{\text{max}} = mv_{\text{max}}$, where v_{max} is the maximum speed calculated in the Newtonian analysis of the oscillator. Derive an expression for p_{max} in terms of $n, \hbar, k'$, and m. (b) Derive an expression for the classical amplitude A in terms of $n, \hbar, k'$, and m. (c) If $\Delta x = A/\sqrt{2}$ and $\Delta p_x = p_{\text{max}}/\sqrt{2}$, what is the uncertainty product $\Delta x \Delta p_x$? How does the uncertainty product depend on n?

40.35 • Chemists use infrared absorption spectra to identify chemicals in a sample. In one sample, a chemist finds that light of wavelength 5.8 μm is absorbed when a molecule makes a transition from its ground harmonic oscillator level to its first excited level. (a) Find the energy of this transition. (b) If the molecule can be treated as a harmonic oscillator with mass 5.6×10^{-26} kg, find the force constant.

40.36 • A harmonic oscillator absorbs a photon of wavelength 6.35 μm when it undergoes a transition from the ground state to the first excited state. What is the ground-state energy, in electron volts, of the oscillator?

40.37 •• The ground-state energy of a harmonic oscillator is 5.60 eV. If the oscillator undergoes a transition from its $n = 3$ to $n = 2$ level by emitting a photon, what is the wavelength of the photon?

40.38 •• While undergoing a transition from the $n = 1$ to the $n = 2$ energy level, a harmonic oscillator absorbs a photon of wavelength 6.50 μm. What is the wavelength of the absorbed photon when this oscillator undergoes a transition (a) from the $n = 2$ to the $n = 3$ energy level and (b) from the $n = 1$ to the $n = 3$ energy level? (c) What is the value of $\sqrt{k'/m}$, the angular oscillation frequency of the corresponding Newtonian oscillator?

40.39 •• For the sodium atom of Example 40.8, find (a) the ground-state energy; (b) the wavelength of a photon emitted when the $n = 4$ to $n = 3$ transition occurs; (c) the energy difference for any $\Delta n = 1$ transition.

40.40 •• For the ground-level harmonic oscillator wave function $\psi(x)$ given in Eq. (40.47), $|\psi|^2$ has a maximum at $x = 0$. (a) Compute the ratio of $|\psi|^2$ at $x = +A$ to $|\psi|^2$ at $x = 0$, where A is given by Eq. (40.48) with $n = 0$ for the ground level. (b) Compute the ratio of $|\psi|^2$ at $x = +2A$ to $|\psi|^2$ at $x = 0$. In each case is your result consistent with what is shown in Fig. 40.27?

PROBLEMS

40.41 •• A particle of mass m in a one-dimensional box has the following wave function in the region $x = 0$ to $x = L$:

$$\Psi(x, t) = \frac{1}{\sqrt{2}}\psi_1(x)e^{-iE_1t/\hbar} + \frac{1}{\sqrt{2}}\psi_3(x)e^{-iE_3t/\hbar}$$

Here $\psi_1(x)$ and $\psi_3(x)$ are the normalized stationary-state wave functions for the $n = 1$ and $n = 3$ levels, and E_1 and E_3 are the energies of these levels. The wave function is zero for $x < 0$ and for $x > L$. (a) Find the value of the probability distribution function at $x = L/2$ as a function of time. (b) Find the angular frequency at which the probability distribution function oscillates.

40.42 •• Hydrogen emits radiation with four prominent visible wavelengths—one red, one cyan, one blue, and one violet. The respective frequencies are 656 nm, 486 nm, 434 nm, and 410 nm. We can model the hydrogen atom as an electron in a one-dimensional box, and attempt to match four adjacent emission lines in the predicted spectrum to the visible part of the hydrogen spectrum. (a) Determine the photon energies associated with the visible part of the hydrogen spectrum. (b) The electron-in-a-box emission spectrum is $E_{n_i \to n_f} = (n_i^2 - n_f^2)\epsilon$, where n_i and n_f are the initial and final quantum numbers of the electron when it drops to a lower energy level and ϵ is the energy determined by the Schrödinger equation. What is the smallest possible value of n_i that can accommodate four emission lines? (c) Using the value from part (b) for n_i, estimate the order of magnitude of ϵ by dividing the four photon energies by the relevant differences $n_i^2 - n_f^2$ for transitions in the possible sets of (n_i, n_f) pairings. (d) Using Eq. (40.31) to identify ϵ, and using the mass of the electron, use your result from part (c) to estimate the length L of the box. (e) What is the ratio of your estimate of L to twice the Bohr radius? (*Note:* The hydrogen atom is better modeled using the Coulomb potential rather than as a particle in a box.)

40.43 •• CALC Consider a beam of free particles that move with velocity $v = p/m$ in the x-direction and are incident on a potential-energy step $U(x) = 0$, for $x < 0$, and $U(x) = U_0 < E$, for $x > 0$. The wave function for $x < 0$ is $\psi(x) = Ae^{ik_1x} + Be^{-ik_1x}$, representing incident and reflected particles, and for $x > 0$ is $\psi(x) = Ce^{ik_2x}$, representing transmitted particles. Use the conditions that both ψ and its first derivative must be continuous at $x = 0$ to find the constants B and C in terms of k_1, k_2, and A.

40.44 • CALC A particle is in the ground level of a box that extends from $x = 0$ to $x = L$. (a) What is the probability of finding the particle in the region between 0 and $L/4$? Calculate this by integrating $|\psi(x)|^2\,dx$, where ψ is normalized, from $x = 0$ to $x = L/4$. (b) What is the probability of finding the particle in the region $x = L/4$ to $x = L/2$? (c) How do the results of parts (a) and (b) compare? Explain. (d) Add the probabilities calculated in parts (a) and (b). (e) Are your results in parts (a), (b), and (d) consistent with Fig. 40.12b? Explain.

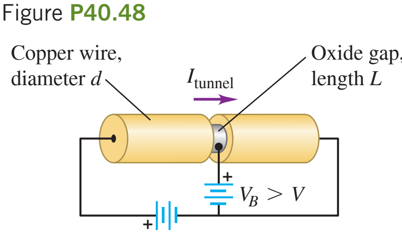
40.45 • Photon in a Dye Laser. An electron in a long, organic molecule used in a dye laser behaves approximately like a particle in a box with width 4.18 nm. What is the wavelength of the photon emitted when the electron undergoes a transition (a) from the first excited level to the ground level and (b) from the second excited level to the first excited level?

40.46 •• Consider a particle in a box with rigid walls at $x = 0$ and $x = L$. Let the particle be in the ground level. Calculate the probability $|\psi|^2 dx$ that the particle will be found in the interval x to $x + dx$ for (a) $x = L/4$; (b) $x = L/2$; (c) $x = 3L/4$.

40.47 •• Repeat Problem 40.46 for a particle in the first excited level.

40.48 •• Suppose we have two copper wires that meet on either side of an oxide gap with length $L = 1$ nm, as shown in **Fig. P40.48**. If the left side is at a potential V relative to the right side, then we can estimate the tunneling current as follows: The wires have diameter $d = 1$ mm, and copper has $n = 8.5 \times 10^{28}$ free electrons per m^3. (a) Assume that the oxide gap provides a rectangular potential barrier of height 12 eV and that the energy of the free electrons incident on this barrier is 5.0 eV. Assume the gap has length $L = 1$ nm. What is the probability T that an electron will tunnel through the barrier? (b) A length x of wire on the left of the barrier will have nAx free electrons, where $A = \pi(d/2)^2$ is the cross-sectional area of the wire. In a similar length of wire on the right side of the barrier, the number of tunneled electrons will be T times that number. If these electrons move with effective velocity $v = dx/dt$, then what will be the tunneling current, expressed symbolically in terms of T, n, A, e, and v? (c) Define their effective speed v by $\frac{1}{2}mv^2 = eV$. The tunneled electrons have energy eV. What is their effective speed v, expressed symbolically in terms of e, V, and m, where m is the mass of an electron? (f) Put these elements together to estimate the tunneling current.

Figure **P40.48**



40.49 •• CALC What is the probability of finding a particle in a box of length L in the region between $x = L/4$ and $x = 3L/4$ when the particle is in (a) the ground level and (b) the first excited level? (*Hint:* Integrate $|\psi(x)|^2\,dx$, where ψ is normalized, between $L/4$ and $3L/4$.) (c) Are your results in parts (a) and (b) consistent with Fig. 40.12b? Explain.

40.50 •• The *penetration distance* η in a finite potential well is the distance at which the wave function has decreased to $1/e$ of the (b) wave function at the classical turning point:

$$\psi(x = L + \eta) = \frac{1}{e}\psi(L)$$

The penetration distance can be shown to be

$$\eta = \frac{\hbar}{\sqrt{2m(U_0 - E)}}$$

The probability of finding the particle beyond the penetration distance is nearly zero. (a) Find η for an electron having a kinetic energy of 13 eV in a potential well with $U_0 = 20$ eV. (b) Find η for a 20.0 MeV proton trapped in a 30.0-MeV-deep potential well.

40.51 •• CALC A fellow student proposes that a possible wave function for a free particle with mass m (one for which the potential-energy function $U(x)$ is zero) is

$$\psi(x) = \begin{cases} e^{+\kappa x}, & x < 0 \\ e^{-\kappa x}, & x \geq 0 \end{cases}$$

where κ is a positive constant. (a) Graph this proposed wave function. (b) Show that the proposed wave function satisfies the Schrödinger equation for $x < 0$ if the energy is $E = -\hbar^2\kappa^2/2m$—that is, if the energy of the particle is *negative*. (c) Show that the proposed wave function also satisfies the Schrödinger equation for $x \geq 0$ with the same energy as in part (b). (d) Explain why the proposed wave function is nonetheless *not* an acceptable solution of the Schrödinger equation for a free particle. (*Hint:* What is the behavior of the function at $x = 0$?) It is in fact impossible for a free particle (one for which $U(x) = 0$) to have an energy less than zero.

40.52 • An electron with initial kinetic energy 5.5 eV encounters a square potential barrier of height 10.0 eV. What is the width of the barrier if the electron has a 0.50% probability of tunneling through the barrier?

40.53 ••• CP Consider a one-dimensional lattice of carbon nuclei with interatomic spacing $d = 0.500$ nm. The potential energy of one nucleus due to electrostatic interaction with its two nearest neighbors is

$$V(x) = \frac{q^2}{4\pi\epsilon_0}\left(\frac{1}{d - x} + \frac{1}{d + x}\right)$$

where $q = 6e$ is the charge of each carbon nucleus and x is its position relative to its equilibrium position. (a) Sketch the potential $V(x)$ between $x = -d$ and $x = d$. (b) At low energy, the nucleus remains near its equilibrium position at $x = 0$ and the potential energy is well approximated as a harmonic oscillator. Use the power series in Appendix B to write $V(x)$ as a quadratic function in x, dropping contributions at higher order in x/d. (c) What is the spring constant k'? (d) What is the energy of the classical ground state, in eV? (e) What is the energy of the lowest-energy photons emitted when this system transitions between energy levels? (f) What is the wavelength of the corresponding radiation?

40.54 • CP A harmonic oscillator consists of a 0.020 kg mass on a spring. The oscillation frequency is 1.50 Hz, and the mass has a speed of 0.480 m/s as it passes the equilibrium position. (a) What is the value of the quantum number n for its energy level? (b) What is the difference in energy between the levels E_n and E_{n+1}? Is this difference detectable?

40.55 • For small amplitudes of oscillation the motion of a pendulum is simple harmonic. For a pendulum with a period of 0.500 s, find the ground-level energy and the energy difference between adjacent energy levels. Express your results in joules and in electron volts. Are these values detectable?

40.56 •• CALC (a) Show by direct substitution in the Schrödinger equation for the one-dimensional harmonic oscillator that the wave function $\psi_1(x) = A_1 x e^{-\alpha^2 x^2/2}$, where $\alpha^2 = m\omega/\hbar$, is a solution with energy corresponding to $n = 1$ in Eq. (40.46). (b) Find the normalization constant A_1. (c) Show that the probability density has a minimum at $x = 0$ and maxima at $x = \pm 1/\alpha$, corresponding to the classical turning points for the ground state $n = 0$.

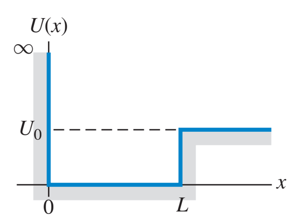
40.57 •• CP (a) The wave nature of particles results in the quantum-mechanical situation that a particle confined in a box can assume only wavelengths that result in standing waves in the box, with nodes at the box walls. Use this to show that an electron confined in a one-dimensional box of length L will have energy levels given by

$$E_n = \frac{n^2 h^2}{8mL^2}$$

(*Hint:* Recall that the relationship between the de Broglie wavelength and the speed of a nonrelativistic particle is $mv = h/\lambda$. The energy of the particle is $\frac{1}{2}mv^2$.) (b) If a hydrogen atom is modeled as a one-dimensional box with length equal to the Bohr radius, what is the energy (in electron volts) of the lowest energy level of the electron?

40.58 ••• Consider a potential well defined as $U(x) = \infty$ for $x < 0$, $U(x) = 0$ for $0 < x < L$, and $U(x) = U_0 > 0$ for $x > L$ (**Fig. P40.58**). Consider a particle with mass m and kinetic energy $E < U_0$ that is trapped in the well. (a) The boundary condition at the infinite wall ($x = 0$) is $\psi(0) = 0$. What must the form of the function $\psi(x)$ for $0 < x < L$ be in order to satisfy both the Schrödinger equation and this boundary condition? (b) The wave function must remain finite as $x \to \infty$. What must the form of the function $\psi(x)$ for $x > L$ be in order to satisfy both the Schrödinger equation and this boundary condition at infinity? (c) Impose the boundary conditions that ψ and $d\psi/dx$ are continuous at $x = L$. Show that the energies of the allowed levels are obtained from solutions of the equation $k \cot kL = -\kappa$, where $k = \sqrt{2mE}/\hbar$ and $\kappa = \sqrt{2m(U_0 - E)}/\hbar$.

Figure **P40.58**



40.59 •• DATA In your research on new solid-state devices, you are studying a solid-state structure that can be modeled accurately as an electron in a one-dimensional infinite potential well (box) of width L. In one of your experiments, electromagnetic radiation is absorbed in transitions in which the initial state is the $n = 1$ ground state. You measure that light of frequency $f = 9.0 \times 10^{14}$ Hz is absorbed and that the next higher absorbed frequency is 16.9×10^{14} Hz. (a) What is quantum number n for the final state in each of the transitions that leads to the absorption of photons of these frequencies? (b) What is the width L of the potential well? (c) What is the longest wavelength in air of light that can be absorbed by an electron if it is initially in the $n = 1$ state?

40.60 •• DATA As an intern at a research lab, you study the transmission of electrons through a potential barrier. You know the height of the barrier, 8.0 eV, but must measure the width L of the barrier. When you measure the tunneling probability T as a function of the energy E of the electron, you get the results shown in the table.

E (eV)	4.0	5.0	6.0	7.0	7.6
T	2.4×10^{-6}	1.5×10^{-5}	1.2×10^{-4}	1.3×10^{-3}	8.1×10^{-3}

(a) For each value of E, calculate the quantities G and κ that appear in Eq. (40.42). Graph $\ln(T/G)$ versus κ. Explain why your data points, when plotted this way, fall close to a straight line. (b) Use the slope of the best-fit straight line to the data in part (a) to calculate L.

40.61 •• DATA When low-energy electrons pass through an ionized gas, electrons of certain energies pass through the gas as if the gas atoms weren't there and thus have transmission coefficients (tunneling probabilities) T equal to unity. The gas ions can be modeled approximately as a rectangular barrier. The value of $T = 1$ occurs when an integral or half-integral number of de Broglie wavelengths of the electron as it passes over the barrier equal the width L of the barrier. You are planning an experiment to measure this effect. To assist you in designing the necessary apparatus, you estimate the electron energies E that will result in $T = 1$. You assume a barrier height of 10 eV and a width of 1.8×10^{-10} m. Calculate the three lowest values of E for which $T = 1$.

40.62 ••• CP A particle is confined to move on a circle with radius R but is otherwise free. We can parameterize points on this circle by using the distance x from a reference point or by using the angle $\theta = x/R$. Since $x = 0$ and $x = 2\pi R$ describe the same point, the wave function must satisfy $\psi(x) = \psi(x + 2\pi R)$ and $\psi'(x) = \psi'(x + 2\pi R)$. (a) Solve the free-particle time-independent Schrödinger equation subject to these boundary conditions. You should find solutions $\psi_n^{\pm}$, where n is a positive integer and where lower n corresponds to lower energy. Express your solutions in terms of θ using unspecified normalization constants A_n^+ and A_n^- corresponding, respectively, to modes that move "counterclockwise" toward higher x and "clockwise" toward lower x. (b) Normalize these functions to determine $A_n^{\pm}$. (c) What are the energy levels E_n? (d) Write the time-dependent wave functions $\Psi_n^{\pm}(x, t)$ corresponding to $\Psi_n^{\pm}(x)$. Use the symbol ω for $E_1/\hbar$. (e) Consider the nonstationary state defined at $t = 0$ by $\Psi(x, 0) = \frac{1}{\sqrt{2}}[\Psi_1^+(x) + \Psi_2^+(x)]$. Determine the probability density $|\Psi(x, t)|^2$ in terms of R, ω, and t. Simplify your result using the identity $1 + \cos\alpha = 2\cos^2(\alpha/2)$. (f) With what angular speed does the density peak move around the circle? (g) If the particle is an electron and the radius is the Bohr radius, then with what speed does its probability peak move?

CHALLENGE PROBLEMS

40.63 ••• CALC The Schrödinger equation for the quantum-mechanical harmonic oscillator in Eq. (40.44) may be written as

$$-\frac{\hbar^2}{2m}\psi'' + \frac{1}{2}m\omega^2 x^2 \psi = E\psi$$

where $\omega = \sqrt{k'/m}$ is the angular frequency for classical oscillations. Its solutions may be cleverly addressed using the following observations: The difference in applying two operations A and B, in opposite order, called a commutator, is denoted by $[A, B] = AB - BA$. For example, if A represents differentiation, so that $Af = \dfrac{df}{dx}$, where $f = f(x)$is a function, and B represents multiplication by x, then $[A, B] = \left[\dfrac{d}{dx}, x\right]f = \dfrac{d}{dx}(xf) - x\dfrac{df}{dx}$. Applying the product rule for differentiation on the first term, we determine $\left[\dfrac{d}{dx}, x\right]f = f$, which we summarize by writing $\left[\frac{d}{dx}, x\right] = 1$. Similarly, consider the two operators a_+ and a_- defined by $a_\pm = 1/\sqrt{2m\hbar\omega}\left(\mp\hbar\frac{d}{dx} + m\omega x\right)$. (a) Determine the commutator $[a_-, a_+]$. (b) Compute $a_+a_-\psi$ in terms of ψ and ψ'', where $\psi = \psi(x)$ is a wave function. Be mindful that $\frac{d}{dx}(x\psi) = \psi + x\psi'$. (c) The Schrödinger equation above may be written as $H\psi = E\psi$, where H is a differential operator. Use your previous result to write H in terms of a_+ and a_-. (d) Determine the commutators $[H, a_\pm]$. (e) Consider a wave function ψ_n with definite energy E_n, whereby $H\psi_n = E_n\psi_n$. If we define a new function $\psi_{n+1} = a_+\psi_n$, then we can readily determine its energy by computing $H\psi_{n+1} = Ha_+\psi_n = (a_+H + [H, a_+])\psi_n$ and comparing the result with $E_{n+1}\psi_{n+1}$. Finish this calculation by inserting what we have determined to find E_{n+1} in terms of E_n and n. (f) We can prove that the lowest-energy wave function ψ_0 is annihilated by a_-, meaning that $a_-\psi_0 = 0$. Using your expression for H in terms of $a_\pm$, determine the ground-state energy E_0. (g) By applying a_+ repeatedly, we can generate all higher states. The energy of the nth excited state can be determined by considering $Ha_+^n\psi_0 = E_0\psi_0$. Using the above results, determine E_n. (*Note:* We did not have to solve any differential equations or take any derivatives to determine the spectrum of this theory.)

40.64 ••• CALC The WKB Approximation. It can be a challenge to solve the Schrödinger equation for the bound-state energy levels of an arbitrary potential well. An alternative approach that can yield good approximate results for the energy levels is the *WKB approximation* (named for the physicists Gregor Wentzel, Hendrik Kramers, and Léon Brillouin, who pioneered its application to quantum mechanics). The WKB approximation begins from three physical statements: (i) According to de Broglie, the magnitude of momentum p of a quantum-mechanical particle is $p = h/\lambda$. (ii) The magnitude of momentum is related to the kinetic energy K by the relationship $K = p^2/2m$. (iii) If there are no nonconservative forces, then in Newtonian mechanics the energy E for a particle is constant and equal at each point to the sum of the kinetic and potential energies at that point: $E = K + U(x)$, where x is the coordinate. (a) Combine these three relationships to show that the wavelength of the particle at a coordinate x can be written as

$$\lambda(x) = \frac{h}{\sqrt{2m[E - U(x)]}}$$

Thus we envision a quantum-mechanical particle in a potential well $U(x)$ as being like a free particle, but with a wavelength $\lambda(x)$ that is a function of position. (b) When the particle moves into a region of increasing potential energy, what happens to its wavelength? (c) At a point where $E = U(x)$, Newtonian mechanics says that the particle has zero kinetic energy and must be instantaneously at rest. Such a point is called a *classical turning point,* since this is where a Newtonian particle must stop its motion and reverse direction. As an example, an object oscillating in simple harmonic motion with amplitude A moves back and forth between the points $x = -A$ and $x = +A$; each of these is a classical turning point, since there the potential energy $\frac{1}{2}k'x^2$ equals the total energy $\frac{1}{2}k'A^2$. In the WKB expression for $\lambda(x)$, what is the wavelength at a classical turning point? (d) For a particle in a box with length L, the walls of the box are classical turning points (see Fig. 40.8). Furthermore, the number of wavelengths that fit within the box must be a half-integer (see Fig. 40.10), so that $L = (n/2)\lambda$ and hence $L/\lambda = n/2$, where $n = 1, 2, 3, \ldots$. [Note that this is a restatement of Eq. (40.29).] The WKB scheme for finding the allowed bound-state energy levels of an *arbitrary* potential well is an extension of these observations. It demands that for an allowed energy E, there must be a half-integer number of wavelengths between the classical turning points for that energy. Since the wavelength in the WKB approximation is not a constant but depends on x, the number of wavelengths between the classical turning points a and b for a given value of the energy is the integral of $1/\lambda(x)$ between those points:

$$\int_a^b \frac{dx}{\lambda(x)} = \frac{n}{2} \quad (n = 1, 2, 3, \ldots)$$

Using the expression for $\lambda(x)$ you found in part (a), show that the *WKB condition for an allowed bound-state energy* can be written as

$$\int_a^b \sqrt{2m[E - U(x)]}\, dx = \frac{nh}{2} \quad (n = 1, 2, 3, \ldots)$$

(e) As a check on the expression in part (d), apply it to a particle in a box with walls at $x = 0$ and $x = L$. Evaluate the integral and show that the allowed energy levels according to the WKB approximation are the same as those given by Eq. (40.31). (*Hint:* Since the walls of the box are infinitely high, the points $x = 0$ and $x = L$ are classical turning points for *any* energy E. Inside the box, the potential energy is zero.) (f) For the finite square well shown in Fig. 40.13, show that the WKB expression given in part (d) predicts the *same* bound-state energies as for an infinite square well of the same width. (*Hint:* Assume $E < U_0$. Then the classical turning points are at $x = 0$ and $x = L$.) This shows that the WKB approximation does a poor job when the potential-energy function changes discontinuously, as for a finite potential well. In the next two problems we consider situations in which the potential-energy function changes gradually and the WKB approximation is much more useful.

40.65 ••• CALC The WKB approximation (see Challenge Problem 40.64) can be used to calculate the energy levels for a harmonic oscillator. In this approximation, the energy levels are the solutions to the equation

$$\int_a^b \sqrt{2m[E - U(x)]}\, dx = \frac{nh}{2} \quad n = 1, 2, 3, \ldots$$

Here E is the energy, $U(x)$ is the potential-energy function, and $x = a$ and $x = b$ are the classical turning points (the points at which E is equal to the potential energy, so the Newtonian kinetic energy would be zero). (a) Determine the classical turning points for a harmonic oscillator with energy E and force constant k'. (b) Carry out the integral in the WKB approximation and show that the energy levels in this approximation are $E_n = \hbar\omega$, where $\omega = \sqrt{k'/m}$ and $n = 1, 2, 3, \ldots$. (*Hint:* Recall that $\hbar = h/2\pi$. A useful standard integral is

$$\int \sqrt{A^2 - x^2}\, dx = \tfrac{1}{2}\left[x\sqrt{A^2 - x^2} + A^2 \arcsin\left(\frac{x}{|A|}\right)\right]$$

where arcsin denotes the inverse sine function. Note that the integrand is even, so the integral from $-x$ to x is equal to twice the integral from 0 to x.) (c) How do the approximate energy levels found in part (b) compare with the true energy levels given by Eq. (40.46)? Does the WKB approximation give an underestimate or an overestimate of the energy levels?

40.66 ••• CALC Protons, neutrons, and many other particles are made of more fundamental particles called *quarks* and *antiquarks* (the antimatter equivalent of quarks). A quark and an antiquark can form a bound state with a variety of different energy levels, each of which corresponds to a different particle observed in the laboratory. As an example, the ψ particle is a low-energy bound state of a so-called charm quark and its antiquark, with a rest energy of 3097 MeV; the $\psi(2S)$ particle is an excited state of this same quark–antiquark combination, with a rest energy of 3686 MeV. A simplified representation of the potential energy of interaction between a quark and an antiquark is $U(x) = A|x|$, where A is a positive constant and x represents the distance between the quark and the antiquark. You can use the WKB approximation (see Challenge Problem 40.64) to determine the bound-state energy levels for this potential-energy function. In the WKB approximation, the energy levels are the solutions to the equation

$$\int_a^b \sqrt{2m[E - U(x)]}\,dx = \frac{nh}{2} \quad (n = 1, 2, 3, \ldots)$$

Here E is the energy, $U(x)$ is the potential-energy function, and $x = a$ and $x = b$ are the classical turning points (the points at which E is equal to the potential energy, so the Newtonian kinetic energy would be zero). (a) Determine the classical turning points for the potential $U(x) = A|x|$ and for an energy E. (b) Carry out the above integral and show that the allowed energy levels in the WKB approximation are given by

$$E_n = \frac{1}{2m}\left(\frac{3mAh}{4}\right)^{2/3} n^{2/3} \quad (n = 1, 2, 3, \ldots)$$

(*Hint:* The integrand is even, so the integral from $-x$ to x is equal to twice the integral from 0 to x.) (c) Does the difference in energy between successive levels increase, decrease, or remain the same as n increases? How does this compare to the behavior of the energy levels for the harmonic oscillator? For the particle in a box? Can you suggest a simple rule that relates the difference in energy between successive levels to the shape of the potential-energy function?

MCAT-STYLE PASSAGE PROBLEMS

Quantum Dots. A *quantum dot* is a type of crystal so small that quantum effects are significant. One application of quantum dots is in fluorescence imaging, in which a quantum dot is bound to a molecule or structure of interest. When the quantum dot is illuminated with light, it absorbs photons and then re-emits photons at a different wavelength. This phenomenon is called *fluorescence*. The wavelength that a quantum dot emits when stimulated with light depends on the dot's size, so the synthesis of quantum dots with different photon absorption and emission properties may be possible. We can understand many quantum-dot properties via a model in which a particle of mass M (roughly the mass of the electron) is confined to a two-dimensional rigid square box of sides L. In this model, the quantum-dot energy levels are given by $E_{m,n} = (m^2 + n^2)(\pi^2\hbar^2)/2ML^2$, where m and n are integers 1, 2, 3, ...

40.67 According to this model, which statement is true about the energy-level spacing of dots of different sizes? (a) Smaller dots have equally spaced levels, but larger dots have energy levels that get farther apart as the energy increases. (b) Larger dots have greater spacing between energy levels than do smaller dots. (c) Smaller dots have greater spacing between energy levels than do larger dots. (d) The spacing between energy levels is independent of the dot size.

40.68 When a given dot with side length L makes a transition from its first excited state to its ground state, the dot emits green (550 nm) light. If a dot with side length $1.1L$ is used instead, what wavelength is emitted in the same transition, according to this model? (a) 600 nm; (b) 670 nm; (c) 500 nm; (d) 460 nm.

40.69 Dots that are the same size but made from different materials are compared. In the same transition, a dot of material 1 emits a photon of longer wavelength than the dot of material 2 does. Based on this model, what is a possible explanation? (a) The mass of the confined particle in material 1 is greater. (b) The mass of the confined particle in material 2 is greater. (c) The confined particles make more transitions per second in material 1. (d) The confined particles make more transitions per second in material 2.

40.70 One advantage of the quantum dot is that, compared to many other fluorescent materials, excited states have relatively long lifetimes (10 ns). What does this mean for the spread in the energy of the photons emitted by quantum dots? (a) Quantum dots emit photons of more well-defined energies than do other fluorescent materials. (b) Quantum dots emit photons of less well-defined energies than do other fluorescent materials. (c) The spread in the energy is affected by the size of the dot, not by the lifetime. (d) There is no spread in the energy of the emitted photons, regardless of the lifetime.

ANSWERS

Chapter Opening Question ?

(i) When an electron in one of these particles—called *quantum dots*—makes a transition from an excited level to a lower level, it emits a photon whose energy is equal to the difference in energy between the levels. The smaller the quantum dot, the larger the energy spacing between levels and hence the shorter (bluer) the wavelength of the emitted photons. See Example 40.6 (Section 40.3) for more details.

Key Example VARIATION Problems

VP40.4.1 (a) 0.549 nm (b) $E_2 = 8.00 \times 10^{-19}$ J, $E_3 = 1.80 \times 10^{-18}$ J

VP40.4.2 (a) 3.94×10^{-12} J (b) 6.57×10^{-12} J

VP40.4.3 (a) 3.03×10^{-19} J (b) 0.772 nm (c) 393 nm

VP40.4.4 (a) $\dfrac{d^2\psi(x)}{dx^2} = -\dfrac{2mE}{\hbar^2}A\cos\left(\dfrac{x\sqrt{2mE}}{\hbar} + \phi\right) = -\dfrac{2mE}{\hbar^2}\psi(x)$, so $-\dfrac{\hbar^2}{2m}\dfrac{d^2\psi(x)}{dx^2} = E\psi(x)$ (b) $\phi = \pm\pi/2$ (c) $\dfrac{L\sqrt{2mE}}{\hbar} = n\pi$, so $E = \dfrac{n^2\pi^2\hbar^2}{2mL^2}$, where $n = 1, 2, 3, \ldots$

VP40.6.1 (a) 4.92×10^{-19} J (b) 3.07×10^{-19} J (c) 2.64×10^{-18} J

VP40.6.2 (a) 9.40×10^{-20} J (b) 5.64×10^{-19} J (c) 0.800 nm

VP40.6.3 (a) initial $n = 3$, final $n = 1$ (b) 596 nm for initial $n = 3$, final $n = 2$; 878 nm for initial $n = 2$, final $n = 1$

VP40.6.4 (a) 2.35 eV (b) 0.010 eV

VP40.7.1 (a) 2.90×10^{-5} (b) 1.47×10^{-3}

VP40.7.2 (a) 5.33×10^{-3} (b) 0.704 nm

VP40.7.3 (a) 2.54×10^{-4} (b) 8.59×10^{-4} (c) 3.46×10^{-3}

VP40.7.4 0.509 nm

Bridging Problem

(a) $$|\Psi(x,t)|^2 = \frac{1}{L}\left[\sin^2\frac{\pi x}{L} + \sin^2\frac{2\pi x}{L} + 2\sin\frac{\pi x}{L}\sin\frac{2\pi x}{L}\cos\left(\frac{(E_2 - E_1)t}{\hbar}\right)\right]$$

(b) no

(d) $\dfrac{3\pi^2\hbar}{2mL^2}$

(e) $\dfrac{0.903\pi^2\hbar}{mL^2}$

? Lithium (with three electrons per atom) is a metal that burns spontaneously in water, while helium (with two electrons per atom) is a gas that undergoes almost no chemical reactions. The additional electron makes lithium behave very differently from helium primarily because (i) the third electron is strongly repelled by electric forces from the other two electrons; (ii) the third electron and larger nucleus make the lithium atom more massive than the helium atom; (iii) there is a limit on the number of electrons that can occupy a given quantum-mechanical state; (iv) the lithium nucleus has more positive charge than a helium nucleus has.

41 Quantum Mechanics II: Atomic Structure

LEARNING OUTCOMES

In this chapter, you'll learn...

41.1 How to extend quantum-mechanical calculations to three-dimensional problems.
41.2 How to solve the Schrödinger equation for a particle trapped in a cubical box.
41.3 How to describe the states of a hydrogen atom in terms of quantum numbers.
41.4 How magnetic fields affect the orbital motion of atomic electrons.
41.5 How we know that electrons have their own intrinsic angular momentum.
41.6 How to analyze the structure of many-electron atoms.
41.7 How x rays emitted by atoms reveal their inner structure.
41.8 What happens when the quantum-mechanical states of two particles become entangled.

You'll need to review...

22.3 Gauss's law.
27.7 Magnetic dipole moment.
32.5 Standing electromagnetic waves.
38.2 X-ray production.
39.2, 39.3 Atoms and the Bohr model.
40.1, 40.2, 40.5, 40.6 One-dimensional Schrödinger equation; particle in a box; harmonic-oscillator wave functions; measuring a quantum-mechanical system.

Some physicists claim that all of chemistry is contained in the Schrödinger equation. This is somewhat of an exaggeration, but this equation can teach us a great deal about the chemical behavior of elements, the periodic table, and the nature of chemical bonds.

In order to learn about the quantum-mechanical structure of atoms, we'll first construct a three-dimensional version of the Schrödinger equation. We'll try this equation out by looking at a three-dimensional version of a particle in a box: a particle confined to a cubical volume.

We'll then see that we can learn a great deal about the structure and properties of *all* atoms from the solutions to the Schrödinger equation for the hydrogen atom. These solutions have quantized values of orbital angular momentum; we don't need to impose quantization as we did with the Bohr model. We label the states with a set of quantum numbers, which we'll use later with many-electron atoms as well. We'll find that the electron also has an intrinsic *spin* angular momentum with its own set of quantized values.

We'll also encounter the exclusion principle, a kind of microscopic zoning ordinance that is the key to understanding many-electron atoms. This principle says that no two electrons in an atom can have the same quantum-mechanical state. We'll then use the principles of this chapter to explain the characteristic x-ray spectra of atoms. Finally, we'll end our discussion of quantum mechanics with a look at the curious concept of quantum entanglement and its application to the new science of quantum computing.

41.1 THE SCHRÖDINGER EQUATION IN THREE DIMENSIONS

We have discussed the Schrödinger equation and its applications only for *one-dimensional* problems, the analog of a Newtonian particle moving along a straight line. The straight-line model is adequate for some applications, but to understand atomic structure, we need a three-dimensional generalization.

It's not difficult to guess what the three-dimensional Schrödinger equation should look like. First, the wave function Ψ is a function of time and all three space coordinates (x, y, z). In general, the potential-energy function also depends on all three coordinates and can be written as $U(x, y, z)$. Next, recall from Section 40.1 that the term $-(\hbar^2/2m)\partial^2\Psi/\partial x^2$ in the one-dimensional Schrödinger equation, Eq. (40.20), is related to the kinetic energy of the particle in the state described by the wave function Ψ. For example, if we insert into this term the wave function $\Psi(x, t) = Ae^{ikx}e^{-i\omega t}$ for a free particle with magnitude of momentum $p = \hbar k$ and kinetic energy $K = p^2/2m$, we obtain $-(\hbar^2/2m)(ik)^2Ae^{ikx}e^{-i\omega t} = (\hbar^2k^2/2m)Ae^{ikx}e^{-i\omega t} = (p^2/2m)\Psi(x, t) = K\Psi(x, t)$. If the particle can move in three dimensions, its momentum has three components (p_x, p_y, p_z) and its kinetic energy is

$$K = \frac{{p_x}^2}{2m} + \frac{{p_y}^2}{2m} + \frac{{p_z}^2}{2m} \tag{41.1}$$

These observations, taken together, suggest that the correct generalization of the Schrödinger equation to three dimensions is

$$-\frac{\hbar^2}{2m}\left(\frac{\partial^2\Psi(x, y, z, t)}{\partial x^2} + \frac{\partial^2\Psi(x, y, z, t)}{\partial y^2} + \frac{\partial^2\Psi(x, y, z, t)}{\partial z^2}\right) + U(x, y, z)\Psi(x, y, z, t) = i\hbar\frac{\partial\Psi(x, y, z, t)}{\partial t} \tag{41.2}$$

(general three-dimensional Schrödinger equation)

The three-dimensional wave function $\Psi(x, y, z, t)$ has a similar interpretation as in one dimension. The wave function itself is a complex quantity with both a real part and an imaginary part, but $|\Psi(x, y, z, t)|^2$—the square of its absolute value, equal to the product of $\Psi(x, y, z, t)$ and its complex conjugate $\Psi^*(x, y, z, t)$—is real and either positive or zero at every point in space. We interpret $|\Psi(x, y, z, t)|^2\,dV$ as the *probability* of finding the particle within a small volume dV centered on the point (x, y, z) at time t, so $|\Psi(x, y, z, t)|^2$ is the *probability distribution function* in three dimensions. The *normalization condition* on the wave function is that the probability that the particle is *somewhere* in space is exactly 1. Hence the integral of $|\Psi(x, y, z, t)|^2$ over all space must equal 1:

$$\int |\Psi(x, y, z, t)|^2\,dV = 1 \qquad \text{(normalization condition in three dimensions)} \tag{41.3}$$

If the wave function $\Psi(x, y, z, t)$ represents a state of a definite energy E—that is, a stationary state—we can write it as the product of a spatial wave function $\psi(x, y, z)$ and a function of time $e^{-iEt/\hbar}$:

$$\Psi(x, y, z, t) = \psi(x, y, z)e^{-iEt/\hbar} \qquad \text{(time-dependent wave function for a state of definite energy)} \tag{41.4}$$

(Compare this to Eq. (40.21) for a one-dimensional state of definite energy.) If we substitute Eq. (41.4) into Eq. (41.2), the right-hand side of the equation becomes $i\hbar\psi(x, y, z)(-iE/\hbar)e^{-iEt/\hbar} = E\psi(x, y, z)e^{-iEt/\hbar}$. We can then divide both sides by the factor $e^{-iEt/\hbar}$, leaving the *time-independent* Schrödinger equation in three dimensions for a stationary state:

Time-independent three-dimensional Schrödinger equation:

Planck's constant divided by 2π

Time-independent wave function

$$-\frac{\hbar^2}{2m}\left(\frac{\partial^2\psi(x, y, z)}{\partial x^2} + \frac{\partial^2\psi(x, y, z)}{\partial y^2} + \frac{\partial^2\psi(x, y, z)}{\partial z^2}\right) + U(x, y, z)\psi(x, y, z) = E\psi(x, y, z) \tag{41.5}$$

Particle's mass

Potential-energy function

Energy of state

The probability distribution function for a stationary state is just the square of the absolute value of the spatial wave function: $|\psi(x, y, z)e^{-iEt/\hbar}|^2 = \psi^*(x, y, z)e^{+iEt/\hbar}\psi(x, y, z)e^{-iEt/\hbar} = |\psi(x, y, z)|^2$. Note that this doesn't depend on time. (As we discussed in Section 40.1, that's why we call these states *stationary.*) Hence for a stationary state the wave function normalization condition, Eq. (41.3), becomes

$$\int |\psi(x, y, z)|^2 \, dV = 1 \qquad \text{(normalization condition for a stationary state in three dimensions)} \qquad (41.6)$$

We won't pretend that we have *derived* Eqs. (41.2) and (41.5). Like their one-dimensional versions, these equations have to be tested by comparison of their predictions with experimental results. Happily, Eqs. (41.2) and (41.5) both pass this test with flying colors, so we are confident that they *are* the correct equations.

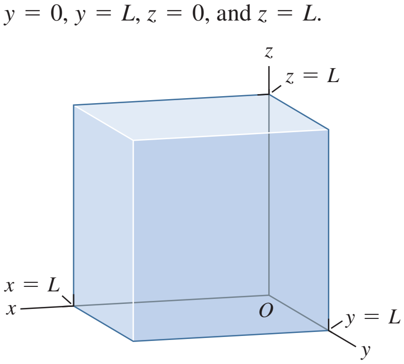
An important topic that we'll address in this chapter is the solutions for Eq. (41.5) for the stationary states of the hydrogen atom. The potential-energy function for an electron in a hydrogen atom is *spherically symmetric;* it depends only on the distance $r = (x^2 + y^2 + z^2)^{1/2}$ from the origin of coordinates. To take advantage of this symmetry, it's best to use *spherical coordinates* rather than the Cartesian coordinates (x, y, z) to solve the Schrödinger equation for the hydrogen atom. Before introducing these new coordinates and investigating the hydrogen atom, it's useful to look at the three-dimensional version of the particle in a box that we considered in Section 40.2. Solving this simpler problem will give us insight into the more complicated stationary states found in atomic physics.

TEST YOUR UNDERSTANDING OF SECTION 41.1 In a certain region of space the potential-energy function for a quantum-mechanical particle is zero. In this region the wave function $\psi(x, y, z)$ for a certain stationary state is real and satisfies $\partial^2\psi/\partial x^2 > 0$, $\partial^2\psi/\partial y^2 > 0$, and $\partial^2\psi/\partial z^2 > 0$. The particle has a definite energy E that is positive. What can you conclude about $\psi(x, y, z)$ in this region? (i) It must be positive; (ii) it must be negative; (iii) it must be zero; (iv) not enough information given to decide.

ANSWER **(ii)** If $U(x, y, z) = 0$ in a certain region of space, we can rewrite the time-independent Schrödinger equation [Eq. (41.5)] for that region as $\partial^2\psi/\partial x^2 + \partial^2\psi/\partial y^2 + \partial^2\psi/\partial z^2 = (-2mE/\hbar^2)\psi$. We are told that all of the second derivatives of $\psi(x, y, z)$ are positive in this region, so the left-hand side of this equation is positive. Hence the right-hand side $(-2mE/\hbar^2)\psi$ must also be positive. Since $E > 0$, the quantity $-2mE/\hbar^2$ is negative, and so $\psi(x, y, z)$ must be negative.

41.2 PARTICLE IN A THREE-DIMENSIONAL BOX

Figure 41.1 A particle is confined in a cubical box with walls at $x = 0$, $x = L$, $y = 0$, $y = L$, $z = 0$, and $z = L$.



Consider a particle enclosed within a cubical box of side L. This could represent an electron that's free to move anywhere within the interior of a solid metal cube but cannot escape the cube. We'll choose the origin to be at one corner of the box, with the x-, y-, and z-axes along edges of the box. Then the particle is confined to the region $0 \le x \le L$, $0 \le y \le L$, $0 \le z \le L$ (**Fig. 41.1**). What are the stationary states of this system?

As for the model of a particle in a one-dimensional box that we considered in Section 40.2, we'll say that the potential energy is zero inside the box but infinite outside. Hence the spatial wave function $\psi(x, y, z)$ must be zero outside the box in order that the term $U(x, y, z)\psi(x, y, z)$ in the time-independent Schrödinger equation, Eq. (41.5), not be infinite. Consequently the probability distribution function $|\psi(x, y, z)|^2$ is zero outside the box, and the probability that the particle will be found there is zero. Inside the box, the spatial wave function for a stationary state obeys the time-independent Schrödinger equation, Eq. (41.5), with $U(x, y, z) = 0$:

$$-\frac{\hbar^2}{2m}\left(\frac{\partial^2\psi(x, y, z)}{\partial x^2} + \frac{\partial^2\psi(x, y, z)}{\partial y^2} + \frac{\partial^2\psi(x, y, z)}{\partial z^2}\right) = E\psi(x, y, z) \qquad (41.7)$$

(particle in a three-dimensional box)

In order for the wave function to be continuous from the inside to the outside of the box, $\psi(x, y, z)$ must equal zero on the walls. Hence our boundary conditions are that $\psi(x, y, z) = 0$ at $x = 0$, $x = L$, $y = 0$, $y = L$, $z = 0$, and $z = L$.

Guessing a solution to a complicated partial differential equation like Eq. (41.7) seems like quite a challenge. To make progress, recall that we wrote the time-*dependent* wave function for a stationary state as the product of one function that depends on only the spatial coordinates x, y, and z and a second function that depends on only the time t: $\Psi(x, y, z, t) = \psi(x, y, z)e^{-iEt/\hbar}$. In the same way, let's try a technique called *separation of variables:* We'll write the spatial wave function $\psi(x, y, z)$ as a product of one function X that depends on only x, a second function Y that depends on only y, and a third function Z that depends on only z:

$$\psi(x, y, z) = X(x)Y(y)Z(z) \tag{41.8}$$

If we substitute Eq. (41.8) into Eq. (41.7), we get

$$\begin{aligned} &-\frac{\hbar^2}{2m}\left(Y(y)Z(z)\frac{d^2X(x)}{dx^2} + X(x)Z(z)\frac{d^2Y(y)}{dy^2} + X(x)Y(y)\frac{d^2Z(z)}{dz^2}\right) \\ &= EX(x)Y(y)Z(z) \end{aligned} \tag{41.9}$$

The partial derivatives in Eq. (41.7) have become ordinary derivatives since they act on functions of a single variable. Now we divide both sides of Eq. (41.9) by the product $X(x)Y(y)Z(z)$:

$$\left(-\frac{\hbar^2}{2m}\frac{1}{X(x)}\frac{d^2X(x)}{dx^2}\right) + \left(-\frac{\hbar^2}{2m}\frac{1}{Y(y)}\frac{d^2Y(y)}{dy^2}\right) + \left(-\frac{\hbar^2}{2m}\frac{1}{Z(z)}\frac{d^2Z(z)}{dz^2}\right) = E \tag{41.10}$$

The right-hand side of Eq. (41.10) is the energy of the stationary state. Since E is a constant that does not depend on the values of x, y, and z, the left-hand side of the equation must also be independent of the values of x, y, and z. Hence the first term in parentheses on the left-hand side of Eq. (41.10) must equal a constant that doesn't depend on x, the second term in parentheses must equal another constant that doesn't depend on y, and the third term in parentheses must equal a third constant that doesn't depend on z. Let's call these constants E_X, E_Y, and E_Z, respectively. We then have a separate equation for each of the three functions $X(x)$, $Y(y)$, and $Z(z)$:

$$-\frac{\hbar^2}{2m}\frac{d^2X(x)}{dx^2} = E_X X(x) \tag{41.11a}$$

$$-\frac{\hbar^2}{2m}\frac{d^2Y(y)}{dy^2} = E_Y Y(y) \tag{41.11b}$$

$$-\frac{\hbar^2}{2m}\frac{d^2Z(z)}{dz^2} = E_Z Z(z) \tag{41.11c}$$

To satisfy the boundary conditions that $\psi(x, y, z) = X(x)Y(y)Z(z)$ be equal to zero on the walls of the box, we demand that $X(x) = 0$ at $x = 0$ and $x = L$, $Y(y) = 0$ at $y = 0$ and $y = L$, and $Z(z) = 0$ at $z = 0$ and $z = L$.

How can we interpret the three constants E_X, E_Y, and E_Z in Eqs. (41.11)? From Eq. (41.10), they are related to the energy E by

$$E_X + E_Y + E_Z = E \tag{41.12}$$

Equation (41.12) should remind you of Eq. (41.1) in Section 41.1, which states that the kinetic energy of a particle is the sum of contributions coming from its x-, y-, and z-components of momentum. Hence the constants E_X, E_Y, and E_Z tell us how much of the particle's energy is due to motion along each of the three coordinate axes. (Inside the box the potential energy is zero, so the particle's energy is purely kinetic.)

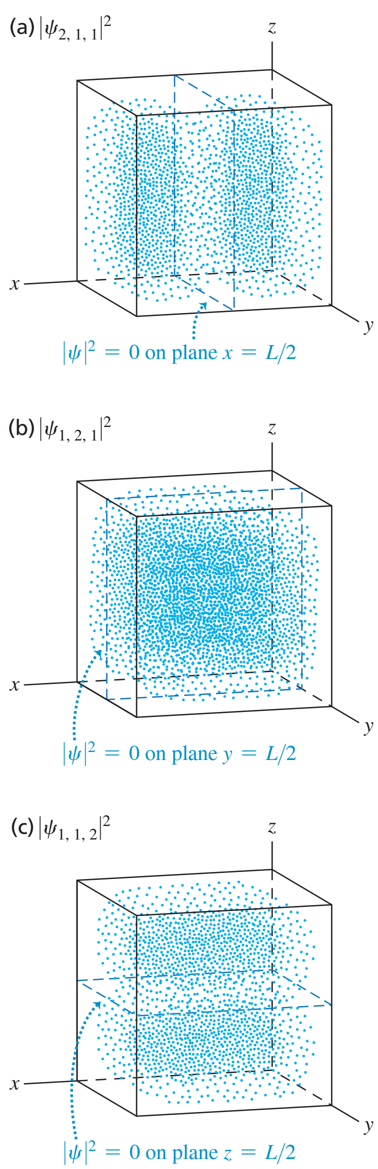
Equations (41.11) represent an enormous simplification; we've reduced the problem of solving a fairly complex *partial* differential equation with three independent variables to the much simpler problem of solving three separate *ordinary* differential equations with one independent variable each. What's more, each of these ordinary differential equations is the same as the time-independent Schrödinger equation for a particle in a *one-dimensional* box, Eq. (40.25), and with exactly the same boundary conditions at 0 and L. (The only differences are that some of the quantities are labeled by different symbols.) By comparing with our work in Section 40.2, you can see that the solutions to Eqs. (41.11) are

$$X_{n_X}(x) = C_X \sin\frac{n_X \pi x}{L} \quad (n_X = 1, 2, 3, \ldots) \tag{41.13a}$$

$$Y_{n_Y}(y) = C_Y \sin\frac{n_Y \pi y}{L} \quad (n_Y = 1, 2, 3, \ldots) \tag{41.13b}$$

$$Z_{n_Z}(z) = C_Z \sin\frac{n_Z \pi z}{L} \quad (n_Z = 1, 2, 3, \ldots) \tag{41.13c}$$

Figure **41.2** Probability distribution function $|\psi_{n_X,n_Y,n_Z}(x, y, z)|^2$ for (n_X, n_Y, n_Z) equal to (a) (2, 1, 1), (b) (1, 2, 1), and (c) (1, 1, 2). The value of $|\psi|^2$ is proportional to the density of dots. The wave function is zero on the walls of the box and on a midplane of the box, so $|\psi|^2 = 0$ at these locations.



where C_X, C_Y, and C_Z are constants. The corresponding values of E_X, E_Y, and E_Z are

$$E_X = \frac{n_X{}^2 \pi^2 \hbar^2}{2mL^2} \quad (n_X = 1, 2, 3, \ldots) \tag{41.14a}$$

$$E_Y = \frac{n_Y{}^2 \pi^2 \hbar^2}{2mL^2} \quad (n_Y = 1, 2, 3, \ldots) \tag{41.14b}$$

$$E_Z = \frac{n_Z{}^2 \pi^2 \hbar^2}{2mL^2} \quad (n_Z = 1, 2, 3, \ldots) \tag{41.14c}$$

There is only one quantum number n for the one-dimensional particle in a box, but *three* quantum numbers n_X, n_Y, and n_Z for the three-dimensional box. If we substitute Eqs. (41.13) back into Eq. (41.8) for the total spatial wave function, $\psi(x, y, z) = X(x)Y(y)Z(z)$, we get the following stationary-state wave functions for a particle in a three-dimensional cubical box:

$$\psi_{n_X,n_Y,n_Z}(x, y, z) = C \sin\frac{n_X \pi x}{L} \sin\frac{n_Y \pi y}{L} \sin\frac{n_Z \pi z}{L} \tag{41.15}$$
$$(n_X = 1, 2, 3, \ldots;\ n_Y = 1, 2, 3, \ldots;\ n_Z = 1, 2, 3, \ldots)$$

where $C = C_X C_Y C_Z$. The value of the constant C is determined by the normalization condition, Eq. (41.6).

In Section 40.2 we saw that the stationary-state wave functions for a particle in a one-dimensional box were analogous to standing waves on a string. In a similar way, the *three*-dimensional wave functions given by Eq. (41.15) are analogous to standing electromagnetic waves in a cubical cavity like the interior of a microwave oven (see Section 32.5). In a microwave oven there are "dead spots" where the wave intensity is zero, corresponding to the nodes of the standing wave. (The moving platform in a microwave oven ensures even cooking by making sure that no part of the food sits at any "dead spot.") In a similar fashion, the probability distribution function corresponding to Eq. (41.15) can have "dead spots" where there is zero probability of finding the particle. As an example, consider the case $(n_X, n_Y, n_Z) = (2, 1, 1)$. From Eq. (41.15), the probability distribution function for this case is

$$|\psi_{2,1,1}(x, y, z)|^2 = |C|^2 \sin^2\frac{2\pi x}{L} \sin^2\frac{\pi y}{L} \sin^2\frac{\pi z}{L}$$

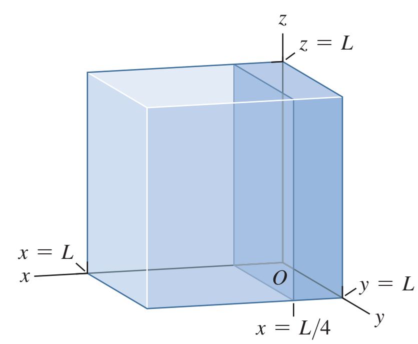
As **Fig. 41.2a** shows, this probability distribution function is zero on the plane $x = L/2$, where $\sin^2(2\pi x/L) = \sin^2 \pi = 0$. The particle is most likely to be found near where all three of the sine-squared functions are greatest, at $(x, y, z) = (L/4, L/2, L/2)$ or $(x, y, z) = (3L/4, L/2, L/2)$. Figures 41.2b and 41.2c show the similar cases $(n_X, n_Y, n_Z) = (1, 2, 1)$ and $(n_X, n_Y, n_Z) = (1, 1, 2)$. For higher values of the quantum numbers n_X, n_Y, and n_Z there are additional planes on which the probability distribution function equals zero, just as the probability distribution function $|\psi(x)|^2$ for a one-dimensional box has more zeros for higher values of n (see Fig. 40.12).

EXAMPLE 41.1 Probability in a three-dimensional box

WITH VARIATION PROBLEMS

(a) Find the value of the constant C that normalizes the wave function of Eq. (41.15). (b) Find the probability that the particle will be found somewhere in the region $0 \le x \le L/4$ (**Fig. 41.3**) for the cases (i) $(n_X, n_Y, n_Z) = (1, 2, 1)$, (ii) $(n_X, n_Y, n_Z) = (2, 1, 1)$, and (iii) $(n_X, n_Y, n_Z) = (3, 1, 1)$.

Figure **41.3** What is the probability that the particle is in the dark-colored quarter of the box?



IDENTIFY and SET UP Equation (41.6) tells us that to normalize the wave function, we have to choose the value of C so that the integral of the probability distribution function $|\psi_{n_X, n_Y, n_Z}(x, y, z)|^2$ over the volume within the box equals 1. (The integral is actually over *all* space, but the particle-in-a-box wave functions are zero outside the box.)

The probability of finding the particle within a certain volume within the box equals the integral of the probability distribution function over that volume. Hence in part (b) we'll integrate $|\psi_{n_X, n_Y, n_Z}(x, y, z)|^2$ for the given values of (n_X, n_Y, n_Z) over the volume $0 \le x \le L/4$, $0 \le y \le L$, $0 \le z \le L$.

EXECUTE (a) From Eq. (41.15),

$$|\psi_{n_X,n_Y,n_Z}(x, y, z)|^2 = |C|^2 \sin^2 \frac{n_X \pi x}{L} \sin^2 \frac{n_Y \pi y}{L} \sin^2 \frac{n_Z \pi z}{L}$$

Hence the normalization condition is

$$\begin{aligned}
&\int |\psi_{n_X,n_Y,n_Z}(x, y, z)|^2\, dV \\
&= |C|^2 \int_{x=0}^{x=L} \int_{y=0}^{y=L} \int_{z=0}^{z=L} \sin^2 \frac{n_X \pi x}{L} \sin^2 \frac{n_Y \pi y}{L} \sin^2 \frac{n_Z \pi z}{L}\, dx\, dy\, dz \\
&= |C|^2 \left(\int_{x=0}^{x=L} \sin^2 \frac{n_X \pi x}{L}\, dx \right) \left(\int_{y=0}^{y=L} \sin^2 \frac{n_Y \pi y}{L}\, dy \right) \\
&\quad \times \left(\int_{z=0}^{z=L} \sin^2 \frac{n_Z \pi z}{L}\, dz \right) \\
&= 1
\end{aligned}$$

We can use the identity $\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta)$ and the variable substitution $\theta = n_X \pi x / L$ to show that

$$\begin{aligned}
\int \sin^2 \frac{n_X \pi x}{L}\, dx &= \frac{L}{2 n_X \pi} \left[\frac{n_X \pi x}{L} - \frac{1}{2} \sin\left(\frac{2 n_X \pi x}{L} \right) \right] \\
&= \frac{x}{2} - \frac{L}{4 n_X \pi} \sin\left(\frac{2 n_X \pi x}{L} \right)
\end{aligned}$$

If we evaluate this integral between $x = 0$ and $x = L$, the result is $L/2$ (recall that $\sin 0 = 0$ and $\sin 2n_X\pi = 0$ for any integer n_X). The y- and z-integrals each yield the same result, so the normalization condition is

$$|C|^2 \left(\frac{L}{2} \right) \left(\frac{L}{2} \right) \left(\frac{L}{2} \right) = |C|^2 \left(\frac{L}{2} \right)^3 = 1$$

or $|C|^2 = (2/L)^3$. If we choose C to be real and positive, then $C = (2/L)^{3/2}$.

(b) We have the same y- and z-integrals as in part (a), but now the limits of integration on the x-integral are $x = 0$ and $x = L/4$:

$$\begin{aligned}
P &= \int_{0 \le x \le L/4} |\psi_{n_X,n_Y,n_Z}|^2\, dV \\
&= |C|^2 \left(\int_{x=0}^{x=L/4} \sin^2 \frac{n_X \pi x}{L}\, dx \right) \left(\int_{y=0}^{y=L} \sin^2 \frac{n_Y \pi y}{L}\, dy \right) \\
&\quad \times \left(\int_{z=0}^{z=L} \sin^2 \frac{n_Z \pi z}{L}\, dz \right)
\end{aligned}$$

The x-integral is

$$\begin{aligned}
\int_{x=0}^{x=L/4} \sin^2 \frac{n_X \pi x}{L}\, dx &= \left[\frac{x}{2} - \frac{L}{4 n_X \pi} \sin\left(\frac{2 n_X \pi x}{L} \right) \right] \Bigg|_{x=0}^{x=L/4} \\
&= \frac{L}{8} - \frac{L}{4 n_X \pi} \sin\left(\frac{n_X \pi}{2} \right)
\end{aligned}$$

Hence the probability of finding the particle somewhere in the region $0 \le x \le L/4$ is

$$\begin{aligned}
P &= \left(\frac{2}{L} \right)^3 \left[\frac{L}{8} - \frac{L}{4 n_X \pi} \sin\left(\frac{n_X \pi}{2} \right) \right] \left(\frac{L}{2} \right) \left(\frac{L}{2} \right) \\
&= \frac{1}{4} - \frac{1}{2 n_X \pi} \sin\left(\frac{n_X \pi}{2} \right)
\end{aligned}$$

This depends only on the value of n_X, not on n_Y or n_Z. For each of the three cases we have

Continued

(i) $n_X = 1$: $P = \frac{1}{4} - \dfrac{1}{2(1)\pi}\sin\left(\dfrac{\pi}{2}\right) = \frac{1}{4} - \dfrac{1}{2\pi}(1)$

$= \frac{1}{4} - \dfrac{1}{2\pi} = 0.091$

(ii) $n_X = 2$: $P = \frac{1}{4} - \dfrac{1}{2(2)\pi}\sin\left(\dfrac{2\pi}{2}\right) = \frac{1}{4} - \dfrac{1}{4\pi}\sin\pi$

$= \frac{1}{4} - 0 = 0.250$

(iii) $n_X = 3$: $P = \frac{1}{4} - \dfrac{1}{2(3)\pi}\sin\left(\dfrac{3\pi}{2}\right) = \frac{1}{4} - \dfrac{1}{6\pi}(-1)$

$= \frac{1}{4} + \dfrac{1}{6\pi} = 0.303$

EVALUATE You can see why the probabilities in part (b) are different by looking at part (b) of Fig. 40.12, which shows $\sin^2 n_X\pi x/L$ for $n_X = 1$, 2, and 3. For $n_X = 2$ the area under the curve between $x = 0$ and $x = L/4$ (equal to the integral between these two points) is exactly $\frac{1}{4}$ of the total area between $x = 0$ and $x = L$. For $n_X = 1$ the area between $x = 0$ and $x = L/4$ is less than $\frac{1}{4}$ of the total area, and for $n_X = 3$ it is greater than $\frac{1}{4}$ of the total area.

KEYCONCEPT If a particle is in a three-dimensional stationary state described by wave function $\psi(x, y, z)$, the probability of finding that particle in an infinitesimal volume dV centered on (x, y, z) is $|\psi(x, y, z)|^2 dV$. The probability of finding the particle somewhere in a finite volume equals the integral $\int |\psi(x, y, z)|^2 dV$ evaluated over that volume.

Energy Levels, Degeneracy, and Symmetry

From Eqs. (41.12) and (41.14), the allowed energies for a particle of mass m in a cubical box of side L are

Quantum numbers n_X, n_Y, n_Z can each equal 1, 2, 3, ...

Planck's constant divided by 2π

Energy levels, particle in a three-dimensional cubical box

$$E_{n_X, n_Y, n_Z} = \frac{(n_X^2 + n_Y^2 + n_Z^2)\pi^2\hbar^2}{2mL^2} \tag{41.16}$$

Particle's mass

Length of each side of box

Figure 41.4 shows the six lowest energy levels given by Eq. (41.16). Note that most energy levels correspond to more than one set of quantum numbers (n_X, n_Y, n_Z) and hence to more than one quantum state. Having two or more distinct quantum states with the same energy is called **degeneracy,** and states with the same energy are said to be **degenerate.** For example, Fig. 41.4 shows that the states $(n_X, n_Y, n_Z) = (2, 1, 1)$, $(1, 2, 1)$, and $(1, 1, 2)$ are degenerate. By comparison, for a particle in a one-dimensional box there is just one state for each energy level (see Fig. 40.11a) and no degeneracy.

The reason the cubical box exhibits degeneracy is that it is *symmetric:* All sides of the box have the same dimensions. As an illustration, Fig. 41.2 shows the probability distribution functions for the three states $(n_X, n_Y, n_Z) = (2, 1, 1)$, $(1, 2, 1)$, and $(1, 1, 2)$. You can

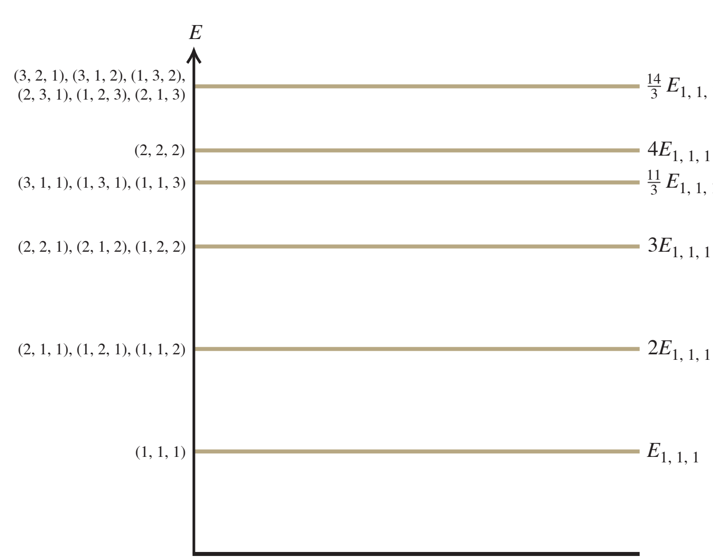


Figure **41.4** Energy-level diagram for a particle in a three-dimensional cubical box. We label each level with the quantum numbers of the states (n_X, n_Y, n_Z) with that energy. Several of the levels are degenerate (more than one state has the same energy). The lowest (ground) level, $(n_X, n_Y, n_Z) = (1, 1, 1)$, has energy $E_{1,1,1} = (1^2 + 1^2 + 1^2)\pi^2\hbar^2/2mL^2 = 3\pi^2\hbar^2/2mL^2$; we show the energies of the other levels as multiples of $E_{1,1,1}$.

transform any one of these three states into a different one by simply rotating the cubical box by 90°. This rotation doesn't change the energy, so the three states are degenerate.

Since degeneracy is a consequence of symmetry, we can remove the degeneracy by making the box asymmetric. We do this by giving the three sides of the box different lengths L_X, L_Y, and L_Z. If we repeat the steps that we followed to solve the time-independent Schrödinger equation, we find that the energy levels are given by

$$E_{n_X,n_Y,n_Z} = \left(\frac{n_X^2}{L_X^2} + \frac{n_Y^2}{L_Y^2} + \frac{n_Z^2}{L_Z^2}\right)\frac{\pi^2\hbar^2}{2m} \qquad \begin{pmatrix} n_X = 1, 2, 3, \ldots; \\ n_Y = 1, 2, 3, \ldots; \\ n_Z = 1, 2, 3, \ldots \end{pmatrix} \tag{41.17}$$

(energy levels, particle in a three-dimensional box with sides of length L_X, L_Y, and L_Z)

If L_X, L_Y, and L_Z are all different, the states $(n_X, n_Y, n_Z) = (2, 1, 1)$, $(1, 2, 1)$, and $(1, 1, 2)$ have different energies and hence are no longer degenerate. Note that Eq. (41.17) reduces to Eq. (41.16) if $L_X = L_Y = L_Z = L$.

Returning to a particle in a three-dimensional cubical box, let's summarize the differences from the one-dimensional case that we examined in Section 40.2:

- We can write the wave function for a three-dimensional stationary state as a product of three functions, one for each spatial coordinate. Only a single function of the coordinate x is needed in one dimension.
- In the three-dimensional case, three quantum numbers are needed to describe each stationary state. Only one quantum number is needed in the one-dimensional case.
- Most of the energy levels for the three-dimensional case are degenerate: More than one stationary state has this energy. There is no degeneracy in the one-dimensional case.
- For a stationary state of the three-dimensional case, there are surfaces on which the probability distribution function $|\psi|^2$ is zero. In the one-dimensional case, there are positions on the x-axis where $|\psi|^2$ is zero.

We'll see these same features in the following section for a three-dimensional situation that's more realistic than a particle in a cubical box: a hydrogen atom in which a negatively charged electron orbits a positively charged nucleus.

TEST YOUR UNDERSTANDING OF SECTION 41.2 Rank the following states of a particle in a cubical box of side L in order from highest to lowest energy: (i) $(n_X, n_Y, n_Z) = (2, 3, 2)$; (ii) $(n_X, n_Y, n_Z) = (4, 1, 1)$; (iii) $(n_X, n_Y, n_Z) = (2, 2, 3)$; (iv) $(n_X, n_Y, n_Z) = (1, 3, 3)$.

ANSWER

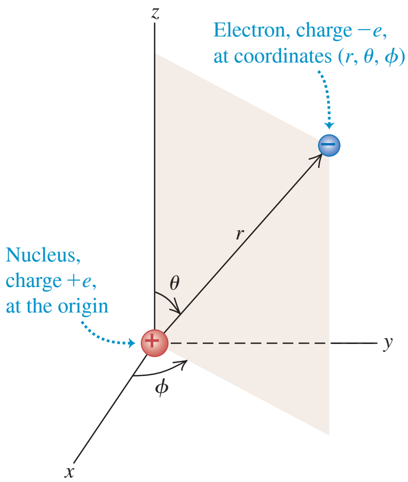
(iv), (ii), (i) and (iii) (tie) Equation (41.16) shows that the energy levels for a cubical box are proportional to the quantity $n_X^2 + n_Y^2 + n_Z^2$. Hence ranking in order of this quantity is the same as ranking in order of energy. For the four cases we are given, we have (i) $n_X^2 + n_Y^2 + n_Z^2 = 2^2 + 3^2 + 2^2 = 17$; (ii) $n_X^2 + n_Y^2 + n_Z^2 = 4^2 + 1^2 + 1^2 = 18$; (iii) $n_X^2 + n_Y^2 + n_Z^2 = 2^2 + 2^2 + 3^2 = 17$; and (iv) $n_X^2 + n_Y^2 + n_Z^2 = 1^2 + 3^2 + 3^2 = 19$. The states $(n_X, n_Y, n_Z) = (2, 3, 2)$ and $(n_X, n_Y, n_Z) = (2, 2, 3)$ have the same energy (they are degenerate).

41.3 THE HYDROGEN ATOM

Let's continue the discussion of the hydrogen atom that we began in Chapter 39. In the Bohr model, electrons move in circular orbits like Newtonian particles, but with quantized values of angular momentum. While this model gave the correct energy levels of the hydrogen atom, as deduced from spectra, it had many conceptual difficulties. It mixed classical physics with new and seemingly contradictory concepts. It provided no insight into the process by which photons are emitted and absorbed. It could not be generalized to atoms with more than one electron. It predicted the wrong magnetic properties for the hydrogen atom. And perhaps most important, its picture of the electron as a localized point particle was inconsistent with the more general view we developed in Chapters 39 and 40. To go beyond the Bohr model, let's apply the Schrödinger equation to find the wave functions for stationary states (states of definite energy) of the hydrogen atom. As in Section 39.3, we include the motion of the nucleus by simply replacing the electron mass m with the reduced mass m_r.

The Schrödinger Equation for the Hydrogen Atom

Figure 41.5 The Schrödinger equation for the hydrogen atom can be solved most readily by using spherical coordinates.



We discussed the three-dimensional version of the Schrödinger equation in Section 41.1. The potential-energy function is *spherically symmetric:* It depends only on the distance $r = (x^2 + y^2 + z^2)^{1/2}$ from the origin of coordinates:

$$U(r) = -\frac{1}{4\pi\epsilon_0}\frac{e^2}{r} \tag{41.18}$$

The hydrogen-atom problem is best formulated in spherical coordinates (r, θ, ϕ), shown in **Fig. 41.5**; the spherically symmetric potential-energy function depends only on r, not on θ or ϕ. The Schrödinger equation with this potential-energy function can be solved exactly; the solutions are combinations of familiar functions. Without going into a lot of detail, we can describe the most important features of the procedure and the results.

First, we find the solutions by using the same method of separation of variables that we employed for a particle in a cubical box in Section 41.2. We express the wave function $\psi(r, \theta, \phi)$ as a product of three functions, each one a function of only one of the three coordinates:

$$\psi(r, \theta, \phi) = R(r)\Theta(\theta)\Phi(\phi) \tag{41.19}$$

That is, the function $R(r)$ depends on only r, $\Theta(\theta)$ depends on only θ, and $\Phi(\phi)$ depends on only ϕ. Just as for a particle in a three-dimensional box, when we substitute Eq. (41.19) into the Schrödinger equation, we get three separate ordinary differential equations. One equation involves only r and $R(r)$, a second involves only θ and $\Theta(\theta)$, and a third involves only ϕ and $\Phi(\phi)$:

$$-\frac{\hbar^2}{2m_r r^2}\frac{d}{dr}\left(r^2\frac{dR(r)}{dr}\right) + \left(\frac{\hbar^2 l(l+1)}{2m_r r^2} + U(r)\right)R(r) = ER(r) \tag{41.20a}$$

$$\frac{1}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d\Theta(\theta)}{d\theta}\right) + \left(l(l+1) - \frac{m_l{}^2}{\sin^2\theta}\right)\Theta(\theta) = 0 \tag{41.20b}$$

$$\frac{d^2\Phi(\phi)}{d\phi^2} + m_l{}^2\Phi(\phi) = 0 \tag{41.20c}$$

In Eqs. (41.20) E is the energy of the stationary state and l and m_l are constants that we'll discuss later.

CAUTION **Two uses of the symbol *m*** Don't confuse the constant m_l in Eqs. (41.20b) and (41.20c) with the similar symbol m_r for the reduced mass of the electron and nucleus (see Section 39.3). The constant m_l is a dimensionless number; the reduced mass m_r has units of kilograms.

We won't attempt to solve this set of three equations, but we can describe how it's done. As for the particle in a cubical box, the physically acceptable solutions of these three equations are determined by boundary conditions. The radial function $R(r)$ in Eq. (41.20a) must approach zero at large r, because we are describing *bound states* of the electron that are localized near the nucleus. This is analogous to the requirement that the harmonic-oscillator wave functions (see Section 40.5) must approach zero at large x. The angular functions $\Theta(\theta)$ and $\Phi(\phi)$ in Eqs. (41.20b) and (41.20c) must be *finite* for all relevant values of the angles. For example, there are solutions of the Θ equation that become infinite at $\theta = 0$ and $\theta = \pi$; these are unacceptable, since $\psi(r, \theta, \phi)$ must be normalizable. Furthermore, the angular function $\Phi(\phi)$ in Eq. (41.20c) must be *periodic*. For example, (r, θ, ϕ) and $(r, \theta, \phi + 2\pi)$ describe the same point, so $\Phi(\phi + 2\pi)$ must equal $\Phi(\phi)$.

The allowed radial functions $R(r)$ turn out to be an exponential function $e^{-\alpha r}$ (where α is positive) multiplied by a polynomial in r. The functions $\Theta(\theta)$ are polynomials containing various powers of $\sin\theta$ and $\cos\theta$, and the functions $\Phi(\phi)$ are simply proportional to $e^{im_l\phi}$, where $i = \sqrt{-1}$ and m_l is an integer that may be positive, zero, or negative.

In the process of finding solutions that satisfy the boundary conditions, we also find the corresponding energy levels. We denote the energies of these levels [E in Eq. (41.20a)] by E_n ($n = 1, 2, 3, \ldots$). These turn out to be *identical* to those from the Bohr model, as given by Eq. (39.15), with the electron rest mass m replaced by the reduced mass m_r. Rewriting that equation with $\hbar = h/2\pi$, we have

Energy levels of hydrogen

$$E_n = -\frac{1}{(4\pi\epsilon_0)^2}\frac{m_r e^4}{2n^2\hbar^2} = -\frac{13.60\text{ eV}}{n^2} \tag{41.21}$$

Reduced mass (m_r); Magnitude of electron charge (e); Electric constant (ϵ_0); Principal quantum number ($n = 1, 2, 3, \ldots$) ; Planck's constant divided by 2π ($\hbar$)

As in Section 39.3, we call n the **principal quantum number.**

Equation (41.21) is an important validation of our Schrödinger-equation analysis of the hydrogen atom. The Schrödinger analysis is quite different from the Bohr model, both mathematically and conceptually, yet both yield the same energy-level scheme—a scheme that agrees with the energies determined from spectra. As we'll see, the Schrödinger analysis can explain many more aspects of the hydrogen atom than can the Bohr model.

Quantization of Orbital Angular Momentum

The solutions to Eqs. (41.20) that satisfy the boundary conditions mentioned above also have quantized values of *orbital angular momentum*. That is, only certain discrete values of the magnitude and components of orbital angular momentum are permitted. In discussing the Bohr model in Section 39.3, we mentioned that quantization of angular momentum was a result with little fundamental justification. With the Schrödinger equation it appears automatically.

The possible values of the magnitude L of orbital angular momentum $\vec{L}$ are determined by the requirement that the $\Theta(\theta)$ function in Eq. (41.20b) must be finite at $\theta = 0$ and $\theta = \pi$. In a level with energy E_n and principal quantum number n, the possible values of L are

Magnitude of orbital angular momentum, hydrogen atom

$$L = \sqrt{l(l+1)}\,\hbar \qquad (l = 0, 1, 2, \ldots, n-1) \tag{41.22}$$

Orbital quantum number (l); Planck's constant divided by 2π ($\hbar$); Principal quantum number ($n = 1, 2, 3, \ldots$)

The **orbital quantum number** l in Eq. (41.22) is the same l that appears in Eqs. (41.20a) and (41.20b). In the Bohr model, each energy level corresponded to a single value of angular momentum. Equation (41.22) shows that in fact there are n different possible values of L for the nth energy level.

An interesting feature of Eq. (41.22) is that the orbital angular momentum is *zero* for $l = 0$ states. This result disagrees with the Bohr model, in which the electron always moved in a circle of definite radius and L was never zero. The $l = 0$ wave functions ψ depend only on r; for these states, the functions $\Theta(\theta)$ and $\Phi(\phi)$ are constants. Thus the wave functions for $l = 0$ states are spherically symmetric. There is nothing in their probability distribution $|\psi|^2$ to favor one direction over any other, and there is no orbital angular momentum.

The permitted values of the *component* of $\vec{L}$ in a given direction, say the z-component L_z, are determined by the requirement that the $\Phi(\phi)$ function must equal $\Phi(\phi + 2\pi)$. The possible values of L_z are

***z*-component of orbital angular momentum, hydrogen atom**

$$L_z = m_l\hbar \qquad (m_l = 0, \pm 1, \pm 2, \ldots, \pm l) \tag{41.23}$$

Orbital magnetic quantum number (m_l); Planck's constant divided by 2π ($\hbar$); Orbital quantum number (l)

The quantum number m_l is the same as that in Eqs. (41.20b) and (41.20c). We see that m_l can be zero or a positive or negative integer up to, but no larger in magnitude than, l. That is, $|m_l| \leq l$. For example, if $l = 1$, m_l can equal 1, 0, or -1. For reasons that will emerge later, we call m_l the *orbital magnetic quantum number,* or **magnetic quantum number** for short.

The component L_z can never be quite as large as L (unless both are zero). For example, when $l = 2$, the largest possible value of m_l is also 2; then Eqs. (41.22) and (41.23) give

$$L = \sqrt{2(2+1)}\hbar = \sqrt{6}\hbar = 2.45\hbar$$

$$L_z = 2\hbar$$

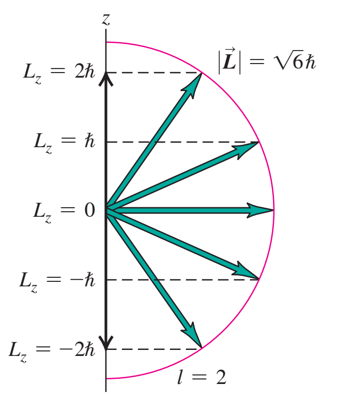
Figure 41.6 shows the situation. The minimum value of the angle θ_L between the vector $\vec{L}$ and the z-axis is

$$\theta_L = \arccos\frac{L_z}{L}$$

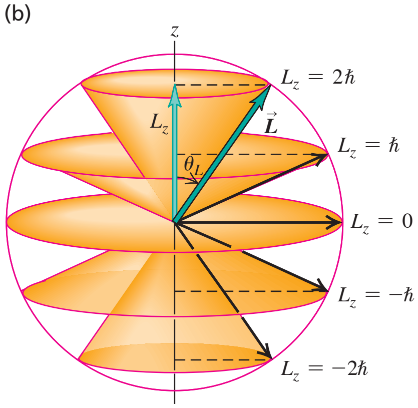
$$= \arccos\frac{2}{2.45} = 35.3°$$

Figure 41.6 (a) When $l = 2$, the magnitude of the angular momentum vector $\vec{L}$ is $\sqrt{6}\hbar = 2.45\hbar$, but $\vec{L}$ does not have a definite direction. In this semiclassical vector picture, $\vec{L}$ makes an angle of 35.3° with the z-axis when the z-component has its maximum value of $2\hbar$. (b) These cones show the possible directions of $\vec{L}$ for different values of L_z.

(a)



(b)



That $|L_z|$ is always less than L is also required by the uncertainty principle. Suppose we could know the precise *direction* of the orbital angular momentum vector. Then we could let that be the direction of the z-axis, and L_z would equal L. This corresponds to a particle moving in the xy-plane only, in which case the z-component of the linear momentum $\vec{p}$ would be zero with no uncertainty Δp_z. Then the uncertainty principle $\Delta z \Delta p_z \geq \hbar$ requires infinite uncertainty Δz in the coordinate z. This is impossible for a localized state; we conclude that we can't know the direction of $\vec{L}$ precisely. Thus, as we've already stated, the component of $\vec{L}$ in a given direction can never be quite as large as its magnitude L. Also, if we can't know the direction of $\vec{L}$ precisely, we can't determine the components L_x and L_y precisely. Thus we show *cones* of possible directions for $\vec{L}$ in Fig. 41.6b.

You may wonder why we have singled out the z-axis. We can't determine all three components of orbital angular momentum with certainty, so we arbitrarily pick one as the component we want to measure. When we discuss interactions of the atom with a magnetic field, we'll consistently choose the positive z-axis to be in the direction of $\vec{B}$.

Quantum Number Notation

The wave functions for the hydrogen atom are determined by the values of three quantum numbers n, l, and m_l. (Compare this to the particle in a three-dimensional box that we considered in Section 41.2. There, too, three quantum numbers were needed to describe each stationary state.) The energy E_n is determined by the principal quantum number n according to Eq. (41.21). The magnitude of orbital angular momentum is determined by the orbital quantum number l, as in Eq. (41.22). The component of orbital angular momentum in a specified axis direction (customarily the z-axis) is determined by the magnetic quantum number m_l, as in Eq. (41.23). The energy does not depend on the values of l or m_l (**Fig. 41.7**), so for each energy level E_n given by Eq. (41.21), there is more than one distinct state having the same energy but different quantum numbers. That is, these states are *degenerate,* just like most of the states of a particle in a three-dimensional box. As for the three-dimensional box, degeneracy arises because the hydrogen atom is symmetric: If you rotate the atom through any angle, the potential-energy function at a distance r from the nucleus has the same value.

States with various values of the orbital quantum number l are often labeled with letters, according to the following scheme:

$l = 0$: s states $\quad$ $l = 3$: f states
$l = 1$: p states $\quad$ $l = 4$: g states
$l = 2$: d states $\quad$ $l = 5$: h states

and so on alphabetically. This seemingly irrational choice of the letters s, p, d, and f originated in the early days of spectroscopy and has no fundamental significance. In an important form of *spectroscopic notation* that we'll use often, a state with $n = 2$ and $l = 1$ is called a $2p$ state; a state with $n = 4$ and $l = 0$ is a $4s$ state; and so on. Only s states $(l = 0)$ are spherically symmetric.

Here's another bit of notation. The radial extent of the wave functions increases with the principal quantum number n, and we can speak of a region of space associated with a particular value of n as a **shell.** Especially in discussions of many-electron atoms, these shells are denoted by capital letters:

$n = 1$: K shell
$n = 2$: L shell
$n = 3$: M shell
$n = 4$: N shell

and so on alphabetically. For each n, different values of l correspond to different *subshells*. For example, the L shell $(n = 2)$ contains the $2s$ and $2p$ subshells.

Table 41.1 shows some of the possible combinations of the quantum numbers n, l, and m_l for hydrogen-atom wave functions. The spectroscopic notation and the shell notation for each are also shown.

Figure **41.7** The energy for an orbiting satellite such as the Hubble Space Telescope depends on the average distance between the satellite and the center of the earth. It does *not* depend on whether the orbit is circular (with a large orbital angular momentum L) or elliptical (in which case L is smaller). In the same way, the energy of a hydrogen atom does not depend on the orbital angular momentum.

TABLE 41.1 Quantum States of the Hydrogen Atom

n	l	m_l	Spectroscopic Notation	Shell
1	0	0	$1s$	K
2	0	0	$2s$	L
2	1	$-1, 0, 1$	$2p$	
3	0	0	$3s$	M
3	1	$-1, 0, 1$	$3p$	
3	2	$-2, -1, 0, 1, 2$	$3d$	
4	0	0	$4s$	N
and so on				

PROBLEM-SOLVING STRATEGY 41.1 Atomic Structure

IDENTIFY *the relevant concepts:* Many problems in atomic structure can be solved simply by reference to the quantum numbers n, l, and m_l that describe the total energy E, the magnitude of the orbital angular momentum $\vec{L}$, the z-component of $\vec{L}$, and other properties of an atom.

SET UP *the problem:* Identify the target variables and choose the appropriate equations, which may include Eqs. (41.21), (41.22), and (41.23).

EXECUTE *the solution* as follows:

1. Be sure you understand the possible values of the quantum numbers n, l, and m_l for the hydrogen atom. They are all integers; n is always greater than zero, l can be zero or positive up to $n - 1$, and m_l can range from $-l$ to l. You should know how to count the number of (n, l, m_l) states in each shell (K, L, M, and so on) and subshell ($3s$, $3p$, $3d$, and so on). Be able to *construct* Table 41.1, not just to write it from memory.
2. Solve for the target variables.

EVALUATE *your answer:* It helps to be familiar with typical magnitudes in atomic physics. For example, the electric potential energy of a proton and electron 0.10 nm apart (typical of atomic dimensions) is about -15 eV. Visible light has wavelengths around 500 nm and frequencies around 5×10^{14} Hz. Problem-Solving Strategy 39.1 (Section 39.1) gives other typical magnitudes.

EXAMPLE 41.2 Counting hydrogen states

WITH VARIATION PROBLEMS

How many distinct (n, l, m_l) states of the hydrogen atom with $n = 3$ are there? What are their energies?

IDENTIFY and SET UP This problem uses the relationships among the principal quantum number n, orbital quantum number l, magnetic quantum number m_l, and energy of a state for the hydrogen atom. We use the rule that l can have n integer values, from 0 to $n - 1$, and that m_l can have $2l + 1$ values, from $-l$ to l. Equation (41.21) gives the energy of any particular state.

EXECUTE When $n = 3$, l can be 0, 1, or 2. When $l = 0$, m_l can be only 0 (1 state). When $l = 1$, m_l can be -1, 0, or 1 (3 states). When $l = 2$, m_l can be $-2, -1, 0, 1,$ or 2 (5 states). The total number of (n, l, m_l) states with $n = 3$ is therefore $1 + 3 + 5 = 9$. (In Section 41.5 we'll find that the total number of $n = 3$ states is in fact twice this, or 18, because of electron spin.)

The energy of a hydrogen-atom state depends only on n, so all 9 of these states have the same energy. From Eq. (41.21),

$$E_3 = \frac{-13.60\text{ eV}}{3^2} = -1.51\text{ eV}$$

EVALUATE For a given value of n, the total number of (n, l, m_l) states turns out to be n^2. In this case $n = 3$ and there are $3^2 = 9$ states. Remember that the ground level of hydrogen has $n = 1$ and $E_1 = -13.6$ eV; the $n = 3$ excited states have a higher (less negative) energy.

KEYCONCEPT Because the potential-energy function for the hydrogen atom is spherically symmetric, its energy levels are degenerate. Stationary states with the same principal quantum number n have the same energy, even though they have different values of the orbital quantum number l and the orbital magnetic quantum number m_l.

EXAMPLE 41.3 Angular momentum in an excited level of hydrogen

WITH VARIATION PROBLEMS

Consider the $n = 4$ states of hydrogen. (a) What is the maximum magnitude L of the orbital angular momentum? (b) What is the maximum value of L_z? (c) What is the minimum angle between $\vec{L}$ and the z-axis? Give your answers to parts (a) and (b) in terms of $\hbar$.

IDENTIFY and SET UP We again need to relate the principal quantum number n and the orbital quantum number l for a hydrogen atom. We also need to relate the value of l and the magnitude and possible directions of the orbital angular momentum vector. We'll use Eq. (41.22) in part (a) to determine the maximum value of L; then we'll use Eq. (41.23) in part (b) to determine the maximum value of L_z. The angle between $\vec{L}$ and the z-axis is minimum when L_z is maximum (so that $\vec{L}$ is most nearly aligned with the positive z-axis).

EXECUTE (a) When $n = 4$, the maximum value of the orbital quantum number l is $(n - 1) = (4 - 1) = 3$; from Eq. (41.22),

$$L_{max} = \sqrt{3(3+1)}\,\hbar = \sqrt{12}\,\hbar = 3.464\hbar$$

(b) For $l = 3$ the maximum value of m_l is 3. From Eq. (41.23),

$$(L_z)_{max} = 3\hbar$$

(c) The *minimum* allowed angle between $\vec{L}$ and the z-axis corresponds to the *maximum* allowed values of L_z and m_l (Fig. 41.6b shows an $l = 2$ example). For the state with $l = 3$ and $m_l = 3$,

$$\theta_{min} = \arccos\frac{(L_z)_{max}}{L} = \arccos\frac{3\hbar}{3.464\hbar} = 30.0°$$

EVALUATE As a check, you can verify that θ is greater than 30.0° for all states with smaller values of l.

KEYCONCEPT For each value of the principal quantum number $n = 1, 2, 3, \ldots$ for the stationary states of the hydrogen atom, there are n possible values of the orbital quantum number l that range from 0 to $n - 1$. For each value of l, there are $2l + 1$ possible values of the orbital magnetic quantum number m_l that range from $-l$ to l.

Electron Probability Distributions

Rather than picturing the electron as a point particle moving in a precise circle, the Schrödinger equation gives an electron *probability distribution* surrounding the nucleus. The hydrogen-atom probability distributions are three-dimensional, so they are harder to visualize than the two-dimensional circular orbits of the Bohr model. It's helpful to look at the *radial probability distribution function* $P(r)$—that is, the probability per radial length for the electron to be found at various distances from the proton. From Section 41.1 the probability for finding the electron in a small volume element dV is $|\psi|^2\,dV$. (We assume that ψ is normalized in accordance with Eq. (41.6)—that is, that the integral of $|\psi|^2\,dV$ over all space equals unity so that there is 100% probability of finding the electron somewhere in the universe.) Let's take as our volume element a thin spherical shell with inner radius r and outer radius $r + dr$. The volume dV of this shell is approximately its area $4\pi r^2$ multiplied by its thickness dr:

$$dV = 4\pi r^2\,dr \tag{41.24}$$

We denote by $P(r)\,dr$ the probability of finding the particle within the radial range dr; then, using Eq. (41.24), we get

$$\underbrace{\overbrace{P(r)}^{\text{Radial probability distribution function}}\,dr}_{\text{Probability that the electron is between } r \text{ and } r+dr} = \overbrace{|\underbrace{\psi}_{\text{Wave function}}|^2}^{\text{Probability distribution function}}\,\underbrace{dV}_{\text{Volume of spherical shell}} = |\psi|^2\underbrace{4\pi r^2\,dr}_{\text{with inner radius } r \text{, outer radius } r+dr} \qquad (41.25)$$

For wave functions that depend on θ and ϕ as well as r, we use the value of $|\psi|^2$ averaged over all angles in Eq. (41.25).

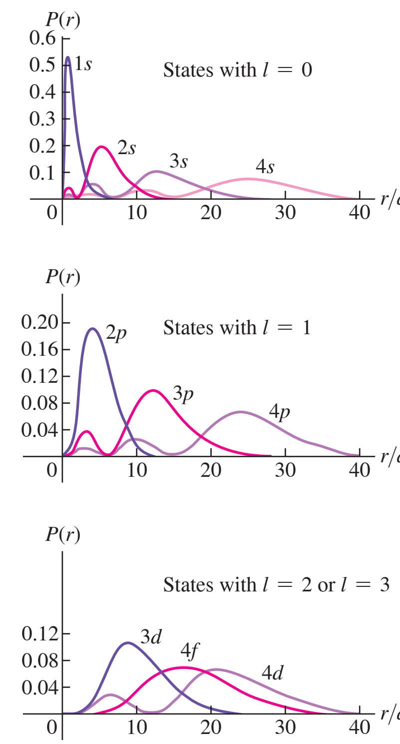
Figure 41.8 shows graphs of $P(r)$ for several hydrogen-atom wave functions. The r scales are labeled in multiples of a, the smallest distance between the electron and the nucleus in the Bohr model:

$$\underbrace{a}_{\text{Radius of smallest orbit in Bohr model}} = \frac{\overbrace{\epsilon_0}^{\text{Electric constant}}\overbrace{h^2}^{\text{Planck's constant}}}{\pi \underbrace{m_r}_{\text{Reduced mass}} \underbrace{e^2}_{\text{Magnitude of electron charge}}} = \frac{4\pi\epsilon_0 \overbrace{\hbar^2}^{\text{Planck's constant divided by } 2\pi}}{m_r e^2} = 5.29 \times 10^{-11}\ \text{m} \qquad (41.26)$$

As for a particle in a cubical box (see Section 41.2), there are some locations where the probability is zero. These surfaces are planes for a particle in a box; for a hydrogen atom these are spherical surfaces (that is, surfaces of constant r). Note that for the states that have the largest possible l for each n (such as $1s$, $2p$, $3d$, and $4f$ states), $P(r)$ has a single maximum at n^2a. For these states, the electron is most likely to be found at the distance from the nucleus that is predicted by the Bohr model, $r = n^2a$.

Figure 41.8 shows *radial* probability distribution functions $P(r) = 4\pi r^2|\psi|^2$, which indicate the relative probability of finding the electron within a thin spherical shell of radius r. By contrast, **Figs. 41.9** and **41.10** (next page) show the *three-dimensional* probability distribution functions $|\psi|^2$, which indicate the relative probability of finding the electron within a small box at a given position. The darker the blue "cloud," the greater the value of $|\psi|^2$. (These are similar to the "clouds" shown in Fig. 41.2.) Figure 41.9 shows cross sections of the spherically symmetric probability clouds for the lowest three s subshells, for which $|\psi|^2$ depends only on the radial coordinate r. Figure 41.10 shows cross sections of the clouds for other electron states for which $|\psi|^2$ depends on both r and θ. For these states the probability distribution function is zero for certain values of θ as well as for certain values of r. In *any* stationary state of the hydrogen atom, $|\psi|^2$ is independent of ϕ.

Figure **41.8** Radial probability distribution functions $P(r)$ for several hydrogen-atom wave functions, plotted as functions of the ratio r/a [see Eq. (41.26)]. For each function, the number of maxima is $(n - l)$. The curves for which $l = n - 1$ $(1s, 2p, 3d, \ldots)$ have only one maximum, located at $r = n^2a$.



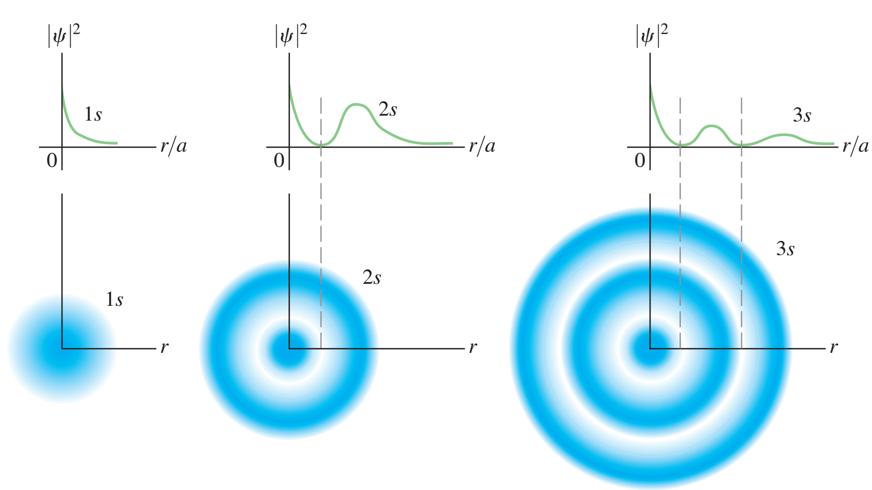
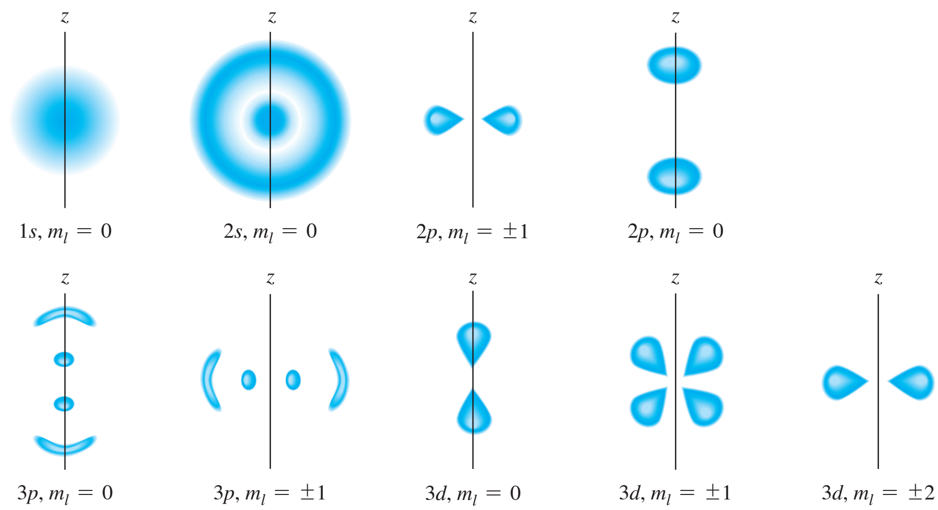


Figure **41.9** Three-dimensional probability distribution functions $|\psi|^2$ for the spherically symmetric $1s$, $2s$, and $3s$ hydrogen-atom wave functions.

Figure 41.10 Cross sections of three-dimensional probability distributions for a few quantum states of the hydrogen atom. They are not to the same scale. Mentally rotate each drawing about the z-axis to obtain the three-dimensional representation of $|\psi|^2$. For example, the $2p$, $m_l = \pm 1$ probability distribution looks like a fuzzy donut.



EXAMPLE 41.4 A hydrogen wave function

WITH VARIATION PROBLEMS

The ground-state wave function for hydrogen (a $1s$ state) is

$$\psi_{1s}(r) = \frac{1}{\sqrt{\pi a^3}} e^{-r/a}$$

(a) Verify that this function is normalized. (b) What is the probability that the electron will be found at a distance less than a from the nucleus?

IDENTIFY and SET UP This example is similar to Example 41.1 in Section 41.2. We need to show that this wave function satisfies the condition that the probability of finding the electron *somewhere* is 1. We then need to find the probability that it will be found in the region $r < a$. In part (a) we'll carry out the integral $\int |\psi|^2 \, dV$ over all space; if it is equal to 1, the wave function is normalized. In part (b) we'll carry out the same integral over a spherical volume that extends from the origin (the nucleus) out to a distance a from the nucleus.

EXECUTE (a) Since the wave function depends only on the radial coordinate r, we can choose our volume elements to be spherical shells of radius r, thickness dr, and volume dV given by Eq. (41.24). We then have

$$\int_{\text{all space}} |\psi_{1s}|^2 \, dV = \int_0^\infty \frac{1}{\pi a^3} e^{-2r/a}(4\pi r^2 \, dr) = \frac{4}{a^3}\int_0^\infty r^2 e^{-2r/a} \, dr$$

You can find the following indefinite integral in a table of integrals or by integrating by parts:

$$\int r^2 e^{-2r/a} \, dr = \left(-\frac{ar^2}{2} - \frac{a^2 r}{2} - \frac{a^3}{4}\right)e^{-2r/a}$$

Evaluating this between the limits $r = 0$ and $r = \infty$ is simple; it is zero at $r = \infty$ because of the exponential factor, and at $r = 0$ only the last term in the parentheses survives. Thus the value of the definite integral is $a^3/4$. Putting it all together, we find

$$\int_0^\infty |\psi_{1s}|^2 \, dV = \frac{4}{a^3}\int_0^\infty r^2 e^{-2r/a} \, dr = \frac{4}{a^3}\frac{a^3}{4} = 1$$

The wave function *is* normalized.

(b) To find the probability P that the electron is found within $r < a$, we carry out the same integration but with the limits 0 and a. We'll leave the details to you. From the upper limit we get $-5e^{-2}a^3/4$; the final result is

$$P = \int_0^a |\psi_{1s}|^2 \, 4\pi r^2 \, dr = \frac{4}{a^3}\left(-\frac{5a^3 e^{-2}}{4} + \frac{a^3}{4}\right)$$
$$= (-5e^{-2} + 1) = 1 - 5e^{-2} = 0.323$$

EVALUATE Our results tell us that in a ground state we expect to find the electron at a distance from the nucleus less than a about $\frac{1}{3}$ of the time and at a greater distance about $\frac{2}{3}$ of the time. It's hard to tell, but in Fig. 41.8, about $\frac{2}{3}$ of the area under the $1s$ curve is at distances greater than a (that is, $r/a > 1$).

KEYCONCEPT If a particle is in a stationary state described by the wave function ψ, the probability of finding the particle at a radial coordinate between r and $r + dr$ is $P(r)dr$, where $P(r) = 4\pi r^2 |\psi|^2$. If the wave function depends on the angle as well as on the radial coordinate, you must first average $|\psi|^2$ over all angles.

Hydrogenlike Atoms

Two generalizations that we discussed with the Bohr model in Section 39.3 are equally valid in the Schrödinger analysis. First, if the "atom" is not composed of a single proton and a single electron, using the reduced mass m_r of the system in Eqs. (41.21) and (41.26) will lead to changes that are substantial for some exotic systems. One example is *positronium,* in which a positron and an electron orbit each other; another is a *muonic atom,* in which the electron is replaced by an unstable particle called a muon that has the same charge as an electron but is 207 times more massive. Second, our analysis is applicable to single-electron ions, such as He^+, Li^{2+}, and so on. For such ions we replace e^2 by Ze^2 in Eqs. (41.21) and (41.26), where Z is the number of protons (the **atomic number**).

TEST YOUR UNDERSTANDING OF SECTION 41.3 Rank the following states of the hydrogen atom in order from highest to lowest probability of finding the electron in the vicinity of $r = 5a$: (i) $n = 1, l = 0, m_l = 0$; (ii) $n = 2, l = 1, m_l = +1$; (iii) $n = 2, l = 1, m_l = 0$.

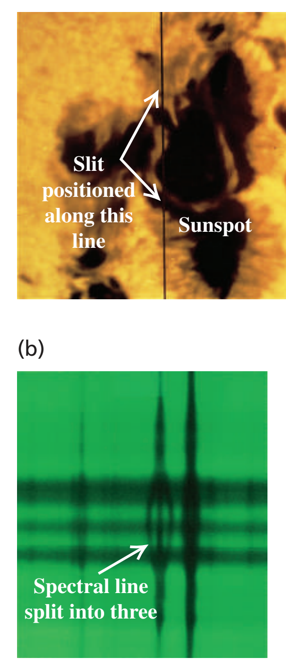
ANSWER

(ii) and (iii) (tie), (i) An electron in a state with principal quantum number n is most likely to be found at $r = n^2a$. This result is independent of the values of the quantum numbers l and m_l. Hence an electron with $n = 2$ (most likely to be found at $r = 4a$) is more likely to be found near $r = 5a$ than an electron with $n = 1$ (most likely to be found at $r = a$).

Figure **41.11** Magnetic effects on the spectrum of sunlight. (a) The slit of a spectrograph is positioned along the black line crossing a portion of a sunspot. (b) The 0.4 T magnetic field in the sunspot (a thousand times greater than the earth's field) splits the middle spectral line into three lines.

(a)

(b)



41.4 THE ZEEMAN EFFECT

The **Zeeman effect** is the splitting of atomic energy levels and the associated spectral lines when the atoms are placed in a magnetic field (**Fig. 41.11**). This effect confirms experimentally the quantization of angular momentum. In this section we'll assume that the only angular momentum is the *orbital* angular momentum of a single electron and learn why we call m_l the magnetic quantum number.

Atoms contain charges in motion, so it should not be surprising that magnetic forces cause changes in that motion and in the energy levels. In 1896 the Dutch physicist Pieter Zeeman was the first to show that in the presence of a magnetic field, some spectral lines were split into groups of closely spaced lines (**Fig. 41.12**). This effect now bears his name.

Magnetic Moment of an Orbiting Electron

Let's begin our analysis of the Zeeman effect by reviewing the concept of *magnetic dipole moment* or *magnetic moment,* introduced in Section 27.7. A plane current loop with vector area $\vec{A}$ carrying current I has a magnetic moment $\vec{\boldsymbol{\mu}}$ given by

$$\vec{\boldsymbol{\mu}} = I\vec{A} \tag{41.27}$$

When a magnetic dipole of moment $\vec{\boldsymbol{\mu}}$ is placed in a magnetic field $\vec{B}$, the field exerts a torque $\vec{\boldsymbol{\tau}} = \vec{\boldsymbol{\mu}} \times \vec{B}$ on the dipole. The potential energy U associated with this interaction is given by Eq. (27.27):

$$U = -\vec{\boldsymbol{\mu}} \cdot \vec{B} \tag{41.28}$$

Now let's use Eqs. (41.27) and (41.28) and the Bohr model to look at the interaction of a hydrogen atom with a magnetic field. The orbiting electron with speed v is equivalent to a current loop with radius r and area πr^2. The average current I is the average charge per unit time that passes a given point of the orbit. This is equal to the charge magnitude e divided by the time T for one revolution, given by $T = 2\pi r/v$. Thus $I = ev/2\pi r$, and from Eq. (41.27) the magnitude μ of the magnetic moment is

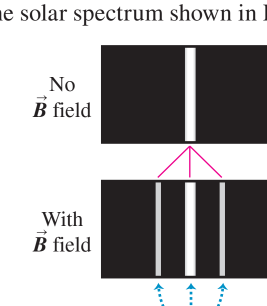
$$\mu = IA = \frac{ev}{2\pi r}\pi r^2 = \frac{evr}{2} \tag{41.29}$$

We can also express this in terms of the magnitude L of the orbital angular momentum. From Eq. (10.28) the angular momentum of a particle in a circular orbit is $L = mvr$, so Eq. (41.29) becomes

$$\mu = \frac{e}{2m}L \tag{41.30}$$

The ratio of the magnitude of $\vec{\boldsymbol{\mu}}$ to the magnitude of $\vec{L}$ is $\mu/L = e/2m$ and is called the *gyromagnetic ratio.*

Figure **41.12** The normal Zeeman effect. Compare this to the magnetic splitting in the solar spectrum shown in Fig. 41.11b.



In the Bohr model, $L = nh/2\pi = n\hbar$, where $n = 1, 2, \ldots$. For an $n = 1$ state (a ground state), Eq. (41.30) becomes $\mu = (e/2m)\hbar$. This quantity is a natural unit for magnetic moment; it is called one **Bohr magneton,** denoted by μ_B:

$$\mu_B = \frac{e\hbar}{2m} \quad \text{(definition of the Bohr magneton)} \tag{41.31}$$

(We defined this quantity in Section 28.8.) Evaluating Eq. (41.31) gives

$$\mu_B = 5.788 \times 10^{-5}\ \text{eV/T} = 9.274 \times 10^{-24}\ \text{J/T or A} \cdot \text{m}^2$$

Note that the units J/T and $\text{A} \cdot \text{m}^2$ are equivalent.

While the Bohr model suggests that the orbital motion of an atomic electron gives rise to a magnetic moment, this model does *not* give correct predictions about magnetic interactions. As an example, the Bohr model predicts that an electron in a hydrogen-atom ground state has an orbital magnetic moment of magnitude μ_B. But the Schrödinger picture tells us that such a ground-state electron is in an s state with zero angular momentum, so the orbital magnetic moment must be *zero!* To get the correct results, we must describe the states by using Schrödinger wave functions.

It turns out that in the Schrödinger formulation, electrons have the same ratio of μ to L (gyromagnetic ratio) as in the Bohr model—namely, $e/2m$. Suppose the magnetic field $\vec{B}$ is directed along the $+z$-axis. From Eq. (41.28) the interaction energy U of the atom's magnetic moment with the field is

$$U = -\mu_z B \tag{41.32}$$

where μ_z is the z-component of the vector $\vec{\mu}$.

Now we use Eq. (41.30) to find μ_z, recalling that e is the *magnitude* of the electron charge and that the actual charge is $-e$. Because the electron charge is negative, the orbital angular momentum and magnetic moment vectors are opposite. We find

$$\mu_z = -\frac{e}{2m} L_z \tag{41.33}$$

CAUTION Again, two uses of the symbol m As in Section 41.3, the symbol m is used in two ways in Eq. (41.34). Don't confuse the electron mass m with the orbital magnetic quantum number m_l.

For the Schrödinger wave functions, $L_z = m_l\hbar$, with $m_l = 0, \pm1, \pm2, \ldots, \pm l$, so Eq. (41.33) becomes

$$\mu_z = -\frac{e}{2m} L_z = -m_l \frac{e\hbar}{2m} \tag{41.34}$$

Finally, using Eq. (41.31) for the Bohr magneton, we can express the interaction energy from Eq. (41.32) as

Orbital magnetic interaction energy

Magnetic dipole component in direction of $\vec{B}$

Magnitude of electron charge

Planck's constant divided by 2π

Bohr magneton

$$U = -\mu_z B = m_l \frac{e\hbar}{2m} B = m_l \mu_B B \tag{41.35}$$

Magnetic-field magnitude

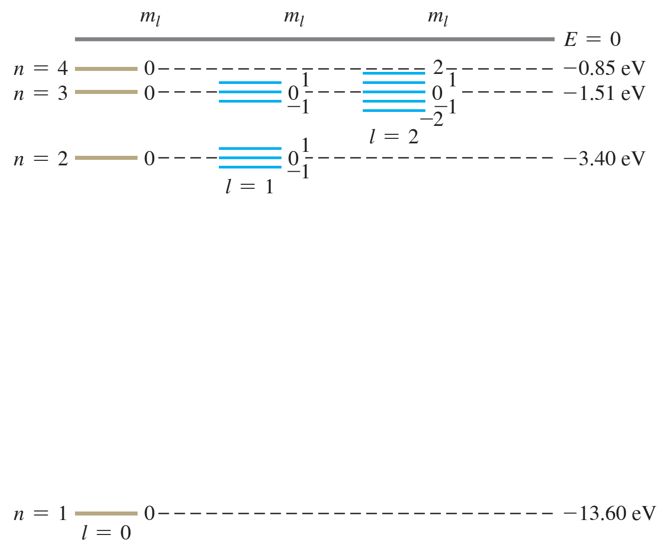
Electron mass

Orbital magnetic quantum number $= 0, \pm1, \pm2, \ldots, \pm l$

The magnetic field shifts the energy of each orbital state by an amount U. The interaction energy U depends on the value of m_l because m_l determines the orientation of the orbital magnetic moment relative to the magnetic field. This dependence is the reason m_l is called the magnetic quantum number.

The values of m_l range from $-l$ to $+l$ in steps of one, so an energy level with a particular value of the orbital quantum number l contains $(2l + 1)$ different orbital states. Without a magnetic field these states all have the same energy; that is, they are degenerate. The magnetic field removes this degeneracy. In the presence of a magnetic field they are split into $2l + 1$ distinct energy levels; adjacent levels differ in energy by $(e\hbar/2m)B = \mu_B B$. We can understand this in terms of the connection between degeneracy and symmetry. With a

Figure **41.13** This energy-level diagram for hydrogen shows how the levels are split when the electron's orbital magnetic moment interacts with an external magnetic field. The values of m_l are shown adjacent to the various levels. The relative magnitudes of the level splittings are exaggerated for clarity. The $n = 4$ splittings are not shown; can you draw them in?



magnetic field applied along the z-axis, the atom is no longer completely symmetric under rotation: There is a preferred direction in space. By removing the symmetry, we remove the degeneracy of states.

Figure 41.13 shows the effect on the energy levels of hydrogen. Spectral lines corresponding to transitions from one set of levels to another set are correspondingly split and appear as a series of three closely spaced spectral lines replacing a single line. As the following example shows, the splitting of spectral lines is quite small because the value of $\mu_B B$ is small even for substantial magnetic fields.

EXAMPLE 41.5 An atom in a magnetic field

An atom in a state with $l = 1$ emits a photon with wavelength 600.000 nm as it decays to a state with $l = 0$. If the atom is placed in a magnetic field with magnitude $B = 2.00$ T, what are the shifts in the energy levels and in the wavelength that result from the interaction between the atom's orbital magnetic moment and the magnetic field?

IDENTIFY and SET UP This problem concerns the splitting of atomic energy levels by a magnetic field (the Zeeman effect). We use Eq. (41.35) to determine the energy-level shifts. The relationship $E = hc/\lambda$ between the energy and wavelength of a photon then lets us calculate the wavelengths emitted during transitions from the $l = 1$ states to the $l = 0$ state.

EXECUTE The energy of a 600 nm photon is

$$E = \frac{hc}{\lambda} = \frac{(4.14 \times 10^{-15}\ \text{eV} \cdot \text{s})(3.00 \times 10^{8}\ \text{m/s})}{600 \times 10^{-9}\ \text{m}} = 2.07\ \text{eV}$$

If there is no external magnetic field, that is the difference in energy between the $l = 0$ and $l = 1$ levels.

With a 2.00 T field present, Eq. (41.35) shows that there is no shift of the $l = 0$ state (which has $m_l = 0$). For the $l = 1$ states, the splitting of levels is given by

$$\begin{aligned} U &= m_l \mu_B B = m_l(5.788 \times 10^{-5}\ \text{eV/T})(2.00\ \text{T}) \\ &= m_l(1.16 \times 10^{-4}\ \text{eV}) = m_l(1.85 \times 10^{-23}\ \text{J}) \end{aligned}$$

The possible values of m_l for $l = 1$ are -1, 0, and $+1$, and the three corresponding levels are separated by equal intervals of 1.16×10^{-4} eV. This is a small fraction of the 2.07 eV photon energy:

$$\frac{\Delta E}{E} = \frac{1.16 \times 10^{-4}\ \text{eV}}{2.07\ \text{eV}} = 5.60 \times 10^{-5}$$

The corresponding *wavelength* shifts are about $(5.60 \times 10^{-5}) \times (600\ \text{nm}) = 0.034$ nm. The original 600.000 nm line is split into a triplet with wavelengths 599.966, 600.000, and 600.034 nm.

EVALUATE Even though 2.00 T would be a strong field in most laboratories, the wavelength splittings are extremely small. Nonetheless, modern spectrographs have more than enough chromatic resolving power to measure these splittings (see Section 36.5).

KEYCONCEPT An electron in an atom has a nonzero magnetic moment if it is in a state with a nonzero orbital angular momentum. If this atom is in a magnetic field $\vec{B}$, the electron energy depends on the electron's orbital magnetic quantum number m_l, which describes how the magnetic moment vector is oriented with respect to $\vec{B}$.

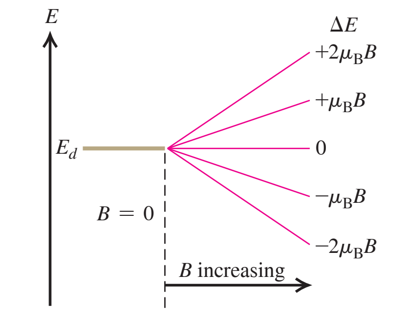


Figure **41.14** This figure shows how the splitting of the energy levels of a d state $(l = 2)$ depends on the magnitude B of an external magnetic field, assuming only an orbital magnetic moment.

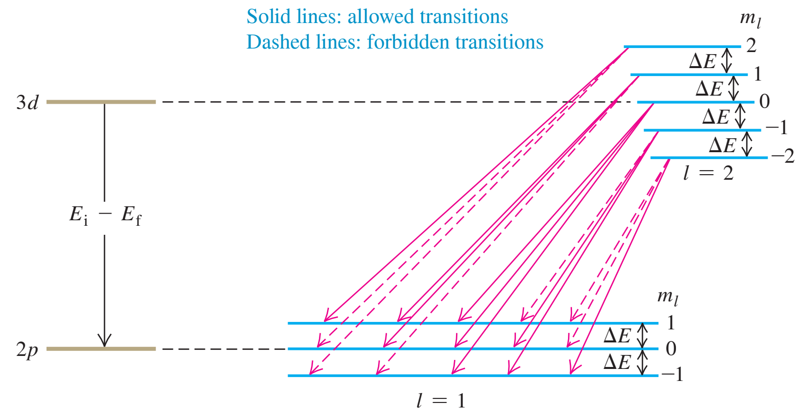


Figure **41.15** The cause of the normal Zeeman effect. The magnetic field splits the levels, but selection rules allow transitions with only three different energy changes, giving three different photon frequencies and wavelengths.

Selection Rules

Figure 41.14 shows what happens to a set of d states $(l = 2)$ as the magnetic field increases. With zero field the five states $m_l = -2, -1, 0, 1,$ and 2 are degenerate (have the same energy), but the applied field spreads the states out. **Figure 41.15** shows the splittings of both the 3d and 2p states. Equal energy differences $(e\hbar/2m)B = \mu_B B$ separate adjacent levels. In the absence of a magnetic field, a transition from a 3d to a 2p state would yield a single spectral line with photon energy $E_i - E_f$. With the levels split as shown, it might seem that there are five possible photon energies.

In fact, there are only three possibilities. Not all combinations of initial and final levels are possible because of a restriction associated with conservation of angular momentum. The photon ordinarily carries off one unit $(\hbar)$ of angular momentum, which leads to the requirements that in a transition l must change by 1 and m_l must change by 0 or ± 1. These requirements are called **selection rules.** Transitions that obey these rules are called *allowed transitions;* those that don't are *forbidden transitions.* In Fig. 41.15 we show the allowed transitions by solid arrows. You should count the possible transition energies to convince yourself that the nine solid arrows give only three possible energies; the zero-field value $E_i - E_f$, and that value plus or minus $\Delta E = (e\hbar/2m)B = \mu_B B$. Figure 41.12 shows the corresponding spectral lines.

What we have described is called the *normal* Zeeman effect. It is based entirely on the orbital angular momentum of the electron. However, it leaves out a very important consideration: the electron *spin* angular momentum, the subject of the next section.

TEST YOUR UNDERSTANDING OF SECTION 41.4 In this section we assumed that the magnetic field points in the positive z-direction. Would the results be different if the magnetic field pointed in the positive x-direction?

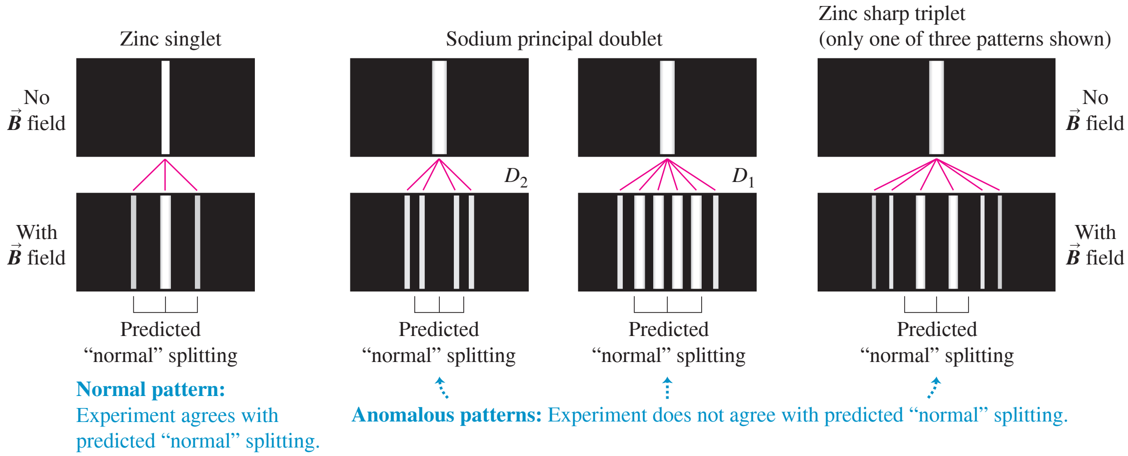
ANSWER

no All that matters is the component of the electron's orbital magnetic moment along the direction of $\vec{B}$. We called this quantity μ_z in Eq. (41.32) because we *defined* the positive z-axis to be in the direction of $\vec{B}$. In reality, the names of the axes are arbitrary.

41.5 ELECTRON SPIN

Despite the success of the Schrödinger equation in predicting the energy levels of the hydrogen atom, experimental observations indicate that it doesn't tell the whole story of the behavior of electrons in atoms. First, spectroscopists have found magnetic-field splitting into other than the three equally spaced lines that we explained in Section 41.4 (see Fig. 41.12). Before this effect was understood, it was called the *anomalous* Zeeman

Figure 41.16 Illustrations of the normal and anomalous Zeeman effects for two elements, zinc and sodium. The brackets under each illustration show the "normal" splitting predicted by ignoring the effect of electron spin.



effect to distinguish it from the "normal" effect discussed in the preceding section. **Figure 41.16** shows both kinds of splittings.

Second, some energy levels show splittings that resemble the Zeeman effect even when there is *no* external magnetic field. For example, when the lines in the hydrogen spectrum are examined with a high-resolution spectrograph, some lines are found to consist of sets of closely spaced lines called *multiplets*. Similarly, the orange-yellow line of sodium, corresponding to the transition $4p \longrightarrow 3s$ of the outer electron, is found to be a doublet $(\lambda = 589.0,\ 589.6\ \text{nm})$, suggesting that the $4p$ level might in fact be two closely spaced levels. The Schrödinger equation in its original form didn't predict any of this.

The Stern–Gerlach Experiment

Similar anomalies appeared in 1922 in atomic-beam experiments performed in Germany by Otto Stern and Walter Gerlach. When they passed a beam of neutral atoms through a nonuniform magnetic field (**Fig. 41.17**), atoms were deflected according to the orientation of their magnetic moments with respect to the field. These experiments demonstrated the quantization of angular momentum in a very direct way. If there were only orbital angular momentum, the deflections would split the beam into an odd number $(2l + 1)$ of different components. However, some atomic beams were split into an *even* number of components. If we use a different symbol j for an angular momentum quantum number, setting

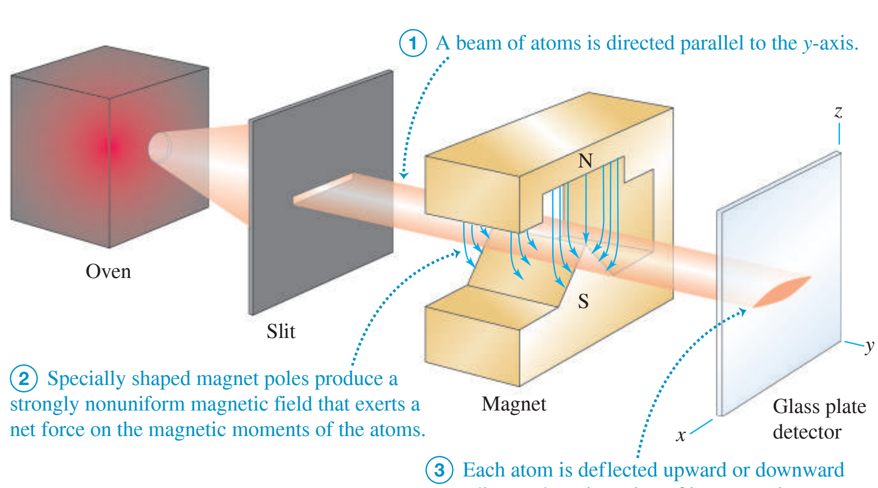


Figure 41.17 The Stern–Gerlach experiment.

$2j + 1$ equal to an even number gives $j = \frac{1}{2}, \frac{3}{2}, \frac{5}{2}, \ldots$, suggesting a half-integer angular momentum. This can't be understood on the basis of the Bohr model and similar pictures of atomic structure.

In 1925 two graduate students in the Netherlands, Samuel Goudsmit and George Uhlenbeck, proposed that the electron might have some additional motion. Using a semiclassical model, they suggested that the electron might behave like a spinning sphere of charge instead of a particle. If so, it would have an additional *spin* angular momentum and magnetic moment. If these were quantized in much the same way as *orbital* angular momentum and magnetic moment, they might help explain the observed energy-level anomalies.

An Analogy for Electron Spin

To introduce the concept of **electron spin,** let's start with an analogy. The earth travels in a nearly circular orbit around the sun, and at the same time it *rotates* on its axis. Each motion has its associated angular momentum, which we call the *orbital* and *spin* angular momentum, respectively. The total angular momentum of the earth is the vector sum of the two. If we were to model the earth as a single point, it would have no moment of inertia about its spin axis and thus no spin angular momentum. But when our model includes the finite size of the earth, spin angular momentum becomes possible.

In the Bohr model, suppose the electron is not just a point charge but a small spinning sphere that orbits the nucleus. Then the electron has not only orbital angular momentum but also spin angular momentum associated with the rotation of its mass about its axis. The sphere carries an electric charge, so the spinning motion leads to current loops and to a magnetic moment, as we discussed in Section 27.7. In a magnetic field, the *spin* magnetic moment has an interaction energy in addition to that of the *orbital* magnetic moment (the normal Zeeman-effect interaction that we discussed in Section 41.4). We should see additional Zeeman shifts due to the spin magnetic moment.

As we mentioned, such shifts *are* indeed observed in precise spectroscopic analysis. This and a variety of other experimental evidence have shown conclusively that the electron *does* have a spin angular momentum and a spin magnetic moment that do not depend on its orbital motion but are intrinsic to the electron itself. The origin of this spin angular momentum is fundamentally quantum-mechanical, so it's not correct to model the electron as a spinning charged sphere. But just as the Bohr model can be a useful conceptual picture for the motion of an electron in an atom, the spinning-sphere analogy can help you visualize the intrinsic spin angular momentum of an electron.

Spin Quantum Numbers

Like orbital angular momentum, the spin angular momentum of an electron (denoted by $\vec{S}$) is found to be quantized. Suppose we have an apparatus that measures a particular component of $\vec{S}$, say the z-component S_z. We find that the only possible values are

$$\underbrace{S_z}_{\substack{z\text{-component of}\\ \text{spin angular momentum}\\ \text{of electron}}} = \overbrace{m_s}^{\text{Spin magnetic quantum number } = \pm\frac{1}{2}}\underbrace{\hbar}_{\substack{\text{Planck's constant}\\ \text{divided by } 2\pi}} \tag{41.36}$$

This relationship is reminiscent of the expression $L_z = m_l\hbar$ for the z-component of orbital angular momentum, except that $|S_z|$ is *one-half* of $\hbar$ instead of an *integer* multiple. In analogy to the orbital magnetic quantum number m_l, we call the quantum number m_s the **spin magnetic quantum number.** Since m_s has only two possible values, $+\frac{1}{2}$ and $-\frac{1}{2}$, it follows that the spin angular momentum vector $\vec{S}$ can have only two orientations in space relative to the z-axis: "*spin up*" with a z-component of $+\frac{1}{2}\hbar$ and "*spin down*" with a z-component of $-\frac{1}{2}\hbar$.

Equation (41.36) also suggests that the magnitude S of the spin angular momentum is given by an expression analogous to Eq. (41.22) with the orbital quantum number l replaced by the **spin quantum number** $s = \frac{1}{2}$:

Maximum value of spin magnetic quantum number $= \frac{1}{2}$

Magnitude of spin angular momentum of electron

$$S = \sqrt{\tfrac{1}{2}\left(\tfrac{1}{2} + 1\right)}\hbar = \sqrt{\tfrac{3}{4}}\hbar \qquad (41.37)$$

Planck's constant divided by 2π

The electron is often called a "spin-one-half particle" or "spin-$\frac{1}{2}$ particle."

We see that to label the state of the electron in a hydrogen atom completely, we need *four* quantum numbers: n, l, and m_l (described in Section 41.3) to specify the electron's motion relative to the nucleus, plus the spin magnetic quantum number m_s to specify the electron spin orientation.

To visualize the quantized spin of an electron in a hydrogen atom, think of the electron probability distribution function $|\psi|^2$ as a cloud surrounding the nucleus like those shown in Figs. 41.9 and 41.10. Then imagine many tiny spin arrows distributed throughout the cloud, either all with components in the $+z$-direction or all with components in the $-z$-direction. But don't take this picture too seriously.

Just as the orbital magnetic moment of the electron is proportional to its orbital angular momentum $\vec{L}$ (see Section 41.4), the electron's spin magnetic moment is proportional to its spin angular momentum $\vec{S}$. The z-component of the spin magnetic moment (μ_z) turns out to be related to S_z by

$$\mu_z = -(2.00232)\frac{e}{2m}S_z \qquad (41.38)$$

where $-e$ and m are (as usual) the charge and mass of the electron. When the atom is placed in a magnetic field, the interaction energy $-\vec{\mu} \cdot \vec{B}$ of the spin magnetic dipole moment with the field causes further splittings in energy levels and in the corresponding spectral lines.

Equation (41.38) shows that the gyromagnetic ratio for electron spin is approximately *twice* as great as the value $e/2m$ for *orbital* angular momentum and magnetic dipole moment. This result has no classical analog. But in 1928 Paul Dirac developed a relativistic generalization of the Schrödinger equation for electrons. His equation gave a spin gyromagnetic ratio of exactly $2(e/2m)$. It took another two decades to develop the area of physics called *quantum electrodynamics,* abbreviated QED, which predicts the value we've given to "only" six significant figures as 2.00232. QED now predicts a value that agrees with the currently accepted experimental value of 2.00231930436182(52), making QED the most precise theory in all science.

BIO APPLICATION Electron Spins and Dating Human Origins In many atoms, the net spin of all of the electrons is zero (as many electrons are "spin up" as are "spin down"). If these atoms are ionized and lose an electron, however, the net spin of the ion that remains is nonzero. This happens naturally in tooth enamel, where ionization is caused by radioactivity in the environment. The longer a tooth is exposed, the more ions are present. To find the age of fossil teeth, such as those in this skull of *Homo neanderthalensis*, a sample of the enamel is placed in a strong magnetic field. The ion spins align opposite to this field (become "spin down"). The sample is then illuminated with microwave photons of just the right energy to flip the spins to the higher-energy configuration aligned with the field ("spin up"). The amount of microwave energy absorbed in this process (called *electron spin resonance*) indicates the number of ions present and hence the age of the enamel.

EXAMPLE 41.6 Energy of electron spin in a magnetic field

WITH VARIATION PROBLEMS

Calculate the interaction energy for an electron in an $l = 0$ state in a magnetic field with magnitude 2.00 T.

IDENTIFY and SET UP For $l = 0$ the electron has zero orbital angular momentum and zero orbital magnetic moment. Hence the only magnetic interaction is that between the $\vec{B}$ field and the spin magnetic moment $\vec{\mu}$. From Eq. (41.28), the interaction energy is $U = -\vec{\mu} \cdot \vec{B}$. As in Section 41.4, we take $\vec{B}$ to be in the positive z-direction so that $U = -\mu_z B$ [Eq. (41.32)]. Equation (41.38) gives μ_z in terms of S_z, and Eq. (41.36) gives S_z.

EXECUTE Combining Eqs. (41.36) and (41.38), we have

$$\begin{aligned}\mu_z &= -(2.00232)\left(\frac{e}{2m}\right)\left(\pm\tfrac{1}{2}\hbar\right)\\ &= \mp\tfrac{1}{2}(2.00232)\left(\frac{e\hbar}{2m}\right) = \mp(1.00116)\mu_B\\ &= \mp(1.00116)(9.274 \times 10^{-24}\ \text{J/T})\\ &= \mp 9.285 \times 10^{-24}\ \text{J/T} = \mp 5.795 \times 10^{-5}\ \text{eV/T}\end{aligned}$$

Then from Eq. (41.32),

$$\begin{aligned}U = -\mu_z B &= \pm(9.285 \times 10^{-24}\ \text{J/T})(2.00\ \text{T})\\ &= \pm 1.86 \times 10^{-23}\ \text{J} = \pm 1.16 \times 10^{-4}\ \text{eV}\end{aligned}$$

The positive value of U and the negative value of μ_z correspond to $S_z = +\frac{1}{2}\hbar$ (spin up); the negative value of U and the positive value of μ_z correspond to $S_z = -\frac{1}{2}\hbar$ (spin down).

EVALUATE Let's check the *signs* of our results. If the electron is spin down, $\vec{S}$ points generally opposite to $\vec{B}$. Then the magnetic moment $\vec{\mu}$ (which is opposite to $\vec{S}$ because the electron charge is negative)

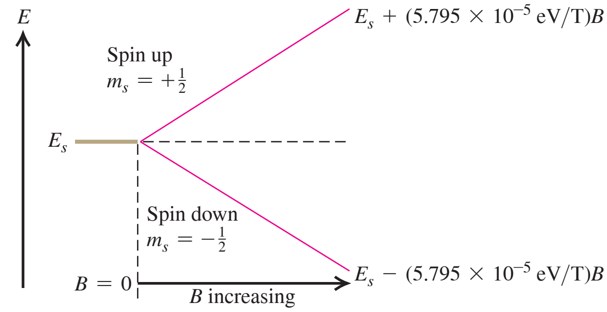
Continued

points generally parallel to $\vec{B}$, and μ_z is positive. From Eq. (41.28), $U = -\vec{\mu} \cdot \vec{B}$, the interaction energy is negative if $\vec{\mu}$ and $\vec{B}$ are parallel. Our results show that U is indeed negative in this case. We can similarly confirm that U must be positive and μ_z negative for a spin-up electron.

The red lines in **Fig. 41.18** show how the interaction energies for the two spin states vary with the magnetic-field magnitude B. The graphs are straight lines because, from Eq. (41.32), U is proportional to B.

KEYCONCEPT In addition to any orbital angular momentum, an electron has an intrinsic angular momentum $\vec{S}$ called spin. The component of $\vec{S}$ along a given axis can have only two possible values, $+\frac{1}{2}\hbar$ ("spin up") or $-\frac{1}{2}\hbar$ ("spin down"). An electron also has a spin magnetic moment $\vec{\mu}$ directed opposite to $\vec{S}$, so when it is placed in a magnetic field $\vec{B}$, the electron's energy depends on the component of $\vec{S}$ along the direction of $\vec{B}$.

Figure 41.18 An $l = 0$ level of a single electron is split by interaction of the spin magnetic moment with an external magnetic field. The greater the magnitude B of the magnetic field, the greater the splitting. The quantity 5.795×10^{-5} eV/T is just $(1.00116)\mu_B$.



Spin-Orbit Coupling

We mentioned earlier that the spin magnetic dipole moment also gives splitting of energy levels even when there is *no* external field. One cause involves the orbital motion of the electron. In the Bohr model, observers moving with the electron would see the positively charged nucleus revolving around them (just as to earthbound observers the sun seems to be orbiting the earth). This apparent motion of charge causes a magnetic field at the location of the electron, as measured in the electron's moving frame of reference. The resulting interaction with the spin magnetic moment causes a twofold splitting of this level, corresponding to the two possible orientations of electron spin.

Discussions based on the Bohr model can't be taken too seriously, but a similar result can be derived from the Schrödinger equation. The interaction energy U can be expressed in terms of the scalar product of the angular momentum vectors $\vec{L}$ and $\vec{S}$. This effect is called **spin-orbit coupling;** it is responsible for the small energy difference between the two closely spaced, lowest excited levels of sodium shown in Fig. 39.19a and for the corresponding doublet (589.0, 589.6 nm) in the spectrum of sodium.

EXAMPLE 41.7 An effective magnetic field

WITH VARIATION PROBLEMS

To six significant figures, the wavelengths of the two spectral lines that make up the sodium doublet are $\lambda_1 = 588.995$ nm and $\lambda_2 = 589.592$ nm. Calculate the effective magnetic field experienced by the electron in the $3p$ levels of the sodium atom.

IDENTIFY and SET UP The two lines in the sodium doublet result from transitions from the two $3p$ levels, which are split by spin-orbit coupling, to the $3s$ level, which is *not* split because it has $L = 0$. We picture the spin-orbit coupling as an interaction between the electron spin magnetic moment and an effective magnetic field due to the nucleus. This example is like Example 41.6 in reverse: There we were given B and found the difference between the energies of the two spin states, while here we use the energy difference to find the target variable B. The difference in energy between the two $3p$ levels is equal to the difference in energy between the two photons of the sodium doublet. We use this relationship and the results of Example 41.6 to determine B.

EXECUTE The energies of the two photons are $E_1 = hc/\lambda_1$ and $E_2 = hc/\lambda_2$. Here $E_1 > E_2$ because $\lambda_1 < \lambda_2$, so the difference in their energies is

$$\Delta E = \frac{hc}{\lambda_1} - \frac{hc}{\lambda_2} = hc\left(\frac{\lambda_2 - \lambda_1}{\lambda_2\lambda_1}\right)$$
$$= (4.136 \times 10^{-15}\ \text{eV}\cdot\text{s})(2.998 \times 10^{8}\ \text{m/s})$$
$$\times \frac{(589.592 \times 10^{-9}\ \text{m}) - (588.995 \times 10^{-9}\ \text{m})}{(589.592 \times 10^{-9}\ \text{m})(588.995 \times 10^{-9}\ \text{m})}$$
$$= 0.00213\ \text{eV} = 3.41 \times 10^{-22}\ \text{J}$$

This equals the energy difference between the two $3p$ levels. The spin-orbit interaction raises one level by 1.70×10^{-22} J (one-half of 3.41×10^{-22} J) and lowers the other by 1.70×10^{-22} J. From Example 41.6, the amount each state is raised or lowered is $|U| = (1.00116)\mu_B B$, so

$$B = \left|\frac{U}{(1.00116)\mu_B}\right| = \frac{1.70 \times 10^{-22}\ \text{J}}{9.28 \times 10^{-24}\ \text{J/T}} = 18.0\ \text{T}$$

EVALUATE The electron experiences a *very* strong effective magnetic field. To produce a steady, macroscopic field of this magnitude in the laboratory requires state-of-the-art electromagnets.

KEYCONCEPT An atomic electron with an orbital angular momentum around the nucleus experiences an effective magnetic field. This field interacts with the spin magnetic moment of the electron, so the electron energy depends on the relative orientation of its spin angular momentum and orbital angular momentum. This is called spin-orbit coupling.

Combining Orbital and Spin Angular Momenta

The orbital and spin angular momenta ($\vec{L}$ and $\vec{S}$, respectively) can combine in various ways. The vector sum of $\vec{L}$ and $\vec{S}$ is the *total* angular momentum $\vec{J}$:

$$\vec{J} = \vec{L} + \vec{S} \tag{41.39}$$

The possible values of the magnitude J are given in terms of a quantum number j, called the **total angular momentum quantum number:**

$$J = \sqrt{j(j+1)}\hbar \tag{41.40}$$

We can then have states in which $j = \left|l \pm \frac{1}{2}\right|$. The $l + \frac{1}{2}$ states correspond to the case in which the vectors $\vec{L}$ and $\vec{S}$ have parallel z-components; for the $l - \frac{1}{2}$ states, $\vec{L}$ and $\vec{S}$ have antiparallel z-components. For example, when $l = 1$, j can be $\frac{1}{2}$ or $\frac{3}{2}$. In another spectroscopic notation these p states are labeled $^2P_{1/2}$ and $^2P_{3/2}$, respectively. The superscript is the number of possible spin orientations, the letter P (now capitalized) indicates states with $l = 1$, and the subscript is the value of j. We used this scheme to label the energy levels of the sodium atom in Fig. 39.19a.

In addition to shifts in energy levels due to magnetic effects within the atom, there are shifts of the same magnitude due to relativistic corrections to the kinetic energy of the electron. (In the Bohr model, an electron in the $n = 1$ orbit of hydrogen moves at about 1% of the speed of light.) The term "fine structure" refers to the energy-level shifts caused by magnetic and relativistic effects together, as well as to the line splittings that result from these shifts. Including these effects, the energy levels of the hydrogen atom are

Energy levels of hydrogen, including fine structure

$$E_{n,j} = -\frac{13.60\text{ eV}}{n^2}\left[1 + \frac{\alpha^2}{n^2}\left(\frac{n}{j+\frac{1}{2}} - \frac{3}{4}\right)\right] \tag{41.41}$$

Fine-structure constant (α)

Principal quantum number ($n = 1, 2, 3, \ldots$)

Total angular momentum quantum number (j)

The *fine-structure constant* α that appears in Eq. (41.41) is a dimensionless number:

$$\alpha = \frac{1}{4\pi\epsilon_0}\frac{e^2}{\hbar c} = 7.2973525664(17) \times 10^{-3} \quad \text{(fine-structure constant)} \tag{41.42}$$

To five significant figures, $\alpha = 7.2974 \times 10^{-3} = 1/137.04$.

In Section 41.3 we found that the energy levels of the hydrogen atom are degenerate: All states that have the same principal quantum number n have the same energy. Our more complete treatment including fine structure shows that this degeneracy is removed: States with the same n but different values of the total angular momentum quantum number j have different energies. Example 41.8 illustrates this for the $n = 2$ levels of hydrogen.

EXAMPLE 41.8 Fine structure and spectral-line splitting

WITH VARIATION PROBLEMS

For an electron with orbital quantum number $l = 0$, the angular momentum is due to spin alone and the only possible value of the total angular momentum quantum number is $j = \frac{1}{2}$. If $l = 1$, two values are possible: $j = \frac{3}{2}$ (the spin and orbital angular momentum vectors are in roughly the same direction and so add together) and $j = \frac{1}{2}$ (the spin and orbital angular momentum vectors are in roughly opposite directions and so partially cancel). (a) Find the energies of a state of the electron in a hydrogen atom with $n = 2$, $l = 1$, $j = \frac{3}{2}$ (a $^2P_{3/2}$ state) and a state with $n = 2$, $l = 1$, $j = \frac{1}{2}$ (a $^2P_{1/2}$ state), and calculate the difference between the two energies. Which state has the higher energy? (b) Find the difference in wavelengths between (i) a photon emitted in a transition from a state with $n = 2$, $l = 1$, $j = \frac{3}{2}$ to a state with $n = 1$, $l = 0$, $j = \frac{1}{2}$ and (ii) a photon emitted in a transition from a state with $n = 2$, $l = 1$, $j = \frac{1}{2}$ to a state with $n = 1$, $l = 0$, $j = \frac{1}{2}$. Which photon has the longer wavelength?

IDENTIFY and SET UP In part (a) we use Eq. (41.41) to find the difference in energy between these two states, which have the same n value but different j values. The difference between the two energies is due to fine structure, so we expect this difference to be small. In part (b) both transitions end in the same state with $n = 1$, so we recognize from Section 39.3 that both are members of the Lyman series. If there were no fine structure, the two initial states would have the same energies and both photons would have the same energy E and hence the same wavelength $\lambda = hc/E$. But because the two initial states differ slightly in energy, the photons in the two transitions will have slightly different wavelengths.

Continued

EXECUTE (a) From Eq. (41.41), the energies of the two states are

$$E_{n=2,j=3/2} = -\frac{13.60\text{ eV}}{2^2}\left[1 + \frac{\alpha^2}{2^2}\left(\frac{2}{\frac{3}{2}+\frac{1}{2}} - \frac{3}{4}\right)\right]$$
$$= -3.40\text{ eV}\left(1 + \frac{\alpha^2}{16}\right)$$
$$E_{n=2,j=1/2} = -\frac{13.60\text{ eV}}{2^2}\left[1 + \frac{\alpha^2}{2^2}\left(\frac{2}{\frac{1}{2}+\frac{1}{2}} - \frac{3}{4}\right)\right]$$
$$= -3.40\text{ eV}\left(1 + \frac{5\alpha^2}{16}\right)$$

The fine-structure terms involving α^2 cause both states to have lower (more negative) energies than in the Bohr model, in which both states would have energy $E_2 = -3.40$ eV. The fine-structure term is five times greater for the $j = \frac{1}{2}$ state, so the $j = \frac{3}{2}$ state has the higher (less negative) energy. Using the value of the fine-structure constant α from Eq. (41.42), we get the difference in energy between the two states:

$$E_{n=2,j=3/2} - E_{n=2,j=1/2}$$
$$= \left[-3.40\text{ eV}\left(1 + \frac{\alpha^2}{16}\right)\right] - \left[-3.40\text{ eV}\left(1 + \frac{5\alpha^2}{16}\right)\right]$$
$$= 3.40\text{ eV}\left(\frac{4\alpha^2}{16}\right) = (3.40\text{ eV})\left(\frac{4}{16}\right)\left(\frac{1}{137.04}\right)^2$$
$$= 4.53 \times 10^{-5}\text{ eV}$$

(b) The photon energy in each case equals the difference between the energies of the initial and final states of the electron. The final electron state for both transitions has $n = 1$ and $j = \frac{1}{2}$, which from Eq. (41.41) has energy

$$E_{n=1,j=1/2} = -\frac{13.60\text{ eV}}{1^2}\left[1 + \frac{\alpha^2}{1^2}\left(\frac{1}{\frac{1}{2}+\frac{1}{2}} - \frac{3}{4}\right)\right]$$
$$= -13.60\text{ eV}\left(1 + \frac{\alpha^2}{4}\right)$$

Note that as for the two $n = 2$ states in part (a), the fine-structure correction to the $n = 1$ state makes the energy more negative. The photon energies for the two transitions are then

$$E_{\text{photon}}\left(n = 2, l = 1, j = \tfrac{3}{2} \text{ to } n = 1, l = 0, j = \tfrac{1}{2}\right)$$
$$= E_{n=2,j=3/2} - E_{n=1,j=1/2}$$
$$= \left[-3.40\text{ eV}\left(1 + \frac{\alpha^2}{16}\right)\right] - \left[-13.60\text{ eV}\left(1 + \frac{\alpha^2}{4}\right)\right]$$
$$= 10.20\text{ eV} + (3.40\text{ eV})\left(\frac{15\alpha^2}{16}\right) = 10.20\text{ eV} + 1.70 \times 10^{-4}\text{ eV}$$

$$E_{\text{photon}}\left(n = 2, l = 1, j = \tfrac{1}{2} \text{ to } n = 1, l = 0, j = \tfrac{1}{2}\right)$$
$$= E_{n=2,j=1/2} - E_{n=1,j=1/2}$$
$$= \left[-3.40\text{ eV}\left(1 + \frac{5\alpha^2}{16}\right)\right] - \left[-13.60\text{ eV}\left(1 + \frac{\alpha^2}{4}\right)\right]$$
$$= 10.20\text{ eV} + (3.40\text{ eV})\left(\frac{11\alpha^2}{16}\right) = 10.20\text{ eV} + 1.24 \times 10^{-4}\text{ eV}$$

The photon emitted when the initial state is $n = 2, l = 1, j = \frac{1}{2}$ has a lower energy E_{photon} and hence will have a longer wavelength, as given by the equation $\lambda = hc/E_{\text{photon}}$. If you plug the two photon energies into this equation, your calculator will tell you that $\lambda = 1.216 \times 10^{-7}$ m $= 121.6$ nm in both cases because the energy difference is so small. To find the wavelength difference $\Delta\lambda$, we instead take the differential of both sides of $\lambda = hc/E_{\text{photon}}$:

$$d\lambda = d\left(\frac{hc}{E_{\text{photon}}}\right) = -\frac{hc}{(E_{\text{photon}})^2}dE_{\text{photon}}$$
$$= -\left(\frac{hc}{E_{\text{photon}}}\right)\left(\frac{1}{E_{\text{photon}}}\right)dE_{\text{photon}} = -\frac{\lambda}{E_{\text{photon}}}dE_{\text{photon}}$$

The minus sign means that a *decrease* in photon energy corresponds to an *increase* in photon wavelength. Replacing $d\lambda$ with $\Delta\lambda$ (the wavelength difference that we seek) and dE_{photon} with ΔE_{photon}, we get the difference between the two photon wavelengths:

$$\Delta\lambda = -\frac{\lambda}{E_{\text{photon}}}\Delta E_{\text{photon}}$$

To four significant digits, we have $\lambda = 121.6$ nm and $E_{\text{photon}} = 10.20$ eV. We find the photon energy difference ΔE_{photon} from the two expressions above, subtracting the larger energy from the smaller one so that $\Delta\lambda$ is positive:

$$\Delta\lambda = -\frac{121.6\text{ nm}}{10.20\text{ eV}}\left\{\left[10.20\text{ eV} + (3.40\text{ eV})\left(\frac{11\alpha^2}{16}\right)\right] - \left[10.20\text{ eV} + (3.40\text{ eV})\left(\frac{15\alpha^2}{16}\right)\right]\right\}$$
$$= -\frac{121.6\text{ nm}}{10.20\text{ eV}}(3.40\text{ eV})\left(-\frac{4\alpha^2}{16}\right)$$
$$= \frac{121.6\text{ nm}}{10.20\text{ eV}}(3.40\text{ eV})\left(\frac{4}{16}\right)\left(\frac{1}{137.04}\right)^2$$
$$= 5.40 \times 10^{-4}\text{ nm}$$

EVALUATE This line splitting is very small, as predicted. Fine structure is fine indeed! It is nonetheless observable with a diffraction grating that has a sufficient number of lines (see Section 36.5). The measured wavelengths are 121.567364 nm for the transition that begins in the $j = \frac{1}{2}$ state and 121.566824 nm for the transition that begins in the $j = \frac{3}{2}$ state. These are ultraviolet wavelengths.

There are also states of the hydrogen atom with $n = 2, l = 0, j = \frac{1}{2}$. [From Eq. (41.41), these states have the same energy as those with $n = 2, l = 1, j = \frac{1}{2}$; the energy $E_{n,j}$ depends on n and j but not on l.] However, an electron in an $n = 2, l = 0, j = \frac{1}{2}$ state *cannot* emit a photon and transition to an $n = 1, l = 0, j = \frac{1}{2}$ state. Such a transition is forbidden by the selection rule that l must change by 1 when a photon is emitted (see Section 41.4).

KEYCONCEPT When both magnetic and relativistic effects are included, the energy levels of the hydrogen atom [Eq. (41.41)] depend on both the principal quantum number $n = 1, 2, 3, \ldots$ and the quantum number j associated with the electron's total angular momentum (the vector sum of the orbital and spin angular momenta). If the electron has orbital quantum number l, the value of j is either $l + \frac{1}{2}$ or $\left|l - \frac{1}{2}\right|$. Transitions between states that involve emitting or absorbing a photon are allowed only if l changes by 1 in the transition.

(a) Galaxies in visible light (negative image; galaxies appear dark)

(b) Radio image of the same galaxies at wavelength 21 cm

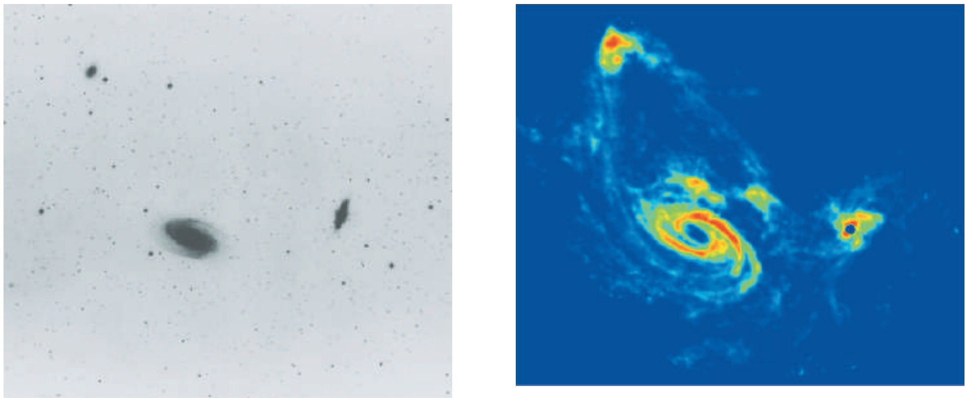


Figure **41.19** (a) In a visible-light image, these three distant galaxies appear to be unrelated. But in fact these galaxies are connected by immense streamers of hydrogen gas. This is revealed in (b), the false-color image made with a radio telescope tuned to the 21 cm wavelength emitted by hydrogen atoms.

Additional, much smaller splittings are associated with the fact that the *nucleus* of the atom has a magnetic dipole moment that interacts with the orbital and/or spin magnetic dipole moments of the electrons. These effects are called *hyperfine structure*. For example, the ground level of hydrogen is split into two states, separated by only 5.9×10^{-6} eV. The photon that is emitted in the transitions between these states has a wavelength of 21 cm. Radio astronomers use this wavelength to map clouds of interstellar hydrogen gas that are too cold to emit visible light (**Fig. 41.19**).

TEST YOUR UNDERSTANDING OF SECTION 41.5 In which of the following situations is the magnetic moment of an electron perfectly aligned with a magnetic field that points in the positive z-direction? (i) $m_s = +\frac{1}{2}$; (ii) $m_s = -\frac{1}{2}$; (iii) both (i) and (ii); (iv) neither (i) nor (ii).

ANSWER

(iv) For the magnetic moment to be perfectly aligned with the z-direction, the z-component of the spin vector $\vec{S}$ would have to have the same absolute value as $\vec{S}$. However, the possible values of S_z are $\pm\frac{1}{2}\hbar$ [Eq. (41.36)], while the magnitude of the spin vector is $S = \sqrt{\frac{3}{4}}\hbar$ [Eq. (41.37)]. Hence $\vec{S}$ can never be perfectly aligned with any one direction in space.

41.6 MANY-ELECTRON ATOMS AND THE EXCLUSION PRINCIPLE

So far our analysis of atomic structure has concentrated on the hydrogen atom. That's natural; neutral hydrogen, with only one electron, is the simplest atom. Let's now take what we've learned about the hydrogen atom and apply that knowledge to the more complicated case of many-electron atoms.

An atom in its normal (electrically neutral) state has Z electrons and Z protons. Recall from Section 41.3 that we call Z the *atomic number*. The total electric charge of such an atom is exactly zero because the neutron has no charge while the proton and electron charges have the same magnitude but opposite sign.

A complete understanding of such a general atom requires that we know the wave function that describes the behavior of all Z of its electrons. This wave function depends on $3Z$ coordinates (three for each electron), so its complexity increases very rapidly with increasing Z. What's more, each of the Z electrons interacts not only with the nucleus but also with every other electron. The potential energy is therefore a complicated function of all $3Z$ coordinates, and the Schrödinger equation contains second derivatives with respect to all of them. Finding exact solutions to such equations is such a complex task that it has not been successfully achieved even for the neutral helium atom, which has only two electrons.

Fortunately, various approximation schemes are available. The simplest approximation is to ignore all interactions between electrons and consider each electron as moving under the action only of the nucleus (considered to be a point charge). In this approximation, we write a separate wave function for each *individual* electron. Each such function is like that for the hydrogen atom, specified by four quantum numbers (n, l, m_l, m_s). The nuclear charge is Ze instead of e, so we replace every factor of e^2 in the wave functions and the energy levels by Ze^2. In particular, the energy levels are given by Eq. (41.21) with e^4 replaced by Z^2e^4:

$$E_n = -\frac{1}{(4\pi\epsilon_0)^2}\frac{m_r Z^2 e^4}{2n^2\hbar^2} = -\frac{Z^2}{n^2}(13.6\text{ eV}) \tag{41.43}$$

This approximation is fairly drastic; when there are many electrons, their interactions with each other are as important as the interaction of each with the nucleus. So this model isn't very useful for quantitative predictions.

The Central-Field Approximation

A less drastic and more useful approximation is to think of all the electrons together as making up a charge cloud that is, on average, *spherically symmetric*. We can then think of each individual electron as moving in the total electric field due to the nucleus and this averaged-out cloud of all the other electrons. There is a corresponding spherically symmetric potential-energy function $U(r)$. This picture is called the **central-field approximation;** it provides a useful starting point for understanding atomic structure.

In the central-field approximation we can again deal with one-electron wave functions. The Schrödinger equation for these functions differs from the equation for hydrogen, which we discussed in Section 41.3, only in that the $1/r$ potential-energy function is replaced by a different function $U(r)$. Now, Eqs. (41.20) show that $U(r)$ does not appear in the differential equations for $\Theta(\theta)$ and $\Phi(\phi)$. So those angular functions are exactly the same as for hydrogen, and the orbital angular momentum *states* are also the same as before. The quantum numbers l, m_l, and m_s have the same meanings as before, and Eqs. (41.22) and (41.23) again give the magnitude and z-component of the orbital angular momentum.

The radial wave functions and probabilities are different than for hydrogen because of the change in $U(r)$, so the energy levels are no longer given by Eq. (41.21). We can still label a state by using the four quantum numbers (n, l, m_l, m_s). In general, the energy of a state now depends on both n and l, rather than just on n as with hydrogen. (Due to fine-structure effects, the energy can also depend on the total angular momentum quantum number j. These effects are generally small, however, so we ignore them for this discussion.) The restrictions on the values of the quantum numbers are the same as before:

Allowed values of quantum numbers for one-electron wave functions:

$$\overset{\text{Principal quantum number}}{n \geq 1} \quad 0 \leq \underset{\text{Orbital quantum number}}{l} \leq n - 1 \quad \overset{\text{Orbital magnetic quantum number}}{|m_l| \leq l} \quad \overset{\text{Spin magnetic quantum number}}{m_s = \pm\tfrac{1}{2}} \tag{41.44}$$

The Exclusion Principle

To understand the structure of many-electron atoms, we need an additional principle, the *exclusion principle*. To see why this principle is needed, let's consider the lowest-energy state, or *ground state,* of a many-electron atom. In the one-electron states of the central-field model, there is a lowest-energy state (corresponding to an $n = 1$ state of hydrogen). We might expect that in the ground state of a complex atom, *all* the electrons should be in this lowest state. If so, then we should see only gradual changes in physical and chemical properties when we look at the behavior of atoms with increasing numbers of electrons (Z).

Such gradual changes are *not* what is observed. Instead, properties of elements vary widely from one to the next, with each element having its own distinct personality. For example, the elements fluorine, neon, and sodium have 9, 10, and 11 electrons, respectively, per atom. Fluorine $(Z = 9)$ is a *halogen;* it tends strongly to form compounds in which each fluorine atom acquires an extra electron. Sodium $(Z = 11)$ is an *alkali metal;* it forms compounds in which each sodium atom *loses* an electron. Neon $(Z = 10)$ is a *noble gas,* forming no naturally occurring compounds at all. Such observations show that in the ground state of a complex atom the electrons *cannot* all be in the lowest-energy states. But why not?

The key to this puzzle, discovered by the Austrian physicist Wolfgang Pauli (**Fig. 41.20**) in 1925, is called the **exclusion principle.** This principle states that **no two electrons can occupy the same quantum-mechanical state** in a given system. That is, **no two electrons in an atom can have the same values of all four quantum numbers (n, l, m_l, m_s).** Each quantum state corresponds to a certain distribution of the electron "cloud" in space. Therefore the principle also says, in effect, that no more than two electrons with opposite values of the quantum number m_s can occupy the same region of space. We shouldn't take this last statement too seriously because the electron probability functions don't have sharp, definite boundaries. But the exclusion principle limits the amount by which electron wave functions can overlap. Think of it as the quantum-mechanical analog of a university rule that allows only one student per desk. This same exclusion principle applies to all spin-$\frac{1}{2}$ particles, not just electrons. (We'll see in Chapter 43 that protons and neutrons are also spin-$\frac{1}{2}$ particles. As a result, the exclusion principle plays an important role in the structure of atomic nuclei.)

Figure **41.20** The key to understanding the periodic table of the elements was the discovery by Wolfgang Pauli (1900–1958) of the exclusion principle. Pauli received the 1945 Nobel Prize in physics for his accomplishment. This photo shows Pauli (on the left) and Niels Bohr watching the physics of a toy top spinning on the floor—a macroscopic analog of a microscopic electron with spin.

CAUTION **The meaning of the exclusion principle** Don't confuse the exclusion principle with the electrical repulsion between electrons. While both effects tend to keep electrons within an atom separated from each other, they are very different in character. Two electrons can always be pushed closer together by adding energy to combat electrical repulsion, but *nothing* can overcome the exclusion principle and force two electrons into the same quantum-mechanical state.

Table 41.2 lists some of the sets of quantum numbers for electron states in an atom. It's similar to Table 41.1 (Section 41.3), but we've added the number of states in each subshell and shell. Because of the exclusion principle, the "number of states" is the same as the *maximum* number of electrons that can be found in those states. For each state, m_s can be either $+\frac{1}{2}$ or $-\frac{1}{2}$.

As with the hydrogen wave functions, different states correspond to different spatial distributions; electrons with larger values of n are concentrated at larger distances from the nucleus. Figure 41.8 (Section 41.3) shows this effect. When an atom has more than two electrons, they can't all huddle down in the low-energy $n = 1$ states nearest to the nucleus because there are only two of these states; the exclusion principle forbids multiple occupancy of a state. Some electrons are forced into states farther away, with higher energies. Each value of n corresponds roughly to a region of space around the nucleus in the form of a spherical *shell*. Hence we speak of the K shell as the region that is occupied by the electrons in the $n = 1$ states, the L shell as the region of the $n = 2$ states, and so on. States with the same n but different l form *subshells,* such as the $3p$ subshell.

TABLE 41.2 Quantum States of Electrons in the First Four Shells

n	l	m_l	Spectroscopic Notation	Number of States		Shell
1	0	0	$1s$	2		K
2	0	0	$2s$	2	8	L
2	1	$-1, 0, 1$	$2p$	6		
3	0	0	$3s$	2	18	M
3	1	$-1, 0, 1$	$3p$	6		
3	2	$-2, -1, 0, 1, 2$	$3d$	10		
4	0	0	$4s$	2	32	N
4	1	$-1, 0, 1$	$4p$	6		
4	2	$-2, -1, 0, 1, 2$	$4d$	10		
4	3	$-3, -2, -1, 0, 1, 2, 3$	4f	14		

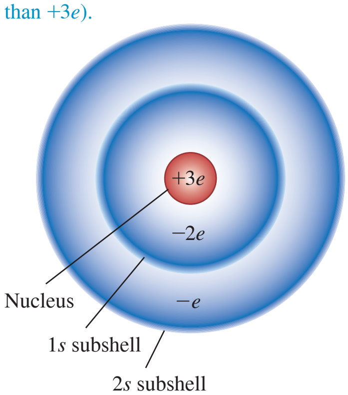
The Periodic Table

We can use the exclusion principle to derive the most important features of the structure and chemical behavior of multielectron atoms, including the periodic table of the elements. Let's imagine constructing a neutral atom by starting with a bare nucleus with Z protons and then adding Z electrons, one by one. To obtain the ground state of the atom as a whole, we fill the lowest-energy electron states (those closest to the nucleus, with the smallest values of n and l) first, and we use successively higher states until all the electrons are in place. The chemical properties of an atom are determined principally by interactions involving the outermost, or *valence,* electrons, so we particularly want to learn how these electrons are arranged.

Let's look at the ground-state electron configurations for the first few atoms (in order of increasing Z). For hydrogen the ground state is $1s$; the single electron is in a state $n = 1$, $l = 0$, $m_l = 0$, and $m_s = \pm\frac{1}{2}$. In the helium atom $(Z = 2)$, *both* electrons are in $1s$ states, with opposite spins; one has $m_s = -\frac{1}{2}$ and the other has $m_s = +\frac{1}{2}$. We denote the helium ground state as $1s^2$. (The superscript 2 is not an exponent; the notation $1s^2$ tells us that there are two electrons in the $1s$ subshell. Also, the superscript 1 is understood, as in $2s$.) For helium the K shell is completely filled, and all others are empty. Helium is a noble gas; it has no tendency to gain or lose an electron, and it forms no naturally occurring compounds.

Figure 41.21 Schematic representation of the charge distribution in a lithium atom. The nucleus has a charge of $+3e$.

On average, the $2s$ electron is considerably farther from the nucleus than the $1s$ electrons. Therefore, it experiences a net nuclear charge of approximately $+3e - 2e = +e$ (rather than $+3e$).



Lithium $(Z = 3)$ has three electrons. In its ground state, two are in $1s$ states and one is in a $2s$ state, so we denote the lithium ground state as $1s^2 2s$. On average, the $2s$ electron is considerably farther from the nucleus than are the $1s$ electrons (**Fig. 41.21**). According to Gauss's law, the *net* charge Q_{encl} attracting the $2s$ electron is nearer to $+e$ than to the value $+3e$ it would have without the two $1s$ electrons present. As a result, the $2s$ electron is loosely bound; only 5.4 eV is required to remove it, compared with the 30.6 eV given by Eq. (41.43) with $Z = 3$ and $n = 2$. Chemically, lithium is an *alkali metal*. It forms ionic compounds in which each lithium atom loses an electron and has a valence of $+1$.

Next is beryllium $(Z = 4)$; its ground-state configuration is $1s^2 2s^2$, with its two valence electrons filling the s subshell of the L shell. Beryllium is the first of the *alkaline earth* elements, forming ionic compounds in which the valence of the atoms is $+2$.

Table 41.3 shows the ground-state electron configurations of the first 30 elements. The L shell can hold eight electrons. At $Z = 10$, both the K and L shells are filled, and there are no electrons in the M shell. We expect this to be a particularly stable configuration, with little tendency to gain or lose electrons. This element is neon, a noble gas with no naturally occurring compounds. The next element after neon is sodium $(Z = 11)$, with filled K and L shells and one electron in the M shell. Its "noble-gas-plus-one-electron" structure resembles that of lithium; both are alkali metals. The element *before* neon is fluorine, with $Z = 9$. It has a vacancy in the L shell and has an affinity for an extra electron to fill the shell. Fluorine forms ionic compounds in which it has a valence of -1. This behavior is characteristic of the *halogens* (fluorine, chlorine, bromine, iodine, and astatine), all of which have "noble-gas-minus-one" configurations (**Fig. 41.22**).

Figure 41.22 Salt (sodium chloride, NaCl) dissolves readily in water, making seawater salty. This is due to the electron configurations of sodium and chlorine: Sodium can easily lose an electron to form an Na^+ ion, and chlorine can easily gain an electron to form a Cl^- ion. These ions are held in solution because they are attracted to the polar ends of water molecules (see Fig. 21.30a).

Proceeding down the list, we can understand the regularities in chemical behavior displayed by the **periodic table of the elements** (Appendix D) on the basis of electron configurations. The similarity of elements in each *group* (vertical column) of the periodic table is the result of similarity in outer-electron configuration. All the noble gases (helium, neon, argon, krypton, xenon, and radon) have filled-shell or filled-shell plus filled p subshell configurations. All the alkali metals (lithium, sodium, potassium, rubidium, cesium, and francium) have "noble-gas-plus-one" configurations. All the alkaline earth metals (beryllium, magnesium, calcium, strontium, barium, and radium) have "noble-gas-plus-two" configurations, and, as we just mentioned, all the halogens (fluorine, chlorine, bromine, iodine, and astatine) have "noble-gas-minus-one" structures.

A slight complication occurs with the M and N shells because the $3d$ and $4s$ subshell levels ($n = 3$, $l = 2$ and $n = 4$, $l = 0$, respectively) have similar energies. (We'll discuss in the next subsection why this happens.) Argon $(Z = 18)$ has all the $1s$, $2s$, $2p$, $3s$, and $3p$ subshells filled, but in potassium $(Z = 19)$ the additional electron goes into a $4s$ energy state rather than a $3d$ state (because the $4s$ state has slightly lower energy).

TABLE 41.3 Ground-State Electron Configurations

Element	Symbol	Atomic Number (Z)	Electron Configuration
Hydrogen	H	1	$1s$
Helium	He	2	$1s^2$
Lithium	Li	3	$1s^2 2s$
Beryllium	Be	4	$1s^2 2s^2$
Boron	B	5	$1s^2 2s^2 2p$
Carbon	C	6	$1s^2 2s^2 2p^2$
Nitrogen	N	7	$1s^2 2s^2 2p^3$
Oxygen	O	8	$1s^2 2s^2 2p^4$
Fluorine	F	9	$1s^2 2s^2 2p^5$
Neon	Ne	10	$1s^2 2s^2 2p^6$
Sodium	Na	11	$1s^2 2s^2 2p^6 3s$
Magnesium	Mg	12	$1s^2 2s^2 2p^6 3s^2$
Aluminum	Al	13	$1s^2 2s^2 2p^6 3s^2 3p$
Silicon	Si	14	$1s^2 2s^2 2p^6 3s^2 3p^2$
Phosphorus	P	15	$1s^2 2s^2 2p^6 3s^2 3p^3$
Sulfur	S	16	$1s^2 2s^2 2p^6 3s^2 3p^4$
Chlorine	Cl	17	$1s^2 2s^2 2p^6 3s^2 3p^5$
Argon	Ar	18	$1s^2 2s^2 2p^6 3s^2 3p^6$
Potassium	K	19	$1s^2 2s^2 2p^6 3s^2 3p^6 4s$
Calcium	Ca	20	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$
Scandium	Sc	21	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d$
Titanium	Ti	22	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^2$
Vanadium	V	23	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^3$
Chromium	Cr	24	$1s^2 2s^2 2p^6 3s^2 3p^6 4s 3d^5$
Manganese	Mn	25	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^5$
Iron	Fe	26	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^6$
Cobalt	Co	27	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^7$
Nickel	Ni	28	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^8$
Copper	Cu	29	$1s^2 2s^2 2p^6 3s^2 3p^6 4s 3d^{10}$
Zinc	Zn	30	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10}$

The next several elements have one or two electrons in the $4s$ subshell and increasing numbers in the $3d$ subshell. These elements are all metals with rather similar chemical and physical properties; they form the first *transition series,* starting with scandium $(Z = 21)$ and ending with zinc $(Z = 30)$, for which all the $3d$ and $4s$ subshells are filled.

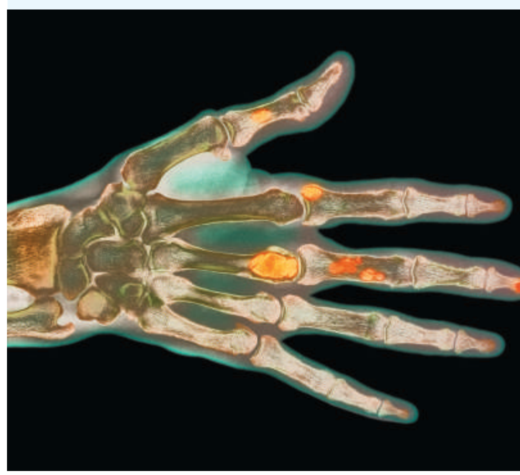
Something similar happens with $Z = 57$ through $Z = 71$, which have one or two electrons in the $6s$ subshell but only partially filled $4f$ and $5d$ subshells. These *rare earth* elements all have very similar physical and chemical properties. Another such series, called the *actinide* series, starts with $Z = 91$.

Screening

We have mentioned that in the central-field model, the energy levels depend on l as well as n. Let's take sodium $(Z = 11)$ as an example. If 10 of its electrons fill its K and L shells, the energies of some of the states for the remaining electron are found experimentally to be

$3s$ states: -5.138 eV
$3p$ states: -3.035 eV
$3d$ states: -1.521 eV
$4s$ states: -1.947 eV

BIO APPLICATION Electron Configurations and Bone Cancer Radiotherapy The orange spots in this colored x-ray image are bone cancer tumors. One method of treating bone cancer is to inject a radioactive isotope of strontium (^{89}Sr) into a patient's vein. Strontium is chemically similar to calcium because in both atoms the two outer electrons are in an s state (the structures are $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 5s^2$ for strontium and $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2$ for calcium). Hence the strontium is readily taken up by the tumors, where calcium turnover is more rapid than in healthy bone. Radiation from the strontium helps destroy the tumors.



The 3*s* states are the lowest (most negative); one is the ground state for the 11th electron in sodium. The energy of the 3*d* states is quite close to the energy of the $n = 3$ state in hydrogen. The surprise is that the 4*s* state energy is 0.426 eV *below* the 3*d* state, even though the 4*s* state has larger *n*.

We can understand these results by using Gauss's law (Section 22.3). For any spherically symmetric charge distribution, the electric-field magnitude at a distance *r* from the center is $Q_{encl}/4\pi\epsilon_0 r^2$, where Q_{encl} is the total charge enclosed within a sphere with radius *r*. Mentally remove the outer (valence) electron atom from a sodium atom. What you have left is a spherically symmetric collection of 10 electrons (filling the *K* and *L* shells) and 11 protons, so $Q_{encl} = -10e + 11e = +e$. If the 11th electron is completely outside this collection of charges, it is attracted by an effective charge of $+e$, not $+11e$. This is a more extreme example of the effect depicted in Fig. 41.21.

This effect is called **screening;** the 10 electrons *screen* 10 of the 11 protons in the sodium nucleus, leaving an effective net charge of $+e$. From the viewpoint of the 11th electron, this is equivalent to reducing the number of protons in the nucleus from $Z = 11$ to a smaller *effective atomic number* Z_{eff}. If the 11th electron is *completely* outside the charge distribution of the other 10 electrons, then $Z_{eff} = 1$. Since the probability distribution of the 11th electron does extend somewhat into those of the other electrons, in fact Z_{eff} is greater than 1 (but still much less than 11). In general, an electron that spends all its time completely outside a positive charge $Z_{eff}e$ has energy levels given by the hydrogen expression with e^2 replaced by $Z_{eff}e^2$. From Eq. (41.43) this is

Effective (screened) atomic number

Energy levels of an electron with screening

$$E_n = -\frac{Z_{eff}^{\,2}}{n^2}(13.6\text{ eV}) \qquad (41.45)$$

Principal quantum number

CAUTION **Different equations for different atoms** Equations (41.21), (41.43), and (41.45) all give values of E_n in terms of $(13.6\text{ eV})/n^2$, but they don't apply in general to the same atoms. Equation (41.21) is for hydrogen *only*. Equation (41.43) is for only the case in which there is no interaction with any other electron (and is thus accurate only when the atom has just one electron). Equation (41.45) is useful when one electron is screened from the nucleus by other electrons.

Now let's use the radial probability functions shown in Fig. 41.8 to explain why the energy of a sodium 3*d* state is approximately the same as the $n = 3$ value of hydrogen, -1.51 eV. The distribution for the 3*d* state (for which *l* has the maximum value $n - 1$) has one peak, and its most probable radius is *outside* the positions of the electrons with $n = 1$ or 2. (Those electrons also are pulled closer to the nucleus than in hydrogen because they are less effectively screened from the positive charge 11*e* of the nucleus.) Thus in sodium a 3*d* electron spends most of its time well outside the $n = 1$ and $n = 2$ states (the *K* and *L* shells). The 10 electrons in these shells screen about ten-elevenths of the charge of the 11 protons, leaving a net charge of about $Z_{eff}e = (1)e$. Then, from Eq. (41.45), the corresponding energy is approximately $-(1)^2(13.6\text{ eV})/3^2 = -1.51$ eV. This approximation is very close to the experimental value of -1.521 eV.

Looking again at Fig. 41.8, we see that the radial probability density for the 3*p* state (for which $l = n - 2$) has two peaks and that for the 3*s* state $(l = n - 3)$ has three peaks. For sodium the first small peak in the 3*p* distribution gives a 3*p* electron a higher probability (compared to the 3*d* state) of being *inside* the charge distributions for the electrons in the $n = 2$ states. That is, a 3*p* electron is less completely screened from the nucleus than is a 3*d* electron because it spends some of its time within the filled *K* and *L* shells. Thus for the 3*p* electrons, Z_{eff} is greater than unity. From Eq. (41.45) the 3*p* energy is lower (more negative) than the 3*d* energy of -1.521 eV. The actual value is -3.035 eV. A 3*s* electron spends even more time within the inner electron shells than a 3*p* electron does, giving an even larger Z_{eff} and an even more negative energy.

This discussion shows that the energy levels given by Eq. (41.45) depend on both the principal quantum number n and the orbital quantum number l. That's because the value of Z_{eff} is different for the 3s state ($n = 3, l = 0$), the 3p state ($n = 3, l = 1$), and the 3d state ($n = 3, l = 2$).

EXAMPLE 41.9 Determining Z_{eff} experimentally

The measured energy of a 3s state of sodium is -5.138 eV. Calculate the value of Z_{eff}.

IDENTIFY and SET UP Sodium has a single electron in the M shell outside filled K and L shells. The ten K and L electrons partially screen the single M electron from the $+11e$ charge of the nucleus; our goal is to determine the extent of this screening. We are given $n = 3$ and $E_n = -5.138$ eV, so we can use Eq. (41.45) to determine Z_{eff}.

EXECUTE Solving Eq. (41.45) for Z_{eff}, we have

$$Z_{\text{eff}}{}^2 = -\frac{n^2 E_n}{13.6\text{ eV}} = -\frac{3^2(-5.138\text{ eV})}{13.6\text{ eV}} = 3.40$$

$$Z_{\text{eff}} = 1.84$$

EVALUATE The effective charge attracting a 3s electron is 1.84e. Sodium's 11 protons are screened by an average of $11 - 1.84 = 9.16$ electrons instead of 10 electrons because the 3s electron spends some time within the inner (K and L) shells.

Each alkali metal (lithium, sodium, potassium, rubidium, and cesium) has one more electron than the corresponding noble gas (helium, neon, argon, krypton, and xenon). This extra electron is mostly outside the other electrons in the filled shells and subshells. Therefore all the alkali metals behave similarly to sodium.

KEYCONCEPT If a many-electron atom has just one outer valence electron, the valence electron's energy levels are like those of a single-electron atom but with the atomic number Z of the nucleus replaced by an effective atomic number Z_{eff}. The value of Z_{eff} is less than Z because the inner electrons partially "screen" the electric field of the nucleus.

EXAMPLE 41.10 Energies for a valence electron

The valence electron in potassium has a 4s ground state. Calculate the approximate energy of the $n = 4$ state having the smallest Z_{eff}, and discuss the relative energies of the 4s, 4p, 4d, and 4f states.

IDENTIFY and SET UP The state with the smallest Z_{eff} is the one in which the valence electron spends the most time outside the inner filled shells and subshells, so that it is most effectively screened from the charge of the nucleus. Once we have determined which state has the smallest Z_{eff}, we can use Eq. (41.45) to determine the energy of this state.

EXECUTE A 4f state has $n = 4$ and $l = 3 = 4 - 1$. Thus it is the state of greatest orbital angular momentum for $n = 4$, and thus the state in which the electron spends the most time outside the electron charge clouds of the inner filled shells and subshells. This makes Z_{eff} for a 4f state close to unity. Equation (41.45) then gives

$$E_4 = -\frac{Z_{\text{eff}}{}^2}{n^2}(13.6\text{ eV}) = -\frac{1}{4^2}(13.6\text{ eV}) = -0.85\text{ eV}$$

This approximation agrees with the measured energy of the sodium 4f state to the precision given.

An electron in a 4d state spends a bit more time within the inner shells, and its energy is therefore a bit more negative (measured to be -0.94 eV). For the same reason, a 4p state has an even lower energy (measured to be -2.73 eV) and a 4s state has the lowest energy (measured to be -4.339 eV).

EVALUATE We can extend this analysis to the *singly ionized alkaline earth elements:* Be^+, Mg^+, Ca^+, Sr^+, and Ba^+. For any allowed value of n, the highest-l state ($l = n - 1$) of the one remaining outer electron sees an effective charge of almost $+2e$, so for these states, $Z_{\text{eff}} = 2$. A 3d state for Mg^+, for example, has an energy of about $-2^2(13.6\text{ eV})/3^2 = -6.0$ eV.

KEYCONCEPT For a many-electron atom with just one outer valence electron, if that electron is in a state of high orbital angular momentum it "sees" the effective atomic number Z_{eff} of the nucleus to be 1. In such a state the valence electron is almost always outside of the other $Z - 1$ electrons, which "screen" $(Z - 1)e$ of the nuclear charge $+Ze$.

TEST YOUR UNDERSTANDING OF SECTION 41.6 If electrons did *not* obey the exclusion principle, would it be easier or more difficult to remove the first electron from sodium?

ANSWER

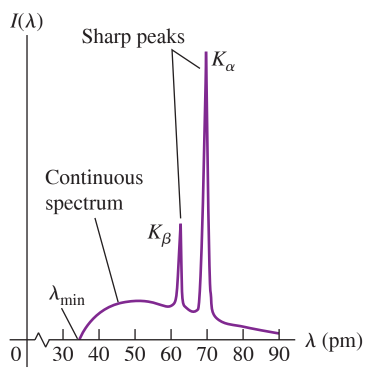
| **more difficult** If there were no exclusion principle, all 11 electrons in the sodium atom would be in the level of lowest energy (the 1s level) and the configuration would be $1s^{11}$. Consequently, it would be more difficult to remove the first electron. (In a real sodium atom the valence electron is in a screened 3s state, which has a comparatively high energy.)

41.7 X-RAY SPECTRA

X-ray spectra provide an example of the richness and power of the model of atomic structure that we derived in the preceding section. In Section 38.2 we discussed how x-ray photons are produced when electrons strike a metal target (see Fig. 38.7). In this section we'll see how the spectrum of x rays produced in this way depends on the type of metal used in the target and how the ideas of atomic energy levels and screening help us understand this dependence.

Characteristic X Rays and Atomic Energy Levels

Figure 41.23 Graph of intensity per unit wavelength as a function of wavelength for x rays produced with an accelerating voltage of 35 kV and a molybdenum target. The curve is a smooth function similar to the bremsstrahlung spectra in Fig. 38.8 (Section 38.2), but with two sharp spikes corresponding to part of the characteristic x-ray spectrum for molybdenum.



X-ray diffraction techniques (see Section 36.6) make it possible to measure x-ray wavelengths quite precisely (to within 0.1% or less). **Figure 41.23** shows the spectrum of x rays produced when fast-moving electrons strike a target of the metal molybdenum. This spectrum has two features:

1. There is a *continuous* spectrum of wavelengths (see Fig. 38.8 in Section 38.2), with a minimum wavelength (corresponding to a maximum frequency and a maximum photon energy) that is determined by the voltage V_{AC} used to accelerate the electrons. As we saw in Section 38.2, this continuous spectrum is due to *bremsstrahlung,* in which the electrons slow down as they interact with the metal atoms in the target and convert their kinetic energy into photons. The minimum wavelength λ_{min} corresponds to all of the kinetic energy eV_{AC} of the electron being converted into the energy of a single photon of energy hc/λ_{min}, so

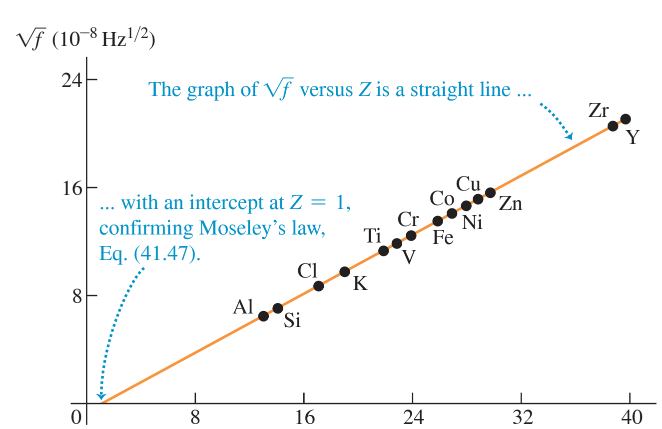
$$\lambda_{min} = \frac{hc}{eV_{AC}} \tag{41.46}$$

   This continuous-spectrum radiation is nearly independent of the target material in the x-ray tube.

2. Depending on the accelerating voltage, sharp peaks may be superimposed on the continuous bremsstrahlung spectrum, as in Fig. 41.23. These peaks are caused when the target atoms are struck by high-energy electrons and emit x rays of very definite wavelengths. Unlike the continuous spectrum, the wavelengths of the peaks are *different* for different target elements; they form what is called a *characteristic x-ray spectrum* for each target element.

In 1913 the English physicist Henry G. J. Moseley undertook a careful experimental study of characteristic x-ray spectra. He found that the most intense short-wavelength line in the characteristic x-ray spectrum from a particular target element, called the K_α line, varied smoothly with that element's atomic number Z (**Fig. 41.24**). This is in sharp contrast to optical spectra, in which elements with adjacent Z-values have spectra that often bear no resemblance to each other.

Figure 41.24 The square root of Moseley's measured frequencies of the K_α line for 14 elements.



Moseley found that the relationship could be expressed in terms of x-ray frequencies f by a simple formula called *Moseley's law:*

Frequency of K_α line in characteristic x-ray spectrum of an element

Moseley's law: $$f = (2.48 \times 10^{15}\ \text{Hz})(Z - 1)^2 \tag{41.47}$$

Atomic number of element

Moseley went far beyond this empirical relationship; he showed how characteristic x-ray spectra could be understood on the basis of energy levels of atoms in the target. His analysis was based on the Bohr model, published in the same watershed year of 1913. We'll recast it somewhat, using the ideas of atomic structure that we discussed in Section 41.6. First recall that the *outer* electrons of an atom are responsible for optical spectra. Their excited states are usually only a few electron volts above their ground state. In transitions from excited states to the ground state, they usually emit photons in or near the visible region.

Characteristic x rays, by contrast, are emitted in transitions involving the *inner* shells of a complex atom. In an x-ray tube the electrons may strike the target with enough energy to knock electrons out of the inner shells of the target atoms. These inner electrons are much closer to the nucleus than are the electrons in the outer shells; they are much more tightly bound, and hundreds or thousands of electron volts may be required to remove them.

Suppose one electron is knocked out of the K shell. This process leaves a vacancy, which we'll call a *hole*. (One electron remains in the K shell.) The hole can then be filled by an electron falling in from one of the outer shells, such as the $L, M, N, \ldots$ shell. This transition is accompanied by a decrease in the energy of the atom (because *less* energy would be needed to remove an electron from an $L, M, N, \ldots$ shell), and an x-ray photon is emitted with energy equal to this decrease. Each state has definite energy, so the emitted x rays have definite wavelengths; the emitted spectrum is a *line* spectrum.

We can estimate the energy and frequency of K_α x-ray photons by using the concept of screening from Section 41.6. A K_α x-ray photon is emitted when an electron in the L shell $(n = 2)$ drops down to fill a hole in the K shell $(n = 1)$. As the electron drops down, it is attracted by the Z protons in the nucleus screened by the one remaining electron in the K shell. We therefore approximate the energy by Eq. (41.45), with $Z_{\text{eff}} = Z - 1$, $n_{\text{i}} = 2$, and n_{f}. The energy before the transition is

$$E_i \approx -\frac{(Z-1)^2}{2^2}(13.6\ \text{eV}) = -(Z-1)^2(3.4\ \text{eV})$$

and the energy after the transition is

$$E_{\text{f}} \approx -\frac{(Z-1)^2}{1^2}(13.6\ \text{eV}) = -(Z-1)^2(13.6\ \text{eV})$$

$E_{K_\alpha} = E_{\text{i}} - E_{\text{f}} \approx (Z-1)^2(-3.4\ \text{eV} + 13.6\ \text{eV})$ is the energy of the K_α x-ray photon. That is,

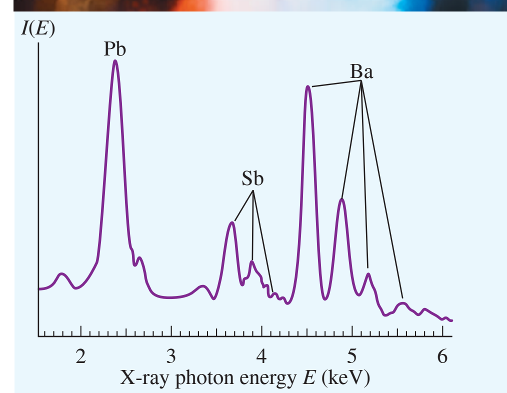
$$E_{K_\alpha} \approx (Z-1)^2(10.2\ \text{eV}) \tag{41.48}$$

The frequency of the photon is its energy divided by Planck's constant:

$$f = \frac{E_{K_\alpha}}{h} \approx \frac{(Z-1)^2(10.2\ \text{eV})}{4.136 \times 10^{-15}\ \text{eV}\cdot\text{s}} = (2.47 \times 10^{15}\ \text{Hz})(Z-1)^2$$

This relationship agrees almost exactly with Moseley's experimental law, Eq. (41.47). Indeed, considering the approximations we have made, the agreement is better than we have a right to expect. But our calculation does show how Moseley's law can be understood on the bases of screening and transitions between energy levels.

APPLICATION X Rays in Forensic Science X-ray spectra are an important tool in crime analysis. When a handgun is fired, a cloud of gunshot residue (GSR) is ejected from the barrel. The x-ray emission spectrum of GSR includes characteristic peaks from lead (Pb), antimony (Sb), and barium (Ba). If a sample taken from a suspect's skin or clothing has an x-ray emission spectrum with these characteristics, it indicates that the suspect recently fired a gun.



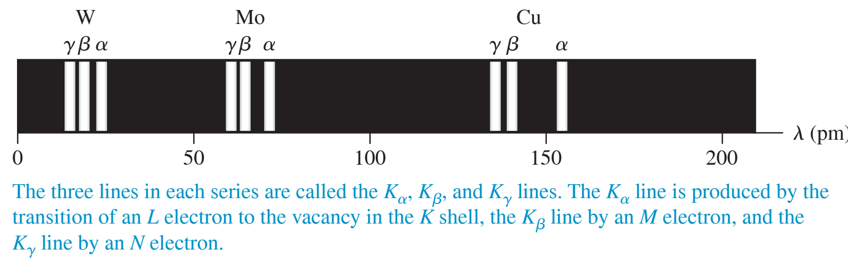


Figure 41.25 Wavelengths of the K_α, K_β, and K_γ lines of tungsten (W), molybdenum (Mo), and copper (Cu).

The hole in the K shell may also be filled by an electron falling from the M or N shell, assuming that these are occupied. If so, the x-ray spectrum of a large group of atoms of a single element shows a series, named the K series, of three lines, called the K_α, K_β, and K_γ lines. These three lines result from transitions in which the K-shell hole is filled by an L, M, or N electron, respectively. **Figure 41.25** shows the K series for tungsten $(Z = 74)$, molybdenum $(Z = 42)$, and copper $(Z = 29)$.

There are other series of x-ray lines, called the L, M, and N series, that are produced after the ejection of electrons from the L, M, and N shells rather than the K shell. Electrons in these outer shells are farther away from the nucleus and are not held as tightly as are those in the K shell, so removing these outer electrons requires less energy. Hence the x-ray photons that are emitted when these vacancies are filled have lower energy than those in the K series.

EXAMPLE 41.11 Chemical analysis by x-ray emission

You measure the K_α wavelength for an unknown element, obtaining the value 0.0709 nm. What is the element?

IDENTIFY and SET UP To determine which element this is, we need to know its atomic number Z. We can find this by using Moseley's law, which relates the frequency of an element's K_α x-ray emission line to that element's atomic number Z. We'll use the relationship $f = c/\lambda$ to calculate the frequency for the K_α line, and then use Eq. (41.47) to find the corresponding value of the atomic number Z. We'll then consult the periodic table (Appendix D) to determine which element has this atomic number.

EXECUTE The frequency is

$$f = \frac{c}{\lambda} = \frac{3.00 \times 10^8\ \text{m/s}}{0.0709 \times 10^{-9}\ \text{m}} = 4.23 \times 10^{18}\ \text{Hz}$$

Solving Moseley's law for Z, we get

$$Z = 1 + \sqrt{\frac{f}{2.48 \times 10^{15}\ \text{Hz}}} = 1 + \sqrt{\frac{4.23 \times 10^{18}\ \text{Hz}}{2.48 \times 10^{15}\ \text{Hz}}} = 42.3$$

We know that Z has to be an integer; we conclude that $Z = 42$, corresponding to the element molybdenum.

EVALUATE If you're worried that our calculation did not give an integer for Z, remember that Moseley's law is an empirical relationship. There are slight variations from one atom to another due to differences in the structure of the electron shells. Nonetheless, this example suggests the power of Moseley's law.

Niels Bohr commented that it was Moseley's observations, not the alpha-particle scattering experiments of Rutherford, Geiger, and Marsden (see Section 39.2), that truly convinced physicists that the atom consists of a positive nucleus surrounded by electrons in motion. Unlike Bohr or Rutherford, Moseley did not receive a Nobel Prize for his important work; these awards are given to living scientists only, and Moseley (at age 27) was killed in combat during the First World War.

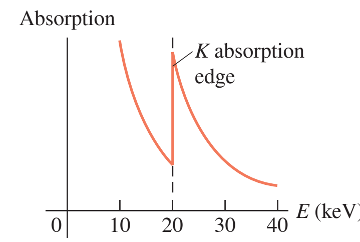
KEYCONCEPT If an electron is kicked out of the innermost, or K, shell of an atom of atomic number Z, then an electron from the next shell (L) drops down to the K shell to fill in the vacancy and emits an x-ray photon in the process. The frequency of this photon is proportional to $(Z - 1)^2$ (Moseley's law).

X-Ray Absorption Spectra

We can also observe x-ray *absorption* spectra. Unlike optical spectra, the absorption wavelengths are usually not the same as those for emission, especially in many-electron atoms, and do not give simple line spectra. For example, the K_α emission line results from a transition from the L shell to a hole in the K shell. The reverse transition doesn't occur in atoms with $Z \geq 10$ because in the atom's ground state, there is no vacancy in the L shell. To be absorbed, a photon must have enough energy to move an electron to an empty state. Since empty states are only a few electron volts in energy below the free-electron continuum, the minimum absorption energies in many-electron atoms are about the same as the minimum energies that are needed to remove an electron from its shell. Experimentally, if we gradually increase the accelerating voltage and hence the maximum photon energy, we observe sudden increases in absorption when we reach these minimum energies. These sudden jumps of absorption are called *absorption edges* (**Fig. 41.26**).

Characteristic x-ray spectra provide a very useful analytical tool. Satellite-borne x-ray spectrometers are used to study x-ray emission lines from highly excited atoms in distant astronomical sources. X-ray spectra are also used in air-pollution monitoring and in studies of the abundance of various elements in rocks.

Figure **41.26** When a beam of x rays is passed through a slab of molybdenum, the extent to which the beam is absorbed depends on the energy E of the x-ray photons. A sharp increase in absorption occurs at the K absorption edge at 20 keV. The increase occurs because photons with energies above this value can excite an electron from the K shell of a molybdenum atom into an empty state.



TEST YOUR UNDERSTANDING OF SECTION 41.7 A beam of photons is passed through a sample of high-temperature atomic hydrogen. At what photon energy would you expect there to be an absorption edge like that shown in Fig. 41.26? (i) 13.60 eV; (ii) 3.40 eV; (iii) 1.51 eV; (iv) all of these; (v) none of these.

ANSWER

(iv) An absorption edge appears if the photon energy is just high enough to remove an electron in a given energy level from the atom. In a sample of high-temperature hydrogen we expect to find atoms whose electron is in the ground level $(n = 1)$, the first excited level $(n = 2)$, and the second excited level $(n = 3)$. From Eq. (41.21) these levels have energies $E_n = (-13.60\text{ eV})/n^2 = -13.60\text{ eV}$, -3.40 eV, and -1.51 eV (see Fig. 39.24b).

41.8 QUANTUM ENTANGLEMENT

We've seen that quantum mechanics is very successful at correctly predicting the results of experiments. As we'll see in the remaining chapters, quantum mechanics is the basis of all electronic devices and is essential to our understanding of atomic nuclei and subatomic particles. But even though the central ideas of quantum mechanics have been established for decades, some aspects of the theory continue to baffle physicists and remain topics of ongoing research. We close this chapter with a discussion of one of these topics, *quantum entanglement*.

The Wave Function for Two Identical Particles

To understand what is meant by "quantum entanglement," let's consider how to write the wave function for two identical particles, such as two electrons (the electrons in the neutral helium atom, for instance). We'll use the subscripts 1 and 2 to refer to these particles.

As we discussed in Section 41.6, a wave function that describes both particles is a function of both the coordinates (x_1, y_1, z_1) of particle 1 and the coordinates (x_2, y_2, z_2) of particle 2. As a shorthand, we'll use the position vectors $\vec{r}_1 = x_1\hat{\imath} + y_1\hat{\jmath} + z_1\hat{k}$ and $\vec{r}_2 = x_2\hat{\imath} + y_2\hat{\jmath} + z_2\hat{k}$. In terms of these vectors, we can write the time-dependent two-particle wave function as $\Psi(\vec{r}_1, \vec{r}_2, t)$. Just as for a single particle, if the system of two particles has total energy E, we can write this time-dependent wave function as the product of a time-*independent* two-particle wave function $\psi(\vec{r}_1, \vec{r}_2)$ and a factor that depends on time only:

$$\Psi(\vec{r}_1, \vec{r}_2, t) = \psi(\vec{r}_1, \vec{r}_2)e^{-iEt/\hbar} \tag{41.49}$$

We saw in Section 41.2 that a technique called *separation of variables* is useful for expressing a wave function that depends on several variables as a product of functions of the individual variables. Let's see whether we can express the time-independent two-particle wave function $\psi(\vec{r}_1, \vec{r}_2)$ in Eq. (41.49) as a product of a function of $\vec{r}_1$ and a function of $\vec{r}_2$. We interpret the function of $\vec{r}_1$ to be the *single*-particle wave function for particle 1 and the function of $\vec{r}_2$ to be the single-particle wave function for particle 2. Suppose particle 1 is in a state A, for which the single-particle wave function is ψ_A, and particle 2 is in a state B, for which the single-particle wave function is ψ_B. [In the helium atom, with two electrons, one electron could be in the spin-up state ($n = 1$, $l = 0$, $m_l = 0$, and $m_s = +\frac{1}{2}$) and the other could be in the spin-down state ($n = 1$, $l = 0$, $m_l = 0$, and $m_s = -\frac{1}{2}$).] Using separation of variables, we would write

$$\psi(\vec{r}_1, \vec{r}_2) = \psi_A(\vec{r}_1)\psi_B(\vec{r}_2) \tag{41.50}$$

(first guess at the two-particle wave function)

However, Eq. (41.50) *cannot* be correct, because particles 1 and 2 are *identical* and *indistinguishable*. We may be able to state with confidence that one particle is in state A and the other is in state B, but it's impossible to specify which particle is in which state. (There's no way even in principle to "tag" the particles.)

To account for this, let's make an improved guess for the two-particle wave function: a combination of two terms like Eq. (41.50)—one term for which particle 1 is in state A and particle 2 is in state B, and one term for which particle 1 is in state B and particle 2 is in state A. Our improved guess is then

$$\psi(\vec{r}_1, \vec{r}_2) = \frac{1}{\sqrt{2}}[\psi_A(\vec{r}_1)\psi_B(\vec{r}_2) \pm \psi_B(\vec{r}_1)\psi_A(\vec{r}_2)] \tag{41.51}$$

(second guess at the two-particle wave function)

The factor $1/\sqrt{2}$ in Eq. (41.51) ensures that if ψ_A and ψ_B are normalized, then $\psi(\vec{r}_1, \vec{r}_2)$ will be normalized as well. Note that the terms $\psi_A(\vec{r}_1)\psi_B(\vec{r}_2)$ and $\psi_B(\vec{r}_1)\psi_A(\vec{r}_2)$ appear with equal magnitudes, so that the two possibilities (particle 1 in A and particle 2 in B, or particle 1 in B and particle 2 in A) are equally probable.

How can we decide whether the $\pm$ sign in Eq. (41.51) should be a plus or a minus? If the particles are two electrons, or two of any other type of spin-$\frac{1}{2}$ particle, the Pauli exclusion principle (Section 41.6) tells us that we must use the minus sign:

$$\psi(\vec{r}_1, \vec{r}_2) = \frac{1}{\sqrt{2}}[\psi_A(\vec{r}_1)\psi_B(\vec{r}_2) - \psi_B(\vec{r}_1)\psi_A(\vec{r}_2)] \tag{41.52}$$

(two-particle wave function, spin-$\frac{1}{2}$ particles)

To check this, suppose we demand that *both* particles be in the same state A. Then we would replace ψ_B in Eq. (41.52) with ψ_A:

$$\psi(\vec{r}_1, \vec{r}_2) = \frac{1}{\sqrt{2}}[\psi_A(\vec{r}_1)\psi_A(\vec{r}_2) - \psi_A(\vec{r}_1)\psi_A(\vec{r}_2)] = 0$$

The zero value of the wave function says that our demand cannot be met. This is in agreement with the Pauli exclusion principle: No two electrons, and indeed no two identical spin-$\frac{1}{2}$ particles of any kind, can occupy the same quantum-mechanical state.

Measurement and "Spooky Action at a Distance"

The wave function in Eq. (41.52) shares features with that of the quantum-mechanical particle in a box that we discussed in Section 40.6. That particle's wave function is also a combination of two terms of equal magnitude representing different situations: one in which the momentum of the particle is in the $+x$-direction, so $p_x > 0$, and one in which its momentum is in the $-x$-direction, so $p_x < 0$. We saw in Section 40.6 that if we *measured* the momentum of the particle, we would get either $p_x > 0$ or $p_x < 0$. Making such a measurement causes the wave function to *collapse,* and only the term corresponding to the measured value of p_x survives. The other term disappears from the wave function.

The same sort of wave-function collapse happens in our two-particle system. Suppose we make a measurement of one particle—call it particle 1—and determine that it is in state A. The measurement causes the wave function in Eq. (41.52) to collapse to $\psi(\vec{r}_1, \vec{r}_2) = \psi_A(\vec{r}_1)\psi_B(\vec{r}_2)$. This wave equation corresponds to particle 1 being in state A but also corresponds to particle 2 being in state B. It follows that, after the measurement, particle 2 *must* be in state B, even though we have not directly measured the state of particle 2. In other words, making a measurement on *one* particle affects the state of the *other* particle. Schrödinger described this situation by saying that the two particles are *entangled*.

If the two entangled particles are the two electrons within a helium atom, the idea that their states are entangled may not seem troubling. After all, these electrons are in very close proximity (a helium atom is only about 0.1 nm in diameter) and exert substantial electric forces on each other. You might imagine that when we measure electron 1 to be in state *A* (say, spin up with $m_s = +\frac{1}{2}$), it exerts forces on electron 2 that require electron 2 to be in state *B* (say, spin down with $m_s = -\frac{1}{2}$).

But suppose we arrange for two identical particles to be in an entangled state in which the particles are *not* close to each other, so they cannot exert forces on each other. When the same kind of measurement experiment is done on such a distant pair of entangled particles, the result is the same as if they are close together: If we measure particle 1 to be in state *A* and subsequently make a measurement on particle 2, we always find that particle 2 is in state *B*. If instead we measure particle 1 to be in state *B* and then make a measurement on particle 2, we always find that particle 2 is in state *A*. So measuring the state of one particle affects the state of the other particle, even when the two particles *cannot* exert forces on each other (**Fig. 41.27**). This finding has been confirmed with entangled particles that are *more than 1200 km apart!* (These experiments with very large distances are done with photons rather than electrons. Like electrons, photons have spin, and the "spin-up" and "spin-down" states correspond to left and right circular polarization. The only difference is that photons are spin-1 particles, not spin-$\frac{1}{2}$, and do not obey the Pauli exclusion principle. As a result, we must use the plus sign rather than the minus sign in Eq. (41.51) to describe two entangled photons. The rest of the physics is identical, however.)

These results contradict the idea of *locality*—the notion that a particle responds to forces or fields that act at its position only, not at some other point in space. We used locality in Chapters 4 and 13 when we expressed the gravitational force on a particle of mass m as $\vec{F}_g = m\vec{g}$, where $\vec{g}$ is the acceleration due to gravity at the point in space where the particle is located. We used locality again in Chapters 21 and 27 when we wrote the electric force $\vec{F}_E$ and the magnetic force $\vec{F}_B$ on a particle of charge q moving with velocity $\vec{v}$ as $\vec{F}_E = q\vec{E}$ and $\vec{F}_B = q\vec{v} \times \vec{B}$, where $\vec{E}$ and $\vec{B}$, respectively, are the electric and magnetic fields at the position of the particle. But the interaction between two entangled, widely separated particles seems *not* to obey locality. For this reason, Albert Einstein referred to the results of an experiment like that shown in Fig. 41.27 as "spooky action at a distance." Spooky or not, quantum mechanics appears to be intrinsically nonlocal.

What makes these results even more striking is that no matter how far apart the two entangled particles are, there appears to be *zero delay* between the time that we make a measurement on one particle and the time that the state of the other particle changes as a result. At first glance this seems to violate a key idea of Einstein's special theory of

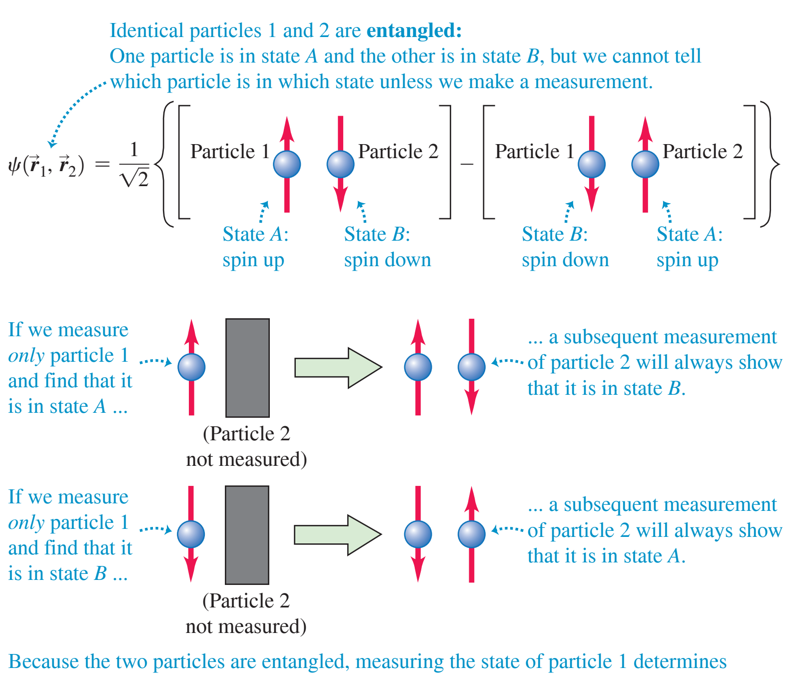


Figure **41.27** If two particles are in an entangled state, making a measurement of one particle determines the result of a subsequent measurement of the other particle.

relativity: that signals of any kind—radio waves, light signals, or beams of particles—cannot travel faster than the speed of light in vacuum, c. If measuring the state of particle 1 in Fig. 41.27 causes the state of particle 2 to change instantaneously, couldn't we make a "quantum radio" that sends signals faster than c, with particle 1 as the transmitter and particle 2 as the receiver?

The answer is no. The "message" in our quantum radio would be the result of a measurement of particle 1 by a physicist (call her Primo) at that particle's position. Primo's measurement collapses the wave function of the two particles, and her result would be that particle 1 is in either state A or state B. Another physicist (call him Secondo) at the position of particle 2 would measure particle 2 to be in state B if Primo measured particle 1 to be in state A, and to be in state A if Primo measured particle 1 to be in state B. But Secondo would have no way of knowing whether his result was caused by Primo making a measurement first or by *Secondo* himself making an independent measurement of particle 2 *without* Primo having made any measurement. (Secondo could determine this later by, for instance, sending text messages back and forth with Primo. But that method of communication involves signals that travel at the speed of light, not instantaneously.) So our quantum radio would transmit no information at all and would not allow us to communicate at speeds faster than c.

A remarkable practical application of quantum entanglement is *quantum computing*. In a conventional ("classical") computer, the memory is made up of *bits*. Each bit has only two possible values (say, 0 or 1), so a computer memory with N bits can have any of 2^N different configurations. (This is analogous to coins that can be either heads up or tails up. Figure 20.21 in Section 20.8 shows the possible configurations of four coins; the number of possibilities is $2^4 = 16$.) In a quantum computer, bits are replaced with *qubits* (short for "quantum bits"). An example is a spin-$\frac{1}{2}$ electron that can be in a spin-up state $\left(m_s = +\frac{1}{2}\right)$ or a spin-down state $\left(m_s = -\frac{1}{2}\right)$, as usual, but can also be in *any combination* of these states. The wave function of N entangled qubits can correspond to any of 2^N configurations (like ordinary bits or coins that can be heads up or tails up) or to an entangled state in which the qubits are in any combination of these configurations. So, unlike a classical computer memory, which can be in only one of its 2^N configurations at a time, a quantum computer memory can essentially be in *all* of these configurations simultaneously. This holds the promise of the ability to do certain types of computations, such as those involved in breaking codes in cryptography, much more rapidly than a classical computer could. As of this writing, the quest to build a fully quantum computer is still in its early stages, but intensive research is under way and rapid progress is being made.

TEST YOUR UNDERSTANDING OF SECTION 41.8 Particle 1 is an electron that can be in state C or D. Particle 2 is a proton that can be in state E or F. Is $\psi(\vec{r}_1, \vec{r}_2) = (1/\sqrt{2})[\psi_C(\vec{r}_1)\psi_E(\vec{r}_2) + \psi_C(\vec{r}_1)\psi_F(\vec{r}_2)]$ a possible wave function for this two-particle system? If so, does it represent an entangled state?

ANSWER

yes; no This wave function says that it is equally possible that the electron (particle 1) is in state C and the proton (particle 2) is in state E or that particle 1 is in state C and particle 2 is in state F. Since particles 1 and 2 are not identical and are distinguishable, there is no reason we can't know that particle 1 is in state C independent of which state particle 2 is in. So this is a valid wave function for the system, and the two particles are not entangled. If we measure the state of the electron alone, we are guaranteed to get C as a result; a subsequent measurement of the proton's state will give either E or F with equal probability, the same as if we had not first measured the state of the electron. Similarly, if we first measure the state of the proton, the results of that measurement will not affect a subsequent measurement of the state of the electron (for which the result is guaranteed to be C).

CHAPTER 41 SUMMARY

Three-dimensional problems: The time-independent Schrödinger equation for three-dimensional problems is given by Eq. (41.5).

$$-\frac{\hbar^2}{2m}\left(\frac{\partial^2\psi(x,y,z)}{\partial x^2}+\frac{\partial^2\psi(x,y,z)}{\partial y^2}+\frac{\partial^2\psi(x,y,z)}{\partial z^2}\right)+U(x,y,z)\psi(x,y,z)$$
$$=E\psi(x,y,z) \tag{41.5}$$

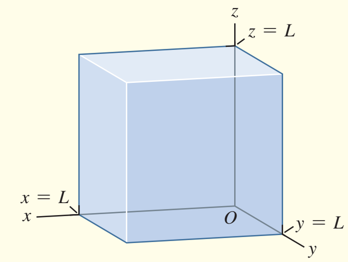
(three-dimensional time-independent Schrödinger equation)

Particle in a three-dimensional box: The wave function for a particle in a cubical box is the product of a function of x only, a function of y only, and a function of z only. Each stationary state is described by three quantum numbers (n_X, n_Y, n_Z). Most of the energy levels given by Eq. (41.16) exhibit degeneracy: More than one quantum state has the same energy. (See Example 41.1.)

$$E_{n_X,n_Y,n_Z}=\frac{(n_X{}^2+n_Y{}^2+n_Z{}^2)\pi^2\hbar^2}{2mL^2} \tag{41.16}$$

$(n_X = 1, 2, 3, \ldots;$
$n_Y = 1, 2, 3, \ldots;$
$n_Z = 1, 2, 3, \ldots)$

(energy levels, particle in a three-dimensional cubical box)



The hydrogen atom: The Schrödinger equation for the hydrogen atom gives the same energy levels as the Bohr model. If the nucleus has charge Ze, there is an additional factor of Z^2 in the numerator of Eq. (41.21). The possible magnitudes L of orbital angular momentum are given by Eq. (41.22), and the possible values of the z-component of orbital angular momentum are given by Eq. (41.23). (See Examples 41.2 and 41.3.)

The probability that an atomic electron is between r and $r + dr$ from the nucleus is $P(r)\,dr$, given by Eq. (41.25). Atomic distances are often measured in units of a, the smallest distance between the electron and the nucleus in the Bohr model. (See Example 41.4.)

$$E_n=-\frac{1}{(4\pi\epsilon_0)^2}\frac{m_r e^4}{2n^2\hbar^2}=-\frac{13.60\text{ eV}}{n^2} \tag{41.21}$$

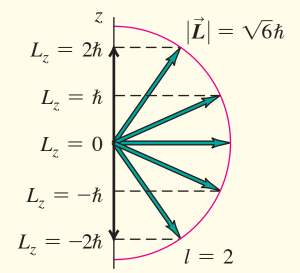
(energy levels of hydrogen)

$$L=\sqrt{l(l+1)}\,\hbar \quad (l=0,1,2,\ldots,n-1) \tag{41.22}$$

$$L_z=m_l\hbar \quad (m_l=0,\pm1,\pm2,\ldots,\pm l) \tag{41.23}$$

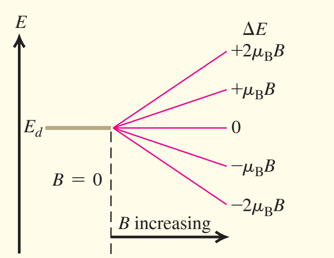
$$P(r)\,dr=|\psi|^2\,dV=|\psi|^2\,4\pi r^2\,dr \tag{41.25}$$

$$a=\frac{\epsilon_0 h^2}{\pi m_r e^2}=\frac{4\pi\epsilon_0\hbar^2}{m_r e^2}=5.29\times10^{-11}\text{ m} \tag{41.26}$$



The Zeeman effect: The interaction energy of an electron (mass m) with magnetic quantum number m_l in a magnetic field $\vec{B}$ along the $+z$-direction is given by Eq. (41.35), where $\mu_B = e\hbar/2m$ is called the Bohr magneton. (See Example 41.5.)

$$U=-\mu_z B=m_l\frac{e\hbar}{2m}B=m_l\mu_B B \quad (m_l=0,\pm1,\pm2,\ldots,\pm l) \tag{41.35}$$



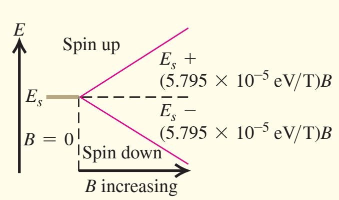
Electron spin: An electron has an intrinsic spin angular momentum of magnitude S, given by Eq. (41.37). The possible values of the z-component of the spin angular momentum are $S_z = m_s\hbar$, where $m_s = \pm\frac{1}{2}$. (See Examples 41.6 and 41.7.)

An orbiting electron experiences an interaction between its spin and the effective magnetic field produced by the relative motions of electron and nucleus. This spin-orbit coupling, along with relativistic effects, splits the energy levels according to their total angular momentum quantum number j. (See Example 41.8.)

$$S=\sqrt{\tfrac{1}{2}\left(\tfrac{1}{2}+1\right)}\,\hbar=\sqrt{\tfrac{3}{4}}\,\hbar \tag{41.37}$$

$$S_z=m_s\hbar \quad \left(m_s=\pm\tfrac{1}{2}\right) \tag{41.36}$$

$$E_{n,j}=-\frac{13.60\text{ eV}}{n^2}\left[1+\frac{\alpha^2}{n^2}\left(\frac{n}{j+\frac{1}{2}}-\frac{3}{4}\right)\right] \tag{41.41}$$

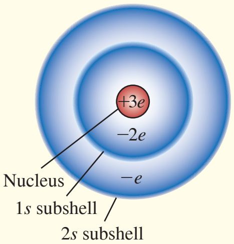


Continued

Many-electron atoms: In a hydrogen atom, the quantum numbers n, l, m_l, and m_s of the electron have certain allowed values given by Eq. (41.44). In a many-electron atom, the allowed quantum numbers for each electron are the same as in hydrogen, but the energy levels depend on both n and l because of screening, the partial cancellation of the field of the nucleus by the inner electrons. If the effective (screened) charge attracting an electron is $Z_{\text{eff}}e$, the energies of the levels are given approximately by Eq. (41.45). (See Examples 41.9 and 41.10.)

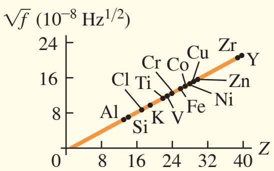
$$n \geq 1 \quad 0 \leq l \leq n-1$$
$$|m_l| \leq l \quad m_s = \pm\tfrac{1}{2} \qquad (41.44)$$

$$E_n = -\frac{Z_{\text{eff}}{}^2}{n^2}(13.6\text{ eV}) \qquad (41.45)$$

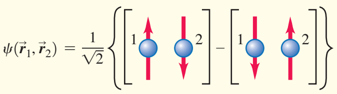


X-ray spectra: Moseley's law states that the frequency of a K_α x ray from a target with atomic number Z is given by Eq. (41.47). Characteristic x-ray spectra result from transitions to a hole in an inner energy level of an atom. (See Example 41.11.)

$$f = (2.48 \times 10^{15}\text{ Hz})(Z-1)^2 \qquad (41.47)$$



Quantum entanglement: The wave function of two identical particles can be such that neither particle is itself in a definite state. For example, the wave function could be a combination of one term with particle 1 in state A and particle 2 in state B and one term with particle 1 in state B and particle 2 in state A. The two particles are said to be entangled, since measuring the state of one particle automatically determines the results of subsequent measurements of the other particle.



Chapter 41 Media Assets

GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review EXAMPLE 41.1 (Section 41.2) before attempting these problems.

VP41.1.1 A particle in the three-dimensional box shown in Fig. 41.1 is in the state $n_X = 2, n_Y = 1, n_Z = 3$. Find (a) the planes (other than the walls of the box) on which the probability distribution function is zero and (b) the probability that the particle will be found somewhere in the region $0 \leq x \leq L/3$.

VP41.1.2 Half of the volume of the three-dimensional box shown in Fig. 41.1 is in the region $L/4 \leq x \leq 3L/4$. Find the probability that a particle in the box will be found in this region if the state of the particle is (a) $n_X = 1, n_Y = 1, n_Z = 1$; (b) $n_X = 2, n_Y = 1, n_Z = 2$; (c) $n_X = 3, n_Y = 2, n_Z = 3$; (d) $n_X = 4, n_Y = 1, n_Z = 1$.

VP41.1.3 The region $0 \leq x \leq L/4, 0 \leq y \leq L/4$ makes up $\frac{1}{16} = 0.0625$ of the volume of the three-dimensional box shown in Fig. 41.1. Find the probability that a particle in the box will be found in this region if the state of the particle is (a) $n_X = 1, n_Y = 1, n_Z = 1$; (b) $n_X = 2, n_Y = 1, n_Z = 2$; (c) $n_X = 3, n_Y = 2, n_Z = 3$; (d) $n_X = 4, n_Y = 1, n_Z = 1$.

VP41.1.4 Consider the cubical region given by $L/4 \leq x \leq 3L/4$, $L/4 \leq y \leq 3L/4$, $L/4 \leq z \leq 3L/4$ at the center of the three-dimensional box shown in Fig. 41.1. (a) What fraction of the total volume of the box is inside this cubical region? (b) If a particle in the box is in the state $n_X = 1, n_Y = 1, n_Z = 1$, find the probability that it will be found somewhere in this cubical region at the center of the box.

Be sure to review EXAMPLES 41.2, 41.3, and 41.4 (Section 41.3) before attempting these problems.

VP41.4.1 Consider the $n = 6$ states of the hydrogen atom. (a) How many distinct (l, m_l) states are there? (b) In terms of $\hbar$, what is the maximum magnitude of the orbital angular momentum L? (c) In terms of $\hbar$, what is the maximum value of the z-component of orbital angular momentum?

VP41.4.2 (a) List all the possible combinations of values of l and m_l for the $n = 3$ states of the hydrogen atom. (b) For which of these states is the angle between the orbital angular momentum vector and the negative z-axis a minimum, and what is that angle?

VP41.4.3 A photon is emitted when a hydrogen atom transitions from one energy level to a lower energy level. Find the energy of this photon, in eV, for each transition: (a) $n = 3, l = 2, m_l = -2$ to $n = 2, l = 1, m_l = -1$; (b) $n = 4, l = 2, m_l = 1$ to $n = 2, l = 1, m_l = 0$; (c) $n = 2, l = 1, m_l = 1$ to $n = 1, l = 0, m_l = 0$.

VP41.4.4 The wave function for an electron in a 1s state in a hydrogen atom is $\psi_{1s}(r) = 1/\sqrt{\pi a^3}\, e^{-r/a}$, where r is the distance from the nucleus. Find the probability that the electron will be found in the region (a) $0 \leq r \leq 2a$; (b) $a \leq r \leq 3a$; (c) $r \geq 4a$.

Be sure to review EXAMPLES 41.6, 41.7, and 41.8 (Section 41.5) before attempting these problems.

VP41.8.1 An isolated electron is placed in a magnetic field $\vec{B} = (3.14\text{ T})\hat{k}$. (a) Find the difference in energy between the $S_z = +\frac{1}{2}\hbar$ and $S_z = -\frac{1}{2}\hbar$ states of the electron. (b) Which state has the higher energy?

VP41.8.2 The outermost electron in a potassium atom is in an $l = 0$ state. If you place a potassium atom in a magnetic field of magnitude 2.36 T and illuminate it with monochromatic electromagnetic radiation, what must be the frequency and wavelength of the radiation to cause a transition between the spin-up and spin-down states of the outermost electron?

VP41.8.3 When the outermost electron in a potassium atom makes a transition from a $4p$ level to a $4s$ level, the wavelength of the photon it emits can be either 766.490 nm or 769.896 nm depending on the initial spin orientation of the electron. Find (a) the energy difference between the two $4p$ levels and (b) the effective magnetic field that the electron experiences in the $4p$ levels.

VP41.8.4 (a) Find the energy in terms of α of a state of the electron in a hydrogen atom with $n = 3$, $l = 1$, $j = \frac{3}{2}$ and the energy in terms of α of a state with $n = 3$, $l = 1$, $j = \frac{1}{2}$. (b) Find the difference between these energies, in eV. Which state has the higher energy? (c) Find the difference in wavelengths of the photons emitted in transitions from each of the states in part (a) to the state $n = 2$, $l = 0$, $j = \frac{1}{2}$. For which initial state is the wavelength longer?

BRIDGING PROBLEM A Many-Electron Atom in a Box

An atom of titanium (Ti) has 22 electrons and has a radius of 1.47×10^{-10} m. As a simple model of this atom, imagine putting 22 electrons into a cubical box that has the same volume as a titanium atom. (a) What is the length of each side of the box? (b) What will be the configuration of the 22 electrons? (c) Find the energies of each of the levels occupied by the electrons. (Ignore the electric forces that the electrons exert on each other.) (d) You remove one of the electrons from the lowest level. As a result, one of the electrons from the highest occupied level drops into the lowest level to fill the hole, emitting a photon in the process. What is the energy of this photon? How does this compare to the energy of the K_α photon for titanium as predicted by Moseley's law?

SOLUTION GUIDE

IDENTIFY and SET UP

1. In this problem you'll use ideas from Section 41.2 about a particle in a cubical box. You'll also apply the exclusion principle from Section 41.6 to find the electron configuration of this cubical "atom." The ideas about x-ray spectra from Section 41.7 are also important.
2. The target variables are (a) the dimensions of the box, (b) the electron configurations (like those given in Table 41.3 for real atoms), (c) the occupied energy levels of the cubical box, and (d) the energy of the emitted photon.

EXECUTE

3. Use your knowledge of geometry to find the length of each side of the box.
4. Each electron state is described by four quantum numbers: n_X, n_Y, and n_Z as described in Section 41.2 and the spin magnetic quantum number m_s described in Section 41.5. Use the exclusion principle to determine the quantum numbers of each of the 22 electrons in the "atom." (*Hint:* Figure 41.4 in Section 41.2 shows the first several energy levels of a cubical box relative to the ground level $E_{1,1,1}$.)
5. Use your results from steps 3 and 4 to find the energies of each of the occupied levels.
6. Use your result from step 5 to find the energy of the photon emitted when an electron makes a transition from the highest occupied level to the ground level. Compare this to the energy that we calculated for titanium by using Moseley's law.

EVALUATE

7. Is this cubical "atom" a useful model for titanium? Why or why not?
8. In this problem you ignored the electrical interactions between electrons. To estimate how large these are, find the electrostatic potential energy of two electrons separated by half the length of the box. How does this compare to the energy levels you calculated in step 5? Is it a good approximation to ignore these interactions?

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q41.1 Particle A is described by the wave function $\psi(x, y, z)$. Particle B is described by the wave function $\psi(x, y, z)e^{i\phi}$, where ϕ is a real constant. How does the probability of finding particle A within a volume dV around a certain point in space compare with the probability of finding particle B within this same volume?

Q41.2 What are the most significant differences between the Bohr model of the hydrogen atom and the Schrödinger analysis? What are the similarities?

Q41.3 For an object orbiting the sun, such as a planet, comet, or asteroid, is there any restriction on the z-component of its orbital angular momentum such as there is with the z-component of the electron's orbital angular momentum in hydrogen? Explain.

Q41.4 Why is the analysis of the helium atom much more complex than that of the hydrogen atom, either in a Bohr type of model or using the Schrödinger equation?

Q41.5 The Stern–Gerlach experiment is always performed with beams of *neutral* atoms. Wouldn't it be easier to form beams using *ionized* atoms? Why won't this work?

Q41.6 (a) If two electrons in hydrogen atoms have the same principal quantum number, can they have different orbital angular momenta? How? (b) If two electrons in hydrogen atoms have the same orbital quantum number, can they have different principal quantum numbers? How?

Q41.7 In the Stern–Gerlach experiment, why is it essential for the magnetic field to be *inhomogeneous* (that is, nonuniform)?

Q41.8 In the ground state of the helium atom one electron must have "spin down" and the other "spin up." Why?

Q41.9 An electron in a hydrogen atom is in an s level, and the atom is in a magnetic field $\vec{B} = B\hat{k}$. Explain why the "spin up" state $\left(m_s = +\frac{1}{2}\right)$ has a higher energy than the "spin down" state $\left(m_s = -\frac{1}{2}\right)$.

Q41.10 The central-field approximation is more accurate for alkali metals than for transition metals such as iron, nickel, or copper. Why?

Q41.11 Table 41.3 shows that for the ground state of the potassium atom, the outermost electron is in a 4*s* state. What does this tell you about the relative energies of the 3*d* and 4*s* levels for this atom? Explain.
Q41.12 Do gravitational forces play a significant role in atomic structure? Explain.
Q41.13 Why do the transition elements ($Z = 21$ to 30) all have similar chemical properties?
Q41.14 Use Table 41.3 to help determine the ground-state electron configuration of the neutral gallium atom (Ga) as well as the ions Ga^+ and Ga^-. Gallium has an atomic number of 31.
Q41.15 On the basis of the Pauli exclusion principle, the structure of the periodic table of the elements shows that there must be a fourth quantum number in addition to n, l, and m_l. Explain.
Q41.16 A small amount of magnetic-field splitting of spectral lines occurs even when the atoms are not in a magnetic field. What causes this?
Q41.17 The ionization energies of the alkali metals (that is, the lowest energy required to remove one outer electron when the atom is in its ground state) are about 4 or 5 eV, while those of the noble gases are in the range from 11 to 25 eV. Why is there a difference?
Q41.18 For magnesium, the first ionization potential is 7.6 eV. The second ionization potential (additional energy required to remove a second electron) is almost twice this, 15 eV, and the third ionization potential is much larger, about 80 eV. How can these numbers be understood?
Q41.19 What is the "central-field approximation" and why is it only an approximation?
Q41.20 The nucleus of a gold atom contains 79 protons. How does the energy required to remove a 1*s* electron completely from a gold atom compare with the energy required to remove the electron from the ground level in a hydrogen atom? In what region of the electromagnetic spectrum would a photon with this energy for each of these two atoms lie?
Q41.21 (a) Can you show that the orbital angular momentum of an electron in any given direction (e.g., along the *z*-axis) is *always* less than or equal to its total orbital angular momentum? In which cases would the two be equal to each other? (b) Is the result in part (a) true for a classical object, such as a spinning top or planet?
Q41.22 An atom in its ground level absorbs a photon with energy equal to the *K* absorption edge. Does absorbing this photon ionize this atom? Explain.
Q41.23 Can a hydrogen atom emit x rays? If so, how? If not, why not?
Q41.24 A system of two electrons has the wave function $\psi(\vec{r}_1, \vec{r}_2) = (1/\sqrt{2})[\psi_\alpha(\vec{r}_1)\psi_\beta(\vec{r}_2) - \psi_\beta(\vec{r}_1)\psi_\alpha(\vec{r}_2)]$, where ψ_α is a normalized wave function for a state with $S_z = +\frac{1}{2}\hbar$ and ψ_β is a normalized wave function for a state with $S_z = -\frac{1}{2}\hbar$. (a) If S_z for electron 1 is measured, what are the possible results? What is the probability of each result? (b) If S_z for electron 2 is measured, what are the possible results? What is the probability of each result? (c) If measurement of S_z for electron 1 yields the value $\frac{1}{2}\hbar$, what are the possible results of a subsequent measurement of S_z for electron 2? What is the probability of each result being obtained? Explain.
Q41.25 Repeat Discussion Question Q41.24 for the wave function $\psi(\vec{r}_1, \vec{r}_2) = \psi_\alpha(\vec{r}_1)\psi_\alpha(\vec{r}_2)$.

EXERCISES

Section 41.2 Particle in a Three-Dimensional Box

41.1 • For a particle in a three-dimensional cubical box, what is the degeneracy (number of different quantum states with the same energy) of the energy levels (a) $3\pi^2\hbar^2/2mL^2$ and (b) $9\pi^2\hbar^2/2mL^2$?
41.2 • **CP** Model a hydrogen atom as an electron in a cubical box with side length L. Set the value of L so that the volume of the box equals the volume of a sphere of radius $a = 5.29 \times 10^{-11}$ m, the Bohr radius. Calculate the energy separation between the ground and first excited levels, and compare the result to this energy separation calculated from the Bohr model.
41.3 • **CP** A photon is emitted when an electron in a three-dimensional cubical box of side length 8.00×10^{-11} m makes a transition from the $n_X = 2, n_Y = 2, n_Z = 1$ state to the $n_X = 1, n_Y = 1, n_Z = 1$ state. What is the wavelength of this photon?
41.4 • For each of the following states of a particle in a three-dimensional cubical box, at what points is the probability distribution function a maximum: (a) $n_X = 1, n_Y = 1, n_Z = 1$ and (b) $n_X = 2, n_Y = 2, n_Z = 1$?
41.5 • A photon with wavelength 8.00 nm is absorbed when an electron in a three-dimensional cubical box makes a transition from the ground state to the second excited state. What is the side length L of the box?
41.6 • What is the energy difference between the two lowest energy levels for a proton in a cubical box with side length 1.00×10^{-14} m, the approximate diameter of a nucleus?
41.7 • A particle is in a three-dimensional cubical box that has side length L. For the state $n_X = 3, n_Y = 2$, and $n_Z = 1$, for what planes (in addition to the walls of the box) is the probability distribution function zero?

Section 41.3 The Hydrogen Atom

41.8 • (a) A hydrogen atom is in a state with quantum number $n = 3$. In the quantum-mechanical description of the atom, what is the largest possible magnitude of the orbital angular momentum? What is the magnitude of the orbital angular momentum of the electron in the Bohr-model description of the atom? What is the percentage difference between these two values? (b) Answer the same questions as in part (a) for $n = 30$.
41.9 •• Consider an electron in the *N* shell. (a) What is the smallest orbital angular momentum it could have? (b) What is the largest orbital angular momentum it could have? Express your answers in terms of $\hbar$ and in SI units. (c) What is the largest orbital angular momentum this electron could have in any chosen direction? Express your answers in terms of $\hbar$ and in SI units. (d) What is the largest spin angular momentum this electron could have in any chosen direction? Express your answers in terms of $\hbar$ and in SI units. (e) For the electron in part (c), what is the ratio of its spin angular momentum in the *z*-direction to its orbital angular momentum in the *z*-direction?
41.10 • An electron is in the hydrogen atom with $n = 5$. (a) Find the possible values of L and L_z for this electron, in units of $\hbar$. (b) For each value of L, find all the possible angles between $\vec{L}$ and the *z*-axis. (c) What are the maximum and minimum values of the magnitude of the angle between $\vec{L}$ and the *z*-axis?
41.11 • The orbital angular momentum of an electron has a magnitude of 4.716×10^{-34} kg · m^2/s. What is the angular momentum quantum number l for this electron?
41.12 • Consider states with angular momentum quantum number $l = 2$. (a) In units of $\hbar$, what is the largest possible value of L_z? (b) In units of $\hbar$, what is the value of L? Which is larger: L or the maximum possible L_z? (c) For each allowed value of L_z, what angle does the vector $\vec{L}$ make with the $+z$-axis? How does the minimum angle for $l = 2$ compare to the minimum angle for $l = 3$ calculated in Example 41.3?
41.13 •• In a particular state of the hydrogen atom, the angle between the angular momentum vector $\vec{L}$ and the *z*-axis is $\theta = 26.6°$. If this is the smallest angle for this particular value of the orbital quantum number l, what is l?
41.14 •• A hydrogen atom is in a state that has $L_z = 2\hbar$. In the semiclassical vector model, the angular momentum vector $\vec{L}$ for this state makes an angle $\theta_L = 63.4°$ with the $+z$-axis. (a) What is the l quantum number for this state? (b) What is the smallest possible n quantum number for this state?
41.15 • Consider the seventh excited level of the hydrogen atom. (a) What is the energy of this level? (b) What is the largest magnitude of the orbital angular momentum? (c) What is the largest angle between the orbital angular momentum and the *z*-axis?

41.16 • (a) Make a chart showing all possible sets of quantum numbers l and m_l for the states of the electron in the hydrogen atom when $n = 4$. How many combinations are there? (b) What are the energies of these states?

41.17 •• (a) How many different $5g$ states does hydrogen have? (b) Which of the states in part (a) has the largest angle between $\vec{L}$ and the z-axis, and what is that angle? (c) Which of the states in part (a) has the smallest angle between $\vec{L}$ and the z-axis, and what is that angle?

41.18 •• **CALC** (a) What is the probability that an electron in the $1s$ state of a hydrogen atom will be found at a distance less than $a/2$ from the nucleus? (b) Use the results of part (a) and of Example 41.4 to calculate the probability that the electron will be found at distances between $a/2$ and a from the nucleus.

41.19 • Show that $\Phi(\phi) = e^{im_l\phi} = \Phi(\phi + 2\pi)$ (that is, show that $\Phi(\phi)$ is periodic with period 2π) if and only if m_l is restricted to the values $0, \pm 1, \pm 2, \ldots$. (*Hint:* Euler's formula states that $e^{i\phi} = \cos\phi + i\sin\phi$.)

41.20 • (a) The radial probability distribution function for a hydrogen atom state has one peak, at $r = 0.476$ nm. What is the nl spectroscopic notation of this state? (b) What is r for the one peak of a $4f$ state?

Section 41.4 The Zeeman Effect

41.21 • A hydrogen atom in a $3p$ state is placed in a uniform external magnetic field $\vec{B}$. Consider the interaction of the magnetic field with the atom's orbital magnetic dipole moment. (a) What field magnitude B is required to split the $3p$ state into multiple levels with an energy difference of 2.71×10^{-5} eV between adjacent levels? (b) How many levels will there be?

41.22 • A hydrogen atom is in a d state. In the absence of an external magnetic field, the states with different m_l values have (approximately) the same energy. Consider the interaction of the magnetic field with the atom's orbital magnetic dipole moment. (a) Calculate the splitting (in electron volts) of the m_l levels when the atom is put in a 0.800 T magnetic field that is in the $+z$-direction. (b) Which m_l level will have the lowest energy? (c) Draw an energy-level diagram that shows the d levels with and without the external magnetic field.

41.23 • A hydrogen atom in the $5g$ state is placed in a magnetic field of 0.600 T that is in the z-direction. (a) Into how many levels is this state split by the interaction of the atom's orbital magnetic dipole moment with the magnetic field? (b) What is the energy separation between adjacent levels? (c) What is the energy separation between the level of lowest energy and the level of highest energy?

41.24 •• **CP** A hydrogen atom undergoes a transition from a $2p$ state to the $1s$ ground state. In the absence of a magnetic field, the wavelength of the photon emitted is 122 nm. The atom is then placed in a strong magnetic field in the z-direction. Ignore spin effects; consider only the interaction of the magnetic field with the atom's orbital magnetic moment. (a) How many different photon wavelengths are observed for the $2p \rightarrow 1s$ transition? What are the m_l values for the initial and final states for the transition that leads to each photon wavelength? (b) One observed wavelength is exactly the same with the magnetic field as without. What are the initial and final m_l values for the transition that produces a photon of this wavelength? (c) One observed wavelength with the field is longer than the wavelength without the field. What are the initial and final m_l values for the transition that produces a photon of this wavelength? (d) Repeat part (c) for the wavelength that is shorter than the wavelength in the absence of the field.

Section 41.5 Electron Spin

41.25 •• **CP** **Classical Electron Spin.** (a) If you treat an electron as a classical spherical object with a radius of 1.0×10^{-17} m, what angular speed is necessary to produce a spin angular momentum of magnitude $\sqrt{\frac{3}{4}}\hbar$? (b) Use $v = r\omega$ and the result of part (a) to calculate the speed v of a point at the electron's equator. What does your result suggest about the validity of this model?

41.26 • **CP** The hyperfine interaction in a hydrogen atom between the magnetic dipole moment of the proton and the spin magnetic dipole moment of the electron splits the ground level into two levels separated by 5.9×10^{-6} eV. (a) Calculate the wavelength and frequency of the photon emitted when the atom makes a transition between these states, and compare your answer to the value given at the end of Section 41.5. In what part of the electromagnetic spectrum does this lie? Such photons are emitted by cold hydrogen clouds in interstellar space; by detecting these photons, astronomers can learn about the number and density of such clouds. (b) Calculate the effective magnetic field experienced by the electron in these states (see Fig. 41.18). Compare your result to the effective magnetic field due to the spin-orbit coupling calculated in Example 41.7.

41.27 • Calculate the energy difference between the $m_s = \frac{1}{2}$ ("spin up") and $m_s = -\frac{1}{2}$ ("spin down") levels of a hydrogen atom in the $1s$ state when it is placed in a 1.45 T magnetic field in the *negative* z-direction. Which level, $m_s = \frac{1}{2}$ or $m_s = -\frac{1}{2}$, has the lower energy?

41.28 •• A hydrogen atom in the $n = 1, m_s = -\frac{1}{2}$ state is placed in a magnetic field with a magnitude of 1.60 T in the $+z$-direction. (a) Find the magnetic interaction energy (in electron volts) of the electron with the field. (b) Is there any orbital magnetic dipole moment interaction for this state? Explain. Can there be an orbital magnetic dipole moment interaction for $n \neq 1$?

Section 41.6 Many-Electron Atoms and the Exclusion Principle

41.29 • Make a list of the four quantum numbers n, l, m_l, and m_s for each of the 10 electrons in the ground state of the neon atom. Do *not* refer to Table 41.2 or 41.3.

41.30 • For germanium (Ge, $Z = 32$), make a list of the number of electrons in each subshell ($1s, 2s, 2p, \ldots$). Use the allowed values of the quantum numbers along with the exclusion principle; do *not* refer to Table 41.3.

41.31 •• (a) Write out the ground-state electron configuration ($1s^2, 2s^2, \ldots$) for the beryllium atom. (b) What element of next-larger Z has chemical properties similar to those of beryllium? Give the ground-state electron configuration of this element. (c) Use the procedure of part (b) to predict what element of next-larger Z than in (b) will have chemical properties similar to those of the element you found in part (b), and give its ground-state electron configuration.

41.32 •• (a) Write out the ground-state electron configuration ($1s^2, 2s^2, \ldots$) for the carbon atom. (b) What element of next-larger Z has chemical properties similar to those of carbon? Give the ground-state electron configuration for this element.

41.33 • The $5s$ electron in rubidium (Rb) sees an effective charge of $2.771e$. Calculate the ionization energy of this electron.

41.34 • The energies of the $4s$, $4p$, and $4d$ states of potassium are given in Example 41.10. Calculate Z_{eff} for each state. What trend do your results show? How can you explain this trend?

41.35 • (a) The doubly charged ion N^{2+} is formed by removing two electrons from a nitrogen atom. What is the ground-state electron configuration for the N^{2+} ion? (b) Estimate the energy of the least strongly bound level in the L shell of N^{2+}. (c) The doubly charged ion P^{2+} is formed by removing two electrons from a phosphorus atom. What is the ground-state electron configuration for the P^{2+} ion? (d) Estimate the energy of the least strongly bound level in the M shell of P^{2+}.

Section 41.7 X-Ray Spectra

41.36 • A K_α x ray emitted from a sample has an energy of 7.46 keV. Of which element is the sample made?

41.37 • Calculate the frequency, energy (in keV), and wavelength of the K_α x ray for the elements (a) calcium (Ca, $Z = 20$); (b) cobalt (Co, $Z = 27$); (c) cadmium (Cd, $Z = 48$).

41.38 •• The energies for an electron in the K, L, and M shells of the tungsten atom are $-69{,}500$ eV, $-12{,}000$ eV, and -2200 eV, respectively. Calculate the wavelengths of the K_α and K_β x rays of tungsten.

PROBLEMS

41.39 • In terms of the ground-state energy $E_{1,1,1}$, what is the energy of the highest level occupied by an electron when 10 electrons are placed into a cubical box?

41.40 •• An electron is in a three-dimensional box with side lengths $L_X = 0.600$ nm and $L_Y = L_Z = 2L_X$. What are the quantum numbers n_X, n_Y, and n_Z and the energies, in eV, for the four lowest energy levels? What is the degeneracy of each (including the degeneracy due to spin)?

41.41 •• CALC A particle is in the three-dimensional cubical box of Section 41.2. (a) Consider the cubical volume defined by $0 \le x \le L/4$, $0 \le y \le L/4$, and $0 \le z \le L/4$. What fraction of the total volume of the box is this cubical volume? (b) If the particle is in the ground state $(n_X = 1, n_Y = 1, n_Z = 1)$, calculate the probability that the particle will be found in the cubical volume defined in part (a). (c) Repeat the calculation of part (b) when the particle is in the state $n_X = 2, n_Y = 1, n_Z = 1$.

41.42 ••• An electron is in a three-dimensional box. The x- and z-sides of the box have the same length, but the y-side has a different length. The two lowest energy levels are 2.24 eV and 3.47 eV, and the degeneracy of each of these levels (including the degeneracy due to the electron spin) is two. (a) What are the n_X, n_Y, and n_Z quantum numbers for each of these two levels? (b) What are the lengths L_X, L_Y, and L_Z for each side of the box? (c) What are the energy, the quantum numbers, and the degeneracy (including the spin degeneracy) for the next higher energy state?

41.43 •• CALC A particle in the three-dimensional cubical box of Section 41.2 is in the ground state, where $n_X = n_Y = n_Z = 1$. (a) Calculate the probability that the particle will be found somewhere between $x = 0$ and $x = L/2$. (b) Calculate the probability that the particle will be found somewhere between $x = L/4$ and $x = L/2$. Compare your results to the result of Example 41.1 for the probability of finding the particle in the region $x = 0$ to $x = L/4$.

41.44 •• CP CALC **A Three-Dimensional Isotropic Harmonic Oscillator.** An isotropic harmonic oscillator has the potential-energy function $U(x, y, z) = \frac{1}{2}k'(x^2 + y^2 + z^2)$. (*Isotropic* means that the force constant k' is the same in all three coordinate directions.) (a) Show that for this potential, a solution to Eq. (41.5) is given by $\psi = \psi_{n_x}(x)\psi_{n_y}(y)\psi_{n_z}(z)$. In this expression, $\psi_{n_x}(x)$ is a solution to the one-dimensional harmonic-oscillator Schrödinger equation, Eq. (40.44), with energy $E_{n_x} = \left(n_x + \frac{1}{2}\right)\hbar\omega$. The functions $\psi_{n_y}(y)$ and $\psi_{n_z}(z)$ are analogous one-dimensional wave functions for oscillations in the y- and z-directions. Find the energy associated with this ψ. (b) From your results in part (a) what are the ground-level and first-excited-level energies of the three-dimensional isotropic oscillator? (c) Show that there is only one state (one set of quantum numbers n_x, n_y, and n_z) for the ground level but three states for the first excited level.

41.45 •• CP CALC **Three-Dimensional Anisotropic Harmonic Oscillator.** An oscillator has the potential-energy function $U(x, y, z) = \frac{1}{2}k_1'(x^2 + y^2) + \frac{1}{2}k_2'z^2$, where $k_1' > k_2'$. This oscillator is called *anisotropic* because the force constant is not the same in all three coordinate directions. (a) Find a general expression for the energy levels of the oscillator (see Problem 41.44). (b) From your results in part (a), what are the ground-level and first-excited-level energies of this oscillator? (c) How many states (different sets of quantum numbers n_x, n_y, and n_z) are there for the ground level and for the first excited level? Compare to part (c) of Problem 41.44.

41.46 •• A particle is in a three-dimensional box. The y length of the box is twice the x length, and the z length is one-third of the y length. (a) What is the energy difference between the first excited level and the ground level? (b) Is the first excited level degenerate? (c) In terms of the x length, where is the probability distribution the greatest in the lowest-energy level?

41.47 •• (a) Show that the total number of atomic states (including different spin states) in a shell of principal quantum number n is $2n^2$. [*Hint:* The sum of the first N integers $1 + 2 + 3 + \cdots + N$ is equal to $N(N + 1)/2$.] (b) Which shell has 50 states?

41.48 •• When our sun exhausts its nuclear fuel, it will ultimately shrink due to gravity and become a white dwarf, with a radius of approximately 7000 km. (a) Using the mass of the sun, $M_{sun} = 2.0 \times 10^{30}$ kg, and the mass of a proton, $M_{proton} = 1.7 \times 10^{-27}$ kg, estimate the number of electrons in the sun. (b) From the radius given, estimate the average volume to be occupied by each electron in the eventual white dwarf. (c) The white dwarf will consist of mostly carbon. Since there are six electrons in each carbon atom, multiply the volume of an electron by 6 to obtain the volume of each carbon atom. (d) Model the atomic arrangement as a cubical lattice and use your earlier estimate to determine the distance L between adjacent carbon nuclei. (e) View each atom as an $L \times L \times L$ box filled with six electrons. Since no two electrons can be in the same quantum state, what quantum numbers (n_X, n_Y, n_Z, m_z) are associated with these six electrons, where m_z is the quantum number for the component of the electron spin along the z-axis? (f) Ignoring contributions from spin couplings and Coulomb interactions between the electrons, what would be the energy of the higher-energy electrons?

41.49 •• CALC Consider a hydrogen atom in the $1s$ state. (a) For what value of r is the potential energy $U(r)$ equal to the total energy E? Express your answer in terms of a. This value of r is called the *classical turning point,* since this is where a Newtonian particle would stop its motion and reverse direction. (b) For r greater than the classical turning point, $U(r) > E$. Classically, the particle cannot be in this region, since the kinetic energy cannot be negative. Calculate the probability of the electron being found in this classically forbidden region.

41.50 • CALC For a hydrogen atom, the probability $P(r)$ of finding the electron within a spherical shell with inner radius r and outer radius $r + dr$ is given by Eq. (41.25). For a hydrogen atom in the $1s$ ground state, at what value of r does $P(r)$ have its maximum value? How does your result compare to the distance between the electron and the nucleus for the $n = 1$ state in the Bohr model, Eq. (41.26)?

41.51 •• CALC The normalized radial wave function for the $2p$ state of the hydrogen atom is $R_{2p} = (1/\sqrt{24a^5})re^{-r/2a}$. After we average over the angular variables, the radial probability function becomes $P(r)\,dr = (R_{2p})^2r^2\,dr$. At what value of r is $P(r)$ for the $2p$ state a maximum? Compare your results to the radius of the $n = 2$ state in the Bohr model.

41.52 •• Interstellar hydrogen emits characteristic microwave radiation that can be understood as follows: (a) Model the magnetic moment of a proton using $\mu_p = (e/2m_p)\,S_z$, where $m_p = 1.7 \times 10^{-27}$ kg and $S_z = \pm\hbar/2$. Model the proton as a current loop with radius $R_p = 0.85$ fm. What current would produce the expected magnetic moment? (b) The magnetic field a distance x along the axis of a current loop is given by Eq. (28.15). Use this equation to estimate the magnetic field B at a distance $x = 0.40a_0$, where $a_0 \gg R_p$ is the Bohr radius. (c) The potential energy associated with the electron spin coupled to this magnetic field is $U_{hyperfine} = \pm\mu_z B$. Use Eq. (41.38) to show that $\mu_z = \mu_B$ where μ_B is the Bohr magneton. The sign depends on whether the electron spin is aligned or antialigned to that of the proton. Estimate the magnitude of this energy. (d) If an electron's spin were to flip, it would radiate a photon with energy $2|U_{hyperfine}|$. What would be the wavelength of such a photon? (e) Note that 21 cm radiation is observed in hydrogen clouds. Could electron spin flips explain this radiation?

41.53 •• **CP** The hydrogen spectrum includes four visible lines. Of these, the blue line corresponds to a transition from the $n = 5$ shell to the $n = 2$ shell and has a wavelength of 434 nm. If we look closer, this line is broadened by fine structure due to spin-orbit coupling and relativistic effects. (a) How many different sets of l and j quantum numbers are there for the $n = 5$ shell and for the $n = 2$ shell? (b) How many different energy levels are there for $n = 5$ and for $n = 2$? For each of these levels, what is their energy difference in eV from $-(13.6\text{ eV})/n^2$? (c) In a transition that emits a photon the quantum number l must change by ± 1. Which transition in the fine structure of the hydrogen blue line emits a photon of the shortest wavelength? For this photon what is the shift in wavelength due to the fine structure? (d) Which transition in the fine structure emits a photon of the longest wavelength? For this photon what is the shift in wavelength due to the fine structure? (e) By what total extent, in nm, is the wavelength of the blue line broadened around the 434 nm value?

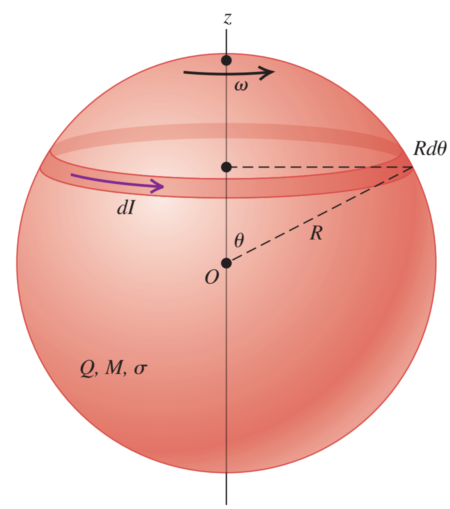
41.54 •• An atom in a $3d$ state emits a photon of wavelength 475.082 nm when it decays to a $2p$ state. (a) What is the energy (in electron volts) of the photon emitted in this transition? (b) Use the selection rules described in Section 41.4 to find the allowed transitions if the atom is now in an external magnetic field of 3.500 T. Ignore the effects of the electron's spin. (c) For the case in part (b), if the energy of the $3d$ state was originally -8.50000 eV with no magnetic field present, what will be the energies of the states into which it splits in the magnetic field? (d) What are the allowed wavelengths of the light emitted during transition in part (b)?

41.55 •• **CALC** **Spectral Analysis.** While studying the spectrum of a gas cloud in space, an astronomer magnifies a spectral line that results from a transition from a p state to an s state. She finds that the line at 575.050 nm has actually split into three lines, with adjacent lines 0.0462 nm apart, indicating that the gas is in an external magnetic field. (Ignore effects due to electron spin.) What is the strength of the external magnetic field?

41.56 •• **CP** **Stern–Gerlach Experiment.** In a Stern–Gerlach experiment, the deflecting force on the atom is $F_z = -\mu_z(dB_z/dz)$, where μ_z is given by Eq. (41.38) and dB_z/dz is the magnetic-field gradient. In a particular experiment, the magnetic-field region is 50.0 cm long; assume the magnetic-field gradient is constant in that region. A beam of silver atoms enters the magnetic field with a speed of 375 m/s. What value of dB_z/dz is required to give a separation of 1.0 mm between the two spin components as they exit the field? (*Note:* The magnetic dipole moment of silver is the same as that for hydrogen, since its valence electron is in an $l = 0$ state.)

41.57 ••• **CP** **CALC** The magnetic moment $\vec{\mu}$ of a spinning object with charge Q and mass M is proportional to its angular momentum $\vec{L}$ according to $\vec{\mu} = g(Q/2M)\vec{L}$. The dimensionless coefficient g is known as the g-factor of the object. Consider a spherical shell with radius R and uniform charge density σ spinning with angular velocity ω around the z-axis. (a) What is the differential area da of the ring at latitude θ, with width $R\,d\theta$, as shown in **Fig. P41.57**? (b) The current dI carried by the ring is its charge $\sigma\,da$ divided by the period of rotation. Determine dI in terms of R, σ, ω, and θ. (c) Determine the magnetic moment $d\vec{\mu}$ of the ring by multiplying dI by the area enclosed by the ring. (d) Determine the magnetic moment of the spherical shell by integrating over the sphere. Express your result in terms of the total charge $Q = 4\pi R^2\sigma$. (e) Now consider a solid sphere of radius R with volume charge density $\rho = \rho(r)$, where r is the distance from the center, spinning with angular velocity ω about the z-axis. Determine its magnetic moment by integrating $\vec{\mu} = \int d\vec{\mu}$, where $d\vec{\mu}$ is now the magnetic moment associated with the shell at radius r with differential width dr. Express your answer in terms of an integral $\int_0^R r^4\rho(r)dr$. (f) The angular momentum of the sphere is $\vec{L} = I\vec{\omega}$, where $I = cMR^2$ is its moment of inertia, and c is a dimensionless factor determined by the mass distribution. Determine the g-factor in terms of the ρ integral and the value of c. (g) If the charge is uniformly distributed so that $\rho = Q/(\frac{4}{3}\pi R^3)$ and if the mass is uniformly distributed so that $c = \frac{2}{5}$, then what is the value of g?

Figure **P41.57**



41.58 •• **Effective Magnetic Field.** An electron in a hydrogen atom is in the $2p$ state. In a simple model of the atom, assume that the electron circles the proton in an orbit with radius r equal to the Bohr-model radius for $n = 2$. Assume that the speed v of the orbiting electron can be calculated by setting $L = mvr$ and taking L to have the quantum-mechanical value for a $2p$ state. In the frame of the electron, the proton orbits with radius r and speed v. Model the orbiting proton as a circular current loop, and calculate the magnetic field it produces at the location of the electron.

41.59 •• **Weird Universe.** In another universe, the electron is a spin-$\frac{3}{2}$ rather than a spin-$\frac{1}{2}$ particle, but all other physics are the same as in our universe. In this universe, (a) what are the atomic numbers of the lightest two inert gases? (b) What is the ground-state electron configuration of sodium?

41.60 •• A lithium atom has three electrons, and the $^2S_{1/2}$ ground-state electron configuration is $1s^22s$. The $1s^22p$ excited state is split into two closely spaced levels, $^2P_{3/2}$ and $^2P_{1/2}$, by the spin-orbit interaction (see Example 41.7 in Section 41.5). A photon with wavelength 67.09608 μm is emitted in the $^2P_{3/2} \rightarrow {}^2S_{1/2}$ transition, and a photon with wavelength 67.09761 μm is emitted in the $^2P_{1/2} \rightarrow {}^2S_{1/2}$ transition. Calculate the effective magnetic field seen by the electron in the $1s^22p$ state of the lithium atom. How does your result compare to that for the $3p$ level of sodium found in Example 41.7?

41.61 •• A hydrogen atom in an $n = 2$, $l = 1$, $m_l = -1$ state emits a photon when it decays to an $n = 1$, $l = 0$, $m_l = 0$ ground state. (a) In the absence of an external magnetic field, what is the wavelength of this photon? (b) If the atom is in a magnetic field in the $+z$-direction and with a magnitude of 2.20 T, what is the shift in the wavelength of the photon from the zero-field value? Does the magnetic field increase or decrease the wavelength? Disregard the effect of electron spin. [*Hint:* Use the result of Problem 39.74(c).]

41.62 •• **CP** **Electron Spin Resonance.** Electrons in the lower of two spin states in a magnetic field can absorb a photon of the right frequency and move to the higher state. (a) Find the magnetic-field magnitude B required for this transition in a hydrogen atom with $n = 1$ and $l = 0$ to be induced by microwaves with wavelength λ. (b) Calculate the value of B for a wavelength of 4.20 cm.

41.63 •• DATA While working in a magnetics lab, you conduct an experiment in which a hydrogen atom in the $n = 1$ state is in a magnetic field of magnitude B. A photon of wavelength λ (in air) is absorbed in a transition from the $m_s = -\frac{1}{2}$ to the $m_s = +\frac{1}{2}$ state. The wavelengths λ as a function of B are given in the table.

B (T)	0.51	0.74	1.03	1.52	2.02	2.48	2.97
λ (mm)	21.4	14.3	10.7	7.14	5.35	4.28	3.57

(a) Graph the data in the table as photon frequency f versus B, where $f = c/\lambda$. Find the slope of the straight line that gives the best fit to the data. (b) Use your results of part (a) to calculate $|\mu_z|$, the magnitude of the spin magnetic moment. (c) Let $\gamma = |\mu_z|/|S_z|$ denote the gyromagnetic ratio for electron spin. Use your result of part (b) to calculate γ. What is the value of $\gamma/(e/2m)$ given by your experimental data?

41.64 •• A hydrogen atom initially in an $n = 3$, $l = 1$ state makes a transition to the $n = 2$, $l = 0$, $j = \frac{1}{2}$ state. Find the difference in wavelength between the following two photons: one emitted in a transition that starts in the $n = 3$, $l = 1$, $j = \frac{3}{2}$ state and one that starts instead in the $n = 3$, $l = 1$, $j = \frac{1}{2}$ state. Which photon has the longer wavelength?

41.65 •• DATA In studying electron screening in multielectron atoms, you begin with the alkali metals. You look up experimental data and find the results given in the table.

Element	Li	Na	K	Rb	Cs	Fr
Ionization energy (kJ/mol)	520.2	495.8	418.8	403.0	375.7	380

The ionization energy is the minimum energy required to remove the least-bound electron from a ground-state atom. (a) The units kJ/mol given in the table are the minimum energy in kJ required to ionize 1 mol of atoms. Convert the given values for ionization energy to the energy in eV required to ionize one atom. (b) What is the value of the nuclear charge Z for each element in the table? What is the n quantum number for the least-bound electron in the ground state? (c) Calculate Z_{eff} for this electron in each alkali-metal atom. (d) The ionization energies decrease as Z increases. Does Z_{eff} increase or decrease as Z increases? Why does Z_{eff} have this behavior?

41.66 •• DATA You are studying the absorption of electromagnetic radiation by electrons in a crystal structure. The situation is well described by an electron in a cubical box of side length L. The electron is initially in the ground state. (a) You observe that the longest-wavelength photon that is absorbed has a wavelength in air of $\lambda = 624$ nm. What is L? (b) You find that $\lambda = 234$ nm is also absorbed when the initial state is still the ground state. What is the value of n^2 for the final state in the transition for which this wavelength is absorbed, where $n^2 = n_X^2 + n_Y^2 + n_Z^2$? What is the degeneracy of this energy level (including the degeneracy due to electron spin)?

CHALLENGE PROBLEMS

41.67 ••• Spin-$\frac{3}{2}$ particles have four distinct spin states corresponding to $S_z = m_z\hbar$, where m_z may be $\pm\frac{1}{2}$ or $\pm\frac{3}{2}$. We can write the corresponding normalized wave functions as $\psi_{m_z}(\vec{r})$. (a) What is the magnitude of such a particle's spin $\vec{S}$? (b) Suppose we have three identical entangled spin-$\frac{3}{2}$ particles, each sharing an equal probability for exhibiting $m_z = \pm\frac{1}{2}$ or $m_z = -\frac{3}{2}$ but a zero probability for being in the state $m_z = +\frac{3}{2}$. The three-particle wave function is a sum of products such as $\psi_{-1/2}(\vec{r}_1)\psi_{+1/2}(\vec{r}_2)\psi_{-3/2}(\vec{r}_3)$, which we abbreviate as $\psi_{-1/2}\psi_{+1/2}\psi_{-3/2}$. (It is understood that the nth factor corresponds to the position $\vec{r}_n$.) Write the normalized three-particle wave function for the entangled state. Be mindful of the Pauli exclusion principal. (c) The three entangled particles are sent off in three different directions toward magnetic containment facilities A, B, and C located, respectively, at $\vec{r}_{1,2,3}$, where they are captured and retained in circular orbits. After an hour, the particle in facility A is sent through a Stern–Gerlach magnet, where it is determined that its state is $m_z = +\frac{1}{2}$. What is the three-particle wave function after this measurement? (d) At this time, what is the probability that a measurement in facility B will show that its particle is in each of the states $m_z = -\frac{1}{2}, +\frac{1}{2}, -\frac{3}{2}$, and $+\frac{3}{2}$? (e) Subsequently, the particle in facility C is measured to have $m_z = -\frac{3}{2}$. What is the wave function after this measurement? (f) At this time, what is the probability that a measurement in facility B will show that its particle is in each of the states $m_z = -\frac{1}{2}, +\frac{1}{2}, -\frac{3}{2}$, and $+\frac{3}{2}$?

41.68 ••• Each of $2N$ electrons (mass m) is free to move along the x-axis. The potential-energy function for each electron is $U(x) = \frac{1}{2}k'x^2$, where k' is a positive constant. The electric and magnetic interactions between electrons can be ignored. Use the exclusion principle to show that the minimum energy of the system of $2N$ electrons is $\hbar N^2\sqrt{k'/m}$. (*Hint:* See Section 40.5 and the hint given in Problem 41.47.)

41.69 ••• CP Consider a simple model of the helium atom in which two electrons, each with mass m, move around the nucleus (charge $+2e$) in the same circular orbit. Each electron has orbital angular momentum $\hbar$ (that is, the orbit is the smallest-radius Bohr orbit), and the two electrons are always on opposite sides of the nucleus. Ignore the effects of spin. (a) Determine the radius of the orbit and the orbital speed of each electron. [*Hint:* Follow the procedure used in Section 39.3 to derive Eqs. (39.8) and (39.9). Each electron experiences an attractive force from the nucleus and a repulsive force from the other electron.] (b) What is the total kinetic energy of the electrons? (c) What is the potential energy of the system (the nucleus and the two electrons)? (d) In this model, how much energy is required to remove both electrons to infinity? How does this compare to the experimental value of 79.0 eV?

MCAT-STYLE PASSAGE PROBLEMS

BIO Atoms of Unusual Size. In photosynthesis in plants, light is absorbed in light-harvesting complexes that consist of protein and pigment molecules. The absorbed energy is then transported to a specialized complex called the *reaction center*. Quantum-mechanical effects may play an important role in this energy transfer. In a recent experiment, researchers cooled rubidium atoms to a very low temperature to study a similar energy-transfer process in the lab. Laser light was used to excite an electron in each atom to a state with large n. This highly excited electron behaves much like the single electron in a hydrogen atom, with an effective (screened) atomic number $Z_{eff} = 1$. Because n is so large, though, the excited electron is quite far from the atomic nucleus, with an orbital radius of approximately 1 μm, and is weakly bound. Using these so-called *Rydberg atoms,* the researchers were able to study the way energy is transported from one atom to the next. This process may be a model for understanding energy transport in photosynthesis. (Source: "Observing the Dynamics of Dipole-Mediated Energy Transport by Interaction Enhanced Imaging," by G. Günter et al., *Science* 342(6161): 954–956, Nov. 2013.)

41.70 In the Bohr model, what is the principal quantum number n at which the excited electron is at a radius of 1 μm? (a) 140; (b) 400; (c) 20; (d) 81.

41.71 Take the size of a Rydberg atom to be the diameter of the orbit of the excited electron. If the researchers want to perform this experiment with the rubidium atoms in a gas, with atoms separated by a distance 10 times their size, the density of atoms per cubic centimeter should be about (a) 10^5 atoms/cm^3; (b) 10^8 atoms/cm^3; (c) 10^{11} atoms/cm^3; (d) 10^{21} atoms/cm^3.

41.72 Assume that the researchers place an atom in a state with $n = 100$, $l = 2$. What is the magnitude of the orbital angular momentum $\vec{L}$ associated with this state? (a) $\sqrt{2}\,\hbar$; (b) $\sqrt{6}\,\hbar$; (c) $\sqrt{200}\,\hbar$; (d) $\sqrt{10{,}100}\,\hbar$.

41.73 How many different possible electron states are there in the $n = 100$, $l = 2$ subshell? (a) 2; (b) 100; (c) 10,000; (d) 10.

ANSWERS

Chapter Opening Question ?

(iii) The Pauli exclusion principle is responsible. Helium is inert because its two electrons fill the K shell; lithium is very reactive because its third electron must go into the L shell and is loosely bound. See Section 41.6 for more details.

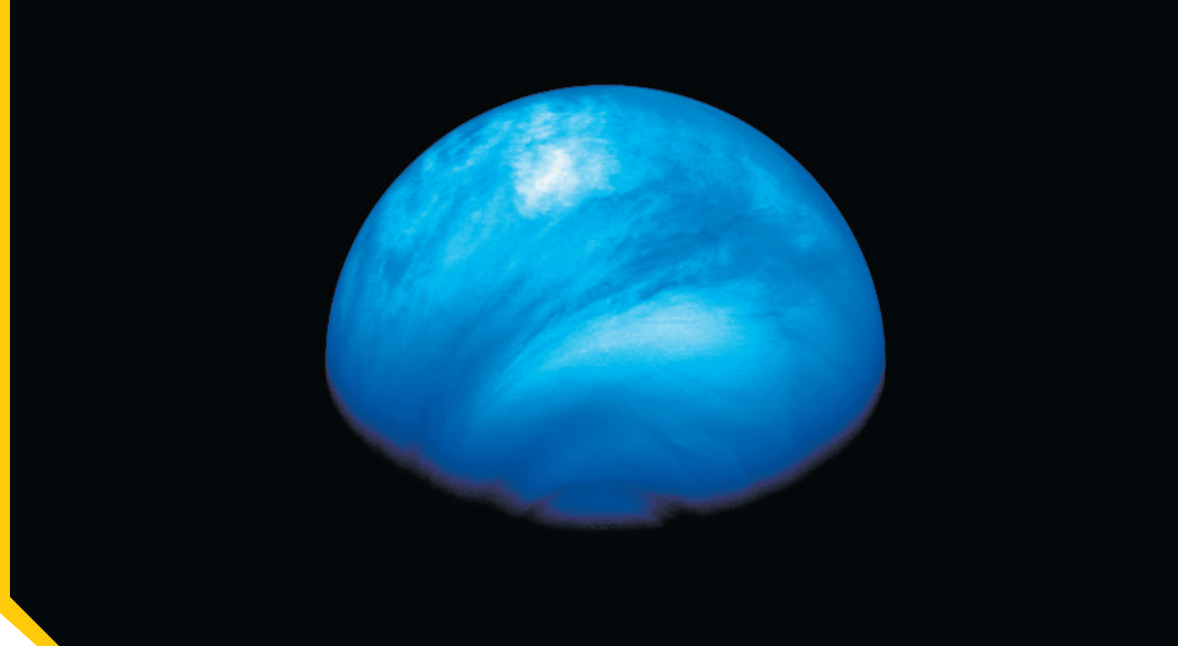
Key Example VARIATION Problems

VP41.1.1 (a) $x = L/2, z = L/3, z = 2L/3$ (b) 0.402
VP41.1.2 (a) 0.818 (b) 0.500 (c) 0.394 (d) 0.500
VP41.1.3 (a) 8.25×10^{-3} (b) 0.0227 (c) 0.0758 (d) 0.0227
VP41.1.4 (a) 0.125 (b) 0.548
VP41.4.1 (a) 36 (b) $\sqrt{30}\hbar = 5.477\hbar$ (c) $5\hbar$
VP41.4.2 (a) $(l, m_l) = (0, 0); (1, 1); (1, 0); (1, -1); (2, 2); (2, 1); (2, 0); (2, -1); (2, -2)$ (b) $(l, m_l) = (2, -2)$, $\theta_{\min} = 35.3°$
VP41.4.3 (a) 1.89 eV (b) 2.55 eV (c) 10.2 eV
VP41.4.4 (a) 0.762 (b) 0.615 (c) 0.0138
VP41.8.1 (a) 3.64×10^{-4} eV (b) $S_z = +\frac{1}{2}\hbar$
VP41.8.2 $f = 6.61 \times 10^{10}$ Hz, $\lambda = 4.53$ mm
VP41.8.3 (a) 0.00716 eV (b) 61.8 T
VP41.8.4 (a) $E_{n=3,j=3/2} = -1.511\ \text{eV}\left(1 + \dfrac{\alpha^2}{12}\right)$, $E_{n=3,j=1/2} = -1.511\ \text{eV}\left(1 + \dfrac{\alpha^2}{4}\right)$ (b) $\Delta E = 1.34 \times 10^{-5}$ eV, the $j = \frac{3}{2}$ state (c) 4.66×10^{-3} nm, the $j = \frac{1}{2}$ state

Bridging Problem

(a) 2.37×10^{-10} m
(b) Values of (n_X, n_Y, n_Z, m_s) for the 22 electrons: $(1, 1, 1, +\frac{1}{2})$, $(1, 1, 1, -\frac{1}{2})$, $(2, 1, 1, +\frac{1}{2})$, $(2, 1, 1, -\frac{1}{2})$, $(1, 2, 1, +\frac{1}{2})$, $(1, 2, 1, -\frac{1}{2})$, $(1, 1, 2, +\frac{1}{2})$, $(1, 1, 2, -\frac{1}{2})$, $(2, 2, 1, +\frac{1}{2})$, $(2, 2, 1, -\frac{1}{2})$, $(2, 1, 2, +\frac{1}{2})$, $(2, 1, 2, -\frac{1}{2})$, $(1, 2, 2, +\frac{1}{2})$, $(1, 2, 2, -\frac{1}{2})$, $(3, 1, 1, +\frac{1}{2})$, $(3, 1, 1, -\frac{1}{2})$, $(1, 3, 1, +\frac{1}{2})$, $(1, 3, 1, -\frac{1}{2})$, $(1, 1, 3, +\frac{1}{2})$, $(1, 1, 3, -\frac{1}{2})$, $(2, 2, 2, +\frac{1}{2})$, $(2, 2, 2, -\frac{1}{2})$
(c) 20.1 eV, 40.2 eV, 60.3 eV, 73.7 eV, and 80.4 eV
(d) 60.3 eV versus 4.52×10^3 eV

? Although Venus is almost twice as far as Mercury is from the sun, it has a higher surface temperature: 735 K (462°C = 863°F). The reason is that Venus has a thick, cloud-shrouded atmosphere (shown here in false color) that is 96.5% carbon dioxide (CO_2). Molecules of CO_2 are a potent agent for raising Venus's temperature because (i) they absorb infrared radiation in vibrational transitions; (ii) they absorb infrared radiation in electronic transitions; (iii) they absorb ultraviolet radiation in vibrational transitions; (iv) they absorb ultraviolet radiation in electronic transitions; (v) more than one of these.



42 Molecules and Condensed Matter

LEARNING OUTCOMES

In this chapter, you'll learn...

42.1 The various types of bonds that hold atoms together.
42.2 How molecular spectra reveal the rotational and vibrational dynamics of molecules.
42.3 How and why atoms form into crystalline structures.
42.4 How to use the energy-band concept to explain the electrical properties of solids.
42.5 A model that explains many of the physical properties of metals.
42.6 How a small amount of an impurity can radically affect the character of a semiconductor.
42.7 Some of the technological applications of semiconductor devices.
42.8 Why certain materials become superconductors at low temperature.

You'll need to review...

10.5 Rotational kinetic energy.
17.7 Greenhouse effect.
18.3–18.5 Ideal gases; equipartition of energy; Maxwell–Boltzmann distribution.
23.1 Electric potential energy.
24.4, 24.5 Dielectrics.
25.2 Resistivity.
39.3 Reduced mass.
40.5 Quantum-mechanical harmonic oscillators.
41.2–41.4, 41.6 Particle in a 3-D box; rotational energy levels; selection rules; exclusion principle.

Isolated atoms, which we studied in Chapter 41, are the exception; usually we find atoms combined to form molecules or more extended structures we call condensed matter (liquid or solid). In this chapter we'll study the attractive forces, called molecular bonds, that cause atoms to combine into molecules. We'll see that just as atoms have quantized energies determined by the quantum-mechanical state of their electrons, so molecules have quantized energies determined by their rotational and vibrational states.

The same physical principles behind molecular bonds also apply to the study of condensed matter, in which various types of bonding occur. We'll explore the concept of energy bands and see how it helps us understand the properties of solids. Then we'll look more closely at the properties of a special class of solids called semiconductors. Devices using semiconductors are found in every mobile phone, TV, and computer used today.

42.1 TYPES OF MOLECULAR BONDS

We can use our discussion of atomic structure in Chapter 41 as a basis for exploring the nature of *molecular bonds,* the interactions that hold atoms together to form stable structures such as molecules and solids.

Ionic Bonds

The **ionic bond** is an interaction between oppositely charged *ionized* atoms. The most familiar example is sodium chloride (NaCl), in which the sodium (Na) atom loses its one 3*s* electron, which fills the vacancy in the 3*p* subshell of the chlorine (Cl) atom.

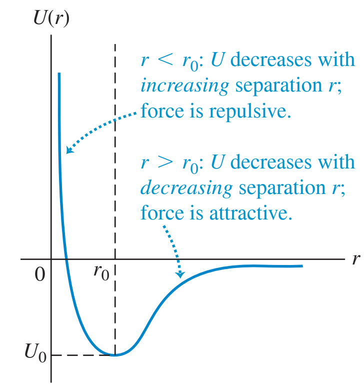
Let's look at the energy balance in this transaction. Removing the 3*s* electron from a neutral Na atom requires 5.138 eV of energy; this is called the *ionization energy* of Na. The neutral Cl atom can attract an extra electron into the vacancy in the 3*p* subshell, where it is incompletely screened by the other electrons and therefore is attracted to the nucleus. This state has 3.613 eV lower energy than a state with a neutral Cl atom and a distant free electron; 3.613 eV is the magnitude of the *electron affinity* of chlorine. Thus creating an Na^+ ion and a Cl^- ion that are well separated from each other (so they do not interact) requires a net investment of only $5.138 \text{ eV} - 3.613 \text{ eV} = 1.525 \text{ eV}$.

When their mutual attraction brings the Na^+ and Cl^- ions together, the magnitude of their negative potential energy is determined by their separation r (**Fig. 42.1**). The exclusion principle (Section 41.6), which states that only one electron can occupy a given quantum-mechanical state, limits how small this separation can be. As r decreases, the exclusion principle distorts the charge clouds, so the ions no longer interact like point charges and the interaction eventually becomes repulsive.

The minimum electric potential energy for NaCl turns out to be -5.7 eV at a separation of 0.24 nm. The net energy released in creating the ions and letting them come together to the equilibrium separation of 0.24 nm is $5.7 \text{ eV} - 1.525 \text{ eV} = 4.2 \text{ eV}$. Thus, if we ignore the kinetic energy of the ions, 4.2 eV is the *binding energy* of the NaCl molecule, the energy that is needed to dissociate the molecule into separate neutral atoms.

Ionic bonds can involve more than one electron per atom. For instance, alkaline earth elements form ionic compounds in which an atom loses *two* electrons; an example is magnesium chloride, or $Mg^{2+}(Cl^-)_2$. Ionic bonds that involve a loss of more than two electrons are relatively rare. Instead, a different kind of bond, the *covalent* bond, comes into operation. We'll discuss this type of bond below.

Figure **42.1** When the separation r between two oppositely charged ions is large, the potential energy $U(r)$ is proportional to $1/r$ as for point charges and the force is attractive. As r decreases, the charge clouds of the two atoms overlap and the force becomes less attractive. If r is less than the equilibrium separation r_0, the force is repulsive.



EXAMPLE 42.1 Electric potential energy of the NaCl molecule

Find the electric potential energy of an Na^+ ion and a Cl^- ion separated by 0.24 nm. Consider the ions as point charges.

IDENTIFY and SET UP Equation (23.9) in Section 23.1 tells us that the electric potential energy of two point charges q and q_0 separated by a distance r is $U = qq_0/4\pi\epsilon_0 r$.

EXECUTE We have $q = +e$ (for Na^+), $q_0 = -e$ (for Cl^-), and $r = 0.24 \text{ nm} = 0.24 \times 10^{-9}$ m. From Eq. (23.9),

$$U = -\frac{1}{4\pi\epsilon_0}\frac{e^2}{r_0} = -(9.0 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2)\frac{(1.6 \times 10^{-19} \text{ C})^2}{0.24 \times 10^{-9} \text{ m}} = -9.6 \times 10^{-19} \text{ J} = -6.0 \text{ eV}$$

EVALUATE This result agrees fairly well with the observed value of -5.7 eV. The reason for the difference is that when the two ions are at their equilibrium separation of 0.24 nm, the outer regions of their electron clouds overlap. Hence the two ions don't behave exactly like point charges.

KEYCONCEPT In an ionic bond, the electric potential energy of the two ions (one positive and one negative) is negative and roughly the same as if we consider the two ions as point charges.

Covalent Bonds

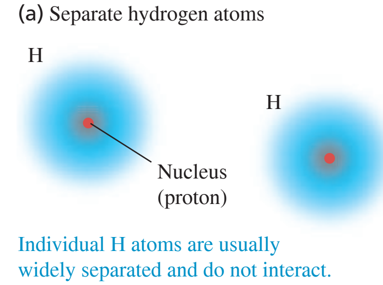
Unlike the transaction that occurs in an ionic bond, in a **covalent bond** there is no net transfer of electrons from one atom to another. The simplest covalent bond is found in the hydrogen molecule, a structure containing two protons and two electrons. As the separate atoms (**Fig. 42.2a**) come together, the electron wave functions are distorted and become more concentrated in the region between the two protons (Fig. 42.2b). The net attraction of the electrons for each proton more than balances the repulsion of the two protons and of the two electrons.

The attractive interaction is then supplied by a *pair* of electrons, one contributed by each atom, with charge clouds that are concentrated primarily in the region between the two atoms. The energy of the covalent bond in the hydrogen molecule H_2 is -4.48 eV.

As we saw in Section 41.6, the exclusion principle permits two electrons to occupy the same region of space (that is, to be in the same spatial quantum state) only when they have opposite spins. Hence the two electrons in the H_2 covalent bond (Fig. 42.2b) must have opposite spins, since both occupy the same region between the two nuclei. Opposite spins are an essential requirement for a covalent bond, and no more than two electrons can participate in such a bond.

Figure **42.2** Covalent bond in a hydrogen molecule.

(a) Separate hydrogen atoms



(b) H_2 molecule

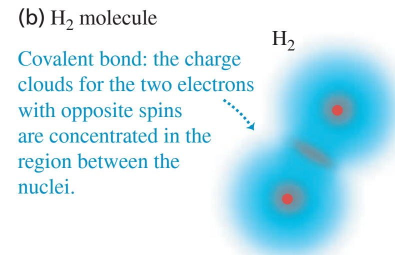
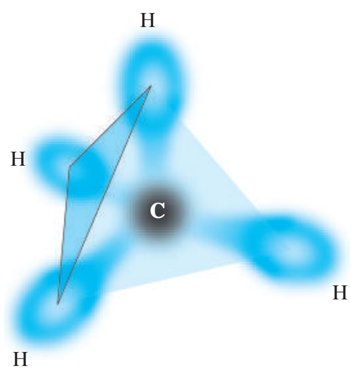


Figure 42.3 Schematic diagram of the methane (CH_4) molecule. The carbon atom is at the center of a regular tetrahedron and forms four covalent bonds with the hydrogen atoms at the corners. Each covalent bond includes two electrons with opposite spins, forming a charge cloud that is concentrated between the carbon atom and a hydrogen atom.



However, an atom with several electrons in its outermost shell can form several covalent bonds. The bonding of carbon and hydrogen atoms, of central importance in organic chemistry, is an example. In the *methane* molecule (CH_4) the carbon atom is at the center of a regular tetrahedron, with a hydrogen atom at each corner. The carbon atom has four electrons in its L shell, and each of these four electrons forms a covalent bond with one of the four hydrogen atoms (**Fig. 42.3**). Similar patterns occur in more complex organic molecules.

Covalent bonds are highly directional. In the methane molecule the wave function for each of carbon's four valence electrons is a combination of the $2s$ and $2p$ wave functions called a *hybrid wave function*. The probability distribution for each one has a lobe protruding toward a corner of a tetrahedron. This symmetric arrangement minimizes the overlap of wave functions for the electron pairs, which in turn minimizes the positive potential energy associated with repulsion between the pairs.

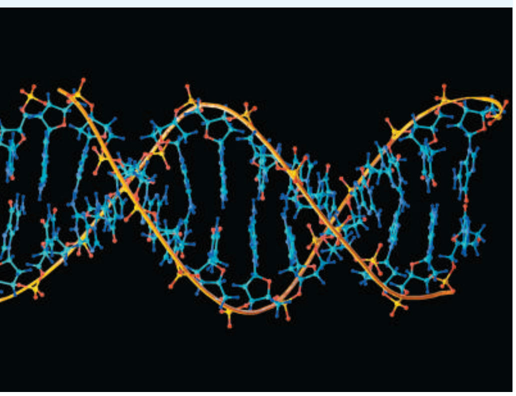
Ionic and covalent bonds represent two extremes in molecular bonding, but there is no sharp division between the two types. Often there is a *partial* transfer of one or more electrons from one atom to another. As a result, many molecules that have dissimilar atoms have electric dipole moments—that is, a preponderance of positive charge at one end and of negative charge at the other. Such molecules are called *polar* molecules. Water molecules have large electric dipole moments; these are responsible for the exceptionally large dielectric constant of liquid water (see Sections 24.4 and 24.5).

Van der Waals Bonds

Ionic and covalent bonds, with typical bond energies of 1 to 5 eV, are called *strong bonds*. There are also two types of weaker bonds. One of these, the **van der Waals bond,** is an interaction between the electric dipole moments of atoms or molecules; typical energies are 0.1 eV or less. The bonding of water molecules in the liquid and solid states results partly from dipole–dipole interactions.

No atom has a permanent electric dipole moment, nor do many molecules. However, fluctuating charge distributions can lead to fluctuating dipole moments; these in turn can induce dipole moments in neighboring structures. Overall, the resulting dipole–dipole interaction is attractive, giving a weak bonding of atoms or molecules. The interaction potential energy drops off very quickly with distance r between molecules, usually as $1/r^6$. The liquefaction and solidification of the inert gases and of molecules such as H_2, O_2, and N_2 are due to induced-dipole van der Waals interactions. Not much thermal-agitation energy is needed to break these weak bonds, so such substances usually exist in the liquid and solid states only at very low temperatures.

BIO APPLICATION Molecular Zipper
A DNA molecule functions like a twisted zipper. Each of the two strands of the "zipper" consists of an outer backbone and inward-facing nucleotide "teeth"; hydrogen bonds between facing teeth "zip" the strands together. The covalent bonds that hold together the atoms of each strand are strong, whereas the hydrogen bonds are relatively weak, so that the cell's biochemical machinery can easily separate the strands for reading or copying.



Hydrogen Bonds

In the other type of weak bond, the **hydrogen bond,** a proton (H^+ ion) gets between two atoms, polarizing them and attracting them by means of the induced dipoles. This bond is unique to hydrogen-containing compounds because only hydrogen has a singly ionized state with no remaining electron cloud; the hydrogen ion is a bare proton, much smaller than any other singly ionized atom. The bond energy is usually less than 0.5 eV. The hydrogen bond is responsible for the cross-linking of long-chain organic molecules such as polyethylene (used in plastic bags). Hydrogen bonding also plays a role in the structure of ice.

All these bond types hold the atoms together in *solids* as well as in molecules. Indeed, a solid is in many respects a giant molecule. Still another type of bonding, the *metallic bond,* comes into play in the structure of metallic solids. We'll return to this subject in Section 42.3.

TEST YOUR UNDERSTANDING OF SECTION 42.1 If electrons obeyed the exclusion principle but did *not* have spin, how many electrons could participate in a covalent bond? (i) One; (ii) two; (iii) three; (iv) more than three.

ANSWER

(i) The exclusion principle states that only one electron can be in a given state. Real electrons have spin, so two electrons (one spin up, one spin down) can be in a given *spatial* state and hence two can participate in a given covalent bond between two atoms. If electrons obeyed the exclusion principle but did not have spin, that state of an electron would be completely described by its spatial distribution and only *one* electron could participate in a covalent bond. (We'll learn in Chapter 44 that this situation is wholly imaginary: There are subatomic particles without spin, but they do *not* obey the exclusion principle.)

42.2 MOLECULAR SPECTRA

Molecules have energy levels that are associated with rotation of a molecule as a whole and with vibration of the atoms relative to each other. Just as transitions between energy levels in atoms lead to atomic spectra, transitions between rotational and vibrational levels in molecules lead to *molecular spectra*.

Rotational Energy Levels

In this discussion we'll concentrate mostly on *diatomic* molecules, to keep things as simple as possible. In **Fig. 42.4** we picture a diatomic molecule as a rigid dumbbell (two point masses m_1 and m_2 separated by a constant distance r_0) that can *rotate* about axes through its center of mass, perpendicular to the line joining them. What are the energy levels associated with this motion?

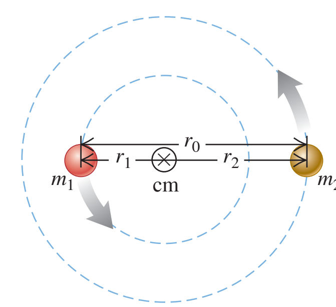


Figure 42.4 A diatomic molecule modeled as two point masses m_1 and m_2 separated by a distance r_0. The distances of the masses from the center of mass are r_1 and r_2, where $r_1 + r_2 = r_0$.

We showed in Section 10.5 that when a rigid body rotates with angular speed ω about a perpendicular axis through its center of mass, the magnitude L of its angular momentum is given by Eq. (10.28), $L = I\omega$, where I is its moment of inertia for that axis. Its kinetic energy is given by Eq. (9.17), $K = \frac{1}{2}I\omega^2$. Combining these two equations, we find $K = L^2/2I$. There is no potential energy U, so the kinetic energy K is equal to the total mechanical energy E:

$$E = \frac{L^2}{2I} \tag{42.1}$$

Zero potential energy means that U does not depend on the angular coordinate of the molecule. But the potential-energy function U for the hydrogen atom (see Section 41.3) also has no dependence on angular coordinates. Thus the angular solutions to the Schrödinger equation for rigid-body rotation are the same as for the hydrogen atom, and the angular momentum is quantized in the same way. As in Eq. (41.22),

$$L = \sqrt{l(l+1)}\hbar \qquad (l = 0, 1, 2, \dots) \tag{42.2}$$

Combining Eqs. (42.1) and (42.2), we obtain the *rotational energy levels:*

Rotational quantum number ($l = 0, 1, 2, \dots$)

Rotational energy levels of a diatomic molecule

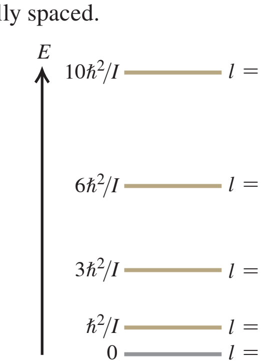
$$E_l = l(l+1)\frac{\hbar^2}{2I} \tag{42.3}$$

Planck's constant divided by 2π

Moment of inertia for axis through molecule's cm

Figure 42.5 is an energy-level diagram showing these rotational levels. The $l = 0$ ground level has zero angular momentum (no rotation and zero rotational energy E). The spacing of adjacent levels increases with increasing l.

Figure 42.5 The ground level and first four excited rotational energy levels for a diatomic molecule. The levels are not equally spaced.



We can express the moment of inertia I in Eqs. (42.1) and (42.3) in terms of the *reduced mass* m_r of the molecule:

Reduced mass of a diatomic molecule

$$m_r = \frac{m_1 m_2}{m_1 + m_2} \qquad (42.4)$$

Mass of atom 1, Mass of atom 2

We introduced the reduced mass in Section 39.3 to accommodate the finite nuclear mass of the hydrogen atom. In Fig. 42.4 the distances r_1 and r_2 are the distances from the center of mass to the centers of the atoms. By the definition of the center of mass, $m_1 r_1 = m_2 r_2$, and the figure also shows that $r_0 = r_1 + r_2$. Solving these equations for r_1 and r_2, we find

$$r_1 = \frac{m_2}{m_1 + m_2} r_0 \qquad r_2 = \frac{m_1}{m_1 + m_2} r_0 \qquad (42.5)$$

The moment of inertia is $I = m_1 r_1^2 + m_2 r_2^2$; substituting Eq. (42.5), we find

$$I = m_1 \frac{m_2^2}{(m_1 + m_2)^2} r_0^2 + m_2 \frac{m_1^2}{(m_1 + m_2)^2} r_0^2 = \frac{m_1 m_2}{m_1 + m_2} r_0^2 \quad \text{or}$$

Moment of inertia of a diatomic molecule, axis through molecule's cm

$$I = m_r r_0^2 \qquad (42.6)$$

Reduced mass; Distance between centers of molecule's two atoms

The moment of inertia is the same as that of an equivalent *single* point mass m_r that orbits the axis in a circle of radius r_0.

To conserve angular momentum and account for the angular momentum of the emitted or absorbed photon, the allowed transitions between rotational states must satisfy the same selection rule that we discussed in Section 41.4 for allowed transitions between the states of an atom: l must change by exactly one unit; that is, $\Delta l = \pm 1$.

EXAMPLE 42.2 Rotational spectrum of carbon monoxide

WITH VARIATION PROBLEMS

The two nuclei in the carbon monoxide (CO) molecule are 0.1128 nm apart. The mass of the most common carbon atom is 1.993×10^{-26} kg; that of the most common oxygen atom is 2.656×10^{-26} kg. (a) Find the energies of the lowest three rotational energy levels of CO. Express your results in meV ($1\text{ meV} = 10^{-3}\text{ eV}$). (b) Find the wavelength of the photon emitted in the transition from the $l = 2$ to the $l = 1$ level.

IDENTIFY and SET UP This problem uses the ideas developed in this section about the rotational energy levels of molecules. We are given the distance r_0 between the atoms and their masses m_1 and m_2. We find the reduced mass m_r from Eq. (42.4), the moment of inertia I from Eq. (42.6), and the energies E_l from Eq. (42.3). The energy E of the emitted photon is equal to the difference in energy between the $l = 2$ and $l = 1$ levels. (This transition obeys the $\Delta l = \pm 1$ selection rule, since $\Delta l = 1 - 2 = -1$.) We determine the photon wavelength by using $E = hc/\lambda$.

EXECUTE (a) From Eqs. (42.4) and (42.6), the reduced mass and moment of inertia of the CO molecule are:

$$m_r = \frac{m_1 m_2}{m_1 + m_2}$$
$$= \frac{(1.993 \times 10^{-26}\text{ kg})(2.656 \times 10^{-26}\text{ kg})}{(1.993 \times 10^{-26}\text{ kg}) + (2.656 \times 10^{-26}\text{ kg})} = 1.139 \times 10^{-26}\text{ kg}$$
$$I = m_r r_0^2$$
$$= (1.139 \times 10^{-26}\text{ kg})(0.1128 \times 10^{-9}\text{ m})^2 = 1.449 \times 10^{-46}\text{ kg} \cdot \text{m}^2$$

The rotational levels are given by Eq. (42.3):

$$E_l = l(l+1)\frac{\hbar^2}{2I} = l(l+1)\frac{(1.0546 \times 10^{-34}\text{ J} \cdot \text{s})^2}{2(1.449 \times 10^{-46}\text{ kg} \cdot \text{m}^2)}$$
$$= l(l+1)(3.838 \times 10^{-23}\text{ J}) = l(l+1)0.2395\text{ meV}$$

($1\text{ meV} = 10^{-3}\text{ eV}$.) Substituting $l = 0, 1, 2$, we find

$$E_0 = 0 \qquad E_1 = 0.479\text{ meV} \qquad E_2 = 1.437\text{ meV}$$

(b) The photon energy and wavelength are

$$E = E_2 - E_1 = 0.958\text{ meV}$$
$$\lambda = \frac{hc}{E} = \frac{(4.136 \times 10^{-15}\text{ eV} \cdot \text{s})(3.00 \times 10^8\text{ m/s})}{0.958 \times 10^{-3}\text{ eV}}$$
$$= 1.29 \times 10^{-3}\text{ m} = 1.29\text{ mm}$$

EVALUATE The differences between the first few rotational energy levels of CO are very small (about $1\text{ meV} = 10^{-3}\text{ eV}$) compared to the differences between atomic energy levels (typically a few eV). Hence a photon emitted by a CO molecule in a transition from the $l = 2$ to the $l = 1$ level has very low energy and a very long wavelength compared to the visible light emitted by excited atoms. Photon wavelengths for rotational transitions in molecules are typically in the microwave and far infrared regions of the spectrum.

In this example we were given the equilibrium separation between the atoms, also called the *bond length,* and we used it to calculate one of the wavelengths emitted by excited CO molecules. In experiments, scientists work this problem backward: By measuring the long-wavelength emissions of a sample of diatomic molecules, they determine the moment of inertia of the molecule and hence the bond length.

KEYCONCEPT Because angular momentum is quantized (quantum number l), the rotational kinetic energy of a diatomic molecule is quantized as well. This energy is proportional to $l(l+1)$ and inversely proportional to the moment of inertia of the molecule for an axis through the molecule's center of mass. Transitions between rotational states that involve photon emission or absorption are allowed only if l changes by 1.

Vibrational Energy Levels

Molecules are never completely rigid. In a more realistic model of a diatomic molecule we represent the connection between atoms not as a rigid rod but as a spring (**Fig. 42.6**). Then, in addition to rotating, the atoms of the molecule can *vibrate* about their equilibrium positions along the line joining them. For small oscillations the restoring force can be taken as proportional to the displacement from the equilibrium separation r_0 (like a spring that obeys Hooke's law with a force constant k'), and the system is a harmonic oscillator. We discussed the quantum-mechanical harmonic oscillator in Section 40.5. The energy levels are given by Eq. (40.46) with the mass m replaced by the reduced mass m_r:

Vibrational energy levels of a diatomic molecule

$$E_n = \left(n + \tfrac{1}{2}\right)\hbar\omega = \left(n + \tfrac{1}{2}\right)\hbar\sqrt{\frac{k'}{m_r}} \qquad (42.7)$$

Vibrational quantum number ($n = 0, 1, 2, \ldots$)
Planck's constant divided by 2π
Oscillation angular frequency
Reduced mass
Force constant

The spacing in energy between any two adjacent vibrational levels is

$$\Delta E = \hbar\omega = \hbar\sqrt{\frac{k'}{m_r}} \qquad (42.8)$$

Figure 42.7 is an energy-level diagram showing these vibrational levels. As an example, for the CO molecule of Example 42.2 the spacing $\hbar\omega$ between levels is 0.2690 eV. From Eq. (42.8) this corresponds to a force constant of 1.90×10^3 N/m, which is a fairly loose spring. (To stretch a macroscopic spring with this value of k' by 1.0 cm would require a force of only 19 N, or about 4 lb.) Force constants for diatomic molecules are typically about 100 to 2000 N/m.

CAUTION Watch out for k, k', and K As in Section 40.5 we're using k' for the force constant, this time to minimize confusion with Boltzmann's constant k, the gas constant per molecule (introduced in Section 18.3). Besides the quantities k and k', we also use the absolute temperature unit $1\ \mathrm{K} = 1$ kelvin.

Rotation and Vibration Combined

Visible-light photons have energies between 1.65 eV and 3.26 eV. The 0.2690 eV energy difference between vibrational levels for carbon monoxide (CO) corresponds to a photon of wavelength 4.613 μm, in the infrared region of the spectrum. This is much closer to the visible region than is the photon in the rotational transition in Example 42.2. In general the energy differences for molecular *vibration* are much smaller than those that produce atomic spectra, but much larger than the energy differences for molecular *rotation*.

When we include *both* rotational and vibrational energies, the energy levels for our diatomic molecule are

$$E_{nl} = l(l+1)\frac{\hbar^2}{2I} + \left(n + \tfrac{1}{2}\right)\hbar\sqrt{\frac{k'}{m_r}} \qquad (42.9)$$

Figure 42.8 shows the energy-level diagram. For each value of n there are many values of l, forming a series of closely spaced levels.

The red arrows in Fig. 42.8 show several possible transitions in which a molecule goes from a level with $n = 2$ to a level with $n = 1$ by emitting a photon. As we mentioned, these transitions must obey the selection rule $\Delta l = \pm 1$ to conserve angular momentum. Another selection rule states that if the vibrational level changes, the vibrational quantum number n in Eq. (42.9) must increase by 1 ($\Delta n = 1$) if a photon is absorbed or decrease by 1 ($\Delta n = -1$) if a photon is emitted.

Illustrating these selection rules, Fig. 42.8 shows that a molecule in the $n = 2$, $l = 4$ level can emit a photon and drop into the $n = 1$, $l = 5$ level ($\Delta n = -1$, $\Delta l = +1$) or the $n = 1$, $l = 3$ level ($\Delta n = -1$, $\Delta l = -1$), but is forbidden from making a $\Delta n = -1$, $\Delta l = 0$ transition into the $n = 1$, $l = 4$ level.

Figure 42.6 A diatomic molecule modeled as two point masses m_1 and m_2 connected by a spring with force constant k'.

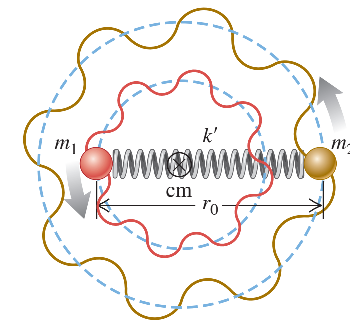


Figure 42.7 The ground level and first three excited vibrational levels for a diatomic molecule, assuming small displacements from equilibrium so we can treat the oscillations as simple harmonic. (Compare Fig. 40.25.)

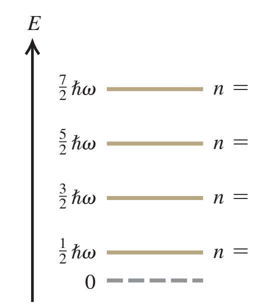


Figure 42.8 Energy-level diagram for vibrational and rotational energy levels of a diatomic molecule. For each vibrational level (n) there is a series of more closely spaced rotational levels (l). Several transitions corresponding to a single band in a band spectrum are shown. These transitions obey the selection rule $\Delta l = \pm 1$.

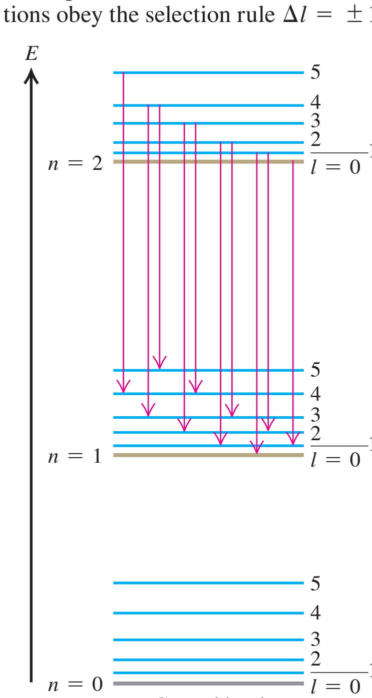


Figure **42.9** A typical molecular band spectrum.

Transitions between states with various pairs of n-values give different series of spectrum lines, and the resulting spectrum has a series of *bands*. Each band corresponds to a particular vibrational transition, and each individual line in a band represents a particular rotational transition, with the selection rule $\Delta l = \pm 1$. **Figure 42.9** shows a typical *band spectrum*.

All molecules can have excited states of the *electrons* in addition to the rotational and vibrational states that we have described. In general, these lie at higher energies than the rotational and vibrational states, and there is no simple rule relating them. When there is a transition between electronic states, the $\Delta n = \pm 1$ selection rule for the vibrational levels no longer holds.

EXAMPLE 42.3 Vibration-rotation spectrum of carbon monoxide

WITH VARIATION PROBLEMS

Consider again the CO molecule of Example 42.2. Find the wavelength of the photon emitted by a CO molecule when its vibrational energy changes and its rotational energy is (a) initially zero and (b) finally zero.

IDENTIFY and SET UP We need to use the selection rules for the vibrational and rotational transitions of a diatomic molecule. Since a photon is emitted as the vibrational energy changes, the selection rule $\Delta n = -1$ tells us that the vibrational quantum number n decreases by 1 in both parts (a) and (b). In part (a) the initial value of l is zero; the selection rule $\Delta l = \pm 1$ tells us that the *final* value of l is 1, so the rotational energy increases in this case. In part (b) the *final* value of l is zero; $\Delta l = \pm 1$ then tells us that the *initial* value of l is 1, and the rotational energy decreases.

The energy E of the emitted photon is the difference between the initial and final energies of the molecule, accounting for the change in both vibrational and rotational energies. In part (a) E equals the difference $\hbar\omega$ between adjacent vibrational energy levels *minus* the rotational energy that the molecule *gains;* in part (b) E equals $\hbar\omega$ *plus* the rotational energy that the molecule *loses*. Example 42.2 tells us that the difference between the $l = 0$ and $l = 1$ rotational energy levels is $0.479 \text{ meV} = 0.000479 \text{ eV}$, and we learned above that the vibrational energy-level separation for CO is $\hbar\omega = 0.2690 \text{ eV}$. We use $E = hc/\lambda$ to determine the corresponding wavelengths (our target variables).

EXECUTE (a) The CO molecule loses $\hbar\omega = 0.2690 \text{ eV}$ of vibrational energy and gains 0.000479 eV of rotational energy. Hence the energy E that goes into the emitted photon equals 0.2690 eV *less* 0.000479 eV, or 0.2685 eV. The photon wavelength is

$$\lambda = \frac{hc}{E} = \frac{(4.136 \times 10^{-15} \text{ eV} \cdot \text{s})(2.998 \times 10^{8} \text{ m/s})}{0.2685 \text{ eV}}$$

$$= 4.618 \times 10^{-6} \text{ m} = 4.618 \ \mu\text{m}$$

(b) Now the CO molecule loses $\hbar\omega = 0.2690 \text{ eV}$ of vibrational energy and also loses 0.000479 eV of rotational energy, so the energy that goes into the photon is $E = 0.2690 \text{ eV} + 0.000479 \text{ eV} = 0.2695 \text{ eV}$. The wavelength is

$$\lambda = \frac{hc}{E} = \frac{(4.136 \times 10^{-15} \text{ eV} \cdot \text{s})(2.998 \times 10^{8} \text{ m/s})}{0.2695 \text{ eV}}$$

$$= 4.601 \times 10^{-6} \text{ m} = 4.601 \ \mu\text{m}$$

EVALUATE In part (b) the molecule loses more energy than it does in part (a), so the emitted photon must have greater energy and a shorter wavelength. That's just what our results show.

KEYCONCEPT In addition to rotational motion around its center of mass (quantum number l), a diatomic molecule can have vibrational motion of its atoms. The vibrational energy levels (quantum number n) are evenly spaced in energy. Transitions between vibrational states that involve photon emission or absorption are allowed only if n and l both change by 1.

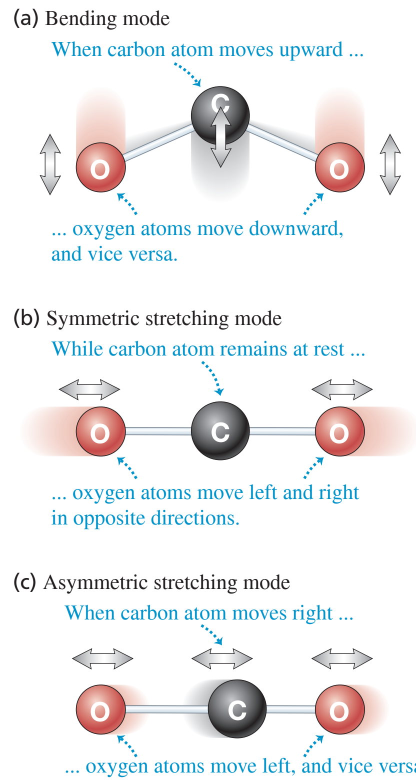
Complex Molecules

We can apply these same principles to more complex molecules. A molecule with three or more atoms has several different kinds or *modes* of vibratory motion. Each mode has its own set of energy levels, related to its frequency by Eq. (42.7). In nearly all cases the associated radiation lies in the infrared region of the electromagnetic spectrum.

Infrared spectroscopy has proved to be an extremely valuable analytical tool. It provides information about the strength, rigidity, and length of molecular bonds and the structure of complex molecules. Also, because every molecule (like every atom) has its characteristic spectrum, infrared spectroscopy can be used to identify unknown compounds.

One molecule that can readily absorb and emit infrared radiation is carbon dioxide (CO_2). **Figure 42.10** shows the three possible modes of vibration of a CO_2 molecule. A number of transitions are possible between excited levels of the same vibrational mode as well as between levels of different vibrational modes. The energy differences are less than 1 eV in all of these transitions, and so involve infrared photons of wavelength longer than 1 μm. Hence a gas of CO_2 can readily absorb light at a number of different infrared wavelengths. This makes CO_2 primarily responsible for the greenhouse effect (Section 17.7) on the earth, even though CO_2 is only 0.04% of our atmosphere by volume. On Venus, however, the atmosphere has more than 90 times the total mass of our atmosphere and is almost entirely CO_2. The resulting greenhouse effect is tremendous: The surface temperature on Venus is more than 400 kelvins higher than what it would be if the planet had no atmosphere at all.

Figure **42.10** The carbon dioxide molecule can vibrate in three different modes. For clarity, the atoms are not shown to scale: The separation between atoms is actually comparable to their diameters.



TEST YOUR UNDERSTANDING OF SECTION 42.2 A rotating diatomic molecule emits a photon when it makes a transition from level (n, l) to level $(n - 1, l - 1)$. If the value of l increases but n is unchanged, does the wavelength of the emitted photon (i) increase, (ii) decrease, or (iii) remain unchanged?

ANSWER

(ii) Figure 42.5 shows that the difference in energy between adjacent rotational levels increases with increasing l. Hence, as l increases, the energy E of the emitted photon increases and the wavelength $\lambda = hc/E$ decreases.

42.3 STRUCTURE OF SOLIDS

The term *condensed matter* includes both solids and liquids. In both states, the interactions between atoms or molecules are strong enough to give the material a definite volume that changes relatively little with applied stress. In condensed matter, adjacent atoms attract one another until their outer electron charge clouds begin to overlap significantly. Thus the distances between adjacent atoms in condensed matter are about the same as the diameters of the atoms themselves, typically 0.1 to 0.5 nm. Also, when we speak of the distances between atoms, we mean the center-to-center (nucleus-to-nucleus) distances.

Ordinarily, we think of a liquid as a material that can flow and of a solid as a material with a definite shape. However, if you heat a horizontal glass rod in the flame of a burner, you'll find that the rod begins to sag (flow) more and more easily as its temperature rises. Glass has no definite transition from solid to liquid, and no definite melting point. On this basis, we can consider glass at room temperature as being an extremely viscous liquid. Tar and butter show similar behavior.

What is the microscopic difference between materials like glass or butter and solids like ice or copper, which do have definite melting points? Ice and copper are examples of *crystalline solids* in which the atoms have *long-range order,* a recurring pattern of atomic positions that extends over many atoms. This pattern is called the *crystal structure*. In contrast, glass at room temperature is an example of an *amorphous* solid, one that has no long-range order but only *short-range order* (correlations between neighboring atoms or molecules). Liquids also have only short-range order. The boundaries between crystalline solid, amorphous solid, and liquid may be sometimes blurred. Some solids, crystalline when perfect, can form with so many imperfections in their structure that they have almost no long-range order. (Yet another kind of order is found in a *liquid crystal,* which is made up of rod-shaped or disc-shaped molecules of an organic compound. The positions of the molecules in the liquid are not fixed, but there is *orientational* order; the axes of the molecules tend to align with each other. This ordering can extend over a distance of many molecules.)

Nearly everything we know about crystal structure was learned from diffraction experiments with x rays, electrons, or neutrons. A typical distance between atoms is of the order of 0.1 nm. You can show that 12.4 keV x rays, 150 eV electrons, and 0.0818 eV neutrons all have wavelengths $\lambda = 0.1$ nm.

BIO APPLICATION Using Crystals to Determine Protein Structure Protein molecules can form crystals, such as these crystals of insulin (a protein composed of 51 amino acids). All of the molecules within a single crystal of a protein have the same orientation; how the crystal diffracts x rays or neutrons depends on the shape and size of the molecules. By analyzing these diffraction patterns, scientists have deduced the molecular structures of more than 100,000 types of proteins.

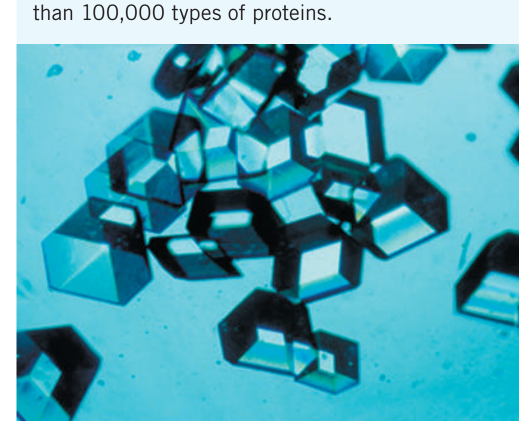
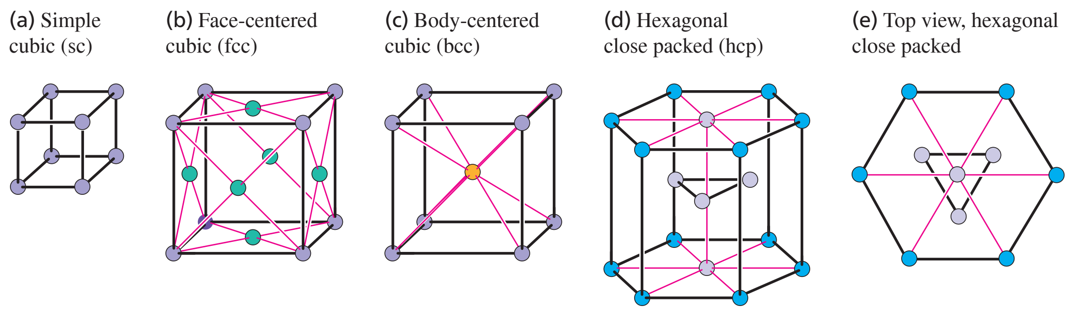


Figure **42.11** Portions of some common types of crystal lattices.



Crystal Lattices and Structures

An essential part of understanding crystals is the idea of a *crystal lattice,* which is a repeating pattern of mathematical points that extends throughout space. There are 14 general types of such patterns; **Fig. 42.11** shows small portions of some common examples. The *simple cubic lattice* (sc) has a lattice point at each corner of a cubic array (Fig. 42.11a). The *face-centered cubic lattice* (fcc) is like the simple cubic but with an additional lattice point at the center of each cube face (Fig. 42.11b). The *body-centered cubic lattice* (bcc) is like the simple cubic but with an additional point at the center of each cube (Fig. 42.11c). The *hexagonal close-packed lattice* (hcp) has layers of lattice points in hexagonal patterns, each hexagon made up of six equilateral triangles (Figs. 42.11d and 42.11e).

CAUTION A perfect crystal lattice is infinitely large Figure 42.11 shows just enough lattice points so that you can easily visualize the pattern; the lattice, a mathematical abstraction, extends throughout space. Thus the lattice points shown repeat endlessly in all directions.

Figure **42.12** (a) The bcc *structure* is composed of a bcc *lattice* with a basis of one atom for each lattice point. (b) The fcc *structure* is composed of an fcc *lattice* with a basis of one atom for each lattice point. These structures repeat precisely to make up perfect crystals.

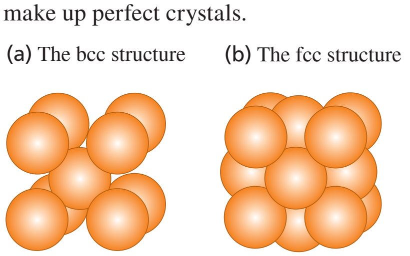
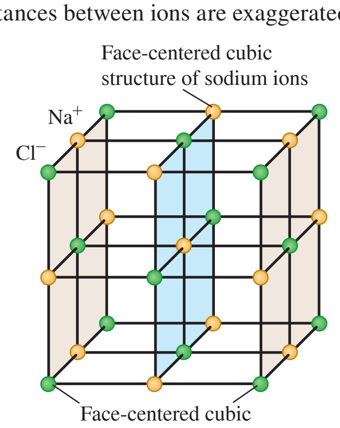


Figure **42.13** Representation of part of the sodium chloride crystal structure. The distances between ions are exaggerated.



In a crystal structure, a single atom or a group of atoms is associated with each lattice point. The group may contain the same or different kinds of atoms. This atom or group of atoms is called a *basis.* Thus a complete description of a crystal structure includes both the lattice and the basis. We initially consider *perfect crystals,* or *ideal single crystals,* in which the crystal structure extends uninterrupted throughout space.

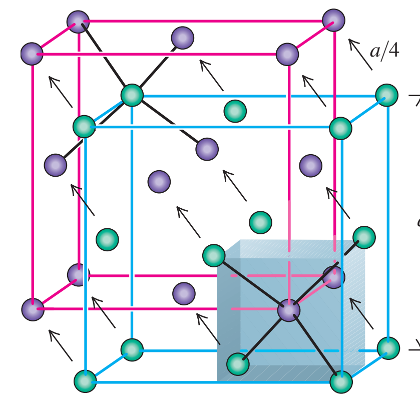
The bcc and fcc structures are two common simple crystal structures. The alkali metals have a bcc structure—that is, a bcc lattice with a basis of one atom at each lattice point. Each atom in a bcc structure has eight nearest neighbors (**Fig. 42.12a**). The elements Al, Ca, Cu, Ag, and Au have an fcc structure—that is, an fcc lattice with a basis of one atom at each lattice point. Each atom in an fcc structure has 12 nearest neighbors (Fig. 42.12b).

Figure 42.13 shows a representation of the structure of sodium chloride (NaCl, ordinary salt). It may look like a simple cubic structure, but it isn't. The sodium and chloride ions each form an fcc structure, so we can think roughly of the sodium chloride structure as being composed of two interpenetrating fcc structures. More correctly, the sodium chloride crystal structure of Fig. 42.13 has an fcc lattice with one chloride ion at each lattice point and one sodium ion half a cube length above it. That is, its basis consists of one chloride and one sodium ion.

Another example is the *diamond structure;* it's called that because it is the crystal structure of carbon in the diamond form. It's also the crystal structure of silicon, germanium, and gray tin. The diamond lattice is fcc; the basis consists of one atom at each lattice point and a second *identical* atom displaced a quarter of a cube length in each of the three cube-edge directions. **Figure 42.14** will help you visualize this. The shaded volume in Fig. 42.14 shows the bottom right front eighth of the basic cube; the four atoms at alternate corners of this cube are at the corners of a regular tetrahedron, and there is an additional atom at the center. Thus each atom in the diamond structure is at the center of a regular tetrahedron with four nearest-neighbor atoms at the corners.

In the diamond structure, both the purple and green spheres in Fig. 42.14 represent *identical* atoms—for example, both carbon or both silicon. In the cubic zinc sulfide structure, the purple spheres represent one type of atom and the green spheres represent a *different* type. For example, in zinc sulfide (ZnS) each zinc atom (purple in Fig. 42.14) is at the center of a regular tetrahedron with four sulfur atoms (green in Fig. 42.14) at its corners, and vice versa. Gallium arsenide (GaAs) and similar compounds have this same structure.

Figure **42.14** The diamond structure, shown as two interpenetrating face-centered cubic structures with distances between atoms exaggerated. Relative to the corresponding green atom, each purple atom is shifted up, back, and to the left by a distance $a/4$.



Bonding in Solids

The forces that are responsible for the regular arrangement of atoms in a crystal are the same as those involved in molecular bonds, plus one additional type discussed below. Not surprisingly, *ionic* and *covalent* molecular bonds are found in ionic and covalent crystals, respectively. The most familiar *ionic crystals* are the alkali halides, such as ordinary salt (NaCl). The positive sodium ions and the negative chloride ions occupy adjacent positions in the crystal (see Fig. 42.13). The attractive forces are the familiar Coulomb's-law forces between charged particles. These forces have no preferred direction, and the arrangement in which the material crystallizes is partly determined by the relative sizes of the two ions. Such a structure is *stable* in the sense that it has lower total energy than the separated ions (see the following example). The negative potential energies of pairs of opposite charges are greater in absolute value than the positive energies of pairs of like charges because the pairs of unlike charges are closer together, on average.

EXAMPLE 42.4 Potential energy of an ionic crystal

Imagine a one-dimensional ionic crystal consisting of a very large number of alternating positive and negative ions with charges e and $-e$, with equal spacing a along a line. Show that the total interaction potential energy is negative, which means that such a "crystal" is stable.

IDENTIFY and SET UP We treat each ion as a point charge and use our results from Section 23.1 for the electric potential energy of a collection of point charges. Equations (23.10) and (23.11) tell us to consider the electric potential energy U of each pair of charges. The total potential energy of the system is the sum of the values of U for every possible pair; we take the number of pairs to be infinite.

EXECUTE Let's pick an ion somewhere in the middle of the line and add the potential energies of its interactions with all the ions to one side of it. From Eq. (23.11), that sum is

$$\begin{aligned}\Sigma U &= -\frac{e^2}{4\pi\epsilon_0}\frac{1}{a} + \frac{e^2}{4\pi\epsilon_0}\frac{1}{2a} - \frac{e^2}{4\pi\epsilon_0}\frac{1}{3a} + \cdots \\ &= -\frac{e^2}{4\pi\epsilon_0 a}\left(1 - \tfrac{1}{2} + \tfrac{1}{3} - \tfrac{1}{4} + \cdots\right)\end{aligned}$$

You may notice that the series in parentheses resembles the Taylor series for the function $\ln(1 + x)$:

$$\ln(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots$$

When $x = 1$ this becomes the series in parentheses above, so

$$\Sigma U = -\frac{e^2}{4\pi\epsilon_0 a}\ln 2$$

This is certainly a negative quantity. The atoms on the other side of the ion we're considering make an equal contribution to the potential energy. And if we include the potential energies of all pairs of atoms, the sum is certainly negative.

EVALUATE We conclude that this one-dimensional ionic "crystal" is stable: It has lower energy than the zero electric potential energy that is obtained when all the ions are infinitely far apart from each other.

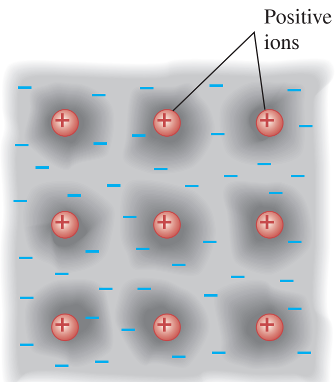
KEYCONCEPT An ionic crystal is stable because the net electric potential energy of its ions is less than if the crystal were taken apart and the ions were moved far from each other.

Types of Crystals

Carbon, silicon, germanium, and tin in the diamond structure are simple examples of *covalent crystals.* These elements are in Group IV of the periodic table, meaning that each atom has four electrons in its outermost shell. Each atom forms a covalent bond with each of four adjacent atoms at the corners of a tetrahedron (Fig. 42.14). These bonds are strongly directional because of the asymmetric electron distributions dictated by the exclusion principle (see Fig. 42.3), and the result is the tetrahedral diamond structure.

A third crystal type, less directly related to the chemical bond than are ionic or covalent crystals, is the **metallic crystal.** In this structure, one or more of the outermost

Figure 42.15 In a metallic solid, one or more electrons are detached from each atom and are free to wander around the crystal, forming an "electron gas." The wave functions for these electrons extend over many atoms. The positive ions vibrate around fixed locations in the crystal.



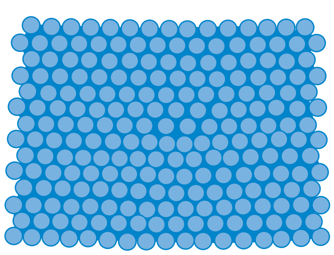
electrons in each atom become detached from the parent atom (leaving a positive ion). The detached electrons are free to move through the crystal and are not localized near the ion from which they originated. So we can picture a metallic crystal as an array of positive ions immersed in a sea of freed electrons whose attraction for the positive ions holds the crystal together (**Fig. 42.15**).

This sea of electrons, which gives metals their high electrical and thermal conductivities, has many of the properties of a gas. Indeed, we speak of the *electron-gas model* of metallic solids. The simplest version of this model is the *free-electron model,* which ignores interactions with the ions completely (except at the surface). We'll return to this model in Section 42.5.

In a metallic crystal the freed electrons are shared among *many* atoms. This gives a bonding that is neither localized nor strongly directional. The crystal structure is determined primarily by considerations of *close packing*—that is, the maximum number of atoms that can fit into a given volume. The two most common metallic crystal lattices are the face-centered cubic and hexagonal close-packed (see Figs. 42.11b, 42.11d, and 42.11e). In structures composed of these lattices with a basis of one atom, each atom has 12 nearest neighbors.

Hydrogen bonds and van der Waals forces also play a role in the structure of some solids. In polyethylene and similar polymers, covalent bonding of atoms forms long-chain molecules, and hydrogen bonding forms cross-links between adjacent chains. In solid water, both hydrogen bonds and van der Waals forces are significant in determining the crystal structures of ice.

Figure 42.16 An edge dislocation in two dimensions. In three dimensions an edge dislocation would look like an extra plane of atoms slipped partway into the crystal.



You can see the irregularity most easily by viewing the figure from various directions at a grazing angle with the page.

Our discussion has centered on perfect crystals. Real crystals show a variety of departures from this idealized structure. Materials are often *polycrystalline,* composed of many small single crystals bonded together at *grain boundaries*. There may be *point defects* within a crystal: *Interstitial* atoms may occur in places where they do not belong, and there may be *vacancies,* positions that should be occupied by an atom but are not. A point defect of interest in semiconductors, which we'll discuss in Section 42.6, is the *substitutional impurity,* a foreign atom replacing a regular atom (for example, arsenic in a silicon crystal).

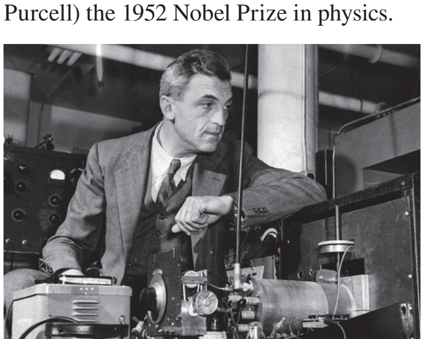
There are several basic types of extended defects called *dislocations*. One type is the *edge dislocation,* shown schematically in **Fig. 42.16**, in which one plane of atoms slips relative to another. The mechanical properties of metallic crystals are influenced strongly by the presence of dislocations. The ductility and malleability of some metals depend on the presence of dislocations that can move through the crystal during plastic deformations. The biggest extended defect of all, present in *all* real crystals, is the surface of the material with its dangling bonds and abrupt change in potential energy.

TEST YOUR UNDERSTANDING OF SECTION 42.3 If a is the distance in an NaCl crystal from an Na^+ ion to one of its nearest-neighbor Cl^- ions, what is the distance from an Na^+ ion to one of its *next-to-nearest*-neighbor Cl^- ions? (i) $a\sqrt{2}$; (ii) $a\sqrt{3}$; (iii) $2a$; (iv) none of these.

ANSWER

❙ **(ii)** In Fig. 42.13 let a be the distance between adjacent Na^+ and Cl^- ions. This figure shows that the Cl^- ion that is the next nearest neighbor to an Na^+ ion is on the opposite corner of a cube of side a. The distance between these two ions is $\sqrt{a^2 + a^2 + a^2} = \sqrt{3a^2} = a\sqrt{3}$.

Figure 42.17 The concept of energy bands was first developed by the Swiss-American physicist Felix Bloch (1905–1983) in his doctoral thesis. Our modern understanding of electrical conductivity stems from that landmark work. Bloch's work in nuclear physics brought him (along with Edward Purcell) the 1952 Nobel Prize in physics.

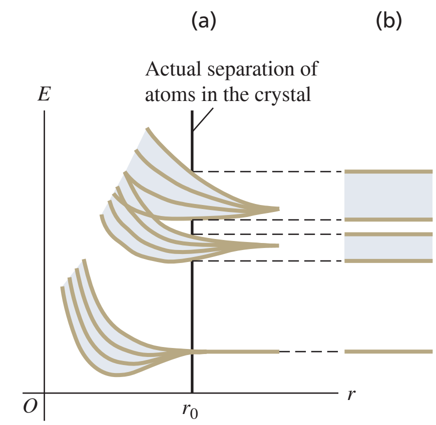


42.4 ENERGY BANDS

The **energy-band** concept, introduced in 1928 (**Fig. 42.17**), is a great help in understanding several properties of solids. To introduce the idea, suppose we have a large number N of identical atoms, far enough apart that their interactions are negligible. Every atom has the same energy-level diagram. We can draw an energy-level diagram for the *entire system*. It looks just like the diagram for a single atom, but the exclusion principle, applied to the entire system, permits each state to be occupied by N electrons (one per atom) instead of just one.

Now we begin to push the atoms uniformly closer together. Because of the electrical interactions and the exclusion principle, the wave functions begin to distort, especially those of the outer, or *valence,* electrons. The corresponding energies also shift, some upward and some downward, by varying amounts, as the valence electron wave functions become less localized and extend over more and more atoms. (The inner electrons in an atom are affected much less by nearby atoms than are the valence electrons, and their energy levels remain relatively sharp.) Thus the valence states that formerly gave the *system* a state with a sharp energy level that could accommodate N electrons now give a *band* containing N closely spaced levels (**Fig. 42.18**). Ordinarily, N is somewhere near the order of Avogadro's number (10^{24}), so we can accurately treat the levels as forming a *continuous* distribution of energies within a band. Between adjacent energy bands are gaps where there are *no* allowed energy levels.

Figure 42.18 Origin of energy bands in a solid. (a) As the distance r between atoms decreases, the energy levels spread into bands. The vertical line at r_0 shows the actual atomic spacing in the crystal. (b) Symbolic representation of energy bands.



Insulators, Semiconductors, and Conductors

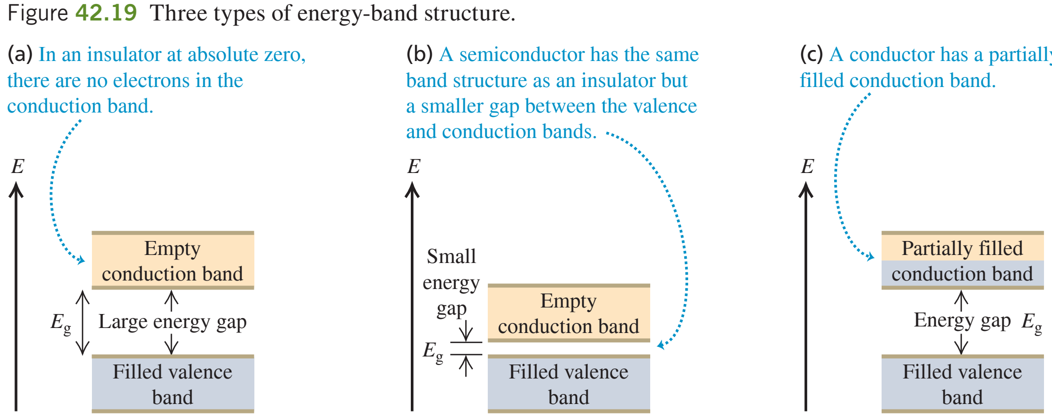
The nature of the energy bands determines whether the material is an electrical insulator, a semiconductor, or a conductor. In particular, what matters are the extent to which the states in each band are occupied and the spacing, called the *band gap* or *energy gap,* between adjacent bands.

In an *insulator* at absolute zero temperature, the highest band that is completely filled, called the **valence band,** is also the highest band that has *any* electrons in it. The next higher band, called the **conduction band,** is completely empty; there are no electrons in its states (**Fig. 42.19a**). Imagine what happens if an electric field is applied to a material of this kind. To move in response to the field, an electron would have to go into a different quantum state with a slightly different energy. It can't do that, however, because all the neighboring states are already occupied. The only way such an electron can move is to jump across the energy gap into the conduction band, where there are plenty of nearby unoccupied states. At any temperature above absolute zero there is some probability this jump can happen, because an electron can gain energy from thermal motion. In an insulator, however, the energy gap between the valence and conduction bands can be 5 eV or more, and that much thermal energy is not ordinarily available. Hence little or no current flows in response to an applied electric field, and the electrical conductivity (Section 25.2) is low. The thermal conductivity (Section 17.7), which also depends on mobile electrons, is likewise low.

We saw in Section 24.4 that an insulator becomes a conductor if it is subjected to a large enough electric field; this is called *dielectric breakdown*. If the electric field is of the order of 10^{10} V/m, there is a potential difference of a few volts over a distance comparable to atomic sizes. In this case the field can do enough work on a valence electron to boost it across the energy gap and into the conduction band. (In practice dielectric breakdown occurs for fields much less than 10^{10} V/m, because imperfections in the structure of an insulator provide some more accessible energy states *within* the energy gap.)

As in an insulator, a *semiconductor* at absolute zero has an empty conduction band above the full valence band. The difference is that in a semiconductor the energy gap

Figure 42.19 Three types of energy-band structure.



between these bands is relatively small and electrons can more readily jump into the conduction band (Fig. 42.19b). As the temperature of a semiconductor increases, the population in the conduction band increases very rapidly, as does the electrical conductivity. For example, in a semiconductor near room temperature with an energy gap of 1 eV, the number of conduction electrons doubles when the temperature rises by just 10°C. We'll use the concept of energy bands to explore semiconductors in more depth in Section 42.6.

In a *conductor* such as a metal, there are electrons in the conduction band even at absolute zero (Fig. 42.19c). The metal sodium is an example. An analysis of the atomic energy-level diagram for sodium (see Fig. 39.19a) shows that for an isolated sodium atom, the six lowest excited states (all 3*p* states) are about 2.1 eV above the two 3*s* ground states. In solid sodium, however, the atoms are so close together that the 3*s* and 3*p* *bands* spread out and overlap into a single band. Each sodium atom contributes one electron to the band, leaving an Na^+ ion behind. Each atom also contributes eight *states* to that band (two 3*s*, six 3*p*), so the band is only one-eighth occupied. We call this structure a *conduction* band because it is only partially occupied. Electrons near the top of the filled portion of the band have many adjacent unoccupied states available, and they can easily gain or lose small amounts of energy in response to an applied electric field. Therefore these electrons are mobile, giving solid sodium its high electrical and thermal conductivity. A similar description applies to other conducting materials.

EXAMPLE 42.5 Photoconductivity in germanium

At room temperature, pure germanium has an almost completely filled valence band separated by a 0.67 eV gap from an almost completely empty conduction band. It is a poor electrical conductor, but its conductivity increases greatly when it is irradiated with electromagnetic waves of a certain maximum wavelength. What is that wavelength?

IDENTIFY and SET UP The conductivity of a semiconductor increases greatly when electrons are excited from the valence band into the conduction band. In germanium, the excitation occurs when an electron absorbs a photon with an energy of at least $E_{\min} = 0.67$ eV. From $E = hc/\lambda$, the *maximum* wavelength $\lambda_{\max}$ (our target variable) corresponds to this *minimum* photon energy.

EXECUTE The wavelength of a photon with energy $E_{\min} = 0.67$ eV is

$$\lambda_{\max} = \frac{hc}{E_{\min}} = \frac{(4.136 \times 10^{-15}\ \text{eV}\cdot\text{s})(3.00 \times 10^{8}\ \text{m/s})}{0.67\ \text{eV}}$$
$$= 1.9 \times 10^{-6}\ \text{m} = 1.9\ \mu\text{m} = 1900\ \text{nm}$$

EVALUATE This wavelength is in the infrared part of the spectrum, so visible-light photons (which have shorter wavelength and higher energy) will also induce conductivity in germanium. As we'll see in Section 42.7, semiconductor crystals are widely used as photocells and for many other applications.

KEYCONCEPT The electron energy levels in a semiconductor form a valence band that is essentially filled, plus a conduction band that is empty and separated from the valence band by a small energy gap. The conductivity of a semiconductor can be increased by using photons to excite electrons from the valence band into the conduction band. To do this, the photon energy must be at least as great as the band gap; lower-energy photons will not be absorbed.

TEST YOUR UNDERSTANDING OF SECTION 42.4 One type of thermometer works by measuring the temperature-dependent electrical resistivity of a sample. Which of the following types of material displays the greatest change in resistivity for a given temperature change? (i) Insulator; (ii) semiconductor; (iii) conductor.

ANSWER

(ii) A small temperature change causes a substantial increase in the population of electrons in a semiconductor's conduction band and a comparably substantial increase in conductivity. The conductivity of conductors and insulators varies more gradually with temperature.

42.5 FREE-ELECTRON MODEL OF METALS

Studying the energy states of electrons in metals can give us a lot of insight into their electrical and magnetic properties, the electron contributions to heat capacities, and other behavior. As we discussed in Section 42.3, one of the distinguishing features of a metal is that one or more valence electrons are detached from their home atom and can move freely within the metal, with wave functions that extend over many atoms.

The **free-electron model** assumes that these electrons don't interact at all with the ions or with each other, but that there are infinite potential-energy barriers at the surfaces. The idea is that a typical electron moves so rapidly within the metal that it "sees" the effect of the ions and other electrons as a uniform potential-energy function, whose value we can choose to be zero.

We can represent the surfaces of the metal by the same cubical box that we analyzed in Section 41.2 (the three-dimensional version of the particle in a box studied in Section 40.2). If the box has sides of length L (**Fig. 42.20**), the energies of the stationary states (quantum states of definite energy) are

$$E_{n_X,n_Y,n_Z} = \frac{(n_X{}^2 + n_Y{}^2 + n_Z{}^2)\pi^2\hbar^2}{2mL^2} \qquad \begin{pmatrix} n_X = 1, 2, 3, \ldots; \\ n_Y = 1, 2, 3, \ldots; \\ n_Z = 1, 2, 3, \ldots \end{pmatrix} \tag{42.10}$$

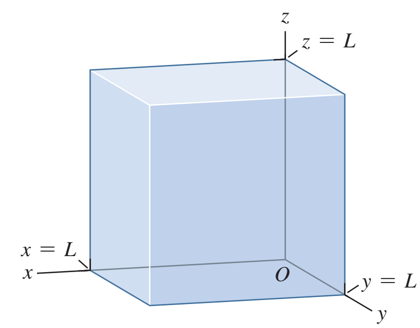


Figure **42.20** A cubical box with side length L. We studied this three-dimensional version of the infinite square well in Section 41.2. The energy levels for a particle in this box are given by Eq. (42.10).

Each state is labeled by the three positive-integer quantum numbers (n_X, n_Y, n_Z).

Density of States

Later we'll need to know the *number dn* of quantum states that have energies in a given range dE. The number of states per unit energy range dn/dE is called the **density of states,** denoted by $g(E)$. We'll begin by working out an expression for $g(E)$. Think of a three-dimensional space with coordinates (n_X, n_Y, n_Z) (**Fig. 42.21**). The radius n_{rs} of a sphere centered at the origin in that space is $n_{rs} = (n_X{}^2 + n_Y{}^2 + n_Z{}^2)^{1/2}$. Each point with integer coordinates in that space represents one spatial quantum state. Thus each point corresponds to one unit of volume in the space, and the total number of points with integer coordinates inside a sphere equals the volume of the sphere, $\frac{4}{3}\pi n_{rs}{}^3$. Because all our n's are positive, we must take only one *octant* of the sphere, with $\frac{1}{8}$ the total volume, or $(\frac{1}{8})(\frac{4}{3}\pi n_{rs}{}^3) = \frac{1}{6}\pi n_{rs}{}^3$. The particles are electrons, so each point corresponds to *two* states with opposite spin components $(m_s = \pm\frac{1}{2})$, and the total number n of electron states corresponding to points inside the octant is twice $\frac{1}{6}\pi n_{rs}{}^3$, or

$$n = \frac{\pi n_{rs}{}^3}{3} \tag{42.11}$$

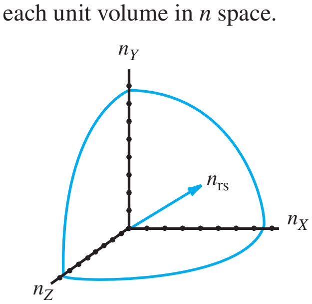


Figure **42.21** The allowed values of n_X, n_Y, and n_Z are positive integers for the electron states in the free-electron gas model. Including spin, there are two states for each unit volume in n space.

The energy E of states at the surface of the sphere can be expressed in terms of n_{rs}. Equation (42.10) becomes

$$E = \frac{n_{rs}{}^2\pi^2\hbar^2}{2mL^2} \tag{42.12}$$

We can combine Eqs. (42.11) and (42.12) to get a relationship between E and n that doesn't contain n_{rs}. We'll leave the details for you to work out; the total number of states with energies of E or less is

$$n = \frac{(2m)^{3/2}VE^{3/2}}{3\pi^2\hbar^3} \tag{42.13}$$

where $V = L^3$ is the volume of the box.

To get the number of states dn in an energy interval dE, we treat n and E as continuous variables and take differentials of both sides of Eq. (42.13):

$$dn = \frac{(2m)^{3/2}VE^{1/2}}{2\pi^2\hbar^3}\,dE \tag{42.14}$$

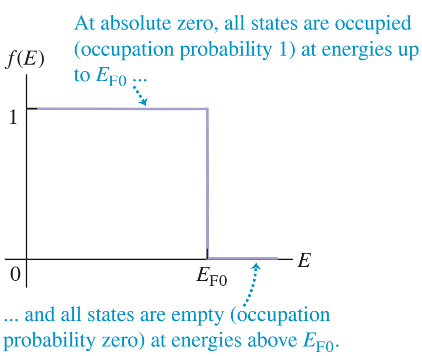
The density of states $g(E)$ is equal to dn/dE, so from Eq. (42.14) we get

Density of states, free-electron model:

Number of states per unit energy range near E; Electron mass; Volume; Electron energy; Planck's constant divided by 2π

$$g(E) = \frac{(2m)^{3/2}V}{2\pi^2\hbar^3}E^{1/2} \qquad (42.15)$$

Figure **42.22** The probability distribution for occupation of free-electron energy states at absolute zero.



Fermi–Dirac Distribution

Let's now see how the electrons are distributed among the various quantum states at any given temperature. The Maxwell–Boltzmann distribution states that the average number of particles in a state of energy E is proportional to $e^{-E/kT}$ (see Sections 18.5 and 39.4). However, there are two reasons why it wouldn't be right to use the Maxwell–Boltzmann distribution. The first reason is the exclusion principle. At absolute zero the Maxwell–Boltzmann function predicts that *all* the electrons would go into the two ground states of the system, with $n_X = n_Y = n_Z = 1$ and $m_s = \pm\frac{1}{2}$. But the exclusion principle allows only one electron in each state. At absolute zero the electrons can fill up the lowest *available* states, but they cannot *all* go into the lowest states. Thus at absolute zero the distribution function is as shown in **Fig. 42.22**. All states with energies E less than some value E_{F0} are occupied, so the occupation probability $f(E) = 1$; all states with energies greater than this value are unoccupied, so $f(E) = 0$.

The second reason we can't use the Maxwell–Boltzmann distribution is more subtle. That distribution assumes that the particles are *distinguishable*. But, as we discussed in Section 41.8, electrons are *indistinguishable;* it's impossible to "tag" electrons to know which is which. If one electron is in state A and the other is in state B, there's no way to tell whether electron 1 is in state A and electron 2 is in state B, or electron 1 is in state B and electron 2 is in state A.

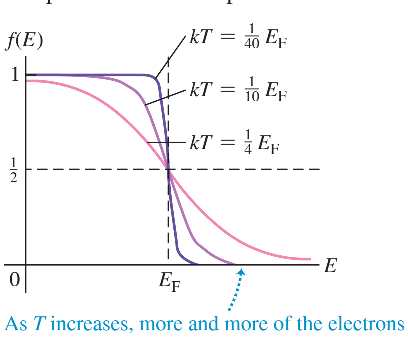
The statistical distribution function that emerges from the exclusion principle and the indistinguishability requirement is called (after its inventors) the **Fermi–Dirac distribution.** This distribution gives the probability $f(E)$ that at temperature T, a particular state of energy E is occupied by an electron:

Fermi–Dirac distribution:

Probability that a given state is occupied by an electron; Absolute temperature; Energy of state; Fermi energy; Boltzmann constant

$$f(E) = \frac{1}{e^{(E - E_F)/kT} + 1} \qquad (42.16)$$

Figure **42.23** Graphs of the Fermi–Dirac distribution function for various values of kT, assuming that the Fermi energy E_F is independent of the temperature T.



The energy E_F is called the **Fermi energy** or the *Fermi level;* we'll discuss its significance below. We use E_{F0} for its value at absolute zero $(T = 0)$ and E_F for other temperatures. We can accurately let $E_F = E_{F0}$ for metals because the Fermi energy does not change much with temperature for solid conductors. However, it is not safe to assume that $E_F = E_{F0}$ for semiconductors, in which the Fermi energy usually does change with temperature.

Figure 42.23 shows graphs of Eq. (42.16) for three temperatures. The $kT = \frac{1}{40}E_F$ curve shows that for $kT \ll E_F$, $f(E)$ approaches the shape of the absolute-zero distribution function shown in Fig. 42.22.

Note that when $E = E_F$, the exponent $(E - E_F)/kT$ in Eq. (42.16) is zero, so $f(E) = f(E_F) = \frac{1}{2}$ for any temperature T. That is, the probability is $\frac{1}{2}$ that a state at the Fermi energy contains an electron. This means that at $E = E_F$, half the states are filled (and half are empty). For states with $E < E_F$, the exponent is negative and the occupation probability $f(E)$ is greater than $\frac{1}{2}$; for states with $E > E_F$, $f(E)$ is less than $\frac{1}{2}$ and approaches zero for E much larger than kT.

EXAMPLE 42.6 Probabilities in the free-electron model

WITH VARIATION PROBLEMS

For free electrons in a solid at temperature T, at what energy is the probability that a particular state is occupied equal to (a) 0.01 and (b) 0.99?

IDENTIFY and SET UP This problem asks us to explore the Fermi–Dirac distribution. Equation (42.16) gives the occupation probability $f(E)$ for a given energy E. If we solve this equation for E, we get an expression for the energy that corresponds to a given occupation probability—which is just what we need.

EXECUTE Using Eq. (42.16), you can show that

$$E = E_F + kT \ln\left(\frac{1}{f(E)} - 1\right)$$

(a) When $f(E) = 0.01$,

$$E = E_F + kT \ln\left(\frac{1}{0.01} - 1\right) = E_F + 4.6kT$$

The probability that a state $4.6kT$ above the Fermi level is occupied is only 0.01, or 1%.

(b) When $f(E) = 0.99$,

$$E = E_F + kT \ln\left(\frac{1}{0.99} - 1\right) = E_F - 4.6kT$$

The probability that a state $4.6kT$ below the Fermi level is occupied is 0.99, or 99%.

EVALUATE At very low temperatures, $4.6kT$ is much less than E_F. Then the occupation probability of levels even slightly below E_F is nearly 1 (100%), and that for levels even slightly above E_F is nearly zero (see Fig. 42.23). In general, if the probability is P that a state with an energy ΔE *above* E_F is occupied, then the probability is $1 - P$ that a state ΔE *below* E_F is occupied. We leave the proof to you.

KEYCONCEPT In the free-electron model of a metal, the Fermi–Dirac distribution gives the probability that a state of a given energy E is occupied at a temperature T. The probability is greater than $\frac{1}{2}$ for energies less than the Fermi energy E_F and less than $\frac{1}{2}$ for energies greater than E_F.

Electron Concentration and Fermi Energy

Equation (42.16) gives the probability that any specific state with energy E is occupied at a temperature T. To get the actual number of electrons in any energy range dE, we have to multiply this probability by the number dn of states in that range $g(E)\,dE$. Thus the number dN of electrons with energies in the range dE is

$$dN = g(E)f(E)\,dE = \frac{(2m)^{3/2}VE^{1/2}}{2\pi^2\hbar^3}\frac{1}{e^{(E-E_F)/kT} + 1}\,dE \tag{42.17}$$

The Fermi energy E_F is determined by the total number N of electrons; at any temperature the electron states are filled up to a point at which all electrons are accommodated. At absolute zero there is a simple relationship between E_{F0} and N. *All* states below E_{F0} are filled; in Eq. (42.13) we set n equal to the total number of electrons N and E to the Fermi energy at absolute zero E_{F0}:

$$N = \frac{(2m)^{3/2}V{E_{F0}}^{3/2}}{3\pi^2\hbar^3} \tag{42.18}$$

Solving Eq. (42.18) for E_{F0}, we get

$$E_{F0} = \frac{3^{2/3}\pi^{4/3}\hbar^2}{2m}\left(\frac{N}{V}\right)^{2/3} \tag{42.19}$$

The quantity N/V is the number of free electrons per unit volume. It is called the *electron concentration* and is usually denoted by n.

If we replace N/V with n, Eq. (42.19) becomes

$$E_{F0} = \frac{3^{2/3}\pi^{4/3}\hbar^2 n^{2/3}}{2m} \tag{42.20}$$

CAUTION **Electron concentration and number of electrons** Don't confuse the electron concentration n with any quantum number n. Furthermore, the number of states is *not* in general the same as the total number of electrons N.

EXAMPLE 42.7 The Fermi energy in copper

WITH VARIATION PROBLEMS

At low temperatures, copper has a free-electron concentration $n = 8.45 \times 10^{28}\ \text{m}^{-3}$. Using the free-electron model, find the Fermi energy for solid copper, and find the speed of an electron with a kinetic energy equal to the Fermi energy.

IDENTIFY and SET UP This problem uses the relationship between Fermi energy and free-electron concentration. Because copper is a solid conductor, its Fermi energy changes very little with temperature and we can use the expression for the Fermi energy at absolute zero, Eq. (42.20). We'll use the nonrelativistic formula $E_F = \frac{1}{2}mv_F^2$ to find the *Fermi speed* v_F that corresponds to kinetic energy E_F.

EXECUTE Using the given value of n, we solve for E_F and v_F:

$$E_F = \frac{3^{2/3}\pi^{4/3}(1.055 \times 10^{-34}\ \text{J}\cdot\text{s})^2(8.45 \times 10^{28}\ \text{m}^{-3})^{2/3}}{2(9.11 \times 10^{-31}\ \text{kg})}$$

$$= 1.126 \times 10^{-18}\ \text{J} = 7.03\ \text{eV}$$

$$v_F = \sqrt{\frac{2E_F}{m}} = \sqrt{\frac{2(1.126 \times 10^{-18}\ \text{J})}{9.11 \times 10^{-31}\ \text{kg}}} = 1.57 \times 10^6\ \text{m/s}$$

EVALUATE Our values of E_F and v_F are within the ranges of typical values for metals, 1.6–14 eV and $0.8\text{–}2.2 \times 10^6\ \text{m/s}$, respectively. Note that the calculated Fermi speed is far less than the speed of light $c = 3.00 \times 10^8\ \text{m/s}$, which justifies our use of the nonrelativistic formula $\frac{1}{2}mv_F^2 = E_F$.

Our calculated Fermi energy is much larger than kT at ordinary temperatures. (At room temperature $T = 20°\text{C} = 293\ \text{K}$, the quantity kT equals $(1.381 \times 10^{-23}\ \text{J/K})(293\ \text{K}) = 4.04 \times 10^{-21}\ \text{J} = 0.0254\ \text{eV}$.) So it is a good approximation to take almost all the states below E_F as completely full and almost all those above E_F as completely empty (see Fig. 42.22).

We can also use Eq. (42.15) to find $g(E)$ if E and V are known. You can show that if $E = 7.03\ \text{eV}$ and $V = 1\ \text{cm}^3$, $g(E)$ is about 2×10^{22} states/eV. This huge number shows why we were justified in treating n and E as continuous variables in our density-of-states derivation.

KEYCONCEPT The Fermi energy in the free-electron model of a metal depends on the electron concentration (number of electrons per unit volume). In a typical metal at room temperature T, the Fermi energy is much greater than kT.

Average Free-Electron Energy

We can calculate the *average* free-electron energy in a metal at absolute zero by using the same ideas that we used to find E_{F0}. From Eq. (42.17) the number dN of electrons with energies in the range dE is $g(E)f(E)\,dE$. The energy of these electrons is $E\,dN = Eg(E)f(E)\,dE$. At absolute zero we substitute $f(E) = 1$ from $E = 0$ to $E = E_{F0}$ and $f(E) = 0$ for all other energies. Therefore the total energy E_{tot} of all the N electrons is

$$E_{tot} = \int_0^{E_{F0}} Eg(E)(1)\,dE + \int_{E_{F0}}^{\infty} Eg(E)(0)\,dE = \int_0^{E_{F0}} Eg(E)\,dE$$

The simplest way to evaluate this expression is to compare Eqs. (42.15) and (42.19). You'll see that

$$g(E) = \frac{3NE^{1/2}}{2E_{F0}^{3/2}}$$

Substituting this expression into the integral and using $E_{av} = E_{tot}/N$, we get

$$E_{av} = \frac{3}{2E_{F0}^{3/2}}\int_0^{E_{F0}} E^{3/2}\,dE = \tfrac{3}{5}E_{F0} \qquad (42.21)$$

At absolute zero the average free-electron energy equals $\frac{3}{5}$ of the Fermi energy.

EXAMPLE 42.8 Free-electron gas versus ideal gas

(a) Find the average energy of the free electrons in copper at absolute zero (see Example 42.7). (b) What would be the average kinetic energy of electrons if they behaved like an ideal gas at room temperature, 20°C (see Section 18.3)? What would be the speed of an electron with this kinetic energy? Compare these ideal-gas values with the (correct) free-electron values.

IDENTIFY and SET UP Free electrons in a metal behave like a kind of gas. In part (a) we use Eq. (42.21) to determine the average kinetic energy of free electrons in terms of the Fermi energy at absolute zero, which we know for copper from Example 42.7. In part (b) we treat electrons as an ideal gas at room temperature: Eq. (18.16) then gives the average kinetic energy per electron as $E_{av} = \frac{3}{2}kT$, and $E_{av} = \frac{1}{2}mv^2$ gives the corresponding electron speed v.

EXECUTE (a) From Example 42.7, the Fermi energy in copper at absolute zero is $1.126 \times 10^{-18}\ \text{J} = 7.03\ \text{eV}$. According to Eq. (42.21), the average energy is $\frac{3}{5}$ of this, or $6.76 \times 10^{-19}\ \text{J} = 4.22\ \text{eV}$.

(b) In Example 42.7 we found that $kT = 4.04 \times 10^{-21}$ J = 0.0254 eV at room temperature $T = 20°C = 293$ K. If electrons behaved like an ideal gas at this temperature, the average kinetic energy per electron would be $\frac{3}{2}$ of this, or 6.07×10^{-21} J = 0.0379 eV. The speed of an electron with this kinetic energy is

$$v = \sqrt{\frac{2E_{av}}{m}} = \sqrt{\frac{2(6.07 \times 10^{-21}\ \text{J})}{9.11 \times 10^{-31}\ \text{kg}}}$$
$$= 1.15 \times 10^5\ \text{m/s}$$

EVALUATE The ideal-gas model predicts an average energy that is about 1% of the value given by the free-electron model, and a speed that is about 7% of the free-electron Fermi speed $v_F = 1.57 \times 10^6$ m/s that we found in Example 42.7. Thus temperature plays a *very* small role in determining the properties of electrons in metals; their average energies are determined almost entirely by the exclusion principle.

A similar analysis allows us to determine the contributions of electrons to the heat capacities of a solid metal. If there is one conduction electron per atom, the principle of equipartition of energy (see Section 18.4) would predict that the kinetic energies of these electrons contribute $3R/2$ to the molar heat capacity at constant volume C_V. But when kT is much smaller than E_F, which is usually the situation in metals, only those few electrons near the Fermi level can find empty states and change energy appreciably when the temperature changes. The number of such electrons is proportional to kT/E_F, so we expect that the electron molar heat capacity at constant volume is proportional to $(kT/E_F)(3R/2) = (3kT/2E_F)R$. A more detailed analysis shows that the actual electron contribution to C_V for a solid metal is $(\pi^2 kT/2E_F)R$, not far from our prediction. You can verify that if $T = 293$ K and $E_F = 7.03$ eV, the electron contribution to C_V is $0.018R$, which is only 1.2% of the (incorrect) $3R/2$ prediction of the equipartition principle. Because the electron contribution is so small, the overall heat capacity of most solid metals is due primarily to vibration of the atoms in the crystal structure (see Fig. 18.18 in Section 18.4).

KEYCONCEPT The exclusion principle, not the effects of temperature, is the dominant reason for the distribution of electron energies in a metal at room temperature.

TEST YOUR UNDERSTANDING OF SECTION 42.5 An ideal gas obeys the relationship $pV = nRT$ (see Section 18.1): For a given volume V and a number of moles n, as the temperature T decreases, the pressure p decreases proportionately and tends to zero as T approaches absolute zero. Is this also true of the free-electron gas in a solid metal?

ANSWER

no The kinetic-molecular model of an ideal gas (Section 18.3) shows that the gas pressure is proportional to the average translational kinetic energy E_{av} of the particles that make up the gas. In a classical ideal gas, E_{av} is directly proportional to the average temperature T, so the pressure decreases as T decreases. In a free-electron gas, the average kinetic energy per electron is *not* related simply to T; as Example 42.8 shows, for the free-electron gas in a metal, E_{av} is almost completely a consequence of the exclusion principle at room temperature and colder. Hence the pressure of a free-electron gas in a solid metal does *not* change appreciably between room temperature and absolute zero.

42.6 SEMICONDUCTORS

A **semiconductor** has an electrical resistivity that is intermediate between those of good conductors and those of good insulators. The tremendous importance of semiconductors in present-day electronics stems in part from the fact that their electrical properties are very sensitive to very small concentrations of impurities. We'll discuss the basic concepts, using the semiconductor elements silicon (Si) and germanium (Ge) as examples.

Silicon and germanium are in Group IV of the periodic table. Both have four electrons in the outermost atomic subshells ($3s^2 3p^2$ for silicon, $4s^2 4p^2$ for germanium), and both crystallize in the covalently bonded diamond structure discussed in Section 42.3 (see Fig. 42.14). Because all four of the outer electrons are involved in the bonding, at absolute zero the band structure (see Section 42.4) has a completely empty conduction band (see Fig. 42.19b). As we discussed in Section 42.4, at very low temperatures electrons cannot jump from the filled valence band into the conduction band. This property makes these materials insulators at very low temperatures; their electrons have no nearby states available into which they can move in response to an applied electric field.

However, in semiconductors the energy gap E_g between the valence and conduction bands is small in comparison to the gap of 5 eV or more for many insulators; room-temperature values are 1.12 eV for silicon and only 0.67 eV for germanium. Thus even at room temperature a substantial number of electrons can gain enough energy to jump the gap to the conduction band, where they are dissociated from their parent atoms and are free to move about the crystal. The number of these electrons increases rapidly with temperature.

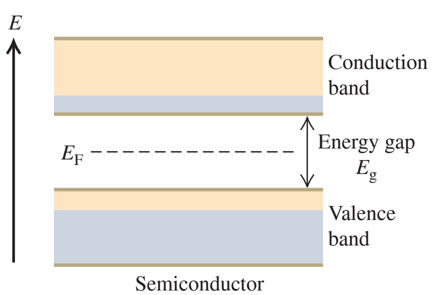
EXAMPLE 42.9 Jumping a band gap

WITH VARIATION PROBLEMS

Consider a material with the band structure described above, with its Fermi energy in the middle of the gap (**Fig. 42.24**). Find the probability that a state at the bottom of the conduction band is occupied at $T = 300$ K, and compare that with the probability at $T = 310$ K, for band gaps of (a) 0.200 eV; (b) 1.00 eV; (c) 5.00 eV.

IDENTIFY and SET UP The Fermi–Dirac distribution function gives the probability that a state of energy E is occupied at temperature T. Figure 42.24 shows that the state of interest at the bottom of the conduction band has an energy $E = E_F + E_g/2$ that is greater than the Fermi energy E_F, with $E - E_F = E_g/2$. Figure 42.23 shows that the higher the temperature, the larger the fraction of electrons with energies greater than the Fermi energy.

Figure **42.24** Band structure of a semiconductor. At absolute zero a completely filled valence band is separated by a narrow energy gap E_g of 1 eV or so from a completely empty conduction band. At ordinary temperatures, a number of electrons are excited to the conduction band.



EXECUTE (a) When $E_g = 0.200$ eV,

$$\frac{E - E_F}{kT} = \frac{E_g}{2kT} = \frac{0.100\text{ eV}}{(8.617 \times 10^{-5}\text{ eV/K})(300\text{ K})} = 3.87$$

$$f(E) = \frac{1}{e^{3.87} + 1} = 0.0205$$

For $T = 310$ K, the exponent is 3.74 and $f(E) = 0.0231$, a 13% increase in probability for a temperature rise of 10 K.

(b) For $E_g = 1.00$ eV, both exponents are five times as large as in part (a), namely 19.3 and 18.7; the values of $f(E)$ are 4.0×10^{-9} and 7.4×10^{-9}. In this case the (low) probability nearly doubles with a temperature rise of 10 K.

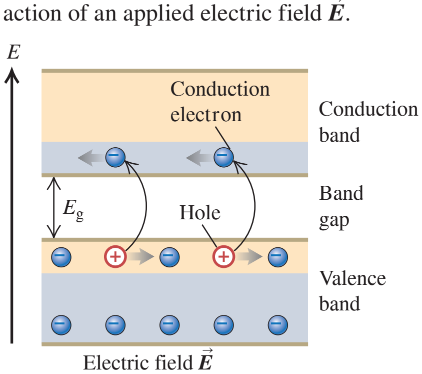
(c) For $E_g = 5.0$ eV, the exponents are 96.7 and 93.6; the values of $f(E)$ are 1.0×10^{-42} and 2.3×10^{-41}. The (extremely low) probability increases by a factor of 23 for a 10 K temperature rise.

EVALUATE This example illustrates two important points. First, the probability of finding an electron in a state at the bottom of the conduction band is extremely sensitive to the width of the band gap. At room temperature, the probability is about 2% for a 0.200 eV gap, a few in a thousand million for a 1.00 eV gap, and essentially zero for a 5.00 eV gap. (Pure diamond, with a 5.47 eV band gap, has essentially no electrons in the conduction band and is an excellent insulator.) Second, for any given band gap the probability depends strongly on temperature, and even more strongly for large gaps than for small ones.

KEYCONCEPT The properties of the Fermi–Dirac distribution explain why the number of electrons in the conduction band of a semiconductor is sensitive to both the temperature and the size of the band gap.

In principle, we could continue the calculation in Example 42.9 to find the actual density $n = N/V$ of electrons in the conduction band at any temperature. To do this, we would have to evaluate the integral $\int g(E)f(E)\,dE$ from the bottom of the conduction band to its top. [To do this we would need to know the density of states function $g(E)$. This function for a semiconductor is more complicated than the free-electron model expression given by Eq. (42.15).] Once we know n, we can *begin* to determine the resistivity of the material (and its temperature dependence) by using the analysis of Section 25.2, which you may want to review. But next we'll see that the electrons in the conduction band don't tell the whole story about conduction in semiconductors.

Figure **42.25** Motion of electrons in the conduction band and of holes in the valence band of a semiconductor under the action of an applied electric field $\vec{E}$.



Holes

When an electron is removed from a covalent bond, it leaves a vacancy behind. An electron from a neighboring atom can move into this vacancy, leaving the neighbor with the vacancy. In this way the vacancy, called a **hole,** can travel through the material and serve as an additional current carrier. It's like describing the motion of a bubble in a liquid. In a pure, or *intrinsic,* semiconductor, valence-band holes and conduction-band electrons are always present in equal numbers. When an electric field is applied, they move in opposite directions (**Fig. 42.25**). Thus a hole in the valence band behaves like a positively charged particle, even though the moving charges in that band are electrons. The conductivity that we just described for a pure semiconductor is called *intrinsic conductivity.* Another kind of conductivity, to be discussed in the next subsection, is due to impurities.

An analogy helps to picture conduction in an intrinsic semiconductor. The valence band at absolute zero is like a floor of a parking garage that's filled bumper to bumper with cars (which represent electrons). No cars can move because there is nowhere for them to go. But if one car is moved to the vacant floor above, it can move freely, just as electrons can move freely in the conduction band. Also, the empty space that it leaves permits cars to move on the nearly filled floor, thereby moving the empty space just as holes move in the normally filled valence band.

Impurities

Suppose we mix into melted germanium $(\mathrm{Ge}, Z = 32)$ a small amount of arsenic $(\mathrm{As}, Z = 33)$, the next element after germanium in the periodic table. We then allow the mixture to cool and crystallize. This deliberate addition of impurity elements is called *doping*. Arsenic is in Group V; it has *five* valence electrons compared to the four valence electrons of germanium. When one of these five valence electrons is removed from an arsenic atom, the remaining electron structure is essentially identical to that of germanium. The only difference is that it is smaller; the arsenic nucleus has a charge of $+33e$ rather than $+32e$, and it pulls the electrons in a little more. An arsenic atom can comfortably take the place of a germanium atom as a substitutional impurity. Four of its five valence electrons form the necessary nearest-neighbor covalent bonds.

The fifth valence electron in arsenic is very loosely bound (**Fig. 42.26a**); it doesn't participate in the covalent bonds, and it is screened from the nuclear charge of $+33e$ by the 32 electrons, leaving a net effective charge of about $+e$. We might guess that the binding energy would be of the same order of magnitude as the energy of the $n = 4$ level in hydrogen—that is, $(\frac{1}{4})^2(13.6\text{ eV}) = 0.85\text{ eV}$. In fact, it is much smaller than this, only about 0.01 eV, because the electron probability distribution actually extends over many atomic diameters and the polarization of intervening atoms provides additional screening.

The energy level of this fifth electron corresponds in the band picture to an isolated energy level lying in the gap, about 0.01 eV below the bottom of the conduction band (Fig. 42.26b). This level is called a *donor level,* and the impurity atom that is responsible for it is simply called a *donor.* All Group V elements, including N, P, As, Sb, and Bi, can serve as donors. At room temperature, kT is about 0.025 eV. This is substantially greater than 0.01 eV, so at ordinary temperatures, most electrons can gain enough energy to jump from donor levels into the conduction band, where they are free to wander through the material. The remaining ionized donor stays at its site in the structure and does not participate in conduction.

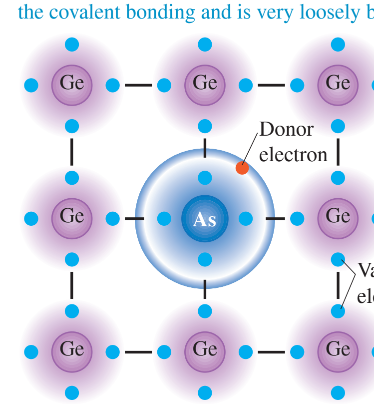
Example 42.9 shows that at ordinary temperatures and with a band gap of 1.0 eV, only a very small fraction (of the order of 10^{-9}) of the states at the bottom of the conduction band in a pure semiconductor contain electrons to participate in intrinsic conductivity. Thus we expect the conductivity of such a semiconductor to be about 10^{-9} as great as that of good metallic conductors, and measurements bear out this prediction. However, a concentration of donors as small as one part in 10^8 can increase the conductivity so drastically that conduction due to impurities becomes by far the dominant mechanism. In this case the conductivity is due almost entirely to *negative* charge (electron) motion. We call the material an ***n*-type semiconductor,** with n-type impurities.

Adding atoms of an element in Group III (B, Al, Ga, In, Tl), with only *three* valence electrons, has an analogous effect. An example is gallium $(\mathrm{Ga}, Z = 31)$; as a substitutional impurity in germanium, the gallium atom would like to form four covalent bonds, but it has only three outer electrons. It can, however, steal an electron from a neighboring germanium atom to complete the required four covalent bonds (**Fig. 42.27a**, next page). The resulting atom has the same electron configuration as germanium but is somewhat larger because gallium's nuclear charge is smaller, $+31e$ instead of $+32e$.

This theft leaves the neighboring atom with a *hole,* or missing electron. The hole acts as a positive charge that can move through the crystal just as with intrinsic conductivity. The stolen electron is bound to the gallium atom in a level called an *acceptor level* about 0.01 eV above the top of the valence band (Fig. 42.27b). The gallium atom, called an *acceptor,* thus accepts an electron to complete its desire for four covalent bonds. This extra electron gives the previously neutral gallium atom a net charge of $-e$. The resulting gallium ion is

Figure **42.26** An n-type semiconductor: germanium (Ge) with an arsenic (As) impurity.

(a) A donor (n-type) impurity atom has a fifth valence electron that does not participate in the covalent bonding and is very loosely bound.



(b) Energy-band diagram for an n-type semiconductor at a low temperature. One donor electron has been excited from the donor levels into the conduction band.

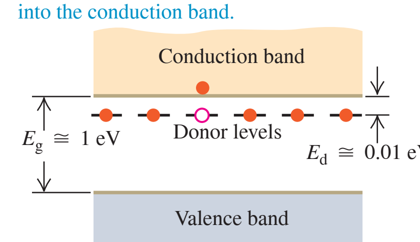
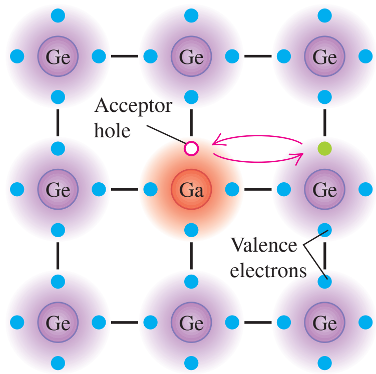


Figure 42.27 A p-type semiconductor: germanium (Ge) with a gallium (Ga) impurity.

(a) An acceptor (p-type) impurity atom has only three valence electrons, so it can borrow an electron from a neighboring atom. The resulting hole is free to move about the crystal.



(b) Energy-band diagram for a p-type semiconductor at a low temperature. One acceptor level has accepted an electron from the valence band, leaving a hole behind.

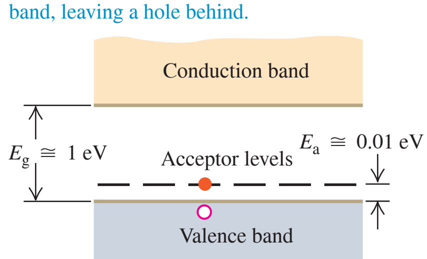
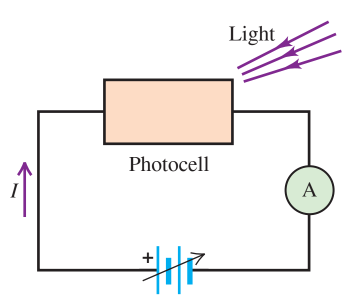


Figure 42.28 A semiconductor photocell in a circuit. The more intense the light falling on the photocell, the greater the conductivity of the photocell and the greater the current measured by the ammeter (A).



not free to move. In a semiconductor that is doped with acceptors, we consider the conductivity to be almost entirely due to *positive* charge (hole) motion. We call the material a ***p*-type semiconductor,** with p-type impurities. Some semiconductors are doped with *both* n- and p-type impurities. Such materials are called *compensated* semiconductors.

CAUTION The meaning of "*p*-type" and "*n*-type" It's a common misconception that a p-type semiconductor has a net positive charge and an n-type semiconductor has a net negative charge. In fact, both types of semiconductors have a net *zero* charge because they are made up of neutral atoms. Instead, "p-type" means that the majority of the mobile carriers of charge within the semiconductor are positive (holes), and "n-type" means that the majority of mobile carriers are negative (electrons).

We can verify the assertion that the current in n- and p-type semiconductors really *is* carried by electrons and holes, respectively, by using the Hall effect (see Section 27.9). The sign of the Hall emf is opposite in the two cases. Hall-effect devices constructed from semiconductor materials are used in probes to measure magnetic fields and the currents that cause those fields.

TEST YOUR UNDERSTANDING OF SECTION 42.6 Would there be any advantage to adding n-type or p-type impurities to copper?

ANSWER

no Pure copper is already an excellent conductor since it has a partially filled conduction band (Fig. 42.19c). Furthermore, copper forms a metallic crystal (Fig. 42.15) as opposed to the covalent crystals of silicon or germanium, so the scheme of using an impurity to donate or accept an electron does not work for copper. In fact, adding impurities to copper *decreases* the conductivity because an impurity tends to scatter electrons, impeding the flow of current.

42.7 SEMICONDUCTOR DEVICES

Semiconductor devices play an indispensable role in contemporary electronics. In the early days of radio and television, transmitting and receiving equipment relied on vacuum tubes, but these have been replaced by solid-state devices, including transistors, diodes, integrated circuits, and other semiconductor devices. All modern consumer electronic devices use semiconductor devices of various kinds.

One simple semiconductor device is the *photocell* (**Fig. 42.28**). When a thin slab of semiconductor is irradiated with an electromagnetic wave whose photons have at least as much energy as the band gap between the valence and conduction bands, an electron in the valence band can absorb a photon and jump to the conduction band, where it and the hole it left behind contribute to the conductivity (see Example 42.5 in Section 42.4). The conductivity therefore increases with wave intensity, thus increasing the current I in the photocell circuit of Fig. 42.28. Hence the ammeter reading indicates the intensity of the light.

Detectors for charged particles operate on the same principle. An external circuit applies a voltage across a semiconductor. An energetic charged particle passing through the semiconductor collides inelastically with valence electrons, exciting them from the valence to the conduction band and creating pairs of holes and conduction electrons. The conductivity increases momentarily, causing a pulse of current in the external circuit. Solid-state detectors are widely used in nuclear and high-energy physics research.

The *p-n* Junction

In many semiconductor devices the essential principle is the fact that the conductivity of the material is controlled by impurity concentrations, which can be varied within wide limits from one region of a device to another. An example is the ***p-n* junction** at the boundary between one region of a semiconductor with p-type impurities and another region containing n-type impurities. One way of fabricating a p-n junction is to deposit some n-type material on the *very* clean surface of some p-type material. (We can't just stick p- and n-type pieces together and expect the junction to work properly because of the impossibility of matching their surfaces at the atomic level.)

When a *p-n* junction is connected to an external circuit, as in **Fig. 42.29a**, and the potential difference $V_p - V_n = V$ across the junction is varied, the current I varies as shown in Fig. 42.29b. In striking contrast to the symmetric behavior of resistors that obey Ohm's law and give a straight line on an I–V graph, a *p-n* junction conducts much more readily in the direction from p to n than the reverse. Such a (mostly) one-way device is called a **diode rectifier.** Later we'll discuss a simple model of *p-n* junction behavior that predicts a current–voltage relationship in the form

$$I = I_S\left(e^{eV/kT} - 1\right) \qquad (42.22)$$

Current through a *p-n* junction: I
Saturation current: I_S
Voltage: V
Absolute temperature: T
Boltzmann constant: k
Magnitude of electron charge: e (in the exponent)
$e = 2.71828\ldots$ (base of the exponential)

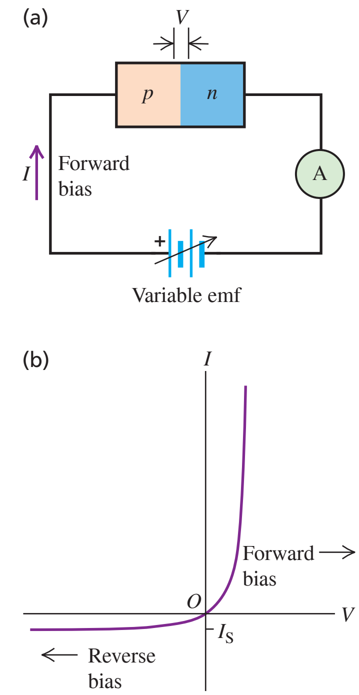


Figure **42.29** (a) A semiconductor *p-n* junction in a circuit. (b) Graph showing the asymmetric current–voltage relationship. The curve is described by Eq. (42.22).

CAUTION **Two different uses of *e*** In $e^{eV/kT}$ the base of the exponent also uses the symbol e, standing for the base of the natural logarithms, 2.71828 . . . . This e is quite different from $e = 1.602 \times 10^{-19}$ C in the exponent.

Equation (42.22) is valid for both positive and negative values of V; note that V and I always have the same sign. As V becomes very negative, I approaches the value $-I_S$. The magnitude I_S (always positive) is called the *saturation current.*

Currents Through a *p-n* Junction

We can understand the behavior of a *p-n* junction diode qualitatively on the basis of the mechanisms for conductivity in the two regions. Suppose, as in Fig. 42.29a, you connect the positive terminal of the battery to the p region and the negative terminal to the n region. Then the p region is at higher potential than the n region, corresponding to positive V in Eq. (42.22), and the resulting electric field is in the direction p to n. This is called the *forward* direction, and the positive potential difference is called *forward bias*. Holes, plentiful in the p region, flow easily across the junction into the n region, and free electrons, plentiful in the n region, easily flow into the p region; these movements of charge constitute a *forward* current. Connecting the battery with the opposite polarity gives *reverse bias,* and the field tends to push electrons from p to n and holes from n to p. But there are very few free electrons in the p region and very few holes in the n region. As a result, the current in the *reverse* direction is much smaller than that with the same potential difference in the forward direction.

Suppose you have a box with a barrier separating the left and right sides: You fill the left side with oxygen gas and the right side with nitrogen gas. What happens if the barrier leaks? Oxygen diffuses to the right, and nitrogen diffuses to the left. A similar diffusion occurs across a *p-n* junction. First consider the equilibrium situation with no applied voltage (**Fig. 42.30**). The many holes in the p region act like a hole gas that diffuses across the junction into the n region. Once there, the holes recombine with some of the many free electrons. Similarly, electrons diffuse from the n region to the p region and fall into some of the many holes there. The hole and electron diffusion currents lead to a

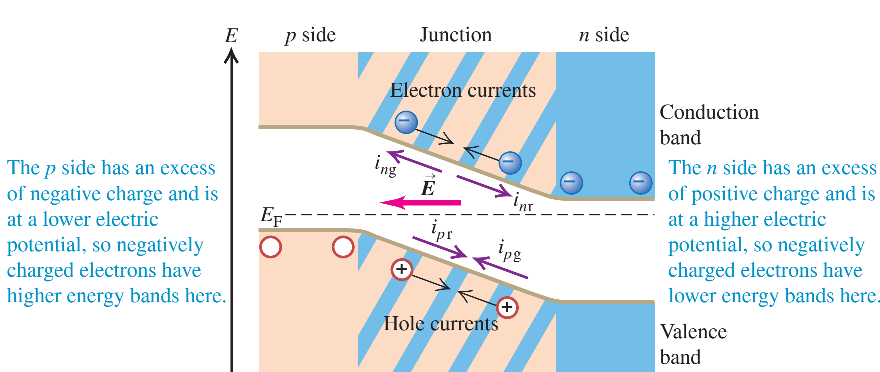


Figure **42.30** A *p-n* junction in equilibrium, with no externally applied field or potential difference. The generation currents (subscript g) and recombination currents (subscript r) exactly balance. The Fermi energy E_F is the same on both sides of the junction. The excess positive and negative charges on the n and p sides produce an electric field $\vec{E}$ in the direction shown.

net positive charge in the n region and a net negative charge in the p region, causing an electric field in the direction from n to p at the junction. The potential energy associated with this field raises the electron energy levels in the p region relative to the same levels in the n region.

There are four currents across the junction, as shown. The diffusion processes lead to *recombination currents* of holes and electrons, labeled $i_{p\mathrm{r}}$ and $i_{n\mathrm{r}}$ in Fig. 42.30. At the same time, electron–hole pairs are generated in the junction region by thermal excitation. The electric field described above sweeps these electrons and holes out of the junction; electrons are swept opposite the field to the n side, and holes are swept in the same direction as the field to the p side. The corresponding currents, called *generation currents,* are labeled $i_{p\mathrm{g}}$ and $i_{n\mathrm{g}}$. At equilibrium the magnitudes of the generation and recombination currents are equal:

$$|i_{p\mathrm{g}}| = |i_{p\mathrm{r}}| \quad \text{and} \quad |i_{n\mathrm{g}}| = |i_{n\mathrm{r}}| \tag{42.23}$$

In thermal equilibrium the Fermi energy is the same at each point across the junction.

Now we apply a forward bias—that is, a positive potential difference V across the junction. A forward bias *decreases* the electric field in the junction region. It also decreases the difference between the energy levels on the p and n sides (**Fig. 42.31**) by an amount $\Delta E = -eV$. It becomes easier for the electrons in the n region to climb the potential-energy hill and diffuse into the p region and for the holes in the p region to diffuse into the n region. This effect increases both recombination currents by the Maxwell–Boltzmann factor $e^{-\Delta E/kT} = e^{eV/kT}$. (We don't have to use the Fermi–Dirac distribution because most of the available states for the diffusing electrons and holes are empty, so the exclusion principle has little effect.) The generation currents don't change appreciably, so the net hole current is

$$\begin{aligned} i_{p\mathrm{tot}} &= i_{p\mathrm{r}} - |i_{p\mathrm{g}}| \\ &= |i_{p\mathrm{g}}|e^{eV/kT} - |i_{p\mathrm{g}}| \\ &= |i_{p\mathrm{g}}|(e^{eV/kT} - 1) \end{aligned} \tag{42.24}$$

The net electron current $i_{n\mathrm{tot}}$ is given by a similar expression, so the total current $I = i_{p\mathrm{tot}} + i_{n\mathrm{tot}}$ is

$$I = I_{\mathrm{S}}(e^{eV/kT} - 1) \tag{42.25}$$

in agreement with Eq. (42.22). We can repeat this entire discussion for reverse bias (negative V and I) with the same result. Therefore Eq. (42.22) is valid for both positive and negative values.

Several effects make the behavior of practical p-n junction diodes more complex than this simple analysis predicts. One effect, *avalanche breakdown,* occurs under large reverse bias. The electric field in the junction is so great that the carriers can gain enough energy between collisions to create electron–hole pairs during inelastic collisions. The electrons and holes then gain energy and collide to form more pairs, and so on. (A similar effect occurs in dielectric breakdown in insulators, discussed in Section 42.4.)

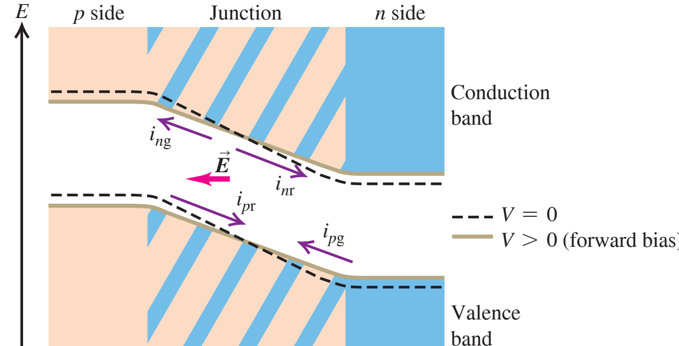
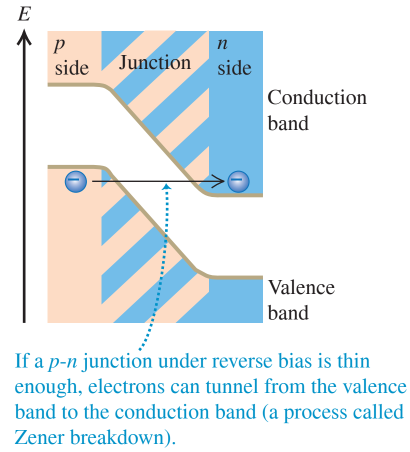


Figure 42.31 A p-n junction under forward-bias conditions. The potential difference between p and n regions is reduced, as is the electric field within the junction. The recombination currents increase but the generation currents are nearly constant, causing a net current from left to right. (Compare Fig. 42.30.)

A second type of breakdown begins when the reverse bias becomes large enough that the top of the valence band in the *p* region is just higher in energy than the bottom of the conduction band in the *n* region (**Fig. 42.32**). If the junction region is thin enough, the probability becomes large that electrons can *tunnel* from the valence band of the *p* region to the conduction band of the *n* region. This process is called *Zener breakdown*. It occurs in Zener diodes, which are used for voltage regulation and protection against voltage surges.

Figure 42.32 Under reverse-bias conditions the potential-energy difference between the *p* and *n* sides of a junction is greater than at equilibrium. If this difference is great enough, the bottom of the conduction band on the *n* side may actually be below the top of the valence band on the *p* side.

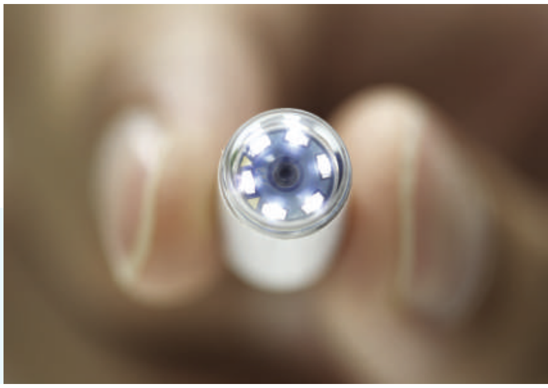


Semiconductor Devices and Light

A *light-emitting diode (LED)* is a *p-n* junction diode that emits light. When the junction is forward biased, many holes are pushed from their *p* region to the junction region, and many electrons are pushed from their *n* region to the junction region. In the junction region the electrons fall into holes (recombine). In recombining, the electron can emit a photon with energy approximately equal to the band gap. This energy (and therefore the photon wavelength and the color of the light) can be varied by using materials with different band gaps. Light-emitting diodes are very energy-efficient light sources and have many applications, including automobile lamps, traffic signals, and flat-screen displays.

The reverse process is called the *photovoltaic effect*. Here the material absorbs photons, and electron–hole pairs are created. Pairs that are created in the *p-n* junction, or close enough to migrate to it without recombining, are separated by the electric field we described above that sweeps the electrons to the *n* side and the holes to the *p* side. We can connect this device to an external circuit, where it becomes a source of emf and power. Such a device is often called a *solar cell,* although sunlight isn't required. *Any* light with photon energies greater than the band gap will do. You might have a calculator powered by such cells. Production of low-cost photovoltaic cells for large-scale solar energy conversion is a very active field of research. The same basic physics is used in charge-coupled device (CCD) image detectors, digital cameras, and video cameras.

BIO APPLICATION Swallow This Semiconductor Device This tiny capsule—designed to be swallowed by a patient—contains a miniature camera with a CCD light detector, plus six LEDs to illuminate the subject. The capsule radios high-resolution images to an external recording unit as it passes painlessly through the patient's stomach and intestines. This technique makes it possible to examine the small intestine, which is not readily accessible with conventional endoscopy.



Transistors

A *bipolar junction transistor* includes two *p-n* junctions in a "sandwich" configuration, which may be either *p-n-p* or *n-p-n*. **Figure 42.33** shows such a *p-n-p* transistor. The three regions are called the emitter, base, and collector, as shown. When there is no current in the left loop of the circuit, there is only a very small current through the resistor *R* because the voltage across the base–collector junction is in the reverse direction. But when a forward bias is applied between emitter and base, as shown, most of the holes traveling from emitter to base travel *through* the base (which is typically both narrow and lightly doped) to the second junction, where they come under the influence of the collector-to-base potential difference and flow on through the collector to give an increased current to the resistor.

In this way the current in the collector circuit is *controlled* by the current in the emitter circuit. Furthermore, V_c may be considerably larger than V_e, so the *power* dissipated in *R* may be much larger than the power supplied to the emitter circuit by the battery V_e. Thus the device functions as a *power amplifier*. If the potential drop across *R* is greater than V_e, it may also be a voltage amplifier.

In this configuration the *base* is the common element between the "input" and "output" sides of the circuit. Another widely used arrangement is the *common-emitter* circuit, shown in **Fig. 42.34** (next page). In this circuit the current in the collector side of the circuit is much larger than that in the base side, and the result is current amplification.

Figure 42.33 Schematic diagram of a *p-n-p* transistor and circuit.

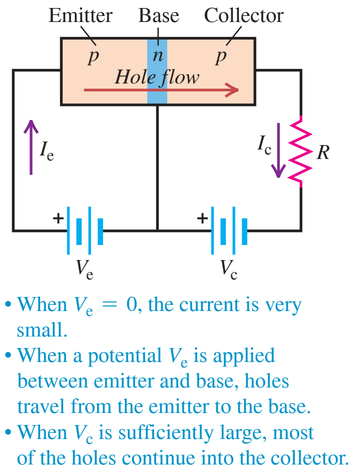
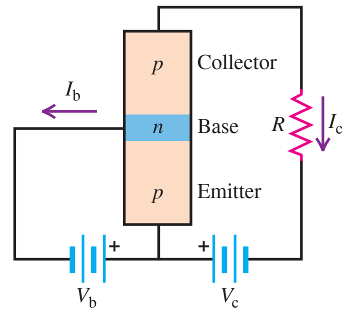
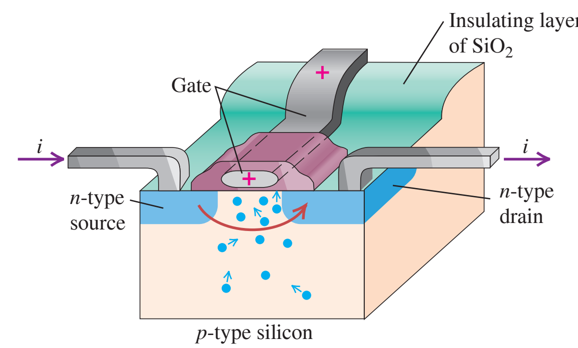


Figure **42.34** A common-emitter circuit.



- When $V_b = 0$, I_c is very small, and most of the voltage V_c appears across the base–collector junction.
- As V_b increases, the base–collector potential decreases, and more holes can diffuse into the collector; thus, I_c increases. Ordinarily, I_c is much larger than I_b.

Figure **42.35** A field-effect transistor. The current from source to drain is controlled by the potential difference between the source and the drain and by the charge on the gate; no current flows through the gate.

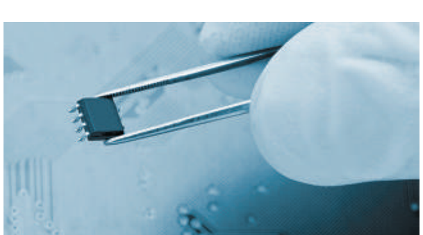


The *field-effect transistor* (**Fig. 42.35**) is an important type. In one variation a slab of *p*-type silicon is made with two *n*-type regions on the top, called the *source* and the *drain;* a metallic conductor is fastened to each. A third electrode called the *gate* is separated from the slab, source, and drain by an insulating layer of SiO_2. When there is no charge on the gate and a potential difference of either polarity is applied between the source and the drain, there is very little current because one of the *p-n* junctions is reverse biased.

Now we place a positive charge on the gate. With dimensions of the order of 10^{-6} m, it takes little charge to provide a substantial electric field. Thus there is very little current into or out of the gate. There aren't many free electrons in the *p*-type material, but there are some, and the effect of the field is to attract them toward the positive gate. The resulting greatly enhanced concentration of electrons near the gate (and between the two junctions) permits current to flow between the source and the drain. The current is very sensitive to the gate charge and potential, and the device functions as an amplifier. The device just described is called an *enhancement-type MOSFET* (metal-oxide-semiconductor field-effect transistor).

Integrated Circuits

Figure **42.36** An integrated circuit chip smaller than your fingertip can contain millions of transistors.



A further refinement in semiconductor technology is the *integrated circuit*. By successively depositing layers of material and etching patterns to define current paths, we can combine the functions of several MOSFETs, capacitors, and resistors on a single square of semiconductor material that may be only a few millimeters on a side. An elaboration of this idea leads to *large-scale integrated circuits*. The resulting integrated circuit chips are the heart of all pocket calculators, mobile devices, and present-day computers (**Fig. 42.36**).

The first semiconductor devices were invented in 1947. Since then, they have completely revolutionized the electronics industry through miniaturization, reliability, speed, energy usage, and cost. They have found applications in communications, computer systems, control systems, and many other areas. In transforming these areas, they have changed, and continue to change, human civilization itself.

TEST YOUR UNDERSTANDING OF SECTION 42.7 Suppose a negative charge is placed on the gate of the MOSFET shown in Fig. 42.35. Will a substantial current flow between the source and the drain?

ANSWER

no A negative charge on the gate will repel, not attract, electrons in the *p*-type silicon. Hence the electron concentration in the region between the two *p-n* junctions will be made even smaller. With so few charge carriers present in this region, very little current will flow between the source and the drain.

42.8 SUPERCONDUCTIVITY

Superconductivity is the complete disappearance of all electrical resistance at low temperatures. We described this property at the end of Section 25.2 and the magnetic properties of type-I and type-II superconductors in Section 29.8. In this section we'll relate superconductivity to the structure and energy-band model of a solid.

Although superconductivity was discovered in 1911, it was not well understood on a theoretical basis until 1957. That year, the American physicists John Bardeen, Leon Cooper, and Robert Schrieffer published the theory of superconductivity, now called the BCS theory, that earned them the Nobel Prize in physics in 1972. (It was Bardeen's second; he shared his first for his work on the development of the transistor.) The key to the BCS theory is an interaction between *pairs* of conduction electrons, called *Cooper pairs,* caused by an interaction with the positive ions of the crystal. Here's a rough picture of what happens. A free electron exerts attractive forces on nearby positive ions, pulling them slightly closer together. The resulting slight concentration of positive charge then exerts an attractive force on another free electron with momentum opposite to the first. At ordinary temperatures this electron-pair interaction is very small in comparison to energies of thermal motion, but at very low temperatures it is significant.

Bound together this way, the pairs of electrons cannot *individually* gain or lose very small amounts of energy, as they would ordinarily be able to do in a partly filled conduction band. Their pairing gives an energy gap in the allowed electron quantum levels, and at low temperatures there is not enough collision energy to jump this gap. Therefore the electrons can move freely through the crystal without any energy exchange through collisions—that is, with zero resistance.

Since 1987 physicists have discovered a number of compounds that remain superconducting at temperatures above 77 K (the boiling point of liquid nitrogen). The original pairing mechanism of the BCS theory cannot explain the properties of these *high-temperature superconductors*. Instead, it appears that electrons in these materials form pairs due to magnetic interactions between their spins.

CHAPTER 42 SUMMARY

Molecular bonds and molecular spectra: The principal types of molecular bonds are ionic, covalent, van der Waals, and hydrogen bonds. In a diatomic molecule the rotational energy levels are given by Eq. (42.3), where I is the moment of inertia of the molecule, m_r is its reduced mass, and r_0 is the distance between the two atoms. The vibrational energy levels are given by Eq. (42.7), where k' is the effective force constant of the interatomic force. (See Examples 42.1–42.3.)

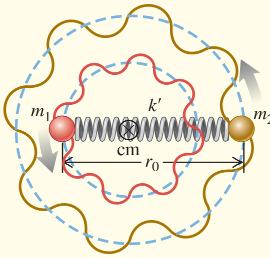
$$E_l = l(l+1)\frac{\hbar^2}{2I} \quad (l = 0, 1, 2, \dots) \tag{42.3}$$

$$I = m_r r_0^2 \tag{42.6}$$

$$m_r = \frac{m_1 m_2}{m_1 + m_2} \tag{42.4}$$

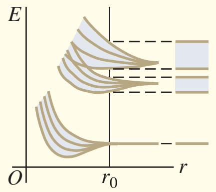
$$E_n = \left(n + \tfrac{1}{2}\right)\hbar\omega = \left(n + \tfrac{1}{2}\right)\hbar\sqrt{\frac{k'}{m_r}} \tag{42.7}$$

$$(n = 0, 1, 2, \dots)$$



Solids and energy bands: Interatomic bonds in solids are of the same types as in molecules plus one additional type, the metallic bond. Associating the basis with each lattice point gives the crystal structure. (See Example 42.4.)

When atoms are bound together in condensed matter, their outer energy levels spread out into bands. At absolute zero, insulators and conductors have a completely filled valence band separated by an energy gap from an empty conduction band. Conductors, including metals, have partially filled conduction bands. (See Example 42.5.)

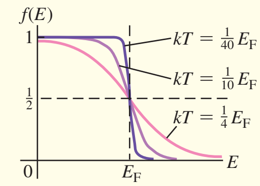


Continued

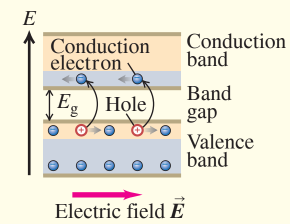
Free-electron model of metals: In the free-electron model of the behavior of conductors, the electrons are treated as completely free particles within the conductor. In this model the density of states is given by Eq. (42.15). The probability that an energy state of energy E is occupied is given by the Fermi–Dirac distribution, Eq. (42.16), which is a consequence of the exclusion principle. In Eq. (42.16), E_F is the Fermi energy. (See Examples 42.6–42.8.)

$$g(E) = \frac{(2m)^{3/2}V}{2\pi^2\hbar^3}E^{1/2} \qquad (42.15)$$

$$f(E) = \frac{1}{e^{(E-E_F)/kT} + 1} \qquad (42.16)$$

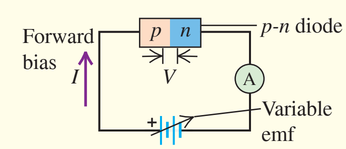


Semiconductors: A semiconductor has an energy gap of about 1 eV between its valence and conduction bands. Its electrical properties may be drastically changed by the addition of small concentrations of donor impurities, giving an n-type semiconductor, or acceptor impurities, giving a p-type semiconductor. (See Example 42.9.)



Semiconductor devices: Many semiconductor devices, including diodes, transistors, and integrated circuits, use one or more p-n junctions. The current–voltage relationship for an ideal p-n junction diode is given by Eq. (42.22).

$$I = I_S(e^{eV/kT} - 1) \qquad (42.22)$$



Chapter 42 Media Assets

GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review EXAMPLES 42.2 and 42.3 (Section 42.2) before attempting these problems.

VP42.3.1 The two nuclei in the nitric oxide (NO) molecule are 0.1154 nm apart. The mass of the most common nitrogen atom is 2.326×10^{-26} kg, and the mass of the most common oxygen atom is 2.656×10^{-26} kg. Find (a) the reduced mass of the NO molecule, (b) the moment of inertia of the NO molecule, and (c) the energies, in meV, of the lowest three rotational energy levels of NO.

VP42.3.2 The $l = 2$ rotational level of the hydrogen chloride (HCl) molecule has energy 7.90 meV. The mass of the most common hydrogen atom is 1.674×10^{-27} kg, and the mass of the most common chlorine atom is 5.807×10^{-26} kg. Find (a) the moment of inertia of the HCl molecule, (b) the reduced mass of the HCl molecule, and (c) the distance between the H and Cl nuclei.

VP42.3.3 The mass of the most common silicon atom is 4.646×10^{-26} kg, and the mass of the most common oxygen atom is 2.656×10^{-26} kg. When a molecule of silicon monoxide (SiO) makes a transition between the $l = 1$ and $l = 0$ rotational levels, it emits a photon of wavelength 6.882 mm. Find (a) the moment of inertia of the SiO molecule, (b) the reduced mass of the SiO molecule, and (c) the distance between the Si and O nuclei.

VP42.3.4 A CO molecule is initially in the $n = 2$ vibrational level. If this molecule loses both vibrational and rotational energy and emits a photon, what are the photon wavelength and frequency if the initial angular momentum quantum number is (a) $l = 3$ and (b) $l = 2$?

Be sure to review EXAMPLES 42.6 and 42.7 (Section 42.5) before attempting these problems.

VP42.7.1 For free electrons in a solid with Fermi energy E_F and temperature T, find the energy for which the probability that a state at that energy is occupied is (a) 0.33 and (b) 0.90.

VP42.7.2 The Boltzmann constant is $k = 8.617 \times 10^{-5}$ eV/K. For a metallic solid at room temperature (293 K), what is the probability that an electron state is occupied if its energy is (a) 0.0250 eV below the Fermi level, (b) 0.0400 eV above the Fermi level, and (c) 0.100 eV above the Fermi level?

VP42.7.3 Silver contains 5.8×10^{28} free electrons per cubic meter. At absolute zero, what are (a) the Fermi energy (in J and eV) of silver, (b) the speed of an electron with this energy, and (c) the density of states (in states/J and states/eV) at the Fermi energy for a block of silver of volume 1.0 cm^3?

VP42.7.4 If kT is small compared to the Fermi energy at absolute zero, the Fermi energy at temperature T is essentially the same as at absolute zero. Use the results of the previous problem to find the electron energy, in eV, for which the probability that an electron state is occupied is (a) 0.92 and (b) 1.0×10^{-4} for silver at room temperature (293 K).

Be sure to review EXAMPLE 42.9 (Section 42.6) before attempting these problems.

VP42.9.1 For a certain semiconductor, the Fermi energy is in the middle of its band gap. If the temperature of the semiconductor is 285 K, find the probability that a state at the bottom of the conduction band is occupied if the band gap is (a) 0.500 eV and (b) 1.50 eV.

VP42.9.2 Find the ratio of the probability that a state at the bottom of the conduction band of a semiconductor is occupied at 315 K to the probability at 295 K if the band gap is (a) 0.400 eV and (b) 0.800 eV. Assume that the Fermi energy is in the middle of the band gap.

VP42.9.3 Consider a material whose Fermi energy is in the middle of its band gap. What band-gap width provides a probability of 0.00190 that a state at the bottom of the conduction band is occupied if the temperature of the material is (a) 305 K and (b) 325 K?

VP42.9.4 In a certain semiconductor, the Fermi energy lies above the top of the valence band by an amount equal to $\frac{3}{4}$ of the band gap. Find the probability that a state at the bottom of the conduction band is occupied if the band gap is 0.300 eV and the temperature is (a) 275 K and (b) 325 K.

BRIDGING PROBLEM Molecular Vibration and Semiconductor Band Gap

At 80 K, the band gap in the semiconductor indium antimonide (InSb) is 0.230 eV. A photon emitted by a hydrogen fluoride (HF) molecule undergoing a vibration-rotation transition from $(n = 1, l = 0)$ to $(n = 0, l = 1)$ is absorbed by an electron at the top of the valence band of InSb. (a) How far above the top of the band gap (in eV) is the final state of the electron? (b) What is the probability that the final state was already occupied? The vibration frequency for HF is 1.24×10^{14} Hz, the mass of a hydrogen atom is 1.67×10^{-27} kg, the mass of a fluorine atom is 3.15×10^{-26} kg, and the equilibrium distance between the two nuclei is 0.092 nm. Assume that the Fermi energy for InSb is in the middle of the gap.

SOLUTION GUIDE

IDENTIFY and SET UP

1. This problem involves what you learned about molecular transitions in Section 42.2, about the Fermi–Dirac distribution in Section 42.5, and about semiconductors in Section 42.6.
2. Equation (42.9) gives the combined vibrational-rotational energy in the initial and final molecular states. The difference between the initial and final molecular energies equals the energy E of the emitted photon, which is in turn equal to the energy gained by the InSb valence electron when it absorbs that photon. The probability that the final state is occupied is given by the Fermi–Dirac distribution, Eq. (42.16).

EXECUTE

3. Before you can use Eq. (42.9), you'll first need to use the data given to calculate the moment of inertia I and the quantity $\hbar\omega$ for the HF molecule. (*Hint:* Be careful not to confuse frequency f and angular frequency ω.)
4. Use your results from step 3 to calculate the initial and final energies of the HF molecule. (*Hint:* Does the vibrational energy increase or decrease? What about the rotational energy?)
5. Use your result from step 4 to find the energy imparted to the InSb electron. Determine the final energy of this electron relative to the bottom of the conduction band.
6. Use your result from step 5 to determine the probability that the InSb final state is already occupied.

EVALUATE

7. Is the molecular transition of the HF molecule allowed? Which is larger: the vibrational energy change or the rotational energy change?
8. Is it likely that the excited InSb electron will be blocked from entering a state in the conduction band?

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q42.1 Van der Waals bonds occur in many molecules, but hydrogen bonds occur only with materials that contain hydrogen. Why is this type of bond unique to hydrogen?

Q42.2 The bonding of gallium arsenide (GaAs) is said to be 31% ionic and 69% covalent. Explain.

Q42.3 The $H_2{}^+$ molecule consists of two hydrogen nuclei and a single electron. What kind of molecular bond do you think holds this molecule together? Explain.

Q42.4 The moment of inertia for an axis through the center of mass of a diatomic molecule calculated from the wavelength emitted in an $l = 19 \rightarrow l = 18$ transition is different from the moment of inertia calculated from the wavelength of the photon emitted in an $l = 1 \rightarrow l = 0$ transition. Explain this difference. Which transition corresponds to the larger moment of inertia?

Q42.5 Analysis of the photon absorption spectrum of a diatomic molecule shows that the vibrational energy levels for small values of n are very nearly equally spaced but the levels for large n are not equally spaced. Discuss the reason for this observation. Do you expect the adjacent levels to move closer together or farther apart as n increases? Explain.

Q42.6 Discuss the differences between the rotational and vibrational energy levels of the deuterium ("heavy hydrogen") molecule D_2 and those of the ordinary hydrogen molecule H_2. A deuterium atom has twice the mass of an ordinary hydrogen atom.

Q42.7 Various organic molecules have been discovered in interstellar space. Why were these discoveries made with radio telescopes rather than optical telescopes?

Q42.8 The air you are breathing contains primarily nitrogen (N_2) and oxygen (O_2). Many of these molecules are in excited rotational energy levels $(l = 1, 2, 3, \ldots)$, but almost all of them are in the vibrational ground level $(n = 0)$. Explain this difference between the rotational and vibrational behaviors of the molecules.

Q42.9 In what ways do atoms in a diatomic molecule behave as though they were held together by a spring? In what ways is this a poor description of the interaction between the atoms?

Q42.10 Individual atoms have discrete energy levels, but certain solids (which are made up of only individual atoms) show energy bands and gaps. What causes the solids to behave so differently from the atoms of which they are composed?

Q42.11 What factors determine whether a material is a conductor of electricity or an insulator? Explain.

Q42.12 Ionic crystals are often transparent, whereas metallic crystals are always opaque. Why?

Q42.13 Speeds of molecules in a gas vary with temperature, whereas speeds of electrons in the conduction band of a metal are nearly independent of temperature. Why are these behaviors so different?

Q42.14 Use the band model to explain how it is possible for some materials to undergo a semiconductor-to-metal transition as the temperature or pressure varies.

Q42.15 An isolated zinc atom has a ground-state electron configuration of filled 1*s*, 2*s*, 2*p*, 3*s*, 3*p*, and 4*s* subshells. How can zinc be a conductor if its valence subshell is full?

Q42.16 The assumptions of the *free-electron model* of metals may seem contrary to reason, since electrons exert powerful electric forces on each other. Give some reasons why these assumptions actually make physical sense.

Q42.17 Why are materials that are good thermal conductors also good electrical conductors? What kinds of problems does this pose for the design of appliances such as clothes irons and electric heaters? Are there materials that do not follow this general rule?

Q42.18 What is the essential characteristic for an element to serve as a donor impurity in a semiconductor such as Si or Ge? For it to serve as an acceptor impurity? Explain.

Q42.19 There are several methods for removing electrons from the surface of a semiconductor. Can holes be removed from the surface? Explain.

Q42.20 A student asserts that silicon and germanium become good insulators at very low temperatures and good conductors at very high temperatures. Do you agree? Explain your reasoning.

Q42.21 The electrical conductivities of most metals decrease gradually with increasing temperature, but the intrinsic conductivity of semiconductors always *increases* rapidly with increasing temperature. What causes the difference?

Q42.22 How could you make compensated silicon that has twice as many acceptors as donors?

Q42.23 The saturation current I_S for a *p-n* junction, Eq. (42.22), depends strongly on temperature. Explain why.

Q42.24 Why does tunneling limit the miniaturization of MOSFETs?

EXERCISES

Section 42.1 Types of Molecular Bonds

42.1 • **An Ionic Bond.** (a) Calculate the electric potential energy for a K^+ ion and a Br^- ion separated by a distance of 0.29 nm, the equilibrium separation in the KBr molecule. Treat the ions as point charges. (b) The ionization energy of the potassium atom is 4.3 eV. Atomic bromine has an electron affinity of 3.5 eV. Use these data and the results of part (a) to estimate the binding energy of the KBr molecule. Do you expect the actual binding energy to be higher or lower than your estimate? Explain your reasoning.

42.2 • During each of these processes, a photon of light is given up. In each process, what wavelength of light is given up, and in what part of the electromagnetic spectrum is that wavelength? (a) A molecule decreases its vibrational energy by 0.198 eV; (b) an atom decreases its energy by 7.80 eV; (c) a molecule decreases its rotational energy by 4.80×10^{-3} eV.

42.3 • For the H_2 molecule the equilibrium spacing of the two protons is 0.074 nm. The mass of a hydrogen atom is 1.67×10^{-27} kg. Calculate the wavelength of the photon emitted in the rotational transition $l = 2$ to $l = 1$.

Section 42.2 Molecular Spectra

42.4 •• The H_2 molecule has a moment of inertia of $4.6 \times 10^{-48}\ \text{kg} \cdot \text{m}^2$. What is the wavelength λ of the photon absorbed when H_2 makes a transition from the $l = 3$ to the $l = 4$ rotational level?

42.5 • A hypothetical NH molecule makes a rotational-level transition from $l = 3$ to $l = 1$ and gives off a photon of wavelength 1.780 nm in doing so. What is the separation between the two atoms in this molecule if we model them as point masses? The mass of hydrogen is 1.67×10^{-27} kg, and the mass of nitrogen is 2.33×10^{-26} kg.

42.6 • Two atoms of cesium (Cs) can form a Cs_2 molecule. The equilibrium distance between the nuclei in a Cs_2 molecule is 0.447 nm. Calculate the moment of inertia about an axis through the center of mass of the two nuclei and perpendicular to the line joining them. The mass of a cesium atom is 2.21×10^{-25} kg.

42.7 •• **CP** The rotational energy levels of CO are calculated in Example 42.2. If the energy of the rotating molecule is described by the classical expression $K = \frac{1}{2}I\omega^2$, for the $l = 1$ level what are (a) the angular speed of the rotating molecule; (b) the linear speed of each atom; (c) the rotational period (the time for one rotation)?

42.8 • The vibrational and rotational energies of the CO molecule are given by Eq. (42.9). Calculate the wavelength of the photon absorbed by CO in each of these vibration-rotation transitions: (a) $n = 0$, $l = 2 \rightarrow n = 1$, $l = 3$; (b) $n = 0$, $l = 3 \rightarrow n = 1$, $l = 2$; (c) $n = 0$, $l = 4 \rightarrow n = 1$, $l = 3$.

42.9 • A lithium atom has mass 1.17×10^{-26} kg, and a hydrogen atom has mass 1.67×10^{-27} kg. The equilibrium separation between the two nuclei in the LiH molecule is 0.159 nm. (a) What is the difference in energy between the $l = 3$ and $l = 4$ rotational levels? (b) What is the wavelength of the photon emitted in a transition from the $l = 4$ to the $l = 3$ level?

42.10 • If a sodium chloride (NaCl) molecule could undergo an $n \rightarrow n - 1$ vibrational transition with no change in rotational quantum number, a photon with wavelength 20.0 μm would be emitted. The mass of a sodium atom is 3.82×10^{-26} kg, and the mass of a chlorine atom is 5.81×10^{-26} kg. Calculate the force constant k' for the interatomic force in NaCl.

42.11 •• When a hypothetical diatomic molecule having atoms 0.8860 nm apart undergoes a rotational transition from the $l = 2$ state to the next lower state, it gives up a photon having energy 8.841×10^{-4} eV. When the molecule undergoes a vibrational transition from one energy state to the next lower energy state, it gives up 0.2560 eV. Find the force constant of this molecule.

Section 42.3 Structure of Solids

42.12 • Potassium bromide (KBr) has a density of $2.75 \times 10^3\ \text{kg/m}^3$ and the same crystal structure as NaCl. The mass of a potassium atom is 6.49×10^{-26} kg, and the mass of a bromine atom is 1.33×10^{-25} kg. (a) Calculate the average spacing between adjacent atoms in a KBr crystal. (b) How does the value calculated in part (a) compare with the spacing in NaCl (see Exercise 42.13)? Is the relationship between the two values qualitatively what you would expect? Explain.

42.13 • **Density of NaCl.** The spacing of adjacent atoms in a crystal of sodium chloride is 0.282 nm. The mass of a sodium atom is 3.82×10^{-26} kg, and the mass of a chlorine atom is 5.89×10^{-26} kg. Calculate the density of sodium chloride.

Section 42.4 Energy Bands

42.14 • The gap between valence and conduction bands in diamond is 5.47 eV. (a) What is the maximum wavelength of a photon that can excite an electron from the top of the valence band into the conduction band? In what region of the electromagnetic spectrum does this photon lie? (b) Explain why pure diamond is transparent and colorless. (c) Most gem diamonds have a yellow color. Explain how impurities in the diamond can cause this color.

42.15 • The maximum wavelength of light that a certain silicon photocell can detect is 1.11 μm. (a) What is the energy gap (in electron volts) between the valence and conduction bands for this photocell? (b) Explain why pure silicon is opaque.

42.16 • The gap between valence and conduction bands in silicon is 1.12 eV. A nickel nucleus in an excited state emits a gamma-ray photon with wavelength 9.31×10^{-4} nm. How many electrons can be excited from the top of the valence band to the bottom of the conduction band by the absorption of this gamma ray?

Section 42.5 Free-Electron Model of Metals

42.17 • Calculate the density of states $g(E)$ for the free-electron model of a metal if $E = 7.0$ eV and $V = 1.0\ \text{cm}^3$. Express your answer in units of states per electron volt.

42.18 • The Fermi energy of sodium is 3.23 eV. (a) Find the average energy E_{av} of the electrons at absolute zero. (b) What is the speed of an electron that has energy E_{av}? (c) At what Kelvin temperature T is kT equal to E_F? (This is called the *Fermi temperature* for the metal. It is approximately the temperature at which molecules in a classical ideal gas would have the same kinetic energy as the fastest-moving electron in the metal.)

42.19 • Consider the density of states in the free-electron model for an energy of 1.60 eV. At what energy will the density of states be (a) doubled and (b) halved?

42.20 • For a metal with kT equal to 25% of the Fermi energy, what is the probability that a state will be occupied by an electron if its energy is (a) 50% of the Fermi energy and (b) 50% greater than the Fermi energy? Are your results consistent with Fig. 42.23?

42.21 • **CP** Silver has a Fermi energy of 5.48 eV. Calculate the electron contribution to the molar heat capacity at constant volume of silver, C_V, at 300 K. Express your result (a) as a multiple of R and (b) as a fraction of the actual value for silver, $C_V = 25.3\ \text{J/mol} \cdot \text{K}$. (c) Is the value of C_V due principally to the electrons? If not, to what is it due? (*Hint:* See Section 18.4.)

42.22 •• At the Fermi temperature T_F, $E_F = kT_F$ (see Exercise 42.18). When $T = T_F$, what is the probability that a state with energy $E = 2E_F$ is occupied?

42.23 •• For a solid metal having a Fermi energy of 8.500 eV, what is the probability, at room temperature, that a state having an energy of 8.520 eV is occupied by an electron?

Section 42.6 Semiconductors

42.24 • Pure germanium has a band gap of 0.67 eV. The Fermi energy is in the middle of the gap. (a) For temperatures of 250 K, 300 K, and 350 K, calculate the probability $f(E)$ that a state at the bottom of the conduction band is occupied. (b) For each temperature in part (a), calculate the probability that a state at the top of the valence band is empty.

42.25 • Germanium has a band gap of 0.67 eV. Doping with arsenic adds donor levels in the gap 0.01 eV below the bottom of the conduction band. At a temperature of 300 K, the probability is 4.4×10^{-4} that an electron state is occupied at the bottom of the conduction band. Where is the Fermi level relative to the conduction band in this case?

Section 42.7 Semiconductor Devices

42.26 •• (a) Suppose a piece of very pure germanium is to be used as a light detector by observing, through the absorption of photons, the increase in conductivity resulting from generation of electron–hole pairs. If each pair requires 0.67 eV of energy, what is the maximum wavelength that can be detected? In what portion of the spectrum does it lie? (b) What are the answers to part (a) if the material is silicon, with an energy requirement of 1.12 eV per pair, corresponding to the gap between valence and conduction bands in that element?

42.27 • **CP** At a temperature of 290 K, a certain *p-n* junction has a saturation current $I_S = 0.500$ mA. (a) Find the current at this temperature when the voltage is (i) 1.00 mV, (ii) −1.00 mV, (iii) 100 mV, and (iv) −100 mV. (b) Is there a region of applied voltage where the diode obeys Ohm's law?

42.28 • For a certain *p-n* junction diode, the saturation current at room temperature (20°C) is 0.950 mA. What is the resistance of this diode when the voltage across it is (a) 85.0 mV and (b) −50.0 mV?

42.29 •• (a) A forward-bias voltage of 15.0 mV produces a positive current of 9.25 mA through a *p-n* junction at 300 K. What does the positive current become if the forward-bias voltage is reduced to 10.0 mV? (b) For reverse-bias voltages of −15.0 mV and −10.0 mV, what is the reverse-bias negative current?

42.30 •• A *p-n* junction has a saturation current of 6.40 mA. (a) At a temperature of 300 K, what voltage is needed to produce a positive current of 40.0 mA? (b) For a voltage equal to the negative of the value calculated in part (a), what is the negative current?

42.31 • The saturation current for a junction diode is 10 microamps. Calculate the resistance of the diode at forward and reverse potential differences of 0.20 V when $T = 17°\text{C}$.

PROBLEMS

42.32 • When a diatomic molecule undergoes a transition from the $l = 2$ to the $l = 1$ rotational state, a photon with wavelength 54.3 μm is emitted. What is the moment of inertia of the molecule for an axis through its center of mass and perpendicular to the line connecting the nuclei?

42.33 •• **CP** (a) The equilibrium separation of the two nuclei in an NaCl molecule is 0.24 nm. If the molecule is modeled as charges $+e$ and $-e$ separated by 0.24 nm, what is the electric dipole moment of the molecule (see Section 21.7)? (b) The measured electric dipole moment of an NaCl molecule is $3.0 \times 10^{-29}\ \text{C} \cdot \text{m}$. If this dipole moment arises from point charges $+q$ and $-q$ separated by 0.24 nm, what is q? (c) A definition of the *fractional ionic character* of the bond is q/e. If the sodium atom has charge $+e$ and the chlorine atom has charge $-e$, the fractional ionic character would be equal to 1. What is the actual fractional ionic character for the bond in NaCl? (d) The equilibrium distance between nuclei in the hydrogen iodide (HI) molecule is 0.16 nm, and the measured electric dipole moment of the molecule is $1.5 \times 10^{-30}\ \text{C} \cdot \text{m}$. What is the fractional ionic character for the bond in HI? How does your answer compare to that for NaCl calculated in part (c)? Discuss reasons for the difference in these results.

42.34 • When a NaF molecule makes a transition from the $l = 3$ to the $l = 2$ rotational level with no change in vibrational quantum number or electronic state, a photon with wavelength 3.83 mm is emitted. A sodium atom has mass 3.82×10^{-26} kg, and a fluorine atom has mass 3.15×10^{-26} kg. Calculate the equilibrium separation between the nuclei in a NaF molecule. How does your answer compare with the value for NaCl given in Section 42.1? Is this result reasonable? Explain.

42.35 •• **CP** Consider a gas of diatomic molecules (moment of inertia I) at an absolute temperature T. If E_g is a ground-state energy and E_{ex} is the energy of an excited state, then the Maxwell–Boltzmann distribution (see Section 39.4) predicts that the ratio of the numbers of molecules in the two states is $n_{ex}/n_g = e^{-(E_{ex}-E_g)/kT}$. (a) Explain why the ratio of the number of molecules in the lth rotational energy *level* to the number of molecules in the ground-state ($l = 0$) rotational level is

$$\frac{n_l}{n_0} = (2l + 1)e^{-[l(l+1)\hbar^2]/2IkT}$$

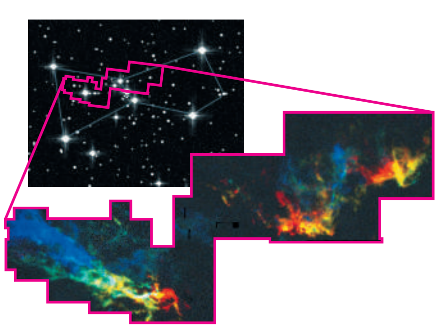
(*Hint:* For each value of l, how many states are there with different values of m_l?) (b) Determine the ratio n_l/n_0 for a gas of CO molecules at 300 K for (i) $l = 1$; (ii) $l = 2$; (iii) $l = 10$; (iv) $l = 20$; (v) $l = 50$. The moment of inertia of the CO molecule is given in Example 42.2 (Section 42.2). (c) Your results in part (b) show that as l is increased, the ratio n_l/n_0 first increases and then decreases. Explain why.

42.36 ••• CALC Part (a) of Problem 42.35 gives an equation for the number of diatomic molecules in the lth rotational level to the number in the ground-state rotational level. (a) Derive an expression for the value of l for which this ratio is the largest. (b) For the CO molecule at $T = 300$ K, for what value of l is this ratio a maximum? (The moment of inertia of the CO molecule is given in Example 42.2.)

42.37 •• Semiconductor devices can radiate heat up to a maximum power rating. If the device dissipates more power than that limit, it will heat up and explode, melt, or simply break down. A typical power rating for a diode is 500 mW. The power dissipated by any two-terminal device is the product of the current through it and the voltage across it. (a) Use Eq. (42.22) to estimate the power dissipated by a diode when it is forward biased at $V = 0.6$ V at a temperature of 300 K. The saturation current is 1.2×10^{-11} A. (b) Estimate to one significant figure the forward voltage at which the device will dissipate 500 mW. (c) Then what is the current?

42.38 •• Our galaxy contains numerous *molecular clouds,* regions many light-years in extent in which the density is high enough and the temperature low enough for atoms to form into molecules. Most of the molecules are H_2, but a small fraction of the molecules are carbon monoxide (CO). Such a molecular cloud in the constellation Orion is shown in **Fig. P42.38**. The upper image was made with an ordinary visible-light telescope; the lower image shows the molecular cloud in Orion as imaged with a radio telescope tuned to a wavelength emitted by CO in a rotational transition. The different colors in the radio image indicate regions of the cloud that are moving either toward us (blue) or away from us (red) relative to the motion of the cloud as a whole, as determined by the Doppler shift of the radiation. (Since a molecular cloud has about 10,000 hydrogen molecules for each CO molecule, it might seem more reasonable to tune a radio telescope to emissions from H_2 than to emissions from CO. Unfortunately, it turns out that the H_2 molecules in molecular clouds do not radiate in either the radio or visible portions of the electromagnetic spectrum.) (a) Using the data in Example 42.2 (Section 42.2), calculate the energy and wavelength of the photon emitted by a CO molecule in an $l = 1 \rightarrow l = 0$ rotational transition. (b) As a rule, molecules in a gas at temperature T will be found in a certain excited rotational energy level, provided the energy of that level is no higher than kT (see Problem 42.35). Use this rule to explain why astronomers can detect radiation from CO in molecular clouds even though the typical temperature of a molecular cloud is a very low 20 K.

Figure **P42.38**



42.39 • The force constant for the internuclear force in a hydrogen molecule (H_2) is $k' = 576$ N/m. A hydrogen atom has mass 1.67×10^{-27} kg. Calculate the zero-point vibrational energy for H_2 (that is, the vibrational energy the molecule has in the $n = 0$ ground vibrational level). How does this energy compare in magnitude with the H_2 bond energy of -4.48 eV?

42.40 • When an OH molecule undergoes a transition from the $n = 0$ to the $n = 1$ vibrational level, its internal vibrational energy increases by 0.463 eV. Calculate the frequency of vibration and the force constant for the interatomic force. (The mass of an oxygen atom is 2.66×10^{-26} kg, and the mass of a hydrogen atom is 1.67×10^{-27} kg.)

42.41 • The hydrogen iodide (HI) molecule has equilibrium separation 0.160 nm and vibrational frequency 6.93×10^{13} Hz. The mass of a hydrogen atom is 1.67×10^{-27} kg, and the mass of an iodine atom is 2.11×10^{-25} kg. (a) Calculate the moment of inertia of HI about a perpendicular axis through its center of mass. (b) Calculate the wavelength of the photon emitted in each of the following vibration-rotation transitions: (i) $n = 1, l = 1 \rightarrow n = 0, l = 0$; (ii) $n = 1, l = 2 \rightarrow n = 0, l = 1$; (iii) $n = 2, l = 2 \rightarrow n = 1, l = 3$.

42.42 • Suppose the hydrogen atom in HF (see the Bridging Problem for this chapter) is replaced by an atom of deuterium, an isotope of hydrogen with a mass of 3.34×10^{-27} kg. The force constant is determined by the electron configuration, so it is the same as for the normal HF molecule. (a) What is the vibrational frequency of this molecule? (b) What wavelength of light corresponds to the energy difference between the $n = 1$ and $n = 0$ levels? In what region of the spectrum does this wavelength lie?

42.43 •• Compute the Fermi energy of potassium by making the simple approximation that each atom contributes one free electron. The density of potassium is 851 kg/m^3, and the mass of a single potassium atom is 6.49×10^{-26} kg.

42.44 •• CALC The one-dimensional calculation of Example 42.4 (Section 42.3) can be extended to three dimensions. For the three-dimensional fcc NaCl lattice, the result for the potential energy of a pair of Na^+ and Cl^- ions due to the electrostatic interaction with all of the ions in the crystal is $U = -\alpha e^2/4\pi\epsilon_0 r$, where $\alpha = 1.75$ is the *Madelung constant*. Another contribution to the potential energy is a repulsive interaction at small ionic separation r due to overlap of the electron clouds. This contribution can be represented by A/r^8, where A is a positive constant, so the expression for the total potential energy is

$$U_{tot} = -\frac{\alpha e^2}{4\pi\epsilon_0 r} + \frac{A}{r^8}$$

(a) Let r_0 be the value of the ionic separation r for which U_{tot} is a minimum. Use this definition to find an equation that relates r_0 and A, and use this to write U_{tot} in terms of r_0. For NaCl, $r_0 = 0.281$ nm. Obtain a numerical value (in electron volts) of U_{tot} for NaCl. (b) The quantity $-U_{tot}$ is the energy required to remove an Na^+ ion and a Cl^- ion from the crystal. Forming a pair of neutral atoms from this pair of ions involves the release of 5.14 eV (the ionization energy of Na) and the expenditure of 3.61 eV (the electron affinity of Cl). Use the result of part (a) to calculate the energy required to remove a pair of neutral Na and Cl atoms from the crystal. The experimental value for this quantity is 6.39 eV; how well does your calculation agree?

42.45 ••• Metallic lithium has a bcc crystal structure. Each unit cell is a cube of side length $a = 0.35$ nm. (a) For a bcc lattice, what is the number of atoms per unit volume? Give your answer in terms of a. (*Hint:* How many atoms are there per unit cell?) (b) Use the result of part (a) to calculate the zero-temperature Fermi energy E_{F0} for metallic lithium. Assume there is one free electron per atom.

42.46 •• DATA To determine the equilibrium separation of the atoms in the HCl molecule, you measure the rotational spectrum of HCl. You find that the spectrum contains these wavelengths (among others): 60.4 μm, 69.0 μm, 80.4 μm, 96.4 μm, and 120.4 μm. (a) Use your measured wavelengths to find the moment of inertia of the HCl molecule about an axis through the center of mass and perpendicular to the line joining the two nuclei. (b) The value of l changes by ± 1 in rotational transitions. What value of l for the upper level of the transition gives rise to each of these wavelengths? (c) Use your result of part (a) to calculate the equilibrium separation of the atoms in the HCl molecule. The mass of a chlorine atom is 5.81×10^{-26} kg, and the mass of a hydrogen atom is 1.67×10^{-27} kg. (d) What is the longest-wavelength line in the rotational spectrum of HCl?

42.47 •• DATA The table gives the occupation probabilities $f(E)$ as a function of the energy E for a solid conductor at a fixed temperature T.

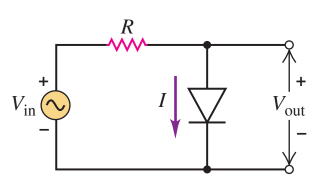
$f(E)$	0.064	0.173	0.390	0.661	0.856	0.950
E (eV)	3.0	2.5	2.0	1.5	1.0	0.5

To determine the Fermi energy of the solid material, you are asked to analyze this information in terms of the Fermi–Dirac distribution. (a) Graph the values in the table as E versus $\ln\{[1/f(E)] - 1\}$. Find the slope and y-intercept of the best-fit straight line for the data points when they are plotted this way. (b) Use your results of part (a) to calculate the temperature T and the Fermi energy of the material.

42.48 •• DATA A p-n junction is part of the control mechanism for a wind turbine that is used to generate electricity. The turbine has been malfunctioning, so you are running diagnostics. You can remotely change the bias voltage V applied to the junction and measure the current through the junction. With a forward-bias voltage of $+5.00$ mV, the current is $I_f = 0.407$ mA. With a reverse-bias voltage of -5.00 mV, the current is $I_r = -0.338$ mA. Assume that Eq. (42.22) accurately represents the current–voltage relationship for the junction, and use these two results to calculate the temperature T and saturation current I_S for the junction. [*Hint:* In your analysis, let $x = e^{eV/kT}$. Apply Eq. (42.22) to each measurement and obtain a quadratic equation for x.]

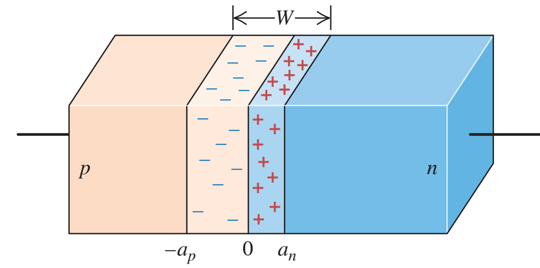
42.49 •• The circuit shown in **Fig. P42.49** is called a half-wave rectifier. The triangular symbol with the flat lower edge represents a diode, such that a forward bias drives the current downward. Use Eq. (42.22) to study the behavior of this circuit. Use a typical value for the saturation current: $I_S = 5 \times 10^{-13}$ A. Use a temperature of 300 K. (a) Any predicted current less than 1 μA can be considered a zero current. Determine the minimum voltage V_{out} for which the current I exceeds 1 μA. (b) The current increases dramatically for diode voltages that exceed that value. At a given diode voltage V_{out}, the diode has an effective resistance given by $R_D = V_{out}/I$. Use Eq. (42.22) to estimate the diode voltage $V_{out} = V_{max}$ for which $R_D = 100\ \Omega$. (c) What is the current I in that case? (d) If the resistance R is greater than 10 kΩ, then $R_D \ll R$ for larger currents, and there is a negligible additional voltage drop across the diode. This effectively pins $V_{out} = V_{max}$ for all input voltages $V_{in} > V_{max}$. With this perspective, estimate how this circuit would respond to a sinusoidal input voltage with amplitude 5 V.

Figure **P42.49**



42.50 ••• CP CALC A p-n junction includes p-type silicon, with donor atom density N_D, adjacent to n-type silicon, with acceptor atom density N_A. Near the junction, free electrons from the n side have diffused into the p side while free holes from the p side have diffused into the n side, leaving a "depletion region" of width W where there are no free charge carriers (**Fig. P42.50**). The width of the depletion region on the n side is a_n, and the width of the depletion region on the p side is a_p. Each depleted region has a constant charge density with magnitude equal to the corresponding donor atom density. Assume there is no net charge outside the depleted regions. Define an x-axis pointing from the p side toward the n side with the origin at the center of the junction. An electric field $\vec{E} = -E\hat{\imath}$ has developed in the depletion region, stabilizing the diffusion. (a) Use Gauss's law to determine the electric field on the p side of the junction, for $-a_p \le x \le 0$. (*Hint:* Use a cylindrical Gaussian surface parallel to the x-axis, with the left end outside the depletion region (where $\vec{E} = 0$) and the right side inside the region.) Note that the dielectric constant of silicon is $K = 11.7$. (b) Determine the electric field on the n side of the junction, for $0 \le x \le a_n$.

Figure **P42.50**



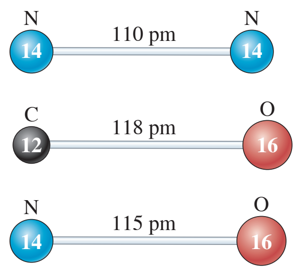
(c) Use continuity at $x = 0$ to determine a relationship between the depths a_p and a_n and in terms of N_A and N_D. (d) The electric potential $V(x)$ may be obtained by integrating the electric field, as shown in Eq. (23.18). Determine $V(x)$ in the region $-a_p \le x \le 0$ using the convention that $V(x) = 0$ for $x < -a_p$. (e) Similarly, determine $V(x)$ in the region $0 \le x \le a_n$. (f) What is the "barrier potential" $V_b = V(a_n) - V(-a_p)$? (g) The p side is doped with boron atoms with density $N_A = 1.00 \times 10^{16}\ \text{cm}^{-3}$. The n side is doped with arsenic atoms with density $N_D = 5.00 \times 10^{16}\ \text{cm}^{-3}$. The n side depletion depth is $a_n = 55.0$ nm. What is the p side depletion depth a_p? (h) What is the peak magnitude of the electric field, at $x = 0$? (i) What is the value of the barrier potential V_b?

42.51 ••• CP Consider the CO_2 molecule shown in Fig. 42.10c. The oxygen molecules have mass M_O and the carbon atom has mass M_C. Parameterize the positions of the left oxygen atom, the carbon atom, and the right oxygen atom using x_1, x_2, and x_3 as the respective rightward deviations from equilibrium. Treat the bonds as Hooke's-law springs with common spring constant k'. (a) Use Newton's second law to obtain expressions for $M_i \ddot{x}_i = M_i\, d^2x/dt^2$ in each case $i = 1, 2, 3$, where $M_{1,2,3} = (M_O, M_C, M_O)$. (*Note:* We represent time derivatives using dots.) Assume $X_C = 0$ at $t = 0$. (b) To ascertain the motion of the asymmetric stretching mode, set $x_1 = x_3 \equiv X_O$ and set $x_2 \equiv X_C$. Write the two independent equations that remain from your previous result. (c) Eliminate the sum $X_O + X_C$ from your equations. Use what remains to ascertain X_C in terms of X_O. (d) Substitute your expression for X_C into your equation for X_O to derive a harmonic oscillator equation $M_{eff}\ddot{X}_O = -k\ddot{X}_O$. What is M_{eff}? (e) This equation has the solution $X_O(t) = A\cos(\omega t)$. What is the angular frequency ω? (f) Using the experimentally determined spring constant $k' = 1860$ N/m and the atomic masses $M_C = 12$ u and $M_O = 16$ u, where $\text{u} = 1.6605 \times 10^{-27}$ kg, to determine the oscillation frequency $f = \omega/2\pi$.

CHALLENGE PROBLEMS

42.52 ••• A colorless, odorless gas in a tank needs to be identified. The gas is nitrogen (N_2), carbon monoxide (CO), or nitric oxide (NO). (See **Fig. P42.52**.) White light is passed through a sample to obtain an absorption spectrum. Among others, the following wavelengths present dark bands on the spectrum: $\lambda = 4.2680\ \mu$m, $4.2753\ \mu$m, $4.2826\ \mu$m, $4.2972\ \mu$m, and $4.3046\ \mu$m. These represent transitions from the lowest rotational states. (a) What photon energies correspond to these wavelengths? Specify your answers to five significant figures. Note that the speed of light is 2.9979×10^8 m/s and $h = 4.1357 \times 10^{-15}$ eV · s. (b) Which one of the absorbed energies corresponds to transitions *from* an $l = 0$ state of the gas? (*Hint:* Use Eq. (42.9) along with the transition rules as a guide.) (c) Which one of the absorbed energies corresponds to a transition *to* an $l = 0$ state? (d) Determine the value of $\hbar\sqrt{k'/m_r}$, where k' is the effective spring constant of the molecule and m_r is its reduced mass. (*Hint:* Use the previous two results and Eq. (42.9).) (e) Determine the molecule's moment of inertia I. (f) The atomic masses and the bond

Figure **P42.52**



lengths for the three molecule options are shown in Fig. P42.52, where u $= 1.6605 \times 10^{-27}$ kg is one atomic mass unit. Based on the data, what is the mystery gas? (g) What is the reduced mass m_r in terms of u? (h) What is the spring constant k'?

42.53 ••• **CALC** Consider a system of N free electrons within a volume V. Even at absolute zero, such a system exerts a pressure p on its surroundings due to the motion of the electrons. To calculate this pressure, imagine that the volume increases by a small amount dV. The electrons will do an amount of work $p\,dV$ on their surroundings, which means that the total energy E_{tot} of the electrons will change by an amount $dE_{tot} = -p\,dV$. Hence $p = -dE_{tot}/dV$. (a) Show that the pressure of the electrons at absolute zero is

$$p = \frac{3^{2/3}\pi^{4/3}\hbar^2}{5m}\left(\frac{N}{V}\right)^{5/3}$$

(b) Evaluate this pressure for copper, which has a free-electron concentration of $8.45 \times 10^{28}\ \text{m}^{-3}$. Express your result in pascals and in atmospheres. (c) The pressure you found in part (b) is extremely high. Why, then, don't the electrons in a piece of copper simply explode out of the metal?

42.54 ••• **CALC** When the pressure p on a material increases by an amount Δp, the volume of the material will change from V to $V + \Delta V$, where ΔV is negative. The *bulk modulus* B of the material is defined to be the ratio of the pressure change Δp to the absolute value $|\Delta V/V|$ of the fractional volume change. The greater the bulk modulus, the greater the pressure increase required for a given fractional volume change, and the more incompressible the material (see Section 11.4). Since $\Delta V < 0$, the bulk modulus can be written as $B = -\Delta p/(\Delta V/V_0)$. In the limit that the pressure and volume changes are very small, this becomes

$$B = -V\frac{dp}{dV}$$

(a) Use the result of Challenge Problem 42.53 to show that the bulk modulus for a system of N free electrons in a volume V at low temperatures is $B = \frac{5}{3}p$. (*Hint:* The quantity p in the expression $B = -V(dp/dV)$ is the *external* pressure on the system. Can you explain why this is equal to the *internal* pressure of the system itself, as found in Challenge Problem 42.53?) (b) Evaluate the bulk modulus for the electrons in copper, which has a free-electron concentration of $8.45 \times 10^{28}\ \text{m}^{-3}$. Express your result in pascals. (c) The actual bulk modulus of copper is 1.4×10^{11} Pa. Based on your result in part (b), what fraction of this is due to the free electrons in copper? (This result shows that the free electrons in a metal play a major role in making the metal resistant to compression.) What do you think is responsible for the remaining fraction of the bulk modulus?

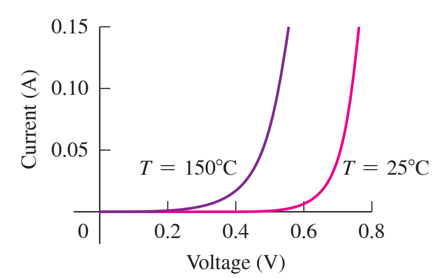
42.55 •• In the discussion of free electrons in Section 42.5, we assumed that we could ignore the effects of relativity. This is not a safe assumption if the Fermi energy is greater than about $\frac{1}{100}mc^2$ (that is, more than about 1% of the rest energy of an electron). (a) Assume that the Fermi energy at absolute zero, as given by Eq. (42.19), is equal to $\frac{1}{100}mc^2$. Show that the electron concentration is

$$\frac{N}{V} = \frac{2^{3/2}m^3c^3}{3000\pi^2\hbar^3}$$

and determine the numerical value of N/V. (b) Is it a good approximation to ignore relativistic effects for electrons in a metal such as copper, for which the electron concentration is $8.45 \times 10^{28}\ \text{m}^{-3}$? Explain. (c) A *white dwarf star* is what is left behind by a star like the sun after it has ceased to produce energy by nuclear reactions. (Our own sun will become a white dwarf star in another 8×10^9 years or so.) A typical white dwarf has mass 2×10^{30} kg (comparable to the sun) and radius 6000 km (comparable to that of the earth). The gravitational attraction of different parts of the white dwarf for each other tends to compress the star; what prevents it from compressing is the pressure of free electrons within the star (see Challenge Problem 42.53). Use both of the following assumptions to estimate the electron concentration within a typical white dwarf star: (i) the white dwarf star is made of carbon, which has a mass per atom of 1.99×10^{-26} kg; and (ii) all six of the electrons from each carbon atom are able to move freely throughout the star. (d) Is it a good approximation to ignore relativistic effects in the structure of a white dwarf star? Explain.

MCAT-STYLE PASSAGE PROBLEMS

Diode Temperature Sensor. The current–voltage characteristics of a forward-biased *p-n* junction diode depend strongly on temperature, as shown in the figure. As a result, diodes can be used as temperature sensors. In actual operation, the voltage is adjusted to keep the current through the diode constant at a specified value, such as 100 mA, and the temperature is determined from a measurement of the voltage at that current.



42.56 The sensitivity of a diode thermometer depends on how much the voltage changes for a given temperature change, with the current remaining constant. What is the sensitivity for this diode thermometer, operated at 100 mA, for a temperature change from 25°C to 150°C? (a) +0.2 mV/°C; (b) +2.0 mV/°C; (c) −0.2 mV/°C; (d) −2.0 mV/°C.

42.57 Which statement best explains the temperature dependence of the current–voltage characteristics that the graph shows? At higher temperatures: (a) The band gap is larger, so the electron–hole pairs have more energy, which causes the current at a given voltage to be larger. (b) More electrons can move to the conduction band, which causes the current at a given voltage to be larger. (c) All of the electrons in the valence band move to the conduction band, and the diode behaves like a metal and follows Ohm's law. (d) The acceptor and donor impurity atoms are free to move through the material, which causes the current at a given voltage to be larger.

42.58 If the voltage rather than the current is kept constant, what happens as the temperature increases from 25°C to 150°C? (a) At first the current increases, then it decreases. (b) The current increases. (c) The current decreases, eventually approaching zero. (d) The current does not change unless the voltage also changes.

ANSWERS

Chapter Opening Question ?

(i) Venus must radiate energy into space at the same rate that it receives energy in the form of sunlight. However, carbon dioxide (CO_2) molecules in the atmosphere absorb infrared radiation emitted by the surface of Venus and re-emit it toward the ground. This involves a transition between vibrational states of the CO_2 molecule (see Section 42.2). To compensate for this and to maintain the balance between emitted and received energy, the surface temperature of Venus and hence the rate of blackbody radiation from the surface increase.

Key Example VARIATION Problems

VP42.3.1 (a) 1.240×10^{-26} kg (b) 1.651×10^{-46} kg·m^2 (c) 0, 0.4203 meV, 1.261 meV
VP42.3.2 (a) 2.64×10^{-47} kg·m^2 (b) 1.627×10^{-27} kg (c) 0.127 nm
VP42.3.3 (a) 3.853×10^{-46} kg·m^2 (b) 1.690×10^{-26} kg (c) 0.1510 nm
VP42.3.4 (a) $f = 6.538 \times 10^{13}$ Hz, $\lambda = 4.585\ \mu$m (b) $f = 6.527 \times 10^{13}$ Hz, $\lambda = 4.593\ \mu$m
VP42.7.1 (a) $E_F + 0.71kT$ (b) $E_F - 2.2kT$
VP42.7.2 (a) 0.729 (b) 0.170 (c) 0.0187
VP42.7.3 (a) 8.8×10^{-19} J $= 5.5$ eV (b) 1.4×10^6 m/s (c) 9.9×10^{40} states/J $= 1.6 \times 10^{22}$ states/eV
VP42.7.4 (a) 5.4 eV (b) 5.7 eV
VP42.9.1 (a) 3.79×10^{-5} (b) 5.46×10^{-14}
VP42.9.2 (a) 1.65 (b) 2.72
VP42.9.3 (a) 0.329 eV (b) 0.351 eV
VP42.9.4 (a) 0.0405 (b) 0.0643

Bridging Problem

(a) 0.278 eV
(b) 1.74×10^{-25}

? This sculpture of a woolly mammoth, just 3.7 cm (1.5 in.) in length, was carved from a mammoth's ivory tusk by an artist who lived in southwestern Germany 35,000 years ago. It is possible to date biological specimens such as this because (i) ancient materials were more radioactive than modern ones; (ii) ancient materials were less radioactive than modern ones; (iii) biological specimens continue to take in radioactive substances after they die; (iv) biological specimens no longer take in radioactive substances after they die; (v) more than one of these.

43 Nuclear Physics

LEARNING OUTCOMES

In this chapter, you'll learn...

43.1 Some key properties of atomic nuclei, including radii, densities, spins, and magnetic moments.
43.2 How the binding energy of a nucleus depends on the numbers of protons and neutrons that it contains.
43.3 The most important ways in which unstable nuclei undergo radioactive decay.
43.4 How the decay rate of a radioactive substance depends on time.
43.5 Some of the biological hazards and medical uses of radiation.
43.6 How to analyze some important types of nuclear reactions.
43.7 What happens in a nuclear fission chain reaction, and how it can be controlled.
43.8 The nuclear reactions that allow the sun to shine.

You'll need to review...

5.5 The strong interaction.
18.3 Kinetic energy of gas molecules.
21.1 Proton and neutron.
26.4 Discharging capacitor.
37.8 Rest mass and rest energy.
39.2 Discovery of the nucleus.
40.3, 40.4 Square-well potential; tunneling.
41.4–41.6 Magnetic moments; spin-$\frac{1}{2}$ particles; central-field approximation.

Every atom contains at its center an extremely dense, positively charged *nucleus,* which is much smaller than the overall size of the atom but contains most of its total mass. In this chapter we'll look at several important general properties of nuclei and of the nuclear force that holds protons and neutrons together within a nucleus. The stability or instability of a particular nucleus is determined by the competition between the attractive nuclear force among the protons and neutrons and the repulsive electrical interactions among the protons. Unstable nuclei *decay,* transforming themselves spontaneously into other nuclei by a variety of processes. Nuclear reactions can also be induced by impact on a nucleus of a particle or another nucleus. Two classes of reactions of special interest are *fission* and *fusion*. Fission is the process that takes place within a nuclear reactor used for generating power. We could not survive without the energy released by one nearby fusion reactor, our sun.

43.1 PROPERTIES OF NUCLEI

As we described in Section 39.2, Rutherford found that the nucleus is tens of thousands of times smaller in radius than the atom itself. Since Rutherford's initial experiments, many additional scattering experiments have been performed with high-energy protons, electrons, neutrons, and alpha particles. These experiments show that we can model a nucleus as a sphere with a radius R that depends on the total number of *nucleons* (neutrons and protons) in the nucleus. This number is called the **nucleon number** A. The radii of most nuclei are represented quite well by the equation

Experimentally determined constant $= 1.2 \times 10^{-15}$ m $= 1.2$ fm

Radius of an atomic nucleus

$$R = R_0 A^{1/3} \tag{43.1}$$

Nucleon number = total number of protons and neutrons

The nucleon number A in Eq. (43.1) is also called the **mass number** because it is the nearest whole number to the mass of the nucleus measured in unified atomic mass units (u).

(The proton mass and the neutron mass are both approximately 1 u.) The best current value is

$$1\ \text{u} = 1.660539040(20) \times 10^{-27}\ \text{kg}$$

In Section 43.2 we'll discuss the masses of nuclei in more detail.

CAUTION Nuclear masses are rest masses When we speak of the masses of the proton, the neutron, and nuclei, we always mean their *rest* masses (see Section 37.7). We'll use this same convention when we discuss other subatomic particles in Chapter 44.

Nuclear Density

The volume V of a sphere is equal to $4\pi R^3/3$, so Eq. (43.1) shows that the *volume* of a nucleus is approximately proportional to A. Since the proton and the neutron both have about the same mass, the *mass* of a nucleus is also approximately proportional to the total number A of protons and neutrons. When we divide the mass by the volume to calculate the density, the factors of A cancel. Thus *all nuclei have approximately the same density.* We'll see in Section 43.2 that this observation is important in understanding nuclear structure.

EXAMPLE 43.1 Calculating nuclear properties

The most common kind of iron nucleus has mass number $A = 56$. Find the approximate radius, mass, and density of the nucleus.

IDENTIFY and SET UP Equation (43.1) tells us how the nuclear radius R depends on the mass number A. The mass of the nucleus in atomic mass units is approximately equal to the value of A, and the density ρ is mass divided by volume.

EXECUTE The radius and mass are

$$R = R_0 A^{1/3} = (1.2 \times 10^{-15}\ \text{m})(56)^{1/3}$$
$$= 4.6 \times 10^{-15}\ \text{m} = 4.6\ \text{fm}$$
$$m \approx (56\ \text{u})(1.66 \times 10^{-27}\ \text{kg/u}) = 9.3 \times 10^{-26}\ \text{kg}$$

The volume V of the nucleus (which we treat as a sphere of radius R) and its density ρ are

$$V = \tfrac{4}{3}\pi R^3 = \tfrac{4}{3}\pi R_0^3 A = \tfrac{4}{3}\pi(4.6 \times 10^{-15}\ \text{m})^3$$
$$= 4.1 \times 10^{-43}\ \text{m}^3$$
$$\rho = \frac{m}{V} \approx \frac{9.3 \times 10^{-26}\ \text{kg}}{4.1 \times 10^{-43}\ \text{m}^3} = 2.3 \times 10^{17}\ \text{kg/m}^3$$

EVALUATE As we mentioned above, *all* nuclei have approximately this same density. The density of solid iron is about 7000 kg/m^3; the iron nucleus is more than 10^{13} times as dense as iron in bulk. Such densities are also found in *neutron stars,* which are similar to gigantic nuclei made almost entirely of neutrons. A 1 cm cube of material with this density would have a mass of 2.3×10^{11} kg, or 230 million metric tons!

KEYCONCEPT The radius of a nucleus is approximately proportional to the $\frac{1}{3}$ power of the number of nucleons A that the nucleus contains. The volume is approximately proportional to A.

Nuclides and Isotopes

The building blocks of the nucleus are the proton and the neutron. In a neutral atom, there is one electron for every proton in the nucleus. We introduced these particles in Section 21.1; we'll recount the discovery of the neutron and proton in Chapter 44. The masses of these particles are

Proton: $m_p = 1.007276\ \text{u} = 1.672622 \times 10^{-27}\ \text{kg}$

Neutron: $m_n = 1.008665\ \text{u} = 1.674927 \times 10^{-27}\ \text{kg}$

Electron: $m_e = 0.000548580\ \text{u} = 9.10938 \times 10^{-31}\ \text{kg}$

The number of protons in a nucleus is the **atomic number** Z. The number of neutrons is the **neutron number** N. The nucleon number or mass number A is the sum of the number of protons Z and the number of neutrons N:

$$A = Z + N \tag{43.2}$$

A single nuclear species having specific values of both Z and N is called a **nuclide. Table 43.1** (next page) lists values of A, Z, and N for some nuclides. The electron structure of an atom, which is responsible for its chemical properties, is determined by the charge Ze of the nucleus. The table shows some nuclides that have the same number of protons Z

APPLICATION Using Isotopes to Measure Ancient Climate This sample of ice from Antarctica was deposited tens of thousands of years ago. The deeper the sample, the further in the past the ice was deposited. Most of the water molecules (H_2O) in the ice contain the oxygen isotope $^{16}_{8}O$, but a small percentage contain the heavier isotope $^{18}_{8}O$. Water molecules that contain the lighter isotope evaporate more readily, but condense less readily, than water molecules that include the heavier isotope, and these processes vary with temperature. Measuring the ratio of $^{18}_{8}O$ to $^{16}_{8}O$ in an ancient ice sample thus allows scientists to determine the average ocean temperature at the time the sample was deposited. Scientists also measure the amount of atmospheric carbon dioxide (CO_2) that was trapped in the ice when it was deposited. These observations have helped confirm the idea that high atmospheric CO_2 concentrations go hand in hand with high temperatures, a key principle for understanding 21st-century climate change (see Section 17.7).

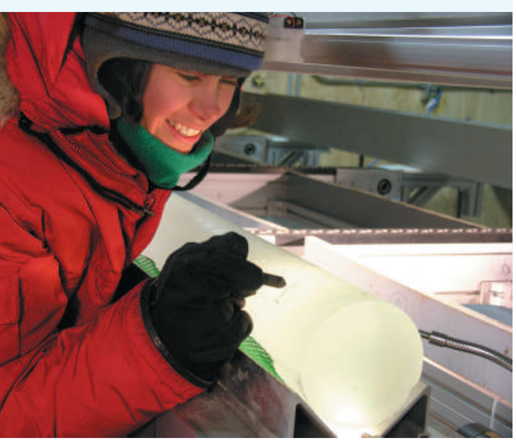


TABLE 43.1 Compositions of Some Common Nuclides

Z = atomic number (number of protons)
N = neutron number
$A = Z + N$ = mass number (total number of nucleons)

Nucleus	Z	N	$A = Z + N$
$^{1}_{1}H$	1	0	1
$^{2}_{1}H$	1	1	2
$^{4}_{2}He$	2	2	4
$^{6}_{3}Li$	3	3	6
$^{7}_{3}Li$	3	4	7
$^{9}_{4}Be$	4	5	9
$^{10}_{5}B$	5	5	10
$^{11}_{5}B$	5	6	11
$^{12}_{6}C$	6	6	12
$^{13}_{6}C$	6	7	13
$^{14}_{7}N$	7	7	14
$^{16}_{8}O$	8	8	16
$^{23}_{11}Na$	11	12	23
$^{65}_{29}Cu$	29	36	65
$^{200}_{80}Hg$	80	120	200
$^{235}_{92}U$	92	143	235
$^{238}_{92}U$	92	146	238

but a different number of neutrons N. These nuclides are called **isotopes** of that element. A familiar example is chlorine (Cl, $Z = 17$). About 76% of chlorine nuclei have $N = 18$; the other 24% have $N = 20$. Different isotopes of an element usually have slightly different physical properties such as melting and boiling temperatures and diffusion rates. The two common isotopes of uranium with $A = 235$ and 238 are usually separated industrially by taking advantage of the different diffusion rates of gaseous uranium hexafluoride (UF_6) containing the two isotopes.

Table 43.1 also shows the usual notation for individual nuclides: the symbol of the element, with a pre-subscript equal to Z and a pre-superscript equal to the mass number A. The general format for an element El is $^{A}_{Z}El$. The isotopes of chlorine mentioned above, with $A = 35$ and 37, are written $^{35}_{17}Cl$ and $^{37}_{17}Cl$ and pronounced "chlorine-35" and "chlorine-37," respectively. This name of the element determines the atomic number Z, so the pre-subscript Z is sometimes omitted, as in ^{35}Cl.

Table 43.2 gives the masses of some common atoms, including their electrons. Note that this table gives masses of *neutral* atoms (with Z electrons) rather than masses of *bare* nuclei, because it is much more difficult to measure masses of bare nuclei with high precision. The mass of a neutral carbon-12 atom is exactly 12 u; that's how the unified atomic mass unit is defined. The masses of other atoms are *approximately* equal to A atomic mass units, as we stated earlier. In fact, the atomic masses are *less* than the sum of the masses of their parts (the Z protons, the Z electrons, and the N neutrons). We'll explain this very important mass difference in the next section.

Nuclear Spins and Magnetic Moments

Like electrons, nucleons (protons and neutrons) are spin-$\frac{1}{2}$ particles with spin angular momenta given by the same equations as in Section 41.5. The magnitude of the spin angular momentum $\vec{S}$ of a nucleon is

$$S = \sqrt{\tfrac{1}{2}\left(\tfrac{1}{2} + 1\right)}\hbar = \sqrt{\tfrac{3}{4}}\hbar \tag{43.3}$$

TABLE 43.2 Neutral Atomic Masses for Some Light Nuclides

Element and Isotope	Atomic Number, Z	Neutron Number, N	Atomic Mass (u)	Mass Number, A
Hydrogen (^1_1H)	1	0	1.007825	1
Deuterium (^2_1H)	1	1	2.014102	2
Tritium (^3_1H)	1	2	3.016049	3
Helium (^3_2He)	2	1	3.016029	3
Helium (^4_2He)	2	2	4.002603	4
Lithium (^6_3Li)	3	3	6.015123	6
Lithium (^7_3Li)	3	4	7.016003	7
Beryllium (^9_4Be)	4	5	9.012183	9
Boron ($^{10}_5B$)	5	5	10.012937	10
Boron ($^{11}_5B$)	5	6	11.009305	11
Carbon ($^{12}_6C$)	6	6	12.000000	12
Carbon ($^{13}_6C$)	6	7	13.003355	13
Nitrogen ($^{14}_7N$)	7	7	14.003074	14
Nitrogen ($^{15}_7N$)	7	8	15.000109	15
Oxygen ($^{16}_8O$)	8	8	15.994915	16
Oxygen ($^{17}_8O$)	8	9	16.999132	17
Oxygen ($^{18}_8O$)	8	10	17.999160	18

Source: International Atomic Energy Agency.

and the z-component is

$$S_z = \pm \tfrac{1}{2}\hbar \tag{43.4}$$

In addition to its spin angular momentum, a nucleon may have *orbital* angular momentum $\vec{L}$ associated with its motion within the nucleus. The values of $\vec{L}$ and of its z-component L_z for a nucleon are quantized in the same way as for an electron in an atom.

The *total* angular momentum $\vec{J}$ of the nucleus is the vector sum of the individual spin and orbital angular momenta of all the nucleons. It has magnitude

$$J = \sqrt{j(j+1)}\,\hbar \tag{43.5}$$

and z-component

$$J_z = m_j\hbar \qquad (m_j = -j, -j+1, \ldots, j-1, j) \tag{43.6}$$

The total nuclear angular momentum quantum number j is usually called the *nuclear spin,* even though in general it refers to a combination of the orbital and spin angular momenta of the nucleons that make up the nucleus. When the total number of nucleons A is *even,* j is an integer; when it is *odd,* j is a half-integer. All nuclides for which both Z and N are even have $J = 0$. As we'll see, this happens because nucleons tend to form pairs with opposite spin components.

Associated with nuclear angular momentum is a *magnetic moment.* When we discussed *electron* magnetic moments in Section 41.4, we introduced the Bohr magneton $\mu_B = e\hbar/2m_e$ as a natural unit of magnetic moment. We found that the magnitude of the z-component of the electron spin magnetic moment is almost exactly equal to μ_B; that is, $|\mu_{sz}|_{\text{electron}} \approx \mu_B$. In discussing *nuclear* magnetic moments, we can define an analogous quantity, the **nuclear magneton** μ_n:

$$\mu_n = \frac{e\hbar}{2m_p} = 5.05078 \times 10^{-27}\ \text{J/T} = 3.15245 \times 10^{-8}\ \text{eV/T} \qquad \text{(nuclear magneton)} \tag{43.7}$$

The proton mass m_p is 1836 times larger than the electron mass m_e, so the nuclear magneton μ_n is 1836 times smaller than the Bohr magneton μ_B.

We might expect the magnitude of the z-component of the spin magnetic moment of the proton to be approximately μ_n. Instead, it turns out to be

$$|\mu_{sz}|_{\text{proton}} = 2.7928\mu_n \tag{43.8}$$

Even more surprising, the neutron, which has zero charge, has a spin magnetic moment; its z-component has magnitude

$$|\mu_{sz}|_{\text{neutron}} = 1.9130\mu_n \tag{43.9}$$

The proton has a positive charge; as expected, its spin magnetic moment $\vec{\boldsymbol{\mu}}$ is parallel to its spin angular momentum $\vec{\boldsymbol{S}}$. However, $\vec{\boldsymbol{\mu}}$ and $\vec{\boldsymbol{S}}$ are opposite for a neutron, as would be expected for a *negative* charge distribution. These *anomalous* magnetic moments arise because the proton and neutron aren't really fundamental particles but are made of simpler particles called *quarks*. We'll discuss quarks in some detail in Chapter 44.

The magnetic moment of an entire nucleus is typically a few nuclear magnetons. When a nucleus is placed in an external magnetic field $\vec{\boldsymbol{B}}$, there is an interaction energy $U = -\vec{\boldsymbol{\mu}} \cdot \vec{\boldsymbol{B}} = -\mu_z B$ just as with atomic magnetic moments. The components of the magnetic moment in the direction of the field μ_z are quantized, so a series of energy levels results from this interaction.

EXAMPLE 43.2 Proton spin flips

Protons are placed in a 2.30 T magnetic field that points in the positive z-direction. (a) What is the energy difference between states with the z-component of proton spin angular momentum parallel and antiparallel to the field? (b) A proton can make a transition from one of these states to the other by emitting or absorbing a photon with the appropriate energy. Find the frequency and wavelength of such a photon.

IDENTIFY and SET UP The proton is a spin-$\frac{1}{2}$ particle with a magnetic moment $\vec{\boldsymbol{\mu}}$ in the same direction as its spin $\vec{\boldsymbol{S}}$, so its energy depends on the orientation of its spin relative to an applied magnetic field $\vec{\boldsymbol{B}}$. If the z-component of $\vec{\boldsymbol{S}}$ is aligned with $\vec{\boldsymbol{B}}$, then μ_z is equal to the positive value given in Eq. (43.8). If the z-component of $\vec{\boldsymbol{S}}$ is opposite $\vec{\boldsymbol{B}}$, then μ_z is the negative of this value. The interaction energy in either case is $U = -\mu_z B$; the difference between these energies is our target variable in part (a). We find the photon frequency and wavelength by using $E = hf = hc/\lambda$.

EXECUTE (a) When the z-components of $\vec{\boldsymbol{S}}$ and $\vec{\boldsymbol{\mu}}$ are parallel to $\vec{\boldsymbol{B}}$, the interaction energy is

$$U = -|\mu_z|B = -(2.7928)(3.152 \times 10^{-8}\ \text{eV/T})(2.30\ \text{T})$$
$$= -2.025 \times 10^{-7}\ \text{eV}$$

When the z-components of $\vec{\boldsymbol{S}}$ and $\vec{\boldsymbol{\mu}}$ are antiparallel to the field, the energy is $+2.025 \times 10^{-7}$ eV. Hence the energy *difference* between the states is

$$\Delta E = 2(2.025 \times 10^{-7}\ \text{eV}) = 4.05 \times 10^{-7}\ \text{eV}$$

(b) The corresponding photon frequency and wavelength are

$$f = \frac{\Delta E}{h} = \frac{4.05 \times 10^{-7}\ \text{eV}}{4.136 \times 10^{-15}\ \text{eV}\cdot\text{s}} = 9.79 \times 10^{7}\ \text{Hz} = 97.9\ \text{MHz}$$

$$\lambda = \frac{c}{f} = \frac{3.00 \times 10^{8}\ \text{m/s}}{9.79 \times 10^{7}\ \text{s}^{-1}} = 3.06\ \text{m}$$

EVALUATE This frequency is in the middle of the FM radio band. When a hydrogen specimen is placed in a 2.30 T magnetic field and irradiated with radio waves of this frequency, proton *spin flips* can be detected by the absorption of energy from the radiation.

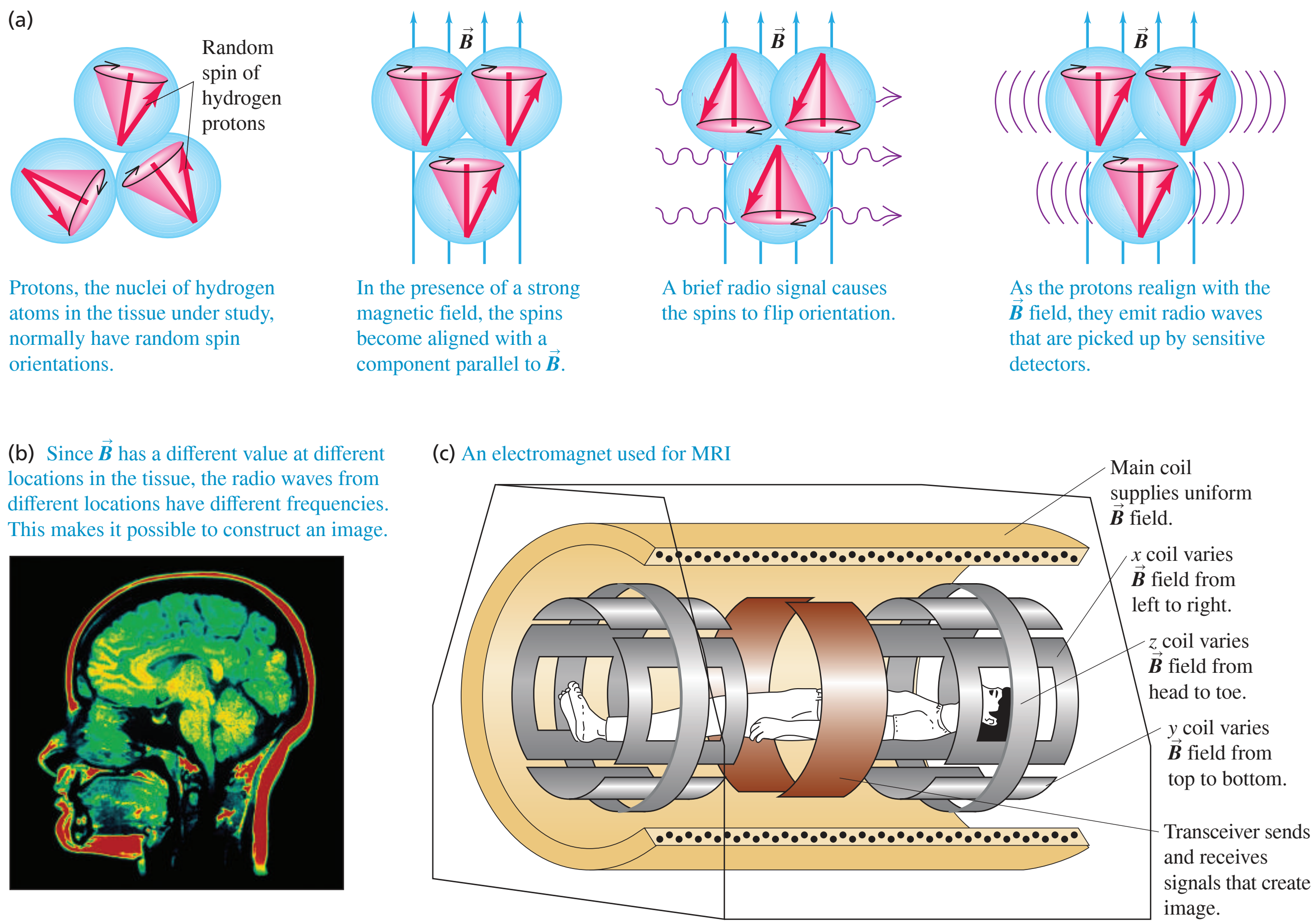
KEYCONCEPT Both the proton and the neutron have a magnetic moment $\vec{\boldsymbol{\mu}}$ associated with its spin. When placed in a magnetic field $\vec{\boldsymbol{B}}$, a proton or neutron has two states of different energy: one where the component of $\vec{\boldsymbol{\mu}}$ in the direction of $\vec{\boldsymbol{B}}$ is positive and one where it is negative.

Nuclear Magnetic Resonance and MRI

Spin-flip experiments of the sort referred to in Example 43.2 are called *nuclear magnetic resonance* (NMR). They have been carried out with many different nuclides. Frequencies and magnetic fields can be measured very precisely, so this technique permits precise measurements of nuclear magnetic moments. An elaboration of this basic idea leads to *magnetic resonance imaging* (MRI), a noninvasive medical imaging technique that discriminates among various types of body tissues on the basis of the differing environments of protons in the tissues (**Fig. 43.1**).

The magnetic moment of a nucleus is also the *source* of a magnetic field. In an atom the interaction of an electron's magnetic moment with the field of the nucleus's magnetic

Figure **43.1** Magnetic resonance imaging (MRI).



moment causes additional splittings in atomic energy levels and spectra. We called this effect *hyperfine structure* in Section 41.5. Measurements of the hyperfine structure may be used to directly determine the nuclear spin.

TEST YOUR UNDERSTANDING OF SECTION 43.1 (a) By what factor must the mass number of a nucleus increase to double its volume? (i) $\sqrt[3]{2}$; (ii) $\sqrt{2}$; (iii) 2; (iv) 4; (v) 8. (b) By what factor must the mass number increase to double the radius of the nucleus? (i) $\sqrt[3]{2}$; (ii) $\sqrt{2}$; (iii) 2; (iv) 4; (v) 8.

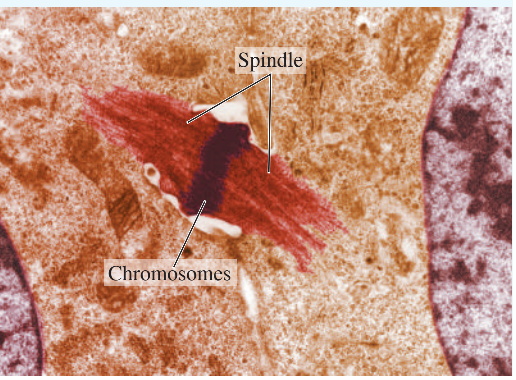
ANSWER

(a) **(iii)**, (b) **(v)** The radius R is proportional to the cube root of the mass number A, while the volume is proportional to R^3 and hence to $(A^{1/3})^3 = A$. Therefore, doubling the volume requires increasing the mass number by a factor of 2; doubling the radius implies increasing both the volume and the mass number by a factor of $2^3 = 8$.

43.2 NUCLEAR BINDING AND NUCLEAR STRUCTURE

Because energy must be added to a nucleus to separate it into its individual protons and neutrons, the total rest energy E_0 of the separated nucleons is greater than the rest energy of the nucleus. The minimum energy that must be added to separate the nucleons is called the **binding energy** E_B; it is the magnitude of the energy by which the nucleons are bound together. Thus the rest energy of the nucleus is $E_0 - E_B$. The binding energy is defined as

BIO APPLICATION Deuterium and Heavy Water Toxicity A crucial step in plant and animal cell division is the formation of a spindle, which separates the two sets of daughter chromosomes. If a plant is given only heavy water—in which one or both of the hydrogen atoms in an H_2O molecule are replaced with a deuterium atom—cell division stops and the plant stops growing. The reason is that deuterium is more massive than ordinary hydrogen, so the O–H bond in heavy water has a slightly different binding energy and heavy water has slightly different properties as a solvent. The biochemical reactions that occur during cell division are very sensitive to these solvent properties, so a spindle never forms and the cell cannot reproduce.



Binding energy of a nucleus with Z protons, N neutrons; Atomic number; Neutron number; (Speed of light in vacuum)^2 = 931.5 MeV/u; Mass of hydrogen atom; Neutron mass; Mass of neutral atom containing nucleus

$$E_B = (ZM_H + Nm_n - {}^A_ZM)c^2 \qquad (43.10)$$

Note that Eq. (43.10) does not include Zm_p, the mass of Z protons. Rather, it contains ZM_H, the mass of Z protons and Z electrons combined as Z neutral ${}^1_1H$ atoms, to balance the Z electrons included in ${}^A_ZM$, the mass of the neutral atom.

The simplest nucleus is that of hydrogen, a single proton. Next comes ${}^2_1H$, the isotope of hydrogen with mass number 2, usually called *deuterium*. Its nucleus consists of a proton and a neutron bound together to form a particle called the *deuteron*. By using values from Table 43.2 in Eq. (43.10), we find that the binding energy of the deuteron is

$$\begin{aligned} E_B &= (1.007825\text{ u} + 1.008665\text{ u} - 2.014102\text{ u})(931.5\text{ MeV/u}) \\ &= 2.224\text{ MeV} \end{aligned}$$

This much energy would be required to pull the deuteron apart into a proton and a neutron. An important measure of how tightly a nucleus is bound is the *binding energy per nucleon*, E_B/A. At $(2.224\text{ MeV})/(2\text{ nucleons}) = 1.112\text{ MeV}$ per nucleon, ${}^2_1H$ has the lowest binding energy per nucleon of all nuclides. The nuclide with the highest value of E_B/A is ${}^{62}_{28}Ni$ (see Example 43.3).

Using the equivalence of rest mass and energy (Section 37.8), we see that the mass of a nucleus is always *less* than the total mass of its nucleons by an amount $\Delta M = E_B/c^2$, called the *mass defect*. For example, the mass defect of ${}^2_1H$ is $\Delta M = E_B/c^2 = (2.224\text{ MeV})/(931.5\text{ MeV/u}) = 0.002388\text{ u}$.

PROBLEM-SOLVING STRATEGY 43.1 Nuclear Properties

IDENTIFY *the relevant concepts:* The key properties of a nucleus are its mass, radius, binding energy, mass defect, binding energy per nucleon, and angular momentum.

SET UP *the problem:* Once you have identified the target variables, assemble the equations needed to solve the problem from this section and Section 43.1.

EXECUTE *the solution:* Solve for the target variables. Binding-energy calculations that use Eq. (43.10) often involve subtracting two nearly equal quantities. To get enough precision in the difference, you may need to carry as many as nine significant figures.

EVALUATE *your answer:* It's useful to be familiar with the following benchmark magnitudes. Protons and neutrons are about 1840 times as massive as electrons. Nuclear radii are of the order of 10^{-15} m. The electric potential energy of two protons in a nucleus is roughly 10^{-13} J or 1 MeV, so nuclear interaction energies are typically a few MeV rather than a few eV as with atoms. The binding energy per nucleon is about 1% of the nucleon rest energy. (The ionization energy of the hydrogen atom is only 0.003% of the electron's rest energy.) Angular momenta are determined only by the value of $\hbar$, so they are of the same order of magnitude in both nuclei and atoms. Nuclear magnetic moments, however, are about a factor of 1000 *smaller* than those of electrons in atoms because nuclei are so much more massive than electrons.

EXAMPLE 43.3 The most strongly bound nuclide

WITH VARIATION PROBLEMS

Find the mass defect, total binding energy, and binding energy per nucleon of ${}^{62}_{28}Ni$, which has neutral atomic mass 61.928345 u.

IDENTIFY and SET UP The mass defect ΔM is the difference between the mass of the nucleus and the combined mass of its constituent nucleons. The binding energy E_B is this quantity multiplied by c^2, and the binding energy per nucleon is E_B divided by the mass number A. We use Eq. (43.10), $\Delta M = ZM_H + Nm_n - {}^A_ZM$, to determine both the mass defect and the binding energy.

EXECUTE With $Z = 28$, $M_H = 1.007825$ u, $N = A - Z = 62 - 28 = 34$, $m_n = 1.008665$ u, and ${}^A_ZM = 61.928345$ u, Eq. (43.10) gives $\Delta M = 0.585365$ u. The binding energy is then

$$E_B = (0.585365\text{ u})(931.5\text{ MeV/u}) = 545.3\text{ MeV}$$

The binding energy *per nucleon* is $E_B/A = (545.3\text{ MeV})/62$, or 8.795 MeV per nucleon. This value is the highest of any nuclide (**Fig. 43.2**).

EVALUATE Our result means that it would take a minimum of 545.3 MeV to pull a ${}^{62}_{28}\text{Ni}$ completely apart into 28 protons and 34 neutrons. The mass defect of ${}^{62}_{28}\text{Ni}$ is about 1% of the atomic (or the nuclear) mass. The binding energy is therefore about 1% of the rest energy of the nucleus, and the binding energy per nucleon is about 1% of the rest energy of a nucleon. Note that the mass defect is more than half the mass of a nucleon, which suggests how tightly bound nuclei are.

KEYCONCEPT The mass defect of a nucleus is the difference between the sum of the masses of its nucleons and the mass of the nucleus itself. We calculate this using the mass of a hydrogen atom rather than the mass of a proton, and using the mass of a neutral atom that contains the nucleus rather than the nuclear mass. To find the binding energy, multiply the mass defect by c^2; to then find the binding energy per nucleon, divide by the nucleon number A.

Figure **43.2** Approximate binding energy per nucleon as a function of mass number A (the total number of nucleons) for stable nuclides.

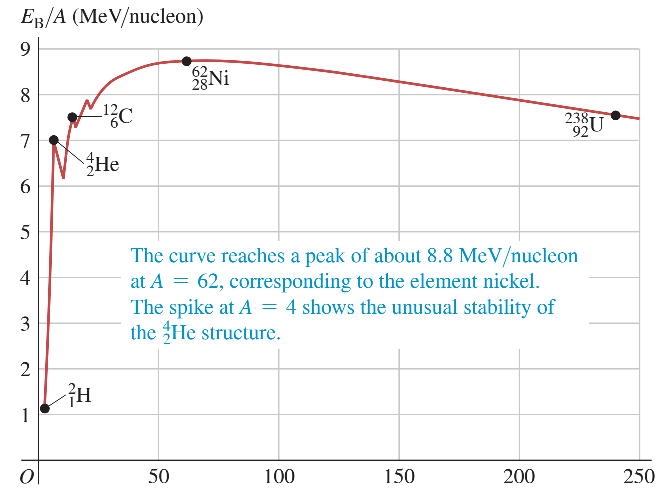


Figure 43.2 is a graph of binding energy per nucleon as a function of the mass number A. This graph shows that nearly all stable nuclides have binding energies per nucleon in the range 7–9 MeV. Note the spike at $A = 4$, showing the unusually large binding energy per nucleon of the ${}^{4}_{2}\text{He}$ nucleus (alpha particle) relative to its neighbors. To explain this curve, we must consider the interactions among the nucleons.

The Nuclear Force

The force that binds protons and neutrons together in the nucleus, despite the electrical repulsion of the protons, is an example of the *strong interaction* that we mentioned in Section 5.5. In the context of nuclear structure, this interaction is called the *nuclear force*. Here are some of its characteristics:

1. The nuclear force does not depend on charge; the binding is the same for both neutrons and protons. It also has a short range, of the order of nuclear dimensions—that is, 10^{-15} m. (If it had a longer range, a nucleus would grow by pulling in additional protons and neutrons.) Within its range, the attractive nuclear force is much stronger than the repulsive electric forces between protons; otherwise, the nucleus could never be stable. Unfortunately, there is no simple equation like Coulomb's law for how the nuclear force depends on the distance between nucleons.
2. The observation that nuclei all have about the same density and about the same binding energy per nucleon, E_B/A, shows that a particular nucleon cannot interact simultaneously through the nuclear force with *all* the other nucleons in a nucleus. (If it could, then in larger nuclei the nucleons would crowd closer together, making the nucleus denser and increasing the value of E_B/A.) Instead, a nucleon interacts with only the few other nucleons in its immediate vicinity. (This is different from electric forces; *every* proton in the nucleus exerts an electric repulsion on every other proton.) This limit on the number of interactions via the nuclear force is called *saturation*.
3. The nuclear force favors the binding of *pairs* of protons or neutrons with opposite spins and of *pairs of pairs*—that is, a pair of protons and a pair of neutrons, with each pair having opposite spins. This explains the spike in binding energy per nucleon shown in Fig. 43.2 for ${}^{4}_{2}\text{He}$ (two protons and two neutrons). We'll see other evidence for pairing effects in nuclei in the next subsection. (In Section 42.8 we described an analogous pairing that binds opposite-spin electrons in a superconductor.)

Analyzing nuclear structure is a challenge because the nature of the nuclear force is so complex and because electric forces are also involved. Even so, we can gain some insight into nuclear structure by the use of simple models. We'll discuss briefly two rather different but successful models, the *liquid-drop model* and the *shell model*.

The Liquid-Drop Model

The **liquid-drop model,** first proposed in 1928 by the Russian physicist George Gamow and later expanded by Niels Bohr, is suggested by the observation that all nuclei have nearly the same density. The individual nucleons are analogous to molecules of a liquid, held together by short-range interactions and surface-tension effects. We can use this simple picture to derive a formula for the estimated total binding energy of a nucleus. We'll include five contributions:

1. An individual nucleon interacts with only a few of its nearest neighbors (saturation). If every nucleon in a nucleus were surrounded by nearest neighbors, the binding energy per nucleon, E_B/A, would have the same constant value for all nucleons. If we call this constant C_1, then the binding energy of a nucleus with A nucleons would be $C_1 A$.
2. The nucleons on the surface of the nucleus have no neighbors outside the surface and so are less tightly bound than those in the interior. This decrease in the binding energy gives a *negative* energy term proportional to the surface area $4\pi R^2$. Because R is proportional to $A^{1/3}$, this term is proportional to $A^{2/3}$; we write it as $-C_2 A^{2/3}$, where C_2 is another constant.
3. Every one of the Z protons repels every one of the $(Z - 1)$ other protons. The total repulsive electric potential energy is proportional to $Z(Z - 1)$ and inversely proportional to the radius R and thus to $A^{1/3}$. This energy term is negative because the nucleons are less tightly bound than they would be without the electrical repulsion. We write this correction as $-C_3 Z(Z - 1)/A^{1/3}$.
4. Observations show that nuclei are most tightly bound if N is close to Z for small A and N is greater than Z (but not too much greater) for larger A. We need a negative energy term corresponding to the difference $|N - Z|$. The best agreement with observed binding energies is obtained if this term is proportional to $(N - Z)^2/A$. If we use $N = A - Z$ to express this energy in terms of A and Z, this correction is $-C_4(A - 2Z)^2/A$.
5. Finally, the nuclear force favors *pairing* of protons and of neutrons. This energy term is positive (more binding) if both Z and N are even, negative (less binding) if both Z and N are odd, and zero otherwise. The best fit to the data occurs with the form $\pm C_5 A^{-4/3}$ for this term.

   The total estimated binding energy E_B is the sum of these five terms:

$$E_B = C_1 A - C_2 A^{2/3} - C_3 \frac{Z(Z-1)}{A^{1/3}} - C_4 \frac{(A-2Z)^2}{A} \pm C_5 A^{-4/3} \quad (43.11)$$

(nuclear binding energy)

The constants C_1, C_2, C_3, C_4, and C_5, chosen to make this formula best fit the observed binding energies of nuclides, are

$$\begin{aligned} C_1 &= 15.75 \text{ MeV} \\ C_2 &= 17.80 \text{ MeV} \\ C_3 &= 0.7100 \text{ MeV} \\ C_4 &= 23.69 \text{ MeV} \\ C_5 &= 39 \text{ MeV} \end{aligned}$$

The constant C_1 is the binding energy per nucleon due to the saturated nuclear force. This energy is almost 16 MeV per nucleon, about double the *total* binding energy per nucleon in most nuclides. The other terms in Eq. (43.11) are responsible for the difference. The negative terms with C_3 and C_4 become more important for larger nuclides that have more protons and have A greater than $2Z$; that explains why the binding energy per nucleon gradually decreases beyond $A = 62$, as Fig. 43.2 shows.

If we use Eq. (43.11) to estimate the binding energy E_B, we can solve Eq. (43.10) to use it to estimate the mass of any neutral atom:

$$ {}^{A}_{Z}M = ZM_{H} + Nm_{n} - \frac{E_B}{c^2} \qquad \text{(semiempirical mass formula)} \qquad (43.12)$$

Equation (43.12) is called the *semiempirical mass formula*. The name is apt; the equation is *empirical* in the sense that the C's have to be determined empirically (experimentally), yet it does have a sound theoretical basis.

EXAMPLE 43.4 Estimating binding energy and mass

WITH VARIATION PROBLEMS

For the nuclide ${}^{62}_{28}\mathrm{Ni}$ of Example 43.3, (a) calculate the five terms in the binding energy and the total estimated binding energy, and (b) find the neutral atomic mass using the semiempirical mass formula.

IDENTIFY and SET UP We use the liquid-drop model of the nucleus and its five contributions to the binding energy, as given by Eq. (43.11), to calculate the total binding energy E_B. We then use Eq. (43.12) to find the neutral atomic mass ${}^{62}_{28}M$.

EXECUTE (a) With $Z = 28$, $A = 62$, and $N = 34$, the five terms in Eq. (43.11) are

1. $C_1A = (15.75\text{ MeV})(62) = 976.5\text{ MeV}$
2. $-C_2A^{2/3} = -(17.80\text{ MeV})(62)^{2/3} = -278.8\text{ MeV}$
3. $-C_3\dfrac{Z(Z-1)}{A^{1/3}} = -(0.7100\text{ MeV})\dfrac{(28)(27)}{(62)^{1/3}} = -135.6\text{ MeV}$
4. $-C_4\dfrac{(A-2Z)^2}{A} = -(23.69\text{ MeV})\dfrac{(62-56)^2}{62} = -13.8\text{ MeV}$
5. $+C_5A^{-4/3} = (39\text{ MeV})(62)^{-4/3} = 0.2\text{ MeV}$

The pairing correction (term 5) is by far the smallest of all the terms; it is positive because both Z and N are even. The sum of all five terms is the total estimated binding energy, $E_B = 548.5$ MeV.

(b) We use $E_B = 548.5$ MeV in Eq. (43.12):

$$ {}^{62}_{28}M = 28(1.007825\text{ u}) + 34(1.008665\text{ u}) - \frac{548.5\text{ MeV}}{931.5\text{ MeV/u}}$$
$$ = 61.925\text{ u}$$

EVALUATE The binding energy of ${}^{62}_{28}\mathrm{Ni}$ calculated in part (a) is only about 0.6% larger than the true value of 545.3 MeV found in Example 43.3, and the mass calculated in part (b) is only about 0.005% smaller than the measured value of 61.928345 u. The semiempirical mass formula can be quite accurate!

KEYCONCEPT You can use the liquid-drop model and its associated semiempirical mass formula to calculate the binding energy and mass of a nuclide.

The liquid-drop model and the mass formula derived from it are quite successful in correlating nuclear masses, and we'll see later that they help in understanding decay processes of unstable nuclides. Other aspects of nuclei, such as angular momentum and excited states, are better approached with different models.

The Shell Model

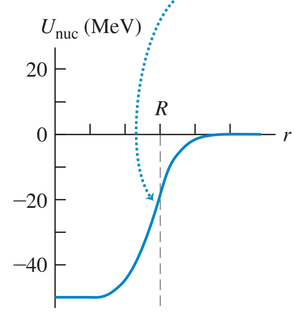
The **shell model** of nuclear structure is analogous to the central-field approximation in atomic physics (see Section 41.6). We picture each nucleon as moving in a potential that represents the averaged-out effect of all the other nucleons. Although this is a very simplified model, in several respects it works out quite well.

The potential-energy function for the nuclear force is the same for protons as for neutrons. **Figure 43.3a** (next page) shows a reasonable assumption for the shape of this function: a spherical version of the square-well potential we discussed in Section 40.3. The function is somewhat rounded because the nucleus doesn't have a sharply defined surface at radius R. For a proton there is an additional electric potential energy due to interactions with the other $Z - 1$ protons, which we collectively treat as a uniformly charged sphere of radius R and total charge $(Z - 1)e$. Figure 43.3b shows the nuclear, electric, and total potential energies for a proton as functions of the distance r from the center of the nucleus.

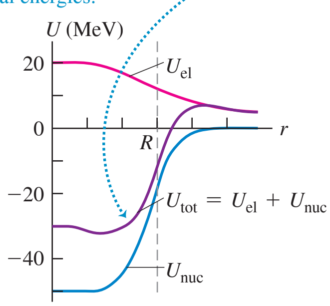
In principle, we could solve the Schrödinger equation for a proton or neutron moving in such a potential. For any spherically symmetric potential energy, the angular-momentum states are the same as for the electrons in the central-field approximation in atomic physics.

Figure 43.3 Approximate potential-energy functions for a nucleon in a nucleus. The approximate nuclear radius is R.

(a) The potential energy U_{nuc} due to the nuclear force is the same for protons and neutrons. For neutrons, it is the *total* potential energy.



(b) For protons, the total potential energy U_{tot} is the sum of the nuclear (U_{nuc}) and electric (U_{el}) potential energies.



In particular, we can use the concept of *filled shells and subshells* and their relationship to stability. As we saw in Section 41.6, the exclusion principle forbids more than one electron from occupying any given quantum-mechanical state in a multielectron atom. This explains why the values $Z = 2$, 10, 18, 36, 54, and 86 (the atomic numbers of the noble gases) correspond to atoms with particularly stable electron arrangements.

A comparable effect occurs in nuclear structure. Like electrons, protons and neutrons are spin-$\frac{1}{2}$ particles. So no more than one proton can be in any given quantum-mechanical state in a nucleus, and likewise for neutrons. Just as for electrons in atoms, there are certain numbers of protons *or* of neutrons, called *magic numbers,* that correspond to particularly stable nuclei—that is, nuclei with particularly high binding energies. The magic numbers are 2, 8, 20, 28, 50, 82, and 126. These numbers are different from those for electrons in atoms because the potential-energy function is different and the nuclear spin-orbit interaction is much stronger and of opposite sign than in atoms. So nuclear subshells fill up in a different order from those for electrons in an atom. Nuclides in which Z is a magic number tend to have an above-average number of stable isotopes. (Nuclides with $Z = 126$ have not been observed in nature.) There are several *doubly magic* nuclides for which both Z and N are magic, including

$$^{4}_{2}\mathrm{He} \qquad ^{16}_{8}\mathrm{O} \qquad ^{40}_{20}\mathrm{Ca} \qquad ^{48}_{20}\mathrm{Ca} \qquad ^{208}_{82}\mathrm{Pb}$$

All these nuclides have substantially higher binding energy per nucleon than do nuclides with neighboring values of N or Z. They also all have zero nuclear spin. The magic numbers correspond to filled-shell or filled-subshell configurations of nucleon energy levels with a relatively large jump in energy to the next allowed level.

TEST YOUR UNDERSTANDING OF SECTION 43.2 Rank the following nuclei in order from largest to smallest value of the binding energy per nucleon. (i) $^{4}_{2}$He; (ii) $^{52}_{24}$Cr; (iii) $^{152}_{62}$Sm; (iv) $^{200}_{80}$Hg; (v) $^{252}_{92}$Cf.

ANSWER

(ii), (iii), (iv), (v), (i) You can find the answers by inspecting Fig. 43.2. The binding energy per nucleon is lowest for very light nuclei such as $^{4}_{2}$He, is greatest around $A = 60$, and then decreases with increasing A.

43.3 NUCLEAR STABILITY AND RADIOACTIVITY

Among about 2500 known nuclides, fewer than 300 are stable. The others are unstable structures that decay to form other nuclides by emitting particles and electromagnetic radiation, a process called **radioactivity.** The time scale of these decay processes ranges from a small fraction of a microsecond to billions of years. The *stable* nuclides are shown by dots on the graph in **Fig. 43.4**, where the neutron number N and proton number (or atomic number) Z for each nuclide are plotted. Such a chart is called a *Segrè chart,* after its inventor, the Italian-American physicist Emilio Segrè (1905–1989).

Each blue line in Fig. 43.4 represents a specific value of the mass number $A = Z + N$. Most lines of constant A pass through only one or two stable nuclides; that is, there is usually a very narrow range of stability for a given mass number. The lines at $A = 20$, 40, 60, and 80 are examples. In four exceptional cases ($A = 94$, 124, 130, and 136), these lines pass through *three* stable nuclides.

Only four stable nuclides have both odd Z and odd N:

$$^{2}_{1}\mathrm{H} \qquad ^{6}_{3}\mathrm{Li} \qquad ^{10}_{5}\mathrm{B} \qquad ^{14}_{7}\mathrm{N}$$

These are called *odd-odd nuclides*. The absence of other odd-odd nuclides shows the importance of pairing in adding to nuclear stability. Also, there is *no* stable nuclide with $A = 5$ or $A = 8$. The doubly magic $^{4}_{2}$He nucleus, with a pair of protons and a pair of neutrons, has no interest in accepting a fifth particle into its structure. Collections of eight nucleons decay to smaller nuclides, with a $^{8}_{4}$Be nucleus immediately splitting into two $^{4}_{2}$He nuclei.

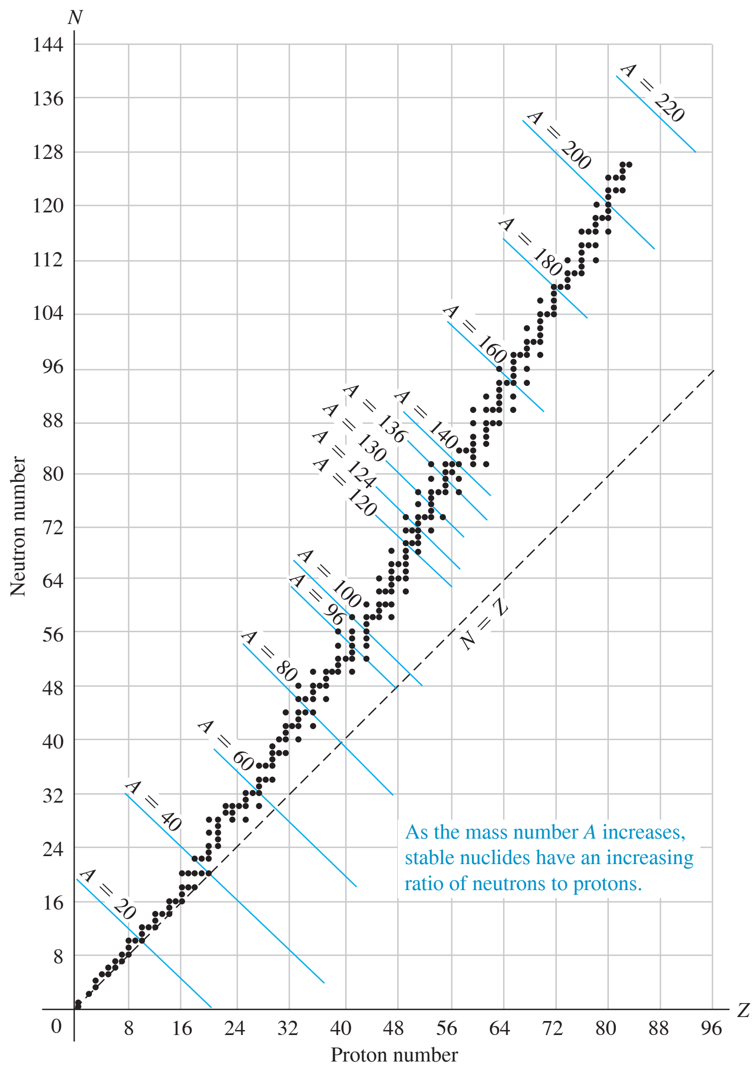
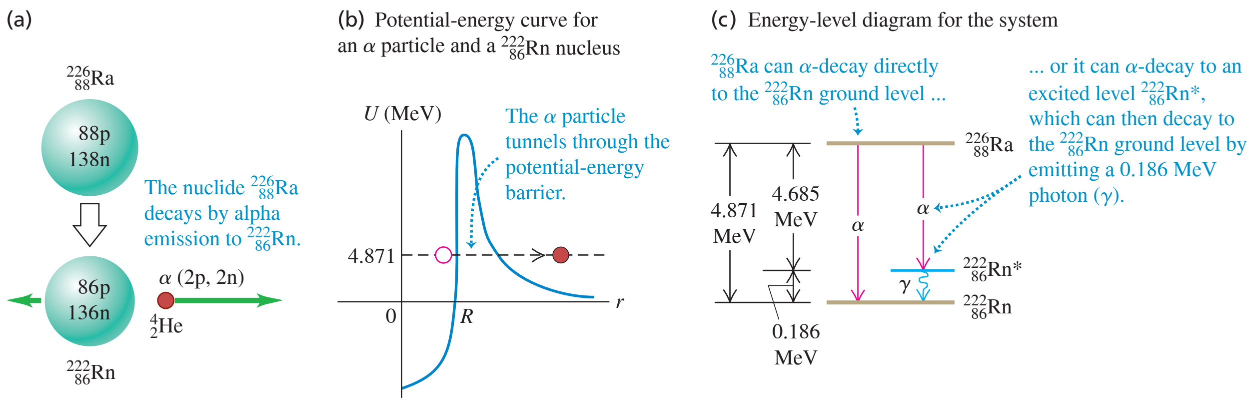


Figure 43.4 Segrè chart showing neutron number and proton number for stable nuclides.

The stable nuclides define a rather narrow region on the Segrè chart. For low mass numbers, the numbers of protons and neutrons are approximately equal, $N \approx Z$. The ratio N/Z increases gradually with A, up to about 1.6 at large mass numbers, because of the increasing influence of the electrical repulsion of the protons. Points to the right of the stability region represent nuclides that have too many protons relative to neutrons. In these cases, repulsion wins, and the nucleus comes apart. To the left are nuclides with too many neutrons relative to protons. In these cases the energy associated with the neutrons is out of balance with that associated with the protons, and the nuclides decay in a process that converts neutrons to protons. The graph also shows that no nuclide with $A > 209$ or $Z > 83$ is stable. A nucleus is unstable if it is too big. Note that there is no stable nuclide with $Z = 43$ (technetium) or 61 (promethium).

Nearly 90% of the 2500 known nuclides are *radioactive;* they are not stable but decay into other nuclides. Many of these radioactive nuclides occur in nature. For example, you are very slightly radioactive because of unstable nuclides such as carbon-14 ($^{14}_{6}C$) and potassium-40 ($^{40}_{19}K$) that are present throughout your body. The study of radioactivity began in 1896, one year after Wilhelm Röntgen discovered x rays (Section 36.6). Henri Becquerel discovered a radiation from uranium salts that seemed similar to x rays. Investigation in the following two decades by Marie and Pierre Curie, Ernest Rutherford, and many others revealed that the emissions consist of positively and negatively charged particles and neutral rays. These particles were given the names *alpha, beta,* and *gamma* because of their differing penetration characteristics.

Figure **43.5** Alpha decay of the unstable radium nuclide $^{226}_{88}\mathrm{Ra}$. The alpha particles used in the Rutherford scattering experiment (Section 39.2) were emitted by this nuclide.



Alpha Decay

When unstable nuclides decay into different nuclides, they usually emit alpha (α) or beta (β) particles. An **alpha particle** is a $^{4}_{2}\mathrm{He}$ nucleus, two protons and two neutrons bound together, with total spin zero. Alpha decay occurs principally with nuclei that are too large to be stable. The nucleus gets rid of the excess nucleons in the form of a $^{4}_{2}\mathrm{He}$ nucleus because this is a particularly stable grouping (see Fig. 43.2). When a nucleus emits an alpha particle, its N and Z values each decrease by 2 and A decreases by 4, moving it closer to stable territory on the Segrè chart.

Figure 43.5a shows the alpha decay of radium-226 ($^{226}_{88}\mathrm{Ra}$). Spontaneous alpha decay of this kind can occur only if energy is released in the process; this released energy goes into the kinetic energy of the emitted α particle and of the nucleus that remains, called the *daughter nucleus*. (For the decay shown in Fig. 43.5a, the daughter nucleus is radon-222, $^{222}_{86}\mathrm{Rn}$.) The original nucleus (in this case, $^{226}_{88}\mathrm{Ra}$) is called the *parent nucleus*. You can use mass-energy conservation to show that

alpha decay is possible whenever the mass of the original neutral atom is greater than the sum of the masses of the final neutral atom and a neutral $^{4}_{2}\mathrm{He}$ atom.

In alpha decay, the α particle tunnels through a potential-energy barrier, as Fig. 43.5b shows. You may want to review the discussion of tunneling in Section 40.4.

Alpha particles are always emitted with definite kinetic energies, determined by conservation of momentum and energy in the alpha-decay process. As Fig. 43.5c shows, an α particle emitted in the decay of $^{226}_{88}\mathrm{Ra}$ can have either of *two* possible energies, depending on the energy level of the $^{222}_{86}\mathrm{Rn}$ daughter nucleus just after the decay. (Later in this section we'll discuss the photon-emission process shown in Fig. 43.5c.)

Alpha particles are emitted at high speeds, typically a few percent of the speed of light (see the following example). Nonetheless, because of their charge and mass, alpha particles can travel only several centimeters in air, or a few tenths or hundredths of a millimeter through solids, before they are brought to rest by collisions.

EXAMPLE 43.5 Alpha decay of radium — WITH VARIATION PROBLEMS

Show that the α-emission process $^{226}_{88}\mathrm{Ra} \rightarrow {}^{222}_{86}\mathrm{Rn} + {}^{4}_{2}\mathrm{He}$ (Fig. 43.5a) is energetically possible, and calculate the kinetic energy of the emitted α particle. The neutral atomic masses are 226.025410 u for $^{226}_{88}\mathrm{Ra}$, 222.017578 u for $^{222}_{86}\mathrm{Rn}$, and 4.002603 u for $^{4}_{2}\mathrm{He}$.

IDENTIFY and SET UP Alpha emission is possible if the mass of the $^{226}_{88}\mathrm{Ra}$ atom is greater than the sum of the atomic masses of $^{222}_{86}\mathrm{Rn}$ and $^{4}_{2}\mathrm{He}$. The mass difference between the initial radium atom and the final radon and helium atoms corresponds (through $E = mc^2$) to the energy E released in the decay. Because momentum is conserved as well as energy, *both* the alpha particle and the $^{222}_{86}\mathrm{Rn}$ atom are in motion after the decay; we'll have to account for this in determining the kinetic energy of the alpha particle.

EXECUTE The difference in mass between the original nucleus and the decay products is

$$226.025410 \text{ u} - (222.017578 \text{ u} + 4.002603 \text{ u}) = +0.005229 \text{ u}$$

Since this is positive, α decay is energetically possible. The energy equivalent of this mass difference is

$$E = (0.005229 \text{ u})(931.5 \text{ MeV/u}) = 4.871 \text{ MeV}$$

In this process the ${}^{222}_{86}\text{Rn}$ nucleus is produced in its ground level (Fig. 43.5c). Thus we expect the decay products to emerge with total kinetic energy 4.871 MeV. Momentum is also conserved; if the parent ${}^{226}_{88}\text{Ra}$ nucleus is at rest, the daughter ${}^{222}_{86}\text{Rn}$ nucleus and the α particle have momenta of equal magnitude p but opposite direction. Kinetic energy is $K = \frac{1}{2}mv^2 = p^2/2m$: Since p is the same for the two particles, the kinetic energy divides inversely as their masses. Hence the α particle gets $222/(222 + 4)$ of the total, or 4.78 MeV.

EVALUATE Experiment shows that ${}^{226}_{88}\text{Ra}$ does emit α particles with a kinetic energy of 4.78 MeV. Check your results by verifying that the alpha particle and the ${}^{222}_{86}\text{Rn}$ nucleus produced in the decay have the same magnitude of momentum $p = mv$. You can calculate the speed v of each of the decay products from its respective kinetic energy [note that the ${}^{222}_{86}\text{Rn}$ nucleus gets $4/(222 + 4)$ of the 4.871 MeV released]. You'll find that the alpha particle moves at $0.0506c = 1.52 \times 10^7$ m/s; if momentum is conserved, you should find that the ${}^{222}_{86}\text{Rn}$ nucleus moves $\frac{4}{222}$ as fast. Does it?

KEYCONCEPT In alpha decay, a large, unstable nucleus with Z protons and N neutrons emits an alpha particle (a ${}^{4}_{2}\text{He}$ nucleus). The daughter nucleus has $Z - 2$ protons and $N - 2$ neutrons. The combined mass of the decay products is less than the mass of the parent; the lost rest energy goes into kinetic energy of the decay products.

Beta Decay

There are three different simple types of *beta decay: beta-minus, beta-plus,* and *electron capture*. A **beta-minus particle** (β^-) is an electron. There are no electrons in the nucleus waiting to be emitted; instead, emission of a β^- involves *transformation* of a neutron into a proton, an electron, and a third particle called an *antineutrino.* In fact, if you freed a neutron from a nucleus, it would decay into a proton, an electron, and an antineutrino in an average time of about 15 minutes.

Beta particles can be identified and their speeds can be measured with techniques that are similar to the Thomson e/m experiment we described in Section 27.5. The speeds of beta particles range up to 0.9995 of the speed of light, so their motion is highly relativistic. They are emitted with a continuous spectrum of energies. This would not be possible if the only two particles were the β^- and the recoiling nucleus, since energy and momentum conservation would then require a definite speed for the β^-. Thus there must be a *third* particle involved. From conservation of charge, it must be neutral, and from conservation of angular momentum, it must be a spin-$\frac{1}{2}$ particle.

This third particle is an antineutrino, the *antiparticle* of a **neutrino.** The symbol for a neutrino is ν_e (the Greek letter nu). Both the neutrino and the antineutrino have zero charge and very small mass and therefore produce very little observable effect when passing through matter. Both evaded detection until 1953, when Frederick Reines and Clyde Cowan succeeded in observing the antineutrino directly. We now know that there are at least three varieties of neutrinos, each with its corresponding antineutrino; one is associated with beta decay and the other two are associated with the decay of two unstable particles, the muon and the tau particle. We'll discuss these particles in more detail in Chapter 44. The antineutrino that is emitted in β^- decay is denoted as $\bar{\nu}_e$. The basic process of β^- decay is

$$\text{n} \rightarrow \text{p} + \beta^- + \bar{\nu}_e \tag{43.13}$$

Beta-minus decay usually occurs with nuclides for which the neutron-to-proton ratio N/Z is too large for stability. In β^- decay, N decreases by 1, Z increases by 1, and A doesn't change. You can use mass-energy conservation to show that

beta-minus decay can occur whenever the mass of the original neutral atom is larger than that of the final atom.

EXAMPLE 43.6 Why cobalt-60 is a beta-minus emitter

WITH VARIATION PROBLEMS

The nuclide ${}^{60}_{27}\mathrm{Co}$, an odd-odd unstable nucleus, is used in medical and industrial applications of radiation. Show that it is unstable relative to β^- decay. The atomic masses you need are 59.933817 u for ${}^{60}_{27}\mathrm{Co}$ and 59.930786 u for ${}^{60}_{28}\mathrm{Ni}$.

IDENTIFY and SET UP Beta-minus decay is possible if the mass of the original neutral atom is greater than that of the final atom. We must first identify the nuclide that will result if ${}^{60}_{27}\mathrm{Co}$ undergoes β^- decay and then compare its neutral atomic mass to that of ${}^{60}_{27}\mathrm{Co}$.

EXECUTE In the presumed β^- decay of ${}^{60}_{27}\mathrm{Co}$, Z increases by 1 from 27 to 28 and A remains at 60, so the final nuclide is ${}^{60}_{28}\mathrm{Ni}$. The neutral atomic mass of ${}^{60}_{27}\mathrm{Co}$ is greater than that of ${}^{60}_{28}\mathrm{Ni}$ by 0.003031 u, so β^- decay *can* occur.

EVALUATE With three decay products in β^- decay—the ${}^{60}_{28}\mathrm{Ni}$ nucleus, the electron, and the antineutrino—the energy can be shared in many different ways that are consistent with conservation of energy and momentum. It's impossible to predict precisely how the energy will be shared for the decay of a particular ${}^{60}_{27}\mathrm{Co}$ nucleus. By contrast, in alpha decay there are just two decay products, and their energies and momenta are determined uniquely (see Example 43.5).

KEYCONCEPT In beta-minus decay, an unstable nucleus with Z protons and N neutrons decays by converting a neutron into a proton and emitting an electron and an antineutrino. The daughter nucleus has $Z + 1$ protons and $N - 1$ neutrons. This is possible only if the neutral parent atom has a greater mass than the neutral daughter atom.

We have noted that β^- decay occurs with nuclides that have too large a neutron-to-proton ratio N/Z. Nuclides for which N/Z is too *small* for stability can emit a *positron,* the electron's antiparticle, which is identical to the electron but with positive charge. (We'll discuss the positron in more detail in Chapter 44.) The basic process, called *beta-plus decay* (β^+), is

$$\mathrm{p} \rightarrow \mathrm{n} + \beta^+ + \nu_\mathrm{e} \qquad (43.14)$$

where β^+ is a positron and ν_e is the electron neutrino.

Beta-plus decay can occur whenever the mass of the original neutral atom is at least two electron masses larger than that of the final atom.

You can show this by using mass-energy conservation.

The third type of beta decay is *electron capture*. There are a few nuclides for which β^+ emission is not energetically possible but in which an orbital electron (usually in the innermost K shell) can combine with a proton in the nucleus to form a neutron and a neutrino. The neutron remains in the nucleus and the neutrino is emitted. The basic process is

$$\mathrm{p} + \beta^- \rightarrow \mathrm{n} + \nu_\mathrm{e} \qquad (43.15)$$

You can use mass-energy conservation to show that

electron capture can occur whenever the mass of the original neutral atom is larger than that of the final atom.

In all types of beta decay, A remains constant. However, in beta-plus decay and electron capture, N increases by 1 and Z decreases by 1 as the neutron-to-proton ratio increases toward a more stable value. The reaction of Eq. (43.15) also helps explain the formation of a neutron star, mentioned in Example 43.1.

CAUTION **Beta decay inside and outside nuclei** The beta-decay reactions given by Eqs. (43.13), (43.14), and (43.15) occur *within* a nucleus. Although the decay of a neutron outside the nucleus proceeds through the reaction of Eq. (43.13), the reaction of Eq. (43.14) is forbidden by mass-energy conservation for a proton outside the nucleus. The reaction of Eq. (43.15) can occur outside the nucleus only with the addition of some extra energy, as in a collision.

EXAMPLE 43.7 Why cobalt-57 is not a beta-plus emitter

WITH VARIATION PROBLEMS

The nuclide ${}^{57}_{27}\text{Co}$ is an odd-even unstable nucleus. Show that it cannot undergo β^+ decay, but that it *can* decay by electron capture. The atomic masses you need are 56.936291 u for ${}^{57}_{27}\text{Co}$ and 56.935394 u for ${}^{57}_{26}\text{Fe}$.

IDENTIFY and SET UP Beta-plus decay is possible if the mass of the original neutral atom is greater than that of the final atom plus two electron masses (0.001097 u). Electron capture is possible if the mass of the original atom is greater than that of the final atom. We must first identify the nuclide that will result if ${}^{57}_{27}\text{Co}$ undergoes β^+ decay or electron capture and then find the corresponding mass difference.

EXECUTE The original nuclide is ${}^{57}_{27}\text{Co}$. In both the presumed β^+ decay and electron capture, Z decreases by 1 from 27 to 26, and A remains at 57, so the final nuclide is ${}^{57}_{26}\text{Fe}$. Its mass is less than that of ${}^{57}_{27}\text{Co}$ by 0.000897 u, a value smaller than 0.001097 u (two electron masses), so β^+ decay *cannot* occur. However, the mass of the original atom is greater than the mass of the final atom, so electron capture *can* occur.

EVALUATE In electron capture there are just two decay products, the final nucleus and the emitted neutrino. As in alpha decay (Example 43.5) but unlike in β^- decay (Example 43.6), the decay products of electron capture have unique energies and momenta. In Section 43.4 we'll see how to relate the probability that electron capture will occur to the *half-life* of this nuclide.

KEYCONCEPT In beta-plus decay, an unstable nucleus with Z protons and N neutrons decays by converting a proton into a neutron and emitting a positron and a neutrino. The daughter nucleus has $Z - 1$ protons and $N + 1$ neutrons. This is possible only if the neutral parent atom has a greater mass than the neutral daughter atom plus two electrons. If this is not the case, electron capture may be possible.

Gamma Decay

The energy of internal motion of a nucleus is quantized. A typical nucleus has a set of allowed energy levels, including a *ground state* (state of lowest energy) and several *excited states*. Because of the great strength of nuclear interactions, excitation energies of nuclei are typically of the order of 1 MeV, compared with a few eV for atomic energy levels. In ordinary physical and chemical transformations the nucleus always remains in its ground state. When a nucleus is placed in an excited state, either by bombardment with high-energy particles or by a radioactive transformation, it can decay to the ground state by emission of one or more photons called **gamma rays** or *gamma-ray photons,* with typical energies of 10 keV to 5 MeV. This process is called *gamma* (γ) *decay*. For example, alpha particles emitted from ${}^{226}\text{Ra}$ have two possible kinetic energies, either 4.784 MeV or 4.602 MeV. Including the recoil energy of the resulting ${}^{222}\text{Rn}$ nucleus, these correspond to a total released energy of 4.871 MeV or 4.685 MeV, respectively (see Fig. 43.5c). When an alpha particle with the smaller energy is emitted, the ${}^{222}\text{Rn}$ nucleus is left in an excited state. It then decays to its ground state by emitting a gamma-ray photon with energy $(4.871 - 4.685)\text{ MeV} = 0.186\text{ MeV}$.

CAUTION Comparing α, β, and γ decays In both α and β decay, the Z value of a nucleus changes and the nucleus of one element becomes the nucleus of a different element. In γ decay, the element does *not* change; the nucleus merely goes from an excited state to a less excited state. Remember that in each of these decays, the initial unstable nucleus does not simply disappear; it's just replaced by a different one (or, in the case of γ decay, a less-excited version of the initial nucleus).

Radioactive Decay Series

When a radioactive nucleus decays, the daughter nucleus may also be unstable. In this case a *series* of successive decays occurs until a stable configuration is reached. One of the most abundant radioactive nuclides found on earth is the uranium isotope ${}^{238}\text{U}$, which undergoes a series of 14 decays, including eight α emissions and six β^- emissions, terminating at a stable isotope of lead, ${}^{206}\text{Pb}$ (**Fig. 43.6**).

Radioactive decay series can be represented on a Segrè chart, as in **Fig. 43.7** (next page). The neutron number N is plotted vertically, and the atomic number Z is plotted horizontally. In alpha emission, both N and Z decrease by 2. In β^- emission, N decreases by 1 and Z increases by 1. The decays can also be represented in equation form; the first two decays in the series are written as

$$ {}^{238}\text{U} \rightarrow {}^{234}\text{Th} + \alpha \qquad \text{and} \qquad {}^{234}\text{Th} \rightarrow {}^{234}\text{Pa} + \beta^- + \bar{\nu}_e $$

Figure 43.6 This earthquake damage to a road was caused in part by the radioactive decay of ${}^{238}\text{U}$ in the earth's interior. These decays release energy that helps produce convection currents in the earth's interior. Such currents drive the motions of the earth's crust, including the sudden sharp motions that we call earthquakes. Other radioactive nuclides whose decays heat our planet's interior include ${}^{235}\text{U}$, ${}^{232}\text{Th}$, and ${}^{40}\text{K}$.

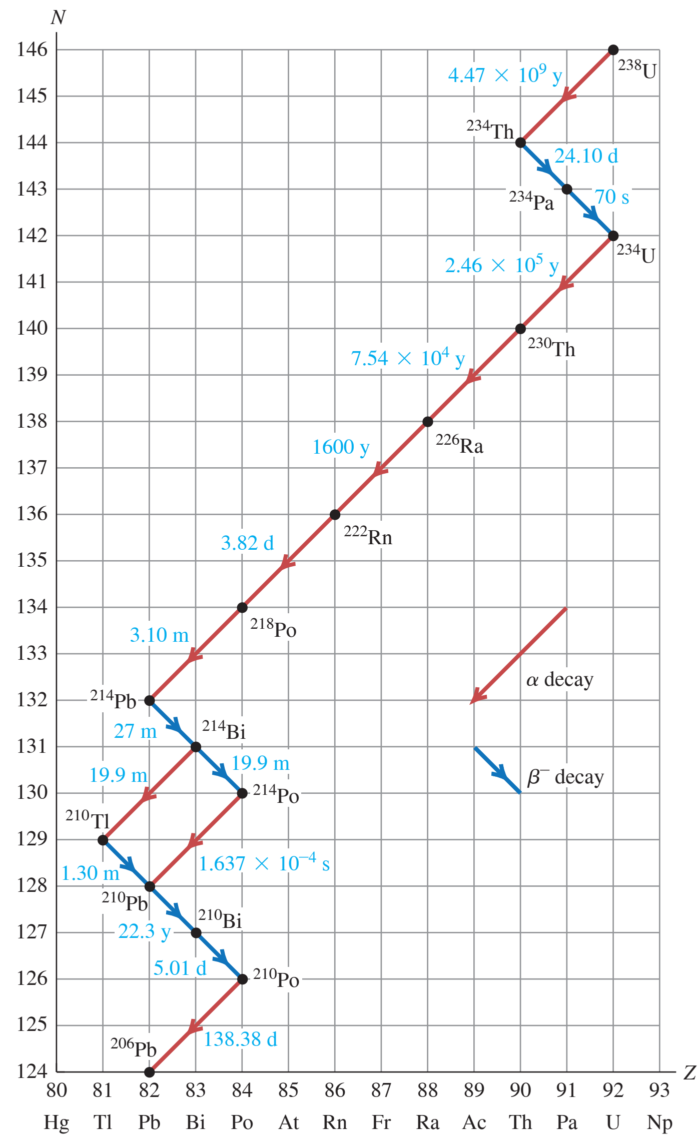


Figure **43.7** Segrè chart showing the uranium ^{238}U decay series, terminating with the stable nuclide ^{206}Pb. The times are half-lives (discussed in the next section), given in years (y), days (d), hours (h), minutes (m), or seconds (s).

or more briefly as

$$^{238}\text{U} \xrightarrow{\alpha} {}^{234}\text{Th} \qquad \text{and} \qquad {}^{234}\text{Th} \xrightarrow{\beta^-} {}^{234}\text{Pa}$$

In the second process, the beta decay leaves the daughter nucleus ^{234}Pa in an excited state, from which it decays to the ground state by emitting a gamma-ray photon. An excited state is denoted by an asterisk, so we can represent the γ emission as

$$^{234}\text{Pa}^* \longrightarrow {}^{234}\text{Pa} + \gamma \qquad \text{or} \qquad {}^{234}\text{Pa}^* \xrightarrow{\gamma} {}^{234}\text{Pa}$$

Note the branching of the ^{238}U decay series that occurs at ^{214}Bi. This nuclide decays to ^{210}Pb by emission of an α and a β^-, which can occur in either order. Note also that the series includes unstable isotopes of several elements that also have stable isotopes, including thallium (Tl), lead (Pb), and bismuth (Bi). These unstable isotopes of these elements all have too many neutrons to be stable.

Many other decay series are known. Two of these occur in nature, one starting with the uncommon isotope ^{235}U and ending with ^{207}Pb, the other starting with thorium (^{232}Th) and ending with ^{208}Pb.

TEST YOUR UNDERSTANDING OF SECTION 43.3 A nucleus with atomic number Z and neutron number N undergoes two decay processes. The result is a nucleus with atomic number $Z - 3$ and neutron number $N - 1$. Which decay processes may have taken place? (i) Two β^- decays; (ii) two β^+ decays; (iii) two α decays; (iv) an α decay and a β^- decay; (v) an α decay and a β^+ decay.

ANSWER

(v) Two protons and two neutrons are lost in an α decay, so Z and N each decrease by 2. A β^+ decay changes a proton to a neutron, so Z decreases by 1 and N increases by 1. The net result is that Z decreases by 3 and N decreases by 1.

43.4 ACTIVITIES AND HALF-LIVES

Suppose you have a certain number of nuclei of a particular radioactive nuclide. If no more are produced, that number decreases in a simple manner as the nuclei decay. This decrease is a statistical process; there is no way to predict when any individual nucleus will decay. No change in physical or chemical environment, such as chemical reactions or heating or cooling, greatly affects most decay rates. The rate varies over an extremely wide range for different nuclides.

Radioactive Decay Rates

Let $N(t)$ be the (very large) number of radioactive nuclei in a sample at time t, and let $dN(t)$ be the (negative) change in that number during a short time interval dt. (We'll use $N(t)$ to minimize confusion with the neutron number N.) The number of decays during the interval dt is $-dN(t)$. The rate of change of $N(t)$ is the negative quantity $dN(t)/dt$; thus $-dN(t)/dt$ is called the *decay rate* or the **activity** of the specimen. The larger the number of nuclei in the specimen, the more nuclei decay during any time interval. That is, the activity is directly proportional to $N(t)$; it equals a constant λ multiplied by $N(t)$:

$$-\frac{dN(t)}{dt} = \lambda N(t) \tag{43.16}$$

The constant λ is called the **decay constant,** and it has different values for different nuclides. A large value of λ corresponds to rapid decay; a small value corresponds to slower decay. Solving Eq. (43.16) for λ shows us that λ is the ratio of the number of decays per time to the number of remaining radioactive nuclei; λ can then be interpreted as the *probability per unit time* that any individual nucleus will decay.

The situation is reminiscent of a discharging capacitor, which we studied in Section 26.4. Equation (43.16) has the same form as the negative of Eq. (26.15), with q and $1/RC$ replaced by $N(t)$ and λ. Then we can make the same substitutions in Eq. (26.16), with the initial number of nuclei $N(0) = N_0$, to find

Number of remaining nuclei at time *t* in sample of radioactive element

Number of nuclei at $t = 0$

$$N(t) = N_0 e^{-\lambda t} \tag{43.17}$$

Time

Decay constant

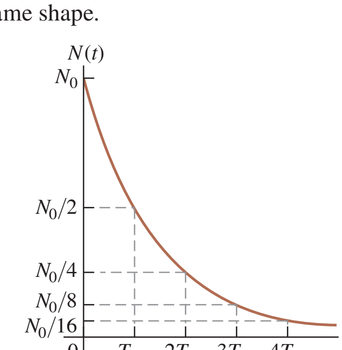
Figure 43.8 is a graph of this function.

The **half-life** $T_{1/2}$ is the time required for the number of radioactive nuclei to decrease to one-half the original number N_0. Then half of the remaining radioactive nuclei decay during a second interval $T_{1/2}$, and so on. The numbers remaining after successive half-lives are $N_0/2$, $N_0/4$, $N_0/8$, ...

To get the relationship between the half-life $T_{1/2}$ and the decay constant λ, we set $N(t)/N_0 = \frac{1}{2}$ and $t = T_{1/2}$ in Eq. (43.17), obtaining

$$\tfrac{1}{2} = e^{-\lambda T_{1/2}}$$

Figure 43.8 The number of nuclei in a sample of a radioactive element as a function of time. The sample's activity has an exponential decay curve with the same shape.



We take logarithms of both sides and solve for $T_{1/2}$:

$$T_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda} \tag{43.18}$$

The mean lifetime T_{mean}, generally called the *lifetime,* of an unstable nucleus or particle is the time for $N(t)$ to decrease by $1/e$. It is proportional to the half-life $T_{1/2}$:

Lifetime of unstable nucleus or particle; Half-life of nucleus or particle; Decay constant of nucleus or particle

$$T_{\text{mean}} = \frac{1}{\lambda} = \frac{T_{1/2}}{\ln 2} = \frac{T_{1/2}}{0.693} \tag{43.19}$$

In particle physics the life of an unstable particle is usually described by the lifetime, not the half-life.

Because the activity $-dN(t)/dt$ at any time equals $\lambda N(t)$, Eq. (43.17) tells us that the activity also depends on time as $e^{-\lambda t}$. Thus the graph of activity versus time has the same shape as Fig. 43.8. Also, after successive half-lives, the activity is one-half, one-fourth, one-eighth, and so on of the original activity.

CAUTION **A half-life may not be enough** It is sometimes implied that any radioactive sample will be safe after a half-life has passed. That's wrong. If your radioactive waste initially has ten times too much activity for safety, it is not safe after one half-life, when it still has five times too much. Even after three half-lives it still has 25% more activity than is safe. The number of radioactive nuclei and the activity approach zero only as t approaches infinity.

A common unit of activity is the **curie,** abbreviated Ci, which is defined to be 3.70×10^{10} decays per second. This is approximately equal to the activity of one gram of radium-226. The SI unit of activity is the *becquerel,* abbreviated Bq. One becquerel is one decay per second, so

$$1 \text{ Ci} = 3.70 \times 10^{10} \text{ Bq} = 3.70 \times 10^{10} \text{ decays/s}$$

EXAMPLE 43.8 Activity of ^{57}Co

WITH VARIATION PROBLEMS

The isotope ^{57}Co decays by electron capture to ^{57}Fe with a half-life of 272 d. The ^{57}Fe nucleus is produced in an excited state, and it almost instantaneously emits gamma rays that we can detect. (a) Find the mean lifetime and decay constant for ^{57}Co. (b) If the activity of a ^{57}Co radiation source is now 2.00 μCi, how many ^{57}Co nuclei does the source contain? (c) What will be the activity after one year?

IDENTIFY and SET UP This problem uses the relationships among decay constant λ, lifetime T_{mean}, and activity $-dN(t)/dt$. In part (a) we use Eq. (43.19) to find λ and T_{mean} from $T_{1/2}$. In part (b), we use Eq. (43.16) to calculate the number of nuclei $N(t)$ from the activity. Finally, in part (c) we use Eqs. (43.16) and (43.17) to find the activity after one year.

EXECUTE (a) It's convenient to convert the half-life from days to seconds:

$$\begin{aligned} T_{1/2} &= (272 \text{ d})(86{,}400 \text{ s/d}) \\ &= 2.35 \times 10^7 \text{ s} \end{aligned}$$

From Eq. (43.19), the mean lifetime and the decay constant are

$$\begin{aligned} T_{\text{mean}} &= \frac{T_{1/2}}{\ln 2} = \frac{2.35 \times 10^7 \text{ s}}{0.693} \\ &= 3.39 \times 10^7 \text{ s} = 392 \text{ days} \end{aligned}$$

$$\lambda = \frac{1}{T_{\text{mean}}} = 2.95 \times 10^{-8} \text{ s}^{-1}$$

(b) The activity $-dN(t)/dt$ is given as 2.00 μCi, so

$$\begin{aligned} -\frac{dN(t)}{dt} &= 2.00\ \mu\text{Ci} = (2.00 \times 10^{-6})(3.70 \times 10^{10} \text{ s}^{-1}) \\ &= 7.40 \times 10^4 \text{ decays/s} \end{aligned}$$

From Eq. (43.16) this is equal to $\lambda N(t)$, so we find

$$\begin{aligned} N(t) &= -\frac{dN(t)/dt}{\lambda} = \frac{7.40 \times 10^4 \text{ s}^{-1}}{2.95 \times 10^{-8} \text{ s}^{-1}} \\ &= 2.51 \times 10^{12} \text{ nuclei} \end{aligned}$$

If you feel we're being too cavalier about the "units" decays and nuclei, you can use decays/(nucleus · s) as the unit for λ.

(c) From Eq. (43.17) the number $N(t)$ of nuclei remaining after one year (3.156×10^7 s) is

$$N(t) = N_0 e^{-\lambda t} = N_0 e^{-(2.95\times 10^{-8}\,\text{s}^{-1})(3.156\times 10^{7}\,\text{s})} = 0.394 N_0$$

The number of nuclei has decreased to 0.394 of the original number. Equation (43.16) says that the activity is proportional to the number of nuclei, so the activity has decreased by this same factor to $(0.394)(2.00\ \mu\text{Ci}) = 0.788\ \mu\text{Ci}$.

EVALUATE The number of nuclei found in part (b) is equivalent to 4.17×10^{-12} mol, with a mass of 2.38×10^{-10} g. This is a far smaller mass than even the most sensitive balance can measure.

After one 272 day half-life, the number of ^{57}Co nuclei has decreased to $N_0/2$; after $2(272\text{ d}) = 544$ d, it has decreased to $N_0/2^2 = N_0/4$. This result agrees with our answer to part (c), which says that after 365 d the number of nuclei is between $N_0/2$ and $N_0/4$.

KEYCONCEPT Because radioactive decay is a statistical process, the number of radioactive nuclei $N(t)$ present in a radioactive sample and the activity $-dN(t)/dt$ of the sample both decay exponentially with time. Both $N(t)$ and $-dN(t)/dt$ decrease by $\frac{1}{2}$ in one half-life and by $1/e$ in one mean lifetime.

Radioactive Dating

An important application of radioactivity is the dating of archaeological and geological specimens by measuring the concentration of radioactive isotopes. The most familiar example is *carbon dating*. The unstable isotope ^{14}C, produced during nuclear reactions in the atmosphere that result from cosmic-ray bombardment, gives a small proportion of ^{14}C in the CO_2 in the atmosphere. Plants that obtain their carbon from this source contain the same proportion of ^{14}C as the atmosphere. When a plant dies, it stops taking in carbon, and its ^{14}C β^- decays to ^{14}N with a half-life of 5730 years. By measuring the proportion of ^{14}C in the remains, we can determine how long ago the organism died.

One difficulty with radiocarbon dating is that the ^{14}C concentration in the atmosphere changes over long time intervals. Corrections can be made on the basis of other data such as measurements of tree rings that show annual growth cycles. Similar radioactive techniques are used with other isotopes for dating geological specimens. Some rocks, for example, contain the unstable potassium isotope ^{40}K, a beta emitter that decays to the stable nuclide ^{40}Ar with a half-life of 2.4×10^8 y. The age of the rock can be determined by comparing the concentrations of ^{40}K and ^{40}Ar.

EXAMPLE 43.9 Radiocarbon dating

WITH VARIATION PROBLEMS

Before 1900 the activity per unit mass of atmospheric carbon due to the presence of ^{14}C averaged about 0.255 Bq per gram of carbon. (a) What fraction of carbon atoms were ^{14}C? (b) In analyzing an archaeological specimen containing 500 mg of carbon, you observe 174 decays in one hour. What is the age of the specimen, assuming that its activity per unit mass of carbon when it died was that average value of the air?

IDENTIFY and SET UP The key idea is that the present-day activity of a biological sample containing ^{14}C is related to both the elapsed time since it stopped taking in atmospheric carbon and its activity at that time. We use Eqs. (43.16) and (43.17) to solve for the age t of the specimen. In part (a) we determine the number of ^{14}C atoms $N(t)$ from the activity $-dN(t)/dt$ by using Eq. (43.16). We find the total number of carbon atoms in 500 mg by using the molar mass of carbon (12.011 g/mol, given in Appendix D), and we use the result to calculate the fraction of carbon atoms that are ^{14}C. The activity decays at the same rate as the number of ^{14}C nuclei; we use this and Eq. (43.17) to solve for the age t of the specimen.

EXECUTE (a) To use Eq. (43.16), we must first find the decay constant λ from Eq. (43.18):

$$T_{1/2} = 5730\text{ y} = (5730\text{ y})(3.156 \times 10^7\text{ s/y}) = 1.808 \times 10^{11}\text{ s}$$

$$\lambda = \frac{\ln 2}{T_{1/2}} = \frac{0.693}{1.808 \times 10^{11}\text{ s}} = 3.83 \times 10^{-12}\text{ s}^{-1}$$

Then, from Eq. (43.16),

$$N(t) = \frac{-dN/dt}{\lambda} = \frac{0.255\text{ s}^{-1}}{3.83 \times 10^{-12}\text{ s}^{-1}} = 6.65 \times 10^{10}\text{ atoms}$$

The *total* number of C atoms in 1 gram (1/12.011 mol) is $(1/12.011)(6.022 \times 10^{23}) = 5.01 \times 10^{22}$. The ratio of ^{14}C atoms to all C atoms is

$$\frac{6.65 \times 10^{10}}{5.01 \times 10^{22}} = 1.33 \times 10^{-12}$$

Only four carbon atoms in every 3×10^{12} are ^{14}C.

Continued

(b) Assuming that the activity per gram of carbon in the specimen when it died ($t = 0$) was $0.255\ \text{Bq/g} = (0.255\ \text{s}^{-1} \cdot \text{g}^{-1}) \times (3600\ \text{s/h}) = 918\ \text{h}^{-1} \cdot \text{g}^{-1}$, the activity of 500 mg of carbon then was $(0.500\ \text{g})(918\ \text{h}^{-1} \cdot \text{g}^{-1}) = 459\ \text{h}^{-1}$. The observed activity now, at time t, is $174\ \text{h}^{-1}$. Since the activity is proportional to the number of radioactive nuclei, the activity ratio $174/459 = 0.379$ equals the number ratio $N(t)/N_0$.

Now we solve Eq. (43.17) for t and insert values for $N(t)/N_0$ and λ:

$$t = \frac{\ln(N(t)/N_0)}{-\lambda} = \frac{\ln 0.379}{-3.83 \times 10^{-12}\ \text{s}^{-1}} = 2.53 \times 10^{11}\ \text{s} = 8020\ \text{y}$$

EVALUATE After 8020 y the ^{14}C activity has decreased from 459 to 174 decays per hour. The specimen died and stopped taking CO_2 out of the air about 8000 years ago.

KEYCONCEPT To find the age of a sample using radioactive dating, measure its present activity and determine what its activity would have been when it formed. The age is proportional to the logarithm of the ratio of these two activities.

Radiation in the Home

A serious health hazard in some areas is the accumulation in houses of ^{222}Rn, an inert, colorless, odorless radioactive gas. Looking at the ^{238}U decay chain in Fig. 43.7, we see that the half-life of ^{222}Rn is 3.82 days. If so, why not just move out of the house for a while and let it decay away? The answer is that ^{222}Rn is continuously being *produced* by the decay of ^{226}Ra, which is found in minute quantities in the rocks and soil on which some houses are built. It's a dynamic equilibrium situation, in which the rate of production equals the rate of decay. The reason ^{222}Rn is a bigger hazard than the other elements in the ^{238}U decay series is that it's a gas. During its short half-life of 3.82 days it can migrate from the soil into your house. If a ^{222}Rn nucleus decays in your lungs, it emits a damaging α particle and its daughter nucleus ^{218}Po, which is *not* chemically inert and is likely to stay in your lungs until it decays, emits another damaging α particle and so on down the ^{238}U decay series.

How much of a hazard is radon? Although reports indicate values as high as 3500 pCi/L, the average activity per volume in the air inside American homes due to ^{222}Rn is about 1.5 pCi/L (over a thousand decays each second in an average-sized room). *If* your environment has this level of activity, it has been estimated that a lifetime exposure would reduce your life expectancy by about 40 days. For comparison, smoking one pack of cigarettes per day reduces life expectancy by 6 years, and it is estimated that the average emission from all the nuclear power plants in the world reduces life expectancy by anywhere from 0.01 day to 5 days. These figures include catastrophes such as the nuclear reactor disasters at Chernobyl, Ukraine (1986), and Fukushima, Japan (2011), for which the *local* effect on life expectancy is much greater.

TEST YOUR UNDERSTANDING OF SECTION 43.4 Which sample contains a greater number of nuclei: a 5.00 μCi sample of ^{240}Pu (half-life 6560 y) or a 4.45 μCi sample of ^{243}Am (half-life 7370 y)? (i) The ^{240}Pu sample; (ii) the ^{243}Am sample; (iii) both have the same number of nuclei.

ANSWER

(iii) The activity $-dN(t)/dt$ of a sample is the product of the number of nuclei in the sample $N(t)$ and the decay constant $\lambda = (\ln 2)/T_{1/2}$. Hence $N(t) = [-dN(t)/dt]T_{1/2}/(\ln 2)$. Taking the ratio of this expression for ^{240}Pu to this same expression for ^{243}Am, the factors of ln 2 cancel and we get

$$\frac{N_{\text{Pu}}}{N_{\text{Am}}} = \frac{(-dN_{\text{Pu}}/dt)T_{1/2\text{-Pu}}}{(-dN_{\text{Am}}/dt)T_{1/2\text{-Am}}} = \frac{(5.00\ \mu\text{Ci})(6560\ \text{y})}{(4.45\ \mu\text{Ci})(7370\ \text{y})} = 1.00$$

The two samples contain *equal* numbers of nuclei. The ^{243}Am sample has a longer half-life and hence a slower decay rate, so it has a lower activity than the ^{240}Pu sample.

43.5 BIOLOGICAL EFFECTS OF RADIATION

The above discussion of radon introduced the interaction of radiation with living organisms, a topic of vital interest and importance. Under *radiation* we include radioactivity (alpha, beta, gamma, and neutrons) and electromagnetic radiation such as x rays. As these particles pass through matter, they lose energy, breaking molecular bonds and creating ions—hence the term *ionizing radiation*. Charged particles interact directly with the electrons in the material. X rays and γ rays interact by the photoelectric effect, in which an electron absorbs a photon and breaks loose from its site, or by Compton scattering (see Section 38.3). Neutrons cause ionization indirectly through collisions with nuclei or absorption by nuclei with subsequent radioactive decay of the resulting nuclei.

These interactions are extremely complex. It is well known that excessive exposure to radiation, including sunlight, x rays, and all the nuclear radiations, can destroy tissues. In mild cases it results in a burn, as with common sunburn. Greater exposure can cause very severe illness or death by a variety of mechanisms, including massive destruction of tissue cells, alterations of genetic material, and destruction of the components in bone marrow that produce red blood cells.

Calculating Radiation Doses

Radiation dosimetry is the quantitative description of the effect of radiation on living tissue. The *absorbed dose* of radiation is defined as the energy delivered to the tissue per unit mass. The SI unit of absorbed dose, the joule per kilogram, is called the *gray* (Gy); 1 Gy = 1 J/kg. Another unit is the *rad,* defined as

$$1 \text{ rad} = 0.01 \text{ J/kg} = 0.01 \text{ Gy}$$

Absorbed dose by itself is not an adequate measure of biological effect because equal energies of different kinds of radiation cause different extents of biological effect. This variation is described by a numerical factor called the **relative biological effectiveness (RBE)**, also called the *quality factor* (QF), of each specific radiation. X rays with 200 keV of energy are defined to have an RBE of unity, and the effects of other radiations can be compared experimentally. **Table 43.3** shows approximate values of RBE for several radiations. All these values depend somewhat on the kind of tissue in which the radiation is absorbed and on the energy of the radiation.

TABLE 43.3 Relative Biological Effectiveness (RBE) for Several Types of Radiation

Radiation	RBE (Sv/Gy or rem/rad)
X rays and γ rays	1
Electrons	1.0–1.5
Slow neutrons	3–5
Protons	10
α particles	20
Heavy ions	20

The biological effect is described by the product of the absorbed dose and the RBE of the radiation; this quantity is called the *biologically equivalent dose,* or simply the equivalent dose. The SI unit of equivalent dose for humans is the sievert (Sv):

$$\text{Equivalent dose (Sv)} = \text{RBE} \times \text{Absorbed dose (Gy)} \tag{43.20}$$

A more common unit, corresponding to the rad, is the rem (an abbreviation of *röntgen equivalent for man*):

$$\text{Equivalent dose (rem)} = \text{RBE} \times \text{Absorbed dose (rad)} \tag{43.21}$$

Thus the unit of the RBE is 1 Sv/Gy or 1 rem/rad, and 1 rem = 0.01 Sv.

EXAMPLE 43.10 Dose from a medical x ray

During a diagnostic x-ray examination a 1.2 kg portion of a broken leg receives an equivalent dose of 0.40 mSv. (a) What is the equivalent dose in mrem? (b) What is the absorbed dose in mrad and in mGy? (c) If the x-ray energy is 50 keV, how many x-ray photons are absorbed?

IDENTIFY and SET UP We are asked to relate the equivalent dose (the biological effect of the radiation, measured in sieverts or rems) to the absorbed dose (the energy absorbed per mass, measured in grays or rads). In part (a) we use the conversion factor 1 rem = 0.01 Sv for equivalent dose. Table 43.3 gives the RBE for x rays; we use this

Continued

value in part (b) to determine the absorbed dose from Eqs. (43.20) and (43.21). Finally, in part (c) we use the mass and the definition of absorbed dose to find the total energy absorbed and the total number of photons absorbed.

EXECUTE (a) The equivalent dose in mrem is

$$\frac{0.40 \text{ mSv}}{0.01 \text{ Sv/rem}} = 40 \text{ mrem}$$

(b) For x rays, RBE = 1 rem/rad or 1 Sv/Gy, so the absorbed dose is

$$\frac{40 \text{ mrem}}{1 \text{ rem/rad}} = 40 \text{ mrad}$$

$$\frac{0.40 \text{ mSv}}{1 \text{ Sv/Gy}} = 0.40 \text{ mGy} = 4.0 \times 10^{-4} \text{ J/kg}$$

(c) The total energy absorbed is

$$(4.0 \times 10^{-4} \text{ J/kg})(1.2 \text{ kg}) = 4.8 \times 10^{-4} \text{ J} = 3.0 \times 10^{15} \text{ eV}$$

The number of x-ray photons is

$$\frac{3.0 \times 10^{15} \text{ eV}}{5.0 \times 10^{4} \text{ eV/photon}} = 6.0 \times 10^{10} \text{ photons}$$

EVALUATE The absorbed dose is relatively large because x rays have a low RBE. If the ionizing radiation had been a beam of α particles, for which RBE = 20, the absorbed dose needed for an equivalent dose of 0.40 mSv would be only 0.020 mGy, corresponding to a smaller total absorbed energy of 2.4×10^{-5} J.

KEYCONCEPT The absorbed dose for a sample of tissue exposed to radiation is the amount of energy delivered per kilogram to the tissue. The equivalent dose equals the absorbed dose multiplied by the relative biological effectiveness (RBE), which depends on the type of radiation used.

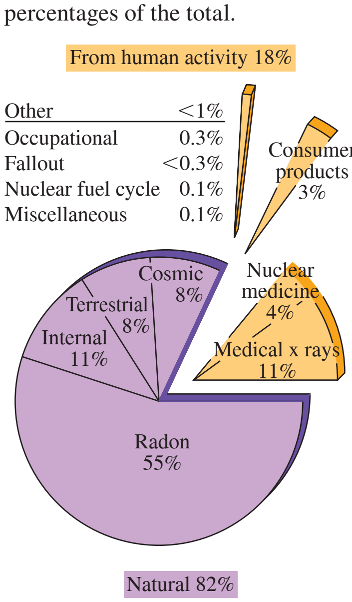
Radiation Hazards

Here are a few numbers for perspective. To convert from Sv to rem, simply multiply by 100. An ordinary chest x-ray exam delivers about 0.20–0.40 mSv to about 5 kg of tissue. Radiation exposure from cosmic rays and natural radioactivity in soil, building materials, and so on is of the order of 2–3 mSv per year at sea level and twice that at an elevation of 1500 m (5000 ft). A whole-body dose of up to about 0.20 Sv causes no immediately detectable effect. A short-term whole-body dose of 5 Sv or more usually causes death within a few days or weeks. A localized dose of 100 Sv causes complete destruction of the exposed tissues.

The long-term hazards of radiation exposure in causing various cancers and genetic defects have been widely publicized, and the question of whether there is any "safe" level of radiation exposure has been hotly debated. U.S. government regulations are based on a maximum *yearly* exposure, from all except natural resources, of 2 to 5 mSv. Workers with occupational exposure to radiation are permitted an average of 20 mSv per year. Recent studies suggest that these limits are too high and that even extremely small exposures carry hazards, but it is very difficult to gather reliable statistics on the effects of low doses. It has become clear that any use of x rays for medical diagnosis should be preceded by a very careful estimation of the relationship of risk to possible benefit.

Another sharply debated question is that of radiation hazards from nuclear power plants. The radiation level from these plants is *not* negligible. However, to make a meaningful evaluation of hazards, we must compare these levels with the alternatives, such as coal-powered plants. The health hazards of coal smoke are serious and well documented, and the natural radioactivity in the smoke from a coal-fired power plant is believed to be roughly 100 times as great as that from a properly operating nuclear plant with equal capacity. But the comparison is not this simple; the possibility of a nuclear accident and the very serious problem of safe disposal of radioactive waste from nuclear plants must also be considered. **Figure 43.9** shows one estimate of the various sources of radiation exposure for the U.S. population. Ionizing radiation is a two-edged sword; it poses very serious health hazards, yet it also provides many benefits to humanity, including the diagnosis and treatments of disease and a wide variety of analytical techniques.

Figure 43.9 Contribution of various sources to the total average radiation exposure in the U.S. population, expressed as percentages of the total.



Beneficial Uses of Radiation

Radiation is widely used in medicine for intentional selective destruction of tissue such as tumors. The hazards are considerable, but if the disease would be fatal without treatment, any hazard may be preferable. Artificially produced isotopes are often used as radiation sources. Such isotopes have several advantages over naturally radioactive isotopes. They may have shorter half-lives and correspondingly greater activity. Isotopes can be chosen that emit the type and energy of radiation desired. Some artificial isotopes have been replaced by photon, proton, and electron beams from linear accelerators.

Nuclear medicine is an expanding field of application. Radioactive isotopes have virtually the same electron configurations and resulting chemical behavior as stable isotopes of the same element. But the location and concentration of radioactive isotopes can easily be detected by measurements of the radiation they emit. A familiar example is the use of radioactive iodine for thyroid studies. Nearly all the iodine ingested is either eliminated or stored in the thyroid, and the body's chemical reactions do not discriminate between the unstable isotope $^{131}\mathrm{I}$ and the stable isotope $^{127}\mathrm{I}$. A minute quantity of $^{131}\mathrm{I}$ is fed or injected into the patient, and the speed with which it becomes concentrated in the thyroid provides a measure of thyroid function. The half-life is 8.02 days, so there are no long-lasting radiation hazards. By use of more sophisticated scanning detectors, one can also obtain a "picture" of the thyroid, which shows enlargement and other abnormalities. This procedure, a type of *autoradiography,* is comparable to photographing the glowing filament of an incandescent light bulb by using the light emitted by the filament itself. If this process discovers cancerous thyroid nodules, they can be destroyed by much larger quantities of $^{131}\mathrm{I}$.

Another useful nuclide for nuclear medicine is technetium-99 $(^{99}\mathrm{Tc})$, which is formed in an excited state by the β^- decay of molybdenum $(^{99}\mathrm{Mo})$. The technetium then decays to its ground state by emitting a γ-ray photon with energy 143 keV. The half-life is 6.01 hours, unusually long for γ emission. (The ground state of $^{99}\mathrm{Tc}$ is also unstable, with a half-life of 2.11×10^5 y; it decays by β^- emission to the stable ruthenium nuclide $^{99}\mathrm{Ru}$.) The chemistry of technetium is such that it can readily be attached to organic molecules that are taken up by various organs of the body. A small quantity of such technetium-bearing molecules is injected into a patient, and a scanning detector or *gamma camera* is used to produce an image, or *scintigram,* that reveals which parts of the body take up these γ-emitting molecules. This technique, in which $^{99}\mathrm{Tc}$ acts as a radioactive *tracer,* plays an important role in locating cancers, embolisms, and other pathologies (**Fig. 43.10**).

Tracer techniques have many other applications. Tritium $(^{3}\mathrm{H})$, a radioactive hydrogen isotope, is used to tag molecules in complex organic reactions; radioactive tags on pesticide molecules, for example, can be used to trace their passage through food chains. In the world of machinery, radioactive iron can be used to study piston-ring wear. Laundry detergent manufacturers have even used radioactive dirt to test the effectiveness of their products.

Many direct effects of radiation are also useful, such as strengthening polymers by cross-linking, sterilizing surgical tools, dispersing unwanted static electricity in the air, and intentionally ionizing the air in smoke detectors. Gamma rays are also used to sterilize and preserve some food products.

TEST YOUR UNDERSTANDING OF SECTION 43.5 Alpha particles have 20 times the relative biological effectiveness of 200 keV x rays. Which would be better to use to radiate tissue deep inside the body? (i) A beam of alpha particles; (ii) a beam of 200 keV x rays; (iii) both are equally effective.

ANSWER

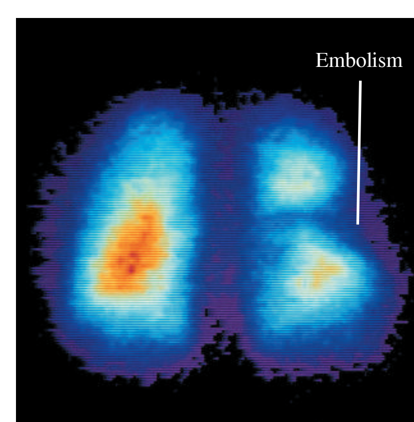
| **(ii)** We saw in Section 43.3 that alpha particles can travel only a very short distance before they are stopped. By contrast, x-ray photons are very penetrating, so they can easily pass into the body.

43.6 NUCLEAR REACTIONS

In the preceding sections we studied the decay of unstable nuclei, especially spontaneous emission of an α or β particle, sometimes followed by γ emission. Nothing needs to be done to initiate this decay, and nothing can be done to control it. This section examines some *nuclear reactions,* rearrangements of nuclear components that result from a bombardment by a particle rather than a spontaneous natural process. Rutherford suggested

BIO APPLICATION A Radioactive Building The United States Capitol building in Washington, DC, is made of granite that contains a small amount of naturally radioactive uranium. As a result, the radiation exposure to someone working in the Capitol is 0.85 mSv per year. The health effects of this are negligible; it is estimated that a person who spent 20 years inside the Capitol would have an extra 0.1% chance of getting cancer, above the 10% chance due to all other causes during the same 20 year period.

Figure **43.10** This colored scintigram shows where a chemical containing radioactive $^{99}\mathrm{Tc}$ was taken up by a patient's lungs. The orange color in the lung on the left indicates strong γ-ray emission by the $^{99}\mathrm{Tc}$, which shows that the chemical was able to pass into this lung through the bloodstream. The lung on the right shows weaker emission, indicating the presence of an embolism (a blood clot or other obstruction in an artery) that is restricting the flow of blood to this lung.



in 1919 that a massive particle with sufficient kinetic energy might be able to penetrate a nucleus. The result would be either a new nucleus with greater atomic number and mass number or a decay of the original nucleus. Rutherford verified this when he bombarded nitrogen $(^{14}\mathrm{N})$ with α particles and obtained an oxygen $(^{17}\mathrm{O})$ nucleus and a proton:

$$^4_2\mathrm{He} + {}^{14}_{7}\mathrm{N} \rightarrow {}^{17}_{8}\mathrm{O} + {}^1_1\mathrm{H} \tag{43.22}$$

Rutherford used alpha particles from naturally radioactive sources. In Chapter 44 we'll describe some of the particle accelerators that are now used to initiate nuclear reactions.

Nuclear reactions are subject to several *conservation laws*. The classical conservation principles for charge, momentum, angular momentum, and energy (including rest energies) are obeyed in all nuclear reactions. An additional conservation law, not anticipated by classical physics, is conservation of the total number of nucleons. The numbers of protons and neutrons need not be conserved separately; in β decay, neutrons and protons change into one another. We'll study the basis of the conservation of nucleon number in Chapter 44.

When two nuclei interact, charge conservation requires that the sum of the initial atomic numbers must equal the sum of the final atomic numbers. Because of conservation of nucleon number, the sum of the initial mass numbers must also equal the sum of the final mass numbers. In general, these are *not* elastic collisions, and the total initial mass does *not* equal the total final mass.

Reaction Energy

The difference between the masses before and after the reaction corresponds to the **reaction energy,** according to the mass–energy relationship $E = mc^2$. If initial particles A and B interact to produce final particles C and D, the reaction energy Q is defined as

$$Q = (M_A + M_B - M_C - M_D)c^2 \quad \text{(reaction energy)} \tag{43.23}$$

To balance the electrons, we use the neutral atomic masses in Eq. (43.23). That is, we use the mass of $^1_1\mathrm{H}$ for a proton, $^2_1\mathrm{H}$ for a deuteron, $^4_2\mathrm{He}$ for an α particle, and so on. When Q is positive, the total mass decreases and the total kinetic energy increases. Such a reaction is called an *exoergic reaction*. When Q is negative, the mass increases and the kinetic energy decreases, and the reaction is called an *endoergic reaction*. The terms *exothermal* and *endothermal,* borrowed from chemistry, are also used. In an endoergic reaction the reaction cannot occur at all unless the initial kinetic energy in the center-of-mass reference frame is at least as great as $|Q|$. That is, there is a **threshold energy,** the minimum kinetic energy to make an endoergic reaction go.

EXAMPLE 43.11 Exoergic and endoergic reactions

(a) When a lithium-7 nucleus is bombarded by a proton, two alpha particles $(^4\mathrm{He})$ are produced. Find the reaction energy. (b) Calculate the reaction energy for the reaction $^4_2\mathrm{He} + {}^{14}_{7}\mathrm{N} \rightarrow {}^{17}_{8}\mathrm{O} + {}^1_1\mathrm{H}$.

IDENTIFY and SET UP The reaction energy Q for any nuclear reaction equals c^2 times the difference between the total initial mass and the total final mass, as in Eq. (43.23). Table 43.2 gives the required masses.

EXECUTE (a) The reaction is $^1_1\mathrm{H} + {}^7_3\mathrm{Li} \rightarrow {}^4_2\mathrm{He} + {}^4_2\mathrm{He}$. The initial and final masses and their respective sums are

A: $^1_1\mathrm{H}$	1.007825 u	C: $^4_2\mathrm{He}$	4.002603 u
B: $^7_3\mathrm{Li}$	7.016003 u	D: $^4_2\mathrm{He}$	4.002603 u
	8.023828 u		8.005206 u

The mass decreases by 0.018622 u. From Eq. (43.23), the reaction energy is

$$Q = (0.018622\ \mathrm{u})(931.5\ \mathrm{MeV/u}) = +17.35\ \mathrm{MeV}$$

(b) The initial and final masses are

A: $^4_2\mathrm{He}$	4.002603 u	C: $^{17}_{8}\mathrm{O}$	16.999132 u
B: $^{14}_{7}\mathrm{N}$	14.003074 u	D: $^1_1\mathrm{H}$	1.007825 u
	18.005677 u		18.006957 u

The mass increases by 0.001280 u, and the corresponding reaction energy is

$$Q = (-0.001280\ \mathrm{u})(931.5\ \mathrm{MeV/u}) = -1.192\ \mathrm{MeV}$$

EVALUATE The reaction in part (a) is *exoergic:* The final total kinetic energy of the two separating alpha particles is 17.35 MeV greater than the initial total kinetic energy of the proton and the lithium nucleus. The reaction in part (b) is *endoergic:* In the center-of-mass system—that is, in a head-on collision with zero total momentum—the minimum total initial kinetic energy required for this reaction to occur is 1.192 MeV.

KEYCONCEPT To find the energy released in a nuclear reaction (the reaction energy), calculate the difference between the total mass of the initial nuclei and the total mass of the final nuclei and then multiply the result by c^2.

Ordinarily, the endoergic reaction of part (b) of Example 43.11 would be produced by bombarding stationary ^{14}N nuclei with alpha particles from an accelerator. In this case an alpha's kinetic energy must be *greater than* 1.192 MeV. If all the alpha's kinetic energy went solely to increasing the rest energy, the final kinetic energy would be zero, and momentum would not be conserved. When a particle with mass m and kinetic energy K collides with a stationary particle with mass M, the total kinetic energy K_{cm} in the center-of-mass coordinate system (the energy available to cause reactions) is

$$K_{cm} = \frac{M}{M + m}K \tag{43.24}$$

This expression assumes that the kinetic energies of the particles and nuclei are much less than their rest energies. We leave the derivation of Eq. (43.24) to you. In part (b) of Example 43.11, $M = 14.003074\text{ u}$ and $m = 4.002603\text{ u}$, so $M/(M + m) = (14.003074\text{ u})/(18.005677\text{ u}) = 0.7777$ and $K_{cm} = 0.7777K$. Since K_{cm} must be at least 1.192 MeV, the α particle's kinetic energy K must be at least $(1.192\text{ MeV})/0.7777 = 1.533\text{ MeV}$.

For a charged particle such as a proton or an α particle to penetrate the nucleus of another atom and cause a reaction, it must usually have enough initial kinetic energy to overcome the potential-energy barrier caused by the repulsive electrostatic forces. In the reaction of part (a) of Example 43.11, if we treat the proton and the ^7Li nucleus as spherically symmetric charges with radii given by Eq. (43.1), their centers will be 3.5×10^{-15} m apart when they touch. The repulsive potential energy of the proton (charge $+e$) and the ^7Li nucleus (charge $+3e$) at this separation r is

$$\begin{aligned} U &= \frac{1}{4\pi\epsilon_0}\frac{(e)(3e)}{r} = (9.0 \times 10^9\text{ N}\cdot\text{m}^2/\text{C}^2)\frac{(3)(1.6 \times 10^{-19}\text{ C})^2}{3.5 \times 10^{-15}\text{ m}} \\ &= 2.0 \times 10^{-13}\text{ J} = 1.2\text{ MeV} \end{aligned}$$

Even though the reaction is exoergic, the proton must have a minimum kinetic energy of about 1.2 MeV for the reaction to occur, unless the proton *tunnels* through the barrier (see Section 40.4).

Neutron Absorption

Absorption of *neutrons* by nuclei forms an important class of nuclear reactions. Heavy nuclei bombarded by neutrons can undergo a series of neutron absorptions alternating with beta decays, in which the mass number A increases by as much as 25. Some of the *transuranic elements,* elements having Z larger than 92, are produced in this way. These elements have not been found in nature. Many transuranic elements, having Z as high as 118, have been identified.

The analytical technique of *neutron activation analysis* uses similar reactions. When bombarded by neutrons, many stable nuclides absorb a neutron to become unstable and then undergo β^- decay. The energies of the β^- and associated γ emissions depend on the unstable nuclide and provide a means of identifying it and the original stable nuclide. Quantities of elements that are far too small for conventional chemical analysis can be detected in this way.

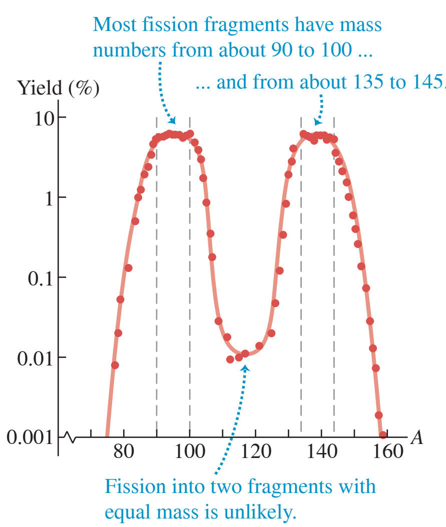
TEST YOUR UNDERSTANDING OF SECTION 43.6 The reaction described in part (a) of Example 43.11 is exoergic. Can it happen naturally when a sample of solid lithium is placed in a flask of hydrogen gas?

ANSWER

no The reaction $^1_1H + ^7_3Li \rightarrow ^4_2He + ^4_2He$ is a *nuclear* reaction that can take place only if a proton (a hydrogen nucleus) comes into contact with a lithium nucleus. If the hydrogen is in atomic form, the interaction between its electron cloud and the electron cloud of a lithium atom keeps the two nuclei from getting close to each other. Even if isolated protons are used, they must be fired at the lithium atoms with enough kinetic energy to overcome the electrical repulsion between the protons and the lithium nuclei. The statement that the reaction is exoergic means that more energy is released by the reaction than had to be put in to make the reaction occur.

43.7 NUCLEAR FISSION

Figure **43.11** Mass distribution of fission fragments from the fission of $^{236}U^*$ (an excited state of ^{236}U), which is produced when ^{235}U absorbs a neutron. The vertical scale is logarithmic.



Nuclear fission is a decay process in which an unstable nucleus splits into two fragments of comparable mass. Fission was discovered in 1938 through the experiments of Otto Hahn and Fritz Strassmann in Germany. Pursuing earlier work by Enrico Fermi, they bombarded uranium $(Z = 92)$ with neutrons. The resulting radiation did not coincide with that of any known radioactive nuclide. Urged on by their colleague Lise Meitner, they used meticulous chemical analysis to reach the astonishing but inescapable conclusion that they had found a radioactive isotope of barium $(Z = 56)$. Later, radioactive krypton $(Z = 36)$ was also found. Meitner and Otto Frisch correctly interpreted these results as showing that uranium nuclei were splitting into two massive fragments called *fission fragments*. Two or three free neutrons usually appear along with the fission fragments and, very occasionally, a light nuclide such as ^3H.

Both the common isotope (99.3%) ^{238}U and the uncommon isotope (0.7%) ^{235}U (as well as several other nuclides) can be easily split by neutron bombardment: ^{235}U by slow neutrons (kinetic energy less than 1 eV) but ^{238}U only by fast neutrons with a minimum of about 1 MeV of kinetic energy. Fission resulting from neutron absorption is called *induced fission*. Some nuclides can also undergo *spontaneous fission* without initial neutron absorption, but this is quite rare. When ^{235}U absorbs a neutron, the resulting nuclide $^{236}U^*$ is in a highly excited state and splits into two fragments almost instantaneously. Strictly speaking, it is $^{236}U^*$, not ^{235}U, that undergoes fission, but it's usual to speak of the fission of ^{235}U.

Over 100 different nuclides, representing more than 20 different elements, have been found among the fission products. **Figure 43.11** shows the distribution of mass numbers for fission fragments from the fission of ^{235}U.

Fission Reactions

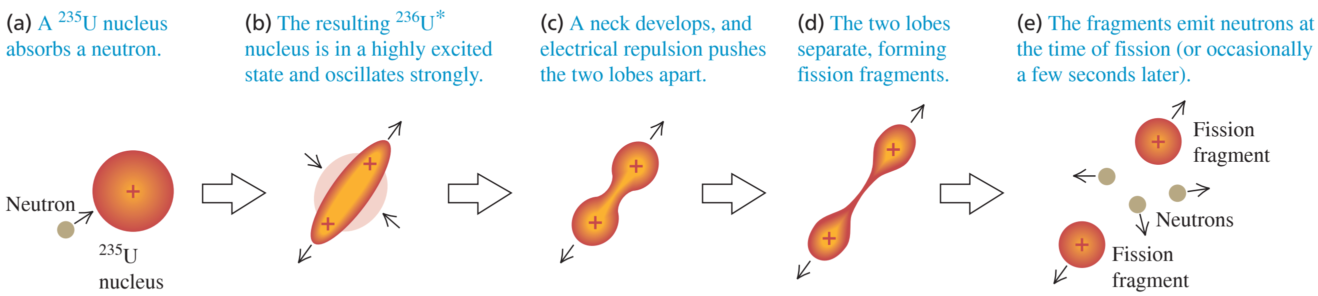
You should check the following two typical fission reactions for conservation of nucleon number and charge:

$$^{235}_{92}U + ^1_0n \rightarrow ^{236}_{92}U^* \rightarrow ^{144}_{56}Ba + ^{89}_{36}Kr + 3^1_0n$$
$$^{235}_{92}U + ^1_0n \rightarrow ^{236}_{92}U^* \rightarrow ^{140}_{54}Xe + ^{94}_{38}Sr + 2^1_0n$$

The total kinetic energy of the fission fragments is enormous, about 200 MeV (compared to typical α and β energies of a few MeV). The reason for this is that nuclides at the high end of the mass spectrum (near $A = 240$) are less tightly bound than those nearer the middle ($A = 90$ to 145). Referring to Fig. 43.2, we see that the average binding energy per nucleon is about 7.6 MeV at $A = 240$ but about 8.5 MeV at $A = 120$. Therefore a rough estimate of the expected *increase* in binding energy during fission is about $8.5\text{ MeV} - 7.6\text{ MeV} = 0.9\text{ MeV}$ per nucleon, or a total of $(235)(0.9\text{ MeV}) \approx 200\text{ MeV}$.

CAUTION Binding energy and rest energy It may seem to be a violation of conservation of energy to have an increase in both the binding energy and the kinetic energy during a fission reaction. But relative to the total rest energy E_0 of the separated nucleons, the rest energy of the nucleus is E_0 *minus* E_B. Thus an *increase* in binding energy corresponds to a *decrease* in rest energy as rest energy is converted to the kinetic energy of the fission fragments.

Figure **43.12** A liquid-drop model of fission.



Fission fragments always have too many neutrons to be stable. We noted in Section 43.3 that the neutron-to-proton ratio (N/Z) for stable nuclides is about 1 for light nuclides but almost 1.6 for the heaviest nuclides because of the increasing influence of the electrical repulsion of the protons. The N/Z value for stable nuclides is about 1.3 at $A = 100$ and 1.4 at $A = 150$. The fragments have about the same N/Z as ^{235}U, about 1.55. They usually respond to this surplus of neutrons by undergoing a series of β^- decays (each of which increases Z by 1 and decreases N by 1) until a stable value of N/Z is reached. A typical example is

$$^{140}_{54}\mathrm{Xe} \xrightarrow{\beta^-} {}^{140}_{55}\mathrm{Cs} \xrightarrow{\beta^-} {}^{140}_{56}\mathrm{Ba} \xrightarrow{\beta^-} {}^{140}_{57}\mathrm{La} \xrightarrow{\beta^-} {}^{140}_{58}\mathrm{Ce}$$

The nuclide ^{140}Ce is stable. This series of β^- decays produces, on average, about 15 MeV of additional kinetic energy. The neutron excess of fission fragments also explains why two or three free neutrons are released during the fission.

Fission appears to set an upper limit on the production of transuranic nuclei, mentioned in Section 43.6, that are relatively stable. There are theoretical reasons to expect that nuclei near $Z = 114$, $N = 184$ or 196, might be stable with respect to spontaneous fission. In the shell model (see Section 43.2), these numbers correspond to filled shells and subshells in the nuclear energy-level structure. Such *superheavy nuclei* would still be unstable with respect to alpha emission. Physicists have found evidence for at least seven isotopes with $Z = 114$, the longest-lived of which has a half-life due to alpha decay of about 2.6 s.

Liquid-Drop Model

We can understand fission qualitatively on the basis of the liquid-drop model of the nucleus (see Section 43.2). The process is shown in **Fig. 43.12** in terms of an electrically charged liquid drop. These sketches shouldn't be taken too literally, but they may help to develop your intuition about fission. A ^{235}U nucleus absorbs a neutron (Fig. 43.12a), becoming a ^{236}U* nucleus with excess energy (Fig. 43.12b). This excess energy causes violent oscillations, during which a neck between two lobes develops (Fig. 43.12c). The electrical repulsion of these two lobes stretches the neck farther (Fig. 43.12d), and finally two smaller fragments are formed (Fig. 43.12e) that move rapidly apart.

This qualitative picture has been developed into a more quantitative theory to explain why some nuclei undergo fission and others don't. **Figure 43.13** shows a hypothetical potential-energy function for two possible fission fragments. If neutron absorption results in an excitation energy greater than the energy barrier height U_B, fission occurs immediately. Even when there isn't quite enough energy to surmount the barrier, fission can take place by quantum-mechanical *tunneling,* discussed in Section 40.4. In principle, many stable heavy nuclei can fission by tunneling. But the probability depends very critically on the height and width of the barrier. For most nuclei this process is so unlikely that it is never observed.

Figure **43.13** Hypothetical potential-energy function for two fission fragments in a fissionable nucleus. At distances r beyond the range of the nuclear force, the potential energy varies approximately as $1/r$. Fission occurs if there is an excitation energy greater than U_B or an appreciable probability for tunneling through the potential-energy barrier.

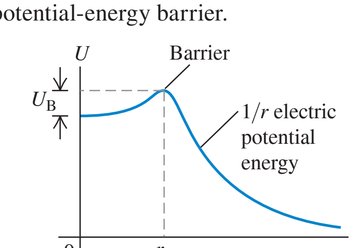
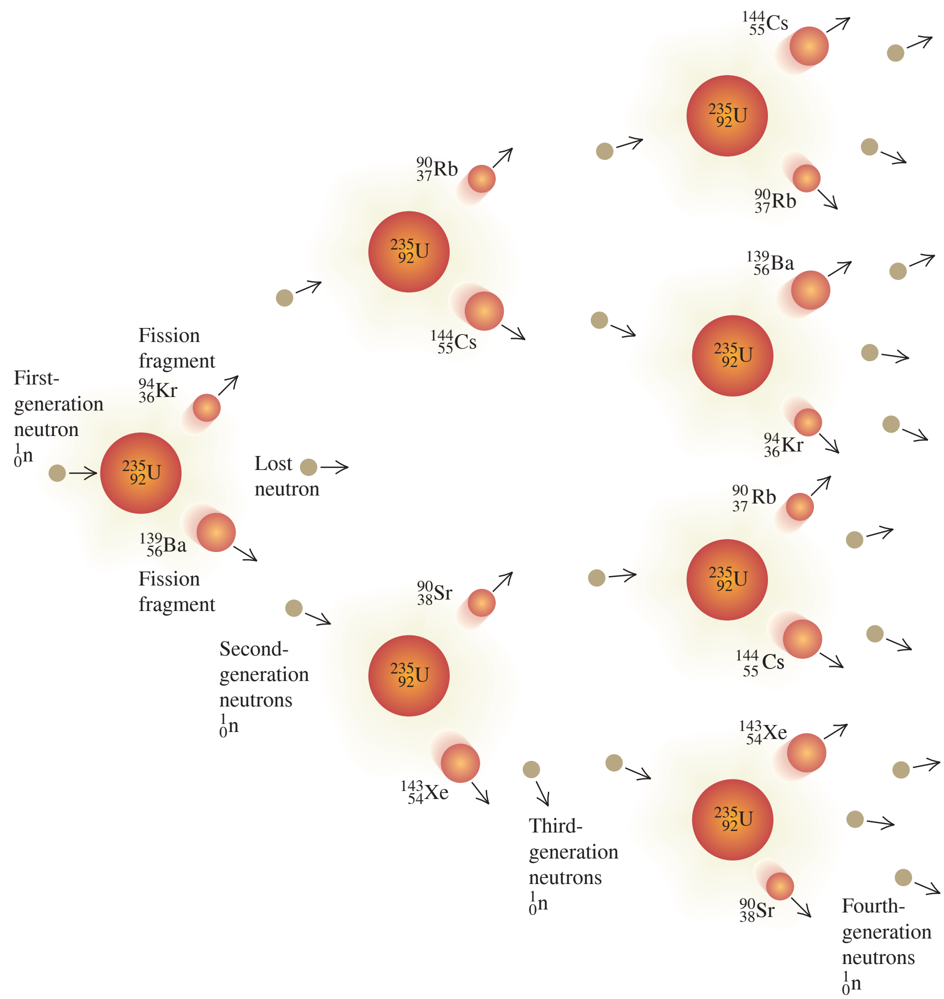
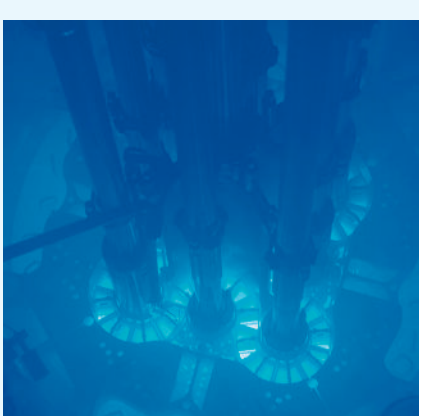


Figure **43.14** Schematic diagram of a nuclear fission chain reaction.



BIO APPLICATION Making Radioactive Isotopes for Medicine The fragments that result from nuclear fission are typically unstable, neutron-rich isotopes. A number of these are useful for medical diagnosis and cancer radiotherapy (see Section 43.5). This photograph shows a nuclear fission reactor used for producing such isotopes. The uranium fuel is kept in a large tank of water for cooling. Some of the neutron-rich fission fragments undergo beta decay and emit electrons that move faster than the speed of light in water (about 0.75*c*). Like an airplane that produces an intense sonic boom when it flies faster than sound (see Section 16.9), these ultrafast electrons produce a "light boom" called *Čerenkov radiation* that has a characteristic blue color.



Chain Reactions

Fission of a uranium nucleus, triggered by neutron bombardment, releases other neutrons that can trigger more fissions, suggesting the possibility of a **chain reaction** (**Fig. 43.14**). The chain reaction may be made to proceed slowly and in a controlled manner in a nuclear reactor or explosively in a bomb. The energy release in a nuclear chain reaction is enormous, far greater than that in any chemical reaction. (In a sense, *fire* is a chemical chain reaction.) For example, when uranium is "burned" to uranium dioxide in the chemical reaction

$$\mathrm{U} + \mathrm{O_2} \rightarrow \mathrm{UO_2}$$

the heat of combustion is about 4500 J/g. Expressed as energy per atom, this is about 11 eV per atom. By contrast, fission liberates about 200 MeV per atom, nearly 20 million times as much energy.

Nuclear Reactors

A *nuclear reactor* is a system in which a controlled nuclear chain reaction is used to liberate energy. In a nuclear power plant, this energy is used to generate steam, which operates a turbine and turns an electrical generator.

On average, each fission of a ^{235}U nucleus produces about 2.5 free neutrons, so 40% of the neutrons are needed to sustain a chain reaction. A ^{235}U nucleus is much more likely to absorb a low-energy neutron (less than 1 eV) than one of the higher-energy neutrons (1 MeV or so) that are liberated during fission. In a nuclear reactor the higher-energy neutrons are slowed down by collisions with nuclei in the surrounding material, called the *moderator,* so they are much more likely to cause further fissions. In nuclear power plants, the moderator is often

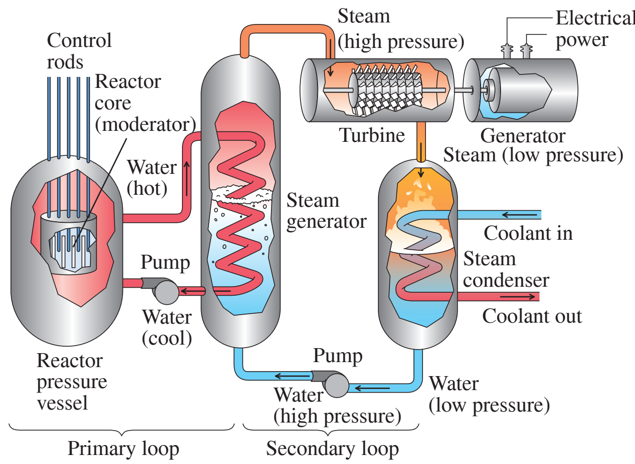


Figure **43.15** Schematic diagram of a nuclear power plant.

water, occasionally graphite. The *rate* of the reaction is controlled by inserting or withdrawing *control rods* made of elements (such as boron or cadmium) whose nuclei *absorb* neutrons without undergoing any additional reaction. The isotope ^{238}U can also absorb neutrons, leading to $^{239}U^*$, but not with high enough probability for it to sustain a chain reaction by itself. Thus uranium that is used in reactors is often "enriched" by increasing the proportion of ^{235}U above the natural value of 0.7%, typically to 3% or so, by isotope-separation processing.

The most familiar application of nuclear reactors is for the generation of electrical power. As was noted above, the fission energy appears as kinetic energy of the fission fragments, and its immediate result is to increase the internal energy of the fuel elements and the surrounding moderator. This increase in internal energy is transferred as heat to generate steam to drive turbines, which spin the electrical generators. **Figure 43.15** is a schematic diagram of a nuclear power plant. The energetic fission fragments heat the water surrounding the reactor core. The steam generator is a heat exchanger that takes heat from this highly radioactive water and generates nonradioactive steam to run the turbines.

A typical nuclear plant has an electric-generating capacity of 1000 MW (or 10^9 W). The turbines are heat engines and are subject to the efficiency limitations imposed by the second law of thermodynamics, discussed in Chapter 20. In modern nuclear plants the overall efficiency is about one-third, so 3000 MW of thermal power from the fission reaction is needed to generate 1000 MW of electrical power.

EXAMPLE 43.12 Uranium consumption in a nuclear reactor

What mass of ^{235}U must undergo fission each day to provide 3000 MW of thermal power?

IDENTIFY and SET UP Fission of ^{235}U liberates about 200 MeV per atom. We use this and the mass of the ^{235}U atom to determine the required amount of uranium.

EXECUTE Each second, we need 3000 MJ or 3000×10^6 J. Each fission provides 200 MeV, or

$$(200\ \text{MeV/fission})(1.6 \times 10^{-13}\ \text{J/MeV}) = 3.2 \times 10^{-11}\ \text{J/fission}$$

The number of fissions needed each second is

$$\frac{3000 \times 10^6\ \text{J}}{3.2 \times 10^{-11}\ \text{J/fission}} = 9.4 \times 10^{19}\ \text{fissions}$$

Each ^{235}U atom has a mass of $(235\ \text{u})(1.66 \times 10^{-27}\ \text{kg/u}) = 3.9 \times 10^{-25}$ kg, so the mass of ^{235}U that undergoes fission each second is

$$(9.4 \times 10^{19})(3.9 \times 10^{-25}\ \text{kg}) = 3.7 \times 10^{-5}\ \text{kg} = 37\ \mu\text{g}$$

In one day (86,400 s), the total consumption of ^{235}U is

$$(3.7 \times 10^{-5}\ \text{kg/s})(86{,}400\ \text{s}) = 3.2\ \text{kg}$$

EVALUATE For comparison, a 1000 MW coal-fired power plant burns 10,600 tons (about 10 million kg) of coal per day!

KEYCONCEPT The thermal power output of a power plant equals the energy released per reaction multiplied by the number of reactions that take place per second. Since the energy released per nuclear fission reaction is very large, a substantial power output can be obtained from relatively few fission reactions per second.

We mentioned above that about 15 MeV of the energy released after fission of a ^{235}U nucleus comes from the β^- decays of the fission fragments. This fact poses a serious problem with respect to control and safety of reactors. Even after the chain reaction has been completely stopped by insertion of control rods into the core, heat continues to be evolved by the β^- decays, which cannot be stopped. For a 3000 MW reactor this heat power is initially very large, about 200 MW. In the event of total loss of cooling water, this power is more than enough to cause a catastrophic meltdown of the reactor core and possible penetration of the containment vessel. The difficulty in achieving a "cold shutdown" following an accident at the Three Mile Island nuclear power plant in Pennsylvania in March 1979 was a result of the continued evolution of heat due to β^- decays.

The catastrophe of April 26, 1986, at Chernobyl reactor No. 4 in Ukraine resulted from a combination of an inherently unstable design and several human errors committed during a test of the emergency core cooling system. Too many control rods were withdrawn to compensate for a decrease in power caused by a buildup of neutron absorbers such as ^{135}Xe. The power level rose from 1% of normal to 100 times normal in 4 seconds; a steam explosion ruptured pipes in the core cooling system and blew the heavy concrete cover off the reactor. The graphite moderator caught fire and burned for several days, and there was a meltdown of the core. The total activity of the radioactive material released into the atmosphere has been estimated as about 10^8 Ci.

TEST YOUR UNDERSTANDING OF SECTION 43.7 The fission of ^{235}U can be triggered by the absorption of a slow neutron by a nucleus. Can a slow *proton* be used to trigger ^{235}U fission?

ANSWER

no Because the neutron has no electric charge, it experiences no electrical repulsion from a ^{235}U nucleus. Hence a slow-moving neutron can approach and enter a ^{235}U nucleus, thereby providing the excitation needed to trigger fission. By contrast, a slow-moving *proton* (charge $+e$) feels a strong electrical repulsion from a ^{235}U nucleus (charge $+92e$). It never gets close to the nucleus, so it cannot trigger fission.

43.8 NUCLEAR FUSION

In a **nuclear fusion** reaction, two or more small light nuclei come together, or *fuse,* to form a larger nucleus. Fusion reactions release energy for the same reason as fission reactions: The binding energy per nucleon after the reaction is greater than before. Referring to Fig. 43.2, we see that the binding energy per nucleon increases with A up to about $A = 60$, so fusion of nearly any two light nuclei to make a nucleus with A less than 60 is likely to be an exoergic reaction. In comparison to fission, we are moving toward the peak of this curve from the opposite side. Another way to express the energy relationships is that the total mass of the products is less than that of the initial particles.

Here are three examples of energy-liberating fusion reactions, written in terms of the neutral atoms:

$$^1_1\text{H} + {}^1_1\text{H} \rightarrow {}^2_1\text{H} + \beta^+ + \nu_e$$

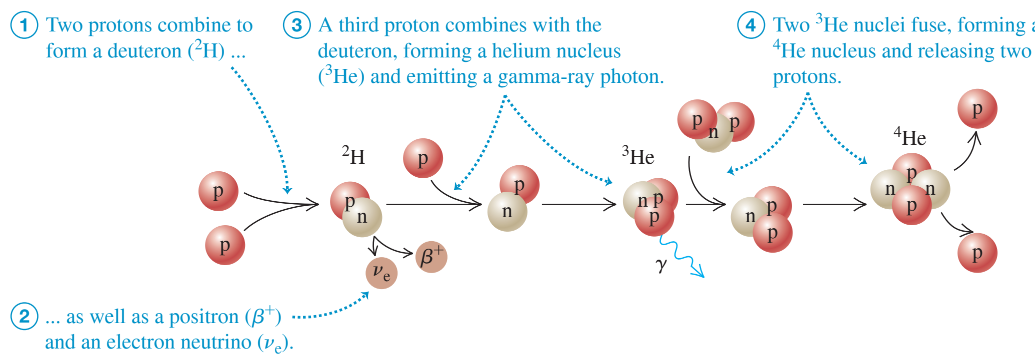
$$^2_1\text{H} + {}^1_1\text{H} \rightarrow {}^3_2\text{He} + \gamma$$

$$^3_2\text{He} + {}^3_2\text{He} \rightarrow {}^4_2\text{He} + {}^1_1\text{H} + {}^1_1\text{H}$$

In the first reaction, two protons combine to form a deuteron (^{2}H), with the emission of a positron (β^+) and an electron neutrino. In the second, a proton and a deuteron combine to form the nucleus of the light isotope of helium, ^{3}He, with the emission of a gamma ray. Now double the first two reactions to provide the two ^{3}He nuclei that fuse in the third reaction to form an alpha particle (^{4}He) and two protons. Together the reactions make up the process called the *proton-proton chain* (**Fig. 43.16**).

The net effect of the chain is the conversion of four protons into one α particle, two positrons, two electron neutrinos, and two γ's. We can calculate the energy release from

Figure **43.16** The proton-proton chain.



this part of the process: The mass of an α particle plus two positrons is the mass of neutral ^{4}He, the neutrinos have zero (or negligible) mass, and the gammas have zero mass.

Mass of four protons	4.029106 u
Mass of ^{4}He	4.002603 u
Mass difference and energy release	0.026503 u and 24.69 MeV

The two positrons that are produced during the first step of the proton-proton chain collide with two electrons; mutual annihilation of the four particles takes place, and their rest energy is converted into $4(0.511\text{ MeV}) = 2.044\text{ MeV}$ of gamma radiation. Thus the total energy released is $(24.69 + 2.044)\text{ MeV} = 26.73\text{ MeV}$. The proton-proton chain takes place in the interior of the sun and other stars (**Fig. 43.17**). Each gram of the sun's mass contains about 4.5×10^{23} protons. If all of these protons were fused into helium, the energy released would be about 130,000 kWh. If the sun were to continue to radiate at its present rate, it would take about 75×10^9 years to exhaust its supply of protons. As we'll soon see, fusion reactions can occur only at extremely high temperatures; in the sun, these temperatures are found only deep within the interior. Hence the sun cannot fuse *all* of its protons and can sustain fusion for only about 10×10^9 years in total. The present age of the solar system (including the sun) is 4.54×10^9 years, so the sun is about halfway through its available store of protons.

Figure **43.17** The energy released as starlight comes from fusion reactions deep within a star's interior. When a star is first formed and for most of its life, it converts the hydrogen in its core into helium. As a star ages, the core temperature can become high enough for additional fusion reactions that convert helium into carbon, oxygen, and other elements.

EXAMPLE 43.13 A fusion reaction

Two deuterons fuse to form a *triton* (a nucleus of tritium, or ^{3}H) and a proton. How much energy is liberated?

IDENTIFY and SET UP This is a nuclear reaction of the type discussed in Section 43.6. We use Eq. (43.23) to find the energy released.

EXECUTE Adding one electron to each nucleus makes each a neutral atom; we find their masses in Table 43.2. Substituting into Eq. (43.23), we find

$$Q = [2(2.014102\text{ u}) - 3.016049\text{ u} - 1.007825\text{ u}](931.5\text{ MeV/u})$$
$$= 4.03\text{ MeV}$$

EVALUATE Thus 4.03 MeV is released in the reaction; the triton and proton together have 4.03 MeV more kinetic energy than the two deuterons had together.

KEYCONCEPT When two light nuclei undergo fusion to form a more massive nucleus, the binding energy per nucleon increases. Hence energy is released in such fusion reactions.

Achieving Fusion

For two nuclei to undergo fusion, they must come together to within the range of the nuclear force, typically of the order of 2×10^{-15} m. To do this, they must overcome the electrical repulsion of their positive charges. For two protons at this distance, the corresponding potential energy is about 1.2×10^{-13} J or 0.7 MeV; this represents the total initial *kinetic* energy that the fusion nuclei must have—for example, 0.6×10^{-13} J each in a head-on collision.

Atoms have this much energy only at extremely high temperatures. The discussion of Section 18.3 showed that the average translational kinetic energy of a gas molecule at temperature T is $\frac{3}{2}kT$, where k is Boltzmann's constant. The temperature at which this is equal to $E = 0.6 \times 10^{-13}$ J is determined by the relationship

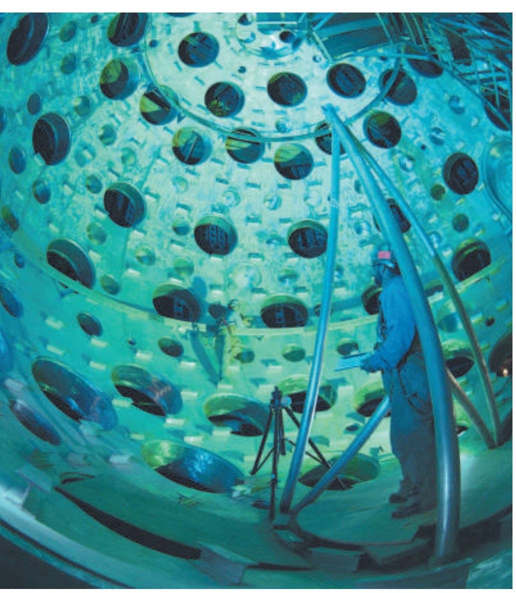
$$E = \tfrac{3}{2}kT$$

$$T = \frac{2E}{3k} = \frac{2(0.6 \times 10^{-13}\ \text{J})}{3(1.38 \times 10^{-23}\ \text{J/K})} = 3 \times 10^{9}\ \text{K}$$

Fusion reactions are possible at lower temperatures because the Maxwell–Boltzmann distribution function (see Section 18.5) gives a small fraction of protons with kinetic energies much higher than the average value. The proton-proton reaction occurs at "only" 1.5×10^{7} K at the center of the sun, making it an extremely low-probability process; but that's why the sun is expected to last so long. At these temperatures the fusion reactions are called *thermonuclear* reactions.

Intensive efforts are under way to achieve controlled fusion reactions, which potentially represent an enormous new resource of energy (see Fig. 24.11). At the temperatures mentioned, light atoms are fully ionized, and the resulting state of matter is called a *plasma*. In one kind of experiment using *magnetic confinement,* a plasma is heated to extremely high temperature by an electrical discharge, while being contained by appropriately shaped magnetic fields. In another, through *inertial confinement,* pellets of the material to be fused are heated by a high-intensity laser beam (see **Fig. 43.18**). Some of the reactions being studied are

Figure 43.18 This target chamber at the National Ignition Facility in California has apertures for 192 powerful laser beams. The lasers deliver 5×10^{14} W of power for a few nanoseconds to a millimeter-sized pellet of deuterium and tritium at the center of the chamber, thus triggering thermonuclear fusion.



$$^{2}_{1}\text{H} + {}^{2}_{1}\text{H} \rightarrow {}^{3}_{1}\text{H} + {}^{1}_{1}\text{H} + 4.0\ \text{MeV} \qquad (1)$$

$$^{3}_{1}\text{H} + {}^{2}_{1}\text{H} \rightarrow {}^{4}_{2}\text{He} + {}^{1}_{0}\text{n} + 17.6\ \text{MeV} \qquad (2)$$

$$^{2}_{1}\text{H} + {}^{2}_{1}\text{H} \rightarrow {}^{3}_{2}\text{He} + {}^{1}_{0}\text{n} + 3.3\ \text{MeV} \qquad (3)$$

$$^{3}_{2}\text{He} + {}^{2}_{1}\text{H} \rightarrow {}^{4}_{2}\text{He} + {}^{1}_{1}\text{H} + 18.3\ \text{MeV} \qquad (4)$$

We considered reaction (1) in Example 43.13; two deuterons fuse to form a triton and a proton. In reaction (2) a triton combines with another deuteron to form an alpha particle and a neutron. The result of these two reactions together is the conversion of three deuterons into an alpha particle, a proton, and a neutron, with 21.6 MeV of energy liberated. Reactions (3) and (4) together achieve the same conversion. In a plasma that contains deuterons, the two pairs of reactions occur with roughly equal probability. As yet, no one has succeeded in producing these reactions under controlled conditions in such a way as to yield a net surplus of usable energy.

Methods of achieving fusion that don't require high temperatures are also being studied; these are called *cold fusion*. One successful scheme uses an unusual hydrogen molecule ion. The usual H_2^+ ion consists of two protons bound by one shared electron; the nuclear spacing is about 0.1 nm. If the protons are replaced by a deuteron (^{2}H) and a triton (^{3}H) and the electron by a *muon,* which is 208 times as massive as the electron, the spacing is reduced by a factor of 208. The probability then becomes appreciable for the two nuclei to tunnel through the narrow repulsive potential-energy barrier and fuse in reaction (2) above. The prospect of making this process, called *muon-catalyzed fusion,* into a practical energy source is still distant.

TEST YOUR UNDERSTANDING OF SECTION 43.8 Are *all* fusion reactions exoergic?

ANSWER

no Fusion reactions between sufficiently light nuclei are exoergic because the binding energy per nucleon E_B/A increases. If the nuclei are too massive, however, E_B/A decreases and fusion is *endoergic* (i.e., it takes in energy rather than releasing it). As an example, imagine fusing together two nuclei of $A = 100$ to make a single nucleus with $A = 200$. From Fig. 43.2, E_B/A is more than 8.5 MeV for the $A = 100$ nuclei but is less than 8 MeV for the $A = 200$ nucleus. Such a fusion reaction is possible, but requires a substantial input of energy.

CHAPTER 43 SUMMARY

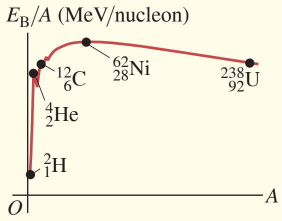
Nuclear properties: A nucleus is composed of A nucleons (Z protons and N neutrons). All nuclei have about the same density. The radius of a nucleus with mass number A is given approximately by Eq. (43.1). A single nuclear species of a given Z and N is called a nuclide. Isotopes are nuclides of the same element (same Z) that have different numbers of neutrons. Nuclear masses are measured in atomic mass units. Nucleons have angular momentum and a magnetic moment. (See Examples 43.1 and 43.2.)

$$R = R_0 A^{1/3} \tag{43.1}$$

$$(R_0 = 1.2 \times 10^{-15}\ \text{m})$$

Nuclear binding and structure: The mass of a nucleus is always less than the mass of the protons and neutrons within it. The mass difference multiplied by c^2 gives the binding energy E_B. The binding energy for a given nuclide is determined by the nuclear force, which is short range and favors pairs of particles, and by the electrical repulsion between protons. A nucleus is unstable if A or Z is too large or if the ratio N/Z is wrong. Two widely used models of the nucleus are the liquid-drop model and the shell model; the latter is analogous to the central-field approximation for atomic structure. (See Examples 43.3 and 43.4.)

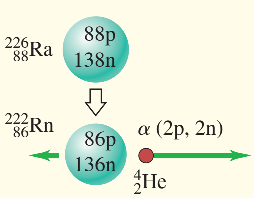
$$E_B = (ZM_H + Nm_n - {}^A_Z M)c^2 \tag{43.10}$$



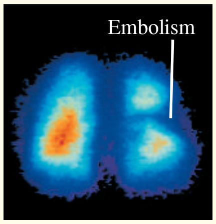
Radioactive decay: Unstable nuclides usually emit an alpha particle (a ${}^4_2$He nucleus) or a beta particle (an electron) in the process of changing to another nuclide, sometimes followed by a gamma-ray photon. The rate of decay of an unstable nucleus is described by the decay constant λ, the half-life $T_{1/2}$, or the lifetime T_{mean}. If the number of nuclei at time $t = 0$ is N_0 and no more are produced, the number at time t is given by Eq. (43.17). (See Examples 43.5–43.9.)

$$N(t) = N_0 e^{-\lambda t} \tag{43.17}$$

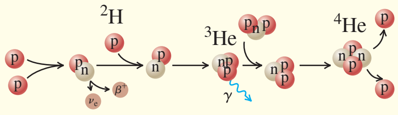
$$T_{mean} = \frac{1}{\lambda} = \frac{T_{1/2}}{\ln 2} = \frac{T_{1/2}}{0.693} \tag{43.19}$$



Biological effects of radiation: The biological effect of any radiation depends on the product of the energy absorbed per unit mass and the relative biological effectiveness (RBE), which is different for different radiations. (See Example 43.10.)



Nuclear reactions: In a nuclear reaction, two nuclei or particles collide to produce two new nuclei or particles. Reactions can be exoergic or endoergic. Several conservation laws, including charge, energy, momentum, angular momentum, and nucleon number, are obeyed. Energy is released by the fission of a heavy nucleus into two lighter, always unstable, nuclei. Energy is also released by the fusion of two light nuclei into a heavier nucleus. (See Examples 43.11–43.13.)



Chapter 43 Media Assets

GUIDED PRACTICE

For assigned homework and other learning materials, go to Mastering Physics.

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review EXAMPLES 43.3 and 43.4 (Section 43.2) before attempting these problems.

VP43.4.1 Find (a) the mass defect, (b) the total binding energy, and (c) the binding energy per nucleon of ${}^{16}_{8}\text{O}$, which has a neutral atomic mass of 15.994915 u.

VP43.4.2 The mass defect of ${}^{63}_{29}\text{Cu}$ is 0.5919378 u. Find (a) its binding energy, (b) its binding energy per nucleon, and (c) its neutral atomic mass.

VP43.4.3 Calculate the binding energy E_B/A for the nuclides (a) ${}^{75}_{33}\text{As}$ (mass defect 0.7005604 u), (b) ${}^{150}_{62}\text{Sm}$ (mass defect 1.330388 u), and (c) ${}^{225}_{88}\text{Ra}$ (mass defect 1.852094 u). (d) Figure 43.2 shows that for nuclei with nucleon numbers $A > 62$, the binding energy per nucleon E_B/A decreases as A increases. Do your results agree with this?

VP43.4.4 Use the semiempirical mass formula to calculate the total estimated binding energy E_B and the binding energy per nucleon E_B/A for the nuclides (a) ruthenium-100 (${}^{100}_{44}\text{Ru}$) and (b) mercury-200 (${}^{200}_{80}\text{Hg}$). (c) Are your results consistent with Fig. 43.2?

Be sure to review EXAMPLES 43.5, 43.6, and 43.7 (Section 43.3) before attempting these problems.

VP43.7.1 Household smoke detectors contain a tiny amount of radioactive americium that decays by α emission to neptunium: ${}^{241}_{95}\text{Am} \rightarrow {}^{237}_{93}\text{Np} + {}^{4}_{2}\text{He}$. The atomic masses are 241.056827 u for ${}^{241}_{95}\text{Am}$, 237.048172 u for ${}^{237}_{93}\text{Np}$, and 4.002603 u for ${}^{4}_{2}\text{He}$. If the ${}^{241}_{95}\text{Am}$ nucleus is originally at rest, find (a) the total energy released in the decay, (b) the kinetic energy of the emitted α particle, and (c) the speed of the emitted α particle.

VP43.7.2 The beryllium isotope ${}^{8}_{4}\text{Be}$ is very unstable, and rapidly decays into two α particles: ${}^{8}_{4}\text{Be} \rightarrow {}^{4}_{2}\text{He} + {}^{4}_{2}\text{He}$. The atomic masses are 8.0053051 u for ${}^{8}_{4}\text{Be}$ and 4.0026033 u for ${}^{4}_{2}\text{He}$. If the ${}^{8}_{4}\text{Be}$ nucleus is originally at rest, find (a) the kinetic energy and (b) the speed of each emitted α particle.

VP43.7.3 State whether each of the following proposed β^- decays is possible or impossible. If it is possible, find the total energy released in the decay. The atomic mass is given for each nuclide. (a) ${}^{12}_{4}\text{Be}$ (12.026922 u) to ${}^{12}_{5}\text{B}$ (12.014353 u); (b) ${}^{33}_{17}\text{Cl}$ (32.977452 u) to ${}^{33}_{18}\text{Ar}$ (32.989926 u); (c) ${}^{82}_{35}\text{Br}$ (81.9168018 u) to ${}^{82}_{36}\text{Kr}$ (81.9134812 u).

VP43.7.4 State whether each of the following proposed β^+ decays is possible or impossible. If it is possible, find the total energy released in the decay; if it is impossible, state whether the decay can occur by electron capture. The atomic mass is given for each nuclide. The combined mass of two electrons is 0.001097 u. (a) ${}^{46}_{23}\text{V}$ (45.960198 u) to ${}^{46}_{22}\text{Ti}$ (45.952627 u); (b) ${}^{134}_{57}\text{La}$ (133.908514 u) to ${}^{134}_{56}\text{Ba}$ (133.904508 u); (c) ${}^{67}_{31}\text{Ga}$ (66.9282024u) to ${}^{67}_{30}\text{Zn}$ (66.9271275 u).

Be sure to review EXAMPLES 43.8 and 43.9 (Section 43.4) before attempting these problems.

VP43.9.1 The sodium isotope ${}^{24}_{11}\text{Na}$ undergoes β^- decay to ${}^{24}_{12}\text{Mg}$ (a stable isotope) with a half-life of 15.0 h. Initially a sample of ${}^{24}_{11}\text{Na}$ has β^- activity of 1.50 μCi. Find (a) the mean lifetime, (b) the decay constant, (c) the initial number of ${}^{24}_{11}\text{Na}$ nuclei in the sample, and (d) the activity of the sample after 24.0 h.

VP43.9.2 You are given a sample that was initially 100% ${}^{95}_{41}\text{Nb}$. This isotope undergoes β^- decay to ${}^{95}_{42}\text{Mo}$ (a stable isotope) with a half-life of 35.0 h. You find that the sample now contains 23.8% ${}^{95}_{41}\text{Nb}$ and 76.2% ${}^{95}_{42}\text{Mo}$ and has a β^- activity of 0.514 μCi. Find (a) the decay constant, (b) the present number of ${}^{95}_{41}\text{Nb}$ nuclei in the sample, (c) the initial number of ${}^{95}_{41}\text{Nb}$ nuclei in the sample, and (d) how much time has elapsed since the sample was 100% ${}^{95}_{41}\text{Nb}$.

VP43.9.3 A certain isotope undergoes γ decay from an excited state to its ground state. Initially a sample of this isotope has an activity of 0.425 μCi; 45.0 s later, the activity has decreased to 0.121 μCi. Find (a) the decay constant and half-life of the excited state and (b) the number of nuclei in the excited state initially and 45.0 s later.

VP43.9.4 Before 1900 the activity per unit mass of atmospheric carbon due to the presence of ${}^{14}\text{C}$ (half-life 5730 y) averaged about 0.255 Bq per gram of carbon. You detect 113 decays of ${}^{14}\text{C}$ in 20.0 min from a sample of carbon of mass 0.510 g. Assuming that its activity per unit mass of carbon when it formed was the pre-1900 average value for the air, find (a) the age of the sample and (b) the current number of ${}^{14}\text{C}$ nuclei in the sample.

BRIDGING PROBLEM Saturation of ${}^{128}\text{I}$ Production

In an experiment, the iodine isotope ${}^{128}\text{I}$ is created by irradiating a sample of ${}^{127}\text{I}$ with a beam of neutrons, yielding 1.50×10^6 ${}^{128}\text{I}$ nuclei per second. Initially no ${}^{128}\text{I}$ nuclei are present. A ${}^{128}\text{I}$ nucleus decays by β^- emission with a half-life of 25.0 min. (a) To what nuclide does ${}^{128}\text{I}$ decay? (b) Could that nuclide decay back to ${}^{128}\text{I}$ by β^+ emission? Why or why not? (c) After the sample has been irradiated for a long time, what is the maximum number of ${}^{128}\text{I}$ atoms that can be present in the sample? What is the maximum activity that can be produced? (This steady-state situation is called *saturation*.) (d) Find an expression for the number of ${}^{128}\text{I}$ atoms present in the sample as a function of time.

SOLUTION GUIDE

IDENTIFY and SET UP

1. What happens to the values of Z, N, and A in β^- decay? What must be true for β^- decay to be possible? For β^+ decay to be possible?
2. You'll need to write an equation for the rate of change dN/dt of the number N of ${}^{128}\text{I}$ atoms in the sample, taking account of both the creation of ${}^{128}\text{I}$ by the neutron irradiation and the decay of any ${}^{128}\text{I}$ present. In the steady state, how do the rates of these two processes compare?
3. List the unknown quantities for each part of the problem and identify your target variables.

EXECUTE

4. Find the values of Z and N of the nuclide produced by the decay of ${}^{128}\text{I}$. What element is this?
5. Decide whether this nuclide can decay back to ${}^{128}\text{I}$.
6. Inspect your equation for dN/dt. What is the value of dN/dt in the steady state? Use this to solve for the steady-state values of N and the activity.
7. Solve your dN/dt equation for the function $N(t)$. (*Hint:* See Section 26.4.)

EVALUATE

8. Your result from step 6 tells you the value of N after a long time (that is, for large values of t). Is this consistent with your result from step 7? What would constitute a "long time" under these conditions?

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q43.1 **BIO** Neutrons have a magnetic dipole moment and can undergo spin flips by absorbing electromagnetic radiation. Why, then, are protons rather than neutrons used in MRI of body tissues? (See Fig. 43.1.)

Q43.2 In Eq. (43.11), as the total number of nucleons becomes larger, the importance of the second term in the equation decreases relative to that of the first term. Does this make physical sense? Explain.

Q43.3 Why aren't the masses of all nuclei integer multiples of the mass of a single nucleon?

Q43.4 The only two stable nuclides with more protons than neutrons are ^1_1H and ^3_2He. Why is $Z > N$ so uncommon?

Q43.5 What are the six known elements for which Z is a magic number? Discuss what properties these elements have as a consequence of their special values of Z.

Q43.6 The binding energy per nucleon for most nuclides doesn't vary much (see Fig. 43.2). Is there similar consistency in the *atomic* energy of atoms, on an "energy per electron" basis? If so, why? If not, why not?

Q43.7 Heavy, unstable nuclei usually decay by emitting an α or a β particle. Why don't they usually emit a single proton or neutron?

Q43.8 As stars age, they use up their supply of hydrogen and eventually begin producing energy by a reaction that involves the fusion of three helium nuclei to form a carbon nucleus. Would you expect the interiors of these old stars to be hotter or cooler than the interiors of younger stars? Explain.

Q43.9 Since lead is a stable element, why doesn't the ^{238}U decay series shown in Fig. 43.7 stop at lead, ^{214}Pb?

Q43.10 In the ^{238}U decay series shown in Fig. 43.7, some nuclides in the series are found much more abundantly in nature than others, even though every ^{238}U nucleus goes through every step in the series before finally becoming ^{206}Pb. Why don't the intermediate nuclides all have the same abundance?

Q43.11 Compared to α particles with the same energy, β particles can much more easily penetrate through matter. Why is this?

Q43.12 If $^A_Z\text{El}_\text{i}$ represents the initial nuclide, what is the decay process or processes if the final nuclide is (a) $_{Z+1}{}^{A}\text{El}_\text{f}$; (b) $^{A-4}_{Z-2}\text{El}_\text{f}$; (c) $_{Z-1}{}^{A}\text{El}_\text{f}$?

Q43.13 In a nuclear decay equation, why can we represent an electron as $^{\;\,0}_{-1}\beta^-$? What are the equivalent representations for a positron, a neutrino, and an antineutrino?

Q43.14 Why is the alpha, beta, or gamma decay of an unstable nucleus unaffected by the *chemical* situation of the atom, such as the nature of the molecule or solid in which it is bound? The chemical situation of the atom can, however, have an effect on the half-life in electron capture. Why is this?

Q43.15 In the process of *internal conversion,* a nucleus decays from an excited state to a ground state by giving the excitation energy directly to an atomic electron rather than emitting a gamma-ray photon. Why can this process also produce x-ray photons?

Q43.16 In Example 43.9 (Section 43.4), the activity of atmospheric carbon *before* 1900 was given. Discuss why this activity may have changed since 1900.

Q43.17 **BIO** One problem in radiocarbon dating of biological samples, especially very old ones, is that they can easily be contaminated with modern biological material during the measurement process. What effect would such contamination have on the estimated age? Why is such contamination a more serious problem for samples of older material than for samples of younger material?

Q43.18 The most common radium isotope found on earth, ^{226}Ra, has a half-life of about 1600 years. If the earth was formed well over 10^9 years ago, why is there any radium left now?

Q43.19 Fission reactions occur only for nuclei with large nucleon numbers, while exoergic fusion reactions occur only for nuclei with small nucleon numbers. Why is this?

Q43.20 When a large nucleus splits during nuclear fission, the daughter nuclei of the fission fly apart with enormous kinetic energy. Why does this happen?

EXERCISES

Section 43.1 Properties of Nuclei

43.1 • How many protons and how many neutrons are there in a nucleus of the most common isotope of (a) silicon, $^{28}_{14}\text{Si}$; (b) rubidium, $^{85}_{37}\text{Rb}$; (c) thallium, $^{205}_{\;81}\text{Tl}$?

43.2 • Hydrogen atoms are placed in an external magnetic field. The protons can make transitions between states in which the nuclear spin component is parallel and antiparallel to the field by absorbing or emitting a photon. What magnetic-field magnitude is required for this transition to be induced by photons with frequency 22.7 MHz?

Section 43.2 Nuclear Binding and Nuclear Structure

43.3 • The most common isotope of boron is $^{11}_{\;5}\text{B}$. (a) Determine the total binding energy of $^{11}_{\;5}\text{B}$ from Table 43.2 in Section 43.1. (b) Calculate this binding energy from Eq. (43.11). (Why is the fifth term zero?) Compare to the result you obtained in part (a). What is the percent difference? Compare the accuracy of Eq. (43.11) for $^{11}_{\;5}\text{B}$ to its accuracy for $^{62}_{28}\text{Ni}$ (see Example 43.4).

43.4 •• The nuclei $^{11}_{\;5}\text{B}$ and $^{11}_{\;6}\text{C}$ are called *mirror nuclei,* because the number of protons in $^{11}_{\;5}\text{B}$ equals the number of neutrons in $^{11}_{\;6}\text{C}$ and the number of neutrons in $^{11}_{\;5}\text{B}$ equals the number of protons in $^{11}_{\;6}\text{C}$. The atomic mass of $^{11}_{\;6}\text{C}$ is 11.011434 u, and the atomic mass of $^{11}_{\;5}\text{B}$ is given in Table 43.2. (a) Calculate the binding energies of $^{11}_{\;5}\text{B}$ and $^{11}_{\;6}\text{C}$. (b) Which of the two nuclei has the larger binding energy? Why is this so?

43.5 • Calculate (a) the total binding energy and (b) the binding energy per nucleon of ^{12}C. (c) What percent of the rest mass of this nucleus is its total binding energy?

43.6 • The most common isotope of uranium, $^{238}_{\;92}\text{U}$, has atomic mass 238.050788 u. Calculate (a) the mass defect; (b) the binding energy (in MeV); (c) the binding energy per nucleon.

43.7 • The binding energy per nucleon for $^{56}_{26}\text{Fe}$ is 8.79 MeV/nucleon. What is the mass, in amu, of a neutral $^{56}_{26}\text{Fe}$ atom? Give your answer to five significant figures.

43.8 •• An alpha particle is strongly bound. The $^{12}_{\;6}\text{C}$ nucleus might be modeled as a composite of three alpha particles. Compare the binding energy of $^{12}_{\;6}\text{C}$ with three times the binding energy of an alpha particle. Which of these quantities is larger, and why might this be so?

43.9 • **CP** A photon with a wavelength of 3.50×10^{-13} m strikes a deuteron, splitting it into a proton and a neutron. (a) Calculate the kinetic energy released in this interaction. (b) Assuming the two particles share the energy equally, and taking their masses to be 1.00 u, calculate their speeds after the photodisintegration.

43.10 • Calculate the mass defect, the binding energy (in MeV), and the binding energy per nucleon of (a) the nitrogen nucleus, $^{14}_{\;7}\text{N}$, and (b) the helium nucleus, ^4_2He. (c) How does the binding energy per nucleon compare for these two nuclei?

Section 43.3 Nuclear Stability and Radioactivity

43.11 • The carbon isotope ${}^{11}_{6}C$ undergoes β^+ (positron) decay. The atomic mass of ${}^{11}_{6}C$ is 11.011433 u. (a) How many protons and how many neutrons are in the daughter nucleus produced by this decay? (b) How much energy, in MeV, is released in the decay of one ${}^{11}_{6}C$ nucleus?

43.12 • (a) Is the decay $n \rightarrow p + \beta^- + \bar{\nu}_e$ energetically possible? If not, explain why not. If so, calculate the total energy released. (b) Is the decay $p \rightarrow n + \beta^+ + \nu_e$ energetically possible? If not, explain why not. If so, calculate the total energy released.

43.13 • What nuclide is produced in the following radioactive decays? (a) α decay of ${}^{239}_{94}Pu$; (b) β^- decay of ${}^{24}_{11}Na$; (c) β^+ decay of ${}^{15}_{8}O$.

43.14 •• CP ${}^{238}U$ decays spontaneously by α emission to ${}^{234}Th$. Calculate (a) the total energy released by this process and (b) the recoil velocity of the ${}^{234}Th$ nucleus. The atomic masses are 238.050788 u for ${}^{238}U$ and 234.043601 u for ${}^{234}Th$.

43.15 • The radioactive isotope ${}^{12}_{7}N$ undergoes β^+ decay. The energy released in the decay of one nucleus is 16.316 MeV. What is the mass, in amu, of a neutral ${}^{12}_{7}N$ atom? Give your answer to eight significant figures.

43.16 • What particle (α particle, electron, or positron) is emitted in the following radioactive decays? (a) ${}^{27}_{14}Si \rightarrow {}^{27}_{13}Al$; (b) ${}^{238}_{92}U \rightarrow {}^{234}_{90}Th$; (c) ${}^{74}_{33}As \rightarrow {}^{74}_{34}Se$.

43.17 • Tritium $({}^{3}_{1}H)$ is an unstable isotope of hydrogen; its mass, including one electron, is 3.016049 u. (a) Show that tritium must be unstable with respect to beta decay because the decay products (${}^{3}_{2}He$ plus an emitted electron) have less total mass than the tritium. (b) Determine the total kinetic energy (in MeV) of the decay products, taking care to account for the electron masses correctly.

Section 43.4 Activities and Half-Lives

43.18 • A sample of radioactive nuclei has N_0 nuclei at time $t = 0$. The half-life of the decay is $T_{1/2}$. In terms of N_0, how many decays occur in the time period between $t = 0$ and $t = 0.500T_{1/2}$?

43.19 • If a 6.13 g sample of an isotope having a mass number of 124 decays at a rate of 0.350 Ci, what is its half-life?

43.20 • BIO Radioactive isotopes used in cancer therapy have a "shelf-life," like pharmaceuticals used in chemotherapy. Just after it has been manufactured in a nuclear reactor, the activity of a sample of ${}^{60}Co$ is 5000 Ci. When its activity falls below 3500 Ci, it is considered too weak a source to use in treatment. You work in the radiology department of a large hospital. One of these ${}^{60}Co$ sources in your inventory was manufactured on October 6, 2011. It is now April 6, 2014. Is the source still usable? The half-life of ${}^{60}Co$ is 5.271 years.

43.21 •• The common isotope of uranium, ${}^{238}U$, has a half-life of 4.47×10^9 years, decaying to ${}^{234}Th$ by alpha emission. (a) What is the decay constant? (b) What mass of uranium is required for an activity of 1.00 curie? (c) How many alpha particles are emitted per second by 10.0 g of uranium?

43.22 •• BIO **Radiation Treatment of Prostate Cancer.** In many cases, prostate cancer is treated by implanting 60 to 100 small seeds of radioactive material into the tumor. The energy released from the decays kills the tumor. One isotope that is used (there are others) is palladium $({}^{103}Pd)$ with a half-life of 17 days. If a typical grain contains 0.250 g of ${}^{103}Pd$, (a) what is its initial activity rate in Bq, and (b) what is the rate 68 days later?

43.23 •• A 12.0 g sample of carbon from living matter decays at the rate of 184 decays/minute due to the radioactive ${}^{14}C$ in it. What will be the decay rate of this sample in (a) 1000 years and (b) 50,000 years?

43.24 •• BIO **Radioactive Tracers.** Radioactive isotopes are often introduced into the body through the bloodstream. Their spread through the body can then be monitored by detecting the appearance of radiation in different organs. One such tracer is ${}^{131}I$, a β^- emitter with a half-life of 8.0 d. Suppose a scientist introduces a sample with an activity of 325 Bq and watches it spread to the organs. (a) Assuming that all of the sample went to the thyroid gland, what will be the decay rate in that gland 24 d (about $3\frac{1}{2}$ weeks) later? (b) If the decay rate in the thyroid 24 d later is measured to be 17.0 Bq, what percentage of the tracer went to that gland? (c) What isotope remains after the I-131 decays?

43.25 • The isotope ${}^{90}_{39}Y$ undergoes β^- decay with a half-life of 64.0 h. You measure an activity of 8.0×10^{16} Bq. (a) How many ${}^{90}_{39}Y$ nuclei are present in the sample at the time you make this measurement? (b) How many will be present after 12.0 days?

43.26 • As a health physicist, you are being consulted about a spill in a radiochemistry lab. The isotope spilled was 400 μCi of ${}^{131}Ba$, which has a half-life of 12 days. (a) What mass of ${}^{131}Ba$ was spilled? (b) Your recommendation is to clear the lab until the radiation level has fallen 1.00 μCi. How long will the lab have to be closed?

43.27 • Measurements on a certain isotope tell you that the decay rate decreases from 8318 decays/min to 3091 decays/min in 4.00 days. What is the half-life of this isotope?

43.28 •• **Radiocarbon Dating.** At an archeological site, a sample from timbers containing 500 g of carbon provides 2690 decays/min. What is the age of the sample?

43.29 • The radioactive nuclide ${}^{199}Pt$ has a half-life of 30.8 minutes. A sample is prepared that has an initial activity of 7.56×10^{11} Bq. (a) How many ${}^{199}Pt$ nuclei are initially present in the sample? (b) How many are present after 30.8 minutes? What is the activity at this time? (c) Repeat part (b) for a time 92.4 minutes after the sample is first prepared.

Section 43.5 Biological Effects of Radiation

43.30 • BIO **Radiation Overdose.** If a person's entire body is exposed to 5.0 J/kg of x rays, death usually follows within a few days. (a) Express this lethal radiation dose in Gy, rad, Sv, and rem. (b) How much total energy does a 70.0 kg person absorb from such a dose? (c) If the 5.0 J/kg came from a beam of protons instead of x rays, what would be the answers to parts (a) and (b)?

43.31 •• BIO A nuclear chemist receives an accidental radiation dose of 5.0 Gy from slow neutrons (RBE = 4.0). What does she receive in rad, rem, and J/kg?

43.32 •• BIO A person exposed to fast neutrons receives a radiation dose of 300 rem on part of his hand, affecting 25 g of tissue. The RBE of these neutrons is 10. (a) How many rad did he receive? (b) How many joules of energy did he receive? (c) Suppose the person received the same rad dosage, but from beta rays with an RBE of 1.0 instead of neutrons. How many rem would he have received?

43.33 • BIO **Food Irradiation.** Food is often irradiated with either x rays or electron beams to help prevent spoilage. A low dose of 5–75 kilorads (krad) helps to reduce and kill inactive parasites, a medium dose of 100–400 krad kills microorganisms and pathogens such as salmonella, and a high dose of 2300–5700 krad sterilizes food so that it can be stored without refrigeration. (a) A dose of 175 krad kills spoilage microorganisms in fish. If x rays are used, what would be the dose in Gy, Sv, and rem, and how much energy would a 220 g portion of fish absorb? (See Table 43.3.) (b) Repeat part (a) if electrons of RBE 1.50 are used instead of x rays.

43.34 • BIO **To Scan or Not to Scan?** It has become popular for some people to have yearly whole-body scans (CT scans, formerly called CAT scans) using x rays, just to see if they detect anything suspicious. A number of medical people have recently questioned the advisability of such scans, due in part to the radiation they impart. Typically, one such scan gives a dose of 12 mSv, applied to the *whole body*. By contrast, a chest x ray typically administers 0.20 mSv to only 5.0 kg of tissue. How many chest x rays would deliver the same *total* amount of energy to the body of a 75 kg person as one whole-body scan?

43.35 •• The radioactive isotope ${}^{90}_{38}\text{Sr}$ is found in the fallout from atmospheric tests of nuclear weapons (see Problem 43.49). It replaces calcium in items such as bone and milk. It has a half-life of 29 years, and the net energy absorbed from each decay averages about 1.1 MeV. (a) Calculate the absorbed dose from 1.0 microgram of ${}^{90}_{38}\text{Sr}$ in one year if the energy is absorbed in 50 kg of body tissue. (b) The radiation from ${}^{90}_{38}\text{Sr}$ involves beta-minus particles and gamma rays. Calculate the equivalent dose in part (a) if you assume the RBE for the electrons is 1.0.

43.36 • BIO In an industrial accident a 65 kg person receives a lethal whole-body equivalent dose of 5.4 Sv from x rays. (a) What is the equivalent dose in rem? (b) What is the absorbed dose in rad? (c) What is the total energy absorbed by the person's body? How does this amount of energy compare to the amount of energy required to raise the temperature of 65 kg of water 0.010 C°?

43.37 •• CP BIO In a diagnostic x-ray procedure, 5.00×10^{10} photons are absorbed by tissue with a mass of 0.600 kg. The x-ray wavelength is 0.0200 nm. (a) What is the total energy absorbed by the tissue? (b) What is the equivalent dose in rem?

Section 43.6 Nuclear Reactions
Section 43.7 Nuclear Fission
Section 43.8 Nuclear Fusion

43.38 •• Calculate the reaction energy Q for the reaction $\text{p} + {}^{3}_{1}\text{H} \rightarrow {}^{2}_{1}\text{H} + {}^{2}_{1}\text{H}$. Is this reaction exoergic or endoergic?

43.39 • Consider the nuclear reaction

$$ {}^{2}_{1}\text{H} + {}^{9}_{4}\text{Be} \rightarrow \text{X} + {}^{4}_{2}\text{He} $$

where X is a nuclide. (a) What are the values of Z and A for the nuclide X? (b) How much energy is liberated? (c) Estimate the threshold energy for this reaction.

43.40 • **Energy from Nuclear Fusion.** Calculate the energy released in the fusion reaction

$$ {}^{3}_{2}\text{He} + {}^{2}_{1}\text{H} \rightarrow {}^{4}_{2}\text{He} + {}^{1}_{1}\text{H} $$

43.41 •• At the beginning of Section 43.7 the equation of a fission process is given in which ${}^{235}\text{U}$ is struck by a neutron and undergoes fission to produce ${}^{144}\text{Ba}$, ${}^{89}\text{Kr}$, and three neutrons. The measured masses of these isotopes are 235.043930 u (${}^{235}\text{U}$), 143.922953 u (${}^{144}\text{Ba}$), 88.917631 u (${}^{89}\text{Kr}$), and 1.0086649 u (neutron). (a) Calculate the energy (in MeV) released by each fission reaction. (b) Calculate the energy released per gram of ${}^{235}\text{U}$, in MeV/g.

43.42 •• Consider the nuclear reaction

$$ {}^{28}_{14}\text{Si} + \gamma \rightarrow {}^{24}_{12}\text{Mg} + \text{X} $$

where X is a nuclide. (a) What are Z and A for the nuclide X? (b) Ignoring the effects of recoil, what minimum energy must the photon have for this reaction to occur? The mass of a ${}^{28}_{14}\text{Si}$ atom is 27.976927 u, and the mass of a ${}^{24}_{12}\text{Mg}$ atom is 23.985042 u.

43.43 • Consider the nuclear reaction

$$ {}^{4}_{2}\text{He} + {}^{7}_{3}\text{Li} \rightarrow \text{X} + {}^{1}_{0}\text{n} $$

where X is a nuclide. (a) What are Z and A for the nuclide X? (b) Is energy absorbed or liberated? How much?

PROBLEMS

43.44 •• CP The nucleus ${}^{10}_{4}\text{Be}$ undergoes β^- decay. The atomic mass of ${}^{10}_{4}\text{Be}$ is 10.013535 u. (a) How much energy, in MeV, is released in this decay? (b) If you assume that all the energy released in the decay appears as kinetic energy of the β^- particle (a very accurate assumption), what is the speed of the emitted β^-? Note that the energy released is nearly the same as the rest energy of the β^-.

43.45 • **Comparison of Energy Released per Gram of Fuel.** (a) When gasoline is burned, it releases 1.3×10^8 J of energy per gallon (3.788 L). Given that the density of gasoline is 737 kg/m^3, express the quantity of energy released in J/g of fuel. (b) During fission, when a neutron is absorbed by a ${}^{235}\text{U}$ nucleus, about 200 MeV of energy is released for each nucleus that undergoes fission. Express this quantity in J/g of fuel. (c) In the proton-proton chain that takes place in stars like our sun, the overall fusion reaction can be summarized as six protons fusing to form one ${}^{4}\text{He}$ nucleus with two leftover protons and the liberation of 26.7 MeV of energy. The fuel is the six protons. Express the energy produced here in units of J/g of fuel. Notice the huge difference between the two forms of nuclear energy, on the one hand, and the chemical energy from gasoline, on the other. (d) Our sun produces energy at a measured rate of 3.86×10^{26} W. If its mass of 1.99×10^{30} kg were all gasoline, how long could it last before consuming all its fuel? (*Historical note:* Before the discovery of nuclear fusion and the vast amounts of energy it releases, scientists were confused. They knew that the earth was at least many millions of years old, but could not explain how the sun could survive that long if its energy came from chemical burning.)

43.46 •• (a) Calculate the minimum energy required to remove one proton from the nucleus ${}^{12}_{6}\text{C}$. This is called the proton-removal energy. (*Hint:* Find the difference between the mass of a ${}^{12}_{6}\text{C}$ nucleus and the mass of a proton plus the mass of the nucleus formed when a proton is removed from ${}^{12}_{6}\text{C}$.) (b) How does the proton-removal energy for ${}^{12}_{6}\text{C}$ compare to the binding energy per nucleon for ${}^{12}_{6}\text{C}$, calculated using Eq. (43.10)?

43.47 •• (a) Calculate the minimum energy required to remove one neutron from the nucleus ${}^{17}_{8}\text{O}$. This is called the neutron-removal energy. (See Problem 43.46.) (b) How does the neutron-removal energy for ${}^{17}_{8}\text{O}$ compare to the binding energy per nucleon for ${}^{17}_{8}\text{O}$, calculated using Eq. (43.10)?

43.48 •• Heat flows to the surface of the earth from its interior. Two-thirds of this heat is radiogenic in origin, which means it comes from the decay of radioactive elements in the earth's mantle, primarily from the isotopes ${}^{238}\text{U}$ and ${}^{232}\text{Th}$. We can estimate the earth's internal energy supply due to the decay of uranium as follows: (a) There are 31 μg of ${}^{238}\text{U}$ in each kg of mantle. Estimate how many ${}^{238}\text{U}$ isotopes there are per kg of mantle. (b) The half-life of ${}^{238}\text{U}$ is 4.47 billion years. Calculate the decay constant in units of s^{-1}. (c) Multiply the results in parts (a) and (b) to obtain the activity in each kg of mantle. (d) Each decay chain ${}^{238}\text{U} \rightarrow {}^{206}\text{Pb}$ releases 52 MeV of energy. Use this value to convert the activity into watts per kg of mantle. (e) The mass of the earth is 6×10^{24} kg, and two-thirds of this is mantle mass. Use these values to estimate the power earth receives from uranium decays. (f) Uranium decays account for 39% of the radiogenic supply of heat. Estimate earth's total heat power related to radioactive decays. (g) Earth also releases heat that remains from its formation, with a power comparable with that due to radioactivity. Estimate the total heat power earth receives from its interior.

43.49 • BIO **Radioactive Fallout.** One of the problems of in-air testing of nuclear weapons (or, even worse, the *use* of such weapons!) is the danger of radioactive fallout. One of the most problematic nuclides in such fallout is strontium-90 (${}^{90}\text{Sr}$), which breaks down by β^- decay with a half-life of 28 years. It is chemically similar to calcium and therefore can be incorporated into bones and teeth, where, due to its rather long half-life, it remains for years as an internal source of radiation. (a) What is the daughter nucleus of the ${}^{90}\text{Sr}$ decay? (b) What percentage of the original level of ${}^{90}\text{Sr}$ is left after 56 years? (c) How long would you have to wait for the original level to be reduced to 6.25% of its original value?

43.50 •• CP Thorium ${}^{230}_{90}\text{Th}$ decays to radium ${}^{226}_{88}\text{Ra}$ by α emission. The masses of the neutral atoms are 230.033134 u for ${}^{230}_{90}\text{Th}$ and 226.025410 u for ${}^{226}_{88}\text{Ra}$. If the parent thorium nucleus is at rest, what is the kinetic energy of the emitted α particle? (Be sure to account for the recoil of the daughter nucleus.)

43.51 •• The atomic mass of $^{25}_{12}\text{Mg}$ is 24.985837 u, and the atomic mass of $^{25}_{13}\text{Al}$ is 24.990428 u. (a) Which of these nuclei will decay into the other? (b) What type of decay will occur? Explain how you determined this. (c) How much energy (in MeV) is released in the decay?

43.52 •• The fusion reactions in the sun produce 26.73 MeV per alpha particle created. Each of these reactions produces two neutrinos. In total, the sun's power is about 3.8×10^{26} W. (a) Estimate how many solar fusion reactions take place every second. (b) Estimate the number of neutrinos that leave the sun each second. (c) The distance between the earth and the sun is 150 million km, and the radius of the earth is 6370 km. Using these figures, estimate the number of solar neutrinos that encounter the earth each second. (d) How many square centimeters does the earth present to the sun? (e) Divide your answer in part (c) by your answer in part (d) to obtain an estimate of the number of solar neutrinos that pass through every square centimeter on earth every second.

43.53 •• BIO Irradiating Ourselves! The radiocarbon in our bodies is one of the naturally occurring sources of radiation. Let's see how large a dose we receive. ^{14}C decays via β^- emission, and 18% of our body's mass is carbon. (a) Write out the decay scheme of carbon-14 and show the end product. (A neutrino is also produced.) (b) Neglecting the effects of the neutrino, how much kinetic energy (in MeV) is released per decay? The atomic mass of ^{14}C is 14.003242 u. (c) How many grams of carbon are there in a 75 kg person? How many decays per second does this carbon produce? (*Hint:* Use data from Example 43.9.) (d) Assuming that all the energy released in these decays is absorbed by the body, how many MeV/s and J/s does the ^{14}C release in this person's body? (e) Consult Table 43.3 and use the largest appropriate RBE for the particles involved. What radiation dose does the person give himself in a year, in Gy, rad, Sv, and rem?

43.54 •• BIO Pion Radiation Therapy. A neutral pion (π^0) has a mass of 264 times the electron mass and decays with a lifetime of 8.4×10^{-17} s to two photons. Such pions are used in the radiation treatment of some cancers. (a) Find the energy and wavelength of these photons. In which part of the electromagnetic spectrum do they lie? What is the RBE for these photons? (b) If you want to deliver a dose of 200 rem (which is typical) in a single treatment to 25 g of tumor tissue, how many π^0 mesons are needed?

43.55 •• Calculate the mass defect for the β^+ decay of $^{11}_{6}\text{C}$. Is this decay energetically possible? Why or why not? The atomic mass of $^{11}_{6}\text{C}$ is 11.011434 u.

43.56 •• BIO A person ingests an amount of a radioactive source that has a very long lifetime and activity 0.52 μCi. The radioactive material lodges in her lungs, where all of the emitted 4.0 MeV α particles are absorbed within a 0.50 kg mass of tissue. Calculate the absorbed dose and the equivalent dose for one year.

43.57 • We Are Stardust. In 1952 spectral lines of the element technetium-99 (^{99}Tc) were discovered in a red giant star. Red giants are very old stars, often around 10 billion years old, and near the end of their lives. Technetium has *no* stable isotopes, and the half-life of ^{99}Tc is 200,000 years. (a) For how many half-lives has the ^{99}Tc been in the red giant star if its age is 10 billion years? (b) What fraction of the original ^{99}Tc would be left at the end of that time? This discovery was extremely important because it provided convincing evidence for the theory (now essentially known to be true) that most of the atoms heavier than hydrogen and helium were made inside of stars by thermonuclear fusion and other nuclear processes. If the ^{99}Tc had been part of the star since it was born, the amount remaining after 10 billion years would have been so minute that it would not have been detectable. This knowledge is what led the late astronomer Carl Sagan to proclaim that "we are stardust."

43.58 • BIO A 70.0 kg person experiences a whole-body exposure to α radiation with energy 4.77 MeV. A total of 7.75×10^{12} α particles are absorbed. (a) What is the absorbed dose in rad? (b) What is the equivalent dose in rem? (c) If the source is 0.0320 g of ^{226}Ra (half-life 1600 y) somewhere in the body, what is the activity of this source? (d) If all of the alpha particles produced are absorbed, what time is required for this dose to be delivered?

43.59 •• Measurements indicate that 27.83% of all rubidium atoms currently on the earth are the radioactive ^{87}Rb isotope. The rest are the stable ^{85}Rb isotope. The half-life of ^{87}Rb is 4.75×10^{10} y. Assuming that no rubidium atoms have been formed since, what percentage of rubidium atoms were ^{87}Rb when our solar system was formed 4.6×10^9 y ago?

43.60 • The nucleus $^{15}_{8}\text{O}$ has a half-life of 122.2 s; $^{19}_{8}\text{O}$ has a half-life of 26.9 s. If at some time a sample contains equal amounts of $^{15}_{8}\text{O}$ and $^{19}_{8}\text{O}$, what is the ratio of $^{15}_{8}\text{O}$ to $^{19}_{8}\text{O}$ (a) after 3.0 min and (b) after 12.0 min?

43.61 •• BIO A ^{60}Co source with activity 2.6×10^{-4} Ci is embedded in a tumor that has mass 0.200 kg. The source emits γ photons with average energy 1.25 MeV. Half the photons are absorbed in the tumor, and half escape. (a) What energy is delivered to the tumor per second? (b) What absorbed dose (in rad) is delivered per second? (c) What equivalent dose (in rem) is delivered per second if the RBE for these γ rays is 0.70? (d) What exposure time is required for an equivalent dose of 200 rem?

43.62 ••• The proton-proton chain described in Section 43.8, known as the p-p I chain, accounts for 83.3% of the helium synthesis in the sun. This process uses six protons to produce one alpha particle, two protons, two neutrinos, and a least six gamma-ray photons. We can determine the energies of each of the products as follows: (a) The first reaction, $2\,^1_1\text{H} \rightarrow {}^2_1\text{H} + \beta^+ + v_e$, is followed immediately by the annihilation of the positron by an electron into two photons, each with energy equal to the rest energy of an electron. What is the remaining energy E_1 carried by the deuteron and the neutrino? (b) Work in the frame where the net momentum of the deuteron and neutrino is zero. Denote the (relativistic) energy of the neutrino by E_v and the (nonrelativistic) deuteron speed by v_d. Denote the mass of the deuteron by m_d. What are the conditions for energy and momentum conservation for these two particles, in terms of E_1? Treat the neutrino as massless. (c) Solve the equations from part (b) and use the deuteron mass in Table 43.2 to determine the percentage of the energy E_1 carried by the neutrino. (d) What is the total energy carried away by neutrinos in the p-p I chain? (*Note:* This chain creates two neutrinos by the above mechanism.) (e) What percentage of the energy generated by the p-p I chain is carried away by neutrinos? (*Note:* The total energy of the p-p I chain is computed in Section 43.8.)

43.63 ••• The sun produces a minority 16.68% of its helium-4 using the p-p II chain, in which four protons and an ambient alpha particle produce two alpha particles, two neutrinos, and at least four gamma-ray photons. The first two steps are the same as the first two steps of the p-p I chain (see Problem 43.62). However, the third step is $^3_2\text{He} + {}^4_2\text{He} \rightarrow {}^7_4\text{Be} + \gamma$, where ^7_4Be is an unstable beryllium nuclide with mass 7.016930 u. (a) What is the energy of the photon? (b) The beryllium nuclide then captures an electron, which changes one of its protons into a neutron, resulting in $^7_4\text{Be} + e^- \rightarrow {}^7_3\text{Li} + v_e$, where ^7_3Li is the common (and stable) nuclide of lithium, with mass 7.016003 u. The excess energy is carried away by the neutrino. What is the energy of that neutrino? (c) A proton then collides with the lithium nucleus, resulting in two alpha particles via $^1_1\text{H} + {}^7_3\text{Li} \rightarrow 2\,^4_2\text{He}$. How much energy is released in this step? (d) If each alpha particle has the same speed, then in the rest frame of the lithium disintegration, what is the speed of each alpha particle? (e) How much energy is released in total by the processes in this p-p II chain? (f) How does this compare with the energy released in the p-p I chain? (g) What percentage of the energy in the p-p II chain is carried away by neutrinos?

43.64 ••• In addition to the p-p I and p-p II chains (see Problems 43.62 and 43.63), the sun produces helium-4 in a relatively rare reaction called the p-p III chain. This produces only 0.1% of the sun's alpha particles but results in neutrinos with substantially higher energy, which have proved crucial to the study of neutrino physics. In the p-p III chain, an ambient ${}^3_2\text{He}$ fuses with ambient ${}^4_2\text{He}$ (alpha particles) to produce ${}^7_4\text{Be}$, precisely as in the p-p II chain. But in the p-p III chain, the ${}^7_4\text{Be}$ then absorbs a proton to produce ${}^8_5\text{B}$. The ${}^8_5\text{B}$ then decays to ${}^8_4\text{Be}$ along with a positron and a neutrino. (a) The total energy released in this process can be ascertained from its net input and output products $4\,{}^1_1\text{H} \rightarrow {}^4_2\text{He}$. How much energy is released? (b) Given the isotope masses 8.024607 u for ${}^8_5\text{B}$ and 8.0053051 u for ${}^8_4\text{Be}$, determine the energy of the neutrino emitted in the ${}^8_5\text{B}$ to ${}^8_4\text{Be}$ decay. (c) What percentage of the energy in the p-p III chain is carried away by neutrinos? (*Note:* This chain also produces a low-energy neutrino.)

43.65 • A bone fragment found in a cave believed to have been inhabited by early humans contains 0.29 times as much ${}^{14}\text{C}$ as an equal amount of carbon in the atmosphere when the organism containing the bone died. (See Example 43.9 in Section 43.4.) Find the approximate age of the fragment.

43.66 •• **BIO** In the 1986 disaster at the Chernobyl reactor in eastern Europe, about $\frac{1}{8}$ of the ${}^{137}\text{Cs}$ present in the reactor was released. The isotope ${}^{137}\text{Cs}$ has a half-life of 30.07 y for β decay, with the emission of a total of 1.17 MeV of energy per decay. Of this, 0.51 MeV goes to the emitted electron; the remaining 0.66 MeV goes to a γ ray. The radioactive ${}^{137}\text{Cs}$ is absorbed by plants, which are eaten by livestock and humans. How many ${}^{137}\text{Cs}$ atoms would need to be present in each kilogram of body tissue if an equivalent dose for one week is 3.5 Sv? Assume that all of the energy from the decay is deposited in 1.0 kg of tissue and that the RBE of the electrons is 1.5.

43.67 •• Consider the fusion reaction ${}^2_1\text{H} + {}^2_1\text{H} \rightarrow {}^3_2\text{He} + {}^1_0\text{n}$. (a) Estimate the barrier energy by calculating the repulsive electrostatic potential energy of the two ${}^2_1\text{H}$ nuclei when they touch. (b) Compute the energy liberated in this reaction in MeV and in joules. (c) Compute the energy liberated *per mole* of deuterium, remembering that the gas is diatomic, and compare with the heat of combustion of hydrogen, about 2.9×10^5 J/mol.

43.68 •• **DATA** As a scientist in a nuclear physics research lab, you are conducting a photodisintegration experiment to verify the binding energy of a deuteron. A photon with wavelength λ in air is absorbed by a deuteron, which breaks apart into a neutron and a proton. The two fragments share the released kinetic energy equally, and the deuteron can be assumed to be initially at rest. You measure the speed of the proton after the disintegration as a function of the wavelength λ of the photon. Your experimental results are given in the table.

λ (10^{-13} m)	3.50	3.75	4.00	4.25	4.50	4.75	5.00
v (10^6 m/s)	11.3	10.2	9.1	8.1	7.2	6.1	4.9

(a) Graph the data as v^2 versus $1/\lambda$. Explain why the data points, when graphed this way, should follow close to a straight line. Find the slope and y-intercept of the straight line that gives the best fit to the data. (b) Assume that h and c have their accepted values. Use your results from part (a) to calculate the mass of the proton and the binding energy (in MeV) of the deuteron.

43.69 •• **DATA** Your company develops radioactive isotopes for medical applications. In your work there, you measure the activity of a radioactive sample. Your results are given in the table.

Time (h)	Decays/s
0	20,000
0.5	14,800
1.0	11,000
1.5	8130
2.0	6020
2.5	4460
3.0	3300
4.0	1810
5.0	1000
6.0	550
7.0	300

(a) Find the half-life of the sample. (b) How many radioactive nuclei were present in the sample at $t = 0$? (c) How many were present after 7.0 h?

43.70 ••• **DATA** In your job as a health physicist, you measure the activity of a mixed sample of radioactive elements. Your results are given in the table.

Time (h)	Decays/s
0	7500
0.5	4120
1.0	2570
1.5	1790
2.0	1350
2.5	1070
3.0	872
4.0	596
5.0	404
6.0	288
7.0	201
8.0	140
9.0	98
10.0	68
12.0	33

(a) What minimum number of different nuclides are present in the mixture? (b) What are their half-lives? (c) How many nuclei of each type are initially present in the sample? (d) How many of each type are present at $t = 5.0$ h?

CHALLENGE PROBLEMS

43.71 ••• **Industrial Radioactivity.** Radioisotopes are used in a variety of manufacturing and testing techniques. Wear measurements can be made using the following method. An automobile engine is produced using piston rings with a total mass of 100 g, which includes 9.4 μCi of ${}^{59}\text{Fe}$ whose half-life is 45 days. The engine is test-run for 1000 hours, after which the oil is drained and its activity is measured. If the activity of the engine oil is 84 decays/s, how much mass was worn from the piston rings per hour of operation?

43.72 ••• Many radioactive decays occur within a sequence of decays—for example, ${}^{234}_{92}\text{U} \rightarrow {}^{230}_{88}\text{Th} \rightarrow {}^{226}_{84}\text{Ra}$. The half-life for the ${}^{234}_{92}\text{U} \rightarrow {}^{230}_{88}\text{Th}$ decay is 2.46×10^5 y, and the half-life for the ${}^{230}_{88}\text{Th} \rightarrow {}^{226}_{84}\text{Ra}$ decay is 7.54×10^4 y. Let 1 refer to ${}^{234}_{92}\text{U}$, 2 to ${}^{230}_{88}\text{Th}$, and 3 to ${}^{226}_{84}\text{Ra}$; let λ_1 be the decay constant for the ${}^{234}_{92}\text{U} \rightarrow {}^{230}_{88}\text{Th}$ decay and λ_2 be the decay constant for the ${}^{230}_{88}\text{Th} \rightarrow {}^{226}_{84}\text{Ra}$ decay. The amount of ${}^{230}_{88}\text{Th}$ present at any time depends on the rate at which it is produced by the decay of ${}^{234}_{92}\text{U}$ and the rate by which it is depleted by its decay to ${}^{226}_{84}\text{Ra}$. Therefore, $dN_2(t)/dt = \lambda_1 N_1(t) - \lambda_2 N_2(t)$. If we start with a sample that contains N_{10} nuclei of ${}^{234}_{92}\text{U}$ and nothing else, then $N(t) = N_{01}e^{-\lambda_1 t}$. Thus $dN_2(t)/dt = \lambda_1 N_{10}e^{-\lambda_1 t} - \lambda_2 N_2(t)$. This differential equation for $N_2(t)$ can be solved as follows. Assume a trial solution of the form $N_2(t) = N_{10}[h_1 e^{-\lambda_1 t} + h_2 e^{-\lambda_2 t}]$, where h_1 and h_2 are constants. (a) Since $N_2(0) = 0$, what must be the relationship between h_1 and h_2? (b) Use the trial solution to calculate $dN_2(t)/dt$, and substitute that into the differential equation for $N_2(t)$. Collect the coefficients of $e^{-\lambda_1 t}$ and $e^{-\lambda_2 t}$. Since the equation must hold at all t, each of these coefficients must be zero. Use this requirement to solve for h_1 and thereby complete the determination of $N_2(t)$. (c) At time $t = 0$, you have a pure sample containing 30.0 g of ${}^{234}_{92}\text{U}$ and nothing else. What mass of ${}^{230}_{88}\text{Th}$ is present at time $t = 2.46 \times 10^5$ y, the half-life for the ${}^{234}_{92}\text{U}$ decay?

MCAT-STYLE PASSAGE PROBLEMS

BIO Radioactive Iodine in Medicine. Iodine in the body is preferentially taken up by the thyroid gland. Therefore, radioactive iodine in small doses is used to image the thyroid and in large doses is used to kill thyroid cells to treat some types of cancer or thyroid disease. The iodine isotopes used have relatively short half-lives, so they must be produced in a nuclear reactor or accelerator. One isotope frequently used for imaging is ${}^{123}\text{I}$; it has a half-life of 13.2 h and emits a 0.16 MeV gamma-ray photon. One method of producing ${}^{123}\text{I}$ is in the nuclear reaction ${}^{123}\text{Te} + \text{p} \rightarrow {}^{123}\text{I} + \text{n}$. The atomic masses relevant to this reaction are ${}^{123}\text{Te}$, 122.904270 u; ${}^{123}\text{I}$, 122.905589 u; n, 1.008665 u; and ${}^{1}\text{H}$, 1.007825 u.

The iodine isotope commonly used for treatment of disease is ${}^{131}\text{I}$, which is produced by irradiating ${}^{130}\text{Te}$ in a nuclear reactor to form ${}^{131}\text{Te}$. The ${}^{131}\text{Te}$ then decays to ${}^{131}\text{I}$. ${}^{131}\text{I}$ undergoes β^- decay with a half-life of 8.04 d, emitting electrons with energies up to 0.61 MeV and gamma-ray photons of energy 0.36 MeV. A typical thyroid cancer treatment might involve administration of 3.7 GBq of ${}^{131}\text{I}$.

43.73 Which reaction produces ${}^{131}\text{Te}$ in the nuclear reactor? (a) ${}^{130}\text{Te} + \text{n} \rightarrow {}^{131}\text{Te}$; (b) ${}^{130}\text{I} + \text{n} \rightarrow {}^{131}\text{Te}$; (c) ${}^{132}\text{Te} + \text{n} \rightarrow {}^{131}\text{Te}$; (d) ${}^{132}\text{I} + \text{n} \rightarrow {}^{131}\text{Te}$.

43.74 Which type of radioactive decay produces ${}^{131}\text{I}$ from ${}^{131}\text{Te}$? (a) Alpha decay; (b) β^- decay; (c) β^+ decay; (d) gamma decay.

43.75 How many ${}^{131}\text{I}$ atoms are administered in a typical thyroid cancer treatment? (a) 4.2×10^{10}; (b) 1.0×10^{12}; (c) 2.5×10^{14}; (d) 3.7×10^{15}.

43.76 In the reaction that produces ${}^{123}\text{I}$, is there a minimum kinetic energy the protons need to make the reaction go? (a) No, because the proton has a smaller mass than the neutron. (b) No, because the total initial mass is smaller than the total final mass. (c) Yes, because the proton has a smaller mass than the neutron. (d) Yes, because the total initial mass is smaller than the total final mass.

43.77 Why might ${}^{123}\text{I}$ be preferred for imaging over ${}^{131}\text{I}$? (a) The atomic mass of ${}^{123}\text{I}$ is smaller, so the ${}^{123}\text{I}$ particles travel farther through tissue. (b) Because ${}^{123}\text{I}$ emits only gamma-ray photons, the radiation dose to the body is lower with that isotope. (c) The beta particles emitted by ${}^{131}\text{I}$ can leave the body, whereas the gamma-ray photons emitted by ${}^{123}\text{I}$ cannot. (d) ${}^{123}\text{I}$ is radioactive, whereas ${}^{131}\text{I}$ is not.

ANSWERS

Chapter Opening Question ?

(iv) When an organism dies, it stops taking in carbon from atmospheric CO_2. Some of this carbon is radioactive ${}^{14}\text{C}$, which decays with a half-life of 5730 years. By measuring the proportion of ${}^{14}\text{C}$ that remains in the specimen, scientists can determine how long ago the organism died. (See Section 43.4.)

Key Example VARIATION Problems

VP43.4.1 (a) 0.137005 u (b) 127.6 MeV (c) 7.976 MeV
VP43.4.2 (a) 551.4 MeV (b) 8.752 MeV (c) 62.929597 u
VP43.4.3 (a) 8.701 MeV (b) 8.262 MeV (c) 7.668 MeV (d) yes
VP43.4.4 (a) $E_B = 868.1$ MeV, $E_B/A = 8.681$ MeV (b) $E_B = 1584$ MeV, $E_B/A = 7.922$ MeV (c) yes
VP43.7.1 (a) 5.637 MeV (b) 5.543 MeV (c) 1.63×10^7 m/s
VP43.7.2 (a) 0.0459 MeV = 45.9 keV (b) 1.49×10^6 m/s
VP43.7.3 (a) possible, 11.71 MeV (b) impossible (c) possible, 3.093 MeV
VP43.7.4 (a) possible, 6.030 MeV (b) possible, 2.710 MeV (c) impossible, yes
VP43.9.1 (a) 7.79×10^4 s (b) $1.28 \times 10^{-5}\ \text{s}^{-1}$ (c) 4.32×10^9 (d) 0.495 μCi
VP43.9.2 (a) $5.50 \times 10^{-6}\ \text{s}^{-1}$ (b) 3.46×10^9 (c) 1.45×10^{10} (d) 72.5 h
VP43.9.3 (a) $\lambda = 0.0279\ \text{s}^{-1}$, $T_{1/2} = 24.8$ s (b) 5.63×10^5 initially, 1.60×10^5 after 45.0 s
VP43.9.4 (a) 2.67×10^3 y (b) 2.46×10^{10}

Bridging Problem

(a) ${}^{128}\text{Xe}$
(b) no; β^+ emission would be endoergic
(c) 3.25×10^9 atoms, 1.50×10^6 Bq
(d) $N(t) = (3.25 \times 10^9\ \text{atoms})(1 - e^{-(4.62\times10^{-4}\ \text{s}^{-1})t})$

? This image shows a portion of the Tarantula Nebula, a region some 170,000 light-years away where new stars are forming. The luminous stars, glowing gas, and opaque dust clouds are all made of "normal" matter—that is, atoms and their constituents. What percentage of the mass and energy in the universe is composed of "normal" matter? (i) 75% to 100%; (ii) 50% to 75%; (iii) 25% to 50%; (iv) 5% to 25%; (v) less than 5%.

44 Particle Physics and Cosmology

What are the most fundamental constituents of matter? How did the universe begin? And what is the fate of our universe? In this chapter we'll explore what physicists and astronomers have learned in their quest to answer these questions.

The chapter title, "Particle Physics and Cosmology," may seem strange. Fundamental particles are the *smallest* things in the universe, and cosmology deals with the *biggest* thing there is—the universe itself. Nonetheless, we'll see in this chapter that physics on the most microscopic scale plays an essential role in determining the nature of the universe on the largest scale.

The development of high-energy accelerators and associated detectors has been crucial in our emerging understanding of particles. We can classify particles and their interactions in several ways in terms of conservation laws and symmetries, some of which are absolute and others of which are obeyed only in certain kinds of interactions. We'll conclude by discussing our present understanding of the nature and evolution of the universe as a whole.

LEARNING OUTCOMES

In this chapter, you'll learn...

44.1 The key varieties of fundamental subatomic particles and how they were discovered.

44.2 How physicists use accelerators and detectors to probe the properties of subatomic particles.

44.3 The four ways in which subatomic particles interact with each other.

44.4 How the structure of protons, neutrons, and other particles can be explained in terms of quarks.

44.5 The standard model of particles and interactions.

44.6 The evidence that the universe is expanding and that the expansion is speeding up.

44.7 The history of the first 380,000 years after the Big Bang.

You'll need to review...

13.3 Escape speed.

27.4 Motion of charged particles in a magnetic field.

32.1 Radiation from accelerated charges.

38.1, 38.3, 38.4 Photons; electron–positron annihilation; uncertainty principle.

39.1, 39.2 Electron waves; discovery of the nucleus.

41.5, 41.6 Electron spin; exclusion principle.

42.6 Valence bands and holes.

43.1, 43.3 Neutrons and protons; β^+ decay.

44.1 FUNDAMENTAL PARTICLES—A HISTORY

The idea that the world is made of fundamental particles has a long history. In about 400 B.C. the Greek philosophers Democritus and Leucippus suggested that matter is made of indivisible particles that they called *atoms,* a word derived from *a-* (not) and *tomos* (cut or divided). This idea lay dormant until about 1804, when the English scientist John Dalton (1766–1844), often called the father of modern chemistry, discovered that many chemical phenomena could be explained if atoms of each element are the basic, indivisible building blocks of matter.

The Electron and the Proton

Toward the end of the 19th century it became clear that atoms are *not* indivisible. The characteristic spectra of elements suggested that atoms have internal structure, and

J. J. Thomson's discovery of the negatively charged *electron* in 1897 showed that atoms could be taken apart into charged particles. Rutherford's experiments in 1910–1911 (see Section 39.2) revealed that an atom's positive charge resides in a small, dense nucleus. In 1919 Rutherford made an additional discovery: When alpha particles are fired into nitrogen, one product is hydrogen gas. He reasoned that the hydrogen nucleus is a constituent of the nuclei of heavier atoms such as nitrogen, and that a collision with a fast-moving alpha particle can dislodge one of those hydrogen nuclei. Thus the hydrogen nucleus is an elementary particle that Rutherford named the *proton*. The following decade saw the blossoming of quantum mechanics, including the Schrödinger equation. Physicists were on their way to understanding the principles that underlie atomic structure.

The Photon

Einstein explained the photoelectric effect in 1905 by assuming that the energy of electromagnetic waves is quantized; that is, it comes in little bundles called *photons* with energy $E = hf$. Atoms and nuclei can emit (create) and absorb (destroy) photons (see Section 38.1). Considered as particles, photons have zero charge and zero rest mass. (Note that any discussions of a particle's mass in this chapter will refer to its rest mass.) In particle physics, a photon is denoted by the symbol γ (the Greek letter gamma).

The Neutron

In 1930 the German physicists Walther Bothe and Herbert Becker observed that when beryllium, boron, or lithium was bombarded by alpha particles, the target material emitted a radiation that had much greater penetrating power than the original alpha particles. Experiments by the English physicist James Chadwick in 1932 showed that the emitted particles were electrically neutral, with mass approximately equal to that of the proton. Chadwick christened these particles *neutrons* (symbol n or ${}^1_0\text{n}$). A typical reaction of the type studied by Bothe and Becker, with a beryllium target, is

$$ {}^4_2\text{He} + {}^9_4\text{Be} \rightarrow {}^{12}_6\text{C} + {}^1_0\text{n} \tag{44.1} $$

Elementary particles are usually detected by their electromagnetic effects—for instance, by the ionization that they cause when they pass through matter. (This is the principle of the cloud chamber, described below.) Because neutrons have no charge, they are difficult to detect directly; they interact hardly at all with electrons and produce little ionization when they pass through matter. However, neutrons can be slowed down by scattering from nuclei, and they can penetrate a nucleus. Hence slow neutrons can be detected by means of a nuclear reaction in which a neutron is absorbed and an alpha particle is emitted. An example is

$$ {}^1_0\text{n} + {}^{10}_5\text{B} \rightarrow {}^7_3\text{Li} + {}^4_2\text{He} \tag{44.2} $$

The ejected alpha particle is easy to detect because it is charged. Later experiments showed that neutrons and protons, like electrons, are spin-$\frac{1}{2}$ particles (see Section 43.1).

The discovery of the neutron cleared up a mystery about the composition of the nucleus. Before 1930 the mass of a nucleus was thought to be due only to protons, but no one understood why the charge-to-mass ratio was not the same for all nuclides. It soon became clear that all nuclides (except ${}^1_1\text{H}$) contain both protons and neutrons. Hence the proton, the neutron, and the electron are the building blocks of atoms. However, that is not the end of the particle story; these are not the only particles, and particles can do more than build atoms.

The Positron

The positive electron, or positron, was discovered by the American physicist Carl D. Anderson in 1932, during an investigation of particles bombarding the earth from space. **Figure 44.1** shows a historic photograph made with a *cloud chamber,* an instrument used to visualize the tracks of charged particles. The chamber contained a supercooled vapor; a charged particle passing through the vapor causes ionization, and the ions trigger the condensation of liquid droplets from the vapor. The droplets make a visible track showing the charged particle's path.

The cloud chamber in Fig. 44.1 is in a magnetic field directed into the plane of the photograph. The particle has passed through a thin lead plate (which extends from left to right in the figure) that lies within the chamber. The track is more tightly curved above the plate than below it, showing that the speed was less above the plate than below it. Therefore the particle had to be moving upward; it could not have gained energy passing through the lead. The thickness and curvature of the track suggested that its mass and the magnitude of its charge equaled those of the electron. But the directions of the magnetic field and the velocity in the magnetic-force equation $\vec{F} = q\vec{v} \times \vec{B}$ showed that the particle had *positive* charge. Anderson christened this particle the *positron.*

To theorists, the appearance of the positron was a welcome development. In 1928 the English physicist Paul Dirac had developed a relativistic generalization of the Schrödinger equation for the electron. In Section 41.5 we discussed how Dirac's ideas helped explain the spin magnetic moment of the electron.

One of the puzzling features of the Dirac equation was that for a *free* electron it predicted not only a continuum of energy states greater than its rest energy $m_e c^2$, as expected, but also a continuum of *negative* energy states *less than* $-m_e c^2$ (**Fig. 44.2a**). That posed a problem. What was to prevent an electron from emitting a photon with energy $2m_e c^2$ or greater and hopping from a positive state to a negative state? It wasn't clear what these negative-energy states meant, and there was no obvious way to get rid of them. Dirac's ingenious interpretation was that all the negative-energy states were filled with electrons, and that these electrons were for some reason unobservable. The exclusion principle (see Section 41.6) would forbid a transition to a state that was already occupied.

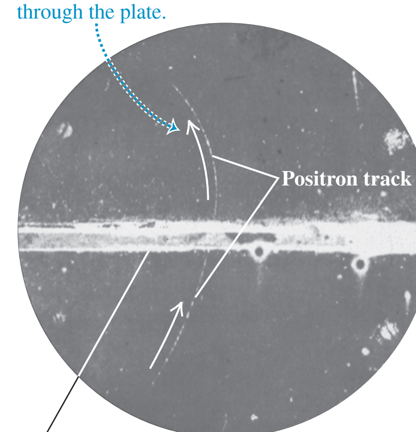
A vacancy in a negative-energy state would act like a positive charge, just as a hole in the valence band of a semiconductor (see Section 42.6) acts like a positive charge. Initially, Dirac tried to argue that such vacancies were protons. But after Anderson's discovery it became clear that the vacancies were observed physically as *positrons.* Furthermore, the Dirac energy-state picture provides a mechanism for the *creation* of positrons. When an electron in a negative-energy state absorbs a photon with energy greater than $2m_e c^2$, it goes to a positive state (Fig. 44.2b), in which it becomes observable. The vacancy that it leaves behind is observed as a positron; the result is the creation of an electron–positron pair. Similarly, when an electron in a positive-energy state falls into a vacancy, both the electron and the vacancy (that is, the positron) disappear, and photons are emitted (Fig. 44.2c). Thus the Dirac theory leads naturally to the conclusion that, like photons, *electrons can be created and destroyed.* While photons can be created and destroyed singly, electrons can be produced or destroyed only in electron–positron pairs or in association with other particles. (Creating or destroying an electron alone would mean creating or destroying an amount of charge $-e$, which would violate the conservation of electric charge.)

In 1949 the American physicist Richard Feynman showed that a positron could be described mathematically as an electron traveling backward in time. His reformulation of the Dirac theory eliminated difficult calculations involving the infinite sea of negative-energy

Figure **44.1** Photograph of the cloud-chamber track made by the first positron ever identified. The photograph was made by Carl D. Anderson in 1932.

The positron follows a curved path owing to the presence of a magnetic field.

The track is more strongly curved above the lead plate, showing that the positron was traveling upward and lost energy and speed as it passed through the plate.



Lead plate (6 mm thick)

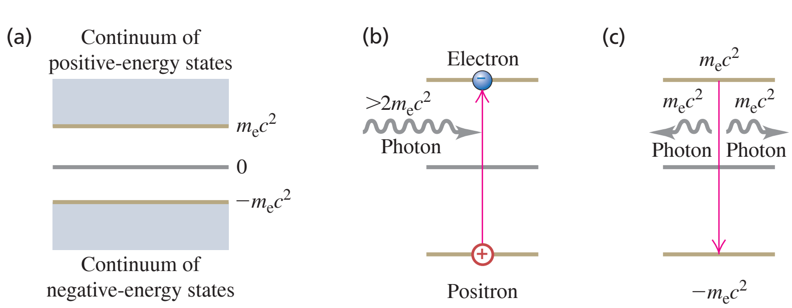
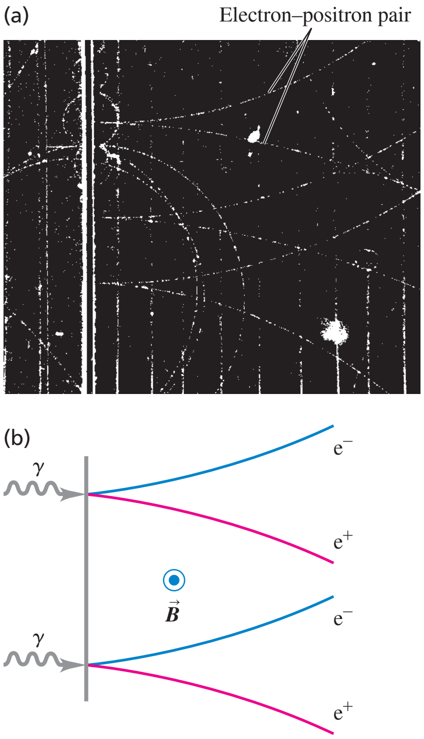


Figure **44.2** (a) Energy states for a free electron predicted by the Dirac equation. (b) Raising an electron from an $E < 0$ state to an $E > 0$ state corresponds to electron–positron pair production. (c) An electron dropping from an $E > 0$ state to a vacant $E < 0$ state corresponds to electron–positron pair annihilation.

Figure 44.3 (a) Photograph of bubble-chamber tracks of electron–positron pairs that are produced when 300 MeV photons strike a lead sheet. A magnetic field directed out of the photograph made the electrons and positrons curve in opposite directions. (b) Diagram showing the pair-production process for two of the photons.



states and put electrons and positrons on the same footing. But the creation and destruction of electron–positron pairs remain. The Dirac theory provides the beginning of a theoretical framework for creation and destruction of all fundamental particles.

Experiment and theory tell us that the masses of the positron and electron are identical and that their charges are equal in magnitude but opposite in sign. The positron's spin angular momentum $\vec{S}$ and magnetic moment $\vec{\mu}$ are parallel; they are opposite for the electron. However, $\vec{S}$ and $\vec{\mu}$ have the same magnitude for both particles because they have the same spin. We use the term **antiparticle** for a particle that is related to another particle as the positron is to the electron. Each kind of particle has a corresponding antiparticle. For a few kinds of particles (necessarily all neutral) the particle and antiparticle are identical, and we can say that they are their own antiparticles. The photon is an example; there is no way to distinguish a photon from an antiphoton. We'll use the standard symbols e^- for the electron and e^+ for the positron. The generic term "electron" often includes both electrons and positrons. Other antiparticles may be denoted by a bar over the particle's symbol; for example, an antiproton is $\overline{p}$. We'll see other examples of antiparticles later.

Positrons do not occur in ordinary matter. Electron–positron pairs are produced during high-energy collisions of charged particles or γ rays with matter. This process is called e^+e^- *pair production* (**Fig. 44.3**). The minimum available energy required for electron–positron pair production equals the rest energy $2m_ec^2$ of the two particles:

$$E_{min} = 2m_ec^2 = 2(9.109 \times 10^{-31}\text{ kg})(2.998 \times 10^8\text{ m/s})^2$$
$$= 1.637 \times 10^{-13}\text{ J} = 1.022\text{ MeV}$$

The inverse process, e^+e^- *pair annihilation,* occurs when a positron and an electron collide (see Example 38.6 in Section 38.3). Both particles disappear, and two (or occasionally three) photons can appear, with total energy of at least $2m_ec^2 = 1.022$ MeV. Decay into a *single* photon is impossible: Such a process could not conserve both energy and momentum.

Positrons also occur in the decay of some unstable nuclei, in which they are called beta-plus particles (β^+). We discussed β^+ decay in Section 43.3.

We'll frequently represent particle masses in terms of the equivalent rest energy by using $m = E/c^2$. Then typical mass units are MeV/c^2; for example, $m = 0.511\text{ MeV}/c^2$ for an electron or positron.

Particles as Force Mediators

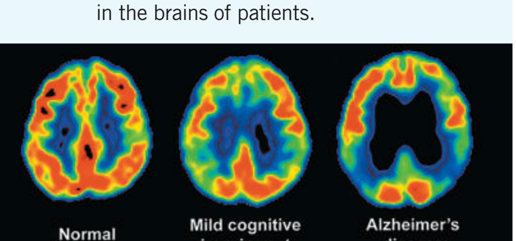
In classical physics we describe the interaction of charged particles in terms of electric and magnetic forces. In quantum mechanics we can describe this interaction in terms of emission and absorption of photons. Two electrons repel each other as one emits a photon and the other absorbs it, just as two skaters can push each other apart by tossing a heavy ball back and forth between them (**Fig. 44.4a**). For an electron and a proton, in which the charges are opposite and the force is attractive, we imagine the skaters trying to grab the ball away from each other (Fig. 44.4b). The electromagnetic interaction between two charged particles is *mediated* or transmitted by photons.

If charged-particle interactions are mediated by photons, where does the energy to create the photons come from? Recall from our discussion of the uncertainty principle (see Sections 38.4 and 39.6) that a state that exists for a short time Δt has an uncertainty ΔE in its energy such that

$$\Delta E\,\Delta t \geq \frac{\hbar}{2} \qquad (44.3)$$

This uncertainty permits the creation of a photon with energy ΔE, provided that it lives no longer than the time Δt given by Eq. (44.3). A photon that can exist for a short time because of this energy uncertainty is called a *virtual photon*. It's as though there were an energy bank; you can borrow energy, provided that you pay it back within the time limit. According to Eq. (44.3), the more you borrow, the sooner you have to pay it back.

BIO APPLICATION Pair Annihilation in Medical Diagnosis A technique called positron emission tomography (PET) can be used to identify the early stages of Alzheimer's disease. A patient is administered a glucose-like compound called FDG in which one oxygen atom is replaced by radioactive ^{18}F. FDG accumulates in active areas of the brain, where glucose metabolism is high. The ^{18}F undergoes β^+ decay (positron emission) with a half-life of 110 minutes, and the emitted positron immediately annihilates with an atomic electron to produce two gamma-ray photons. A scanner detects both photons, then calculates where the annihilation took place—the site of FDG accumulation. These PET images—which show areas of strongest emission, and hence greatest glucose metabolism, in red—reveal changes in the brains of patients.



Mesons

Is there a particle that mediates the *nuclear* force? By the mid-1930s the nuclear force between two nucleons (neutrons or protons) appeared to be described by a potential energy $U(r)$ with the general form

$$U(r) = -f^2\left(\frac{e^{-r/r_0}}{r}\right) \quad \text{(nuclear potential energy)} \tag{44.4}$$

The constant f characterizes the strength of the interaction, and r_0 describes its range. **Figure 44.5** compares the absolute value of this function with the function f^2/r, which would be analogous to the *electric* interaction of two protons:

$$U(r) = \frac{1}{4\pi\epsilon_0}\frac{e^2}{r} \quad \text{(electric potential energy)} \tag{44.5}$$

In 1935 the Japanese physicist Hideki Yukawa suggested that a hypothetical particle that he called a **meson** might mediate the nuclear force. He showed that the range of the force was related to the mass of the particle. Yukawa argued that the particle must live for a time Δt long enough to travel a distance comparable to the range r_0 of the nuclear force. This range was known from the sizes of nuclei and other information to be about 1.5×10^{-15} m $= 1.5$ fm. If we assume that an average particle's speed is comparable to c and travels about half the range, its lifetime Δt must be about

$$\Delta t = \frac{r_0}{2c} = \frac{1.5 \times 10^{-15}\ \text{m}}{2(3.0 \times 10^8\ \text{m/s})} = 2.5 \times 10^{-24}\ \text{s}$$

From Eq. (44.3), the minimum necessary uncertainty ΔE in energy is

$$\Delta E = \frac{\hbar}{2\Delta t} = \frac{1.05 \times 10^{-34}\ \text{J}\cdot\text{s}}{2(2.5 \times 10^{-24}\ \text{s})} = 2.1 \times 10^{-11}\ \text{J} = 130\ \text{MeV}$$

The mass equivalent Δm of this energy is about 250 times the electron mass:

$$\Delta m = \frac{\Delta E}{c^2} = \frac{2.1 \times 10^{-11}\ \text{J}}{(3.00 \times 10^8\ \text{m/s})^2} = 2.3 \times 10^{-28}\ \text{kg} = 130\ \text{MeV}/c^2$$

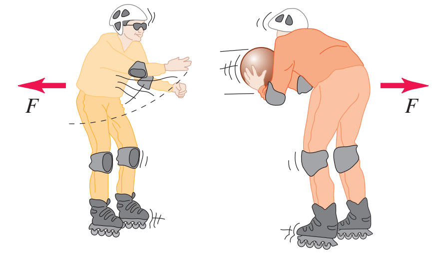
Yukawa postulated that an as yet undiscovered particle with this mass serves as the messenger for the nuclear force.

A year later, Carl Anderson and his colleague Seth Neddermeyer discovered in cosmic radiation two new particles, now called **muons.** The μ^- has charge equal to that of the electron, and its antiparticle the μ^+ has a positive charge with equal magnitude. The two particles have equal mass, about 207 times the electron mass. But it soon became clear that muons were *not* Yukawa's particles because they interacted with nuclei only very weakly.

In 1947 a family of three particles, called π *mesons* or **pions,** were discovered. Their charges are $+e$, $-e$, and zero, and their masses are about 270 times the electron mass. The pions interact strongly with nuclei, and they *are* the particles predicted by Yukawa. Other, heavier mesons, the ω and ρ, evidently also act as shorter-range messengers of the nuclear force. The complexity of this explanation suggests that the nuclear force has simpler underpinnings; these involve the quarks and gluons that we'll discuss in Section 44.4. Before discussing mesons further, we'll describe some particle accelerators and detectors to see how mesons and other particles are created in a controlled fashion and observed.

Figure **44.4** An analogy for how particles act as force mediators.

(a) Two skaters exert repulsive forces on each other by tossing a ball back and forth.



(b) Two skaters exert attractive forces on each other when one tries to grab the ball out of the other's hands.

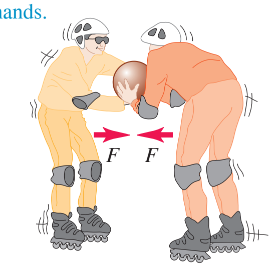
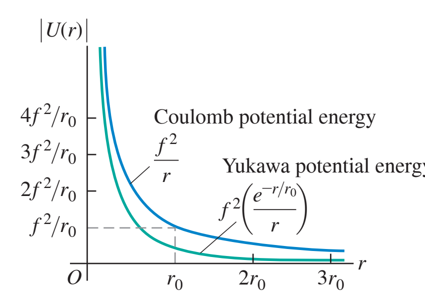


Figure **44.5** Graph of the magnitude of the Yukawa potential-energy function for nuclear forces, $|U(r)| = f^2 e^{-r/r_0}/r$. The function $U(r) = f^2/r$, proportional to the potential energy for Coulomb's law, is also shown. The two functions are similar at small r, but the Yukawa potential energy drops off much more quickly at large r.



TEST YOUR UNDERSTANDING OF SECTION 44.1 Each of the following particles can be exchanged between two protons, two neutrons, or a neutron and a proton as part of the nuclear force. Rank the particles in order of the range of the interaction that they mediate, from largest to smallest range. (i) The π^+ (pi-plus) meson, mass 140 MeV/c^2; (ii) the ρ^+ (rho-plus) meson, mass 776 MeV/c^2; (iii) the η^0 (eta-zero) meson, mass 548 MeV/c^2; (iv) the ω^0 (omega-zero) meson, mass 783 MeV/c^2.

ANSWER

(i), (iii), (ii), (iv) The more massive the virtual particle, the shorter its lifetime and the shorter the distance that it can travel during its lifetime.

44.2 PARTICLE ACCELERATORS AND DETECTORS

Early nuclear physicists used alpha and beta particles from naturally occurring radioactive elements for their experiments, but they were restricted in energy to the few MeV that are available in such random decays. Present-day particle accelerators can produce precisely controlled beams of particles, from electrons and positrons up to heavy ions, with a wide range of energies. These beams have three main uses. First, high-energy particles can collide to produce new particles, just as a collision of an electron and a positron can produce photons. Second, a high-energy particle has a short de Broglie wavelength and so can probe the small-scale interior structure of other particles, just as electron microscopes (see Section 39.1) can give better resolution than optical microscopes. Third, they can be used to produce nuclear reactions of scientific or medical use.

BIO APPLICATION Linear Accelerators in Medicine Electron linear accelerators that provide a kinetic energy of 4–20 MeV are important tools in the treatment of many cancers. The electrons themselves are used to irradiate superficial tumors. Alternatively, the electrons can be directed at a metal target; then bremsstrahlung (see Section 38.2) produces x rays that are used to irradiate tumors that lie deeper inside the patient.

Linear Accelerators

Particle accelerators use electric and magnetic fields to accelerate and guide beams of charged particles. A *linear accelerator* (linac) accelerates particles in a straight line. J. J. Thomson's cathode-ray tubes were early examples of linacs. Modern linacs use a series of electrodes with gaps to give the particles a series of boosts. Most present-day high-energy linear accelerators use a traveling electromagnetic wave; the charged particles "ride" the wave in more or less the way that a surfer rides an incoming ocean wave. In the highest-energy linac in the world today, at the SLAC National Accelerator Laboratory, electrons and positrons can be accelerated to 50 GeV in a tube 3 km long. At this energy their de Broglie wavelengths are 0.025 fm, much smaller than the size of a proton or a neutron.

The Cyclotron

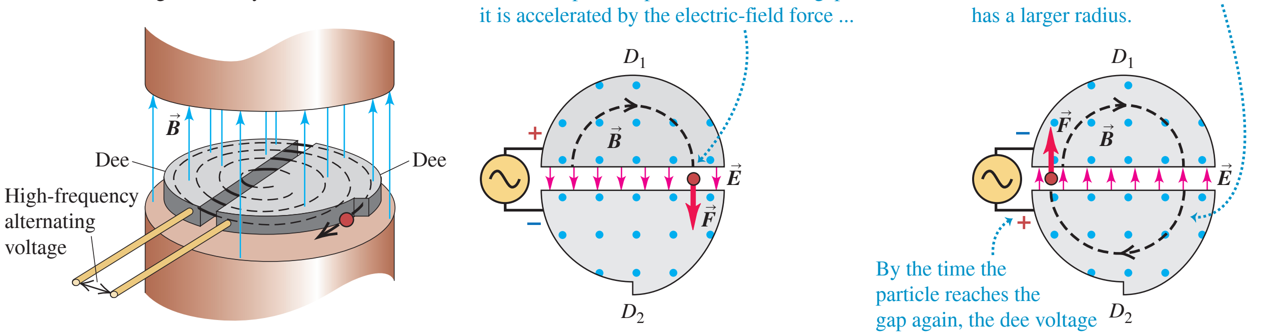
Many accelerators use magnets to deflect the charged particles into circular paths. The first was the *cyclotron,* invented in 1931 by E. O. Lawrence and M. Stanley Livingston at the University of California (**Fig. 44.6a**). Particles with mass m and charge q move inside a vacuum chamber in a uniform magnetic field $\vec{B}$ that is perpendicular to the plane of

Figure 44.6 Layout and operation of a cyclotron.

(a) Schematic diagram of a cyclotron

(b) As the positive particle reaches the gap, it is accelerated by the electric-field force ...

(c) ... and the next semicircular orbit has a larger radius.



their paths. In Section 27.4 we showed that in such a field, a particle with speed v moves in a circular path with radius r given by

$$r = \frac{mv}{|q|B} \tag{44.6}$$

and with angular speed (angular frequency) ω given by

$$\omega = \frac{v}{r} = \frac{|q|B}{m} \tag{44.7}$$

An alternating potential difference is applied between the two hollow electrodes D_1 and D_2 (called *dees*), creating an electric field in the gap between them. The polarity of the potential difference and electric field is changed precisely twice each revolution (Figs. 44.6b and 44.6c), so that the particles get a push each time they cross the gap. The pushes increase their speed and kinetic energy, boosting them into paths of larger radius. The maximum speed v_{max} and kinetic energy K_{max} are determined by the radius R of the largest possible path. Solving Eq. (44.6) for v, we find $v = |q|Br/m$ and $v_{\text{max}} = |q|BR/m$. Assuming nonrelativistic speeds, we have

$$K_{\text{max}} = \tfrac{1}{2}mv_{\text{max}}{}^2 = \frac{q^2B^2R^2}{2m} \tag{44.8}$$

EXAMPLE 44.1 Frequency and energy in a proton cyclotron

WITH VARIATION PROBLEMS

One cyclotron built during the 1930s has a path of maximum radius 0.500 m and a magnetic field of magnitude 1.50 T. If it is used to accelerate protons, find (a) the frequency of the alternating voltage applied to the dees and (b) the maximum particle energy.

IDENTIFY and SET UP The frequency f of the applied voltage must equal the frequency of the proton orbital motion. Equation (44.7) gives the *angular* frequency ω of the proton orbital motion; we find f from $f = \omega/2\pi$. The proton reaches its maximum energy K_{max}, given by Eq. (44.8), when the radius of its orbit equals the radius of the dees.

EXECUTE (a) For protons, $q = 1.60 \times 10^{-19}$ C and $m = 1.67 \times 10^{-27}$ kg. From Eq. (44.7),

$$f = \frac{\omega}{2\pi} = \frac{|q|B}{2\pi m} = \frac{(1.60 \times 10^{-19}\ \text{C})(1.50\ \text{T})}{2\pi(1.67 \times 10^{-27}\ \text{kg})}$$
$$= 2.3 \times 10^7\ \text{Hz} = 23\ \text{MHz}$$

(b) From Eq. (44.8) the maximum kinetic energy is

$$K_{\text{max}} = \frac{(1.60 \times 10^{-19}\ \text{C})^2(1.50\ \text{T})^2(0.50\ \text{m})^2}{2(1.67 \times 10^{-27}\ \text{kg})}$$
$$= 4.3 \times 10^{-12}\ \text{J} = 2.7 \times 10^7\ \text{eV} = 27\ \text{MeV}$$

This proton kinetic energy is much larger than that available from natural radioactive sources.

EVALUATE From Eq. (44.6) or Eq. (44.7), the proton speed is $v = 7.2 \times 10^7$ m/s, which is about 25% of the speed of light. At such speeds, relativistic effects are beginning to become important. Since we ignored these effects in our calculation, the results for f and K_{max} are in error by a few percent; this is why we kept only two significant figures.

KEYCONCEPT In the type of particle accelerator called a cyclotron, particles follow a circular path in a plane perpendicular to a uniform magnetic field. Every one-half orbit the particles pass through a potential difference that increases their kinetic energy and the radius of their circular path. The maximum radius of the cyclotron path determines the maximum kinetic energy attainable.

The maximum energy that can be attained with a cyclotron is limited by relativistic effects. The relativistic version of Eq. (44.7) is

$$\omega = \frac{|q|B}{m}\sqrt{1 - v^2/c^2}$$

As the particles speed up, their angular frequency ω *decreases,* and their motion gets out of phase with the alternating dee voltage. In the *synchrocyclotron* the particles are accelerated in bursts. For each burst, the frequency of the alternating voltage is decreased as the particles speed up, maintaining the correct phase relationship with the particles' motion.

Another limitation of the cyclotron is the difficulty of building very large electromagnets. The largest synchrocyclotron ever built has a vacuum chamber that is about 8 m in diameter and accelerates protons to energies of about 600 MeV.

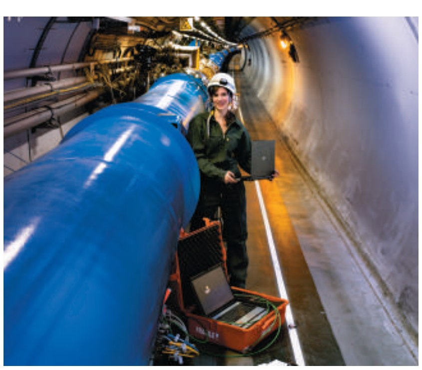
The Synchrotron

To attain higher energies, a type of machine called the *synchrotron* is more practical. Particles move in a vacuum chamber in the form of a thin doughnut called the *accelerating ring*. The particle beam is bent to follow the ring by a series of electromagnets placed around the ring. As the particles speed up, the magnetic field is increased so that the particles retrace the same trajectory over and over. The Large Hadron Collider (LHC) near Geneva, Switzerland, is the highest-energy accelerator in the world (**Fig. 44.7**). It is designed to accelerate protons to a maximum energy of 7 TeV, or 7×10^{12} eV. (As we'll see in Section 44.3, *hadrons* are a class of elementary particles that includes protons and neutrons.)

Figure 44.7 (a) The Large Hadron Collider at the European Organization for Nuclear Research (CERN). The underground accelerating ring (shown by the red circle) is 100 m underground and 8.5 km in diameter, so large that it spans the border between Switzerland and France. (Note the Alps in the background.) When accelerated to 7 TeV, protons travel around the ring more than 11,000 times per second. (b) An engineer working on one of the 9593 superconducting electromagnets around the LHC ring.

(a)

(b)



As we pointed out in Section 32.1, accelerated charges radiate electromagnetic energy. In an accelerator in which the particles move in curved paths, this radiation is often called *synchrotron radiation*. High-energy accelerators are typically constructed underground to provide protection from this radiation. From the accelerator standpoint, synchrotron radiation is undesirable, since the energy given to an accelerated particle is radiated right back out. It can be minimized by making the accelerator radius r large so that the centripetal acceleration v^2/r is small. On the positive side, synchrotron radiation is used as a source of well-controlled high-frequency electromagnetic waves.

Available Energy

When a beam of high-energy particles collides with a stationary target, not all of the energy of the incident particles is *available* to form new particle states. Because momentum must be conserved, the particles emerging from the collision must have some net motion and thus some kinetic energy. The discussion following Example 43.11 (Section 43.6) presented a nonrelativistic example of this principle. The available energy is greatest in the frame of reference in which the total momentum is zero. We call this the *center-of-momentum system;* it is the relativistic generalization of the center-of-mass system that we discussed in Section 8.5. In this system the total kinetic energy after the collision can be zero, so that the maximum amount of the initial energy (the sum of the rest energies and kinetic energies of the colliding particles) becomes available to cause the reaction being studied.

Consider the *laboratory system,* in which a target particle with mass M is initially at rest and is bombarded by a particle with mass m and total energy (including rest energy) E_m. The total available energy E_a in the center-of-momentum system (including rest energies of all the particles) can be shown to be

$$E_a^2 = 2Mc^2E_m + (Mc^2)^2 + (mc^2)^2 \quad \text{(available energy)} \tag{44.9}$$

When the masses of the target and projectile particles are equal, we can simplify:

$$E_a^2 = 2mc^2(E_m + mc^2) \quad \text{(available energy, equal masses)} \tag{44.10}$$

If in addition E_m is much greater than mc^2, we can ignore the second term in the parentheses in Eq (44.10). Then E_a is

$$E_a = \sqrt{2mc^2E_m} \quad \text{(available energy, equal masses, } E_m \gg mc^2\text{)} \tag{44.11}$$

CAUTION Available energy includes rest energies Note that the available energy given by Eqs. (44.9), (44.10), and (44.11) is the energy that goes into *all* of the particles that are present after the collision. If the initial particles survive the collision (see Example 44.2), some of the available energy goes back to these survivors; the rest goes to whatever additional particles are produced in the collision.

The square root in Eq. (44.11) means that with a stationary target, doubling the energy E_m of the bombarding particle increases the available energy E_a by only a factor of $\sqrt{2} = 1.414$. Examples 44.2 and 44.3 explore the limitations of having a stationary target particle.

EXAMPLE 44.2 Threshold energy for pion production

WITH VARIATION PROBLEMS

A proton (rest energy 938 MeV) with kinetic energy K collides with a proton at rest. Both protons survive the collision, and a neutral pion (π^0, rest energy 135 MeV) is produced. What is the threshold energy (minimum value of K) for this process?

IDENTIFY and SET UP The final state includes the two original protons (mass m) and the pion (mass m_π). The threshold energy corresponds to the minimum-energy case in which all three particles are at rest in the center-of-momentum system. The total available energy E_a in that system must be at least the total rest energy, $2mc^2 + m_\pi c^2$. We use this to solve Eq. (44.10) for the total energy E_m of the bombarding proton; the kinetic energy K (our target variable) is then E_m minus the proton rest energy mc^2.

EXECUTE We substitute $E_a = 2mc^2 + m_\pi c^2$ into Eq. (44.10), simplify, and solve for E_m:

$$4m^2c^4 + 4mm_\pi c^4 + m_\pi^2 c^4 = 2mc^2E_m + 2(mc^2)^2$$

$$E_m = mc^2 + m_\pi c^2\left(2 + \frac{m_\pi c^2}{2mc^2}\right) = mc^2 + K$$

$$K = m_\pi c^2\left(2 + \frac{m_\pi c^2}{2mc^2}\right) = (135\text{ MeV})\left(2 + \frac{(135\text{ MeV})}{2(938\text{ MeV})}\right) = 280\text{ MeV}$$

EVALUATE The bombarding proton's kinetic energy must be more than twice the pion rest energy of 135 MeV. By contrast, in Example 37.11 (Section 37.8) we found that a pion can be produced in a head-on collision of two protons, each with only 67.5 MeV of kinetic energy. We discuss the energy advantage of such collisions in the next subsection.

KEYCONCEPT The available energy E_a in a particle collision is the energy available to produce the particles that are present after the collision. This equals the total energy of the colliding particles in the center-of-momentum system, a frame in which the total momentum of the particles is zero. Given E_a, you can calculate the required kinetic energy in a different frame in which one of the particles is initially at rest.

EXAMPLE 44.3 Increasing the available energy

WITH VARIATION PROBLEMS

The Fermilab accelerator in Illinois was designed to bombard stationary targets with 800 GeV protons. (a) What is the available energy E_a in a proton-proton collision? (b) What is E_a if the beam energy is increased to 980 GeV?

IDENTIFY and SET UP Our target variable is the available energy E_a in a stationary-target collision between identical particles. In both parts (a) and (b) the beam energy E_m is much larger than the proton rest energy $mc^2 = 938\text{ MeV} = 0.938\text{ GeV}$, so we can safely use the approximation of Eq. (44.11).

EXECUTE (a) For $E_m = 800$ GeV, Eq. (44.11) gives

$$E_a = \sqrt{2(0.938\text{ GeV})(800\text{ GeV})} = 38.7\text{ GeV}$$

(b) For $E_m = 980$ GeV,

$$E_a = \sqrt{2(0.938\text{ GeV})(980\text{ GeV})} = 42.9\text{ GeV}$$

EVALUATE With a stationary-proton target, increasing the proton beam energy by 180 GeV increases the available energy by only 4.2 GeV! This shows a major limitation of experiments in which one of the colliding particles is initially at rest. Below we describe how physicists can overcome this limitation.

KEYCONCEPT In experiments in which a fast-moving particle strikes a second target particle that is initially stationary, a large increase in the energy of the moving particle produces only a relatively small increase in the available energy.

Colliding Beams

The limitation illustrated by Example 44.3 doesn't apply to *colliding-beam* experiments. In these experiments there is no stationary target; instead, beams of particles moving in opposite directions are tightly focused onto one another so that head-on collisions can occur. Usually the two colliding particles have momenta of equal magnitude and opposite direction, so the total momentum is zero. Hence the laboratory system is also the center-of-momentum system, and the available energy is maximized.

The highest-energy colliding beams available are those at the Large Hadron Collider (see Fig. 44.7). In operation, 2808 bunches of 7 TeV protons circulate around the ring, half in one direction and half in the opposite direction. Each bunch contains about 10^{11} protons. Magnets steer the oppositely moving bunches to collide at interaction points. The available energy E_a in the resulting head-on collisions is the total energy of the two colliding particles: $E_a = 2 \times 7\text{ TeV} = 14\text{ TeV}$. The very large available energy at the Large Hadron Collider makes it possible to produce particles that have never been seen before (see Section 44.5).

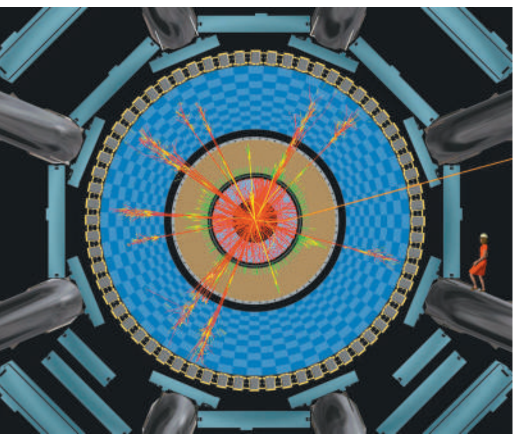


Figure 44.8 This computer-generated image shows the result of a simulated collision between two protons (not shown) in one of the interaction regions at the Large Hadron Collider. The view is along the beampipe. The different color tracks show different types of particles emerging from the collision. A variety of different detectors surround the collision region. (Note the woman in a red dress, drawn for scale.)

Detectors

A wide variety of devices have been designed to measure the properties of subatomic particles. Many detectors use the ionization caused by charged particles as they move through a gas, liquid, or solid. The ions along the particle's path give rise to droplets of liquid in the supersaturated vapor of a cloud chamber (Fig. 44.1) or cause small volumes of vapor in the superheated liquid of a bubble chamber (Fig. 44.3a). In a semiconducting solid the ionization can take the form of electron–hole pairs. We discussed their detection in Section 42.7. *Wire chambers* contain arrays of closely spaced wires that detect the ions. The charge collected and time information from each wire are processed by using computers to reconstruct the particle trajectories. The detectors at the Large Hadron Collider use an array of devices to follow the tracks of particles produced by collisions between protons (**Fig. 44.8**). The giant solenoid in the photo that opens Chapter 28 is at the heart of one these detector arrays. The intense magnetic field of the solenoid helps identify newly produced particles, which curve in different directions and along paths of different radii depending on their charge and energy.

Cosmic-Ray Experiments

Large numbers of particles called *cosmic rays* continually bombard the earth from sources both within and beyond our galaxy. These particles consist mostly of neutrinos, protons, and heavier nuclei, with energies ranging from less than 1 MeV to more than 10^{20} eV. The earth's atmosphere and magnetic field protect us from much of this radiation. This means that cosmic-ray experimentation often must be carried out above all or most of the atmosphere by means of rockets or high-altitude balloons.

In contrast, neutrino detectors are buried below the earth's surface in tunnels or mines or submerged deep in the ocean. This is done to screen out all other types of particles so that only neutrinos, which interact only very weakly with matter, reach the detector. Because neutrino interactions with matter are so weak, neutrino detectors must consist of huge amounts of matter: The Super-Kamiokande detector looks for flashes of light produced when a neutrino interacts in a tank containing 5×10^7 kg of water (see Section 44.5).

Cosmic rays were important in early particle physics, and their study currently brings us important information about the rest of the universe. Although cosmic rays provide a source of high-energy particles that does not depend on expensive accelerators, most particle physicists use accelerators because the high-energy cosmic-ray particles they want are too few and too random.

TEST YOUR UNDERSTANDING OF SECTION 44.2 In a colliding-beam experiment, a 90 GeV electron collides head-on with a 90 GeV positron. The electron and the positron annihilate each other, forming a single virtual photon that then transforms into other particles. Does the virtual photon obey the same relationship $E = pc$ as real photons do?

ANSWER

no In a head-on collision between an electron and a positron of equal energy, the net momentum is zero. Since both momentum and energy are conserved in the collision, the virtual photon also has momentum $p = 0$ but has energy $E = 90\text{ GeV} + 90\text{ GeV} = 180\text{ GeV}$. Hence the relationship $E = pc$ is definitely *not* true for this virtual photon.

44.3 PARTICLES AND INTERACTIONS

We have mentioned the array of subatomic particles that were known as of 1947: photons, electrons, positrons, protons, neutrons, muons, and pions. Since then, literally hundreds of additional particles have been discovered in accelerator experiments. The vast majority of known particles are *unstable* and decay spontaneously into other particles. Particles of all kinds, whether stable or unstable, can be created or destroyed in interactions between particles. Each such interaction involves the exchange of virtual particles, which exist on borrowed energy allowed by the uncertainty principle.

Although the world of subatomic particles and their interactions is complex, some key results bring order and simplicity to the seeming chaos. One key simplification is that there are only four fundamental types of interactions, each mediated or transmitted by the exchange of certain characteristic virtual particles. Furthermore, not all particles respond to all four kinds of interaction. In this section we'll examine the fundamental interactions more closely and see how physicists classify particles in terms of the ways in which they interact.

Four Forces and Their Mediating Particles

In Section 5.5 we first described the four fundamental types of forces or interactions (**Fig. 44.9**). They are, in order of decreasing strength:

1. The strong interaction
2. The electromagnetic interaction
3. The weak interaction
4. The gravitational interaction

Figure **44.9** The ties that bind us together originate in the fundamental interactions of nature. The nuclei within our bodies are held together by the strong interaction. The electromagnetic interaction binds nuclei and electrons together to form atoms, binds atoms together to form molecules, and binds molecules together to form us.

The *electromagnetic* and *gravitational* interactions are familiar from classical physics. Both are characterized by a $1/r^2$ dependence on distance. In this scheme, the mediating particles for both interactions have mass zero and are stable as ordinary particles. The mediating particle for the electromagnetic interaction is the familiar photon, which has spin 1. (That means its spin quantum number is $s = 1$, so the magnitude of its spin angular momentum is $S = \sqrt{s(s+1)}\,\hbar = \sqrt{2}\,\hbar$.) The mediating particle for the gravitational force is the spin-2 *graviton* $(s = 2, S = \sqrt{s(s+1)}\,\hbar = \sqrt{6}\,\hbar)$. The graviton has not yet been observed experimentally because the gravitational force is very much weaker than the electromagnetic force. For example, the gravitational attraction of two protons is smaller than their electrical repulsion by a factor of about 10^{36}. The gravitational force is of primary importance in the structure of stars and the large-scale behavior of the universe, but it is not believed to play a significant role in particle interactions at the energies that are currently attainable.

The other two forces are less familiar. One, usually called the *strong interaction,* is responsible for the nuclear force and also for the production of pions and several other particles in high-energy collisions. At the most fundamental level, the mediating particle for the strong interaction is called a *gluon*. However, the force between nucleons is more easily described in terms of mesons as the mediating particles. We'll discuss the spin-1, massless gluon in Section 44.4.

Equation (44.4) is a possible potential-energy function for the nuclear force. The strength of the interaction is described by the constant f^2, which has units of energy times distance. A better basis for comparison with other forces is the dimensionless ratio $f^2/\hbar c$, called the *coupling constant* for the interaction. (We invite you to verify that this ratio is a pure number and so must have the same value in all systems of units.) The observed behavior of nuclear forces suggests that $f^2/\hbar c \approx 1$. The dimensionless coupling constant for *electromagnetic* interactions is the fine-structure constant, which we introduced in Section 41.5:

$$\frac{1}{4\pi\epsilon_0}\frac{e^2}{\hbar c} = 7.2974 \times 10^{-3} = \frac{1}{137.04} \tag{44.12}$$

Thus the strong interaction is roughly 100 times as strong as the electromagnetic interaction; however, it drops off with distance more quickly than $1/r^2$.

The fourth interaction is called the *weak* interaction. It is responsible for beta decay, such as the conversion of a neutron into a proton, an electron, and an antineutrino. It is also responsible for the decay of many unstable particles (pions into muons, muons into electrons, and so on). Its mediating particles are the short-lived particles W^+, W^-, and Z^0. The existence of these particles was confirmed in 1983 in experiments at CERN, for which Carlo Rubbia and Simon van der Meer were awarded the Nobel Prize in 1984.

TABLE 44.1 Four Fundamental Interactions

Interaction	Relative Strength	Range	Mediating Particle: Name	Mass	Charge	Spin
Strong	1	Short ($\sim$1 fm)	Gluon	0	0	1
Electromagnetic	$\frac{1}{137.04}$	Long ($1/r^2$)	Photon	0	0	1
Weak	10^{-9}	Short ($\sim$0.001 fm)	$W^{\pm}$, Z^0	80.4, 91.2 GeV/c^2	$\pm e$, 0	1
Gravitational	10^{-38}	Long ($1/r^2$)	Graviton	0	0	2

The $W^{\pm}$ and Z^0 have spin 1 like the photon and the gluon, but they are *not* massless. In fact, they have enormous masses, 80.4 GeV/c^2 for the W's and 91.2 GeV/c^2 for the Z^0. With such massive mediating particles the weak interaction has a much shorter range than the strong interaction. It also lives up to its name by being weaker than the strong interaction by a factor of about 10^9.

Table 44.1 compares the main features of these four fundamental interactions.

More Particles

In Section 44.1 we mentioned the discoveries of muons in 1937 and of pions in 1947. The electric charges of the muons and the charged pions have the same magnitude e as the electron charge. The positive muon μ^+ is the antiparticle of the negative muon μ^-. Each has spin $\frac{1}{2}$, like the electron, and a mass of about $207m_e = 106\ \text{MeV}/c^2$. Muons are unstable; each decays with a lifetime of 2.2×10^{-6} s into an electron of the same sign, a neutrino, and an antineutrino.

There are three kinds of pions, all with spin 0; they have *no* spin angular momentum. The π^+ and π^- have masses of $273m_e = 140\ \text{MeV}/c^2$. They are unstable; each $\pi^{\pm}$ decays with a lifetime of 2.6×10^{-8} s into a muon of the same sign along with a neutrino for the π^+ and an antineutrino for the π^-. The π^0 is somewhat less massive, $264m_e = 135\ \text{MeV}/c^2$, and it decays with a lifetime of 8.4×10^{-17} s into two photons. The π^+ and π^- are antiparticles of one another, while the π^0 is its own antiparticle. (That is, there is no distinction between particle and antiparticle for the π^0.)

The existence of the *antiproton* $\overline{\text{p}}$ had been suspected ever since the discovery of the positron. The $\overline{\text{p}}$ was found in 1955, when proton–antiproton $(\text{p}\overline{\text{p}})$ pairs were created by use of a beam of 6 GeV protons from the Bevatron at the University of California, Berkeley. The *antineutron* $\overline{\text{n}}$ was found soon afterward. After 1960, as higher-energy accelerators and more sophisticated detectors were developed, a veritable blizzard of new unstable particles were identified. To describe and classify them, we need a small blizzard of new terms.

Initially, particles were classified by mass into three categories: (1) leptons ("light ones" such as electrons); (2) mesons ("intermediate ones" such as pions); and (3) baryons ("heavy ones" such as nucleons and more massive particles). But this scheme has been superseded by a more useful one in which particles are classified in terms of their *interactions*. For instance, *hadrons* (which include mesons and baryons) have strong interactions, and *leptons* do not.

In the following discussion we'll also distinguish between **fermions,** which have half-integer spins, and **bosons,** which have zero or integer spins. Fermions obey the exclusion principle, on which the Fermi–Dirac distribution function (see Section 42.5) is based. Bosons do not obey the exclusion principle (there is no limit on how many bosons can occupy the same quantum state) and have a different distribution function, the Bose–Einstein distribution.

TABLE 44.2 The Six Leptons

Particle Name	Symbol	Anti-particle	Mass (MeV/c^2)	L_e	L_μ	L_τ	Lifetime (s)	Principal Decay Modes
Electron	e^-	e^+	0.511	+1	0	0	Stable	
Electron neutrino	ν_e	$\bar{\nu}_e$	$<2 \times 10^{-6}$	+1	0	0	Stable	
Muon	μ^-	μ^+	105.7	0	+1	0	2.20×10^{-6}	$e^- \bar{\nu}_e \nu_\mu$
Muon neutrino	ν_μ	$\bar{\nu}_\mu$	<0.19	0	+1	0	Stable	
Tau	τ^-	τ^+	1777	0	0	+1	2.9×10^{-13}	$\mu^- \bar{\nu}_\mu \nu_\tau$ or $e^- \bar{\nu}_e \nu_\tau$
Tau neutrino	ν_τ	$\bar{\nu}_\tau$	<18.2	0	0	+1	Stable	

Note: In addition to the limits on the individual neutrino masses, there is a much more stringent limit on the *sum* of the masses of the three types of neutrinos. Evidence suggests that this sum is less than about 3×10^{-7} MeV/c^2 = 0.3 eV/c^2.

Leptons

The **leptons,** which do not have strong interactions, include six particles: the electron (e^-) and its neutrino (ν_e), the muon (μ^-) and its neutrino (ν_μ), and the tau particle (τ^-) and its neutrino (ν_τ). Each of these has a distinct antiparticle. All leptons have spin $\frac{1}{2}$ and thus are fermions. **Table 44.2** shows the family of leptons. The taus have mass $3478m_e = 1777\ \text{MeV}/c^2$. Taus and muons are unstable; a τ^- decays into a μ^- plus a tau neutrino and a muon antineutrino, or an electron plus a tau neutrino and an electron antineutrino. A μ^- decays into an electron plus a muon neutrino and an electron antineutrino. They have relatively long lifetimes because their decays are mediated by the weak interaction. Despite their zero charge, a neutrino is distinct from an antineutrino; the spin angular momentum of a neutrino has a component that is opposite its linear momentum, while for an antineutrino that component is parallel to its linear momentum. Because neutrinos are so elusive, physicists have only been able to place upper limits on the rest masses of the ν_e, the ν_μ, and the ν_τ. It was thought that the rest masses of the neutrinos were zero; compelling recent evidence indicates that they have small but nonzero masses. We'll return to this point and its implications later.

Leptons obey a *conservation principle.* Corresponding to the three pairs of leptons are three lepton numbers L_e, L_μ, and L_τ. The electron e^- and the electron neutrino ν_e are assigned $L_e = 1$, and their antiparticles e^+ and $\bar{\nu}_e$ are given $L_e = -1$. Corresponding assignments of L_μ and L_τ are made for the μ and τ particles and their neutrinos.

In all interactions, each lepton number is separately conserved.

For example, in the decay of the μ^-, the lepton numbers are

$$\begin{array}{ccccccc} \mu^- & \rightarrow & e^- & + & \bar{\nu}_e & + & \nu_\mu \\ L_\mu = 1 & & L_e = 1 & & L_e = -1 & & L_\mu = 1 \end{array}$$

These conservation principles have no counterpart in classical physics.

EXAMPLE 44.4 Lepton number conservation

Check conservation of lepton numbers for these decay schemes:

(a) $\mu^+ \rightarrow e^+ + \nu_e + \bar{\nu}_\mu$

(b) $\pi^- \rightarrow \mu^- + \bar{\nu}_\mu$

(c) $\pi^0 \rightarrow \mu^- + e^+ + \nu_e$

IDENTIFY and SET UP Lepton number conservation requires that L_e, L_μ, and L_τ (given in Table 44.2) separately have the same sums after the decay as before.

EXECUTE We tabulate L_e and L_μ for each decay scheme. An antiparticle has the opposite lepton number from its corresponding particle listed in Table 44.2. No τ particles or τ neutrinos appear in any of the schemes, so $L_\tau = 0$ both before and after each decay and L_τ is conserved.

(a) $\mu^+ \rightarrow e^+ + \nu_e + \bar{\nu}_\mu$

L_e: $0 = -1 + 1 + 0$

L_μ: $-1 = 0 + 0 + (-1)$

(b) $\pi^- \rightarrow \mu^- + \bar{\nu}_\mu$

L_e: $0 = 0 + 0$

L_μ: $0 = 1 + (-1)$

Continued

(c) $\pi^0 \rightarrow \mu^- + e^+ + \nu_e$

L_e: $0 = 0 + (-1) + 1$

L_μ: $0 \neq 1 + 0 + 0$

EVALUATE Decays (a) and (b) are consistent with lepton number conservation and are observed. Decay (c) violates the conservation of L_μ and has *never* been observed. Physicists used these and other experimental results to deduce the principle that all three lepton numbers must separately be conserved.

KEYCONCEPT There are three types of lepton number: electron lepton number L_e, muon lepton number L_μ, and tau lepton number L_τ. Each of these is separately conserved in particle interactions; no violations of these conservation laws have ever been observed.

Hadrons

Hadrons, the strongly interacting particles, are a more complex family than leptons. Each hadron has an antiparticle, often denoted with an overbar, as with the antiproton $\overline{\text{p}}$. There are two subclasses of hadrons: *mesons* and *baryons*. **Table 44.3** shows some of the many hadrons that are currently known. (We'll explain *strangeness* and *quark content* later in this section and in the next one.)

Mesons include the pions that have already been mentioned, K mesons or *kaons,* η mesons, and others that we'll discuss later. Mesons have spin 0 or 1 and therefore are all bosons. There are no stable mesons; all mesons decay to less massive particles, obeying all the conservation laws for such decays.

Baryons include the nucleons and several particles called *hyperons,* including the Λ, Σ, Ξ, and Ω. These resemble nucleons but are more massive. Baryons have half-integer spin, and therefore all are fermions. The only stable baryon is the proton; a free neutron decays to a proton, and hyperons decay to other hyperons or to nucleons by various processes. Baryons obey the *conservation of baryon number,* analogous to conservation of lepton numbers, again with no counterpart in classical physics. We assign a baryon number $B = 1$ to each baryon (p, n, Λ, Σ, and so on) and $B = -1$ to each antibaryon ($\overline{\text{p}}$, $\overline{\text{n}}$, $\overline{\Lambda}$, $\overline{\Sigma}$, and so on).

In all interactions, the total baryon number is conserved.

This principle is the reason the mass number A was conserved in all of the nuclear reactions that we studied in Chapter 43.

TABLE 44.3 Some Hadrons and Their Properties

Particle	Mass (MeV/c^2)	Charge Ratio, Q/e	Spin	Baryon Number, B	Strangeness, S	Mean Lifetime (s)	Typical Decay Modes	Quark Content
Mesons								
π^0	135.0	0	0	0	0	8.5×10^{-17}	$\gamma\gamma$	$u\overline{u}$, $d\overline{d}$
π^+	139.6	+1	0	0	0	2.60×10^{-8}	$\mu^+\nu_\mu$	$u\overline{d}$
π^-	139.6	−1	0	0	0	2.60×10^{-8}	$\mu^-\overline{\nu}_\mu$	$\overline{u}d$
K^+	493.7	+1	0	0	+1	1.24×10^{-8}	$\mu^+\nu_\mu$	$u\overline{s}$
K^-	493.7	−1	0	0	−1	1.24×10^{-8}	$\mu^-\overline{\nu}_\mu$	$\overline{u}s$
η^0	547.9	0	0	0	0	$\approx 10^{-18}$	$\gamma\gamma$	$u\overline{u}$, $d\overline{d}$, $s\overline{s}$
Baryons								
p	938.3	+1	$\frac{1}{2}$	1	0	Stable	—	uud
n	939.6	0	$\frac{1}{2}$	1	0	880	$pe^-\overline{\nu}_e$	udd
Λ^0	1116	0	$\frac{1}{2}$	1	−1	2.63×10^{-10}	$p\pi^-$ or $n\pi^0$	uds
Σ^+	1189	+1	$\frac{1}{2}$	1	−1	8.02×10^{-11}	$p\pi^0$ or $n\pi^+$	uus
Σ^0	1193	0	$\frac{1}{2}$	1	−1	7.4×10^{-20}	$\Lambda^0\gamma$	uds
Σ^-	1197	−1	$\frac{1}{2}$	1	−1	1.48×10^{-10}	$n\pi^-$	dds
Ξ^0	1315	0	$\frac{1}{2}$	1	−2	2.90×10^{-10}	$\Lambda^0\pi^0$	uss
Ξ^-	1322	−1	$\frac{1}{2}$	1	−2	1.64×10^{-10}	$\Lambda^0\pi^-$	dss
Δ^{++}	1232	+2	$\frac{3}{2}$	1	0	$\approx 10^{-23}$	$p\pi^+$	uuu
Ω^-	1672	−1	$\frac{3}{2}$	1	−3	8.2×10^{-11}	Λ^0K^-	sss
Λ_c^+	2286	+1	$\frac{1}{2}$	1	0	2.0×10^{-13}	$pK^-\pi^+$	udc

EXAMPLE 44.5 Baryon number conservation

Check conservation of baryon number for these reactions:

(a) $\mathrm{n} + \mathrm{p} \rightarrow \mathrm{n} + \mathrm{p} + \mathrm{p} + \overline{\mathrm{p}}$

(b) $\mathrm{n} + \mathrm{p} \rightarrow \mathrm{n} + \mathrm{p} + \overline{\mathrm{n}}$

IDENTIFY and SET UP This example is similar to Example 44.4. We compare the total baryon number before and after each reaction, using data from Table 44.3.

EXECUTE We tabulate the baryon numbers, noting that a baryon has $B = 1$ and an antibaryon has $B = -1$:

(a) $\mathrm{n} + \mathrm{p} \rightarrow \mathrm{n} + \mathrm{p} + \mathrm{p} + \overline{\mathrm{p}}$: $1 + 1 = 1 + 1 + 1 + (-1)$

(b) $\mathrm{n} + \mathrm{p} \rightarrow \mathrm{n} + \mathrm{p} + \overline{\mathrm{n}}$: $1 + 1 \neq 1 + 1 + (-1)$

EVALUATE Reaction (a) is consistent with baryon number conservation. It can occur if enough energy is available in the n + p collision. Reaction (b) violates baryon number conservation and has never been observed.

KEYCONCEPT Baryon number, like the three lepton numbers, is always conserved in particle interactions.

EXAMPLE 44.6 Antiproton creation

What is the minimum proton energy required to produce an antiproton in a collision with a stationary proton?

IDENTIFY and SET UP The reaction must conserve baryon number, charge, and energy. Since the target and bombarding protons are of equal mass and the target is at rest, we determine the minimum energy E_m of the bombarding proton from Eq. (44.10).

EXECUTE Conservation of charge and conservation of baryon number forbid the creation of an antiproton by itself; it must be created as part of a proton–antiproton pair. The complete reaction is

$$\mathrm{p} + \mathrm{p} \rightarrow \mathrm{p} + \mathrm{p} + \mathrm{p} + \overline{\mathrm{p}}$$

For this reaction to occur, the minimum available energy E_a in Eq. (44.10) is the final rest energy $4mc^2$ of three protons and an antiproton. Equation (44.10) then gives

$$(4mc^2)^2 = 2mc^2(E_m + mc^2)$$
$$E_m = 7mc^2$$

EVALUATE The energy E_m of the bombarding proton includes its rest energy mc^2, so its minimum *kinetic* energy must be $6mc^2 = 6(938\text{ MeV}) = 5.63\text{ GeV}$.

The search for the antiproton was a principal reason for the construction of the Bevatron at the University of California, Berkeley, with beam energy of 6 GeV. The search succeeded in 1955, and Emilio Segrè and Owen Chamberlain were later awarded the Nobel Prize for this discovery.

KEYCONCEPT To calculate the minimum energy required to produce a new particle, you must first consider the conservation laws associated with the particle. These laws will tell you which other particles may have to be produced along with the desired one. Once you have done this, you can determine how much available energy is required.

Strangeness

The K mesons and the Λ and Σ hyperons were discovered during the late 1950s. Because of their unusual behavior they were called *strange particles.* They were produced in high-energy collisions such as $\pi^- + \mathrm{p}$, and a K meson and a hyperon were always produced *together.* The relatively high rate of production of these particles suggested that it was a *strong*-interaction process, but their relatively long lifetimes suggested that their decay was a *weak*-interaction process. The K^0 appeared to have *two* lifetimes, one about 9×10^{-11} s and another nearly 600 times longer. Were the K mesons strongly interacting hadrons or not?

The search for the answer to this question led physicists to introduce a new quantity called **strangeness.** The hyperons Λ^0 and $\Sigma^{\pm,0}$ were assigned a strangeness quantum number $S = -1$, and the associated K^0 and K^+ mesons were assigned $S = +1$. The corresponding antiparticles had opposite strangeness, $S = +1$ for $\overline{\Lambda}^0$ and $\overline{\Sigma}^{\pm,0}$ and $S = -1$ for $\overline{\mathrm{K}}^0$ and K^-. Then strangeness was *conserved* in production processes such as

$$\mathrm{p} + \pi^- \rightarrow \Sigma^- + \mathrm{K}^+$$
$$\mathrm{p} + \pi^- \rightarrow \Lambda^0 + \mathrm{K}^0$$

The process

$$\mathrm{p} + \pi^- \rightarrow \mathrm{p} + \mathrm{K}^-$$

does not conserve strangeness and it does not occur.

When strange particles decay individually, strangeness is usually *not* conserved. Typical processes include

$$\Sigma^+ \rightarrow n + \pi^+$$
$$\Lambda^0 \rightarrow p + \pi^-$$
$$K^- \rightarrow \pi^+ + \pi^- + \pi^-$$

In each of these decays, the initial strangeness is 1 or -1, and the final value is zero. All observations of these particles are consistent with the conclusion that *strangeness is conserved in strong interactions but it can change by zero or one unit in weak interactions.* There is no counterpart to the strangeness quantum number in classical physics.

CAUTION **Strangeness vs. spin** Take care not to confuse the symbol S for strangeness with the identical symbol for the magnitude of the spin angular momentum.

Conservation Laws

The decay of strange particles provides our first example of a *conditional conservation law,* one that is obeyed in some interactions and not in others. By contrast, several conservation laws are obeyed in *all* interactions. These include the familiar conservation laws: energy, momentum, angular momentum, and electric charge. These are called *absolute conservation laws.* Baryon number and the three lepton numbers are also conserved in all interactions. Strangeness is conserved in strong and electromagnetic interactions but *not* in all weak interactions.

Two other quantities, which are conserved in some but not all interactions, are useful in classifying particles and their interactions. One is *isospin,* a quantity that is used to describe the charge independence of the strong interactions. The other is *parity,* which describes the comparative behavior of two systems that are mirror images of each other. Isospin is conserved in strong interactions, which are charge independent, but not in electromagnetic or weak interactions. (The electromagnetic interaction is certainly *not* charge independent.) Parity is conserved in strong and electromagnetic interactions but not in weak ones. The Chinese-American physicists T. D. Lee and C. N. Yang received the Nobel Prize in 1957 for laying the theoretical foundations for nonconservation of parity in weak interactions.

This discussion shows that conservation laws provide another basis for classifying particles and their interactions. Each conservation law is also associated with a *symmetry* property of the system. A familiar example is angular momentum. If a system is in an environment that has spherical symmetry, no torque can act on it because the direction of the torque would violate the symmetry. In such a system, total angular momentum is *conserved*. When a conservation law is violated, the interaction may be described as a *symmetry-breaking interaction*.

TEST YOUR UNDERSTANDING OF SECTION 44.3 From conservation of energy, a particle of mass m and rest energy mc^2 can decay only if the decay products have a total mass less than m. (The remaining energy goes into the kinetic energy of the decay products.) Can a proton decay into less massive mesons?

ANSWER

no Mesons all have baryon number $B = 0$, while a proton has $B = 1$. The decay of a proton into one or more mesons would require that baryon number *not* be conserved. No violation of this conservation principle has ever been observed, so the proposed decay is impossible.

44.4 QUARKS AND GLUONS

The leptons form a fairly neat package: three particles and three neutrinos, each with its antiparticle, and a conservation law relating their numbers. Physicists believe that leptons are genuinely fundamental particles. The hadron family, by comparison, is a mess. Table 44.3 (in Section 44.3) contains only a sample of well over 100 hadrons that have been discovered since 1960, and it has become clear that these particles *do not* represent the most fundamental level of the structure of matter.

Our present understanding of the structure of hadrons is based on a proposal made initially in 1964 by the American physicist Murray Gell-Mann and his collaborators. In this proposal, hadrons are not fundamental particles but are composite structures whose constituents are spin-$\frac{1}{2}$ fermions called **quarks.** (The name is from the line "Three quarks for Muster Mark!" from *Finnegans Wake,* by James Joyce.) Each baryon is composed of three quarks (qqq), each antibaryon of three antiquarks $(\overline{qqq})$, and each meson of a quark–antiquark pair $(q\overline{q})$. Table 44.3 gives the quark content of many hadrons. No other compositions seem to be necessary. This scheme requires that quarks have electric charges with magnitudes $\frac{1}{3}$ and $\frac{2}{3}$ of the electron charge e, which had been thought to be the smallest unit of charge. Each quark also has a fractional value $\frac{1}{3}$ for its baryon number B, and each antiquark has a baryon-number value $-\frac{1}{3}$. In a meson, a quark and antiquark combine with net baryon number 0 and can have their spin angular momentum components parallel to form a spin-1 meson or antiparallel to form a spin-0 meson. Similarly, the three quarks in a baryon combine with net baryon number 1 and can form a spin-$\frac{1}{2}$ baryon or a spin-$\frac{3}{2}$ baryon.

The Three Original Quarks

The first (1964) quark theory included three types (called *flavors*) of quarks, labeled u (up), d (down), and s (strange). Their principal properties are listed in **Table 44.4**. The corresponding antiquarks $\overline{u}$, $\overline{d}$, and $\overline{s}$ have opposite values of charge Q, B, and S. Protons, neutrons, π and K mesons, and several hyperons can be constructed from these three quarks. For example, the proton quark content is uud. Checking Table 44.4, we see that the values of Q/e add to 1 and that the values of the baryon number B also add to 1, as we should expect. The neutron is udd, with total $Q = 0$ and $B = 1$. The π^+ meson is $u\overline{d}$, with $Q/e = 1$ and $B = 0$, and the K^+ meson is $u\overline{s}$. Checking the values of S for the quark content, we see that the proton, neutron, and π^+ have strangeness 0 and that the K^+ has strangeness 1, in agreement with Table 44.3. The antiproton is $\overline{p} = \overline{uud}$, the negative pion is $\pi^- = \overline{u}d$, and so on. The quark content can also be used to explain hadron excited states and magnetic moments. **Figure 44.10** shows the quark content of two baryons and two mesons.

Figure **44.10** Quark content of four hadrons. The various color combinations that are needed for color neutrality are not shown.

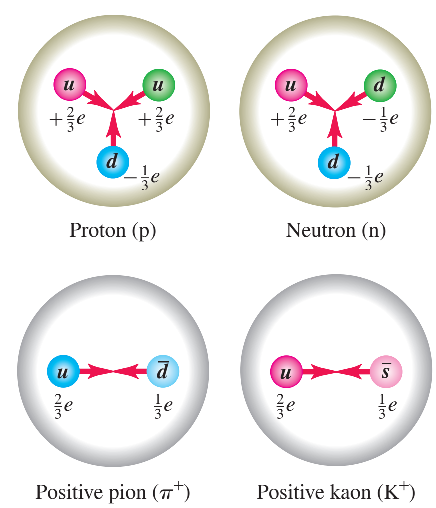


TABLE 44.4 Properties of the Three Original Quarks

Symbol	Q/e	Spin	Baryon Number, B	Strangeness, S	Charm, C	Bottomness, B'	Topness, T
u	$\frac{2}{3}$	$\frac{1}{2}$	$\frac{1}{3}$	0	0	0	0
d	$-\frac{1}{3}$	$\frac{1}{2}$	$\frac{1}{3}$	0	0	0	0
s	$-\frac{1}{3}$	$\frac{1}{2}$	$\frac{1}{3}$	-1	0	0	0

EXAMPLE 44.7 Determining the quark content of baryons

Given that they contain only u, d, s, $\overline{u}$, $\overline{d}$, and/or $\overline{s}$, find the quark content of (a) Σ^+ and (b) $\overline{\Lambda}{}^0$. The Σ^+ and Λ^0 (the antiparticle of the $\overline{\Lambda}{}^0$) are both baryons with strangeness $S = -1$.

IDENTIFY and SET UP We use the idea that the total charge of each baryon is the sum of the individual quark charges, and similarly for the baryon number and strangeness. We use the quark properties given in Table 44.4.

EXECUTE Baryons contain three quarks. If $S = -1$, exactly *one* of the three must be an s quark, which has $S = -1$ and $Q/e = -\frac{1}{3}$.

(a) The Σ^+ has $Q/e = +1$, so the other two quarks must both be u quarks (each of which has $Q/e = +\frac{2}{3}$). Hence the quark content of Σ^+ is uus.

(b) First we find the quark content of the Λ^0. To yield zero total charge, the other two quarks must be u $(Q/e = +\frac{2}{3})$ and d $(Q/e = -\frac{1}{3})$, so the quark content of the Λ^0 is uds. The quark content of the $\overline{\Lambda}{}^0$ is therefore $\overline{u}\,\overline{d}\,\overline{s}$.

EVALUATE Although the Λ^0 and $\overline{\Lambda}{}^0$ are both electrically neutral and have the same mass, they are different particles: Λ^0 has $B = 1$ and $S = -1$, while $\overline{\Lambda}{}^0$ has $B = -1$ and $S = 1$.

KEYCONCEPT All baryons are composed of three quarks, and all antibaryons are composed of three antiquarks. For a baryon or antibaryon, the total electric charge equals the sum of the charges of its constituent quarks or antiquarks; the same is true of the total baryon number and total strangeness.

Motivating the Quark Model

What caused physicists to suspect that hadrons were made up of something smaller? The magnetic moment of the neutron (see Section 43.1) was one of the first reasons. In Section 27.7 we learned that a magnetic moment results from a circulating current (a motion of electric charge). But the neutron has *no* charge or, to be more accurate, no *total* charge. It could be made up of smaller particles whose charges add to zero. The quantum motion of these particles within the neutron would then give its surprising nonzero magnetic moment. To verify this hypothesis by "seeing" inside a neutron, we need a probe with a wavelength that is much smaller than the neutron's size of about a femtometer. This probe should not be affected by the strong interaction, so that it won't interact with the neutron as a whole but will penetrate into it and interact electromagnetically with these supposed smaller charged particles. A probe with these properties is an electron with energy greater than 10 GeV. In experiments carried out at SLAC, such electrons were scattered from neutrons and protons to help show that nucleons are indeed made up of fractionally charged, spin-$\frac{1}{2}$ pointlike particles.

The Eightfold Way

Symmetry considerations play a very prominent role in particle theory. Here are two examples. Consider the eight spin-$\frac{1}{2}$ baryons we've mentioned: the familiar p and n; the strange Λ^0, Σ^+, Σ^0, and Σ^-; and the doubly strange Ξ^0 and Ξ^-. For each we plot the value of strangeness S versus the value of charge Q in **Fig. 44.11**. The result is a hexagonal pattern. A similar plot for the nine spin-0 mesons (six shown in Table 44.3 plus three others not included in that table) is shown in **Fig. 44.12**; the particles fall in exactly the same hexagonal pattern! In each plot, all the particles have masses that are within about $\pm 200\ \text{MeV}/c^2$ of the median mass value of that plot, with variations due to differences in quark masses and internal potential energies.

The symmetries that lead to these and similar patterns are collectively called the **eightfold way.** They were discovered in 1961 by Murray Gell-Mann and independently by Yuval Néeman. (The name is a slightly irreverent reference to the Noble Eightfold Path, a set of principles for right living in Buddhism.) A similar pattern for the spin-$\frac{3}{2}$ baryons contains *ten* particles, arranged in a triangular pattern like pins in a bowling alley. When this pattern was first discovered, one of the particles was missing. But Gell-Mann gave

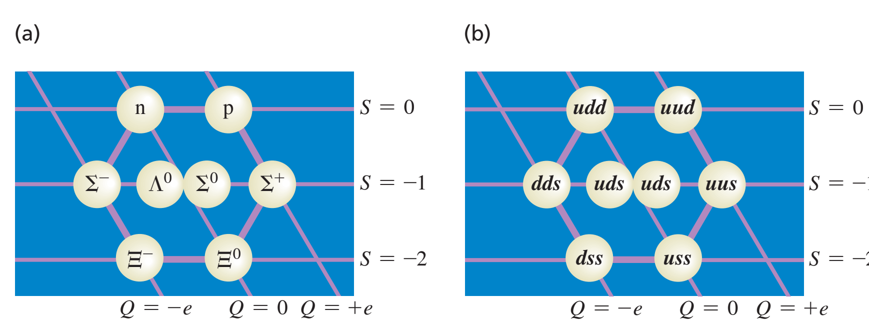


Figure 44.11 (a) Plot of S and Q values for spin-$\frac{1}{2}$ baryons, showing the symmetry pattern of the eightfold way. (b) Quark content of each spin-$\frac{1}{2}$ baryon. The quark contents of the Σ^0 and Λ^0 are the same; the Σ^0 is an excited state of the Λ^0 and can decay into it by photon emission.

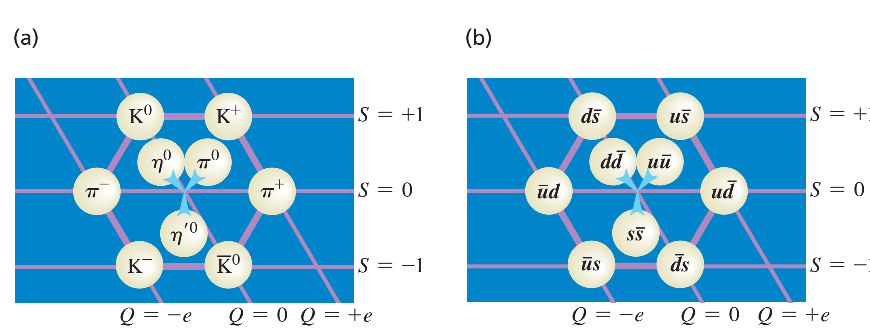


Figure 44.12 (a) Plot of S and Q values for nine spin-0 mesons, showing the symmetry pattern of the eightfold way. Each particle is on the opposite side of the hexagon from its antiparticle; each of the three particles in the center is its own antiparticle. (b) Quark content of each spin-0 meson. The particles in the center are different mixtures of the three quark–antiquark pairs shown.

it a name anyway (Ω^-), predicted the properties it should have, and told experimenters what they should look for. Three years later, the particle was found during an experiment at Brookhaven National Laboratory, a spectacular success for Gell-Mann's theory. The whole series of events is reminiscent of the way in which Mendeleev used gaps in the periodic table of the elements to predict properties of undiscovered elements and to guide chemists in their search for these elements.

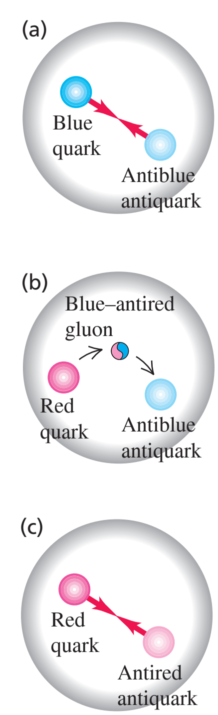
What binds quarks to one another? The attractive interactions among quarks are mediated by massless spin-1 bosons called **gluons** in much the same way that photons mediate the electromagnetic interaction or that pions mediated the nucleon–nucleon force in the old Yukawa theory.

Color

Quarks, having spin $\frac{1}{2}$, are fermions and so are subject to the exclusion principle. This would seem to forbid a baryon having two or three quarks with the same flavor and same spin component. To avoid this difficulty, it is assumed that each quark comes in three varieties, which are whimsically called *colors*. Red, green, and blue are the usual choices. The exclusion principle applies separately to each color. A baryon always contains one red, one green, and one blue quark, so the baryon itself has no net color. Each gluon has a color–anticolor combination (for example, blue–antired) that allows it to transmit color when exchanged, and color is conserved during emission and absorption of a gluon by a quark. The gluon-exchange process changes the colors of the quarks in such a way that there is always one quark of each color in every baryon. The color of an individual quark changes continually as gluons are exchanged.

Similar processes occur in mesons such as pions. The quark–antiquark pairs of mesons have canceling color and anticolor (for example, blue and antiblue), so mesons also have no net color. Suppose a pion initially consists of a blue quark and an antiblue antiquark. The blue quark can become a red quark by emitting a blue–antired virtual gluon. The gluon is then absorbed by the antiblue antiquark, converting it to an antired antiquark (**Fig. 44.13**). Color is conserved in each emission and absorption, but a blue–antiblue pair has become a red–antired pair. Such changes occur continually, so we have to think of a pion as a superposition of three quantum states: blue–antiblue, green–antigreen, and red–antired. On a larger scale, the strong interaction between nucleons was described in Section 44.3 as due to the exchange of virtual mesons. In terms of quarks and gluons, these mediating virtual mesons are quark–antiquark systems bound together by the exchange of gluons.

Figure 44.13 (a) A pion containing a blue quark and an antiblue antiquark. (b) The blue quark emits a blue–antired gluon, changing to a red quark. (c) The gluon is absorbed by the antiblue antiquark, which becomes an antired antiquark. The pion now consists of a red–antired quark–antiquark pair. The actual quantum state of the pion is an equal superposition of red–antired, green–antigreen, and blue–antiblue pairs.



The theory of strong interactions is known as *quantum chromodynamics* (QCD). No one has been able to isolate an individual quark, and indeed QCD predicts that quarks are bound in such a way that it is impossible to obtain a free quark. An impressive body of experimental evidence supports the correctness of the quark model and the idea that quantum chromodynamics is the key to understanding the strong interactions.

Three More Quarks

Before the tau particles were discovered, there were four known leptons. This fact, together with some puzzling decay rates, led to the speculation that there might be a fourth quark flavor. This quark is labeled $\boldsymbol{c}$ (the *charmed* quark); it has $Q/e = \frac{2}{3}$, $B = \frac{1}{3}$, $S = 0$, and a new quantum number **charm** $C = +1$. This was confirmed in 1974 by the observation at both SLAC and the Brookhaven National Laboratory of a meson, now named ψ, with mass $3097\ \mathrm{MeV}/c^2$. This meson was found to have several decay modes, decaying into e^+e^-, $\mu^+\mu^-$, or hadrons. The mean lifetime was found to be about 10^{-20} s. These results are consistent with ψ being a spin-1 $\boldsymbol{c\bar{c}}$ system. Almost immediately after this, similar mesons of greater mass were observed and identified as excited states of the $\boldsymbol{c\bar{c}}$ system. A few years later, individual mesons with a nonzero net charm quantum number, D^0 $(\boldsymbol{c\bar{u}})$ and D^+ $(\boldsymbol{c\bar{d}})$, and a charmed baryon, Λ_c^+ $(\boldsymbol{udc})$, were also observed.

In 1977 a meson with mass $9460\ \mathrm{MeV}/c^2$, called upsilon (Υ), was discovered at Brookhaven. Because it had properties similar to ψ, it was conjectured that the meson was really the bound system of a new quark, $\boldsymbol{b}$ (the *bottom* quark), and its antiquark, $\boldsymbol{\bar{b}}$.

TABLE 44.5 Properties of the Six Quarks

Symbol	Q/e	Spin	Baryon Number, B	Strangeness, S	Charm, C	Bottomness, B'	Topness, T
u	$\frac{2}{3}$	$\frac{1}{2}$	$\frac{1}{3}$	0	0	0	0
d	$-\frac{1}{3}$	$\frac{1}{2}$	$\frac{1}{3}$	0	0	0	0
s	$-\frac{1}{3}$	$\frac{1}{2}$	$\frac{1}{3}$	-1	0	0	0
c	$\frac{2}{3}$	$\frac{1}{2}$	$\frac{1}{3}$	0	$+1$	0	0
b	$-\frac{1}{3}$	$\frac{1}{2}$	$\frac{1}{3}$	0	0	-1	0
t	$\frac{2}{3}$	$\frac{1}{2}$	$\frac{1}{3}$	0	0	0	$+1$

The bottom quark has the value -1 of a new quantum number B' called *bottomness*. Excited states of the Υ were soon observed, as were the B^+ ($\bar{b}u$) and B^0 ($\bar{b}d$) mesons.

With the five flavors of quarks (u, d, s, c, and b) and the six flavors of leptons (e, μ, τ, ν_e, ν_μ, and ν_τ) it was an appealing conjecture that nature is symmetric in its building blocks and that therefore there should be a *sixth* quark. This quark, labeled t (top), would have $Q/e = \frac{2}{3}$, $B = \frac{1}{3}$, and a new quantum number, $T = 1$. In 1995, groups using two different detectors at Fermilab's Tevatron announced the discovery of the top quark. The groups collided 0.9 TeV protons with 0.9 TeV antiprotons, but even with 1.8 TeV of available energy, a top–antitop ($t\bar{t}$) pair was detected in fewer than two of every 10^{11} collisions! **Table 44.5** lists some properties of the six quarks. Each has a corresponding antiquark with opposite values of Q, B, S, C, B', and T.

CAUTION **Bottomness vs. baryon number** Don't confuse the bottomness quantum number B' with baryon number B. For example, the proton (which has zero bottomness and is a baryon) has $B' = 0$ and $B = +1$; the B^+ meson (which includes an antibottom quark but is not a baryon) has $B' = +1$ and $B = 0$.

TEST YOUR UNDERSTANDING OF SECTION 44.4 Is it possible to have a baryon with charge $Q = +e$ and strangeness $S = -2$?

ANSWER

no Only the s quark, with $S = -1$, has nonzero strangeness. For a baryon to have $S = -2$, it must have two s quarks and one quark of a different flavor. Since each s quark has charge $-\frac{1}{3}e$, the nonstrange quark must have charge $+\frac{5}{3}e$ to make the net charge equal to $+e$. But *no* quark has charge $+\frac{5}{3}e$, so the proposed baryon is impossible.

44.5 THE STANDARD MODEL AND BEYOND

The particles and interactions that we've discussed in this chapter provide a reasonably comprehensive picture of the fundamental building blocks of nature. There is enough confidence in the basic correctness of this picture that it is called the **standard model.**

The standard model includes three families of particles: (1) the six leptons, which have no strong interactions; (2) the six quarks, from which all hadrons are made; and (3) the particles that mediate the various interactions. These mediators are gluons for the strong interaction among quarks, photons for the electromagnetic interaction, the $W^{\pm}$ and Z^0 particles for the weak interaction, and the graviton for the gravitational interaction.

Electroweak Unification

Theoretical physicists have long dreamed of combining all the interactions of nature into a single unified theory. As a first step, Einstein spent much of his later life trying to develop a field theory that would unify gravitation and electromagnetism. He was only partly successful.

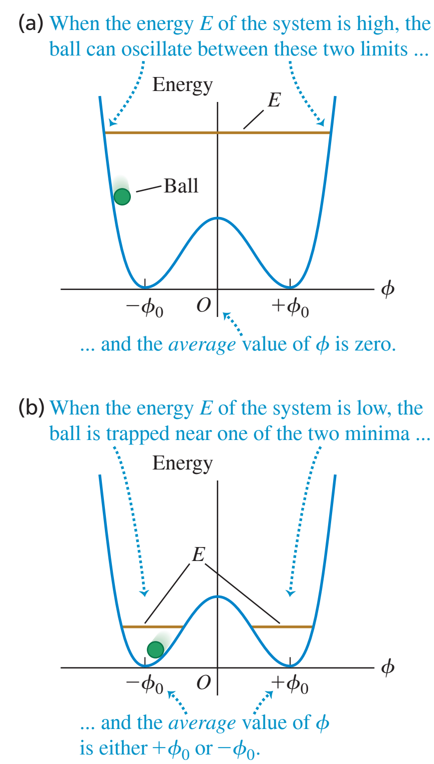
Between 1961 and 1967, Sheldon Glashow, Abdus Salam, and Steven Weinberg developed a theory that unifies the weak and electromagnetic forces. One outcome of their **electroweak theory** is a prediction of the weak-force mediator particles, the $W^{\pm}$ and Z^0

bosons, including their masses. The basic idea is that the mass difference between photons (zero mass) and the weak bosons $(\approx 100\ \mathrm{GeV}/c^2)$ makes the electromagnetic and weak interactions behave quite differently at low energies. At sufficiently high energies (well above 100 GeV), however, the distinction disappears, and the two merge into a single interaction. This prediction was verified in 1983 in experiments with proton-antiproton collisions at CERN. The weak bosons were found, again with the help provided by the theoretical description, and their observed masses agreed with the predictions of the electroweak theory, a wonderful convergence of theory and experiment. The electroweak theory and quantum chromodynamics form the backbone of the standard model. Glashow, Salam, and Weinberg received the Nobel Prize in 1979.

In the electroweak theory photons are massless but the weak bosons are very massive. To account for the broken symmetry among these interaction mediators, a field called the *Higgs field* was proposed by theoretical physicists in the 1960s. We use the symbol $\phi(\vec{r}, t)$ to denote the value of this field at position $\vec{r}$ and time t. (Unlike the electric and magnetic fields, which are vectors, the Higgs field is a scalar quantity.) According to the theory, the mass of the weak bosons is proportional to the absolute value of ϕ_{av}, where ϕ_{av} is the average value of $\phi(\vec{r}, t)$ over space. **Figure 44.14** shows a simplified model of how ϕ_{av} depends on energy. At very high energies, the value of the Higgs field $\phi(\vec{r}, t)$ oscillates between positive and negative values, so its average value is $\phi_{\mathrm{av}} = 0$ (Fig. 44.14a). But at low energies, $\phi(\vec{r}, t)$ oscillates around either a positive average value $\phi_{\mathrm{av}} = +\phi_0$ or a negative average value $\phi_{\mathrm{av}} = -\phi_0$ (Fig. 44.14b). The oscillation is no longer symmetric around $\phi = 0$, so the symmetry has been broken. Hence at low energies, the weak bosons acquire a nonzero mass proportional to $|\phi_{\mathrm{av}}|$, which is equal to ϕ_0 for either of the cases shown in Fig. 44.14b.

This theory also predicts that there should be a particle called the *Higgs boson* associated with the Higgs field itself. (In an analogous way, the photon is the particle associated with the electromagnetic field.) The Higgs boson was predicted to be unstable, have zero charge and spin 0, and have a large mass. An important mission of the Large Hadron Collider at CERN was to produce the Higgs boson from the available energy in proton-proton collisions and thereby verify the existence of the Higgs field. In 2012 the first Higgs bosons were detected in such collisions, with the predicted properties. This suggests that the concept of the Higgs field—that all of space is filled with a field that gives mass to the weak bosons—is indeed correct. The Nobel Prize was awarded in 2013 to François Englert and Peter Higgs, two of the theorists who first proposed the idea of the Higgs field in 1964. Current experiments show that the mass of the Higgs boson is about 125 GeV/c^2, even greater than the masses of the $W^{\pm}$ and Z^0 weak bosons.

Figure **44.14** The Higgs-field value ϕ can oscillate, much like the value of the coordinate of a ball rolling within a trough with two minima. The average value of ϕ determines the masses of the $W^{\pm}$ and Z^0 weak bosons. (a) At high energies, the average value of ϕ is zero and the $W^{\pm}$ and Z^0 are massless (like the photon). (b) At low energies, the symmetry is broken. The average value of ϕ is nonzero, and the $W^{\pm}$ and Z^0 acquire nonzero masses.



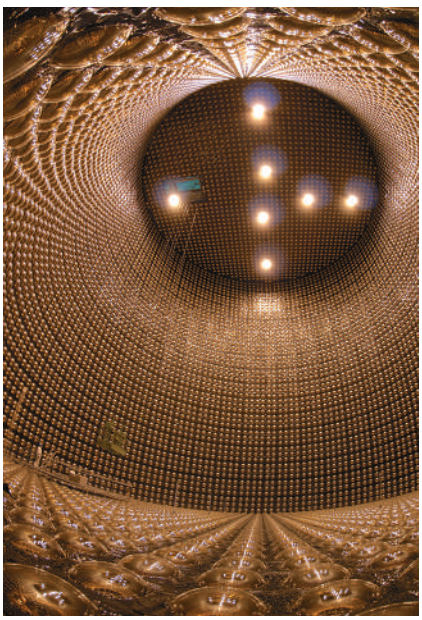
Grand Unified Theories

Perhaps at sufficiently high energies the strong interaction and the electroweak interaction have a convergence similar to that between the electromagnetic and weak interactions. If so, they can be unified to give a comprehensive theory of strong, weak, and electromagnetic interactions. Such schemes, called **grand unified theories** (GUTs), are still speculative.

Some grand unified theories predict the decay of the proton (in violation of conservation of baryon number), with an estimated lifetime of more than 10^{28} years. (For comparison the age of the universe is known to be 1.38×10^{10} years.) With a lifetime of 10^{28} years, six metric tons of protons would be expected to have only one decay per day, so huge amounts of material must be examined. Some of the neutrino detectors that we mentioned in Section 44.2 originally looked for, and failed to find, evidence of proton decay. Current estimates set the proton lifetime well over 10^{33} years. Some GUTs also predict the existence of magnetic monopoles, which we mentioned in Chapter 27. At present there is no confirmed experimental evidence that magnetic monopoles exist.

In the standard model, the neutrinos have zero mass. Nonzero values are controversial because experiments to determine neutrino masses are difficult both to perform

Figure **44.15** This photo shows the interior of the Super-Kamiokande neutrino detector in Japan. When in operation, the detector is filled with 5×10^7 kg of water. A neutrino passing through the detector can produce a faint flash of light, which is detected by the 13,000 photomultiplier tubes lining the detector walls. Data from this detector were the first to indicate that neutrinos have mass.



and to analyze. In most GUTs the neutrinos *must* have nonzero masses. If neutrinos do have mass, transitions called *neutrino oscillations* can occur, in which one type of neutrino (ν_e, ν_μ, or ν_τ) changes into another type. In 1998, scientists using the Super-Kamiokande neutrino detector in Japan (**Fig. 44.15**) reported the discovery of oscillations between muon neutrinos and tau neutrinos. Subsequent measurements at the Sudbury Neutrino Observatory in Canada have confirmed the existence of neutrino oscillations. This discovery is evidence for exciting physics beyond that predicted by the standard model.

The discovery of neutrino oscillations cleared up a long-standing mystery. Since the 1960s, physicists have been using sensitive detectors to look for electron neutrinos produced by nuclear fusion reactions in the sun's core (see Section 43.8). However, the observed flux of solar electron neutrinos is only one-third of the predicted value. The explanation was provided in 2002 by the Sudbury Neutrino Observatory, which can detect neutrinos of all three flavors. The results showed that the combined flux of solar neutrinos of *all* flavors is equal to the theoretical prediction for the flux of *electron* neutrinos. The explanation is that the sun is producing electron neutrinos at the predicted rate, but that two-thirds of these electron neutrinos are transformed into muon or tau neutrinos during their flight from the sun's core to a detector on earth.

Supersymmetric Theories and TOEs

The ultimate dream of theorists is to unify all four fundamental interactions, adding gravitation to the strong and electroweak interactions that are included in GUTs. Such a unified theory is whimsically called a Theory of Everything (TOE). It turns out that an essential ingredient of such theories is a spacetime continuum with more than four dimensions. The additional dimensions are "rolled up" into extremely tiny structures that we ordinarily do not notice. Depending on the scale of these structures, it may be possible for the next generation of particle accelerators to reveal the presence of extra dimensions.

Another ingredient of many theories is *supersymmetry,* which gives every boson and fermion a "superpartner" of the other spin type. For example, the proposed supersymmetric partner of the spin-$\frac{1}{2}$ electron is a spin-0 particle called the *selectron,* and that of the spin-1 photon is a spin-$\frac{1}{2}$ *photino.* As yet, no superpartner particles have been discovered, perhaps because they are too massive to be produced by the present generation of accelerators. Within a few years, new data from the Large Hadron Collider will help us decide whether these intriguing theories have merit.

TEST YOUR UNDERSTANDING OF SECTION 44.5 One aspect of the standard model is that a ***d*** quark can transform into a ***u*** quark, an electron, and an antineutrino by means of the weak interaction. If this happens to a ***d*** quark inside a neutron, what kind of particle remains afterward in addition to the electron and antineutrino? (i) A proton; (ii) a Σ^-; (iii) a Σ^+; (iv) a Λ^0 or a Σ^0; (v) any of these.

ANSWER

(i) If a ***d*** quark in a neutron (quark content ***udd***) undergoes the process $d \rightarrow u + e^- + \bar{\nu}_e$, the remaining baryon has quark content ***uud*** and hence is a proton (see Fig. 44.11). An electron is the same as a β^- particle, so the net result is beta-minus decay: $n \rightarrow p + \beta^- + \bar{\nu}_e$.

44.6 THE EXPANDING UNIVERSE

In the last two sections of this chapter we'll explore briefly the connections between the early history of the universe and the interactions of fundamental particles. It is remarkable that there are such close ties between physics on the smallest scale that we've explored experimentally (the range of the weak interaction, of the order of 10^{-18} m) and physics on the largest scale (the universe itself, of the order of at least 10^{26} m).

Gravitational interactions play an essential role in the large-scale behavior of the universe. We saw in Chapter 13 how the law of gravitation explains the motions of planets in the solar system. Astronomical evidence shows that gravitational forces also dominate in larger systems such as galaxies and clusters of galaxies (**Fig. 44.16**).

Until early in the 20th century it was usually assumed that the universe was *static;* stars might move relative to each other, but there was not thought to be any overall expansion or contraction. But measurements that were begun in 1912 by Vesto Slipher at Lowell Observatory in Arizona, and continued in the 1920s by Edwin Hubble with the help of Milton Humason at Mount Wilson in California, indicated that the universe is *not* static. The motions of galaxies relative to the earth can be measured by observing the shifts in the wavelengths of their spectra. For distant galaxies these shifts are always toward longer wavelength, so they appear to be receding from us and from each other. Astronomers first assumed that these were Doppler shifts and used a relationship between the wavelength λ_0 of light measured now on earth from a source receding at speed v and the wavelength λ_S measured in the rest frame of the source when it was emitted. We can derive this relationship by inverting Eq. (37.25) for the Doppler effect, making subscript changes, and using $\lambda = c/f$; the result is

$$\lambda_0 = \lambda_S\sqrt{\frac{c+v}{c-v}} \tag{44.13}$$

Wavelengths from receding sources are always shifted toward longer wavelengths; this increase in λ is called the **redshift.** We can solve Eq. (44.13) for v:

$$v = \frac{(\lambda_0/\lambda_S)^2 - 1}{(\lambda_0/\lambda_S)^2 + 1}c \tag{44.14}$$

CAUTION **Redshift, not Doppler shift** Equations (44.13) and (44.14) are from the *special* theory of relativity and refer to the Doppler effect. As we'll see, the redshift from *distant* galaxies is caused by an effect that is explained by the *general* theory of relativity and is *not* a Doppler shift. However, as the ratio v/c and the fractional wavelength change $(\lambda_0 - \lambda_S)/\lambda_S$ become small, the general theory's equations approach Eqs. (44.13) and (44.14), and those equations may be used.

Figure 44.16 (a) The galaxy M101 is a larger version of the Milky Way galaxy of which our solar system is a part. Like all galaxies, M101 is held together by the mutual gravitational attraction of its stars, gas, dust, and other matter, all of which orbit around the galaxy's center of mass. M101 is 25 million light-years away. (b) This image shows part of the Coma cluster, an immense grouping of over 1000 galaxies that lies 300 million light-years from us. The galaxies within the cluster are all in motion. Gravitational forces between the galaxies prevent them from escaping from the cluster.

(a)

(b)

EXAMPLE 44.8 Recession speed of a galaxy

WITH VARIATION PROBLEMS

The spectral lines of various elements are detected in light from a galaxy in the constellation Ursa Major. An ultraviolet line from singly ionized calcium ($\lambda_S = 393$ nm) is observed at wavelength $\lambda_0 = 414$ nm, redshifted into the visible portion of the spectrum. At what speed is this galaxy receding from us?

IDENTIFY and SET UP This example uses the relationship between redshift and recession speed for a distant galaxy. We can use the wavelengths λ_S at which the light is emitted and λ_0 that we detect on earth in Eq. (44.14) to determine the galaxy's recession speed v if the fractional wavelength shift is not too great.

EXECUTE The fractional wavelength redshift for this galaxy is $\lambda_0/\lambda_S = (414\text{ nm})/(393\text{ nm}) = 1.053$. This is only a 5.3% increase, so we can use Eq. (44.14) with reasonable accuracy:

$$v = \frac{(1.053)^2 - 1}{(1.053)^2 + 1}c = 0.0516c = 1.55 \times 10^7\text{ m/s}$$

EVALUATE The galaxy is receding from the earth at 5.16% of the speed of light. Rather than going through this calculation, astronomers often just state the *redshift* $z = (\lambda_0 - \lambda_S)/\lambda_S = (\lambda_0/\lambda_S) - 1$. This galaxy has redshift $z = 0.053$.

KEYCONCEPT The redshift of light from distant galaxies shows that they are moving away from us. The recession speed can be calculated from the ratio of the measured wavelength λ_0 of light from the galaxy to the wavelength λ_S measured in the rest frame of the galaxy.

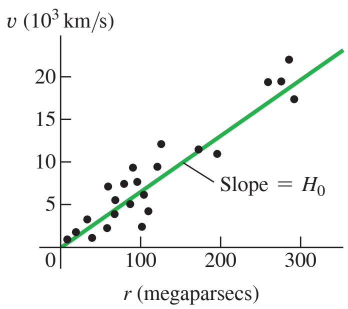


Figure 44.17 Graph of recession speed versus distance for several galaxies. The best-fit straight line illustrates Hubble's law. The slope of the line is the Hubble constant, H_0.

The Hubble Law

Analysis of redshifts from many distant galaxies led Edwin Hubble to a remarkable conclusion: The speed of recession v of a galaxy is proportional to its distance r from us (**Fig. 44.17**). This relationship is now called the **Hubble law;** expressed as an equation,

$$v = H_0 r \tag{44.15}$$

where H_0 is an experimental quantity commonly called the *Hubble constant,* since at any given time it is constant over space. Determining H_0 has been a key goal of the Hubble Space Telescope, which can measure distances to galaxies with unprecedented accuracy. The current best value is about $2.2 \times 10^{-18}\ \text{s}^{-1}$, with an uncertainty of about 2%.

Astronomical distances are often measured in *parsecs* (pc); one parsec is the distance at which there is a one arcsecond $(1/3600°)$ angular separation between two objects 1.50×10^{11} m apart (the average distance from the earth to the sun). A distance of 1 pc is equal to 3.26 *light-years* (ly), where 1 ly $= 9.46 \times 10^{12}$ km is the distance that light travels in one year. The Hubble constant is then commonly expressed in the mixed units (km/s)/Mpc (kilometers per second per megaparsec), where 1 Mpc $= 10^6$ pc:

$$H_0 = (2.2 \times 10^{-18}\ \text{s}^{-1})\left(\frac{9.46 \times 10^{12}\ \text{km}}{1\ \text{ly}}\right)\left(\frac{3.26\ \text{ly}}{1\ \text{pc}}\right)\left(\frac{10^6\ \text{pc}}{1\ \text{Mpc}}\right) = 68\ \frac{\text{km/s}}{\text{Mpc}}$$

EXAMPLE 44.9 Determining distance with the Hubble law — WITH VARIATION PROBLEMS

Use the Hubble law to find the distance from earth to the galaxy in Ursa Major described in Example 44.8.

IDENTIFY and SET UP The Hubble law relates the redshift of a distant galaxy to its distance r from earth. We solve Eq. (44.15) for r and substitute the recession speed v from Example 44.8.

EXECUTE Using $H_0 = 68\ (\text{km/s})/\text{Mpc} = 6.8 \times 10^4\ (\text{m/s})/\text{Mpc}$,

$$r = \frac{v}{H_0} = \frac{1.55 \times 10^7\ \text{m/s}}{6.8 \times 10^4\ (\text{m/s})/\text{Mpc}} = 230\ \text{Mpc}$$
$$= 2.3 \times 10^8\ \text{pc} = 7.5 \times 10^8\ \text{ly} = 7.1 \times 10^{24}\ \text{m}$$

EVALUATE A distance of 230 million parsecs (750 million light-years) is truly stupendous, but many galaxies lie much farther away. To appreciate the immensity of this distance, consider that our farthest-ranging unmanned spacecraft have traveled only about 0.002 ly from our planet.

KEYCONCEPT The Hubble law is the observation that the speed at which a distant galaxy moves away from us is directly proportional to its distance from us. The speed of recession can be determined from the galaxy's redshift, and the distance can then be calculated from the Hubble law.

Another aspect of Hubble's observations was that, *in all directions,* distant galaxies appeared to be receding from us. There is no particular reason to think that our galaxy is at the very center of the universe; if we lived in some other galaxy, every distant galaxy would still seem to be moving away. That is, at any given time, *the universe looks more or less the same, no matter where in the universe we are.* This important idea is called the **cosmological principle.** There are local fluctuations in density, but on average, the universe looks the same from all locations. Thus the Hubble constant is constant in space although not necessarily constant in time, and the laws of physics are the same everywhere.

The Big Bang

The Hubble law suggests that at some time in the past, all the matter in the universe was far more concentrated than it is today. It was then blown apart in a rapid expansion called the **Big Bang,** giving all observable matter more or less the velocities that we observe today. When did this happen? According to the Hubble law, matter at a distance r away from us is traveling with speed $v = H_0 r$. The time t needed to travel a distance r is

$$t = \frac{r}{v} = \frac{r}{H_0 r} = \frac{1}{H_0} = 4.5 \times 10^{17}\ \text{s} = 1.4 \times 10^{10}\ \text{y}$$

By this hypothesis the Big Bang occurred about 14 billion years ago. It assumes that all speeds are *constant* after the Big Bang; that is, it ignores any change in the expansion rate due to gravitational attraction or other effects. We'll return to this point later. For now, however, notice that the age of the earth determined from radioactive dating (see Section 43.4) is 4.54 billion (4.54×10^9) years. It's encouraging that our hypothesis tells us that the universe is older than the earth!

Expanding Space

The general theory of relativity takes a radically different view of the expansion just described. According to this theory, the increased wavelength is *not* caused by a Doppler shift as the universe expands into a previously empty void. Rather, the increase comes from *the expansion of space itself* and everything in intergalactic space, including the wavelengths of light traveling to us from distant sources. This is not an easy concept to grasp, and if you haven't encountered it before, it may sound like doubletalk.

An analogy may help you develop some intuition on this point. Imagine we are all bugs crawling around on a horizontal surface. We can't leave the surface, and we can see in any direction along the surface, but not up or down. We are then living in a two-dimensional world; some writers have called such a world *flatland*. If the surface is a plane, we can locate our position with two Cartesian coordinates (x, y). If the plane extends indefinitely in both the x- and y-directions, we described our space as having *infinite* extent, or as being *unbounded*. No matter how far we go, we never reach an edge or a boundary.

An alternative habitat for us bugs would be the surface of a sphere of radius R. The space would still seem infinite—we could crawl forever and never reach an edge or a boundary. Yet in this case the space is *finite* or *bounded*. To describe the location of a point in this space, we could still use two coordinates: latitude and longitude, or the spherical coordinates θ and ϕ shown in Fig. 41.5.

Now suppose the spherical surface is that of a balloon (**Fig. 44.18**). As we inflate the balloon more and more, increasing the radius R, the coordinates of a point don't change, yet the distance between any two points gets larger and larger. Furthermore, as R increases, the *rate of change* of distance between two points (their recession speed) is proportional to their distance apart. *The recession speed is proportional to the distance,* just as with the Hubble law. For example, the distance from Pittsburgh to Miami is twice as great as the distance from Pittsburgh to Boston. If the earth were to begin to swell, Miami would recede from Pittsburgh twice as fast as Boston would.

Although the quantity R isn't one of the two coordinates giving the position of a point on the balloon's surface, it nevertheless plays an essential role in any discussion of distance. It is the radius of curvature of our two-dimensional universe, and it is also a varying *scale factor* that changes as this universe expands.

Generalizing this picture to three dimensions isn't so easy. We have to think of our three-dimensional space as being embedded in a space with four or more dimensions, just as we visualized the two-dimensional spherical flatland as being embedded in a three-dimensional Cartesian space. Our real three-space is *not Cartesian;* to describe its characteristics in any small region requires at least one additional parameter, the curvature of space, which is analogous to the radius of the sphere. In a sense, this scale factor, which we'll continue to call R, describes the *size* of the universe, just as the radius of the sphere described the size of our two-dimensional spherical universe. We'll return later to the question of whether the universe is bounded or unbounded.

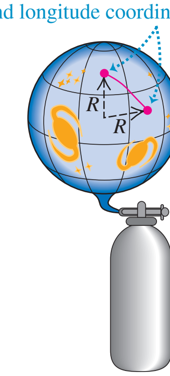
Any length that is measured in intergalactic space is proportional to R, so the wavelength of light traveling to us from a distant galaxy increases along with every other dimension as the universe expands. That is,

$$\frac{\lambda_0}{\lambda} = \frac{R_0}{R} \tag{44.16}$$

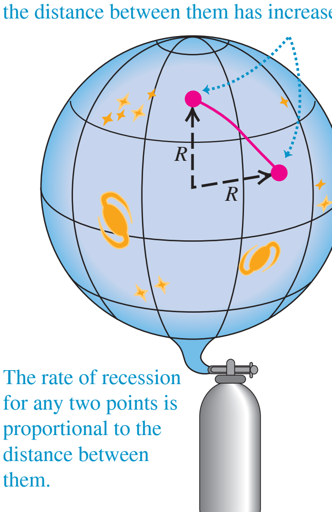
The zero subscripts refer to the values of the wavelength and scale factor *now,* just as H_0 is the current value of the Hubble constant. The quantities λ and R without subscripts are the values at *any* time—past, present, or future. For the galaxy described in Examples 44.8 and 44.9,

Figure **44.18** An inflating balloon as an analogy for an expanding universe.

(a) Points (representing galaxies) on the surface of a balloon are described by their latitude and longitude coordinates.



(b) The radius R of the balloon has increased. The coordinates of the points are the same, but the distance between them has increased.



The rate of recession for any two points is proportional to the distance between them.

we have $\lambda_0 = 414$ nm and $\lambda = \lambda_S = 393$ nm, so Eq. (44.16) gives $R_0/R = 1.053$. That is, the scale factor *now* (R_0) is 5.3% larger than it was 750 million years ago when the light was emitted from that galaxy in Ursa Major. This increase of wavelength with time as the scale factor increases in our expanding universe is called the *cosmological redshift*. The farther away an object is, the longer its light takes to get to us and the greater the change in R and λ. The current largest measured wavelength ratio for galaxies is about 12.1, meaning that the volume of space itself is about $(12.1)^3 \approx 1770$ times larger than it was when the light was emitted. Do *not* attempt to substitute $\lambda_0/\lambda_S = 12.1$ into Eq. (44.14) to find the recession speed; that equation is accurate only for small cosmological redshifts and $v \ll c$. The actual value of v depends on the density of the universe, the value of H_0, and the expansion history of the universe.

Here's a surprise: If the distance from us in the Hubble law is large enough, then the speed of recession is greater than the speed of light! This does *not* violate the special theory of relativity because the recession speed is *not* caused by the motion of the astronomical object relative to some coordinates in its region of space. Rather, $v > c$ when two sets of coordinates move apart fast enough as space itself expands. In other words, there are objects whose coordinates have been moving away from our coordinates so fast that light from them hasn't had enough time in the entire history of the universe to reach us. What we see is just the *observable* universe; we have no direct evidence about what lies beyond its horizon.

CAUTION **The universe isn't expanding into emptiness** The balloon shown in Fig. 44.18 is expanding into the empty space around it. It's a common misconception to picture the universe in the same way as a large but finite collection of galaxies that's expanding into unoccupied space. The reality is quite different! All evidence shows that our universe is *infinite:* It has no edges, so there is nothing "outside" it and it isn't "expanding into" anything. The expansion of the universe simply means that the scale factor of the universe is increasing. A good two-dimensional analogy is to think of the universe as a flat, infinitely large rubber sheet that's stretching and expanding much like the surface of the balloon in Fig. 44.18. In a sense, the infinite universe is becoming more infinite!

Critical Density

In an expanding universe, gravitational attractions between galaxies should slow the initial expansion. But by how much? If these attractions are strong enough, the universe should expand more and more slowly, eventually stop, and then begin to contract, perhaps all the way down to what's been called a *Big Crunch*. On the other hand, if gravitational forces are much weaker, they slow the expansion only a little, and the universe should continue to expand forever.

The situation is analogous to the problem of escape speed of a projectile launched from the earth. We studied this problem in Example 13.5 (Section 13.3). The total energy $E = K + U$ when a projectile of mass m and speed v is at a distance r from the center of the earth (mass m_E) is

$$E = \tfrac{1}{2}mv^2 - \frac{Gmm_E}{r}$$

If E is positive, the projectile has enough kinetic energy to move infinitely far from the earth $(r \rightarrow \infty)$ and have some kinetic energy left over. If E is negative, the kinetic energy $K = \frac{1}{2}mv^2$ becomes zero and the projectile stops when $r = -Gmm_E/E$. In that case, no greater value of r is possible, and the projectile can't escape the earth's gravity.

We can carry out a similar analysis for the universe. Whether the universe continues to expand indefinitely should depend on the average *density* of matter. If matter is relatively dense, there is a lot of gravitational attraction to slow and eventually stop the expansion and make the universe contract again. If not, the expansion should continue indefinitely. We can derive an expression for the *critical density* ρ_c needed to just barely stop the expansion.

Here's a calculation based on Newtonian mechanics; it isn't relativistically correct, but it illustrates the idea. Consider a large sphere with radius R, containing many galaxies (**Fig. 44.19**), with total mass M. Suppose our own galaxy has mass m and is located at the surface of this sphere. According to the cosmological principle, the average distribution of matter within the sphere is uniform. The total gravitational force on our galaxy is just the force due to the mass M inside the sphere. The force on our galaxy and potential energy U due to this spherically symmetric distribution are the same as though m and M were both points, so $U = -GmM/R$, just as in Section 13.3. The net force from all the uniform distribution of mass *outside* the sphere is zero, so we'll ignore it.

The total energy E (kinetic plus potential) for our galaxy is

$$E = \tfrac{1}{2}mv^2 - \frac{GmM}{R} \tag{44.17}$$

If E is *positive,* our galaxy has enough energy to escape from the gravitational attraction of the mass M inside the sphere; in this case the universe should keep expanding forever. If E is negative, our galaxy cannot escape and the universe should eventually pull back together. The crossover between these two cases occurs when $E = 0$, so

$$\tfrac{1}{2}mv^2 = \frac{GmM}{R} \tag{44.18}$$

The total mass M inside the sphere is the volume $4\pi R^3/3$ times the density ρ_c:

$$M = \tfrac{4}{3}\pi R^3 \rho_c$$

We'll assume that the speed v of our galaxy relative to the center of the sphere is given by the Hubble law: $v = H_0 R$. Substituting these expressions for m and v into Eq. (44.18), we get

$$\tfrac{1}{2}m(H_0R)^2 = \frac{Gm}{R}\left(\tfrac{4}{3}\pi R^3 \rho_c\right) \quad \text{or}$$

$$\rho_c = \frac{3H_0{}^2}{8\pi G} \qquad \text{(critical density of the universe)} \tag{44.19}$$

This is the *critical density.* If the average density is less than ρ_c, the universe should continue to expand indefinitely; if it is greater, the universe should eventually stop expanding and begin to contract.

Putting numbers into Eq. (44.19), we find

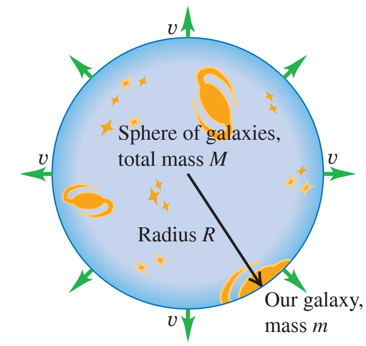
$$\rho_c = \frac{3(2.2 \times 10^{-18}\ \text{s}^{-1})^2}{8\pi(6.67 \times 10^{-11}\ \text{N} \cdot \text{m}^2/\text{kg}^2)} = 8.7 \times 10^{-27}\ \text{kg/m}^3$$

The mass of a hydrogen atom is 1.67×10^{-27} kg, so this density is equivalent to about five hydrogen atoms per cubic meter.

Dark Matter, Dark Energy, and the Accelerating Universe

Astronomers have made extensive studies of the average density of matter in the universe. One way to do so is to count the number of galaxies in a patch of sky. Based on the mass of an average star and the number of stars in an average galaxy, this effort gives an estimate of the average density of *luminous* matter in the universe—that is, matter that emits electromagnetic radiation. (You are made of luminous matter because you emit infrared radiation as a consequence of your temperature; see Sections 17.7 and 39.5.) It's also necessary to take into account other luminous matter within a galaxy, including the tenuous gas and dust between the stars.

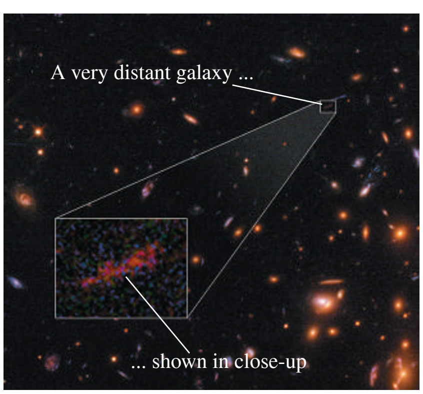
Figure 44.19 An imaginary sphere of galaxies. The net gravitational force exerted on our galaxy (at the surface of the sphere) by the other galaxies is the same as if all of their mass were concentrated at the center of the sphere. (Since the universe is infinite, there's also an infinity of galaxies outside this sphere.)



Another technique is to study the motions of galaxies within clusters of galaxies (see Fig. 44.16b). The motions are so slow that we can't actually see galaxies changing positions within a cluster. However, observations show that different galaxies within a cluster have somewhat different redshifts, which indicates that the galaxies are moving relative to the center of mass of the cluster. The speeds of these motions are related to the gravitational force exerted on each galaxy by the other members of the cluster, which in turn depends on the total mass of the cluster. By measuring these speeds, astronomers can determine the average density of *all* kinds of matter within the cluster, whether or not the matter emits electromagnetic radiation.

Observations using these and other techniques show that the average density of *all* matter in the universe is 31.0% of the critical density, but the average density of *luminous* matter is only 4.9% of the critical density. In other words, most of the matter in the universe is not luminous: It does not emit electromagnetic radiation of *any* kind. At present, the nature of this **dark matter** remains an outstanding mystery. Some proposed candidates for dark matter are WIMPs (weakly interacting massive particles, which are hypothetical subatomic particles far more massive than those produced in accelerator experiments) and MACHOs (massive compact halo objects, which include objects such as black holes that might form "halos" around galaxies). Whatever the true nature of dark matter, it is by far the dominant form of matter in the universe. For every kilogram of the conventional matter that has been our subject for most of this book—including electrons, protons, atoms, molecules, blocks on inclined planes, planets, and stars—there are about *five and one-third* kilograms of dark matter.

Figure 44.20 The bright spots in this image are not stars but entire galaxies. We see the most distant of these, magnified in the inset, as it was 13.3 billion years ago, when the universe was just 500 million years old. At that time the scale factor of the universe was only about 12% as large as it is now. (The red color of this galaxy is due to its very large redshift.) By comparison, we see the relatively nearby Coma cluster (see Fig. 44.16b) as it was 300 million years ago, when the scale factor was 98% of the present-day value.

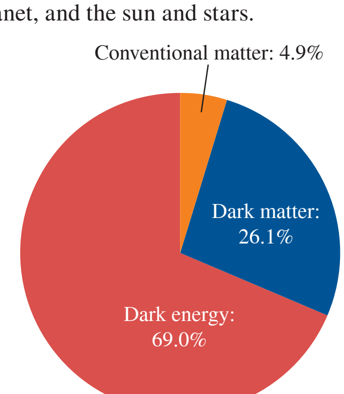


Since the average density of matter in the universe is less than the critical density, it might seem fair to conclude that the universe will continue to expand indefinitely, and that gravitational attraction between matter in different parts of the universe should slow the expansion down (albeit not enough to stop it). One way to test this prediction is to examine the redshifts of extremely distant objects. The more distant a galaxy is, the more time it takes that light to reach us from that galaxy, so the further back in time we look when we observe that galaxy. If the expansion of the universe has been slowing down, the expansion must have been more rapid in the distant past. Thus we would expect very distant galaxies to have *greater* redshifts than predicted by the Hubble law, Eq. (44.15).

Only since the 1990s has it become possible to measure accurately both the distances and the redshifts of extremely distant galaxies. The results have been totally surprising: Very distant galaxies, seen as they were when the universe was a small fraction of its present age (**Fig. 44.20**), have *smaller* redshifts than predicted by the Hubble law! The implication is that the expansion of the universe was slower in the past than it is now, so the expansion has been *speeding up* rather than slowing down.

If gravitational attraction should make the expansion slow down, why is it speeding up instead? Our best explanation is that space is suffused with a kind of energy that has no gravitational effect and emits no electromagnetic radiation, but rather acts as a kind of "antigravity" that produces a universal *repulsion*. This invisible, immaterial energy is called **dark energy.** As the name suggests, the nature of dark energy is poorly understood but is the subject of very active research.

Figure 44.21 The composition of our universe. Conventional matter includes all of the familiar sorts of matter that you see around you, including your body, our planet, and the sun and stars.



Observations show that the *energy* density of dark energy (measured in, say, joules per cubic meter) is 69.0% of the critical density times c^2; that is, it is equal to $0.690\rho_c c^2$. As described above, the average density of matter of all kinds is 31.0% of the critical density. From the Einstein relationship $E = mc^2$, the average *energy* density of matter in the universe is therefore $0.310\rho_c c^2$. Because the energy density of dark energy is nearly three times greater than that of matter, the expansion of the universe will continue to accelerate. This expansion will never stop, and the universe will never contract.

If we account for energy of *all* kinds, the average energy density of the universe is equal to $0.690\rho_c c^2 + 0.310\rho_c c^2 = 1.00\rho_c c^2$. Of this, 69.0% is the mysterious dark energy, 26.1% is the no less mysterious dark matter, and a mere 4.9% is well-understood conventional matter. How little we know about the contents of our universe (**Fig. 44.21**)! When we take account of the density of matter in the universe (which tends to slow the expansion of space) and the density of dark energy (which tends to speed up the expansion), the age of the universe turns out to be 13.8 billion (1.38×10^{10}) years.

What is the significance of the result that within observational error, the average energy density of the universe is equal to $\rho_c c^2$? It tells us that the universe is infinite and unbounded, but just barely so. If the average energy density were even slightly larger than $\rho_c c^2$, the universe would be finite like the surface of the balloon depicted in Fig. 44.18. As of this writing, the observational error in the average energy density is less than 1%, but we can't be totally sure that the universe *is* unbounded. Improving these measurements will be an important task for physicists and astronomers in the years ahead.

TEST YOUR UNDERSTANDING OF SECTION 44.6 Is it accurate to say that your body is made of "ordinary" matter?

ANSWER

yes... and no The material of which your body is made is ordinary to us on earth. But from a cosmic perspective your material is quite *extraordinary*: Only 4.9% of the mass and energy in the universe is in the form of atoms.

BIO APPLICATION A Fossil Both Ancient and Recent This fossil trilobite is an example of a group of marine arthropods that flourished in earth's oceans from 540 to 250 million years ago. (By comparison, the first dinosaurs did not appear until 230 million years ago.) From our perspective, this makes trilobites almost unfathomably ancient. But compared to the time that has elapsed since the Big Bang, 13.8 billion years, even trilobites are a very recent phenomenon: They first appeared when the universe was already 96% of its present age.

44.7 THE BEGINNING OF TIME

What an odd title for the very last section of a book! We'll describe in general terms some of the current theories about the very early history of the universe and their relationship to fundamental particle interactions. We'll find that an astonishing amount happened in the very first second.

Temperatures

The early universe was extremely dense and extremely hot, and the average particle energies were extremely large, all many orders of magnitude beyond anything that exists in the present universe. We can compare particle energy E and absolute temperature T by using the equipartition principle (see Section 18.4):

$$E = \tfrac{3}{2}kT \qquad (44.20)$$

In this equation k is Boltzmann's constant, which we'll often express in eV/K:

$$k = 8.617 \times 10^{-5}\ \text{eV/K}$$

Thus we can replace Eq. (44.20) by $E \approx (10^{-4}\ \text{eV/K})T = (10^{-13}\ \text{GeV/K})T$ when we're discussing orders of magnitude.

EXAMPLE 44.10 Temperature and energy

(a) What is the average kinetic energy E (in eV) of particles at room temperature ($T = 290$ K) and at the surface of the sun ($T = 5800$ K)? (b) What approximate temperature corresponds to the ionization energy of the hydrogen atom and to the rest energies of the electron and the proton?

IDENTIFY and SET UP In this example we are to apply the equipartition principle. We use Eq. (44.20) to relate the target variables E and T.

EXECUTE (a) At room temperature, from Eq. (44.20),

$$E = \tfrac{3}{2}kT = \tfrac{3}{2}(8.617 \times 10^{-5}\ \text{eV/K})(290\ \text{K}) = 0.0375\ \text{eV}$$

The temperature at the sun's surface is higher than room temperature by a factor of $(5800\ \text{K})/(290\ \text{K}) = 20$, so the average kinetic energy there is $20(0.0375\ \text{eV}) = 0.75$ eV.

(b) The ionization energy of hydrogen is 13.6 eV. Using the approximation $E \approx (10^{-4}\ \text{eV/K})T$, we have

$$T \approx \frac{E}{10^{-4}\ \text{eV/K}} = \frac{13.6\ \text{eV}}{10^{-4}\ \text{eV/K}} \approx 10^5\ \text{K}$$

Repeating this calculation for the rest energies of the electron ($E = 0.511$ MeV) and proton ($E = 938$ MeV) gives temperatures of 10^{10} K and 10^{13} K, respectively.

EVALUATE Temperatures in excess of 10^5 K are found in the sun's interior, so most of the hydrogen there is ionized. Temperatures of 10^{10} K or 10^{13} K are not found anywhere in the solar system; as we'll see, temperatures were this high in the very early universe.

KEYCONCEPT The average kinetic energy of particles in a gas is directly proportional to the absolute temperature T. At extremely high temperatures above 10^{10} K, this average kinetic energy can be comparable to the rest energies of subatomic particles.

Uncoupling of Interactions

We've characterized the expansion of the universe by a continual increase of the scale factor R, which we can think of very roughly as characterizing the *size* of the universe, and by a corresponding decrease in average density. As the total gravitational potential energy increased during expansion, there were corresponding *decreases* in temperature and average particle energy. As this happened, the basic interactions became progressively uncoupled.

To understand the uncouplings, recall that the unification of the electromagnetic and weak interactions occurs at energies that are large enough that the differences in mass among the various spin-1 bosons that mediate the interactions become insignificant by comparison. The electromagnetic interaction is mediated by the massless photon, and the weak interaction is mediated by the weak bosons $W^{\pm}$ and Z^0 with masses of the order of $100\ \mathrm{GeV}/c^2$. At energies much *less* than 100 GeV, the two interactions seem quite different. But at energies much *greater* than 100 GeV, they become part of a single interaction, because the $W^{\pm}$ and Z^0 weak bosons become massless like the photon. (This occurs because the average value ϕ_{av} of the Higgs field is zero at high energy, as in Fig. 44.14a.)

The grand unified theories (GUTs) provide a similar behavior for the strong interaction. It becomes unified with the electroweak interaction at energies of the order of 10^{14} GeV, but at lower energies the two appear quite distinct. One of the reasons GUTs are still very speculative is that there is no way to do controlled experiments in this energy range, which is larger by a factor of 10^{11} than energies available with any current accelerator.

Finally, at sufficiently high energies and short distances, it is assumed that gravitation becomes unified with the other three interactions. The distance at which this happens is thought to be of the order of 10^{-35} m. This distance, called the *Planck length* l_P, is determined by the speed of light c and the fundamental constants of quantum mechanics and gravitation, h and G, respectively:

$$l_P = \sqrt{\frac{\hbar G}{c^3}} = 1.616 \times 10^{-35}\ \mathrm{m} \tag{44.21}$$

You should verify that this combination of constants has units of length. The *Planck time* $t_P = l_P/c$ is the time required for light to travel a distance l_P:

$$t_P = \frac{l_P}{c} = \sqrt{\frac{\hbar G}{c^5}} = 0.539 \times 10^{-43}\ \mathrm{s} \tag{44.22}$$

If we mentally go backward in time toward the Big Bang at $t = 0$, we have to stop when we reach $t = 10^{-43}$ s because we have no adequate theory that unifies all four interactions. So as yet we have no way of knowing what happened or how the universe behaved at times earlier than the Planck time or on scales smaller than the Planck length.

The Standard Model of the History of the Universe

The description that follows is called the *standard model* of the history of the universe. The title indicates that there are substantial areas of theory that rest on solid experimental foundations and are quite generally accepted. The figure on pages 1514–1515 is a graphical description of this history, with the characteristic temperature, particle energy, and scale factor at various times. Referring to this figure frequently will help you to understand the following discussion.

In this standard model, the temperature of the universe at time $t = 10^{-43}$ s (the Planck time) was about 10^{32} K, and the average energy per particle was approximately

$$E \approx (10^{-13}\ \mathrm{GeV/K})(10^{32}\ \mathrm{K}) = 10^{19}\ \mathrm{GeV}$$

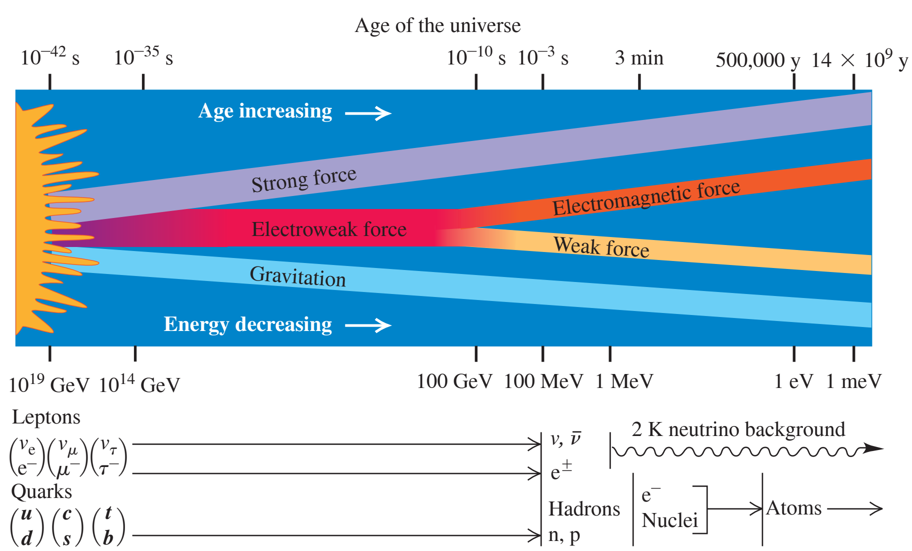
In a totally unified theory this is about the energy below which gravity begins to behave as a separate interaction. This time therefore marked the transition from any proposed TOE to the GUT period.

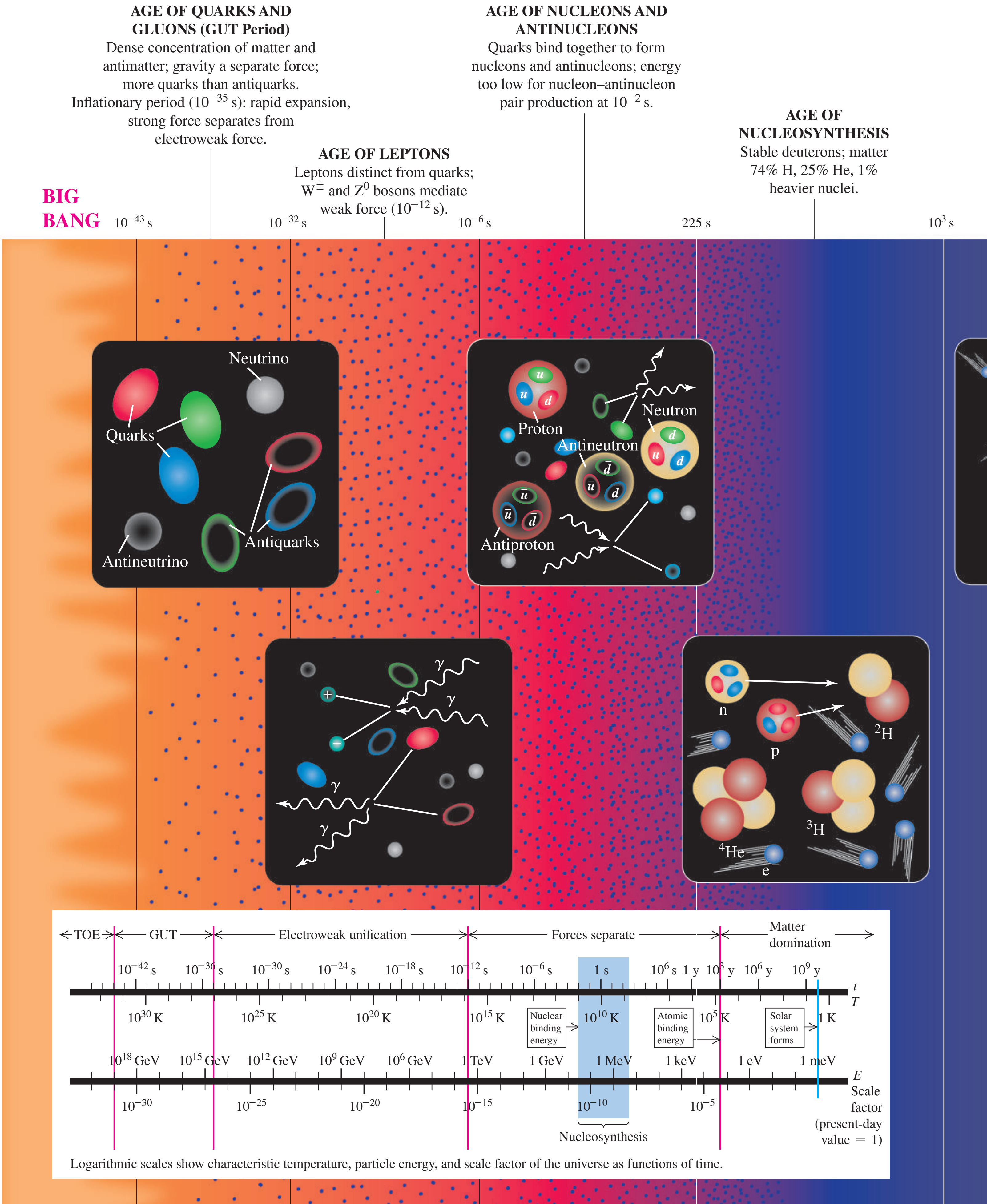
During the GUT period, roughly $t = 10^{-43}$ to 10^{-35} s, the strong and electroweak forces were still unified, and the universe consisted of a soup of quarks and leptons transforming into each other so freely that there was no distinction between the two families of particles. Other, much more massive particles may also have been freely created and destroyed. One important characteristic of GUTs is that at sufficiently high energies, baryon number is not conserved. (We mentioned earlier the proposed decay of the proton, which has not yet been observed.) Thus by the end of the GUT period the numbers of quarks and antiquarks may have been unequal. This point has important implications; we'll return to it at the end of the section.

By $t = 10^{-35}$ s the temperature had decreased to about 10^{27} K and the average energy to about 10^{14} GeV. At this energy the strong force separated from the electroweak force (**Fig. 44.22**), and baryon number and lepton numbers began to be separately conserved. This separation of the strong force was analogous to a phase change such as boiling a liquid, with an associated heat of vaporization. Think of it as being similar to boiling a heavy nucleus, pulling the particles apart beyond the short range of the nuclear force. As a result, the universe underwent a dramatic expansion (far more rapid than the present-day expansion rate) called *cosmic inflation*. In one model, the scale factor R increased by a factor of 10^{50} in 10^{-32} s.

At $t = 10^{-32}$ s the universe was a mixture of quarks, leptons, and the mediating bosons (gluons, photons, and the weak bosons $W^{\pm}$ and Z^0). It continued to expand and cool from the inflationary period to $t = 10^{-6}$ s, when the temperature was about 10^{13} K and typical energies were about 1 GeV (comparable to the rest energy of a nucleon; see Example 44.11). The quarks then began to bind together, forming nucleons and antinucleons. There were still enough photons of sufficient energy to produce nucleon–antinucleon pairs to balance the process of nucleon–antinucleon annihilation. However, by about $t = 10^{-2}$ s, most photon energies fell well below the threshold energy for such pair production. There was a slight excess of nucleons over antinucleons; as a result, virtually all of the antinucleons and most of the nucleons annihilated one another. A similar equilibrium occurred later between the production of electron–positron pairs from photons and the annihilation of such pairs. At about $t = 14$ s the average energy dropped to around 1 MeV, below the threshold for e^+e^- pair production. After pair production ceased, virtually all of the remaining positrons were annihilated, leaving the universe with many more protons and electrons than the antiparticles of each.

Figure **44.22** Schematic diagram showing the times and energies at which the various interactions are thought to have uncoupled. The energy scale is backward because the average energy decreased as the age of the universe increased.




AGE OF QUARKS AND GLUONS (GUT Period)
Dense concentration of matter and antimatter; gravity a separate force; more quarks than antiquarks. Inflationary period (10^{-35} s): rapid expansion, strong force separates from electroweak force.
AGE OF NUCLEONS AND ANTINUCLEONS
Quarks bind together to form nucleons and antinucleons; energy too low for nucleon–antinucleon pair production at 10^{-2} s.
AGE OF LEPTONS
Leptons distinct from quarks; $W^{\pm}$ and Z^0 bosons mediate weak force (10^{-12} s).
AGE OF NUCLEOSYNTHESIS
Stable deuterons; matter 74% H, 25% He, 1% heavier nuclei.
BIG BANG
10^{-43} s
10^{-32} s
10^{-6} s
225 s
10^{3} s
Neutrino
Quarks
Antiquarks
Antineutrino
Proton
Neutron
Antineutron
Antiproton
u
d
$\bar{u}$
$\bar{d}$
γ
n
p
^{2}H
^{3}H
^{4}He
e^-
TOE
GUT
Electroweak unification
Forces separate
Matter domination
10^{-42} s
10^{-36} s
10^{-30} s
10^{-24} s
10^{-18} s
10^{-12} s
10^{-6} s
1 s
10^{6} s
1 y
10^{3} y
10^{6} y
10^{9} y
t
T
10^{30} K
10^{25} K
10^{20} K
10^{15} K
Nuclear binding energy
10^{10} K
Atomic binding energy
10^{5} K
Solar system forms
1 K
10^{18} GeV
10^{15} GeV
10^{12} GeV
10^{9} GeV
10^{6} GeV
1 TeV
1 GeV
1 MeV
1 keV
1 eV
1 meV
E
10^{-30}
10^{-25}
10^{-20}
10^{-15}
10^{-10}
10^{-5}
Scale factor (present-day value = 1)
Nucleosynthesis
Logarithmic scales show characteristic temperature, particle energy, and scale factor of the universe as functions of time.

A Brief History of the Universe

AGE OF IONS
Expanding, cooling gas of ionized H and He.

AGE OF ATOMS
Neutral atoms form; universe becomes transparent to most light.

AGE OF STARS AND GALAXIES
Thermonuclear fusion begins in stars, forming heavier nuclei.

Up until about $t = 1$ s, neutrons and neutrinos could be produced in the endoergic reaction

$$e^- + p \rightarrow n + \nu_e$$

After this time, most electrons no longer had enough energy for this reaction. The average neutrino energy also decreased, and as the universe expanded, equilibrium reactions that involved *absorption* of neutrinos (which occurred with decreasing probability) became inoperative. At this time, in effect, the flux of neutrinos and antineutrinos throughout the universe uncoupled from the rest of the universe. Because of the extraordinarily low probability for neutrino absorption, most of this flux is still present today, although cooled greatly by expansion. The standard model of the universe predicts a present neutrino temperature of about 2 K, but no experiment has yet been able to test this prediction.

Nucleosynthesis

At about $t = 1$ s, the ratio of protons to neutrons was determined by the Boltzmann distribution factor $e^{-\Delta E/kT}$, where ΔE is the difference between the neutron and proton rest energies: $\Delta E = 1.294$ MeV. At a temperature of about 10^{10} K, this distribution factor gives about 4.5 times as many protons as neutrons. However, as we have discussed, free neutrons (with a half-life of 887 s) decay spontaneously to protons. This decay caused the proton-to-neutron ratio to increase until about $t = 225$ s. At this time, the temperature was about 10^9 K, and the average energy was well below 2 MeV.

This energy average was critical because the binding energy of the *deuteron* (a neutron and a proton combined to form a ^{2}H nucleus) is 2.22 MeV (see Section 43.2). A neutron bound in a deuteron does not decay spontaneously. As the average energy decreased, a proton and a neutron could combine to form a deuteron, and there were fewer and fewer photons with 2.22 MeV or more of energy to dissociate the deuterons again. Therefore the combining of protons and neutrons into deuterons halted the decay of free neutrons.

The formation of deuterons starting at about $t = 225$ s marked the beginning of the period of formation of nuclei, or *nucleosynthesis*. At this time, there were about seven protons for each neutron. The deuteron (^2H) can absorb a neutron and form a triton (^3H), or it can absorb a proton and form ^{3}He. Then ^{3}H can absorb a proton and ^{3}He can absorb a neutron, each yielding ^{4}He (the alpha particle). A few ^{7}Li nuclei may also have formed by fusion of ^{3}H and ^{4}He nuclei. According to the theory, essentially all the ^{1}H and ^{4}He in the present universe formed at this time. But then the building of nuclei almost ground to a halt. The reason is that *no* nuclide with mass number $A = 5$ has a half-life greater than 10^{-21} s. Alpha particles simply do not permanently absorb neutrons or protons. The nuclide ^{8}Be that is formed by fusion of two ^{4}He nuclei is unstable, with an extremely short half-life, about 7×10^{-17} s. At this time, the average energy was still much too large for electrons to be bound to nuclei; there were not yet any atoms.

CONCEPTUAL EXAMPLE 44.11 The relative abundance of hydrogen and helium in the universe

Nearly all of the protons and neutrons in the seven-to-one ratio at $t = 225$ s either formed ^{4}He or remained as ^{1}H. After this time, what was the resulting relative abundance of ^{1}H and ^{4}He, by mass?

SOLUTION The ^{4}He nucleus contains two protons and two neutrons. For every two neutrons present at $t = 225$ s there were 14 protons. The two neutrons and two of the 14 protons make up one ^{4}He nucleus, leaving 12 protons (^{1}H nuclei). So there were eventually 12 ^{1}H nuclei for every ^{4}He nucleus. The masses of ^{1}H and ^{4}He are about 1 u and 4 u, respectively, so there were 12 u of ^{1}H for every 4 u of ^{4}He. Therefore the relative abundance, by mass, was 75% ^{1}H and 25% ^{4}He. This result agrees very well with estimates of the present H–He ratio in the universe, an important confirmation of this part of the theory.

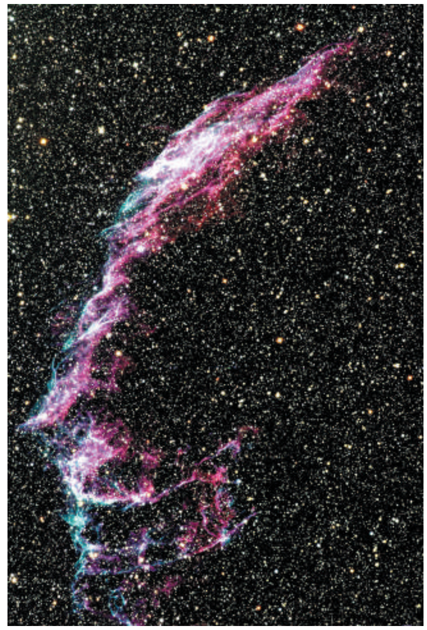
KEYCONCEPT The first atomic nuclei formed about three and a half minutes after the Big Bang. The present-day ratio of hydrogen nuclei to helium nuclei in the universe was determined by the ratio of protons to neutrons at that moment in the history of the universe.

Further nucleosynthesis did not occur until very much later, well after $t = 10^{13}$ s (about 380,000 y). At that time, the temperature was about 3000 K, and the average energy was a few tenths of an electron volt. Because the ionization energies of hydrogen and helium atoms are 13.6 eV and 24.5 eV, respectively, almost all the hydrogen and helium was electrically neutral (not ionized). With the electrical repulsions of the nuclei canceled out, gravitational attraction could slowly pull the neutral atoms together to form clouds of gas and eventually stars. Thermonuclear reactions in stars then produced all of the more massive nuclei. In Section 43.8 we discussed one cycle of thermonuclear reactions in which ^{1}H becomes ^{4}He.

For stars whose mass is 40% of the sun's mass or greater, as the hydrogen is consumed the star's core begins to contract as the inward gravitational pressure exceeds the outward gas and radiation pressure. The gravitational potential energy decreases as the core contracts, so the kinetic energy of nuclei in the core increases. Eventually the core temperature becomes high enough to begin another process, *helium fusion.* First two ^{4}He nuclei fuse to form ^{8}Be, which is highly unstable. But because a star's core is so dense and collisions among nuclei are so frequent, there is a nonzero probability that a third ^{4}He nucleus will fuse with the ^{8}Be nucleus before it can decay. The result is the stable nuclide ^{12}C. This is called the *triple-alpha process,* since three ^{4}He nuclei (that is, alpha particles) fuse to form one carbon nucleus. Then successive fusions with ^{4}He give ^{16}O,^{20}Ne, and ^{24}Mg. All these reactions are exoergic. They release energy to heat up the star, and ^{12}C and ^{16}O can fuse to form elements with higher and higher atomic number.

For nuclides that can be created in this manner, the binding energy per nucleon peaks at mass number $A = 56$ with the nuclide ^{56}Fe, so exoergic fusion reactions stop with Fe. But successive neutron captures followed by beta decays can continue the synthesis of more massive nuclei. If the star is massive enough, it may eventually explode as a *supernova,* sending out into space the heavy elements that were produced by the earlier processes (**Fig. 44.23**; see also Fig. 37.7). In space, the debris and other interstellar matter can gravitationally bunch together to form a new generation of stars and planets. Our sun is one such "second-generation" star. The sun's planets and everything on them (including you) contain matter that was long ago blasted into space by an exploding supernova.

Figure 44.23 The Veil Nebula in the constellation Cygnus is a remnant of a supernova explosion that occurred about 8000 years ago. The gas ejected from the supernova is still moving very rapidly. Collisions between this fast-moving gas and the tenuous material of interstellar space excite the gas and cause it to glow. The portion of the nebula shown here is about 40 ly (12 pc) in length.



Background Radiation

In 1965 Arno Penzias and Robert Wilson, working at Bell Telephone Laboratories in New Jersey on satellite communications, turned a microwave antenna skyward and found a background signal that had no apparent preferred direction. (This signal produces about 1% of the "hash" you see on a TV that's disconnected from cable and tuned to an unused channel.) Further research has shown that the radiation that is received has a frequency spectrum that fits Planck's blackbody radiation law, Eq. (39.24) (Section 39.5). The wavelength of peak intensity is 1.063 mm (in the microwave region of the spectrum), with a corresponding absolute temperature $T = 2.725$ K. Penzias and Wilson contacted physicists at Princeton University who had begun the design of an antenna to search for radiation that was a remnant from the early evolution of the universe. We mentioned above that neutral atoms began to form at about $t = 380{,}000$ years when the temperature was 3000 K. With far fewer charged particles present than previously, the universe became transparent at this time to electromagnetic radiation of long wavelength. The 3000 K blackbody radiation therefore survived, cooling to its present 2.725 K temperature as the universe expanded. The *cosmic background radiation* is among the most clear-cut experimental confirmations of the Big Bang theory. **Figure 44.24** shows a modern map of the cosmic background radiation.

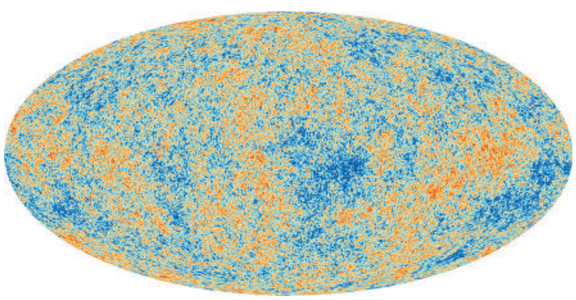


Figure 44.24 This false-color map shows microwave radiation from the entire sky mapped onto an oval. When this radiation was emitted 380,000 years after the Big Bang, the regions shown in blue were slightly cooler and denser than average. Within these cool, dense regions formed galaxies, including the Milky Way galaxy of which our solar system, our earth, and our selves are part.

EXAMPLE 44.12 Expansion of the universe

By approximately what factor has the universe expanded since $t = 380{,}000$ y?

IDENTIFY and SET UP We use the idea that as the universe has expanded, all intergalactic wavelengths have expanded with it. The Wien displacement law, Eq. (39.21), relates the peak wavelength λ_m in blackbody radiation to the temperature T. Given the temperatures of the cosmic background radiation today (2.725 K) and at $t = 380{,}000$ y (3000 K) we can determine the factor by which wavelengths have changed and hence determine the factor by which the universe has expanded.

EXECUTE We rewrite Eq. (39.21) as

$$\lambda_m = \frac{2.90 \times 10^{-3}\ \text{m} \cdot \text{K}}{T}$$

Hence the peak wavelength λ_m is inversely proportional to T. As the universe expands, all intergalactic wavelengths (including λ_m) increase in proportion to the scale factor R. The temperature has decreased by the factor $(3000\ \text{K})/(2.725\ \text{K}) \approx 1100$, so λ_m and the scale factor must both have *increased* by this factor. Thus, between $t = 380{,}000$ y and the present, the universe has expanded by a factor of about 1100.

EVALUATE Our results show that since $t = 380{,}000$ y, any particular intergalactic *volume* has increased by a factor of about $(1100)^3 = 1.3 \times 10^9$. They also show that when the cosmic background radiation was emitted, its peak wavelength was $\frac{1}{1100}$ of the present-day value of 1.063 mm, or 967 nm. This is in the infrared region of the spectrum.

KEYCONCEPT The present-day temperature of the cosmic background radiation tells us how much the universe has expanded since the time 380,000 years after the Big Bang when the first atoms formed and the universe became transparent.

Matter and Antimatter

One of the most remarkable features of our universe is the asymmetry between matter and antimatter. You might think that the universe should have equal numbers of protons and antiprotons and of electrons and positrons, but this doesn't appear to be the case. Theories of the early universe must explain this imbalance.

We've mentioned that most GUTs include violation of conservation of baryon number at energies at which the strong and electroweak interactions have converged. If particle–antiparticle symmetry is also violated, we have a mechanism for making more quarks than antiquarks, more leptons than antileptons, and eventually more matter than antimatter. One serious problem is that any asymmetry that is created in this way during the GUT era might be wiped out by the electroweak interaction after the end of the GUT era. If so, there must be some mechanism that creates particle–antiparticle asymmetry at a much *later* time. The problem of the matter–antimatter asymmetry is still very much an open one.

There are still many unanswered questions at the intersection of particle physics and cosmology. Is the energy density of the universe precisely equal to $\rho_c c^2$, or are there small but important differences? What is dark energy? Has the density of dark energy remained constant over the history of the universe, or has the density changed? What is dark matter? What happened during the first 10^{-43} s after the Big Bang? Can we see evidence that the strong and electroweak interactions undergo a grand unification at high energies? The search for the answers to these and many other questions about our physical world continues to be one of the most exciting adventures of the human mind.

TEST YOUR UNDERSTANDING OF SECTION 44.7 Given a sufficiently powerful telescope, could we detect photons emitted earlier than $t = 380{,}000$ y?

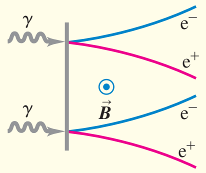
ANSWER

no Prior to $t = 380{,}000$ y the temperature was so high that atoms could not form, so free electrons and protons were plentiful. These charged particles are very effective at scattering photons, so light could not propagate very far and the universe was opaque. The oldest photons that we can detect date from the time $t = 380{,}000$ y when atoms formed and the universe became transparent.

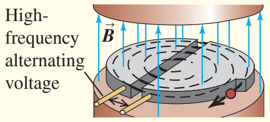
CHAPTER 44 SUMMARY

Fundamental particles: Each particle has an antiparticle; some particles are their own antiparticles. Particles can be created and destroyed, some of them (including electrons and positrons) only in pairs or in conjunction with other particles and antiparticles.

Particles serve as mediators for the fundamental interactions. The photon is the mediator of the electromagnetic interaction. Yukawa proposed the existence of mesons to mediate the nuclear interaction. Mediating particles that can exist only because of the uncertainty principle for energy are called virtual particles.



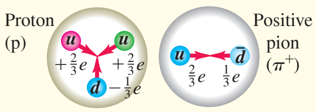
Particle accelerators and detectors: Cyclotrons, synchrotrons, and linear accelerators are used to accelerate charged particles to high energies for experiments with particle interactions. Only part of the beam energy is available to cause reactions with targets at rest. This problem is avoided in colliding-beam experiments. (See Examples 44.1–44.3.)



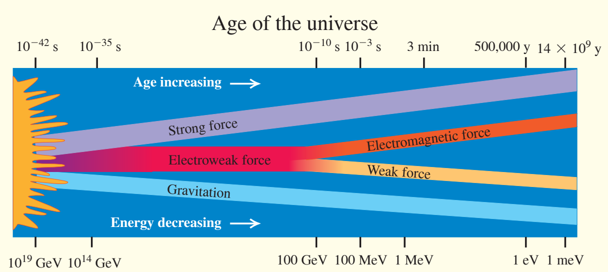
Particles and interactions: Four fundamental interactions are found in nature: the strong, electromagnetic, weak, and gravitational interactions. Particles can be described in terms of their interactions and of quantities that are conserved in all or some of the interactions.

Fermions have half-integer spins; bosons have integer spins. Leptons, which are fermions, have no strong interactions. Strongly interacting particles are called hadrons. They include mesons, which are always bosons, and baryons, which are always fermions. There are conservation laws for three different lepton numbers and for baryon number. Additional quantum numbers, including strangeness and charm, are conserved in some interactions. (See Examples 44.4–44.6.)

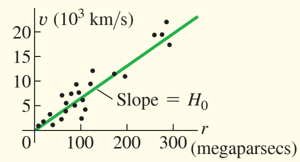
Quarks: Hadrons are composed of quarks. There are thought to be six types of quarks. The interaction between quarks is mediated by gluons. Quarks and gluons have an additional attribute called color. (See Example 44.7.)



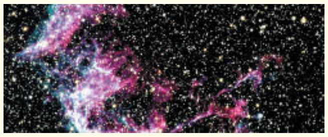
Symmetry and the unification of interactions: Symmetry considerations play a central role in all fundamental-particle theories. The electromagnetic and weak interactions become unified at high energies into the electroweak interaction. In grand unified theories the strong interaction is also unified with these interactions, but at much higher energies.



The expanding universe and its composition: The Hubble law shows that galaxies are receding from each other and that the universe is expanding. Observations show that the rate of expansion is accelerating due to the presence of dark energy, which makes up 69.0% of the energy in the universe. Only 4.9% of the energy in the universe is in the form of conventional matter; the remaining 26.1% is dark matter, whose nature is poorly understood. (See Examples 44.8 and 44.9.)



The history of the universe: In the standard model of the universe, a Big Bang gave rise to the first fundamental particles. They eventually formed into the lightest atoms as the universe expanded and cooled. The cosmic background radiation is a relic of the time when these atoms formed. The heavier elements were manufactured much later by fusion reactions inside stars. (See Examples 44.10–44.12.)



Chapter 44 Media Assets

For assigned homework and other learning materials, go to Mastering Physics.

GUIDED PRACTICE

KEY EXAMPLE VARIATION PROBLEMS

Be sure to review EXAMPLE 44.1 (Section 44.2) before attempting these problems.

VP44.1.1 In a cyclotron the maximum radius for the path of protons is 0.800 m. The magnetic field is 0.600 T. Find (a) the frequency of the alternating voltage applied to the dees, (b) the maximum proton kinetic energy in MeV, and (c) the maximum proton speed in terms of c.

VP44.1.2 A cyclotron at the TRIUMF laboratory in Vancouver, Canada, has a magnetic field of 0.460 T. It accelerates negative hydrogen ions with charge $-e$ and mass 1.67×10^{-27} kg. Find (a) the radius of the path of the ions when they have speed 6.00×10^{6} m/s and (b) the angular frequency of the ions' motion.

VP44.1.3 The first cyclotron (in 1930) had a diameter of 11.4 cm and was used to accelerate negative hydrogen ions (charge $-e$, mass 1.67×10^{-27} kg) to kinetic energies of 80.0 keV. Find (a) the magnetic field used and (b) the frequency of the alternating voltage applied to the dees.

VP44.1.4 A cyclotron is used to accelerate alpha particles (charge $+2e$, mass 6.64×10^{-27} kg). The frequency of the alternating voltage applied to the dees is 5.80 MHz. Find (a) the magnetic field used and (b) the kinetic energy, in MeV, of the alpha particles when the radius of their circular path is 0.650 m.

Be sure to review EXAMPLES 44.2 and 44.3 (Section 44.2) before attempting these problems.

VP44.3.1 A proton with kinetic energy K collides with a proton at rest. Both protons remain after the collision along with an eta-prime (η') particle that has rest energy 958 MeV. In order for this reaction to happen, what must be (a) the minimum available energy in the center-of-momentum system and (b) the minimum value of K?

VP44.3.2 A proton with kinetic energy K collides with a proton at rest. The particles that remain after the collision are one proton and a delta-plus (Δ^{+}) that has rest energy 1232 MeV. In order for this reaction to happen, what must be (a) the minimum available energy in the center-of-momentum system and (b) the minimum value of K?

VP44.3.3 A negative pion (π^{-}) that has rest energy 140 MeV and kinetic energy K collides with a proton at rest. What remains after the collision is a delta-zero (Δ^{0}) that has rest energy 1232 MeV. In order for this reaction to happen, what must be (a) the minimum available energy in the center-of-momentum system and (b) the minimum value of K?

VP44.3.4 The Large Hadron Collider gives each proton a total energy of 7 TeV $= 7.00 \times 10^{3}$ GeV. (a) If a proton that has this energy collides with a second proton that is stationary, what is the available energy? (b) After the collision, both protons remain as well as a neutral particle X. What is the maximum mass of X?

Be sure to review EXAMPLES 44.8 and 44.9 (Section 44.6) before attempting these problems.

VP44.9.1 The red spectral line of hydrogen has a wavelength of 656.3 nm when the hydrogen is at rest. This line is observed to have a wavelength of 725 nm in the spectrum of a galaxy. Find (a) the redshift z and (b) the recession speed of the galaxy in terms of c.

VP44.9.2 A prominent absorption line in the spectrum of the sun (see Fig. 39.9) and many galaxies is due to singly ionized calcium. In the rest frame, this line has an ultraviolet wavelength of 396.9 nm. If a galaxy is receding from us at $0.0711c$, find (a) the wavelength at which we'll observe this line and (b) the redshift z.

VP44.9.3 Find the distances to (a) a galaxy with a recession speed of $0.0992c$ and (b) a galaxy with a recession speed of $0.0711c$. Give your answers in pc.

VP44.9.4 The spectrum of a Type Ia supernova (an exploding star so luminous that it can be seen even in distant galaxies) has an emission line due to singly ionized silicon. If astronomers observe this emission line at 615.0 nm in the laboratory but 666 nm in the spectrum of a Type Ia supernova in a distant galaxy, find (a) the recession speed of the galaxy in m/s and (b) the distance to the galaxy in pc.

BRIDGING PROBLEM Hyperons, Pions, and the Expanding Universe

(a) A Λ^{0} hyperon decays into a neutron and a neutral pion (π^{0}). Find the kinetic energies of the decay products and the fraction of the kinetic energy that is carried off by each particle. (b) A π^{0} is at rest in the galaxy shown in Fig. 44.20. If a physicist on earth detects one of the two photons emitted in the decay of this π^{0}, what is the energy of this detected photon?

SOLUTION GUIDE

IDENTIFY and SET UP

1. Which quantities are conserved in the Λ^{0} decay? In the π^{0} decay?
2. The universe expanded during the time that the photon traveled from the cluster to earth. How does this affect the wavelength and energy of the photon that the physicist detects?
3. List the unknown quantities for each part of the problem and identify the target variables.
4. Select the equations that will allow you to solve for the target variables.

EXECUTE

5. Write the conservation equations for the decay of the Λ^{0}. [*Hint:* It's useful to write the energy E of a particle in terms of its momentum p and mass m with $E = (p^2c^2 + m^2c^4)^{1/2}$.]
6. Solve the conservation equations for the energy of one of the decay products. (*Hint:* Rearrange the energy conservation equation so that one of the $(p^2c^2 + m^2c^4)^{1/2}$ terms is on one side of the equation. Then square both sides.) Then use $K = E - mc^2$.
7. Find the fraction of the total kinetic energy that goes into the neutron and into the pion.
8. Write the conservation equations for the decay of the π^{0} at rest and find the energy of each emitted photon. By what factor does the wavelength of this photon change as it travels from the galaxy to earth? By what factor does the photon *energy* change? (*Hint:* See Fig. 44.20.)

EVALUATE

9. Which of the Λ^{0} decay products should have the greater kinetic energy? Should the detected π^{0} decay photon have more or less energy than when it was emitted?

PROBLEMS

•, ••, •••: Difficulty levels. **CP**: Cumulative problems incorporating material from earlier chapters. **CALC**: Problems requiring calculus. **DATA**: Problems involving real data, scientific evidence, experimental design, and/or statistical reasoning. **BIO**: Biosciences problems.

DISCUSSION QUESTIONS

Q44.1 Is it possible that some parts of the universe contain antimatter whose atoms have nuclei made of antiprotons and antineutrons, surrounded by positrons? How could we detect this condition without actually going there? Can we detect these antiatoms by identifying the light they emit as composed of antiphotons? Explain. What problems might arise if we actually *did* go there?

Q44.2 Given the Heisenberg uncertainty principle, is it possible to create particle–antiparticle pairs that exist for extremely short periods of time before annihilating? Does this mean that empty space is really empty?

Q44.3 When they were first discovered during the 1930s and 1940s, there was confusion as to the identities of pions and muons. What are the similarities and most significant differences?

Q44.4 The gravitational force between two electrons is weaker than the electric force by the order of 10^{-40}. Yet the gravitational interactions of matter were observed and analyzed long before electrical interactions were understood. Why?

Q44.5 When a π^0 decays to two photons, what happens to the quarks of which it was made?

Q44.6 Why can't an electron decay to two photons? To two neutrinos?

Q44.7 According to the standard model of the fundamental particles, what are the similarities between baryons and leptons? What are the most important differences?

Q44.8 According to the standard model of the fundamental particles, what are the similarities between quarks and leptons? What are the most important differences?

Q44.9 The quark content of the neutron is ***udd***. (a) What is the quark content of the antineutron? Explain your reasoning. (b) Is the neutron its own antiparticle? Why or why not? (c) The quark content of the ψ is $c\bar{c}$. Is the ψ its own antiparticle? Explain your reasoning.

Q44.10 Does the universe have a center? Explain.

Q44.11 Does it make sense to ask, "If the universe is expanding, what is it expanding into?"

Q44.12 Assume that the universe has an edge. Placing yourself at that edge in a thought experiment, explain why this assumption violates the cosmological principle.

Q44.13 Explain why the cosmological principle requires that H_0 must have the same value everywhere in space, but does not require that it be constant in time.

EXERCISES

Section 44.1 Fundamental Particles—A History

44.1 • The starship *Enterprise*, of television and movie fame, is powered by combining matter and antimatter. If the entire 400-kg antimatter fuel supply of the *Enterprise* combines with matter, how much energy is released? How does this compare to the U.S. yearly energy use, which is roughly 1.0×10^{20} J?

44.2 •• **CP** Two equal-energy photons collide head-on and annihilate each other, producing a $\mu^+\mu^-$ pair. The muon mass is given in terms of the electron mass in Section 44.1. (a) Calculate the maximum wavelength of the photons for this to occur. If the photons have this wavelength, describe the motion of the μ^+ and μ^- immediately after they are produced. (b) If the wavelength of each photon is half the value calculated in part (a), what is the speed of each muon after they have moved apart? Use correct relativistic expressions for momentum and energy.

44.3 •• A positive pion at rest decays into a positive muon and a neutrino. (a) Approximately how much energy is released in the decay? (Assume the neutrino has zero rest mass. Use the muon and pion masses given in terms of the electron mass in Section 44.1.) (b) Why can't a positive muon decay into a positive pion?

44.4 • A proton and an antiproton annihilate, producing two photons. Find the energy, frequency, and wavelength of each photon (a) if the p and $\bar{p}$ are initially at rest and (b) if the p and $\bar{p}$ collide head-on, each with an initial kinetic energy of 620 MeV.

44.5 •• **CP** For the nuclear reaction given in Eq. (44.2) assume that the initial kinetic energy and momentum of the reacting particles are negligible. Calculate the speed of the α particle immediately after it leaves the reaction region.

44.6 •• Estimate the range of the force mediated by an ω^0 meson that has mass 783 MeV/c^2.

Section 44.2 Particle Accelerators and Detectors

44.7 • In a collision experiment, a proton at rest is struck by an antiproton. (a) What is the minimum kinetic energy of the antiproton if the available energy is 2.00 TeV? (b) If a colliding beam is used instead of a stationary target, what minimum kinetic energy for each beam is required for an available energy of 2.00 TeV?

44.8 • An electron with a total energy of 30.0 GeV collides with a stationary positron. (a) What is the available energy? (b) If the electron and positron are accelerated in a collider, what total energy corresponds to the same available energy as in part (a)?

44.9 • Deuterons in a cyclotron travel in a circle with radius 32.0 cm just before emerging from the dees. The frequency of the applied alternating voltage is 9.00 MHz. Find (a) the magnetic field and (b) the kinetic energy and speed of the deuterons upon emergence.

44.10 • Bubble-chamber photographs taken with a 0.40 T magnetic field show at one point a positron moving perpendicular to the field in a 15 cm circle. What is the magnitude of the positron's momentum at that point?

44.11 • (a) A high-energy beam of alpha particles collides with a stationary helium gas target. What must the total energy of a beam particle be if the available energy in the collision is 16.0 GeV? (b) If the alpha particles instead interact in a colliding-beam experiment, what must the energy of each beam be to produce the same available energy?

44.12 • The first cyclotron used a 1.3 T magnetic field and had an 11 cm radius. (a) Find the maximum kinetic energy of protons accelerated by this cyclotron. Is it accurate to use nonrelativistic expressions in your calculation? (b) When the cyclotron was accelerating protons, at what frequency was it operated?

44.13 •• (a) What is the speed of a proton that has total energy 1000 GeV? (b) What is the angular frequency ω of a proton with the speed calculated in part (a) in a magnetic field of 4.00 T? Use both the nonrelativistic Eq. (44.7) and the correct relativistic expression, and compare the results.

44.14 •• Calculate the minimum beam energy in a proton-proton collider to initiate the $p + p \rightarrow p + p + \eta^0$ reaction. The rest energy of the η^0 is 547.9 MeV (see Table 44.3).

44.15 •• You work for a start-up company that is planning to use antiproton annihilation to produce radioactive isotopes for medical applications. One way to produce antiprotons is by the reaction $p + p \rightarrow p + p + p + \bar{p}$ in proton-proton collisions. (a) You first consider a colliding-beam experiment in which the two proton beams have equal kinetic energies. To produce an antiproton via this reaction, what is the required minimum kinetic energy of the protons in each beam? (b) You then consider the collision of a proton beam with a stationary proton target. For this experiment, what is the required minimum kinetic energy of the protons in the beam?

Section 44.3 Particles and Interactions

44.16 • In about 8% of Ω^- decays, the decay products are a Ξ^- and a π^0. (a) What is the energy released in this decay? (b) In the decay, what is the change in baryon number, and what is the change in strangeness? Your results should show that this decay is allowed for the weak nuclear interaction but not for the strong interaction.

44.17 • A K^+ meson at rest decays into two π mesons. (a) What are the allowed combinations of π^0, π^+, and π^- as decay products? (b) Find the total kinetic energy of the π mesons.

44.18 • How much energy is released when a μ^- muon at rest decays into an electron and two neutrinos? Neglect the small masses of the neutrinos.

44.19 • What is the mass (in kg) of the Z^0? What is the ratio of the mass of the Z^0 to the mass of the proton?

44.20 • The discovery of the Ω^- particle helped confirm Gell-Mann's eightfold way. If an Ω^- decays into a Λ^0 and a K^-, what is the total kinetic energy of the decay products?

44.21 • If a Σ^+ at rest decays into a proton and a π^0, what is the total kinetic energy of the decay products?

44.22 • Which of the following reactions obey the conservation of baryon number? (a) $p + p \rightarrow p + e^+$; (b) $p + n \rightarrow 2e^+ + e^-$; (c) $p \rightarrow n + e^- + \bar{\nu}_e$; (d) $p + \bar{p} \rightarrow 2\gamma$.

44.23 • In which of the following decays are the three lepton numbers conserved? In each case, explain your reasoning. (a) $\mu^- \rightarrow e^- + \nu_e + \bar{\nu}_\mu$; (b) $\tau^- \rightarrow e^- + \bar{\nu}_e + \nu_\tau$; (c) $\pi^+ \rightarrow e^+ + \gamma$; (d) $n \rightarrow p + e^- + \bar{\nu}_e$.

44.24 • In which of the following reactions or decays is strangeness conserved? In each case, explain your reasoning. (a) $K^+ \rightarrow \mu^+ + \nu_\mu$; (b) $n + K^+ \rightarrow p + \pi^0$; (c) $K^+ + K^- \rightarrow \pi^0 + \pi^0$; (d) $p + K^- \rightarrow \Lambda^0 + \pi^0$.

Section 44.4 Quarks and Gluons

44.25 • Determine the electric charge, baryon number, strangeness quantum number, and charm quantum number for the following quark combinations: (a) uds; (b) $c\bar{u}$; (c) ddd; and (d) $d\bar{c}$. Explain your reasoning.

44.26 •• Determine the electric charge, baryon number, strangeness quantum number, and charm quantum number for the following quark combinations: (a) uus, (b) $c\bar{s}$, (c) $\bar{d}\bar{d}\bar{u}$, and (d) $\bar{c}b$.

44.27 • Given that each particle contains only combinations of $u, d, s, \bar{u}, \bar{d}$, and $\bar{s}$, use the method of Example 44.7 to deduce the quark content of (a) a particle with charge $+e$, baryon number 0, and strangeness $+1$; (b) a particle with charge $+e$, baryon number -1, and strangeness $+1$; (c) a particle with charge 0, baryon number $+1$, and strangeness -2.

44.28 • What is the total kinetic energy of the decay products when an upsilon particle at rest decays to $\tau^+ + \tau^-$?

Section 44.5 The Standard Model and Beyond

44.29 • Section 44.5 states that current experiments show that the mass of the Higgs boson is about 125 GeV/c^2. What is the ratio of the mass of the Higgs boson to the mass of a proton?

Section 44.6 The Expanding Universe

44.30 • The spectrum of the sodium atom is detected in the light from a distant galaxy. (a) If the 590.0 nm line is redshifted to 658.5 nm, at what speed is the galaxy receding from the earth? (b) Use the Hubble law to calculate the distance of the galaxy from the earth.

44.31 • A galaxy in the constellation Pisces is 5210 Mly from the earth. (a) Use the Hubble law to calculate the speed at which this galaxy is receding from earth. (b) What redshifted ratio λ_0/λ_S is expected for light from this galaxy?

44.32 • **Redshift.** The definition of the redshift z is given in Example 44.8. (a) Show that Eq. (44.13) can be written as $1 + z = ([1 + \beta]/[1 - \beta])^{1/2}$, where $\beta = v/c$. (b) The observed redshift for a certain galaxy is $z = 0.700$. Find the speed of the galaxy relative to the earth; assume the redshift is described by Eq. (44.14). (c) Use the Hubble law to find the distance of this galaxy from the earth.

Section 44.7 The Beginning of Time

44.33 • Calculate the reaction energy Q (in MeV) for the reaction $e^- + p \rightarrow n + \nu_e$. Is this reaction endoergic or exoergic?

44.34 • Calculate the energy (in MeV) released in the triple-alpha process $3\,{}^4\text{He} \rightarrow {}^{12}\text{C}$.

44.35 • Calculate the reaction energy Q (in MeV) for the nucleosynthesis reaction

$$^{12}_{6}\text{C} + {}^{4}_{2}\text{He} \rightarrow {}^{16}_{8}\text{O}$$

Is this reaction endoergic or exoergic?

PROBLEMS

44.36 •• The strong nuclear force can be crudely modeled as a Hooke's-law spring force, increasing linearly with the quark separation distance. The energy stored in this "spring" corresponds to the energy content of the gluon field. In this picture, as quarks are further separated, increasing energy is stored between them. At a critical separation distance, the energy converts to matter, and a new quark–antiquark pair is generated, guaranteeing that there can never be a free quark. (a) A proton has a diameter of about 1.5 fm. Estimate the repulsive Coulomb force between two up quarks separated by 0.5 fm. (b) Model the strong force as $F_s = ks$, where s is the distance between two quarks. If this force balances the electrostatic repulsion between two up quarks when $s = 0.5$ fm, what is the effective spring constant k, in SI units? (c) Convert k into units of MeV/fm^2. (d) How much energy is stored in the gluon field when $s = 0.5$ fm? The mass of an up quark is thought to be about 2.3 MeV/c^2. (e) How much energy is needed to produce an up quark and an antiup quark? (f) How far would two up quarks need to be separated so that the gluon energy $\frac{1}{2}ks^2$ matches the rest energy of an up–antiup quark pair?

44.37 •• CP BIO **Radiation Therapy with π^- Mesons.** Beams of π^- mesons are used in radiation therapy for certain cancers. The energy comes from the complete decay of the π^- to *stable* particles. (a) Write out the complete decay of a π^- meson to stable particles. What are these particles? (b) How much energy is released from the complete decay of a single π^- meson to stable particles? (You can ignore the very small masses of the neutrinos.) (c) How many π^- mesons need to decay to give a dose of 50.0 Gy to 10.0 g of tissue? (d) What would be the equivalent dose in part (c) in Sv and in rem? Consult Table 43.3 and use the largest appropriate RBE for the particles involved in this decay.

44.38 •• A proton and an antiproton collide head-on with equal kinetic energies. Two γ rays with wavelengths of 0.720 fm are produced. Calculate the kinetic energy of the incident proton.

44.39 •• The 7 TeV proton bunches that circulate in opposite directions around the 27 km ring of the Large Hadron Collider smash together at beam crossings every 25 ns. (a) How many bunches meet every second? (b) Each bunch has 115 billion protons. Of these, typically 20 collide during each crossing. Estimate the fraction of protons that collide per crossing. (c) Estimate how many collisions take place each second. (d) The bunches are 30 cm long and are squeezed to a diameter of 20 μm. Estimate the density of protons in a bunch, in units of protons/mm^3. (e) Estimate the density of hadrons in ordinary matter. (*Hint:* Divide your mass, which is mostly due to hadrons, by the mass of a proton to get the number of hadrons in your body. Then divide by the estimated volume.)

44.40 •• Calculate the threshold kinetic energy for the reaction $\pi^- + \mathrm{p} \rightarrow \Sigma^0 + \mathrm{K}^0$ if a π^- beam is incident on a stationary proton target. The K^0 has a mass of 497.7 MeV/c^2.

44.41 • Each of the following reactions is missing a single particle. Calculate the baryon number, charge, strangeness, and the three lepton numbers (where appropriate) of the missing particle, and from this identify the particle. (a) $\mathrm{p} + \mathrm{p} \rightarrow \mathrm{p} + \Lambda^0 + ?$; (b) $\mathrm{K}^- + \mathrm{n} \rightarrow \Lambda^0 + ?$; (c) $\mathrm{p} + \overline{\mathrm{p}} \rightarrow \mathrm{n} + ?$; (d) $\overline{\nu}_\mu + \mathrm{p} \rightarrow \mathrm{n} + ?$

44.42 •• An η^0 meson at rest decays into three π mesons. (a) What are the allowed combinations of π^0, π^+, and π^- as decay products? (b) Find the total kinetic energy of the π mesons.

44.43 • The ϕ meson has mass 1019.4 MeV/c^2 and a measured energy width of 4.4 MeV/c^2. Using the uncertainty principle, estimate the lifetime of the ϕ meson.

44.44 • Estimate the energy width (energy uncertainty) of the ψ if its mean lifetime is 7.6×10^{-21} s. What fraction is this of its rest energy?

44.45 •• CP BIO One proposed proton decay is $\mathrm{p}^+ \rightarrow \mathrm{e}^+ + \pi^0$, which violates both baryon and lepton number conservation, so the proton lifetime is expected to be very long. Suppose the proton half-life were 1.0×10^{18} y. (a) Calculate the energy deposited per kilogram of body tissue (in rad) due to the decay of the protons in your body in one year. Model your body as consisting entirely of water. Only the two protons in the hydrogen atoms in each H_2O molecule would decay in the manner shown; do you see why? Assume that the π^0 decays to two γ rays, that the positron annihilates with an electron, and that all the energy produced in the primary decay and these secondary decays remains in your body. (b) Calculate the equivalent dose (in rem) assuming an RBE of 1.0 for all the radiation products, and compare with the 0.1 rem due to the natural background and the 5.0 rem guideline for industrial workers. Based on your calculation, can the proton lifetime be as short as 1.0×10^{18} y?

44.46 •• A ϕ meson (see Problem 44.43) at rest decays via $\phi \rightarrow \mathrm{K}^+ + \mathrm{K}^-$. It has strangeness 0. (a) Find the kinetic energy of the K^+ meson. (Assume that the two decay products share kinetic energy equally, since their masses are equal.) (b) Suggest a reason the decay $\phi \rightarrow \mathrm{K}^+ + \mathrm{K}^- + \pi^0$ has not been observed. (c) Suggest reasons the decays $\phi \rightarrow \mathrm{K}^+ + \pi^-$ and $\phi \rightarrow \mathrm{K}^+ + \mu^-$ have not been observed.

44.47 ••• CP About 10,000 cosmic-ray protons, each with hundreds of MeV of energy, impinge on each square meter of our upper atmosphere each second. They collide with atmospheric nitrogen and oxygen to produce secondary showers of newly created particles, including many muons. Muons have a mass of 105.7 MeV/c^2 and an average lifetime of 2.197 μs. Consider a secondary cosmic muon produced at an altitude of 15.00 km aimed directly downward with an energy of 6.000 GeV. With such high energy, the muon can travel a great distance into the earth without slowing down. (a) Determine the speed of this muon. (b) How far would this muon travel in one lifetime if there were no relativistic effects? (c) In the frame of the muon, what distance separates its creation position from the earth's surface? (d) If one lifetime passes in the frame of the muon, how much time passes in the frame of the earth? (e) How far does a muon travel in this time as seen from the earth? (f) Does the muon survive its trip to the surface? How far will it penetrate the earth in its lifetime?

44.48 ••• CP A Ξ^- particle at rest decays to a Λ^0 and a π^-. (a) Find the total kinetic energy of the decay products. (b) What fraction of the energy is carried off by each particle? (Use relativistic expressions for momentum and energy.)

44.49 ••• CP A Σ^- particle moving in the $+x$-direction with kinetic energy 180 MeV decays into a π^- and a neutron. The π^- moves in the $+y$-direction. What is the kinetic energy of the neutron, and what is the direction of its velocity? Use relativistic expressions for energy and momentum.

44.50 ••• CP The K^0 meson has rest energy 497.7 MeV. A K^0 meson moving in the $+x$-direction with kinetic energy 225 MeV decays into a π^+ and a π^-, which move off at equal angles above and below the $+x$-axis. Calculate the kinetic energy of the π^+ and the angle it makes with the $+x$-axis. Use relativistic expressions for energy and momentum.

44.51 •• DATA While tuning up a medical cyclotron for use in isotope production, you obtain the data given in the table.

***B* (T)**	0.10	0.20	0.30	0.40
K_{max} (MeV)	0.068	0.270	0.608	1.080

B is the uniform magnetic field in the cyclotron, and K_{max} is the maximum kinetic energy of the particle being accelerated, which is a proton. The radius R of the proton path at maximum kinetic energy has the same value for each magnetic-field value. (a) Compare the kinetic energy values in the table to the rest energy mc^2 of a proton. Is it necessary to use relativistic expressions in your analysis? Explain. (b) Graph your data as K_{max} versus B^2. Use the slope of the best-fit straight line to your data to find R. (c) What is the maximum kinetic energy for a 0.25 T magnetic field? (d) What is the angular frequency ω of the proton when $B = 0.40$ T?

44.52 •• DATA The decay products from the decay of short-lived unstable particles can provide evidence that these particles have been produced in a collision experiment. As an initial step in designing an experiment to detect short-lived hadrons, you make a literature study of their decays. Table 44.3 gives experimental data for the mass and typical decay modes of the particles Σ^-, Ξ^0, Δ^{++}, and Ω^-. (a) Which of these four particles has the largest mass? The smallest? (b) By the decay modes shown in the table, for which of these particles do the decay products have the greatest total kinetic energy? The least?

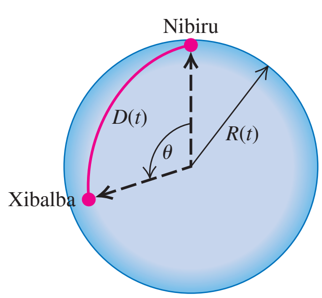
44.53 •• DATA You have entered a graduate program in particle physics and are learning about the use of symmetry. You begin by repeating the analysis that led to the prediction of the Ω^- particle. Nine of the spin-$\frac{3}{2}$ baryons are four Δ particles, each with mass 1232 MeV/c^2, strangeness 0, and charges $+2e$, $+e$, 0, and $-e$; three Σ^* particles, each with mass 1385 MeV/c^2, strangeness -1, and charges $+e$, 0, and $-e$; and two Ξ^* particles, each with mass 1530 MeV/c^2, strangeness -2, and charges 0 and $-e$. (a) Place these particles on a plot of S versus Q. Deduce the Q and S values of the tenth spin-$\frac{3}{2}$ baryon, the Ω^- particle, and place it on your diagram. Also label the particles with their masses. The mass of the Ω^- is 1672 MeV/c^2; is this value consistent with your diagram? (b) Deduce the three-quark combinations (of ***u***, ***d***, and ***s***) that make up each of these ten particles. Redraw the plot of S versus Q from part (a) with each particle labeled by its quark content. What regularities do you see?

44.54 ••• **CP** A positive kaon K^+ which is a bound state of an up quark and an antidown quark, can decay into an antimuon and a muon neutrino via a weak interaction process mediated by a W^+ boson. We can determine how much of the kaon's energy each of the daughter particles obtains using relativistic kinematics. (a) In the rest frame of the kaon, the speed of the antimuon is v and its Lorentz factor is $\gamma = (1 - v^2/c^2)^{-1/2}$. Using the notation E_ν for the energy of the neutrino, M_K for the mass of the kaon, and M_μ for the mass of the antimuon, write the relativistic expression for conservation of energy. Treat the neutrino as massless. (b) Write the relativistic equation for conservation of momentum. (c) Solve these two equations to determine v as a function of the mass ratio $\sigma = M_K/M_\mu$. (d) Using the masses M_K and M_μ found in Tables 44.3 and 44.2, respectively, determine the value of σ. (e) What is the energy of $E_\mu = \gamma M_\mu c^2$ of the antimuon? (f) What is the energy of the neutrino? (g) Do these energies add up to the rest energy of the kaon?

CHALLENGE PROBLEMS

44.55 ••• **CALC** Consider the following hypothetical universe: The Nibiruvians and the Xibalbans are minuscule "flat" societies located at an angular separation of $\theta = 60°$ on the surface of a two-dimensional spherical universe with radius R, as shown in **Fig. P44.55.** (a) What is the distance D between the two societies in terms of R and θ? (b) If R is increasing in time, what is the speed $V = dD/dt$ at which the civilizations are separating, as a function of R, dR/dt, and D? (c) The separation velocity and the distance D are related by $V = BD$. Determine the "Bubble parameter" B in terms of $R(t)$. Note that B is not constant. (Similarly, the Hubble "constant" H_0 is presumed to be changing with time.) (d) The radius of this universe was $R_0 = 500.0$ m upon its creation at time $t = 0$, and it has been increasing at a constant rate of 1.00 μm/s. What was the Bubble parameter exactly four years after creation? (e) What is the distance between Nibiru and Xibalba four years after creation? (f) At what speed are they separating at that time? (g) At that time, the Nibiruvians transmit isotropic ripple waves with their "light speed" of $c = 6.35$ μm/s and wavelength 1.00 nm. How long does it take these waves to reach Xibalba? (*Hint:* In time dt the waves travel an angular distance $d\theta = c\,dt/R(t)$. Integrate to obtain an expression for the total angular distance traveled as a function of time. Solve for the time needed to travel a given angular distance.) (h) At what wavelength will the Xibalbans observe these waves? (*Hint:* Use Eq. (44.16).)

Figure **P44.55**



44.56 ••• **CP** Consider a collision in which a stationary particle with mass M is bombarded by a particle with mass m, speed v_0, and total energy (including rest energy) E_m. (a) Use the Lorentz transformation to write the velocities v_m and v_M of particles m and M in terms of the speed v_{cm} of the center of momentum. (b) Use the fact that the total momentum in the center-of-momentum frame is zero to obtain an expression for v_{cm} in terms of m, M, and v_0. (c) Combine the results of parts (a) and (b) to obtain Eq. (44.9) for the total energy in the center-of-momentum frame.

MCAT-STYLE PASSAGE PROBLEMS

BIO Looking Under the Hood of PET. In the imaging method called positron emission tomography (PET), a patient is injected with molecules containing atoms that have nuclei with an excess of protons. As they decay into neutrons, these protons emit positrons. An emitted positron travels a short distance and slows to near-zero velocity; when it encounters an electron, they may annihilate each other and emit two photons in opposite directions. The patient is enclosed in a circular array of detectors, with the tissue to be imaged centered in the array. If two photons of the proper energy strike two detectors simultaneously (within 10 ns), we can conclude that the photons were produced by positron–electron annihilation somewhere along a line connecting the detectors. By observing many such simultaneous events, we can create a map of the distribution of positron-emitting atoms in the tissue. However, photons can be absorbed or scattered as they pass through tissue. The number of photons remaining after they travel a distance x through tissue is given by $N = N_0 e^{-\mu x}$, where N_0 is the initial number of photons and μ is the attenuation coefficient, which is approximately 0.1 cm^{-1} for photons of this energy. The index of refraction of biological tissue for x rays is 1.

44.57 What is the energy of each photon produced by positron–electron annihilation? (a) $\frac{1}{2}m_e v^2$, where v is the speed of the emitted positron; (b) $m_e v^2$; (c) $\frac{1}{2}m_e c^2$; (d) $m_e c^2$.

44.58 Suppose that positron–electron annihilations occur on the line 3 cm from the center of the line connecting two detectors. Will the resultant photons be counted as having arrived at these detectors simultaneously? (a) No, because the time difference between their arrivals is 100 ms; (b) no, because the time difference is 200 ms; (c) yes, because the time difference is 0.1 ns; (d) yes, because the time difference is 0.2 ns.

44.59 If the annihilation photons come from a part of the body that is separated from the detector by 20 cm of tissue, what percentage of the photons that originally traveled toward the detector remains after they have passed through the tissue? (a) 1.4%; (b) 8.6%; (c) 14%; (d) 86%.

ANSWERS

Chapter Opening Question ?

(v) Only 4.9% of the mass and energy of the universe is in the form of "normal" matter. Of the rest, 26.1% is poorly understood dark matter and 69.0% is even more mysterious dark energy.

Key Example ✓ARIATION Problems

VP44.1.1 (a) 9.15 MHz (b) 11.0 MeV (c) 0.153c
VP44.1.2 (a) 0.136 m (b) 4.41×10^7 rad/s
VP44.1.3 (a) 0.717 T (b) 10.9 MHz
VP44.1.4 (a) 0.756 T (b) 11.6 MeV
VP44.3.1 (a) 2834 MeV (b) 2.41×10^3 MeV $= 2.41$ GeV
VP44.3.2 (a) 2170 MeV (b) 634 MeV
VP44.3.3 (a) 1232 MeV (b) 190 MeV
VP44.3.4 (a) 115 GeV (b) 113 GeV/c^2
VP44.9.1 (a) 0.105 (b) 0.0992c
VP44.9.2 (a) 426 nm (b) 0.0738
VP44.9.3 (a) 4.4×10^8 pc (b) 3.1×10^8 pc
VP44.9.4 (a) 2.38×10^7 m/s (b) 3.5×10^8 pc

Bridging Problem

(a) neutron: 5.78 MeV (0.140 of total);
pion: 35.62 MeV (0.860 of total)
(b) 8.1 MeV

APPENDIX A

THE INTERNATIONAL SYSTEM OF UNITS

The Système International d'Unités, abbreviated SI, is the system developed by the General Conference on Weights and Measures and adopted by nearly all the industrial nations of the world. The following material is adapted from the National Institute of Standards and Technology (http://physics.nist.gov/cuu).

Quantity	Name of unit	Symbol	
	SI base units		
length	meter	m	
mass	kilogram	kg	
time	second	s	
electric current	ampere	A	
thermodynamic temperature	kelvin	K	
amount of substance	mole	mol	
luminous intensity	candela	cd	
	SI derived units		**Equivalent units**
area	square meter	m^2	
volume	cubic meter	m^3	
frequency	hertz	Hz	s^{-1}
mass density (density)	kilogram per cubic meter	kg/m^3	
speed, velocity	meter per second	m/s	
angular velocity	radian per second	rad/s	
acceleration	meter per second squared	m/s^2	
angular acceleration	radian per second squared	rad/s^2	
force	newton	N	$kg \cdot m/s^2$
pressure (mechanical stress)	pascal	Pa	N/m^2
kinematic viscosity	square meter per second	m^2/s	
dynamic viscosity	newton-second per square meter	$N \cdot s/m^2$	
work, energy, quantity of heat	joule	J	$N \cdot m$
power	watt	W	J/s
quantity of electricity	coulomb	C	$A \cdot s$
potential difference, electromotive force	volt	V	J/C, W/A
electric field strength	volt per meter	V/m	N/C
electrical resistance	ohm	Ω	V/A
capacitance	farad	F	$A \cdot s/V$
magnetic flux	weber	Wb	$V \cdot s$
inductance	henry	H	$V \cdot s/A$
magnetic flux density	tesla	T	Wb/m^2
magnetic field strength	ampere per meter	A/m	
magnetomotive force	ampere	A	
luminous flux	lumen	lm	$cd \cdot sr$
luminance	candela per square meter	cd/m^2	
illuminance	lux	lx	lm/m^2
wave number	1 per meter	m^{-1}	
entropy	joule per kelvin	J/K	
specific heat capacity	joule per kilogram-kelvin	$J/kg \cdot K$	
thermal conductivity	watt per meter-kelvin	$W/m \cdot K$	

Quantity	Name of unit	Symbol	Equivalent units
radiant intensity	watt per steradian	W/sr	
activity (of a radioactive source)	becquerel	Bq	s^{-1}
radiation dose	gray	Gy	J/kg
radiation dose equivalent	sievert	Sv	J/kg
	SI supplementary units		
plane angle	radian	rad	
solid angle	steradian	sr	

Definitions of SI Units

meter (m) The *meter* is the length equal to the distance traveled by light, in vacuum, in a time of 1/299,792,458 second.

kilogram (kg) The *kilogram* is the unit of mass; it is defined by taking the value of Planck's constant h to be exactly $6.62607015 \times 10^{-34}\ \mathrm{kg \cdot m^2/s}$.

second (s) The *second* is the duration of 9,192,631,770 periods of the radiation corresponding to the transition between the two hyperfine levels of the ground state of the cesium-133 atom.

ampere (A) The *ampere* is a current of one coloumb per second, where the coulomb is defined in terms of the elementary charge e.

kelvin (K) The *kelvin,* unit of thermodynamic temperature, is defined by taking the value of the Boltzmann constant k to be exactly 1.380649×10^{-23} J/K.

ohm (Ω) The *ohm* is the electric resistance between two points of a conductor when a constant difference of potential of 1 volt, applied between these two points, produces in this conductor a current of 1 ampere, this conductor not being the source of any electromotive force.

coulomb (C) The *coulomb* is the quantity of electricity such that the elementary charge e is exactly $1.602176634 \times 10^{-19}$ C transported in 1 second by a current of 1 ampere.

candela (cd) The *candela* is the luminous intensity, in a given direction, of a source that emits monochromatic radiation of frequency 540×10^{12} hertz and that has a radiant intensity in that direction of 1/683 watt per steradian.

mole (mol) The *mole* is the SI unit of substance. One mole contains exactly $6.02214076 \times 10^{23}$ elementary entities.

newton (N) The *newton* is that force that gives to a mass of 1 kilogram an acceleration of 1 meter per second per second.

joule (J) The *joule* is the work done when the point of application of a constant force of 1 newton is displaced a distance of 1 meter in the direction of the force.

watt (W) The *watt* is the power that gives rise to the production of energy at the rate of 1 joule per second.

volt (V) The *volt* is the difference of electric potential between two points of a conducting wire carrying a constant current of 1 ampere, when the power dissipated between these points is equal to 1 watt.

weber (Wb) The *weber* is the magnetic flux that, linking a circuit of one turn, produces in it an electromotive force of 1 volt as it is reduced to zero at a uniform rate in 1 second.

lumen (lm) The *lumen* is the luminous flux emitted in a solid angle of 1 steradian by a uniform point source having an intensity of 1 candela.

farad (F) The *farad* is the capacitance of a capacitor between the plates of which there appears a difference of potential of 1 volt when it is charged by a quantity of electricity equal to 1 coulomb.

henry (H) The *henry* is the inductance of a closed circuit in which an electromotive force of 1 volt is produced when the electric current in the circuit varies uniformly at a rate of 1 ampere per second.

radian (rad) The *radian* is the plane angle between two radii of a circle that cut off on the circumference an arc equal in length to the radius.

steradian (sr) The *steradian* is the solid angle that, having its vertex in the center of a sphere, cuts off an area of the surface of the sphere equal to that of a square with sides of length equal to the radius of the sphere.

SI Prefixes To form the names of multiples and submultiples of SI units, apply the prefixes listed in Appendix F.

APPENDIX B

USEFUL MATHEMATICAL RELATIONS

Algebra

$a^{-x} = \dfrac{1}{a^x} \qquad a^{(x+y)} = a^x a^y \qquad a^{(x-y)} = \dfrac{a^x}{a^y}$

Logarithms: If $\log a = x$, then $a = 10^x$. $\quad \log a + \log b = \log(ab) \quad \log a - \log b = \log(a/b) \quad \log(a^n) = n \log a$

If $\ln a = x$, then $a = e^x$. $\quad \ln a + \ln b = \ln(ab) \quad \ln a - \ln b = \ln(a/b) \quad \ln(a^n) = n \ln a$

Quadratic formula: If $ax^2 + bx + c = 0$, $\qquad x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

Binomial Theorem

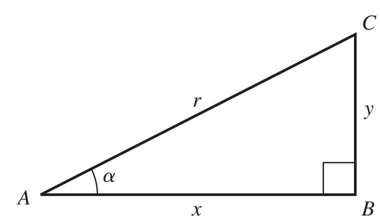
$$(a + b)^n = a^n + na^{n-1}b + \frac{n(n-1)a^{n-2}b^2}{2!} + \frac{n(n-1)(n-2)a^{n-3}b^3}{3!} + \cdots$$

Trigonometry

In the right triangle ABC, $x^2 + y^2 = r^2$.

Definitions of the trigonometric functions:

$\sin\alpha = y/r \qquad \cos\alpha = x/r \qquad \tan\alpha = y/x$



Identities:

$\sin^2\alpha + \cos^2\alpha = 1 \qquad \tan\alpha = \dfrac{\sin\alpha}{\cos\alpha}$

$\sin 2\alpha = 2\sin\alpha\cos\alpha \qquad \cos 2\alpha = \cos^2\alpha - \sin^2\alpha = 2\cos^2\alpha - 1 = 1 - 2\sin^2\alpha$

$\sin\tfrac{1}{2}\alpha = \sqrt{\dfrac{1-\cos\alpha}{2}} \qquad \cos\tfrac{1}{2}\alpha = \sqrt{\dfrac{1+\cos\alpha}{2}}$

$\sin(-\alpha) = -\sin\alpha \qquad \sin(\alpha \pm \beta) = \sin\alpha\cos\beta \pm \cos\alpha\sin\beta$

$\cos(-\alpha) = \cos\alpha \qquad \cos(\alpha \pm \beta) = \cos\alpha\cos\beta \mp \sin\alpha\sin\beta$

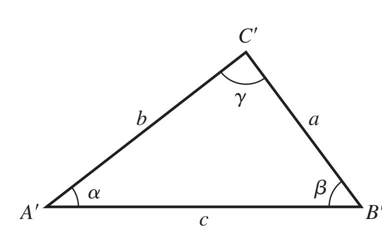
$\sin(\alpha \pm \pi/2) = \pm\cos\alpha \qquad \sin\alpha + \sin\beta = 2\sin\tfrac{1}{2}(\alpha+\beta)\cos\tfrac{1}{2}(\alpha-\beta)$

$\cos(\alpha \pm \pi/2) = \mp\sin\alpha \qquad \cos\alpha + \cos\beta = 2\cos\tfrac{1}{2}(\alpha+\beta)\cos\tfrac{1}{2}(\alpha-\beta)$

For *any* triangle $A'B'C'$ (not necessarily a right triangle) with sides a, b, and c and angles α, β, and γ:

Law of sines: $\dfrac{\sin\alpha}{a} = \dfrac{\sin\beta}{b} = \dfrac{\sin\gamma}{c}$

Law of cosines: $c^2 = a^2 + b^2 - 2ab\cos\gamma$



Geometry

Circumference of circle of radius r: $\quad C = 2\pi r$

Area of circle of radius r: $\quad A = \pi r^2$

Volume of sphere of radius r: $\quad V = 4\pi r^3/3$

Surface area of sphere of radius r: $\quad A = 4\pi r^2$

Volume of cylinder of radius r and height h: $\quad V = \pi r^2 h$

Calculus

Derivatives:

$$\frac{d}{dx}x^n = nx^{n-1} \qquad \frac{d}{dx}\ln ax = \frac{1}{x} \qquad \frac{d}{dx}e^{ax} = ae^{ax}$$

$$\frac{d}{dx}\sin ax = a\cos ax \qquad \frac{d}{dx}\cos ax = -a\sin ax$$

Integrals:

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} \quad (n \neq -1) \qquad \int \frac{dx}{x} = \ln x \qquad \int e^{ax}\,dx = \frac{1}{a}e^{ax}$$

$$\int \sin ax\,dx = -\frac{1}{a}\cos ax \qquad \int \cos ax\,dx = \frac{1}{a}\sin ax \qquad \int \frac{dx}{\sqrt{a^2 - x^2}} = \arcsin\frac{x}{a}$$

$$\int \frac{dx}{\sqrt{x^2 + a^2}} = \ln\left(x + \sqrt{x^2 + a^2}\right) \qquad \int \frac{dx}{x^2 + a^2} = \frac{1}{a}\arctan\frac{x}{a} \qquad \int \frac{dx}{(x^2 + a^2)^{3/2}} = \frac{1}{a^2}\frac{x}{\sqrt{x^2 + a^2}}$$

$$\int \frac{x\,dx}{(x^2 + a^2)^{3/2}} = -\frac{1}{\sqrt{x^2 + a^2}}$$

Power series (convergent for range of x shown):

$$(1 + x)^n = 1 + nx + \frac{n(n-1)x^2}{2!} + \frac{n(n-1)(n-2)}{3!}x^3 + \cdots \; (|x| < 1)$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots \; (\text{all } x)$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots \; (\text{all } x)$$

$$\tan x = x + \frac{x^3}{3} + \frac{2x^5}{15} + \frac{17x^7}{315} + \cdots \; (|x| < \pi/2)$$

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots \; (\text{all } x)$$

$$\ln(1 + x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots \; (|x| < 1)$$

APPENDIX C

THE GREEK ALPHABET

Name	Capital	Lowercase	Name	Capital	Lowercase	Name	Capital	Lowercase
Alpha	A	α	Iota	I	ι	Rho	P	ρ
Beta	B	β	Kappa	K	κ	Sigma	Σ	σ
Gamma	Γ	γ	Lambda	Λ	λ	Tau	T	τ
Delta	Δ	δ	Mu	M	μ	Upsilon	Υ	υ
Epsilon	E	ϵ	Nu	N	ν	Phi	Φ	ϕ
Zeta	Z	ζ	Xi	Ξ	ξ	Chi	X	χ
Eta	H	η	Omicron	O	o	Psi	Ψ	ψ
Theta	Θ	θ	Pi	Π	π	Omega	Ω	ω

APPENDIX D

PERIODIC TABLE OF THE ELEMENTS

Group Period	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
1	1 **H** 1.008																	2 **He** 4.003
2	3 **Li** 6.941	4 **Be** 9.012											5 **B** 10.811	6 **C** 12.011	7 **N** 14.007	8 **O** 15.999	9 **F** 18.998	10 **Ne** 20.180
3	11 **Na** 22.990	12 **Mg** 24.305											13 **Al** 26.982	14 **Si** 28.086	15 **P** 30.974	16 **S** 32.065	17 **Cl** 35.453	18 **Ar** 39.948
4	19 **K** 39.098	20 **Ca** 40.078	21 **Sc** 44.956	22 **Ti** 47.867	23 **V** 50.942	24 **Cr** 51.996	25 **Mn** 54.938	26 **Fe** 55.845	27 **Co** 58.933	28 **Ni** 58.693	29 **Cu** 63.546	30 **Zn** 65.409	31 **Ga** 69.723	32 **Ge** 72.64	33 **As** 74.922	34 **Se** 78.96	35 **Br** 79.904	36 **Kr** 83.798
5	37 **Rb** 85.468	38 **Sr** 87.62	39 **Y** 88.906	40 **Zr** 91.224	41 **Nb** 92.906	42 **Mo** 95.94	43 **Tc** (98)	44 **Ru** 101.07	45 **Rh** 102.906	46 **Pd** 106.42	47 **Ag** 107.868	48 **Cd** 112.411	49 **In** 114.818	50 **Sn** 118.710	51 **Sb** 121.760	52 **Te** 127.60	53 **I** 126.904	54 **Xe** 131.293
6	55 **Cs** 132.905	56 **Ba** 137.327	71 **Lu** 174.967	72 **Hf** 178.49	73 **Ta** 180.948	74 **W** 183.84	75 **Re** 186.207	76 **Os** 190.23	77 **Ir** 192.217	78 **Pt** 195.078	79 **Au** 196.967	80 **Hg** 200.59	81 **Tl** 204.383	82 **Pb** 207.2	83 **Bi** 208.980	84 **Po** (209)	85 **At** (210)	86 **Rn** (222)
7	87 **Fr** (223)	88 **Ra** (226)	103 **Lr** (262)	104 **Rf** (267)	105 **Db** (270)	106 **Sg** (269)	107 **Bh** (270)	108 **Hs** (269)	109 **Mt** (278)	110 **Ds** (281)	111 **Rg** (281)	112 **Cn** (285)	113 **Nh** (286)	114 **Fl** (289)	115 **Mc** (289)	116 **Lv** (293)	117 **Ts** (293)	118 **Og** (294)

Lanthanoids	57 **La** 138.905	58 **Ce** 140.116	59 **Pr** 140.908	60 **Nd** 144.24	61 **Pm** (145)	62 **Sm** 150.36	63 **Eu** 151.964	64 **Gd** 157.25	65 **Tb** 158.925	66 **Dy** 162.500	67 **Ho** 164.930	68 **Er** 167.259	69 **Tm** 168.934	70 **Yb** 173.04
Actinoids	89 **Ac** (227)	90 **Th** (232)	91 **Pa** (231)	92 **U** (238)	93 **Np** (237)	94 **Pu** (244)	95 **Am** (243)	96 **Cm** (247)	97 **Bk** (247)	98 **Cf** (251)	99 **Es** (252)	100 **Fm** (257)	101 **Md** (258)	102 **No** (259)

For each element, the average atomic mass of the mixture of isotopes occurring in nature is shown. For elements having no stable isotope, the approximate atomic mass of the longest-lived isotope is shown in parentheses. All atomic masses are expressed in atomic mass units (1 u $=$ 1.660539040(20) $\times$ 10^{-27} kg), equivalent to grams per mole (g/mol).

APPENDIX E

UNIT CONVERSION FACTORS

Length

1 m $=$ 100 cm $=$ 1000 mm $= 10^6\ \mu$m $= 10^9$ nm

1 km $=$ 1000 m $=$ 0.6214 mi

1 m $=$ 3.281 ft $=$ 39.37 in.

1 cm $=$ 0.3937 in.

1 in. $=$ 2.540 cm

1 ft $=$ 30.48 cm

1 yd $=$ 91.44 cm

1 mi $=$ 5280 ft $=$ 1.609 km

1 Å $= 10^{-10}$ m $= 10^{-8}$ cm $= 10^{-1}$ nm

1 nautical mile $=$ 6080 ft

1 light-year $= 9.461 \times 10^{15}$ m

Area

1 cm^2 $=$ 0.155 in.^{2}

1 m^2 $= 10^4$ cm^2 $=$ 10.76 ft^2

1 in.^{2} $=$ 6.452 cm^2

1 ft^2 $=$ 144 in.^{2} $=$ 0.0929 m^2

Volume

1 liter $=$ 1000 cm^3 $= 10^{-3}$ m^3 $=$ 0.03531 ft^3 $=$ 61.02 in.^{3}

1 ft^3 $=$ 0.02832 m^3 $=$ 28.32 liters $=$ 7.477 gallons

1 gallon $=$ 3.788 liters

Time

1 min $=$ 60 s

1 h $=$ 3600 s

1 d $=$ 86,400 s

1 y $=$ 365.24 d $= 3.156 \times 10^7$ s

Angle

1 rad $= 57.30° = 180°/\pi$

$1° = 0.01745$ rad $= \pi/180$ rad

1 revolution $= 360° = 2\pi$ rad

1 rev/min (rpm) $=$ 0.1047 rad/s

Speed

1 m/s $=$ 3.281 ft/s

1 ft/s $=$ 0.3048 m/s

1 mi/min $=$ 60 mi/h $=$ 88 ft/s

1 km/h $=$ 0.2778 m/s $=$ 0.6214 mi/h

1 mi/h $=$ 1.466 ft/s $=$ 0.4470 m/s $=$ 1.609 km/h

1 furlong/fortnight $= 1.662 \times 10^{-4}$ m/s

Acceleration

1 m/s^2 $=$ 100 cm/s^2 $=$ 3.281 ft/s^2

1 cm/s^2 $=$ 0.01 m/s^2 $=$ 0.03281 ft/s^2

1 ft/s^2 $=$ 0.3048 m/s^2 $=$ 30.48 cm/s^2

1 mi/h $\cdot$ s $=$ 1.467 ft/s^2

Mass

1 kg $= 10^3$ g $=$ 0.0685 slug

1 g $= 6.85 \times 10^{-5}$ slug

1 slug $=$ 14.59 kg

1 u $= 1.661 \times 10^{-27}$ kg

1 kg has a weight of 2.205 lb when $g = 9.80$ m/s^2

Force

1 N $= 10^5$ dyn $=$ 0.2248 lb

1 lb $=$ 4.448 N $= 4.448 \times 10^5$ dyn

Pressure

1 Pa $=$ 1 N/m^2 $= 1.450 \times 10^{-4}$ lb/in.^{2} $=$ 0.0209 lb/ft^2

1 bar $= 10^5$ Pa

1 lb/in.^{2} $=$ 6895 Pa

1 lb/ft^2 $=$ 47.88 Pa

1 atm $= 1.013 \times 10^5$ Pa $=$ 1.013 bar

$=$ 14.7 lb/in.^{2} $=$ 2117 lb/ft^2

1 mm Hg $=$ 1 torr $=$ 133.3 Pa

Energy

1 J $= 10^7$ ergs $=$ 0.239 cal

1 cal $=$ 4.186 J (based on 15° calorie)

1 ft $\cdot$ lb $=$ 1.356 J

1 Btu $=$ 1055 J $=$ 252 cal $=$ 778 ft $\cdot$ lb

1 eV $= 1.602 \times 10^{-19}$ J

1 kWh $= 3.600 \times 10^6$ J

Mass–Energy Equivalence

1 kg $\leftrightarrow 8.988 \times 10^{16}$ J

1 u $\leftrightarrow$ 931.5 MeV

1 eV $\leftrightarrow 1.074 \times 10^{-9}$ u

Power

1 W $=$ 1 J/s

1 hp $=$ 746 W $=$ 550 ft $\cdot$ lb/s

1 Btu/h $=$ 0.293 W

APPENDIX F

NUMERICAL CONSTANTS

Fundamental Physical Constants*

Name	Symbol	Value
Speed of light in vacuum	c	2.99792458×10^{8} m/s
Magnitude of charge of electron	e	$1.602176634 \times 10^{-19}$ C
Gravitational constant	G	$6.67408(31) \times 10^{-11}\ \mathrm{N \cdot m^2/kg^2}$
Planck's constant	h	$6.62607015 \times 10^{-34}\ \mathrm{J \cdot s}$
Boltzmann constant	k	1.380649×10^{-23} J/K
Avogadro's number	N_{A}	$6.02214076 \times 10^{23}$ molecules/mol
Gas constant	$R = N_{\mathrm{A}}K$	$8.314462618\ldots\ \mathrm{J/mol \cdot K}$
Mass of electron	m_{e}	$9.10938356(11) \times 10^{-31}$ kg
Mass of proton	m_{p}	$1.672621898(21) \times 10^{-27}$ kg
Mass of neutron	m_{n}	$1.674927471(21) \times 10^{-27}$ kg
Magnetic constant	μ_0	$1.25663706 \times 10^{-6}\ \mathrm{Wb/A \cdot m}$ (approximate) $\cong 4\pi \times 10^{-7}\ \mathrm{Wb/A \cdot m}$
Electric constant	$\epsilon_0 = 1/\mu_0 c^2$	$8.854187817 \times 10^{-12}\ \mathrm{C^2/N \cdot m^2}$ (approximate)
	$1/4\pi\epsilon_0$	$8.987551787 \times 10^{9}\ \mathrm{N \cdot m^2/C^2}$ (approximate)

Other Useful Constants*

Mechanical equivalent of heat		4.186 J/cal (15° calorie)
Standard atmospheric pressure	1 atm	1.01325×10^{5} Pa
Absolute zero	0 K	−273.15°C
Electron volt	1 eV	$1.6021766209(98) \times 10^{-19}$ J
Atomic mass unit	1 u	$1.660539040(20) \times 10^{-27}$ kg
Electron rest energy	$m_{\mathrm{e}}c^2$	0.5109989461(31) MeV
Volume of ideal gas (0°C and 1 atm)		22.413962(13) liter/mol
Acceleration due to gravity (standard)	g	$9.80665\ \mathrm{m/s^2}$

*Source: National Institute of Standards and Technology (**http://physics.nist.gov/cuu**). Numbers in parentheses show the uncertainty in the final digits of the main number; for example, the number 1.6454(21) means 1.6454 $\pm$ 0.0021. Values shown without uncertainties are exact. The exact values of the magnitude of the charge of the electron, Planck's constant, the Boltzmann constant, Avogadro's number, and the gas constant are from the redefinitions adopted in 2018. As consequences of these redefinitions, the values of the magnetic constant and electric constant now have fractional uncertainties of about 2×10^{-10}. As of this writing (2018) it was expected that updated values of the magnetic and electric constants, as well as of all other constants with uncertainties, were to be announced in May 2019. These updated values will be available at **http://physics.nist.gov/cuu**.

Astronomical Data†

Body	Mass (kg)	Radius (m)	Orbit radius (m)	Orbital period
Sun	1.99×10^{30}	6.96×10^{8}	—	—
Moon	7.35×10^{22}	1.74×10^{6}	3.84×10^{8}	27.3 d
Mercury	3.30×10^{23}	2.44×10^{6}	5.79×10^{10}	88.0 d
Venus	4.87×10^{24}	6.05×10^{6}	1.08×10^{11}	224.7 d
Earth	5.97×10^{24}	6.37×10^{6}	1.50×10^{11}	365.3 d
Mars	6.42×10^{23}	3.39×10^{6}	2.28×10^{11}	687.0 d
Jupiter	1.90×10^{27}	6.99×10^{7}	7.78×10^{11}	11.86 y
Saturn	5.68×10^{26}	5.82×10^{7}	1.43×10^{12}	29.45 y
Uranus	8.68×10^{25}	2.54×10^{7}	2.87×10^{12}	84.02 y
Neptune	1.02×10^{26}	2.46×10^{7}	4.50×10^{12}	164.8 y
Pluto‡	1.30×10^{22}	1.19×10^{6}	5.91×10^{12}	248.0 y

†Source: NASA (**http://solarsystem.nasa.gov/planets/**). For each body, "radius" is its average radius and "orbit radius" is its average distance from the sun or (for the moon) from the earth.
‡In August 2006, the International Astronomical Union reclassified Pluto and similar small objects that orbit the sun as "dwarf planets."

Prefixes for Powers of 10

Power of ten	Prefix	Abbreviation	Pronunciation
10^{-24}	yocto-	y	*yoc*-toe
10^{-21}	zepto-	z	*zep*-toe
10^{-18}	atto-	a	*at*-toe
10^{-15}	femto-	f	*fem*-toe
10^{-12}	pico-	p	*pee*-koe
10^{-9}	nano-	n	*nan*-oe
10^{-6}	micro-	μ	*my*-crow
10^{-3}	milli-	m	*mil*-i
10^{-2}	centi-	c	*cen*-ti
10^{3}	kilo-	k	*kil*-oe
10^{6}	mega-	M	*meg*-a
10^{9}	giga-	G	*jig*-a or *gig*-a
10^{12}	tera-	T	*ter*-a
10^{15}	peta-	P	*pet*-a
10^{18}	exa-	E	*ex*-a
10^{21}	zetta-	Z	*zet*-a
10^{24}	yotta-	Y	*yot*-a

Examples:

1 femtometer $=$ 1 fm $= 10^{-15}$ m

1 picosecond $=$ 1 ps $= 10^{-12}$ s

1 nanocoulomb $=$ 1 nC $= 10^{-9}$ C

1 microkelvin $=$ 1 μK $= 10^{-6}$ K

1 millivolt $=$ 1 mV $= 10^{-3}$ V

1 kilopascal $=$ 1 kPa $= 10^{3}$ Pa

1 megawatt $=$ 1 MW $= 10^{6}$ W

1 gigahertz $=$ 1 GHz $= 10^{9}$ Hz

ANSWERS TO ODD-NUMBERED PROBLEMS

Chapter 1

1.1 (a) 1.61 km b) 3.28×10^3 ft
1.3 1.02 ns
1.5 31.7 y
1.7 (a) 23.4 km/L (b) 1.4 tanks
1.9 9.0 cm
1.11 (a) 72 mm^2 (b) 0.50 (c) 36 mm (d) 6 mm (e) 2.0
1.13 0.45%
1.15 (a) no (b) no (c) no (d) no (e) no
1.17 $300 million
1.19 2×10^5
1.21 7.8 km, 38° north of east
1.23 144 m, 41° south of west.
1.25 $A_x = 0$, $A_y = -8.00$ m, $B_x = 7.50$ m, $B_y = 13.0$ cm, $C_x = -10.9$ cm, $C_y = -5.07$ m, $D_x = -7.99$ m, $D_y = 6.02$ m
1.27 (a) −6.00 m (b) 11.3 m
1.29 (a) 9.01 m, 33.7° (b) 9.01 m, 33.7° (c) 22.3 m, 250° (d) 22.3 m, 70.3°
1.31 2.81 km, 38.5° north of west
1.33 (a) 2.48 cm, 18.4° (b) 4.09 cm, 83.7° (c) 4.09 cm, 264°
1.35 $\vec{A} = -(8.00\text{ m})\,\hat{\jmath}$, $\vec{B} = (7.50\text{ m})\,\hat{\imath} + (+13.0\text{ m})\,\hat{\jmath}$, $\vec{C} = (-10.9\text{ m})\hat{\imath} + (-5.07\text{ m})\hat{\jmath}$, $\vec{D} = (-7.99\text{ m})\hat{\imath} + (6.02\text{ m})\hat{\jmath}$
1.37 (a) $\vec{A} = (1.23\text{ m})\hat{\imath} + (3.38\text{ m})\hat{\jmath}$, $\vec{B} = (-2.08\text{ m})\hat{\imath} + (-1.20\text{ m})\hat{\jmath}$ (b) $\vec{C} = (12.0\text{ m})\hat{\imath} + (14.9\text{ m})\hat{\jmath}$ (c) 19.2 m, 51.2°
1.39 (a) $A = 5.38$, $B = 4.36$ (b) $-5.00\,\hat{\imath} + 2.00\,\hat{\jmath} + 7.00\,\hat{k}$ (c) 8.83, yes
1.41 (a) −104 m^2 (b) −148 m^2 (c) 40.6 m^2
1.43 (a) 165° (b) 28° (c) 90°
1.45 (a) $(-63.9\text{ m}^2)\hat{k}$ (b) $(63.9\text{ m}^2)\hat{k}$
1.47 −2.0 m, $-x$-direction
1.49 (a) 5.49 g/cm^3 (b) 1.1×10^6 g/cm^3 (c) 4.7×10^{14} g/cm^3
1.51 (a) 1.64×10^4 km (b) $2.57 r_E$
1.53 $\approx 6 \times 10^{27}$
1.55 (a) $0.60\,\hat{\imath} - 0.80\,\hat{k}$ (b) $-0.60\,\hat{\imath} + 0.80\,\hat{k}$ (c) $\vec{B}_+ = 0.80\,\hat{\imath} + 0.60\,\hat{k}$ and $\vec{B}_- = -0.80\,\hat{\imath} - 0.60\,\hat{k}$
1.57 144 m, 41° south of west
1.59 361 N/C, 184.73° from the $+x$-axis
1.61 7.55 N
1.63 60.9 km, 33.0° south of west
1.65 28.8 m, 11.4° north of east
1.67 71.9 m, 64.1° north of west
1.69 (a) 818 m, 15.8° west of south
1.71 18.6° east of south, 29.6 m
1.73 28.2 m
1.75 −2.00 J
1.77 124°
1.79 (a) $-bd$, $-bc\hat{\imath} - ad\hat{\jmath} - ac\hat{k}$ (b) $-bd$, ad in the $-y$-direction
1.81 156 m^2
1.83 28.0 m
1.85 $C_x = -8.0$, $C_y = -6.1$
1.87 D, F, B, C, A, E
1.89 (b) (i) 0.9857 AU (ii) 1.3820 AU (iii) 1.695 AU (c) 54.6°
1.91 (a) 76.2 ly (b) 129°
1.93 Choice (a)

Chapter 2

2.1 25.0 m
2.3 55 min
2.5 (a) 0.313 m/s (b) 1.56 m/s
2.7 (a) 12.0 m/s (b) (i) 0 (ii) 15.0 m/s (iii) 12.0 m/s (c) 13.3 m/s
2.9 (a) 2.33 m/s, 2.33 m/s (b) 2.33 m/s, 0.33 m/s
2.11 6.7 m/s, 6.7 m/s, 0, −40.0 m/s, −40.0 m/s, −40.0 m/s, 0
2.13 (a) 2.00 cm/s, 50.0 cm, −0.125 cm/s^2 (b) 16.0 s (c) 32.0 s (d) 6.20 s, 1.23 cm/s; 25.8 s, −1.23 cm/s; 36.4 s, −2.55 cm/s
2.15 (a) 0.500 m/s^2 (b) 0, 1.00 m/s^2
2.17 (a) 2.17 m, 9.60 m/s^2; 15.0 m, −38.4 m/s^2
2.19 (a) 8.33 m/s (b) 1.11 m/s^2
2.21 (a) 675 m/s^2 (b) 0.0667 s
2.23 1.70 m
2.25 0.38 m
2.27 (a) 3.1×10^6 m/s^2 = $3.2 \times 10^5\ g$ (b) 1.6 ms (c) no
2.29 (a) 0, 6.3 m/s^2, −11.2 m/s^2 (b) 100 m, 230 m, 320 m
2.31 (a) 2.94 m/s (b) 0.600 s
2.33 1.67 s
2.35 (a) 33.5 m (b) 15.8 m/s
2.37 (a) upward, downward, decreasing (b) downward, downward, increasing
2.39 (a) $t = \sqrt{2d/g}$ (b) 0.190 s
2.41 (a) 646 m (b) 16.4 s, 112 m/s
2.43 15.9 m
2.45 0.0868 m/s^2
2.47 37.6 m/s
2.49 (a) 467 m (b) 110 m/s
2.51 (a) $x = (0.25\text{ m/s}^3)t^3 - (0.010\text{ m/s}^4)t^4$, $v_x = (0.75\text{ m/s}^3)t^2 - (0.040\text{ m/s}^4)t^3$ (b) 39.1 m/s
2.53 (a) 10.0 m (b) (i) 8.33 m/s (ii) 9.09 m/s (iii) 9.52 m/s
2.55 250 km
2.57 (a) 197 m/s (b) 169 m/s
2.59 (a) 4.24 m/s (b) 7.54 s
2.61 (a) 92.0 m (b) 92.0 m
2.63 67 m
2.65 (a) 7.56 s (b) 37.2 m (c) 25.7 m/s (car), 15.9 m/s (truck)
2.67 3.2 m/s^2
2.69 (a) 2.0 s (b) 16 m/s in the $-x$-direction, −40 m/s^2 in the $-x$-direction
2.71 (a) −4.00 m/s (b) 12.0 m/s
2.73 38.2 m
2.75 (a) 6.69 m/s (b) 4.49 m (c) 1.42 s
2.77 (a) 3.3 s (b) $9H$
2.79 (a) 380 m (b) 184 m
2.81 (a) 0.625 m/s^3 (b) 107 m
2.83 (a) car A (b) 2.27 s, 5.73 s (c) 1.00 s, 4.33 s (d) 2.67 s
2.85 (a) 0.0510 s^2/m (b) lower than (c) no
2.87 4.8
2.89 (a) 8.3 m/s (b) (i) 0.411 m (ii) 1.15 km (c) 9.8 m/s (d) 4.9 m/s
2.91 Choice (b)

Chapter 3

3.1 (a) 1.4 m/s, −1.3 m/s (b) 1.9 m/s, 317°
3.3 (a) 7.1 cm/s, 45° (b) 5.0 cm/s, 90°; 7.1 cm/s, 45°; 11 cm/s, 27°
3.5 (b) −8.67 m/s^2, −2.33 m/s^2 (c) 8.98 m/s^2, 195°
3.7 (b) $\vec{v} = \alpha\,\hat{\imath} - 2\beta t\,\hat{\jmath}$, $\vec{a} = -2\beta\,\hat{\jmath}$ (c) 5.4 m/s, 297°; 2.4 m/s^2, 270° (d) speeding up and turning right
3.9 (a) 1.13 m (b) 0.528 m (c) $v_x = 1.10$ m/s, $v_y = -4.70$ m/s, 4.83 m/s, 76.8° below the horizontal
3.11 2.57 m
3.13 (a) 24.1 m/s (b) 31.0 m/s
3.15 1.28 m/s^2
3.17 (a) 0.683 s, 2.99 s (b) 24.0 m/s, 11.3 m/s; 24.0 m/s, −11.3 m/s (c) 30.0 m/s, 36.9° below the horizontal
3.19 (a) 1.5 m (b) −0.89 m/s
3.21 (a) 13.6 m (b) 34.6 m/s (c) 103 m
3.25 (a) 0.034 m/s^2 = $0.0034g$ (b) 1.4 h
3.27 52.8 m/s^2
3.29 120 m/s, 270 mph
3.31 (a) 2.57 m/s^2 upward (b) 2.57 m/s^2 downward (c) 14.7 s
3.33 (a) 32.9 m/s (b) 27.7 m/s^2 (c) 35.5 rpm
3.35 (a) 14 s (b) 70 s
3.37 0.36 m/s, 52.5° south of west
3.39 (a) 24° west of south (b) 5.5 h
3.41 (a) 4.7 m/s, 25° south of east (b) 120 s (c) 240 m
3.43 (a) 14°, north of west (b) 310 km/h
3.45 $2b/3c$
3.47 (a) 128 m (b) 315 m
3.49 274 m
3.51 (a) 55.5 m south
3.53 33.7 m
3.55 (a) 16.6 m/s (b) 10.9 m/s, 40.5° below the horizontal
3.57 (a) 1.50 m/s (b) 4.66 m
3.59 (a) $v = \sqrt{v_0^2 + 2gH}$ (b) $v = \sqrt{v_0^2 + 2gH}$ (c) $\sqrt{v_0^2 + 2gH}$ (d) stays the same
3.61 (a) 6.91 m (c) no
3.63 (a) 17.8 m/s (b) in the river, 28.4 m horizontally from his launch point
3.65 (a) 49.5 m/s (b) 50 m
3.67 6.25 s
3.69 (a) 13.3 m/s (b) 3.8 m
3.71 (a) 44.7 km/h, 26.6° west of south (b) 10.5° north of west
3.73 7.39 m/s, 12.4° north of east
3.75 3.01 m/s, 33.7° north of east
3.77 (a) graph R^2 versus h (b) 16.4 m/s (c) 23.8 m
3.79 70.5°
3.81 5.15 s
3.83 Choice (b)
3.85 Choice (c)

Chapter 4

4.1 494 N, 31.8°
4.3 3.15 N
4.5 (a) −8.10 N, 3.00 N (b) 8.64 N
4.7 46.7 N, opposite to the motion of the skater
4.9 21.8 kg
4.11 (a) 3.12 m, 3.12 m/s (b) 21.9 m, 6.24 m/s
4.13 (a) 45.0 N, between 2.0 s and 4.0 s (b) between 2.0 s and 4.0 s (c) 0 s, 6.0 s
4.15 (a) $A = 100$ N, $B = 12.5$ N/s^2 (b) (i) 21.6 N, 2.70 m/s^2 (ii) 134 N, 16.8 m/s^2 (c) 26.6 m/s^2

4.17 2940 N
4.19 (a) 4.49 kg (b) 4.49 kg, 8.13 N
4.21 825 N, blocks
4.23 50 N
4.25 7.4×10^{-23} m/s^2
4.27 (b) yes
4.29 (a) yes (b) no
4.33 2.58 s
4.35 (a) 17 N, 90° clockwise from the $+x$-axis
(b) 840 N
4.37 (a) 2.50 m/s^2 (b) 10.0 N (c) to the right, $F > T$ (d) 25.0 N
4.39 (a) 4.4 m (b) 300 m/s (c) (i) 2.7×10^4 N
(ii) 9.0×10^3 N
4.41 (a) 0.603 m/s^2, upward
(b) 1.26 m/s^2, downward
4.43 1630 N away from him
4.45 (a) 4.34 kg (b) 5.30 kg
4.47 309 N
4.49 7.78 m
4.51 (a) largest: Ferrari; smallest: Alpha Romeo and Honda Civic (b) largest: Ferrari; smallest: Volvo (c) 7.5 kN, smaller (d) zero
4.53 (b) 26 kg, 8.3 m/s^2
4.55 (a) 0.593 m, 1.33 s, 4.00 N (b) 2.00 s, 2.00 m/s
4.57 Choice (d)
4.59 Choice (a)

Chapter 5

5.1 (a) 25.0 N (b) 50.0 N
5.3 (a) 990 N, 735 N (b) 926 N
5.5 48°
5.7 (a) $T_A = 0.732w$, $T_B = 0.897w$, $T_C = w$
(b) $T_A = 2.73w$, $T_B = 3.35w$, $T_C = w$
5.9 (a) 574 N (b) 607 N
5.11 (a) 1.10×10^8 N (b) $5w$ (c) 8.4 s
5.13 (a) 4610 m/s^2 = $470g$
(b) 9.70×10^5 N = $471w$ (c) 0.0187 s
5.15 (b) 2.96 m/s^2 (c) 191 N; greater than; less than
5.17 (b) 3.75 m/s^2 (c) 2.48 kg (d) $T <$ weight of the hanging block
5.19 (a) 3.4 m/s (c) $2.2w$
5.21 (a) 14.0 m (b) 18.0 m/s
5.23 50°
5.25 (a) 33 N (b) 3.1 m
5.27 (a) μ_s: 0.710; μ_k: 0.472 (b) 258 N (c) (i) 51.8 N
(ii) 4.97 m/s^2
5.31 (a) 18.3 m/s^2 (b) 2.29 m/s^2
5.33 (a) 57.1 N (b) 146 N up the ramp
5.35 (a) 52.5 m (b) 16.0 m/s
5.37 (a) $\mu_k(m_A + m_B)g$ (b) $\mu_k m_A g$
5.39 (a) 0.218 m/s (b) 11.7 N
5.41 (a) $\dfrac{\mu_k mg}{\cos\theta - \mu_k \sin\theta}$ (b) $1/\tan\theta$
5.43 (b) 8.2 m/s
5.45 (a) 61.8 N (b) 30.4 N
5.47 3.66 s
5.49 (a) 21.0°, no (b) 11,800 N (car), 23,600 N (truck)
5.51 6200 N (horizontal cable), 1410 (upper cable)
5.53 (a) 1.5 rev/min (b) 0.92 rev/min
5.55 (a) $\sqrt{Lg}$ (b) $5mg$
5.57 2.42 m/s
5.59 (a) rope making 60° angle (b) 6400 N
5.61 $T_B = 4960$ N, $T_C = 1200$ N
5.63 3.18 m/s
5.65 (a) 101 N (b) 33.8 N (c) 62.3 N
5.67 762 N
5.69 (a) (i) −3.80 m/s (ii) 24.6 m/s
(b) 4.36 m (c) 2.45 s
5.71 245 N, equal to it
5.73 12.3 m/s
5.75 1.78 m/s
5.77 (a) $m_1(\sin\alpha + \mu_k\cos\alpha)$ (b) $m_1(\sin\alpha - \mu_k\cos\alpha)$
(c) $m_1(\sin\alpha - \mu_s\cos\alpha) \le m_2 \le m_1(\sin\alpha + \mu_s\cos\alpha)$
5.79 (a) 1.44 N (b) 1.80 N
5.81 920 N
5.83 1.30 kg, 2.19 kg
5.85 (a) 294 N (18.0-cm wire), 152 N, 152 N
(b) 40.0 N
5.87 3.0 N
5.89 (a) 12.9 kg (b) $T_{AB} = 47.2$ N, $T_{BC} = 101$ N
5.91 $a_1 = \dfrac{2m_2 g}{4m_1 + m_2}$, $a_2 = \dfrac{m_2 g}{4m_1 + m_2}$
5.93 (a) 7.75 N (b) 0.242
5.95 g/μ_s
5.97 (b) 0.452
5.99 (b) 0.28 (c) no
5.101 (a) 1.42×10^4 N (b) 817 m/s^2 (c) 27.1 m/s^2
5.103 (b) 8.8 N (c) 31.0 N (d) 1.54 m/s^2
5.105 $v = (2mg/k)\left[\dfrac{1}{2} + e^{-(k/m)t}\right]$
5.107 (a) 81.1° (b) no (c) The bead rides at the bottom of the hoop.
5.109 (a) 0.371 (b) 0.290 (c) yes, same slope, less-negative intercept
5.111 (a) 5/8 in (b) 23.9 kN (c) 3.57 kN, smaller
(d) larger; accurate
5.113 $F = (M + m)g\tan\alpha$
5.115 $\cos^2\beta$
5.117 Choice (b)

Chapter 6

6.1 (a) 3.60 J (b) −0.900 J (c) 0 (d) 0
(e) 2.70 J
6.3 (a) 74 N (b) 333 J (c) −330 J (d) 0, 0
(e) 0
6.5 (a) −1750 J (b) no
6.7 (a) (i) 9.00 J (ii) −9.00 J (b) (i) 0
(ii) 9.00 J (iii) −9.00 J (iv) 0
(c) zero for each block
6.9 −6.7 J
6.11 (a) 0° (b) 4
6.13 −572 J
6.15 (a) 120 J (b) −108 J (c) 24.3 J
6.17 (a) 36,000 J (b) 4
6.19 (a) 1.0×10^{16} J (b) 2.4 times
6.21 $4W_1$
6.23 $\sqrt{2gh(1 + \mu_k/\tan\alpha)}$
6.25 48.0 N
6.27 (a) 4.48 m/s (b) 3.61 m/s
6.29 (a) twice as fast (b) both decrease by a factor of $1/\sqrt{3}$
6.31 (a) $v_0^2/2\mu_k g$ (b) (i) $^1/_2$ (ii) 4 (iii) 2
6.33 (b) 13.1 cm (bottom), 14.1 cm (middle), 15.2 cm (top)
6.35 (a) 2.83 m/s (b) 3.46 m/s
6.37 8.5 cm
6.39 (a) 1.76 (b) 0.666 m/s
6.41 (a) 4.0 J (b) 0 (c) −1.0 J (d) 3.0 J (e) −1.0 J
6.43 (a) 2.83 m/s (b) 2.40 m/s
6.45 8.17 m/s
6.47 360,000 J; 100 m/s
6.53 (a) 84.6/min (b) 22.7/min
6.55 29.6 kW
6.57 0.20 W
6.59 (a) 608 J (b) −395 J (c) 0 (d) −189 J
(e) 24 J (f) 1.5 m/s
6.61 (a) −35.5 J (b) decreases (c) −51.6 J
6.63 (a) 5.62 J (20.0-N block), 3.38 J (12.0-N block)
(b) 2.58 J (20.0-N block), 1.54 J (12.0-N block)
6.65 (a) 1.8 m/s = 4.0 mi/h
(b) 180 m/s^2 ≈ $18g$, 900 N
6.67 (a) 5.11 m (b) 0.304 (c) 10.3 m
6.69 (a) $V/\sqrt{2}$, greater than (b) $W_D/4$, less than
6.71 (a) 0.074 N (b) 4.7 N (c) 0.22 J
6.73 6.3×10^4 N/m
6.75 1.1 m
6.77 (a) 2.39 m/s (b) 9.42 m/s, away from the wall
6.79 (a) 0.600 m (b) 1.50 m/s
6.81 0.786
6.83 1.3 m
6.85 (a) 1.10×10^5 J (b) 1.30×10^5 J (c) 3.99 kW
6.87 (a) 47.0 J, −29.6 J, 2.41 m/s
(b) 23.2 J, 29.6 J, −6.4 J
(c) 40.6 J, 47.0 J, −6.4 J
6.89 (a) 1.26×10^5 J (b) 1.46 W
6.91 (b) $v^2 = -\dfrac{k}{m}d^2 + 2d\left[\dfrac{k}{m}(0.400\text{ m}) - \mu_k g\right]$
(c) 1.29 m/s, 0.204 m (d) 12.0 N/m, 0.800
6.93 (a) $Mv^2/6$ (b) 6.1 m/s (c) 3.9 m/s
(d) 0.40 J, 0.60 J
6.95 Choice (a)
6.97 Choice (d)

Chapter 7

7.1 (a) 6.6×10^5 J (b) -7.7×10^5 J
7.3 (a) 610 N (b) (i) 0 (ii) 550 J
7.5 (a) 24.0 m/s (b) 24.0 m/s (c) part (b)
7.7 (a) 2.0 m/s (b) 9.8×10^{-7} J, 2.0 J/kg
(c) 200 m, 63 m/s (d) 5.9 J/kg
(e) in its tensed legs
7.9 (a) (i) 0 (ii) 0.98 J (b) 2.8 m/s
(c) Only gravity is constant. (d) 5.1 N
7.11 −5400 J
7.13 (a) 11.8 J, −15.7 J (b) (0.200 m)T, −(0.200 m)T
(c) 0, −3.9 J, 0.75 m/s
7.15 (a) 52.0 J (b) 3.25 J
7.17 (a) (i) $4U_0$ (ii) $U_0/4$ (b) (i) $x_0\sqrt{2}$
(ii) $x_0/\sqrt{2}$
7.19 (a) 5.48 cm (b) 3.92 cm
7.21 (a) 6.32 cm (b) 12 cm
7.23 (a) 3.03 m/s, as it leaves the spring
(b) 95.9 m/s^2, when the spring has its maximum compression
7.25 (a) 4.46×10^5 N/m (b) 0.128 m
7.27 (a) −5.4 J (b) −5.4 J (c) −10.8 J
(d) nonconservative
7.29 (a) 8.16 m/s (b) 766 J
7.31 1.29 N, $+x$-direction
7.33 130 m/s^2, 132° counterclockwise from the $+x$-axis
7.35 (a) $F(r) = (12a/r^{13}) - (6b/r^7)$
(b) $(2a/b)^{1/6}$, yes (c) $b^2/4a$
(d) $a = 6.67 \times 10^{-138}$ J·m^{12},
$b = 6.41 \times 10^{-78}$ J·m^6
7.37 (a) zero (gravel), 637 N (box) (b) 2.99 m/s
7.39 0.41
7.41 (a) 16.0 m/s (b) 11,500 N
7.43 (a) 20.0 m along the rough bottom (b) −78.4 J
7.45 (a) 22.2 m/s (b) 16.4 m (c) no
7.47 10.8 J
7.49 15.5 m/s
7.51 4.4 m/s
7.53 (a) 7.00 m/s (b) 8.82 N
7.55 48.2°
7.57 (a) 0.392 (b) −0.83 J
7.59 (a) $U(x) = \frac{1}{2}\alpha x^2 + \frac{1}{3}\beta x^3$ (b) 7.85 m/s
7.61 (a) $\alpha/(x + x_0)$ (b) 3.27 m/s

7.63 7.01 m/s
7.65 (a) 0.747 m/s (b) 0.931 m/s
7.67 (a) 0.480 m/s (b) 0.566 m/s
7.69 (a) 3.87 m/s (b) 0.10 m
7.71 0.456 N
7.73 119 J
7.75 (a) −50.6 J (b) −67.5 J (c) nonconservative
7.77 (a) 57.0 m (b) 16.5 m (c) negative work done by air resistance
7.79 (a) yes (b) 0.14 J
(d) −1.0 m, 0, 1.0 m
(e) positive: $-1.5\text{ m} < x < -1.0\text{ m}$ and $0 < x < 1.0\text{ m}$
negative: $-1.0\text{ m} < x < 0$ and $1.0\text{ m} < x < 1.5\text{ m}$ (f) −0.55 m, 0.12 J
7.81 Choice (c)
7.83 Choice (b)

Chapter 8

8.1 (a) 1.20×10^5 kg · m/s (b) (i) 60.0 m/s (ii) 26.8 m/s
8.3 (a) −30 kg · m/s, −55 kg · m/s
(b) 0, 52 kg · m/s (c) 0, −3.0 kg · m/s
8.5 (a) 22.5 kg · m/s, to the left (b) 838 J
8.7 562 N, not significant
8.9 (a) 10.8 m/s, to the right (b) 0.750 m/s, to the left
8.11 (a) 500 N/s^2 (b) 5810 N · s (c) 2.70 m/s
8.13 (a) 2.50 N · s, in the direction of the force
(b) (i) 6.25 m/s, to the right (ii) 3.75 m/s, to the right
8.15 (a) 8.00×10^2 kg · m/s (b) 8.00×10^3 J
(c) 160 N opposite to her velocity
8.17 0.87 kg · m/s, in the same direction as the bullet is traveling
8.19 (a) 6.79 m/s (b) 55.2 J
8.21 (a) 0.790 m/s (b) −0.0023 J
8.23 1.97 m/s
8.25 (a) 0.0559 m/s (b) 0.0313 m/s
8.27 (a) 7.20 m/s, 38.0° from Rebecca's original direction (b) −680 J
8.29 (a) 4.3 m/s (c) 4.3 m/s
8.31 0.31 m/s toward the south
8.33 (a) 0.846 m/s (b) 2.10 J
8.35 13.5 J
8.37 5.9 m/s, 58° north of east
8.39 5.46 m/s, 36.0° south of east
8.41 19.5 m/s (car), 21.9 m/s (truck)
8.43 (a) 2.93 cm (b) 866 J (c) 1.73 J
8.45 13.6 N
8.47 (a) 3.00 J; 0.500 m/s for both
(b) A: −1.00 m/s; B: 1.00 m/s
8.49 (a) $v_1/3$ (b) $K_1/9$ (c) 10
8.51 (a) B (b) 3 times greater (c) 3.0 m/s to the left
8.53 2520 km
8.55 0.488 m
8.57 0.73 m/s
8.59 $F_x = -(1.50\text{ N/s})t$, $F_y = 0.25$ N, $F_z = 0$
8.61 (a) 0.053 kg (b) 5.19 N
8.65 (a) −1.14 N · s, 0.330 N · s
(b) 0.04 m/s, 1.8 m/s
8.67 (a) 5.21 J, −0.0833 m/s
(b) −2.17 m/s (A), 0.333 m/s (B)
8.69 (a) 1.75 m/s, 0.260 m/s (b) −0.092 J
8.71 0.946 m
8.73 1.8 m
8.75 $v_A/2$ for each one
8.77 12 m/s (SUV), 21 m/s (sedan)
8.79 (a) 2.60 m/s (b) 325 m/s
8.81 (a) 5.3 m/s (b) 5.7 m
8.83 (a) 1 (b) $3 - 2\sqrt{2}$, $3 + 2\sqrt{2}$
8.85 (a) 0.0781 (b) 248 J (c) 0.441 J
8.87 (a) 9.35 m/s (b) 3.29 m/s
8.89 4.04 m/s
8.91 1.33 m
8.93 $H/9$
8.95 (a) 0.150 J (b) 15.4 J (c) 0.560 m
8.97 (a) 71.6 m/s (0.28-kg piece), 14.3 m/s (1.40-kg piece) (b) 347 m
8.99 (a) yes (b) no, decreases by 4800 J
8.101 (a) maximum: C, minimum: B (b) 69 N/m
(c) 0.12 m
8.103 (a) $g/3$ (b) 14.7 m (c) 29.4 g
8.105 0, $4a/3\pi$
8.107 Choice (b)
8.109 Choice (b)

Chapter 9

9.1 (a) 0.600 rad, 34.4° (b) 6.27 cm (c) 1.05 m
9.3 (a) rad/s, rad/s^3 (b) (i) 0 (ii) 15.0 rad/s^2
(c) 9.50 rad
9.5 (a) $\omega_z = \gamma + 3\beta t^2$ (b) 0.400 rad/s
(c) 1.30 rad/s, 0.700 rad/s
9.7 (a) $\pi/4$ rad, 2.00 rad/s, −0.139 rad/s^3
(b) 0 (c) 19.5 rad, 9.36 rad/s
9.9 (a) 2.00 rad/s (b) 4.38 rad
9.11 (a) 24.0 s (b) 68.8 rev
9.13 3.00 rad/s
9.15 (a) 300 rpm (b) 75.0 s, 312 rev
9.17 9.00 rev
9.21 (a) 15.1 m/s^2 (b) 15.1 m/s^2
9.25 0.107 m, no
9.27 5.97 rad/s^2
9.29 0.180 m/s^2, 0.432 m/s^2, 0.468 m/s^2
9.31 (a) (i) 0.469 kg · m^2 (ii) 0.117 kg · m^2 (iii) 0
(b) (i) 0.0433 kg · m^2 (ii) 0.0722 kg · m^2
(c) (i) 0.0288 kg · m^2 (ii) 0.0144 kg · m^2
9.33 (a) 1.93 kg · m^2 (b) 6.53 kg · m^2
(c) 0 (d) 1.15 kg · m^2
9.35 0.193 kg · m^2
9.37 0.208 s
9.39 $\omega_1\sqrt{3/5}$
9.41 6.49 m/s
9.43 (a) wheel B (b) $\dfrac{3}{16}$
9.45 7.35×10^4 J
9.47 (a) 0.673 m (b) 45.5%
9.49 46.5 kg
9.51 an axis that is parallel to a diameter and is $0.516R$ from the center
9.53 $M(a^2 + b^2)/3$
9.55 $\frac{1}{2}MR^2$
9.57 (a) $\gamma L^2/2$ (b) $ML^2/2$ (c) $ML^2/6$
9.59 7.68 m
9.61 (a) 0.600 m/s^3 (b) $\alpha = (2.40\text{ rad/s}^3)t$ (c) 3.54 s
(d) 17.7 rad
9.63 1.22 kg
9.65 8.44 kg
9.67 2.99 cm
9.69 68.4 J
9.71 4.65 kg · m^2
9.73 (a) −0.882 J (b) 5.42 rad/s (c) 5.42 m/s
(d) 5.42 m/s compared to 4.43 m/s
9.75 1.46 m/s
9.77 $\sqrt{\dfrac{2gd(m_B - \mu_k m_A)}{m_A + m_B + I/R^2}}$
9.79 (a) 2.25×10^{-3} kg · m^2 (b) 3.40 m/s
(c) 4.95 m/s
9.81 13.9 m
9.85 (a) 55.3 kg (b) 0.804 kg · m^2
9.87 (a) 4.0 rev, no (b) 15 rad/s (c) 9.5 rad/s
9.89 (a) yes (b) 3.15 m/s (c) 0.348 kg · m^2
(d) 36.4 N
9.91 (a) $s(\theta) = r_0\theta + \dfrac{\beta}{2}\theta^2$
(b) $\theta(t) = \dfrac{1}{\beta}\left(\sqrt{r_0^2 + 2\beta vt} - r_0\right)$
(c) $\omega_z(t) = \dfrac{v}{\sqrt{r_0^2 + 2\beta vt}}$,
$\alpha_z(t) = -\dfrac{\beta v^2}{(r_0^2 + 2\beta vt)^{3/2}}$, no
(d) 25.0 mm, 0.247 μm/rad, 2.13×10^4 rev
9.93 Choice (d)
9.95 Choice (d)

Chapter 10

10.1 (a) 40.0 N · m, out of the page (b) 34.6 N · m, out of the page (c) 20.0 N · m, out of the page
(d) 17.3 N · m, into the page (e) 0 (f) 0
10.3 2.50 N · m, out of the page
10.5 (b) $-\hat{\boldsymbol{k}}$ (c) $(-1.05\text{ N}\cdot\text{m})\hat{\boldsymbol{k}}$
10.7 (a) 2.56 N · m (b) 4.25 N · m, perpendicular to handle
10.9 8.38 N · m
10.11 (a) 14.8 rad/s^2 (b) 1.52 s
10.13 (a) 7.5 N (at book on table), 18.2 N (at hanging book) (b) 0.16 kg · m^2
10.15 0.255 kg · m^2
10.17 0.0131 kg · m^2
10.19 (a) 1.56 m/s (b) 5.35 J
(c) (i) 3.12 m/s to the right (ii) 0 (iii) 2.21 m/s at 45° below the horizontal
(d) (i) 1.56 m/s to the right (ii) 1.56 m/s to the left
(iii) 1.56 m/s downward
10.21 (a) $\frac{1}{3}$ (b) $\frac{2}{7}$ (c) $\frac{2}{5}$ (d) $\frac{5}{13}$
10.23 (a) 0.613 (b) no (c) no slipping
10.25 14.0 m
10.29 (a) 3.76 m (b) 8.58 m/s
10.31 (a) 67.9 rad/s (b) 8.35 J
10.33 (a) 0.309 rad/s (b) 100 J (c) 6.67 W
10.35 (a) 0.704 N · m (b) 157 rad (c) 111 J (d) 111 J
10.37 (a) 115 kg · m^2/s into the page (b) 125 kg · m^2/s^2 out of the page
10.39 4.71×10^{-6} kg · m^2/s
10.41 (a) A: rad/s^2; B: rad/s^4 (b) (i) 59.0 kg · m^2/s
(ii) 56.1 N · m
10.43 4600 rad/s
10.45 1.14 rev/s
10.47 (a) 1.38 rad/s (b) 1080 J, 495 J
10.49 (a) 0.120 rad/s (b) 3.20×10^{-4} J (c) work done by the bug
10.51 4.82 m/s^2
10.53 $\dfrac{m_{\text{rock}}v}{\left(m + \dfrac{M}{2}\right)R}$
10.55 2.4×10^{-12} N · m
10.59 0.483
10.61 (a) 16.3 rad/s^2 (b) no, decrease (c) 5.70 rad/s
10.63 0.921 m/s^2, 7.68 rad/s^2, 35.5 N (at A), 21.4 N (at B)
10.65 $\dfrac{7v_{\text{cm}}^2}{10g}$, $\dfrac{5v_{\text{cm}}^2}{6g}$, hollow
10.67 68.0 N
10.69 270 N
10.71 $a = \dfrac{2g}{2 + (R/b)^2}$, $\alpha = \dfrac{2g}{2b + R^2/b}$,
$T = \dfrac{2mg}{2(b/R)^2 + 1}$
10.73 (a) $3H_0/5$
10.75 29.0 m/s
10.77 (a) 26.0 m/s (b) no change
10.79 $g/3$

10.81 (a) $\frac{6}{19}v/L$ (b) $\frac{3}{19}$

10.83 3200 J

10.85 (a) 2.00 rad/s (b) 6.58 rad/s

10.87 0.776 rad/s

10.89 (a) *A*: solid sphere, *B*: solid cylinder, *C*: hollow sphere, *D*: hollow cylinder (b) same (c) *D* (d) 0.350

10.91 (a) $mv_1^2r_1^2/r^3$ (b) $\frac{mv_1^2}{2}r_1^2\left(\frac{1}{r_2^2}-\frac{1}{r_1^2}\right)$ (c) same

10.93 (a) 39.2 N upward, 39.2 N upward (b) 60.0 N upward, 18.4 N upward (c) 165 N upward, 86.2 N downward (d) 0.0940 rev/s

10.95 Choice (c)

10.97 Choice (a)

Chapter 11

11.1 29.8 cm

11.3 1.35 m

11.5 9.70 cm

11.7 6.6 kN

11.9 (a) 1000 N, 0.800 m from end where 600-N force is applied (b) 800 N, 0.75 m from end where 600-N force is applied

11.11 (a) 550 N (b) 0.614 m from *A*

11.13 (a) 1920 N (b) 1140 N

11.15 (a) $T = 2.60w$; $3.28w$, 37.6° (b) $T = 4.10w$; $5.39w$, 48.8°

11.17 (a) 3410 N (b) 3410 N, 7600 N

11.19 (b) 530 N (c) 600 N, 270 N; downward

11.21 220 N (left), 255 N (right), 42°

11.23 (a) 0.800 m (b) clockwise (c) 0.800 m, clockwise

11.25 6.12 kg

11.27 1.9 mm

11.29 2.0×10^{11} Pa

11.31 2.0×10^3 atm

11.33 (a) 150 atm (b) 1.5 km, no

11.35 4.8×10^9 Pa, 2.1×10^{-10} Pa^{-1}

11.37 (a) 2.4×10^{-2} (b) 2.4×10^{-3} m

11.39 $1.4T_{\text{steel}}$

11.41 $0.19\Delta p_1$

11.43 3.41×10^7 Pa

11.47 20.0 kg

11.49 (a) 525 N (b) 222 N, 328 N (c) 1.48

11.51 (a) 140 N (b) 6 cm to the right

11.53 (a) 409 N (b) 161 N

11.55 49.9 cm

11.57 (a) $\arctan(mg/2T)$ (b) $\sqrt{\frac{3g\sin\theta}{L}}$

11.59 (a) $\arctan\left[\left(\frac{1}{\alpha}-1\right)\tan\theta\right]$ (b)1/2 (c) 0

11.61 5500 N

11.63 (b) 2000 N = $2.72mg$ (c) 4.4 mm

11.65 (a) 1.39 m (b) 1.8×10^{-3} m

11.67 (a) 175 N at each hand, 200 N at each foot (b) 91 N at each hand and at each foot

11.69 (a) 1150 N (b) 1940 N (c) 918 N (d) 0.473

11.71 590 N (person above), 1370 N (person below); person above

11.73 (a) $\frac{T_{\max}hD}{L\sqrt{h^2+D^2}}$ (b) $\frac{T_{\max}h}{L\sqrt{h^2+D^2}}\left(1-\frac{D^2}{h^2+D^2}\right)$, positive

11.75 (a) 375 N (b) 325 N (c) 512 N

11.77 (a) 0.424 N (*A*), 1.47 N (*B*), 0.424 N (*C*) (b) 0.848 N

11.79 (a) 27° to tip, 31° to slip, tips first (b) 27° to tip, 22° to slip, slips first

11.81 (a) 80 N (*A*), 870 N (*B*) (b) 1.92 m

11.83 (a) 0.70 m from *A* (b) 0.60 m from *A*

11.85 (a) 4.2×10^4 N (b) 65 m

11.87 (b) $x = 1.50\text{ m} + \frac{(1.30\text{ m})m_1 - (0.38\text{ m})M}{m_2}$ (c) 1.59 kg (d) 1.50 m

11.89 (a) 391 N (4.00-m ladder), 449 N (3.00-m ladder) (b) 322 N (c) 334 N (d) 937 N

11.91 (a) 0.66 mm (b) 0.022 J (c) 8.35×10^{-3} J (d) -3.04×10^{-2} J (e) 3.04×10^{-2} J

11.93 Choice (a)

11.95 Choice (d)

Chapter 12

12.1 no (41.8 N)

12.3 7020 kg/m^3; yes

12.5 1.6

12.7 (a) 1.86×10^6 Pa (b) 184 m

12.9 0.581 m

12.11 828 kg/m^3

12.13 2.8 m

12.15 6.0×10^4 Pa

12.17 2.27×10^5 N

12.19 (a) 636 Pa (b) (i) 1170 Pa (ii) 1170 Pa

12.21 10.9

12.23 10.3 m

12.25 0.0898 m^3, 1020 kg/m^3

12.27 0.122 m

12.29 6.43×10^{-4} m^3, 2.78×10^3 kg/m^3

12.31 10.5 N

12.33 (a) 116 Pa (b) 921 Pa (c) 0.822 kg, 822 kg/m^3

12.35 1640 kg/m^3

12.37 501 N

12.39 9.6 m/s

12.41 (a) 17.0 m/s (b) 0.317 m

12.43 (a) 0.536 mm (b) 30.8 s, yes

12.45 28.4 m/s

12.47 1.47×10^5 Pa

12.49 2.03×10^4 Pa

12.51 2.25×10^5 Pa

12.53 $1.19D$

12.55 (a) 5.9×10^5 N (b) 1.8×10^5 N

12.57 2.61×10^4 N·m

12.59 0.800 kg

12.61 0.964 cm, rise

12.63 (a) 1470 Pa (b) 13.9 cm

12.65 (a) 0.0500 m^3 (b) 10.0 kg

12.67 9.8×10^6 kg, yes

12.71 (a) 16.5 cm (b) 1.75 m

12.73 (a) 5.07 m/s, 1.28 (b) 32.4 min, 2.08

12.75 (a) $1-\frac{\rho_B}{\rho_L}$ (b) $\left(\frac{\rho_L-\rho_B}{\rho_L-\rho_w}\right)L$ (c) 4.60 cm

12.77 (a) $2\sqrt{h(H-h)}$ (b) h

12.79 5.47 m

12.81 (a) 0.200 m^3/s (b) 6.97×10^4 Pa

12.83 $3h_1$

12.85 (b) no

12.87 (a) 2.5×10^{-4} m^2/Pa (slope), 16 m^2 (intercept) (b) 8.2 m, 800 kg/m^3

12.89 Choice (b)

12.91 Choice (a)

Chapter 13

13.1 2.18

13.3 (a) 1.2×10^{-11} m/s^2 (b) 15 days (c) no, increase

13.5 2.1×10^{-9} m/s^2, downward

13.7 (a)2.4×10^{-3} N (b)$F_{\text{moon}}/F_{\text{earth}} = 3.5 \times 10^{-6}$

13.9 (a) 0.634 m from $3m$ (b) (i) unstable (ii) stable

13.11 1.38×10^7 m

13.13 (a) 0.37 m/s^2 (b) 1700 kg/m^3

13.15 (a) 12.3 m/s^2 (b) 80.0% the density of earth

13.17 (a) 5020 m/s (b) 60,600 m/s

13.19 9.03 m/s^2

13.21 2.40×10^2 N

13.23 -2.00×10^6 J, -4.00×10^6 J

13.25 (a) 7410 m/s (b) 1.71 h

13.27 7330 m/s

13.29 (a) 4.1 m/s = 9.1 mph, yes (b) 2.6 h

13.31 (a) 82,700 m/s (b) 14.5 days

13.33 (a) 7.84×10^9 s = 248 y (b) 4.44×10^{12} m, 7.38×10^{12} m

13.35 (a) 0 (b) 1.07×10^{-12} N (c) 1.25×10^{-12} N

13.37 (a) (i) 5.31×10^{-9} N (ii) 2.67×10^{-9} N

13.39 (a) $-\frac{GmM}{\sqrt{x^2+a^2}}$ (b) $-GmM/x$ (c) $\frac{GmMx}{(x^2+a^2)^{3/2}}$, toward the ring (d) GmM/x^2 (e) $U = -GmM/a$, $F_x = 0$

13.41 (a) 33.7 N (b) 32.8 N

13.43 (a) 4.3×10^{37} kg = $(2.1 \times 10^7)M_S$ (b) no (c) 6.32×10^{10} m, yes

13.45 (a) 9.67×10^{-12} N, at 45° above $+x$-axis (b) 3.02×10^{-5} m/s

13.47 (b) (i) 1.49×10^{-5} m/s (50.0-kg sphere), 7.46×10^{-6} m/s (100.0-kg sphere) (ii) 2.24×10^{-5} m/s (c) 26.6 m

13.49 (a) 3.59×10^7 m

13.51 177 m/s

13.53 (a) 7.36 h (b) 2.47 h

13.55 1.83×10^{27} kg

13.57 22.8 m

13.59 $4\pi GM$

13.61 $v = \sqrt{\frac{2Gm_E h}{R_E(R_E+h)}}$

13.63 (a) $GM^2/4R^2$ (b) $v = \sqrt{GM/4R}$, $T = 4\pi\sqrt{R^3/GM}$ (c) $GM^2/4R$

13.65 6.8×10^4 m/s

13.67 (a) 7910 s (b) 1.53 (c) 8430 m/s (perigee), 5510 m/s (apogee) (d) 2410 m/s; 3250 m/s; perigee

13.69 5.36×10^9 J

13.71 9.36 m/s^2

13.73 $GmMx/(a^2+x^2)^{3/2}$

13.75 (a) $U(r) = \frac{Gm_E m}{2R_E^3}r^2$ (b) 7.90×10^3 m/s

13.77 (a) It is considerable and shows no apparent pattern. (b) Earth (5500 kg/m^3), Mercury (5400 kg/m^3), Venus (5300 kg/m^3), Mars (3900 kg/m^3), Neptune (1600 kg/m^3), Uranus (1200 kg/m^3), Jupiter (1200 kg/m^3), Saturn (530 kg/m^3) (c) no effect (d) 90 m/s^2

13.79 (a) opposite; opposite (b) 259 days (c) 44.1°

13.81 $\frac{2GMm}{a^2}\left(1-\frac{x}{\sqrt{a^2+x^2}}\right)$

13.83 Choice (c)

Chapter 14

14.1 (a) 2.15 ms, 2930 rad/s (b) 2.00×10^4 Hz, 1.26×10^5 rad/s (c) 1.3×10^{-15} s $\leq T \leq 2.3 \times 10^{-15}$ s, 4.3×10^{14} Hz $\leq f \leq 7.5 \times 10^{14}$ Hz (d) 2.0×10^{-7} s, 3.1×10^7 rad/s

14.3 5530 rad/s, 1.14 ms

14.5 0.0625 s

14.7 (a) 0.80 s (b) 1.25 Hz (c) 7.85 rad/s (d) 3.0 cm (e) 148 N/m

14.9 (a) 0.167 s (b) 37.7 rad/s (c) 0.0844 kg

14.11 (a) 0.150 s (b) 0.0750 s

14.13 (a) 0.98 m (b) $\pi/2$ rad (c) $x = (-0.98\text{ m})\sin[(12.2\text{ rad/s})t]$

14.15 (a) $T/6$ (b) $T/12$ (c) no

14.17 120 kg

14.19 (a) 0.253 kg (b) 1.21 cm (c) 3.03 N
14.21 (a) 1.51 s (b) 26.0 N/m (c) 30.8 cm/s (d) 1.92 N (e) −0.0125 m, 30.4 cm/s, 0.216 m/s^2 (f) 0.324 N
14.23 (a) $\omega^3 A \sin \omega t$ (b) 0 (c) 0 (d) $\pm A$ (e) 1.3 s
14.25 92.2 m/s^2
14.27 (a) 0.0336 J (b) 0.0150 m (c) 0.669 m/s
14.29 (a) 1.20 m/s (b) 1.11 m/s (c) 36 m/s^2 (d) 13.5 m/s^2 (e) 0.36 J
14.31 $\frac{\sqrt{3}}{2}A$, larger
14.33 0.240 m
14.35 (a) 0.376 m (b) 59.3 m/s^2 (c) 119 N
14.37 (a) 4.06 cm (b) 1.21 m/s (c) 29.8 rad/s
14.39 (a) 0, 0, 3.92 J, 3.92 J (b) 3.92 J, 0, 0, 3.92 J (c) 0.98 J, 0.98 J, 1.96 J, 3.92 J
14.41 (a) $2.7 \times 10^{-8}\ \text{kg} \cdot \text{m}^2$ (b) $4.3 \times 10^{-6}\ \text{N} \cdot \text{m/rad}$
14.43 (a) 0.25 s (b) 0.25 s
14.45 0.407 swings per second
14.47 10.7 m/s^2
14.49 (a) $g \sin\theta$ (b) $g\sqrt{\cos^2(\theta/2) + 4[\cos(\theta/2) - \cos\theta]^2}$ (c) $2g(1 - \cos\theta)$, $\sqrt{2gL(1 - \cos\theta)}$
14.51 A: $2\pi\sqrt{\frac{L}{g}}$, B: $\frac{2\sqrt{2}}{3}\left(2\pi\sqrt{\frac{L}{g}}\right)$; pendulum A
14.53 A: $2\pi\sqrt{\frac{L}{g}}$, B: $\sqrt{\frac{11}{10}}\left(2\pi\sqrt{\frac{L}{g}}\right)$, pendulum B
14.55 $\frac{\sqrt{3}}{2}\omega$
14.57 (a) 0.393 Hz (b) 1.73 kg/s
14.59 (a) $A_1/3$ (b) $2A_1$
14.61 (a) 12 (b) 72
14.63 0.353 m
14.65 (a) $16k_1$ (b) double it
14.67 599 N/m
14.69 (a) 24.4 cm (b) 0.221 s (c) 1.19 m/s
14.71 2.00 m
14.73 $0.921\left(\frac{1}{2\pi}\sqrt{\frac{g}{L}}\right)$
14.75 (a) 0.784 s (b) -1.12×10^{-4} s per s; shorter (c) 0.419 s
14.77 (a) 0.150 m/s (b) 0.112 m/s^2 downward (c) 0.700 s (d) 4.38 m
14.79 (a) 2.6 m/s (b) 0.21 m (c) 0.49 s
14.81 1.17 s
14.83 0.421 s
14.85 0.705 Hz, 14.5°
14.87 $2\pi\sqrt{\frac{M}{3k}}$
14.89 (a) 1.60 s (b) 0.625 HZ (c) 3.93 rad/s (d) 5.1 cm; 0.4 s, 1.2 s, 1.8 s (e) 79 cm/s^2; 0.4 s, 1.2 s, 1.8 s (f) 4.9 kg
14.91 (b) The angular amplitude increases as L decreases. (c) about 53°
14.93 (a) $Mv^2/6$ (c) $\omega = \sqrt{3k/M}$, $M' = M/3$
14.95 Choice (a)

Chapter 15

15.1 (a) 0.439 m, 1.28 ms (b) 0.219 m
15.3 220 m/s = 800 km/h
15.5 (a) 1.7 cm to 17 m (b) 4.3×10^{14} Hz to 7.5×10^{14} Hz (c) 1.5 cm (d) 6.4 cm
15.7 (a) 25.0 Hz, 0.0400 s, 19.6 rad/m (b) $y(x, t) = (0.0700\ \text{m}) \cos[(19.6\ \text{m}^{-1})x + (157\ \text{rad/s})t]$ (c) 4.95 cm (d) 0.0050 s
15.9 (a) yes (b) yes (c) no (d) $v_y = \omega A \cos(kx + \omega t)$, $a_y = -\omega^2 A \sin(kx + \omega t)$
15.11 (a) 4 mm (b) 0.040 s (c) 0.14 m, 3.6 m/s (d) 0.24 m, 6.0 m/s (e) no
15.13 (b) $+x$-direction
15.15 (a) 17.5 m/s (b) 0.146 m (c) both would increase by a factor of $\sqrt{2}$
15.17 0.337 kg
15.19 (a) 9.53 N (b) 20.8 m/s
15.23 4.10 mm
15.25 (a) 95 km (b) 0.25 μW/m^2 (c) 110 kW
15.27 (a) 0.050 W/m^2 (b) 22 kJ
15.35 (a) (1.33 m)n, $n = 0, 1, 2, \ldots$ (b) (1.33 m)$(n + \frac{1}{2})$, $n = 0, 1, 2, \ldots$
15.37 (a) 96.0 m/s (b) 461 N (c) 1.13 m/s, 4.26 m/s^2
15.39 (b) 2.80 cm (c) 277 cm (d) 185 cm, 7.96 Hz, 0.126 s, 1470 cm/s (e) 280 cm/s (f) $y(x, t) = (5.60\ \text{cm}) \times \sin[(0.0906\ \text{rad/cm})x] \sin[(133\ \text{rad/s})t]$
15.41 (a) 0.240 m (b) 0.0600 m
15.43 (a) 45.0 cm (b) no
15.45 1.80×10^3 m/s, 1.06×10^{-4} m
15.47 (a) 20.0 Hz, 126 rad/s, 3.49 rad/m (b) $y(x, t) = (2.50 \times 10^{-3}\ \text{m}) \cos[(3.49\ \text{rad/m})x - (126\ \text{rad/s})t]$ (c) $y(0, t) = (2.50 \times 10^{-3}\ \text{m}) \cos[(126\ \text{rad/s})t]$ (d) $y(1.35\ m, t) = (2.50 \times 10^{-3}\ \text{m}) \times \cos[(126\ \text{rad/s})t - 3\pi/2\ \text{rad}]$ (e) 0.315 m/s (f) -2.50×10^{-3} m, 0
15.49 6.00×10^{-2} kg
15.51 (a) 62.1 m
15.53 13.7 Hz, 25.0 m
15.55 1.83 m
15.57 360 Hz (copper), 488 Hz (aluminum)
15.59 (a) 18.8 cm (b) 0.0169 kg
15.61 (a) 7.07 cm (b) 0.400 kW
15.63 (a) 0.0673 kg/m (b) 14.3 N (c) 14.6 m/s (d) 19.5 Hz (e) 1, 26, 51, 76, 101 (f) 0.0615 m (g) $(n-1)(15.0\ \text{mm})$ (h) 3.64 m/s
15.65 (a) 2.22 g (b) 2.24×10^4 m/s^2
15.67 233 N
15.69 1780 kg/m^3
15.71 (a) 148 N (b) 26%
15.73 (c) 47.5 Hz (d) 138 g
15.75 (a) because $P_{av} = 2\pi^2\sqrt{\mu F}A^2 f^2$ (b) 280 m/s (c) 1100 rad/s
15.77 (a) 392 N (b) 392 N + (7.70 N/m)x (c) 3.89 s
15.79 Choice (b)

Chapter 16

16.1 (a) 0.344 m (b) 1.2×10^{-5} m (c) 6.9 m, 50 Hz
16.3 (a) 7.78 Pa (b) 77.8 Pa (c) 778 Pa
16.5 0.156 s
16.7 90.8 m
16.9 81.4°C
16.11 (a) 5.5×10^{-15} J (b) 0.074 mm/s
16.13 15.0 cm
16.15 (a) 4.14 Pa (b) 0.0208 W/m^2 (c) 103 dB
16.17 (a) 4.4×10^{-12} W/m^2 (b) 6.4 dB (c) 5.8×10^{-11} m
16.19 14.0 dB
16.21 (a) 2.0×10^{-7} W/m^2 (b) 6.0 m (c) 290 m (d) yes, no
16.23 (a) fundamental: 0.60 m; 0, 1.20 m; first overtone: 0.30 m, 0.90 m; 0, 0.60 m, 1.20 m; second overtone: 0.20 m, 0.60 m, 1.00 m; 0, 0.40 m, 0.80 m, 1.20 m (b) fundamental: 0; 1.20 m; first overtone: 0, 0.80 m; 0.40 m, 1.20 m; second overtone: 0, 0.48 m, 0.96 m; 0.24 m, 0.72 m, 1.20 m
16.25 506 Hz, 1517 Hz, 2529 Hz
16.27 (a) 35.2 Hz (b) 17.6 Hz
16.29 (a) closed (b) 5$^{\text{th}}$ harmonic (c) 0.453 m (d) 190 Hz
16.31 (a) 614 Hz (b) 1230 Hz
16.33 (a) 172 Hz (b) 86 Hz
16.35 0.125 m
16.37 (a) (820 Hz)n, $n = 1, 2, 3, \ldots$ (b) (410 Hz)$(2n + 1)$, $n = 0, 1, 2, \ldots$
16.39 (a) 433 Hz (b) loosen
16.41 780 m/s
16.43 (a) 375 Hz (b) 371 Hz (c) 4 Hz
16.45 (a) 0.25 m/s (b) 0.91 m
16.47 19.8 m/s
16.49 (a) 355 m/s (b) 0.710 m (c) 500 Hz
16.51 (a) 7.02 m/s, toward (b) 1404 Hz
16.53 (a) 36.0° (b) 2.94 s
16.55 (b) 344 m/s (c) no
16.57 (a) 0.879 s (b) 27.3 Hz
16.59 (a) stopped (b) 7$^{\text{th}}$ and 9$^{\text{th}}$ (c) 0.439 m
16.61 (a) 0.026 m, 0.53 m, 1.27 m, 2.71 m, 9.01 m (b) 0.26 m, 0.86 m, 1.84 m, 4.34 m (c) 86 Hz
16.63 (a) 0.0823 m (b) 120 Hz
16.67 (b) 2.0 m/s
16.69 (a) 38 Hz (b) no
16.71 (a) 375 m/s (b) 1.39 (c) 0.8 cm
16.73 (a) 441 Hz (b) 0.430 m
16.75 (d) 9.69 cm/s, 667 m/s^2
16.77 Choice (b)
16.79 Choice (a)
16.81 Choice (b)

Chapter 17

17.1 (a) −81.0°F (b) 134.1°F (c) 88.0°F
17.3 (a) 27.2 C° (b) −55.6 C°
17.5 (a) −18.0 F° (b) −10.0 C°
17.7 0.964 atm
17.9 (a) −282°C (b) no, 47,600 Pa
17.11 0.39 m
17.13 1.9014 cm; 1.8964 cm
17.15 49.4°C
17.17 $1.7 \times 10^{-5}\ (\text{C}°)^{-1}$
17.19 (a) 1.431 cm^2 (b) 1.436 cm^2
17.21 (a) 6.0 mm (b) -1.0×10^8 Pa
17.23 $\frac{L_{\text{Invar}}}{L_{\text{Al}}} = \frac{1}{0.225} = 4$
17.25 554 kJ
17.27 23 min
17.29 240 J/kg · K
17.31 0.526 C°
17.33 0.0613 C°
17.35 (a) 215 J/kg · K (b) water (c) too small
17.37 0.114 kg
17.39 27.5°C
17.41 150°C
17.43 54.5 kJ, 13.0 kcal, 51.7 Btu
17.45 357 m/s
17.47 3.45 L
17.49 5.05×10^{15} kg
17.51 0.0674 kg
17.53 2370 J/kg · K
17.55 157 W/m · K
17.57 (a) −0.86°C (b) 24 W/m^2
17.61 105.5°C
17.63 (a) 21 kW (b) 6.4 kW
17.65 2.1 cm^2
17.67 (a) 1.61×10^{11} m (b) 5.43×10^6 m
17.69 35.0°C
17.71 $c = \frac{Q\beta}{\rho \Delta V}$
17.73 (a) 35.1°M (b) 39.6 C°
17.75 69.4°C
17.77 23.0 cm (first rod), 7.0 cm (second rod)
17.79 (b) 1.9×10^8 Pa
17.81 (a) 87°C (b) −80°C
17.83 2.00×10^3 J/kg · K

17.85 (a) 83.6 J (b) 1.86 J/mol·K (c) 5.60 J/mol·K
17.87 (a) 4.20×10^7 J (b) 10.7 C° (c) 30.0 C°
17.89 (a) 0.60 kg (b) 0.80 bottles/h
17.91 3.4×10^5 J/kg
17.93 (a) no (b) 0.0°C, 0.156 kg
17.95 (a) 86.1°C (b) no ice, 0.130 kg liquid water, no steam
17.97 (a) 100°C (b) 0.0214 kg steam, 0.219 kg liquid water
17.99 (a) 93.9 W (b) 1.35
17.101 2.9
17.103 (a) 1.04 kW (b) 87.1 W (c) 1.13 kW (d) 28 g (e) 1.1 bottles
17.105 (c) 170 h (d) 1.5×10^{10} s $\approx$ 500 y, no
17.107 5.82 g
17.109 (a) 3.00×10^4 J/kg (b) 1.00×10^3 J/kg·K (liquid), 1.33×10^3 J/kg·K (solid)
17.111 *A:* 216 W/m·K, *B:* 130 W/m·K
17.113 $H = \dfrac{k\pi R_1 R_2}{L}(T_H - T_C)$
17.115 (a) $H = \dfrac{(T_2 - T_1)2\pi kL}{\ln(b/a)}$
(b) $T = T_2 - \dfrac{(T_2 - T_1)\ln(r/a)}{\ln(b/a)}$ (d) 73°C
(e) 49 W
17.117 Choice (a)
17.119 Choice (a)

Chapter 18

18.1 (a) 0.122 mol (b) 14,700 Pa, 0.145 atm
18.3 0.100 atm
18.5 (a) 5.7×10^6 Pa (b) 3.79×10^6 Pa, −25.2%
18.7 503°C
18.9 19.7 kPa
18.11 0.159 L
18.13 $0.0508V$
18.15 0.421 mol
18.17 (a) 6.95×10^{-16} kg (b) 2.32×10^{-13} kg/m^3
18.19 55.6 mol, 3.35×10^{25} molecules
18.21 (a) 2.20×10^6 molecules
(b) 2.44×10^{19} molecules
18.23 (a) 3.45×10^{-9} m (b) about 10 times the diameter of a typical molecule (c) 10 times the spacing of atoms in solids
18.25 (a) 5.83×10^7 J (b) 242 m/s
18.27 6.46×10^{23}
18.29 (a) 1.93×10^6 m/s, no (b) 7.3×10^{10} K
18.31 (a) 6.21×10^{-21} J (b) 2.34×10^5 m^2/s^2
(c) 484 m/s (d) 2.57×10^{-23} kg·m/s
(e) 1.24×10^{-19} N (f) 1.24×10^{-17} Pa
(g) 8.17×10^{21} molecules
(h) 2.45×10^{22} molecules
18.33 3800°C
18.35 1100 m/s
18.37 (a) 1870 J (b) 1120 J
18.39 (a) 741 J/kg·K, $c_w = 5.65c_{N_2}$
(b) 5.65 kg; 4850 L
18.41 (a) 337 m/s (b) 380 m/s (c) 412 m/s
18.43 (a) 1.34 (b) 1.23
18.45 (a) 610 Pa (b) 22.12 MPa
18.49 (a) 11.8 kPa (b) 0.566 L
18.51 272°C
18.53 0.195 kg
18.55 (a) $a = \left(\dfrac{T_{in}}{T_{out}} - 1\right)g$ (b) $T_{in} = T_{out}\left(\dfrac{a}{g} + 1\right)$
18.57 (a) 0.0959 m^3 (b) 17.8 mol/s (c) 42.6 m
18.59 (a) 30.7 cylinders (b) 8420 N (c) 7800 N
18.61 (a) 26.2 m/s (b) 16.1 m/s, 5.44 m/s (c) 1.74 m
18.63 $\approx 5 \times 10^{27}$ atoms
18.65 (a) 4.65×10^{-26} kg (b) 6.11×10^{-21} J
(c) 2.04×10^{24} molecules (d) 12.5 kJ
18.67 (b) r_2 (c) $r_1 = \dfrac{R_0}{2^{1/6}}$, $r_2 = R_0$, $2^{-1/6}$ (d) U_0
18.69 (a) $2R = 16.6$ J/mol·K (b) less than
18.71 (b) 1.40×10^5 K (N_2), 1.01×10^4 K (H_2)
(c) 6370 K (N_2), 459 K (H_2)
18.73 $3kT/m$, same
18.75 (b) $0.0421N$ (c) $(2.94 \times 10^{-21})N$
(d) $0.0297N$, $(2.08 \times 10^{-21})N$
(e) $0.0595N$, $(4.15 \times 10^{-21})N$
18.77 (a) $p_0 + \dfrac{mg}{\pi r^2}$ (b) $-\left(\dfrac{y}{h}\right)(p_0\pi r^2 + mg)$
(c) $\dfrac{1}{2\pi}\sqrt{\dfrac{g}{h}\left(1 + \dfrac{p_0\pi r^2}{mg}\right)}$, no
18.79 (a) 42.6% (b) 3 km (c) 1 km
18.81 (a) 4.1×10^{-46} kg·m^2 (c) $v_f = 0$, $\omega = v_i/r$
(d) 5.5×10^{12} rad/s
18.83 (a) 4.5×10^{11} m (b) 703 m/s, 6.4×10^8 s ($\approx$20 y) (c) 1.4×10^{-14} Pa
(d) 650 m/s, $v_H > v_{esc}$, evaporate
(f) 2×10^5 K, $> 3T_{sun}$, no
18.85 Choice (a)
18.87 Choice (c)

Chapter 19

19.1 (b) 1330 J
19.3 (b) −6180 J
19.5 (a) 1.04 atm
19.7 (a) $(p_1 - p_2)(V_2 - V_1)$ (b) negative of work done in reverse direction
19.9 (a) 34.7 kJ (b) 80.4 kJ (c) no
19.11 (a) 278 K, at *a* (b) 0; 162 J (c) 53 J
19.13 (a) $T_a = 535$ K, $T_b = 9350$ K, $T_c = 15{,}000$ K
(b) 21 kJ done by gas (c) 36 kJ
19.15 (a) $Q = +100$ J, $W = -300$ J (b) $Q = -100$ J, $W = -300$ J (c) $Q = +100$ J, $W = +300$ J
(d) −100 J, $W = +300$ J
19.17 (b) 208 J (c) on the piston (d) 712 J
(e) 920 J (f) 208 J
19.19 (a) 948 K (b) 900 K
19.21 $\frac{2}{5}$
19.23 (a) 747 J (b) 1.30
19.25 (a) −605 J (b) 0 (c) yes, 605 J, liberate
19.27 (a) 476 kPa (b) −10.6 kJ (c) 1.59, heated
19.29 (b) 314 J (c) −314 J
19.31 11.6°C
19.33 $4\sqrt{2}$
19.35 (a) increase (b) 4.8 kJ
19.37 (a) 0.681 mol (b) 0.0333 m^3 (c) 2.23 Kj d) 0
19.39 (a) 45.0 J (b) liberate, 65.0 J (c) 23.0 J, 22.0 J
19.41 (a) the same (b) 4.0 kJ, absorb (c) 8.0 kJ
19.43 (a) 0.80 L (b) 305 K, 1220 K, 1220 K
(c) *ab:* 76 J, into the gas
ca: −107 J, out of the gas
bc: 56 J, into the gas
(d) *ab:* 76 J, increased
bc: 0, no change
ca: −76 J, decreased
19.45 (b) −300 J, out of gas
19.47 (b) 6.00 L, 2.5×10^4 Pa, 75.0 K (c) 95 J
(d) heat it at constant volume
19.49 (b) 11.9 C°
19.51 (a) 0.169 m (b) 198°C (c) 71.3 kJ
19.53 (a) $Q = 450$ J, $\Delta U = 0$ (b) $Q = 0$, $\Delta U = -450$ J (c) $Q = 1125$ J, $\Delta U = 675$ J
19.55 (a) $W = 738$ J, $Q = 2590$ J, $\Delta U = 1850$ J
(b) $W = 0$, $Q = -1850$ J, $\Delta U = -1850$ J
(c) $\Delta U = 0$
19.57 (a) $W = -187$ J, $Q = -654$ J, $\Delta U = -467$ J
(b) $W = 113$ J, $Q = 0$, $\Delta U = -113$ J
(c) $W = 0$, $Q = 580$ J, $\Delta U = 580$ J
19.59 (a) *a:* adiabatic, *b:* isochoric, *c:* isobaric
(b) 28.0°C (c) *a:*−30.0 J, *a:* 0, *a:* 20.0 J (d) *a*
(e) *a:* decrease, *b:* stay the same, *c:* increase
19.61 (a) $53.5\text{ J} + v^{0.40}[(21.4\text{ J})\ln v - 53.5\text{ J}]$
(b) 7 (c) 7 (d) 7 (e) 63.1 J
19.63 Choice (c)
19.65 Choice (d)

Chapter 20

20.1 (a) 6500 J (b) 34%
20.3 (a) 23% (b) 12,400 J (c) 0.350 g
(d) 222 kW = 298 hp
20.5 (a) 12.3 atm (b) 5470 J, *ca* (c) 3723 J, *bc*
(d) 1747 J (e) 31.9%
20.7 (a) 12 (b) 7.4 kJ, 20 kJ
20.9 (a) 58% (b) 1.4%
20.11 1.2 h
20.13 576 K, 504 K
20.15 (a) 215 J (b) 378 K (c) 39.0%
20.17 (a) 38 kJ (b) 590°C
20.19 (a) 492 J (b) 212 W (c) 5.4
20.21 161 K
20.23 (a) 429 J/K (b) −393 J/K (c) 36 J/K
20.25 −6.31 J/K
20.27 (a) 6.05×10^3 J/K (b) about five times greater for vaporization
20.29 10.0 J/K
20.31 (a) 121 J (b) 3800 cycles
20.33 (a) 90.2 J (b) 320 J (c) 45°C (d) 0 (e) 263 g
20.35 −5.8 J/K, decrease
20.37 (b) absorbed: *bc*; rejected: *ab* and *ca*
(c) $T_a = T_b = 241$ K, $T_c = 481$ K
(d) 610 J, 610 J
(e) 8.7%
20.39 (a) 21.0 kJ (enters), 16.6 kJ (leaves)
(b) 4.4 kJ, 21% (c) $e = 0.31e_{max}$
20.41 (a) 7.0% (b) 3.0 MW; 2.8 MW
(c) 6×10^5 kg/h $= 6 \times 10^5$ L/h
20.43 (a) 1: 2.00 atm, 4.00 L; 2: 2.00 atm, 6.00 L; 3: 1.11 atm, 6.00 L; 4: 1.67 atm, 4.00 L
(b) 1 → 2: 1422 J, 405 J; 2 → 3:−1355 J, 0; 3 → 4:−274 J, −274 J; 4 → 1: 339 J, 0
(c) 131 J (d) 7.44%; $e = 0.168e_C$
20.45 $1 - T_C/T_H$, same
20.47 (a) 122 J, −78 J
(b) 5.10×10^{-4} m^3
(c) *b:* 2.32 MPa, 4.81×10^{-5} m^3, 771 K
c: 4.01 MPa, 4.81×10^{-5} m^3, 1332 K
d: 0.147 MPa, 5.10×10^{-4} m^3, 518 K
(d) 61.1%, 77.5%
20.49 6.23
20.53 (a) A: 28.9%, B: 38.3%, C: 53.8%, D: 24.4%
(b) C (c) B > D > A
20.55 (a) 4.83% (b) 4.83% (c) 6.25%
(d) $e = \dfrac{0.80T_d - 200}{12T_d - 2700}$, 6.67%
20.57 (a) 151 K (b) 305 Pa (c) 5.39 kJ (d) 4.39 kJ
20.59 Choice (b)
20.61 Choice (d)

Chapter 21

21.1 (a) 2.00×10^{10} (b) 8.59×10^{-13}
21.3 3.4×10^{18} m/s^2 (proton), 6.3×10^{21} m/s^2 (electron)
21.5 1.3 nC
21.7 3.7 km
21.9 (a) 0.742 μC (b) 0.371 μC, 1.48 μC
21.11 (a) 2.21×10^4 m/s^2
21.13 +0.750 nC
21.15 1.8×10^{-4} N, in the $+x$-direction

21.17 2.58 μN, in the $-y$-direction

21.19 Attractive

21.21 (a) 4.40×10^{-16} N (b) 2.63×10^{11} m/s^2 (c) 2.63×10^5 m/s

21.23 (a) 3.30×10^6 N/C, to the left (b) 14.2 ns (c) 1.80×10^3 N/C, to the right

21.25 (a) $-21.9\ \mu$C (b) 1.02×10^{-7} N/C

21.27 (a) 364 N/C (b) no; 2.73 μm, downward

21.29 (a) $-\hat{\boldsymbol{j}}$ (b) $\dfrac{\sqrt{2}}{2}\hat{\boldsymbol{\imath}} + \dfrac{\sqrt{2}}{2}\hat{\boldsymbol{j}}$ (c) $-0.39\,\hat{\boldsymbol{\imath}} + 0.92\,\hat{\boldsymbol{j}}$

21.31 3.09×10^{-3} N in the $+y$-direction

21.33 20.0 cm

21.35 (a) (i) 574 N/C, $+x$-direction (ii) 268 N/C, $-x$-direction (iii) 404 N/C, $-x$-direction (b) (i) 9.20×10^{-17} N, $-x$-direction (ii) 4.30×10^{-17} N, $+x$-direction (iii) 6.48×10^{-17} N, $+x$-direction

21.37 1.04×10^7 N/C, toward the -2.00-μC charge

21.39 (a) 8740 N/C, to the right (b) 6540 N/C, to the right (c) 1.40×10^{-15} N, to the right

21.41 1.73×10^{-8} N, toward the point midway between the electrons

21.43 (a) $E_x = E_y = E = 0$ (b) $E_x = 2660$ N/C, $E_y = 0$, $E = 2660$ N/C, $+x$-direction (c) $E_x = 129$ N/C, $E_y = -510$ N/C, $E = 526$ N/C, 284° counterclockwise from the $+x$-axis (d) $E_x = 0$, $E_y = 1380$ N/C, $E = 1380$ N/C, $+y$-direction

21.45 113 N/C, $-x$ direction

21.47 (a) 1340 N/C, $+x$-direction (b) 1340 N/C, $-x$-direction (c) 8260 N/C, $+x$-direction (d) 8260 N/C, $-x$-direction

21.49 (a) $(7.0\text{ N/C})\hat{\boldsymbol{\imath}}$ (b) $(1.75 \times 10^{-5}\text{ N})\hat{\boldsymbol{\imath}}$

21.51 (a) 1.4×10^{-11} C·m, from q_1 toward q_2 (b) 860 N/C

21.53 (a) $\vec{\boldsymbol{p}}$ aligned in the same or the opposite direction as $\vec{\boldsymbol{E}}$ (b) stable: $\vec{\boldsymbol{p}}$ aligned in the same direction as $\vec{\boldsymbol{E}}$; unstable: $\vec{\boldsymbol{p}}$ aligned in the opposite direction

21.55 (a) 1680 N, from the $+5.00$-μC charge toward the -5.00-μC charge (b) 22.3 N·m, clockwise

21.57 (b) $\dfrac{Q^2}{8\pi\epsilon_0 L^2}(1 + 2\sqrt{2})$, away from the center of the square

21.59 (a) 8.63×10^{-5} N, -5.52×10^{-5} N (b) 1.02×10^{-4} N, 32.6° below the $+x$-axis

21.61 (b) 2.80 μC (c) 39.5°

21.63 3.41×10^4 N/C, to the left

21.65 (a) 1 nC (b) 6×10^{10}

21.67 between the charges, 0.24 m from the 0.500-nC charge

21.69 at $x = d/3$, $q = -4Q/9$

21.71 (a) $\dfrac{6q^2}{4\pi\epsilon_0 L^2}$, away from the vacant corner (b) $\dfrac{3q^2}{4\pi\epsilon_0 L^2}\left(\sqrt{2} + \dfrac{1}{2}\right)$, toward the center of the square

21.73 (a) 6.0×10^{23} (b) 4.1×10^{-31} N (gravitational), 510 kN (electric) (c) yes (electric), no (gravitational)

21.75 2190 km/s

21.77 (a) $\dfrac{mv_0^2\sin^2\alpha}{2eE}$ (b) $\dfrac{mv_0^2\sin^2\alpha}{eE}$ (d) h_{max}: 0.418 m, d: 2.89 m

21.79 (a) $E_x = \dfrac{Q}{4\pi\epsilon_0 a}\left(\dfrac{1}{x-a} - \dfrac{1}{x}\right)$, $E_y = 0$ (b) $\dfrac{qQ}{4\pi\epsilon_0 a}\left(\dfrac{1}{r} - \dfrac{1}{r+a}\right)\hat{\boldsymbol{\imath}}$

21.81 (a) 1.56 N/C, $+x$-direction (c) smaller (d) 4.7%

21.83 $E_x = E_y = \dfrac{Q}{2\pi^2\epsilon_0 a^2}$

21.85 (a) 6.25×10^4 N/C, 225° counterclockwise from an axis pointing to the right at point P (b) 1.00×10^{-14} N, opposite the electric field direction

21.87 (a) $\pi(R_2^2 - R_1^2)\sigma$ (b) $\dfrac{\sigma}{2\epsilon_0}\left(\dfrac{1}{\sqrt{(R_1/x)^2+1}} - \dfrac{1}{\sqrt{(R_2/x)^2+1}}\right)\dfrac{|x|}{x}\hat{\boldsymbol{\imath}}$ (c) $\dfrac{\sigma}{2\epsilon_0}\left(\dfrac{1}{R_1} - \dfrac{1}{R_2}\right)x\hat{\boldsymbol{\imath}}$; $x \ll R_1$ (d) $\dfrac{1}{2\pi}\sqrt{\dfrac{q\sigma}{2\epsilon_0 m}\left(\dfrac{1}{R_1} - \dfrac{1}{R_2}\right)}$

21.89 (a) $q_1 = 8.00\ \mu$C, $q_2 = 3.00\ \mu$C (b) 7.50 N, in the $-x$-direction (c) $x = 0.248$ m

21.91 (a) $\dfrac{\sigma}{2\epsilon_0}\left[\dfrac{1}{\sqrt{1 + (R/\Delta)^2}}\right]$ downward (b) $q\sigma/2g\epsilon_0$ (c) $-\dfrac{q\sigma}{2\epsilon_0}\left(\sqrt{\Delta^2 + R^2} - R\right)$ (d) $mg\Delta$ (e) $\dfrac{2R\alpha}{1-\alpha^2}$ (f) $\sqrt{2g\left(\Delta - \dfrac{\sqrt{\Delta^2 + R^2} - R}{\alpha}\right)}$ (g) 57.7 g, 10.7 cm

21.93 (b) $-\dfrac{\lambda}{4\pi\epsilon_0 a}\left[\dfrac{y_2 - L}{\sqrt{(y_2 - L)^2 + a^2}} - \dfrac{y_2}{\sqrt{y_2^2 + a^2}}\right]\lambda dy_2$ (c) $\dfrac{Q^2}{2\pi\epsilon_0 L^2}\left(\sqrt{(L/a)^2 + 1} - 1\right)\hat{\boldsymbol{\imath}}$ (e) $\dfrac{Q^2}{2\pi\epsilon_0 L^2}\left[L\ln\left(\dfrac{L + \sqrt{a^2 + L^2}}{a}\right) + a - \sqrt{a^2 + L^2}\right]$ (f) 21.5 m/s

21.95 (b) $q_1 < 0$, $q_2 > 0$ (c) 0.843μC (d) 56.2 N

21.97 (a) $\dfrac{Q}{2\pi\epsilon_0 L}\left(\dfrac{1}{2x+a} - \dfrac{1}{2L+2x+a}\right)$

21.99 Choice (c)

21.101 Choice (b)

Chapter 22

22.1 (a) 1.8 N·m^2/C (b) no (c) (i) 0° (ii) 90°

22.3 (a) 3.53×10^5 N·m^2/C (b) 3.13 μC

22.5 7.70×10^3 N/C

22.7 14.8 cm^2

22.9 (a) 0 (b) 1.22×10^8 N/C, radially inward (c) 3.64×10^7 N/C, radially inward

22.11 0.977 N·m^2/C, inward

22.13 0.0810 N

22.15 (a) 350 N/C (b) 700 N/C (c) 1400 N/C

22.17 (a) 6.47×10^5 N/C, $+y$-direction (b) 7.2×10^4 N/C, $-y$-direction

22.19 (a) 5.73 μC/m^2 (b) 6.47×10^5 N/C (c) -5.65×10^4 N·m^2/C

22.21 (a) 0.260 μC/m^3 (b) 1960 N/C

22.23 (a) 6.56×10^{-21} J (b) 1.20×10^5 m/s

22.25 (a) 6.00 nC (b) -1.00 nC

22.27 (a) $2\pi R\sigma$ (b) $\dfrac{\sigma R}{\epsilon_0 r}$ (c) $\dfrac{\lambda}{2\pi\epsilon_0 r}$

22.29 1.16 km/s

22.31 10.2°

22.33 (a) 750 N·m^2/C (b) 0 (c) 577 N/C, $+x$-direction (d) within and outside

22.35 (a) $\dfrac{Q}{4\pi\epsilon_0 r^2}$, toward the center of the shell (b) 0

22.37 (a) $\dfrac{\lambda}{2\pi\epsilon_0 r}$, radially outward (b) $\dfrac{\lambda}{2\pi\epsilon_0 r}$, radially outward (d) $-\lambda$ (inner), $+\lambda$ (outer)

22.39 (a) $\dfrac{\rho r}{2\epsilon_0}$ (b) $\dfrac{\lambda}{2\pi\epsilon_0 r}$ (c) They are equal.

22.41 (a) $r < R$: 0; $R < r < 2R$: $\dfrac{1}{4\pi\epsilon_0}\dfrac{Q}{r^2}$, radially outward; $r > 2R$: $\dfrac{1}{4\pi\epsilon_0}\dfrac{2Q}{r^2}$, radially outward

22.43 (a) (i) 0 (ii) 0 (iii) $\dfrac{q}{2\pi\epsilon_0 r^2}$, radially outward (iv) 0 (v) $\dfrac{3q}{2\pi\epsilon_0 r^2}$, radially outward (b) (i) 0 (ii) $+2q$ (iii) $-2q$ (iv) $+6q$

22.45 (a) $\dfrac{\alpha}{2\epsilon_0}\left(1 - \dfrac{a^2}{r^2}\right)$ (b) $q = +2\pi\alpha a^2$, $E = \dfrac{\alpha}{2\epsilon_0}$

22.47 (c) $\dfrac{Qr}{4\pi\epsilon_0 R^3}\left(4 - \dfrac{3r}{R}\right)$ (e) $2R/3$, $\dfrac{Q}{3\pi\epsilon_0 R^2}$

22.49 $\dfrac{\alpha r}{\epsilon_0}\left(\dfrac{1}{2} - \dfrac{r}{3R}\right)$, $\dfrac{\alpha R^2}{6\epsilon_0 r}$, yes

22.51 $|x| > d$ (outside the slab): $\dfrac{\rho_0 d}{3\epsilon_0}\dfrac{x}{|x|}\hat{\boldsymbol{\imath}}$; $|x| < d$ (inside the slab): $\dfrac{\rho_0 x^3}{3\epsilon_0 d^2}\hat{\boldsymbol{\imath}}$

22.53 (b) $\dfrac{\rho\vec{\boldsymbol{b}}}{3\epsilon 0}$

22.55 (a) uniform line of charge: A; uniformly charged sphere: B (b) $\lambda = 1.50 \times 10^{-7}$ C/m, $\rho = 2.81 \times 10^{-3}$ C/m^3

22.57 (i) 377 N/C (ii) 653 N/C (iii) 274 N/C (iv) 0

22.59 (a) $\dfrac{\pi\rho_0 R^2}{3}$ (b) $T = 2\pi\left(\dfrac{R_{\text{orbit}}}{R_{\text{cylinder}}}\right)\sqrt{\dfrac{6\epsilon_0 M}{Q\rho_0}}$

22.61 (a) $\dfrac{QL}{2\epsilon_0\sqrt{(L/2)^2 + R^2}}$ (b) $\dfrac{QL}{4\epsilon_0}\left(\dfrac{2}{L} - \dfrac{1}{\sqrt{R^2 + (L/2)^2}}\right)$ (c) $\dfrac{QL}{4\epsilon_0}\left(\dfrac{2}{L} - \dfrac{1}{\sqrt{R^2 + (L/2)^2}}\right)$ (d) Q/ϵ_0

22.63 Choice (a)

22.65 Choice (b)

Chapter 23

23.1 -0.356 J

23.3 3.46×10^{-13} J $= 2.16$ MeV

23.5 (a) 12.5 m/s (b) 0.323 m

23.7 1.94×10^{-5} N

23.9 (a) 13.6 km/s; very long after release (b) 2.45×10^{17} m/s^2; just after release

23.11 89.3 V, point a

23.13 7.42 m/s, faster

23.15 (a) 0 (b) 0.750 mJ (c) -2.06 mJ

23.17 (a) 0 (b) -175 kV (c) -0.875 J

23.19 (a) -737 V (b) -704 V (c) 8.2×10^{-8} J

23.21 (a) negative (b) 171 V/m (c) -17.1 V

23.23 (a) b (b) 800 V/m (c) $-48.0\ \mu$J

23.25 (a) 0.0539 J (b) 3.00 m/s

23.27 (a) (i) 180 V (ii) -270 V (iii) -450 V (b) 719 V, inner shell

23.29 (a) oscillatory (b) 1.67×10^7 m/s

23.31 150 m/s

23.33 (a) 78.2 kV (b) 0

23.35 0.474 J

23.37 (a) 8.00 kV/m (b) 19.2 μN (c) 0.864 μJ (d) $-0.864\ \mu$J

23.39 -760 V

23.41 (a) (i) $V = \dfrac{q}{4\pi\epsilon_0}\left(\dfrac{1}{r_a} - \dfrac{1}{r_b}\right)$
(ii) $V = \dfrac{q}{4\pi\epsilon_0}\left(\dfrac{1}{r} - \dfrac{1}{r_b}\right)$ (iii) $V = 0$
(d) 0 (e) $E = \dfrac{q - Q}{4\pi\epsilon_0 r^2}$
23.43 (a) $E_x = -Ay + 2Bx, E_y = -Ax - C, E_z = 0$
(b) $x = -C/A$, $y = -2BC/A^2$, any value of z
23.45 (a) 0.762 nC
23.47 (a) $-0.360\ \mu$J (b) $x = 0.074$ m
23.49 $+1.90\ 10^5$ V, point A
23.51 4.2×10^6 V
23.53 (a) $-21.5\ \mu$J (b) -2.83 kV (c) 35.4 kV/m
23.55 (a) $7.85 \times 10^4\ \text{V/m}^{4/3}$
(b) $E_x(x) = -(1.05 \times 10^5\ \text{V/m}^{4/3})x^{1/3}$
(c) 3.13×10^{-15} N, toward the anode
23.57 (a) $-\dfrac{1.46q^2}{\pi\epsilon_0 d}$
23.59 47.8 V
23.61 (a) (i) $V = (\lambda/2\pi\epsilon_0)\ln(b/a)$
(ii) $V = (\lambda/2\pi\epsilon_0)\ln(b/r)$
(iii) $V = 0$
(d) $(\lambda/2\pi\epsilon_0)\ln(b/a)$
23.63 (a) 1.76×10^{-16} N, downward
(b) $1.93 \times 10^{14}\ \text{m/s}^2$, downward
(c) 0.822 cm
(d) 15.3° (e) 3.29 cm
23.65 (a) 97.1 kV/m (b) 30.3 pC
23.67 $\frac{3}{5}\left(\dfrac{Q^2}{4\pi\epsilon_0 R}\right)$
23.69 360 kV
23.71 (a) $\dfrac{Q}{4\pi\epsilon_0 a}\ln\left(\dfrac{x + a}{x}\right)$
(b) $\dfrac{Q}{4\pi\epsilon_0 a}\ln\left(\dfrac{a + \sqrt{a^2 + y^2}}{y}\right)$
(c) (a): $\dfrac{Q}{4\pi\epsilon_0 x}$, (b): $\dfrac{Q}{4\pi\epsilon_0 y}$
23.73 (a) $U = \dfrac{1}{2}kd^2 + \dfrac{1}{4\pi\epsilon_0}\dfrac{e^2}{d}$
(b) $d_0 = \left(\dfrac{e^2}{4\pi\epsilon_0 k}\right)^{1/3}$ (c) 2.30×10^{17} N/m
(d) 2.16 MeV (e) 1.44×10^7 m/s
23.75 2.48×10^{-14} m
23.77 (a) $A = -6.0\ \text{V/m}^2$, $B = -4.0\ \text{V/m}^3$,
$C = -2.0\ \text{V/m}^6$, $D = 10$ V, $l = 2.0$, $m = 3.0$,
$n = 6.0$
(b) (0, 0, 0): 10.0 V, 0;
(0.50 m, 0.50 m, 0.50 m):
8.0 V, 6.7 V/m;
(1.00 m, 1.00 m, 1.00 m):
-2.0 V, 21 V/m 23.79
23.79 (c) 4.79×10^{-19} C (drop 1),
1.59×10^{-19} C (drop 2),
8.09×10^{-19} C (drop 3),
3.23×10^{-19} C (drop 4)
(d) 3 (drop 1), 5 (drop 3), 2 (drop 4)
(e) 1.60×10^{-19} C (drop 1),
1.59×10^{-19} C (drop 2),
1.62×10^{-19} C (drops 3 and 4);
1.61×10^{-19} C 23.81
23.81 (a) 4.75×10^{-15} C (b) 94.9 kV/m
(c) 5.81×10^{-15} C (d) 36,300
23.83 1.01×10^{-12} m, 1.11×10^{-13} m, 2.54×10^{-14} m
23.85 Choice (b)

Chapter 24

24.1 (a) 10.0 kV (b) 22.6 cm^2 (c) 8.00 pF
24.3 (a) 604 V (b) 90.8 cm^2 (c) 1840 kV/m
(d) 16.3 $\mu\text{C/m}^2$
24.5 (a) 120 μC (b) 60 μC (c) 480 μC
24.7 (a) 1.05 mm (b) 84.0 V
24.9 (a) 4.35 pF (b) 2.30 V
24.11 (a) 15.0 pF (b) 3.09 cm (c) 31.2 kN/C
24.13 5.57 μF
24.15 (a) series (b) 5000
24.17 (a) $Q_1 = Q_2 = 22.4\ \mu$C, $Q_3 = 44.8\ \mu$C,
$Q_4 = 67.2\ \mu$C
(b) $V_1 = V_2 = 5.6$ V, $V_3 = 11.2$ V, $V_4 = 16.8$ V
(c) 11.2 V
24.19 (a) $Q_1 = 156\ \mu$C, $Q_2 = 260\ \mu$C
(b) $V_1 = V_2 = 52.0$ V
24.21 (a) 19.3 nF (b) 482 nC (c) 162 nC (d) 25 V
24.23 0.0283 J/m^3
24.25 (a) 90.0 pF (b) 0.0152 m^3 (c) 4.5 kV
(d) 1.80 μJ
24.27 (a) $U_p = 4U_s$ (b) $Q_p = 2Q_s$ (c) $E_p = 2E_s$
24.29 (a) 24.2 μC (b) $Q_{35} = 7.7\ \mu$C, $Q_{75} = 16.5\ \mu$C
(c) 2.66 mJ (d) $U_{35} = 0.85$ mJ, $U_{75} = 1.81$ mJ
(e) 220 V
24.31 (a) 3.60 mJ (before), 13.5 mJ (after)
(b) 9.9 mJ, increase
24.33 (a) 0.620 $\mu\text{C/m}^2$ (b) 1.28
24.35 $\dfrac{2KQ_0}{1 + K}$, increases
24.37 0.0135 m^2
24.39 (a) 6.3 μC (b) 6.3 μC (c) none
24.41 (a) 10.1 V (b) 2.25
24.43 (a) $\dfrac{Q}{\epsilon_0 AK}$ (b) $\dfrac{Qd}{\epsilon_0 AK}$ (c) $K\dfrac{\epsilon_0 A}{d} = KC_0$
24.45 (a) 421 J (b) 0.054 F
24.47 (a) 0.531 pF (b) 0.224 mm
24.49 (a) 0.0160 C (b) 533 V (c) 4.26 J (d) 2.14 J
24.51 (a) 158 μJ (b) 72.1 μJ
24.53 (a) 2.5 μF
(b) $Q_1 = 550\ \mu$C, $Q_2 = 370\ \mu$C,
$Q_3 = Q_4 = 180\ \mu$C, $Q_5 = 550\ \mu$C;
$V_1 = 65$ V, $V_2 = 87$ V,
$V_3 = V_4 = 43$ V, $V_5 = 65$ V
24.55 $C_2 = 6.00\ \mu$F, $C_3 = 4.50\ \mu$F
24.57 (a) 76 μC (b) 1.4 mJ (c) 11 V (d) 1.3 mJ
24.59 (a) 2.3 μF (b) $Q_1 = 970\ \mu$C, $Q_2 = 640\ \mu$C
(c) 47 V
24.61 1.67 μF
24.63 (a) $\dfrac{Q^2}{2A\epsilon_0 K}$ (b) $\dfrac{d_0}{2A^2\epsilon_0 YK}$ (c) $7.53 \times 10^7\ \text{m/C}^2$
(d) 95.7 kV (e) 133 kV
24.65 (a) $\dfrac{\pi\epsilon_0 r^2 V^2}{2z^2}$ (b) $\dfrac{\pi\epsilon_0 r^2 V^2}{4d}$ (c) $\dfrac{\pi\epsilon_0 r^2 V^2}{2d}$
(d) $\dfrac{\pi\epsilon_0 r^2 V^2}{4d}$ (e) no
24.67 (a) $C_1 = 6.00\ \mu$F, $C_2 = 3.00\ \mu$F
(b) same charge; C_2 stores more energy
(c) C_1 stores more charge and energy
24.69 (a) first (connected) (b) 144 cm^2
(c) disconnected
24.71 (a) $\dfrac{\epsilon_0 A(K - 1)}{d\ln K}$
24.73 Choice (c)
24.75 Choice (a)

Chapter 25

25.1 1.0 C
25.3 (a) 3.12×10^{19} (b) $1.51 \times 10^6\ \text{A/m}^2$
(c) 0.111 mm/s (d) both (b) and
(c) would increase
25.5 (a) 0.26 mm/s (b) 0.0120 N/C
25.7 (a) 330 C (b) 41 A
25.9 9.0 μA
25.11 $4.02 \times 10^{-8}\ \Omega \cdot \text{m}$
25.13 (a) $1.06 \times 10^{-5}\ \Omega \cdot \text{m}$ (b) 0.00105 $(\text{C}°)^{-1}$
25.15 (a) 0.206 mV (b) 0.176 mV
25.17 0.457
25.19 0.125 Ω
25.21 (a) 11 A (b) 3.1 V (c) 0.28 Ω
25.23 (a) 99.54 Ω (b) 0.0158 Ω
25.25 (a) 27.4 V (b) 12.3 MJ
25.27 (a) 0 (b) 5.0 V (c) 5.0 V
25.29 3.08 V, 0.067 Ω, 1.80 Ω
25.31 (a) 1.41 A, clockwise (b) 13.7 V (c) -1.0 V
25.33 21 W, 21 W
25.35 (a) 144 Ω (b) 240 Ω
(c) 100 W: 0.833 A; 60 W: 0.500 A
25.37 (a) 29.8 W (b) 0.248 A
25.39 (a) 3.1 W (b) 7.2 W (c) 4.1 W
25.41 (a) 300 W (b) 0.90 J
25.43 (a) 2.6 MJ (b) 0.063 L (c) 1.6 h
25.45 12.3%
25.47 (a) 24.0 W (b) 4.0 W (c) 20.0 W
25.49 (a) 1.55×10^{-12} s
25.51 (a) $3.65 \times 10^{-8}\ \Omega \cdot \text{m}$ (b) 172 A
(c) 2.58 mm/s
25.53 0.060 Ω
25.55 (a) 2.5 mA (b) 21.4 μV/m (c) 85.5 μV/m
(d) 0.180 mV
25.57 (a) 80 C° (b) no
25.59 (a) $\dfrac{\rho h}{\pi r_1 r_2}$
25.61 (a) 0.20 Ω (b) 8.7 V
25.63 (a) 1.0 kΩ (b) 100 V (c) 10 W
25.65 (a) \$78.90 (b) \$140.27
25.67 (a) 171$\mu\Omega$ (b) 176 μV/m (c) left: 54.7$\mu\Omega$;
right: 116$\mu\Omega$
25.69 (a) 4.25 ppt (b) 510 pC
(c) 1.33 W (d) 1.60 ppt
25.71 6.67 V
25.73 (b) no (c) yes (d) 9.40 W (e) 4.12 W
25.75 (a) inward (b) $\dfrac{V}{\ln(r_{\text{outer}}/r_{\text{inner}})}$
(c) $\dfrac{\rho}{2\pi L}\ln(r_{\text{outer}}/r_{\text{inner}})$
(d) 616 $\Omega \cdot$ m
25.77 (a) $R = \dfrac{\rho_0 L}{A}\left(1 - \dfrac{1}{e}\right)$, $I = \dfrac{V_0 A}{\rho_0 L\left(1 - \dfrac{1}{e}\right)}$
(b) $E(x) = \dfrac{V_0 e^{-x/L}}{L\left(1 - \dfrac{1}{e}\right)}$
(c) $V(x) = \dfrac{V_0\left(e^{-x/L} - \dfrac{1}{e}\right)}{1 - \dfrac{1}{e}}$
25.79 Choice (c)
25.81 Choice (d)

Chapter 26

26.1 $3R/4$
26.3 22.5 W
26.5 (a) 3.50 A (b) 4.50 A (c) 3.15 A (d) 3.25 A
26.7 0.769 A
26.9 (a) $I = \dfrac{\mathcal{E}}{R}$ (b) $I = \dfrac{\mathcal{E}}{6R}$ (c) parallel
26.11 (a) $I_1 = 8.00$ A, $I_3 = 12.0$ A
(b) 84.0 V
26.13 5.00 Ω; $I_{3.00} = 8.00$ A, $I_{4.00} = 9.00$ A,
$I_{6.00} = 4.00$ A, $I_{12.0} = 3.00$ A
26.15 (a) $I_1 = 1.50$ A, $I_2 = I_3 = I_4 = 0.500$ A
(b) $P_1 = 10.1$ W, $P_2 = P_3 = P_4 = 1.12$ W;
bulb R_1
(c) $I_1 = 1.33$ A, $I_2 = I_3 = 0.667$ A
(d) $P_1 = 8.00$ W, $P_2 = P_3 = 2.00$ W
(e) brighter: R_2 and R_3; less bright: R_1

26.17 18.0 V, 3.00 A

26.19 1010 s

26.21 (a) 0.100 A (b) $P_{400} = 4.0$ W, $P_{800} = 8.0$ W (c) 12.0 W (d) $I_{400} = 0.300$ A, $I_{800} = 0.150$ A (e) $P_{400} = 36.0$ W, $P_{800} = 18.0$ W (f) 54.0 W (g) series: 800-Ω bulb; parallel: 400-Ω bulb (h) parallel

26.23 (a) 2.00 A (b) 5.00 Ω (c) 42.0 V (d) 3.50 A

26.25 (a) 8.00 A (b) $\mathcal{E}_1 = 36.0$ V, $\mathcal{E}_2 = 54.0$ V (c) 9.00 Ω

26.27 (a) 1.60 A (top), 1.40 A (middle), 0.20 A (bottom) (b) 10.4 V

26.29 (a) 36.4 V (b) 0.500 A

26.31 (a) 2.14 V, a (b) 0.050 A, 0; down

26.33 (a) 0.641 Ω (b) 975 Ω

26.35 (a) 17.9 V (b) 22.7 V (c) 21.4%

26.37 (a) 0.849 μF (b) 2.89 s

26.39 (a) 0 (b) 245 V (c) 0 (d) 32.7 mA (e) (a): 245 V; (b): 0; (c): 1.13 mC; (d): 0

26.41 (a) 4.21 ms (b) 0.125 A

26.43 192 μC

26.45 13.6 A

26.47 (a) 0.937 A (b) 0.606 A

26.49 (a) 165 μC (b) 463 Ω (c) 12.6 ms

26.51 900 W

26.53 (a) $U_C = \frac{1}{2}C\mathcal{E}^2$ (b) $U_\mathcal{E} = C\mathcal{E}^2$ (c) $U_R = \frac{1}{2}C\mathcal{E}^2$ (d) 1/2, 1/2

26.55 (a) 2.29 A (b) 3.71 A (c) 1.43 A

26.57 (a) +0.22 V (b) 0.464 A

26.59 $I_1 = 0.848$ A, $I_2 = 2.14$ A, $I_3 = 0.171$ A

26.63 (a) 109 V; no (b) 13.5 s

26.67 (a) −12.0 V (b) 1.71 A (c) 4.21 Ω

26.69 (a) $\epsilon_0\rho K$ (b) 1.28 nC (c) 165 s (d) 2.68 pA

26.71 (a) 11.8 pC (b) 0.917 A (c) 4.92 pJ (d) 703 pJ (e) 1300 pJ (f) 595 pJ

26.73 (a) 114 V (b) 263 V (c) 266 V

26.75 (a) 18.0 V (b) a (c) 6.00 V (d) both decrease by 36.0 μC

26.77 1.7 MΩ, 3.1 μF

26.79 (a) −1.23 ms (slope), 79.5 μC (y-intercept) (b) 247 Ω, 15.9 V (c) 1.22 ms (d) 11.9 V

26.81 (a) $V_{out} = 21\text{V} - 10V_{in}$ (b) 1.35 V (c) −10

26.85 (b) 4 (c) 3.2 MΩ, 4.0×10^{-3} (d) 3.4×10^{-4} (e) 0.88

26.87 Choice (d)

Chapter 27

27.1 (a) $(-6.68 \times 10^{-4}\text{ N})\hat{\boldsymbol{k}}$ (b) $(6.68 \times 10^{-4}\text{ N})\hat{\boldsymbol{\imath}} + (7.27 \times 10^{-4}\text{ N})\hat{\boldsymbol{\jmath}}$

27.3 (a) positive (b) 0.0505 N

27.5 172 m/s, $-x$-direction

27.7 (a) 1.46 T, in the xz-plane at 40° from the $+x$-axis toward the $-z$-axis (b) 7.47×10^{-16} N, in the xz-plane at 50° from the $-x$-axis toward the $-z$-axis

27.9 (a) 3.05 mWb (b) 1.83 mWb (c) 0

27.11 −0.78 mWb

27.13 (a) 0 (b) −0.0115 Wb (c) +0.0115 Wb (d) 0

27.15 (a) 0.160 mT, into the page (b) 0.111 μs

27.17 7.93×10^{-10} N, toward the south

27.19 (a) 2.84×10^6 m/s, negative (b) yes (c) same

27.21 0.838 mT

27.23 (a) 7900 N/C, $\hat{\boldsymbol{\imath}}$ (b) 7900 N/C, $\hat{\boldsymbol{\imath}}$

27.25 0.0445 T, out of the page

27.27 (a) 4.92 km/s (b) 9.96×10^{-26} kg

27.29 0.724 N, 63.4° below the current direction in the upper wire segment

27.31 (a) 817 V (b) 113 m/s^2

27.33 (a) a (b) 3.21 kg

27.35 (a) 55 turns (b) counterclockwise (c) 8.73 N · m, upward

27.37 (b) $F_{cd} = 1.20$ N (c) 0.420 N · m

27.39 (a) A_2 (b) 290 rad/s^2

27.41 (a) $-NIAB\hat{\boldsymbol{\imath}}, 0$ (b) $0, -NIAB$ (c) $+NIAB\hat{\boldsymbol{\imath}}, 0$ (d) $0, +NIAB$

27.43 (a) 1.13 A (b) 3.69 A (c) 98.2 V (d) 362 W

27.45 (a) 4.7 mm/s (b) $+4.5 \times 10^{-3}$ V/m, $+z$-direction (c) 53 μV

27.47 (a) $x = 0.565$ m, $y = 0.825$ m, $z = 0.0900$ m (b) 31.3 m/s

27.49 (a) 8.3×10^6 m/s (b) 0.14 T

27.51 4.81×10^7 C/kg

27.53 1.6 mm

27.55 $\dfrac{Mg\tan\theta}{LB}$, right to left

27.57 (a) 8.46 mT (b) 27.2 cm (c) 2.2 cm; yes

27.59 (a) ILB, to the right (b) $\dfrac{v^2 m}{2ILB}$ (c) 1960 km

27.61 (a) 1.13 N, toward the origin (b) 0.800 N, $+z$-direction

27.63 (a) 48,000 A (b) not feasible (c) 19 G (d) 0.83 N

27.65 0.024 T, $+y$-direction

27.67 (a) $F_{PQ} = 0$; $F_{RP} = 12.0$ N, into the page; $F_{QR} = 12.0$ N, out of the page (b) 0 (c) $\tau_{PQ} = \tau_{RP} = 0$; $\tau_{QR} = 3.60$ N · m (d) 3.60 N · m; yes (e) out

27.69 $-(0.444\text{ N})\hat{\boldsymbol{\jmath}}$

27.71 (b) left: $(B_0LI/2)\hat{\boldsymbol{\imath}}$; top: $-IB_0L\hat{\boldsymbol{\jmath}}$; right: $-(B_0LI/2)\hat{\boldsymbol{\imath}}$; bottom: 0 (c) $-IB_0L\hat{\boldsymbol{\jmath}}$

27.73 (a) $\dfrac{Mg}{2WNB}$ (b) $R(1 + \pi/2)$
(c) $Mg(\sigma - 1)\sqrt{h(2R - h)}$
(d) $\left[\sigma\cos\left(\dfrac{h}{R} - 1\right) - 1\right]MgR$
(e) $(1 - \sigma)Mgh$
(f) $\left\{\dfrac{h}{R} - \sigma\left[\sin\left(\dfrac{h}{R} - 1\right) + 1\right]\right\}(MgR)$
(g) $R[1 + \arccos(1/\sigma)]$ (h) $\sigma < \dfrac{1}{2}\left(1 + \dfrac{\pi}{2}\right)$

27.75 (a) $-IA\hat{\boldsymbol{k}}$ (b) $B_x = \dfrac{3D}{IA}, B_y = \dfrac{4D}{IA}, B_z = -\dfrac{12D}{IA}$

27.77 (b) 1.85×10^{-28} kg (c) 1.20 kV (d) 8.32×10^5 m/s

27.79 (a) $r = R\sin\theta$ (b) $dI = \sigma\omega R^2\sin\theta d\theta$
(c) $d\mu = \pi R^4\sigma\omega\sin^3\theta d\theta$ (d) $\dfrac{Q\omega R^2}{3}\hat{\boldsymbol{k}}$
(e) $\dfrac{Q\omega BR^2}{3}(\sin\alpha\hat{\boldsymbol{\jmath}} - \cos\alpha\hat{\boldsymbol{\imath}})B$

27.81 (a) 5.14 m (b) 1.72 μs (c) 6.08 mm (d) 3.05 cm

27.83 Choice (c)

27.85 Choice (a)

Chapter 28

28.1 (a) $-(19.2\ \mu\text{T})\hat{\boldsymbol{k}}$ (b) 0 (c) $(19.2\ \mu\text{T})\hat{\boldsymbol{\imath}}$ (d) $(6.79\ \mu\text{T})\hat{\boldsymbol{\imath}}$

28.3 (a) 60.0 nT, out of the page at A and B (b) 0.120 μT, out of the page (c) 0

28.5 (a) 0 (b) $-(1.31\ \mu\text{T})\hat{\boldsymbol{k}}$ (c) $-(0.462\ \mu\text{T})\hat{\boldsymbol{k}}$ (d) $(1.31\ \mu\text{T})\hat{\boldsymbol{\jmath}}$

28.7 (a) left (b) 0 (c) 1.21 pT, out of the page (d) 1.21 pT, into the page

28.9 (a) 0.440 μT, out of the page (b) 16.7 nT, out of the page (c) 0

28.11 (a) $(50.0\text{ pT})\hat{\boldsymbol{\jmath}}$ (b) $-(50.0\text{ pT})\hat{\boldsymbol{\imath}}$ (c) $-(17.7\text{ pT})(\hat{\boldsymbol{\imath}} - \hat{\boldsymbol{\jmath}})$ (d) 0

28.13 17.6 μT, into the page

28.15 (a) 0.8 mT (b) 40 μT (20 times larger)

28.17 25 μA

28.19 (a) $-(0.10\ \mu\text{T})\hat{\boldsymbol{\imath}}$ (b) 2.19 μT, at 46.8° from the $+x$-axis to the $+z$-axis (c) $(7.9\ \mu\text{T})\hat{\boldsymbol{\imath}}$

28.21 (a) 0 (b) 6.67 μT, toward the top of the page (c) 7.54 μT, to the left

28.23 (a) 0 (b) 0 (c) 0.40 mT, to the left

28.25 (a) P: 41 μT, into the page; Q: 25 μT, out of the page (b) P: 9.0 μT, out of the page; Q: 9.0 μT, into the page

28.27 $\dfrac{\mu_0 I^2}{2\pi g\lambda}$

28.29 (a) 6.00 μN; repulsive (b) 24.0 μN

28.31 0.38 μA

28.33 $\dfrac{\mu_0|I_1 - I_2|}{4R}$; 0

28.35 18.0 A, counterclockwise

28.37 (a) 305 A (b) -3.83×10^{-4} T · m

28.39 (a) $\mu_0 I/2\pi r$ (b) 0

28.41 (a) $\dfrac{\mu_0 I_1}{2\pi r}$ (b) $\dfrac{\mu_0(I_1 + I_2)}{2\pi r}$

28.43 (a) 1790 turns per meter (b) 63.0 m

28.45 (a) 3.72 MA (b) 124 kA (c) 237 A

28.47 (a) (i) 1.13 mT (ii) 4.68 MA/m (iii) 5.88 T

28.49 (a) 1.00 μT, into the page (b) $(74.9\text{ nN})\hat{\boldsymbol{\jmath}}$

28.51 0.200 m

28.53 (a) $\mu_0 I^2 R/d$ (b) $\dfrac{4B^2}{\mu_0 d}\left(\dfrac{A}{\pi}\right)^{3/2}$ (c) $\dfrac{1}{2}\sqrt{\mu_0 Fd}\left(\dfrac{\pi}{A}\right)^{3/4}$

28.55 (a) 5.7×10^{12} m/s^2, away from the wire (b) 32.5 N/C, away from the wire (c) no

28.59 (a) 2.00 A, out of the page (b) 2.13 μT, to the right (c) 2.06 μT

28.61 23.2 A

28.63 (a) $I_0 = 2\pi b\delta\,(1 - e^{-a/\delta})$, 81.5 A (b) $\dfrac{\mu_0 I_0}{2\pi r}$
(c) $\left(\dfrac{e^{r/\delta} - 1}{e^{a/\delta} - 1}\right)I_0$ (d) $\dfrac{\mu_0 I}{2\pi r}\left(\dfrac{e^{r/\delta} - 1}{e^{a/\delta} - 1}\right)$
(e) $r = \delta$: 175 μT;
$r = a$: 326 μT;
$r = 2a$: 163 μT

28.65 (a) $\dfrac{3I}{2\pi R^3}$ (b) (i) $B = \dfrac{\mu_0 I r^2}{2\pi R^3}$ (ii) $B = \dfrac{\mu_0 I}{2\pi r}$

28.67 (b) $B = \dfrac{\mu_0 I_0}{2\pi r}$ (c) $\dfrac{I_0 r^2}{a^2}\left(2 - \dfrac{r^2}{a^2}\right)$
(d) $B = \dfrac{\mu_0 I_0 r}{2\pi a^2}\left(2 - \dfrac{r^2}{a^2}\right)$

28.69 (a) $B = \mu_0 In/2$, $+x$-direction (b) $B = \mu_0 In/2$, $-x$-direction

28.71 (a) $\dfrac{Q_1\omega_1}{2\pi}$ (b) $\dfrac{\mu_0\omega_1 Q_1}{2\pi H}\hat{\boldsymbol{k}}$
(c) $\dfrac{\mu_0 Q_1 Q_2\omega_1\omega_2 R_2^2}{8\pi H}\sin\theta$
(d) $\dfrac{1}{2}MR_2^2\omega_2$ (e) $\dfrac{\mu_0}{4\pi H}\dfrac{Q_1 Q_2\omega_1\sin\theta}{M}$

28.73 (a) no (c) 64.9 A, 1.19 cm

28.75 (a) nq (b) $n^{2/3}q$ (c) $\dfrac{1}{2}\mu_0 nqvR\,\hat{\boldsymbol{\phi}}$.
(d) $n^{2/3}qvRd\phi$ (e) $\dfrac{\mu_0}{2}n^{5/3}q^2v^2R^2Ld\phi$
(f) $\pi\mu_0 n^{5/3}q^2v^2R^2L, \dfrac{\mu_0}{2}n^{5/3}q^2v^2R$
(g) 1.9×10^{-5} Pa

28.77 (a) $\dfrac{Q\omega}{4\pi}\sin\theta d\theta$ (b) $\dfrac{1}{4}Q\omega R^2\sin^3\theta\, d\theta$, $+z$-direction
(c) $\dfrac{Q\omega R^2}{3}$, $+z$-direction
(d) $\dfrac{2}{3}MR^2\omega$ (e) $\dfrac{Q}{2M}$ (f) 1

28.79 (b) $\dfrac{1}{2g}\left(\dfrac{\mu_0 Q_0^2}{4\pi\lambda RCd}\right)^2$

28.81 Choice (b)

28.83 Choice (c)

Chapter 29

29.1 (a) 17.1 mV (b) 28.5 mA

29.3 (a) Wb/s, Wb/s^3 (b) $\alpha = (0.750\ \text{s}^2)\beta$ (c) -1.20 V

29.5 (a) 34 V (b) counterclockwise

29.7 (a) $\mu_0 i/2\pi r$, into the page (b) $\dfrac{\mu_0 i}{2\pi r}Ldr$ (c) $\dfrac{\mu_0 iL}{2\pi}\ln(b/a)$ (d) $\dfrac{\mu_0 L}{2\pi}\ln(b/a)\dfrac{di}{dt}$ (e) 0.506 μV

29.9 (a) 5.44 mV (b) clockwise

29.11 (a) 12.6 mA (b) 626 μA

29.13 10.4 rad/s

29.15 (a) counterclockwise (b) clockwise (c) no induced current

29.17 (a) C: counterclockwise; A: clockwise (b) toward the wire

29.19 (a) a to b (b) b to a (c) b to a

29.21 (a) clockwise (b) no induced current (c) counterclockwise

29.23 (a) 0.72 V (b) 0

29.25 (a) 0.675 V (b) b (c) 2.25 V/m, b to a (d) b (e) (i) 0 (ii) 0

29.27 (a) 3.00 V (b) b to a (c) 0.800 N, to the right (d) 6.00 W for each

29.29 (a) counterclockwise (b) 42.4 mW

29.31 35.0 m/s, to the right

29.33 (a) 0.225 A, clockwise (b) 0 (c) 0.225 A, counterclockwise

29.35 9.21 A/s

29.37 0.950 mV

29.39 (a) 0.599 nC (b) 6.00 mA (c) 6.00 mA

29.41 (a) 2.2 nA (b) 0.75 s

29.43 (a) 2.5×10^5 C (b) 65 A (c) 11 T

29.45 (a) 3.7 A (b) 1.33 mA (c) counterclockwise

29.47 16.2 μV

29.49 (a) $\dfrac{\mu_0 Iabv}{2\pi r(r+a)}$ (b) clockwise

29.51 (a) 17.9 mV (b) a to b

29.53 $\mu_0 IW/4\pi$

29.55 (a) $\dfrac{\mu_0 Iv}{2\pi}\ln(1 + L/d)$ (b) a (c) 0

29.57 (a) 0.165 V (b) 0.165 V (c) 0; 0.0412 V

29.59 (a) B^2L^2v/R

29.61 a: $\dfrac{qr}{2}\dfrac{dB}{dt}$, to the left; b: $\dfrac{qr}{2}\dfrac{dB}{dt}$, toward the top of the page; c: 0

29.63 5.0 s

29.65 (a) 0.3071 s^{-1} (b) 3.69 T (c) a (d) 2.26 s

29.67 (a) $\vec{v} - (1.26\ \text{m/s})\sin\theta\,\hat{\imath}$ (b) $-1.26\sin\theta\ \text{T}\cdot\text{m/s}\,\hat{\jmath}$ (c) $(10.0\ \text{cm})(\cos\theta\,\hat{\jmath} - \sin\theta\,\hat{k})d\theta$ (d) 4.73 A (e) upward

29.69 (a) a to b (b) $\dfrac{Rmg\tan\phi}{L^2B^2\cos\phi}$ (c) $\dfrac{mg\tan\phi}{LB}$ (d) $\dfrac{Rm^2g^2\tan^2\phi}{L^2B^2}$ (e) $\dfrac{Rm^2g^2\tan^2\phi}{L^2B^2}$; same

29.71 Choice (c)

29.73 Choice (a)

Chapter 30

30.1 (a) 0.270 V; yes (b) 0.270 V

30.3 (a) 1.96 H (b) 7.11 mWb

30.5 (a) 1940 (b) 800 A/s

30.7 (a) 0.250 H (b) 0.450 mWb

30.9 (a) 4.68 mV (b) a

30.11 (b) 0.111 μH

30.13 0.0300 T

30.15 2850

30.17 (a) 0.161 T (b) 10.3 kJ/m^3 (c) 0.129 J (d) 40.2 μH

30.19 91.7 J

30.21 (a) 2.40 A/s (b) 0.800 A/s (c) 0.413 A (d) 0.750 A

30.23 (a) 17.3 μs (b) 30.7 μs

30.25 15.4 V

30.27 (a) 0.250 A (b) 0.137 A (c) 32.9 V; c (d) 0.462 ms

30.29 15.3 V

30.31 (a) 0 (b) 0 (c) $\mathcal{E}^2/4R$

30.33 (a) 443 nC (b) 358 nC

30.35 (a) 105 rad/s, 59.6 ms (b) 0.720 mC (c) 4.32 mJ (d) -0.542 mC (e) -0.050 A, counterclockwise (f) $U_C = 2.45$ mJ, $U_L = 1.87$ mJ

30.37 (a) 7.50 μC (b) 15.9 kHz (c) 21.2 mJ

30.39 (a) 298 rad/s (b) 83.8 Ω

30.43 (a) 59 NH (b) 26.8 μV

30.45 20 km/s; about 30 times smaller

30.47 (a) 5.00 H (b) 31.7 m; no

30.49 222 μF, 9.31 μH

30.51 (a) 111 N (b) 37.1 ms (c) 4.00×10^3 rad/s^2

30.53 (a) 15.0 mC (b) 93.8 μS (c) 2.67 kHz (d) 11.3 J (e) 1.30 ms

30.55 (a) 24.0 mV (b) 1.55 mA (c) 72.1 nJ (d) 5.20 μC, 18.0 nJ

30.57 (a) 0, 20.0 V (b) 0.267 A, 0 (c) 0.147 A, 9.0 V

30.59 (b) 5.0 Ω, 8.5 H (c) 1.7 kJ; 2.0 kW

30.61 (a) 60.0 V (b) a (c) 60.0 V (d) c (e) -96.0 V (f) b (g) 156 V (h) d

30.63 (a) 0; $v_{ac} = 0$, $v_{cb} = 36.0$ V (b) 0.180 A, $v_{ac} = 9.0$ V, $v_{cb} = 27.0$ V (c) $i_0 = (0.180\ \text{A})(1 - e^{-t/(0.020\ \text{s})})$, $v_{ac} = (9.0\ \text{V})(1 - e^{-t/(0.020\ \text{s})})$, $v_{cb} = (9.0\ \text{V})(3.00 + e^{-t/(0.020\ \text{s})})$

30.65 (a) $A_1 = A_4 = 0.455$ A, $A_2 = A_3 = 0$ (b) $A_1 = 0.585$ A, $A_2 = 0.320$ A, $A_3 = 0.160$ A, $A_4 = 0.107$ A

30.67 (a) $v_L = \dfrac{\mathcal{E}}{R + R_L}(R_L + Re^{-(R+R_L)t/L})$ (b) 50.0 V (c) 30.0 V; 3.00 A (d) 6.67 Ω (e) 40.0 mH

30.69 (a) $\dfrac{\mu_0 I(N_1 - N_2)\pi a^2}{\lambda}$ (b) $\dfrac{\mu_0 I\pi}{\lambda}(N_1 b^2 - N_2 a^2)$ (c) $\dfrac{\mu_0\pi}{\lambda}(N_1^2 b^2 - N_2^2 a^2)$ (d) $\dfrac{\mu_0\pi}{\lambda}(N_1^2 b^2 + 2N_1N_2a^2 + N_2^2a^2)$ (e) 10.3 mH (f) 16.0 mH

30.71 (a) $i_1 = \dfrac{\mathcal{E}}{R_1}(1 - e^{-R_1t/L})$, $i_2 = \dfrac{\mathcal{E}}{R_2}e^{-t/R_2C}$, $q_2 = \mathcal{E}C(1 - e^{-t/R_2C})$ (b) $i_1 = 0$, $i_2 = 9.60$ mA (c) $i_1 = 1.92$ A, $i_2 = 0$; $t \gg L/R_1$ and $t \gg R_2C$ (d) 1.6 ms (e) 9.4 mA (f) 0.22 s

30.73 Choice (b)

30.75 Choice (c)

Chapter 31

31.1 1.06 A

31.3 (a) 90°; lead (b) 193 Hz

31.5 (a) 1510 Ω (b) 0.239 H (c) 497 Ω (d) 16.6 μF

31.7 (a) 1700 Ω (b) 106 Ω (c) 150 Hz

31.9 (a) $(12.5\ \text{V})\cos[(480\ \text{rad/s})t]$ (b) 7.17 V

31.11 (a) $i = (0.0253\ \text{A})\cos[(720\ \text{rad/s})t]$ (b) 180 Ω (c) $v_L = -(4.56\ \text{V})\sin[(720\ \text{rad/s})t]$

31.13 (a) 601 Ω (b) 49.9 mA (c) $-70.6°$; lag (d) $V_R = 9.98$ V, $V_L = 4.99$ V, $V_C = 33.3$ V

31.15 50.0 V

31.17 32.8°

31.19 (a) 40.0 W (b) 0.167 A (c) 720 Ω

31.21 (b) 32.4 Ω

31.23 (a) 45.8°, 0.697 (b) 344 Ω (c) 155 V (d) 48.6 W (e) 48.6 W (f) 0 (g) 0

31.25 (a) 0.302 (b) 0.370 W (c) 0.370 W (resistor), 0, 0

31.27 (a) 113 Hz; 15.0 mA (b) 7.61 mA; lag

31.29 (a) 150 V (b) $V_R = 150$ V, $V_L = V_C = 1290$ V (c) 37.5 W

31.31 (a) 1.00 (b) 75.0 W (c) 75.0 W

31.33 (a) 945 rad/s (b) 70.6 Ω (c) $V_L = V_C = 450$ V, $V_R = 120$ V

31.35 (a) 10 (b) 2.40 A (c) 28.8 W (d) 500 Ω

31.37 0.124 H

31.39 230 Ω

31.41 3.59×10^7 rad/s

31.43 (a) inductor (b) 0.133 H

31.45 (a) 0.831 (b) 161 W

31.47 $\dfrac{V_{\text{out}}}{V_s} = \sqrt{\dfrac{R^2 + \omega^2L^2}{R^2 + \left(\omega L - \dfrac{1}{\omega C}\right)^2}}$

31.51 (a) 102 Ω (b) 0.882 A (c) 270 V

31.53 (a) $C = 5\epsilon_0 a^2\theta/g$ (b) 131° (c) 7.41 V (d) 1230 kHz (e) 7.74 V

31.55 (b) 5770 rad/s (c) 2.40 A (d) 2.40 A (e) 0.139 A (f) 0.139 A

31.57 (a) $\omega = 28{,}800$ rad/s so $\phi = 60°$ (b) $P_R = 0.375$ W, $P_L = P_C = 0$; 0.100 A

31.59 (a) 0.750 A (b) 160 Ω (c) 341 Ω, 619 Ω (d) 341 Ω

31.61 (a) $\dfrac{V}{R}$ (b) $\dfrac{V}{R}\sqrt{\dfrac{L}{C}}$ (c) $\dfrac{V}{R}\sqrt{\dfrac{L}{C}}$ (d) $\frac{1}{2}L\dfrac{V^2}{R^2}$ (e) $\frac{1}{2}L\dfrac{V^2}{R^2}$

31.63 20.0 Ω, 0.180 H

31.65 (a) $\dfrac{V_{\text{in}}}{\sqrt{R^2 + (\omega L)^2}}$ (b) $\arctan\left(\dfrac{\omega L}{R}\right)$ (c) $\dfrac{1}{\sqrt{1 + (R/\omega L)^2}}$ (d) $\dfrac{R}{2.00\pi L}$ (e) 1.6 mH

31.67 (a) $\frac{1}{2}V_R I$ (b) 0 (c) 0

31.69 Choice (b)

31.71 Choice (d)

Chapter 32

32.1 (a) 1.28 s (b) 8.15×10^{13} km

32.3 13.3 nT, $+y$-direction

32.5 3.0×10^{18} Hz, 3.3×10^{-19} s, 6.3×10^{10} rad/s

32.7 (a) (i) 60 kHz (ii) 6.0×10^{13} Hz (iii) 6.0×10^{16} Hz (b) (i) 4.62×10^{-14} m $= 4.62 \times 10^{-5}$ nm (ii) 508 m $= 5.08 \times 10^{11}$ nm

32.9 (a) $+y$-direction (b) 0.149 mm (c) $\vec{B} = (1.03\ \text{mT})\cos[(4.22 \times 10^4\ \text{rad/m})y - (1.265 \times 10^{13}\ \text{rad/s})t]\,\hat{\imath}$

32.11 (a) 361 m (b) 0.0174 rad/m (c) 5.22×10^6 rad/s (d) 0.0144 V/m

32.13 (a) 0.381 μm (b) 0.526 μm (c) 1.38 (d) 1.90

32.15 (a) 330 W/m^2 (b) 500 V/m, 1.7 μT

32.17 2.5×10^{25} W

32.19 93.8 W

32.21 12.0 V/m, 40.0 nT

32.23 (a) 0.18 mW (b) 274 V/m, 0.913 μT (c) 0.18 mJ/s (d) 0.010 W/cm^2

32.25 (a) 637 W/m^2 (b) 693 V/m, 2.31 μT
(c) 2.12 $\mu J/m^3$
32.27 (a) 30.5 cm (b) 2.46 GHz (c) 2.11 GHz
32.29 7.50 cm
32.31 (a) 0.375 mJ (b) 4.08 mPa
(c) 604 nm, 3.70×10^{14} Hz
(d) 30.3 kV/m, 101 μT
32.33 (a) 6.02×10^{-3} W/m^2 (b) 2.13 V/m, 7.10 nT
(c) 1.20 pN; no
32.35 (a) at $r = R$: 64 MW/m^2, 0.21 Pa; at $r = R/2$: 260 MW/m^2, 0.85 Pa (b) no
32.37 3.89×10^{-13} rad/s^2
32.39 (a) $\rho I/\pi a^2$, in the direction of the current
(b) $\mu_0 I/2\pi a$, counterclockwise if the current is out of the page
(c) $\dfrac{\rho I^2}{2\pi^2 a^3}$, radially inward
(d) $\dfrac{\rho l I^2}{\pi a^2} = I^2 R$
32.41 (a) 1.364 m (b) 10.91 m
32.43 (a) 9.75×10^{-15} W/m^2
(b) 2.71 $\mu V/m$, 9.03×10^{-15} T, 67.3 ms
(c) 3.25×10^{-23} Pa (d) 0.190 m
32.45 (a) $\dfrac{4\rho G\pi MR^3}{3r^2}$ (b) $\dfrac{LR^2}{4cr^2}$ (c) 0.19 μm; no
32.47 (b) 3.00×10^8 m/s
32.49 (a) $\dfrac{I_0}{c}(2 - e)$ (b) 3.3×10^{-16} N (c) 0.55
32.51 (b) 1.39×10^{-11} (c) 2.54×10^{-8}
32.53 (c) 66.0 μm
32.55 Choice (d)

Chapter 33

33.1 39.4°
33.3 (a) 2.04×10^8 m/s (b) 442 nm
33.5 2.11×10^8 m/s, 417 nm, 5.07×10^{14} Hz
33.7 (a) 47.5° (b) 66.0°
33.9 2.51×10^8 m/s
33.11 (a) 2.34 (b) 82°
33.13 38.5°
33.15 (a) 51.3° (b) 33.6°
33.17 (a) 58.1° (b) 22.8°
33.19 1.77
33.21 (a) 48.9° (b) 28.7°
33.23 0.6°
33.25 (a) red: 1.36, violet: 1.40 (b) red: 2.21×10^8 m/s, violet: 2.14×10^8 m/s
33.27 $0.375I_0$
33.29 (a) A: $I_0/2$, B: $0.125I_0$, C: $0.0938I_0$ (b) 0
33.31 (a) 1.40 (b) 35.5°
33.33 6.38 W/m^2
33.35 (a) $0.374I$ (b) $2.35I$
33.37 (a) 46.7° (b) 13.4°
33.39 72.1°
33.41 1.28
33.43 3.52×10^4
33.45 1.84
33.47 (a) 48.6° (b) 48.6°
33.49 39.1°
33.51 (b) 0.23°; about the same
33.53 23.3°
33.55 (a) A: 1.46, carbon tetrachloride;
B: 1.33, water;
C: 1.63, carbon disulfide;
D: 1.50, benzene
(b) A: 2.13, B: 1.77, C: 2.66, D: 2.25
(c) all: 5.09×10^{14} Hz
33.57 (a) 35° (b) $I_0 = 10.1$ W/m^2, $I_p = 19.9$ W/m^2
33.59 (a) $\theta = \arcsin\left(\dfrac{2R - d}{2R + d}\right)$ (b) $R = \dfrac{d(n_1 + n_2)}{2(n_1 - n_2)}$
(c) 2.07 cm (d) ≈50 cm (e) 1.49 μs
33.61 (a) $\Delta = 2\theta_a^A - 6\arcsin\left(\dfrac{\sin\theta_a^A}{n}\right) + 2\pi$
(b) $\theta_2 = \arccos\sqrt{\dfrac{n^2 - 1}{8}}$
(c) violet: $\theta_2 = 71.55°$, $\Delta = 233.2°$
red: $\theta_2 = 71.94°$, $\Delta = 230.1°$; violet
33.63 Choice (d)

Chapter 34

34.1 39.2 cm to the right of the mirror, 4.85 cm
34.3 9.0 cm; tip of the lead
34.5 (b) 33.0 cm to the left of the vertex, 1.20 cm, inverted, real
34.7 0.213 mm
34.9 18.0 cm from the vertex; 0.50 cm, erect, virtual
34.11 (a) +4.00 (b) 48.0 cm to the right of the mirror; virtual
34.13 (a) concave (b) $f = 2.50$ cm, $R = 5.00$ cm
34.15 (a) 10.0 cm to the left of the shell vertex, 2.20 mm
(b) 4.29 cm to the right of the shell vertex, 0.944 mm
34.19 2.67 cm
34.21 3.30 m
34.23 (a) at the center of the bowl, 1.33 (b) no
34.25 39.5 cm
34.27 (a) 30 cm, 9.60 mm, inverted (b) 60.0 cm, 9.60 mm, upright
34.29 (a) 107 cm to the right of the lens, 17.8 mm; real; inverted (b) the same
34.31 71.2 cm to the right of the lens; −2.97
34.33 3.69 cm; 2.82 cm to the left of the lens
34.35 1.67
34.37 12.0 mm
34.39 −12 cm
34.41 26.3 cm from the lens, 12.4 mm; erect; same side
34.43 (a) 200 cm to the right of the first lens, 4.80 cm
(b) 150 cm to the right of the second lens, 7.20 cm
34.45 (a) 53.0 cm (b) real (c) 2.50 mm; inverted
34.47 (a) $s' = -2f$, virtual, left (b) $3h$, upright (c) yes
34.49 8.69 cm; no
34.51 (a) 12 cm to the right of the converging lens
(b) −8 cm (c) 24 cm to the right of the diverging lens (d) 12 cm to the right of the converging lens, –4 cm, 6 cm to the right of the diverging lens
34.53 (a) 80.0 cm (b) 76.9 cm
34.55 49.4 cm, 2.02 diopters
34.57 (a) far point: 202 cm, near point: 44.6 cm
(b) 56.1 cm
34.59 (a) 6.06 cm (b) 4.12 mm
34.61 (a) −6.33 (b) 1.90 cm (c) 0.127 rad
34.63 (a) 0.661 m (b) 59.1
34.65 16.0 mm
34.67 (a) 20.0 cm (b) 39.0 cm
34.69 51 m/s
34.71 (a) 1.49 cm
34.73 (b) 2.4 cm; −0.133
34.75 2.00
34.77 (a) converging, 52.5 cm from the lens
(b) converging, 17.5 cm from the lens
34.79 converging, +50.2 cm
34.81 (a) 58.7 cm, converging (b) 4.48 mm; virtual
34.83 (a) 6.48 mm (b) no, behind the retina
(c) 19.3 mm from the cornea; in front of the retina
34.85 (a) 6.49 cm (b) 6.76 cm $\le s \le$ 7.05 cm
(c) 66.2 cm (d) 1.36 m/s, downward (e) 24.1 cm
34.87 (a) 30.9 cm (b) 29.2 cm
34.89 (b) first image: (i) 51.3 cm to the right of the lens
(ii) real (iii) inverted
second image: (i) 51.3 cm to the right of the lens
(ii) real (iii) erect
34.91 −26.7 cm
34.93 7.06 cm to the left of the spherical mirror vertex, 0.177 cm tall; 13.3 cm to the left of the spherical mirror vertex, 0.111 cm tall
34.95 134 cm to the left of the object
34.97 4.17 diopters
34.99 (d) 36.0 cm, 21.6 cm; $d = 1.2$ cm
34.101 (a) −16.6 cm (b) 20.0 cm to the right
34.103 (b) 10.0 cm (c) 90 cm in front of the glass, virtual
34.105 (a) $4f$
34.107 (b) 1.74 cm
34.109 Choice (d)
34.111 Choice (b)

Chapter 35

35.1 0.75 m, 2.00 m, 3.25 m, 4.50 m, 5.75 m, 7.00 m, 8.25 m
35.3 (a) 2.0 m (b) constructively
(c) 1.0 m, destructively
35.5 (a) 11 (b) 2.50 m
35.7 1.14 mm
35.9 0.83 mm
35.11 (a) 39 (b) ±73.3°
35.13 12.6 cm
35.15 1200 nm
35.17 (a) $0.750I_0$ (b) 80 nm
35.19 1670 rad
35.21 (a) 4.52 rad (b) $0.404I_0$
35.23 114 nm
35.25 (a) 55.6 nm (b) (i) 2180 nm
(ii) 11.0 wavelengths
35.27 (a) 514 nm; green (b) 603 nm; orange
35.29 0.11 μm
35.31 0.570 mm
35.33 1.57
35.35 (a) 96.0 nm (b) no, no
35.37 (a) 1.58 mm (green), 1.72 mm (orange)
(b) 3.45 mm (violet), 4.74 mm (green), 5.16 mm (orange) (c) 9.57 μm
35.39 1.730
35.41 761 m, 219 m, 90.1 m, 20.0 m
35.43 $6.8 \times 10^{-5}(C°)^{-1}$
35.45 1.33 μm
35.47 600 nm, 467 nm; no
35.49 (a) 1.54 (b) ±15.0°
35.51 (a) 50 MHz (b) 237.0 m
35.53 (a) $\lambda/2$ (b) 6
(c) ±9.64 cm, ±31.7 cm, ±84.1 cm
(d) 2, ±11.3 cm from the center
35.55 14.0
35.57 Choice (d)
35.59 Choice (c)

Chapter 36

36.1 506 nm
36.3 (a) 226 (b) ±83.0°
36.5 9.07 m
36.7 (a) 63.8 cm
(b) ±22.1°, ±34.3°, ±48.8°, ±70.1°
36.9 ±16.0°, ±33.4°, ±55.6°
36.11 (a) 10.9 mm (b) 5.4 mm
36.13 (a) 580 nm (b) 0.128

36.15 (a) $\pm 13.0°, \pm 26.7°, \pm 42.4°, \pm 64.1°$
(b) 2.08 W/m^2
36.17 (a) $d = 2a$ (b) 0.101 W/m^2
36.19 (a) 3 (b) 2
36.21 (a) 0.0627°, 0.125° (b) $0.249I_0$, $0.0256I_0$
36.23 (a) 3.00 mm (b) 6.00 mm
36.25 (a) 4830 lines/cm (b) 4; $\pm 37.7°, \pm 66.5°$
36.27 (a) 2.40 μm (b) 417 slits/mm (c) 1.80 μm
36.29 (a) 2.69×10^4 (b) $m = 3$
36.31 (a) 467 nm (b) 27.8°
36.33 (a) 1820 slits (b) 41.0297°, 0.0137°
36.35 0.232 nm
36.37 92 cm
36.39 1.88 m
36.41 220 m
36.43 (a) 77 m (Hubble), 1100 km (Arecibo)
(b) 1500 km
36.45 30.2 μm
36.47 (a) 78 (b) $\pm 80.8°$ (c) 555 $\mu W/m^2$
36.49 1.68
36.51 (b) 4.49 rad, 7.73 rad (c) 3.14 rad, 6.28 rad, 9.42 rad; no (d) 4.78°, 6.84°, 9.59°
36.53 −0.033 mm; decrease
36.55 360 nm
36.57 second
36.59 (a) 65° (b) 0.16 nm (c) 4.4×10^6 m/s
(d) 2.6×10^{16} Hz (e) 3.3×10^{-34} J·s
(f) 6.6×10^{-34} J·s (g) yes
36.61 (a) 1.03 mm (b) 0.148 mm
36.63 (a) 12.1 μm (b) 10.4 cm, 15.2 cm
36.65 (a) 10.6 μm (b) 532 nm/s (c) 1.60 μm
(d) 1.11 s (e) 92.7% (f) 2.88°
36.69 Choice (d)
36.71 Choice (a)

Chapter 37

37.1 bolt A
37.3 $0.867c$; no
37.5 (a) $0.998c$ (b) 126 m
37.7 (a) 12.0 ms measured by the first officer
(b) $0.997c$
37.9 92.5 m
37.11 (a) 0.66 km (b) 49 μs; 15 km (c) 0.45 km
37.13 (a) 3570 m (b) 90.0 μs (c) 89.2 μs
37.15 (a) $0.806c$ (b) $0.974c$ (c) $0.997c$
37.17 (a) toward (b) $0.385c$
37.19 $0.784c$
37.21 $0.611c$
37.23 (a) $0.159c$ (b) \$172 million
37.25 (a) 2.30 (b) 4.11
37.27 $3.06p_0$
37.29 (a) $0.866c$ (b) $0.608c$
37.31 (a) $0.866c$ (b) $0.986c$
37.33 (a) 0.450 nJ (b) 1.94×10^{-18} kg·m/s
(c) $0.968c$
37.35 (a) 0.867 nJ (b) 0.270 nJ (c) 0.452
37.37 (a) 5.34 pJ (nonrel), 5.65 pJ (rel), 1.06
(b) 67.8 pJ (nonrel), 331 pJ (rel), 4.88
37.39 (a) 2.06 MV (b) 0.330 pJ = 2.06 MeV
37.41 $0.700c$
37.43 42.5 y
37.45 (a) 1.0 kg (b) A: 1.7 kg, B: 1.8 kg
37.47 (a) $\Delta = 9 \times 10^{-9}$ (b) $7000m$
37.49 5.01 ns, clock on plane
37.51 0.168 MeV
37.53 (a) 1.08×10^{14} J (b) 2.70×10^{19} W
(c) 1.10×10^{10} kg
37.55 (a) $IHL\hat{\boldsymbol{j}}$
(b) $IHLB\hat{\boldsymbol{i}}$
(c) $\vec{\mu}\,' = \vec{\mu}/\gamma$
(d) $\vec{B}\,' = B\gamma\hat{\boldsymbol{k}}$ (e) $\vec{B}\,'_\perp = \vec{B}_\perp\gamma$
37.57 $0.357c$; receding
37.59 154 km/h
37.61 2.04×10^{-13} N
37.63 (a) 2.6×10^{-8} s (b) 0.97
37.65 (a) 2.0×10^{-18} kg (b) 4.0×10^4 m/s^2
37.67 (a) (0, 0), $(-L\gamma, Lv\gamma/c^2)$, $(-vT\gamma, T\gamma)$,
$((-L + vT)\gamma, (T + Lv/c^2)\gamma)$
(c) cLT (d) cLT (e) yes
37.69 (a) 2494 MeV (b) 2.526 times
(c) 987.4 MeV, twice as much
37.71 Choice (c)
37.73 Choice (b)

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Astronomical Data†

Body	Mass (kg)	Radius (m)	Orbit radius (m)	Orbital period
Sun	1.99×10^{30}	6.96×10^{8}	—	—
Moon	7.35×10^{22}	1.74×10^{6}	3.84×10^{8}	27.3 d
Mercury	3.30×10^{23}	2.44×10^{6}	5.79×10^{10}	88.0 d
Venus	4.87×10^{24}	6.05×10^{6}	1.08×10^{11}	224.7 d
Earth	5.97×10^{24}	6.37×10^{6}	1.50×10^{11}	365.3 d
Mars	6.42×10^{23}	3.39×10^{6}	2.28×10^{11}	687.0 d
Jupiter	1.90×10^{27}	6.99×10^{7}	7.78×10^{11}	11.86 y
Saturn	5.68×10^{26}	5.82×10^{7}	1.43×10^{12}	29.45 y
Uranus	8.68×10^{25}	2.54×10^{7}	2.87×10^{12}	84.02 y
Neptune	1.02×10^{26}	2.46×10^{7}	4.50×10^{12}	164.8 y
Pluto‡	1.30×10^{22}	1.19×10^{6}	5.91×10^{12}	248.0 y

†Source: NASA (**http://solarsystem.nasa.gov/planets/**). For each body, "radius" is its average radius and "orbit radius" is its average distance from the sun or (for the moon) from the earth.
‡In August 2006, the International Astronomical Union reclassified Pluto and similar small objects that orbit the sun as "dwarf planets."

Prefixes for Powers of 10

Power of ten	Prefix	Abbreviation	Pronunciation
10^{-24}	yocto-	y	*yoc*-toe
10^{-21}	zepto-	z	*zep*-toe
10^{-18}	atto-	a	*at*-toe
10^{-15}	femto-	f	*fem*-toe
10^{-12}	pico-	p	*pee*-koe
10^{-9}	nano-	n	*nan*-oe
10^{-6}	micro-	μ	*my*-crow
10^{-3}	milli-	m	*mil*-i
10^{-2}	centi-	c	*cen*-ti
10^{3}	kilo-	k	*kil*-oe
10^{6}	mega-	M	*meg*-a
10^{9}	giga-	G	*jig*-a or *gig*-a
10^{12}	tera-	T	*ter*-a
10^{15}	peta-	P	*pet*-a
10^{18}	exa-	E	*ex*-a
10^{21}	zetta-	Z	*zet*-a
10^{24}	yotta-	Y	*yot*-a

Examples:

1 femtometer $=$ 1 fm $= 10^{-15}$ m

1 picosecond $=$ 1 ps $= 10^{-12}$ s

1 nanocoulomb $=$ 1 nC $= 10^{-9}$ C

1 microkelvin $=$ 1 μK $= 10^{-6}$ K

1 millivolt $=$ 1 mV $= 10^{-3}$ V

1 kilopascal $=$ 1 kPa $= 10^{3}$ Pa

1 megawatt $=$ 1 MW $= 10^{6}$ W

1 gigahertz $=$ 1 GHz $= 10^{9}$ Hz

Fundamental Physical Constants*

Name	Symbol	Value
Speed of light in vacuum	c	2.99792458×10^{8} m/s
Magnitude of charge of electron	e	$1.602176634 \times 10^{-19}$ C
Gravitational constant	G	$6.6.67408(31) \times 10^{-11}\ \mathrm{N \cdot m^2/kg^2}$
Planck's constant	h	$6.62607015 \times 10^{-34}\ \mathrm{J \cdot s}$
Boltzmann constant	k	1.380649×10^{-23} J/k
Avogadro's number	N_A	$6.02214076 \times 10^{23}$ molecules/mol
Gas constant	R	$8.314462618\ldots\ \mathrm{J/mol \cdot K}$
Mass of electron	m_e	$9.10938356(11) \times 10^{-31}$ kg
Mass of proton	m_p	$1.672621898(21) \times 10^{-27}$ kg
Mass of neutron	m_n	$1.674927471(21) \times 10^{-27}$ kg
Magnetic constant	μ_0	$1.25663706 \times 10^{-6}\ \mathrm{Wb/A \cdot m}$ (approximate) $\cong 4\pi \times 10^{-7}\ \mathrm{Wb/A \cdot m}$
Electric constant	$\epsilon_0 = 1/\mu_0 c^2$	$8.854187817 \times 10^{-12}\ \mathrm{C^2/N \cdot m^2}$ (approximate)
	$1/4\pi\epsilon_0$	$8.987551787 \times 10^{9}\ \mathrm{N \cdot m^2/C^2}$ (approximate)

Other Useful Constants*

Mechanical equivalent of heat		4.186 J/cal (15° calorie)
Standard atmospheric pressure	1 atm	1.01325×10^{5} Pa
Absolute zero	0 K	−273.15°C
Electron volt	1 eV	$1.6021766209(98) \times 10^{-19}$ J
Atomic mass unit	1 u	$1.660539040(20) \times 10^{-27}$ kg
Electron rest energy	$m_e c^2$	0.5109989461(31) MeV
Volume of ideal gas (0°C and 1 atm)		22.413962(13) liter/mol
Acceleration due to gravity (standard)	g	$9.80665\ \mathrm{m/s^2}$

*Source: National Institute of Standards and Technology (**http://physics.nist.gov/cuu**). Numbers in parentheses show the uncertainty in the final digits of the main number; for example, the number 1.6454(21) means 1.6454 ± 0.0021. Values shown without uncertainties are exact. The exact values of the magnitude of the charge of the electron, Planck's constant, the Boltzmann constant, Avogadro's number, and the gas constant are from the redefinitions adopted in 2018. As consequences of these redefinitions, the values of the magnetic constant and electric constant now have fractional uncertainties of about 2×10^{-10}. As of this writing (2018) it was expected that updated values of the magnetic and electric constants, as well as of all other constants with uncertainties, were to be announced in May 2019. These updated values will be available at **http://physics.nist.gov/cuu**.